Seminar on Algebraic Geometry of Bois Marie

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Intersection Theory and the Riemann–Roch Theorem

(SGA 6)

a seminar directed by

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Preface

The present edition of *Séminaire de Géométrie Algébrique 6* reproduces, up to minor changes, the first edition, distributed by the I.H.E.S. in the form of mimeographed exposés. The corrections made concern essentially printing errors, except in Exposés I and III, where slight modifications were made by L. Illusie to a few statements which were incorrect in the primitive version. Footnotes have also been added, especially in Exposé XIV by A. Grothendieck, in order to point out recent progress on questions that were still open at the time of the seminar.

Finally, note that Exposé XI, by D. Ferrand, on determinant theory for perfect complexes, was not written up and therefore will not be published; to avoid errors in references, we have left unchanged the numbering of the following exposés.

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Introduction

The purpose of the present seminar is to develop a global theory of intersections, and a Riemann–Roch formula, for arbitrary schemes. The reader will find a description of the programme of the seminar in Exposé 0. The possibility in principle of proving a Riemann–Roch formula for general regular schemes, along the lines of the well-known Borel–Serre report of 1958, was already clear at that time, at least for a projective morphism. The extension to the general case is less evident; the programme in which it belongs, and which is partly carried out in this seminar, goes back to 1960. As was also the case for the duality theory of coherent sheaves, an essential tool for formulating a satisfactory theory is Verdier's theory of derived categories; familiarity with it is indispensable for understanding the seminar.

Apart from this theory, the study of the seminar requires little more than a general knowledge of the foundations of algebraic geometry as set out in EGA, chapters I, II, III. We shall only occasionally refer to EGA IV, for certain facts concerning flat or smooth morphisms, which for the most part also appear in the first exposés of SGA 1. Finally, to develop some results with the full generality desirable for applications, we sometimes use the notion of a ringed site and of a ringed topos, for which we refer to SGA 4, Exposés I to IV. A reader unfamiliar with the language of sites and topoi may everywhere replace these objects by ordinary topological spaces, the objects of the topos then being replaced by open subsets of these spaces; nevertheless, we recommend that the reader assimilate the language of topoi, which provides an extremely convenient unifying principle.

The fundamental notion for the theory presented here is that of a perfect complex of modules, together with its various variants, developed in Exposés I, II, III. It seems clear that these notions, also important in other contexts in algebraic geometry, notably for formulas of Lefschetz–Verdier type in cohomology with discrete coefficients (SGA 5) or coherent coefficients, will also have a role to play outside algebraic geometry, especially in the formulation of a complex analytic variant of the Riemann–Roch theorem presented here, or of suitable variants of the Atiyah–Singer index theorem. Some indications in this direction will be given in Exposé II. Unfortunately, at present there is still no statement, even heuristic, which would encompass these two theorems, the first of which remains conjectural for the moment. A new idea seems to be missing, just as one is also missing in algebraic geometry for proving Riemann–Roch beyond projective hypotheses (cf. Exposé XIV, no. 2).

We shall make no use in this seminar of the local theory of intersections, set out in J.-P. Serre, *Algèbre locale. Multiplicités* (Lecture Notes in Mathematics no. 11, Springer), and merely point out here that Serre's course had an evident influence on the genesis of the theory in 1957. Nor shall we use the theory of the Chow ring developed in the Chevalley Seminar of 1958, Exposés 2 and 3 (École Normale Supérieure). That theory is essentially tied to nonsingularity hypotheses, whereas the aim of our seminar is, on the contrary, to develop a theory of intersections on general schemes, and even on general locally ringed topoi. On this matter see the comments in Exposé XIV, nos. 4 and 8, which give the relations between our theory and that of the Chow ring and raise the question of a generalization of the latter. Thus one may say that the seminar presents a coherent and self-contained theory of intersections, but one which is by no means exhaustive of the different points of view useful, indeed indispensable, in intersection theory; consequently it should be complemented by the cited exposés of Serre and Chevalley.

We have appended to the seminar, as an appendix to Exposé 0, Grothendieck's 1957 report, "Classes of sheaves and the Riemann–Roch theorem", which served as the basis for the Borel–Serre seminar at Princeton in the same year and for their already cited article. That report sketches two proofs of the Riemann–Roch theorem for nonsingular quasi-projective varieties. The first, valid for the time being only in characteristic zero but giving, in return, a slightly more precise result in the case of an immersion, has not yet appeared in the literature, apart from the work of Borel–Serre and the present seminar. The report presupposes, of course, none of the other exposés of the seminar, and can even serve, like Exposé 0, as an introduction to the study of the latter, especially for a reader not yet familiar with Borel–Serre. The proof just alluded to applies essentially unchanged to the case of a projective morphism of complex analytic spaces, and will doubtless also apply in arbitrary characteristic once the problem of the operations "exterior powers" in the derived category is solved (cf. Exposé XIV, nos. 1, 2, 3). The absence of a study of this question is no doubt the most serious gap in this seminar, from the point of view we have adopted.

One will notice the absence, in the table of contents of the present seminar, of two of the participants mentioned on the cover page. One of them, J.-P. Jouanolou, took an active part in the technical elaboration of the first part of the seminar, but was prevented from taking part in the oral exposés; in particular, he is responsible for the linear-independence assertion contained in the important statement VI 1.1, giving the structure of the ring K° of classes of locally free modules on a projective bundle, which had previously been proved only under the assumption that the base scheme has an ample invertible module. The other, J.-P. Serre, gave two oral exposés which were logically independent of the rest of the seminar, and therefore preferred to publish them in the form of a separate article (cf. J.-P. Serre, "Grothendieck groups of split reductive group schemes", to appear in *Publications Mathématiques*, no. 34).

As in the other parts of the *Séminaire de Géométrie Algébrique du Bois Marie*, the sigla SGA 1 to SGA 6 refer to the different parts of that seminar; the Roman numeral following the siglum indicates the number of the exposé, and the Arabic numerals following it correspond to the internal numbering of the exposé in question. For internal references within the present seminar the siglum SGA 6 is omitted, and references of the form I 4.9 are used, meaning: Exposé I, statement 4.9. The siglum EGA refers to the *Éléments de Géométrie Algébrique* of J. Dieudonné and A. Grothendieck.

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Exposé 0

Outline of a programme for a theory of intersections on general schemes

by A. Grothendieck

The present exposé is introductory in nature, and its reading is not logically indispensable for the study of the seminar. It is addressed more particularly to readers familiar with the provisional version of the Riemann–Roch theorem, as it is presented in the Borel–Serre report or in Grothendieck's report already mentioned in the introduction, cited below as [RRR], and reproduced as an appendix at the end of the present exposé.

1. Reminder on the Riemann–Roch formula

Recall the Riemann–Roch formula for a proper morphism

$$f : X \longrightarrow Y$$

of quasi-projective smooth schemes over a field k, and a coherent sheaf F on X, whose class in the group $K(X)$ of classes of coherent sheaves on X is denoted by $\mathrm{cl}(F)$:

$$\mathrm{Todd}(T_Y)\ \mathrm{ch}_Y\big(f_!(\mathrm{cl}(F))\big) = f_*\big(\mathrm{Todd}(T_X)\ \mathrm{ch}_X(F)\big). \tag{1.1}$$

This formula is valid in $A(Y) \otimes \mathbb{Q}$, where $A(Y)$ is the Chow ring of Y. The f_* on the right hand side is obtained by tensoring with $\mathbb{Q}$ the homomorphism "direct image of cycles"

$$f_* : A(X) \longrightarrow A(Y),$$

while the term on the left is the Euler–Poincaré characteristic of F relative to f:

$$f_!(\mathrm{cl}(F)) = \sum_i (-1)^i \,\mathrm{cl}\big(R^i f_*(F)\big).$$

The symbols ch_X, ch_Y denote the Chern characters on X and on Y, and T_X, T_Y the tangent bundles of X and Y. As is known, $\mathrm{Todd}(-)$ and $\mathrm{ch}(-)$ are universal polynomials with coefficients in $\mathbb{Q}$ in the Chern classes of the argument. Since the constant term of $\mathrm{Todd}(-)$ is 1, it is an invertible element for every value of the argument. Thus, after multiplying by $\mathrm{Todd}(T_Y)^{-1}$, formula (1.1) takes the equivalent form, more convenient for our purposes,

$$\mathrm{ch}_Y\big(f_!(\mathrm{cl}(F))\big) = f_*\big(\mathrm{Todd}(T_f)\ \mathrm{ch}_X(F)\big), \tag{1.2}$$

where

$$T_f = T_X - f^* T_Y \in K(X). \tag{1.3}$$

Thus T_f plays the role of a virtual relative tangent bundle of X over Y. When f is smooth, one simply has $T_f = T_{X/Y}$, whereas when $f : X \to Y$ is an immersion, one obtains

$$T_f = -\check{N}_{X/Y},$$

where $N_{X/Y}$ denotes the normal sheaf of X in Y.

One of the main aims of the seminar is to generalize (1.2) simultaneously in two directions:

a) to get rid of the hypothesis that a base field k exists;

b) to replace the regularity hypotheses on Y, X by a hypothesis of "local regularity" on f.

Finally, along the way, we shall also deal with the following problem:

c) to eliminate the quasi-projectivity hypotheses which, in the absence of a base field, are expressed by the existence of ample invertible modules on X and Y.

2. Removing the base field

Let us first examine generalization a), while retaining the hypotheses of regularity and of the existence of ample invertible modules on X and on Y.

The definitions of $K(X)$, $K(Y)$ and of the homomorphism $f_! : K(X) \to K(Y)$ then present no new problem, thanks to the fact that X and Y are regular. The most natural way to give a meaning to (1.2) would seem to consist in defining Chow rings $A(X), A(Y)$ and a group homomorphism

$$f_* : A(X) \longrightarrow A(Y),$$

in establishing a theory of Chern classes, giving maps

$$c_i : K(X) \longrightarrow A(X),$$

and similarly for Y, and finally in describing an element, the virtual relative tangent bundle,

$$T_f \in K(X).$$

2.1

For the latter, a definition presents itself rather evidently. One uses the fact that the morphism f, thanks to the hypothesis that ample invertible modules exist on X and on Y, may be factored as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y,$$

where i is a closed immersion and f' is smooth; for example, one may take $X' = \mathbb{P}^r_Y$ for suitable r. One then sets

$$T_f = i^* T_{X'/Y} - \check{N}_{X/X'} \in K(X), \tag{2.1}$$

where $\Omega^1_{X'/Y}$ is the locally free module of relative 1-differentials of X' over Y, or the relative cotangent module of X' over Y, and $N_{X/X'}$ is the conormal module of X in X', by definition equal to J/J^2, where J is the ideal on X' defining X. The regularity hypotheses on X and X' also imply that $N_{X/X'}$ is locally free. Finally, $\check{N}$ denotes the dual of N. One verifies without difficulty, in Exposé VIII, that the element T_f defined by (2.1) is independent of the chosen factorization of f.

2.2

As for a definition, under the conditions considered, of a Chow ring and a corresponding theory of Chern classes, although it is plausible that such a theory should be developable on the model of smooth varieties over a field, it seems that this work has not been done at the present time. We shall therefore adopt another viewpoint, by directly defining a ring which replaces the Chow ring, and whose tensor product with $\mathbb{Q}$ is indeed isomorphic to $A(X) \otimes \mathbb{Q}$ in the classical case.

A first natural filtration of $K(X)$, the one considered in [RRR], is the topological filtration by the codimension of the support of coherent sheaves: $\operatorname{Fil}^i K(X)$ is generated by the $\operatorname{cl}(F)$ with F coherent and $\operatorname{codim}(\operatorname{Supp} F, X) \geq i$. A first question is the compatibility of this filtration with the multiplicative structure of $K(X)$. In [RRR], this compatibility is obtained, for X smooth over a field, by means of Chow's moving lemma. Assuming this assertion, one obtains a graded ring $\mathrm{Gr}_{\mathrm{top}}(X)$ and a canonical surjective homomorphism

$$\varphi : A(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}(X),$$

sending the class of a prime cycle Z to $\operatorname{cl}(\mathcal{O}_Z)$; it then turns out, Exposé XIV 4.2, that the kernel is a torsion subgroup. Moreover, [RRR] shows how the images in $\mathrm{Gr}_{\mathrm{top}}(X)$ of the Chern classes $c_i(F)$ can be computed directly in terms of $\operatorname{cl}(F) \in K(X)$ and the exterior-power operations λ^i in $K(X)$.

This shows that, for a noetherian regular scheme X endowed with an ample invertible module, the ring $\mathrm{Gr}_{\mathrm{top}}(X)$ is a convenient substitute for the hypothetical Chow ring $A(X)$. It also has the advantage over the latter that certain computations of [RRR], leading to important formulas in this ring, are performed directly in it–formulas which, even for X projective and smooth over a field, have not at present been established in $A(X)$ itself (Exp. XIV, 4.3 and 6.4). To justify this viewpoint completely, however, one would have to solve the problem mentioned above of the compatibility of the topological filtration of $K(X)$ with its ring structure, a problem which seems essentially connected with Chow's moving lemma, itself the essential technical ingredient of the theory of the Chow ring which we have claimed to be able to avoid.

2.3

To avoid this problem, one endows $K(X)$ with a second filtration, finer than the topological filtration, and describable in purely algebraic terms from the structure of $K(X)$ as an augmented λ-ring. The definition of this filtration is suggested by the expression given in [RRR] for the Chern classes of an element $x \in K(X)$ in $\mathrm{Gr}_{\mathrm{top}}(X)$:

$$c_i(x) = \gamma^i(x - \epsilon(x)) \mod \mathrm{Fil}^{i+1} K(X), \tag{2.3}$$

where γ^i denotes a certain linear combination of the λ^j $(j \leq i)$, made explicit in loc. cit., and where ϵ is the augmentation. One shows that $\gamma^i(x - \epsilon(x))$ is indeed of topological filtration at least i, which gives meaning to formula (2.3). One then defines the λ-filtration of $K(X)$ by the ideals $\mathrm{Fil}^i_\lambda K(X)$, where $\mathrm{Fil}^i_\lambda K(X)$ consists of finite sums of expressions of the form

$$\gamma^{i_1}(x_1 - \epsilon(x_1)) \cdots \gamma^{i_r}(x_r - \epsilon(x_r)), \qquad i_1 + \cdots + i_r \geq i.$$

The associated graded ring for this filtration will be denoted $\mathrm{Gr}^\bullet(X)$; it is this ring which will replace the Chow ring of X in the seminar. In the case under consideration here, one can moreover show that the canonical homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}(X) \tag{2.4}$$

is an isomorphism modulo torsion, justifying the viewpoint that, after tensoring with $\mathbb{Q}$, this ring can play exactly the same role as the Chow ring tensored with $\mathbb{Q}$. On the other hand, exactly what was needed has been done so that formula (2.3), with $\mathrm{Fil}_{\mathrm{top}}$ replaced by Fil_λ, defines a theory of Chern classes with values in $\mathrm{Gr}^\bullet(X)$; the Chern classes in $\mathrm{Gr}_{\mathrm{top}}(X)$ are then deduced by means of the homomorphism (2.4). It is shown in Exposé V, using the fact proved in Exposé VI that $K(X)$ is a special λ-ring in the sense of [RRR], that is, a λ-ring in the sense of Exposé V, that this notion of Chern class satisfies all the usual formal properties reviewed in [RRR].

2.4

One should note a very appreciable advantage of the definition of $\mathrm{Gr}^\bullet(X)$ over that of $A(X)$ and $\mathrm{Gr}_{\mathrm{top}}(X)$: it remains meaningful for an absolutely arbitrary scheme X, and more generally for an arbitrary locally ringed topos X. One then takes for $K(X)$ the group of classes constructed from locally free modules on X, which is again an augmented λ-ring, hence equipped with a corresponding filtration and an associated graded $\mathrm{Gr}^\bullet(X)$. The price of this generality, however, is that when one is not willing to neglect torsion phenomena, i.e. to tensor with $\mathbb{Q}$, the properties of $\mathrm{Gr}^\bullet(X)$ are in several respects less satisfactory than those of its two competitors (Exp. XIV no. 4). Thus, assuming again that $f : X \to Y$ is a proper morphism of noetherian regular schemes endowed with ample invertible modules, it is not true in general that the homomorphism

$$f_* : K(X) \longrightarrow K(Y)$$

maps $\mathrm{Fil}^i_\lambda K(X)$ into $\mathrm{Fil}^{i-d}_\lambda K(Y)$, where d is the relative dimension of X over Y, and hence defines, by passage to associated gradeds, a degree $-d$ homomorphism from $\mathrm{Gr}^\bullet(X)$ to $\mathrm{Gr}^\bullet(Y)$. However, Exposé VII proves that this is true after tensoring with $\mathbb{Q}$, so that one obtains a degree $-d$ homomorphism

$$f_* : \mathrm{Gr}^\bullet(X) \otimes \mathbb{Q} \longrightarrow \mathrm{Gr}^\bullet(Y) \otimes \mathbb{Q}, \tag{2.5}$$

which plays the role of the homomorphism "direct image of cycles" $A(X) \to A(Y)$.

Having said this, we see that we possess all the structural elements needed to give meaning to formula (1.2) when the existence of a base field is not postulated, with $f : X \to Y$ a proper morphism of noetherian regular schemes admitting ample invertible modules.

3. The case of general schemes

We now show how one can still give meaning to formula (1.2) when the regularity hypothesis on X, Y is abandoned and replaced by a local hypothesis on the morphism f, to be specified below.

3.1

A first difficulty comes from the fact that the observation which served as the starting point of the theory [RRR], namely that the group $K(X)$ of classes of sheaves on X is the same whether it is defined in terms of coherent sheaves or in terms of locally free sheaves, is essentially tied to the regularity hypothesis on X. For a general scheme X, say noetherian for simplicity, one must distinguish between these two groups; we shall denote them respectively by $K_\bullet(X)$ and $K^\bullet(X)$. The first is covariant in X for proper morphisms, the second contravariant in X for arbitrary morphisms. Notice that $K^\bullet(X)$ is naturally endowed with a ring structure, indeed with an augmented λ-ring structure, whereas $K_\bullet(X)$ no longer has, in general, a ring structure, but only a module structure over $K^\bullet(X)$. The homomorphism $x \mapsto x \cdot \mathrm{cl}(\mathcal{O}_X)$ from $K^\bullet(X)$ to $K_\bullet(X)$ is no longer an isomorphism in general. For our formulation of the generalized Riemann–Roch theorem, we shall work exclusively with $K^\bullet(X)$ and $K^\bullet(Y)$, to the exclusion of $K_\bullet(X)$ and $K_\bullet(Y)$. As noted in 2.3, these augmented λ-rings, equipped with their natural λ-filtration, give rise to graded rings $\mathrm{Gr}^\bullet(X)$, $\mathrm{Gr}^\bullet(Y)$, and to Chern-class homomorphisms

$$c_i : K^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet(X),$$

and similarly for Y. The difficulty, however, lies in defining a direct-image homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(Y), \tag{3.1}$$

which can no longer be defined by the naive formula

$$f_*(\mathrm{cl}(F)) = \sum_i (-1)^i \,\mathrm{cl}(R^i f_*(F)),$$

because even if F is locally free, the modules $R^i f_*(F)$ are in general no longer locally free, nor even of finite Tor-dimension, and the preceding formula loses its meaning. A solution to this difficulty is furnished by the viewpoint of derived categories. Indeed, whereas the $R^i f_*(F)$ are in general "bad" sheaves, say of infinite Tor-dimension, to which one would not know how to associate a class in $K^\bullet(Y)$, the complex $Rf_*(F)$ on Y, the direct image of F in the sense of derived categories, whose cohomology sheaves are the $R^i f_*(F)$, tends to be excellent. More precisely, suppose that F has finite Tor-dimension over $\mathcal{O}_Y$; this is the case if $\mathcal{O}_X$ has finite Tor-dimension over $\mathcal{O}_Y$ and, in addition, F is locally free. Then one proves (Exp. III) that the complex $Rf_*(F)$ is "perfect", by which we mean here that it has coherent cohomology sheaves (as was already known in any case) and globally finite Tor-dimension; equivalently, on every affine open U of Y, this complex is isomorphic, in the derived category $D(U)$, to a bounded complex of locally free modules of finite type. From the viewpoint of homological algebra, perfect complexes are "as good as" locally free modules, of which they are the natural generalization. They moreover form a "triangulated subcategory" of the derived category $D(Y)$, i.e. a subcategory stable under the cones (or "mapping cylinders") of morphisms. On the other hand, to every triangulated category C one associates a group $K(C)$ of classes, a natural variant of the construction of the group of classes of an abelian category, by taking the free group generated by the objects of C, modulo the relations

$$\mathrm{cl}(K) - \mathrm{cl}(K') - \mathrm{cl}(K'')$$

for every distinguished triangle

$$\begin{tikzcd} & K'' \arrow[dl] & \\ K' \arrow[rr] & & K \arrow[ul] \end{tikzcd}$$

What then replaces the fundamental observation of [RRR] in the present case is the fact that, when the scheme Y admits an ample invertible module, every perfect complex $L_\bullet$ on Y is globally (and not merely locally) isomorphic in $D(Y)$ to a bounded complex L' of locally free modules of finite type (Exp. II). It follows easily (Exp. IV) that the canonical homomorphism

$$K^\bullet(Y) \longrightarrow K(\mathrm{Parf}(Y))$$

is an isomorphism, where $\mathrm{Parf}(Y)$ is the triangulated category of perfect complexes on Y. Thus to every perfect complex $L_\bullet$ on Y one can associate a class

$$\mathrm{cl}(L_\bullet) \in K^\bullet(Y),$$

which is none other than

$$\mathrm{cl}(L_\bullet) = \sum_i (-1)^i \,\mathrm{cl}(L'_i),$$

where L' is as above, so that the L'_i are indeed locally free modules on Y whose classes may be taken in $K^\bullet(Y)$.

3.2

To give a meaning to (1.2), it remains to impose on f a hypothesis making it possible to give a meaning to the homomorphism (2.5), i.e. ensuring that the homomorphism (3.1) just defined is, up to a shift and modulo torsion, compatible with the filtrations of these rings; and finally one must again define an element, the "virtual relative tangent bundle",

$$T_f \in K^\bullet(X).$$

For the latter, returning to the definition proposed in 2.1, one sees that a hypothesis is needed ensuring that, with the notation of loc. cit., the conormal module $N_{X/X'}$ is locally free. The most natural hypothesis to this end is that i be a "regular immersion", i.e. identify X with a subscheme of X' defined by an ideal which locally can be defined by an $\mathcal{O}_{X'}$-regular sequence of sections of $\mathcal{O}_{X'}$. This hypothesis, purely local in nature, in fact does not depend on the choice of the factorization of f into an immersion followed by a smooth morphism (Exp. VIII), and, as above, the same is true for the class T_f defined by formula (2.1). When the hypothesis just considered is satisfied, the morphism f is called a "locally complete intersection" morphism. Such a morphism is also locally of finite Tor-dimension.

We shall also allow ourselves to omit the word "locally" in this terminology, which cannot cause any confusion.

If f is a locally complete intersection morphism, one introduces a natural notion of "virtual relative dimension" $d(f)$ by the formula

$$d(f)(x) = \dim_x(f^{-1}(f(x))) - \operatorname{codim}_x(X, X').$$

It is an integer-valued function, locally constant at the point $x \in X$, hence constant if X is connected; moreover it does not depend on the choice of the factorization (2.1). When this virtual relative dimension d is constant, when f is proper, and when X and Y admit ample invertible modules, one can prove (Exposé VIII), although this is far from a trivial result, that

$$f_*(\operatorname{Fil}^i K^\bullet(X)_{\mathbb{Q}}) \subset \operatorname{Fil}^{i-d} K^\bullet(Y)_{\mathbb{Q}}, \tag{3.4}$$

whence a homomorphism (3.3) of degree $-d$.

3.3

We have therefore assembled all the structural elements needed to give a meaning to formula (1.2) in the case of a proper locally complete intersection morphism of noetherian schemes X, Y, both admitting ample invertible modules. With a little additional care, this formulation in fact retains a meaning independently of any noetherian hypothesis.

In this general form, this Riemann–Roch formula will be proved in Exposé VIII. The reader will observe that the proof given in this seminar combines elements of the two proofs which appear in [RRR] and which were discussed in the Introduction. The need to resort to the methods of the first of these proofs, involving more K-formalism than the second, followed in Borel–Serre, is due to the need for a proof of the inclusion (3.4), a question which did not arise in the more restricted framework of [RRR] and Borel–Serre [2], where one could work with the Chow ring or the ring $\mathrm{Gr}^\bullet_{\mathrm{top}}$. From the second proof, we retain the introduction of the scheme obtained by blowing up Y along X in the case where f is an immersion. This technical artifice will no doubt disappear, and the proof in the case of an immersion will extend to the case where one no longer postulates the existence of an ample invertible module on Y, once the question raised in Exposé XIV, no. 1, concerning the introduction of exterior-power operations in the derived category $D^-(Y)$ has been resolved.

4. Removing the ample invertible module hypothesis

We must finally give a few indications of how one can still give a meaning to (1.2) in the absence of a hypothesis asserting the existence of ample invertible modules on X and Y.

4.1

First note that for an arbitrary scheme X, or more generally an arbitrary locally ringed topos, it seems beyond doubt that the "right" invariant which should replace the ring $K(X)$ of [RRR] is not the ring $K^\bullet(X)$ considered above, defined in terms of locally free modules on X, but rather the group $K(\mathrm{Parf}(X))$ considered in 3.1, defined in terms of perfect complexes on X. We shall denote this group here by $K^\bullet(X)$, and distinguish it from the "naive" group of 3.1, denoted

$K^\bullet_{\text{naive}}(X)$. This point of view is forced by the need to define a direct-image homomorphism (3.1) for a proper morphism

$$f : X \longrightarrow Y$$

of finite Tor-dimension, say between noetherian schemes; the non-noetherian case requires somewhat stronger local finiteness properties on f, cf. Exposé III. With the definition of $K^\bullet(X)$ now adopted, this homomorphism f_* is indeed defined in a trivial way by formula (3.2), where now F is an arbitrary perfect complex on X, which implies that $Rf_*(F)$ is also perfect.

4.2

This new definition of $K^\bullet(X)$ raises, however, a new problem in homological algebra: that of defining a λ-structure on $K^\bullet(X)$ compatible with its natural ring structure coming from tensor product in the derived category. If L and L' are two perfect complexes, then so is their total tensor product $L \otimes^{\mathbf{L}} L'$. More precisely, one would need to define "exterior-power" operations in the derived category $D^-(X)$, transforming perfect complexes into perfect complexes. This question has not yet been clarified at present, and, as was said in the Introduction, it constitutes the most serious gap in the seminar. Once this question is solved, and proceeding as in 2.3, one obtains a natural filtration on $K^\bullet(X)$, allowing one to form the associated graded ring $\mathrm{Gr}^\bullet(X)$, which will play the role of the Chow ring, and to define a theory of Chern classes

$$c_i : K^\bullet(X) \longrightarrow \mathrm{Gr}^i(X).$$

4.3

To give a meaning to formula (1.2) for a proper locally complete intersection morphism

$$f : X \longrightarrow Y$$

of noetherian schemes, one must still include in the statement of this formula the nontrivial inclusions (3.4), where d is the virtual relative dimension, in order to be able to define (3.3); and finally one must define the virtual relative tangent class

$$T_f \in K^\bullet(X).$$

When f admits a factorization (2.1) through an immersion into a relative smooth scheme, the definition of T_f presents no new difficulty. It can moreover be reformulated in that case, in a way more consonant with the spirit of derived categories, by considering the well-known canonical homomorphism

$$N_{X/X'} \longrightarrow \Omega^1_{X'/Y} \otimes_{\mathcal{O}_{X'}} \mathcal{O}_X \tag{4.1}$$

induced by the exterior differential, and the chain complex $L_{X/Y}$ defined from (4.1) by "extending by zeros", with the complex equal in degree 1, respectively 0, to the source, respectively the target, of the homomorphism (4.1). Formula (2.2) then takes the more satisfactory form

$$T_f = \mathrm{cl}(L^\vee_{X/Y}), \tag{4.2}$$

a formula which makes sense in $K^\bullet(X)$ when f is a locally complete intersection morphism, since in that case the complex $L_{X/Y}$ is manifestly perfect. The assertion that T_f is invariant relative to the chosen factorization of f can then be made precise as an invariance assertion for the complex $L_{X/Y}$: up to canonical isomorphism in the derived category $D(X)$, this complex is likewise independent of the chosen factorization (Exposé VIII).

4.4

When $f : X \to Y$ is an arbitrary morphism of schemes, one can still construct a canonical object $L_{X/Y}$ of the derived category $D(X)$, defined up to unique isomorphism, which on each affine open of X lying above an affine open of Y is isomorphic to the complex constructed according to the principle explained in the preceding paragraph. In the general case, however, "smooth" should be replaced by "formally smooth", so that every algebra B over a ring A occurs as a quotient of a formally smooth algebra, for example a polynomial algebra. For such a definition it is not possible to proceed by simple gluing from the affine constructions, since the objects of the derived category $D(X)$ are, essentially, not glueable.

Here is a general construction of $L_{X/Y}$, valid more generally for every morphism

$$f : X \longrightarrow Y$$

of commutatively ringed topoi. Let $B = \mathcal{O}_X$ and $A = f^{-1}\mathcal{O}_Y$, so that A is a ring on the space X and B is an A-algebra. Let $T \to B$ be a homomorphism of sheaves of sets which generates B as an A-algebra, i.e. such that the corresponding homomorphism of A-algebras

$$C = A[T] \longrightarrow B$$

is an epimorphism of sheaves of sets; here $A[T]$ denotes the commutative A-algebra freely generated by T, that is, the sheaf associated with the presheaf

$$U \longmapsto A(U)[T(U)].$$

Let J be the kernel of $C \to B$, and consider, as in 4.3, the chain complex of length 1 defined by the natural homomorphism

$$J/J^2 \longrightarrow \Omega^1_{C/A} \otimes_C B.$$

It is not difficult to see that, up to canonical isomorphism in the category $D(X)$, the complex so constructed is independent of the choice of T and of $T \to B$, so that it deserves the designation of the relative cotangent complex $L_{X/Y}$ of X over Y. One also verifies that, on every affine open of X over an affine open of Y, it is given, up to canonical isomorphism, by the procedure of 4.3.

In the case where f is locally complete intersection, one sees moreover that the result is a perfect complex, which makes it possible again to define an element T_f of $K^\bullet(X)$ by formula (4.2). Here again, in general, it would not have been possible to define T_f as an element of $K^\bullet_{\text{naive}}(X)$, and nothing permits one to suppose that T_f is globally isomorphic in $D(X)$ to a bounded complex with locally free components of finite type. This is therefore an additional justification for the point of view advanced in 4.1 in favour of the "sophisticated" $K^\bullet(X)$, in preference to $K^\bullet_{\text{naive}}(X)$.

4.5

Subject to the question raised in 4.2, we have therefore again assembled all the structural elements needed to give a meaning to (1.2) for a proper locally complete intersection morphism of arbitrary schemes; the noetherian hypothesis implicitly made in the preceding sections can in fact be eliminated without any essential difficulty. Unfortunately, this does not mean that, even subject to this reservation, the formula in question is established, even in the case where Y is the spectrum of an algebraically closed field of characteristic $p > 0$ and X is, say, a proper smooth algebraic scheme of dimension 3. See the comments in Exposé XIV, no. 2.

5. Analytic variant

In the form presented in no. 4, the Riemann–Roch formula (1.2) also has a meaning for a proper locally complete intersection morphism of complex analytic spaces, or of rigid analytic spaces in the sense of Tate. But of course the proof given in [RRR] applies to this case only when one assumes in addition that the morphism f is projective; and, as in the algebraic case, it seems that an essentially new idea is needed to treat the general case. The formula we have in view here is a formula with values in an "analytic Chow ring" $\mathrm{Gr}^\bullet(X)$, or rather $\mathrm{Gr}^\bullet(X)_\mathbb{Q}$, constructed according to the principle explained in 4.2, the difficulty encountered there disappearing, moreover, because one is in characteristic zero, as is noted in Exposé XIV, no. 1.

When X and Y are nonsingular, it is possible to give another version of the Riemann–Roch formula by proceeding as follows. First let X be an arbitrary complex analytic space, and let X^{top} be the same space equipped with the sheaf of rings of continuous complex-valued functions. We therefore have a canonical morphism of ringed spaces

$$X^{\mathrm{top}} \longrightarrow X,$$

which, by the evident contravariant nature of the functor $K^\bullet(X)$ in the variable ringed space X, defines a homomorphism of rings

$$K^\bullet(X) \longrightarrow K^\bullet(X^{\mathrm{top}}). \tag{5.1}$$

For simplicity suppose that X^{top} is compact; then, as in the case of a scheme admitting an ample invertible module, one proves (Exposé II) that every perfect complex on X^{top} is globally isomorphic to a bounded complex, locally free in each degree, i.e. to a complex of complex vector bundles on X^{top}, and hence that $K^\bullet(X^{\mathrm{top}})$ is canonically isomorphic to the well-known group $K(X^{\mathrm{top}})$ of topologists [1], defined in terms of complex vector bundles on the compact space X^{top}. Thus one may interpret (5.1) as a homomorphism from the $K^\bullet(X)$ defined in terms of the analytic structure to the well-known ring $K(X^{\mathrm{top}})$ of topologists. If now

$$f : X \longrightarrow Y$$

is a morphism of compact nonsingular analytic spaces, one knows how to associate to it, thanks to Atiyah–Hirzebruch [1], a group homomorphism

$$f_*^{\mathrm{top}} : K^\bullet(X^{\mathrm{top}}) \longrightarrow K^\bullet(Y^{\mathrm{top}}). \tag{5.2}$$

Using likewise the homomorphism f_* of (3.1), defined by formula (3.2), which here implicitly uses Grauert's finiteness theorem [4] for a proper morphism of analytic spaces, one obtains a square of natural homomorphisms involving only groups of type $K^\bullet$, to the exclusion of rings of Chow-ring or cohomology type:

$$\begin{array}{ccc} K^\bullet(X) & \longrightarrow & K^\bullet(X^{\mathrm{top}}) \\ \downarrow f_* & & \downarrow f_*^{\mathrm{top}} \\ K^\bullet(Y) & \longrightarrow & K^\bullet(Y^{\mathrm{top}}). \end{array} \tag{5.3}$$

The Riemann–Roch type formula to which we are alluding consists in asserting the commutativity of this square. Note the difference in nature between this assertion, which does not neglect torsion phenomena and is situated in a group $K^\bullet(Y^{\mathrm{top}})$ of an essentially discrete nature, and the Riemann–Roch formula of the type considered in no. 4, which is situated in a group of "Chow" type no longer of a discrete nature, but which, on the other hand, must be tensored with $\mathbb{Q}$. Let us note that formula (5.3) is proved at any rate when X and Y are nonsingular projective varieties [7], as is the formula of the type considered in no. 4, proved in this case in

[RRR]. What is involved here is a plausible variant of the Atiyah–Singer index theorem, and of its generalizations due to Shih [7] and Atiyah, which obviously calls for a common generalization. Once the commutativity of (5.3) is proved, one would immediately deduce from it, using the "differentiable Riemann–Roch theorem" of Atiyah–Hirzebruch [1], a cohomological variant of the complex analytic Riemann–Roch formula, namely the commutativity of the diagram

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{x \mapsto \mathrm{ch}(x)\,\mathrm{Todd}(T_X)} & H^{2\bullet}(X,\mathbb{Q}) \\ \downarrow f_* & & \downarrow f_* \\ K^\bullet(Y) & \xrightarrow{y \mapsto \mathrm{ch}(y)\,\mathrm{Todd}(T_Y)} & H^{2\bullet}(Y,\mathbb{Q}), \end{array} \tag{5.4}$$

where the horizontal arrows are deduced from cohomological Chern classes, themselves deduced from the homomorphism (5.1) and from the ordinary Chern-class homomorphism

$$K^\bullet(X^{\mathrm{top}}) \longrightarrow H^{2\bullet}(X,\mathbb{Z}),$$

the first vertical arrow is the same as in (5.3), and the second is the Gysin homomorphism in rational cohomology.

6. The covariant invariant $K_\bullet(X)$

In all that precedes, scarcely anything has been said except about the contravariant invariant $K^\bullet(X)$, to the exclusion of the covariant invariant $K_\bullet(X)$ whose existence was indicated in 3.1. If one regards $K^\bullet(X)$ as giving an analogue of the integral cohomology ring of a topological space, one should regard $K_\bullet(X)$ as giving an analogue of integral homology. The module structure of $K_\bullet(X)$ over $K^\bullet(X)$ then corresponds to the cap-product operation. The fact that, when X is regular, the natural homomorphism $K^\bullet(X) \to K_\bullet(X)$ is an isomorphism should be regarded as the analogue of the Poincaré duality theorem for oriented topological varieties, asserting that cap product with the fundamental homology class defines an isomorphism from cohomology to homology. These analogies moreover suggest the usefulness of also defining, for a finite polyhedron X, say, a covariant invariant $K_\bullet(X)$ whose relations to integral homology – spectral sequence, Chern homomorphism, and so on – would be analogous to those existing between $K^\bullet(X)$ and integral cohomology. Nor does it seem excluded that there might exist, in terms of these groups $K_\bullet(X)$, variants of the differentiable Riemann–Roch theorem valid for spaces more general than compact differentiable manifolds.

Returning to the case where X is an arbitrary noetherian scheme, one should endow $K_\bullet(X)$ with an increasing filtration, $\mathrm{Fil}_i\, K_\bullet(X)$ being generated by classes $\mathrm{cl}_\bullet(F)$, with F a coherent sheaf on X such that $\dim \mathrm{Supp}\, F \le i$. One will prove (Exposé X) the compatibility of this filtration with the module structure and with the filtration of $K^\bullet(X)$, i.e. the formula

$$\mathrm{Fil}^i\, K^\bullet(X) \cdot \mathrm{Fil}_j\, K_\bullet(X) \subset \mathrm{Fil}_{j-i}\, K_\bullet(X), \tag{6.1}$$

which allows the graded group associated with the filtered module $K_\bullet(X)$, denoted $\mathrm{Gr}_\bullet(X)$, to be endowed with a module structure over $\mathrm{Gr}^\bullet(X)$. Its elements may in fact be interpreted as classes of algebraic cycles on X, for an equivalence relation less fine than rational equivalence when X is of finite type over a base field k. If

$$f : X \longrightarrow Y$$

is a proper morphism of noetherian schemes, then

$$f_* : K_\bullet(X) \longrightarrow K_\bullet(Y) \tag{6.2}$$

is compatible with the filtrations and defines a homomorphism of graded groups

$$f_* : \mathrm{Gr}_\bullet(X) \longrightarrow \mathrm{Gr}_\bullet(Y), \tag{6.3}$$

giving rise to a "projection formula", i.e. a formula which says that it is $\mathrm{Gr}^\bullet(Y)$-linear; this formula is obtained by passage to associated gradeds from the analogous projection formula for the homomorphism (6.2), which is $K^\bullet(Y)$-linear.

From the point of view proposed in the seminar, a manageable theory of intersections on noetherian schemes X, not necessarily regular, should consist in the combined study of the contravariant invariants $K^\bullet(X), \mathrm{Gr}^\bullet(X)$ and the covariant invariants $K_\bullet(X), \mathrm{Gr}_\bullet(X)$. These play roles of essentially different nature, and their properties complement each other. Thus the covariant groups $K_\bullet(X), \mathrm{Gr}_\bullet(X)$ are easier to compute for certain nonproper fibrations, thanks to the "homotopy exact sequence" which one can establish for them ([RRR] and Exposé IX). For example, one proves (Exposé IX) canonical isomorphisms such as

$$K_\bullet(X[t]) \simeq K_\bullet(X), \qquad \mathrm{Gr}_\bullet(X[t]) \simeq \mathrm{Gr}_\bullet(X),$$

which fail in general for $K^\bullet$ when X is a nonregular scheme. On the other hand, $K^\bullet(X)$ and $\mathrm{Gr}^\bullet(X)$ lend themselves well to computations, thanks to their ring structures.

When one considers schemes X proper over a field, the projection

$$\pi_X : X \longrightarrow (\text{point})$$

defines a homomorphism, the "Euler–Poincaré characteristic",

$$\chi_X = \pi_{X*} : K_\bullet(X) \longrightarrow K_\bullet(\text{point}) = \mathbb{Z}, \tag{6.4}$$

which induces on the subgroup $\mathrm{Gr}_0(X) = \mathrm{Fil}_0\, K_\bullet(X)$ the homomorphism "degree of zero-cycles"

$$\deg_X : \mathrm{Gr}_0(X) \longrightarrow \mathbb{Z}. \tag{6.5}$$

The multiplication homomorphism

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathrm{Gr}_0(X),$$

composed with the degree homomorphism (6.5), then furnishes a pairing

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathbb{Z}, \tag{6.6}$$

which explains that, in certain respects, $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}_\bullet(X)$ play dual roles, just as cohomology and homology do. Thus, if $f : X \to Y$ is a morphism of proper algebraic schemes, the two homomorphisms

$$f^* : \mathrm{Gr}^\bullet(Y) \longrightarrow \mathrm{Gr}^\bullet(X), \qquad f_* : \mathrm{Gr}_\bullet(X) \longrightarrow \mathrm{Gr}_\bullet(Y)$$

are formally transposes of one another, in the sense of the pairings (6.6).

The homomorphisms (6.4) and (6.5), and the pairing (6.6) deduced from them, may be regarded as the source of numerical relations in intersection theory on proper algebraic schemes over a given field k. This point of view will be developed in some detail in Exposé X, which is essentially independent of Exposés VI, VII, VIII, centred on the proof of the Riemann–Roch theorem, and of Exposé IX. Some notions concerning the behaviour of $K^\bullet(X)$ and $\mathrm{Gr}^\bullet(X)$ under specialization will also be given there.

7. Picard groups and the determinant functor

As in every intersection theory, it is important to specify the place occupied in the theory by the Picard group, or group of classes of divisors in the classical terminology. Subject to a slight restriction, automatically satisfied when X is for instance quasi-compact, one establishes in Exposé X a canonical isomorphism

$$c_1 : \mathrm{Pic}(X) \xrightarrow{\sim} \mathrm{Gr}^1(X), \tag{7.1}$$

which shows the exact role of the Picard group in the theory we are proposing. This role justifies a more detailed study of the Picard group in Exposés XII and XIII, which are logically independent of the text of the seminar. Kleiman, in Exposé XIII, studies numerical equivalence and certain finiteness theorems for the Picard group, announced without proof in [6], using purely numerical techniques, not relying on the theory of the Picard functor and its representability, due to himself and Mumford, and markedly simpler than Grothendieck's initial proofs. The only result refractory to Kleiman's methods, and which he is obliged to use, is [6, C-07 (i)]; its proof and certain refinements, including certain representability theorems for the Picard functor, occupy Exposé XII. The latter is moreover of a spirit and technique entirely foreign to the rest of the seminar, and is connected with [5]; this exposé is logically independent of all the others.

Finally, in Exposé XI, of a nature appreciably more general than XII and XIII and closer to the general spirit of the seminar, one studies, on an arbitrary locally ringed topos, a remarkable functor called the "determinant functor":

$$\overset{\bullet}{\det} : \mathrm{Parf}_{\mathrm{is}}(X) \longrightarrow \mathrm{Inv}(X) \tag{7.2}$$

from the category of perfect complexes on X, whose morphisms are only the isomorphisms in the sense of the derived category $D(X)$, to the category of invertible modules on X. This functor is described, up to very little, by the requirement that it be of "local nature" and that it be given, for a bounded complex L with locally free components, by the formula

$$\overset{\bullet}{\det}(L) = \bigotimes_i \det(L_i)^{(-1)^i}, \tag{7.3}$$

where $\det(L_i)$ denotes the exterior power of maximal degree of the locally free module L_i. Although this exposé is also logically independent of the rest of the seminar, except for Exposé I, it nevertheless constitutes an important element of the general "yoga" proposed in the seminar and developed essentially in Exposés I to XI.

Bibliography

[1] M. Atiyah and F. Hirzebruch, *Vector bundles and homogeneous spaces*, Symposia in Pure Mathematics, vol. 3, Differential Geometry, pp. 7–38, 1961.

[2] A. Borel and J.-P. Serre, *Le théorème de Riemann–Roch*, Bull. Soc. math. France, t. 86, pp. 97–136 (1958).

[3] H. Grauert, *Ein Theorem der analytischen Garbentheorie*, Pub. Math. no. 5, pp. 5–64 (1960).

[4] A. Grothendieck, *Classes de faisceaux et théorème de Riemann–Roch*, mimeographed notes, Princeton 1957, cited as [RRR] and reproduced as an appendix to Exposé 0 of the seminar.

[5] A. Grothendieck, *Technique de descente et théorèmes d'existence en Géométrie Algébrique*, in *Fondements de la Géométrie Algébrique*, extracts from the Bourbaki Seminar 1957–1962, Secrétariat Mathématique, Paris.

[6] W. Shih, Séminaire de Topologie à l'I.H.E.S., 1966/67.

[7] M. Atiyah and F. Hirzebruch, *The Riemann–Roch theorem for analytic embeddings*, Topology, vol. 1, pp. 151–166 (1962).

Appendix to Exposé 0: RRR

Classes of Sheaves and the Riemann–Roch Theorem

by A. Grothendieck[1]

Demonstration of the Riemann–Roch theorem (1 November 1957).

Chapter I. λ-rings (formal preliminaries)

Chapter II. Classes of coherent algebraic sheaves and Chern classes

Chapter I. λ-rings (formal preliminaries)

1. Definitions

A *λ-ring*[2] means a commutative ring K with unit, provided with a family of maps

$$\lambda^i : K \longrightarrow K \qquad (i \text{ an integer } \geq 0) \tag{1.1}$$

satisfying the following conditions:

$$\lambda_x^0 = 1, \qquad \lambda_x^1 = x, \qquad \lambda^n(x+y) = \sum_{i=0}^{n} \lambda_x^i \lambda_y^{n-i}. \tag{1.2}$$

For every $x \in K$ put

$$\lambda_t(x) = \sum_{n\geq 0} \lambda^n(x) t^n \in K[[t]]. \tag{1.3}$$

[1]This is the literal reproduction of the report cited in the Introduction to the present seminar. The footnotes indicated by the signs (*), (**) were added in October 1967.

[2]In this seminar we rather say *pre-λ-ring* (V 2.1), reserving the name λ-ring for those satisfying the supplementary condition of page 3 (cf. V 2.4).

The preceding relations then say that $x \mapsto \lambda_t(x)$ is a homomorphism from the additive group K to the multiplicative group $1 + K[[t]]^+$ of formal series over K with augmentation 1, lifting the canonical homomorphism

$$1 + \sum_{i>0} x_i t^i \longmapsto x_1.$$

Thus on the ring $\mathbb{Z}$ there exists a unique λ-structure compatible with its ring structure and such that

$$\lambda_t(1) = 1 + t. \tag{1.4}$$

One then has

$$\lambda_t(n) = (1+t)^n, \tag{1.5}$$

whence

$$\lambda^i(n) = \binom{n}{i}.$$

The notion of a λ-homomorphism of λ-rings is clear; an *augmented λ-ring* is a λ-ring provided with a λ-homomorphism to $\mathbb{Z}$.

To define the notion of a *special* λ-ring, the following considerations are useful. Let k be a commutative ring with unit and let

$$K = 1 + k[[t]]^+$$

be the group of formal series with coefficients in k and augmentation 1. We shall introduce on it a λ-ring law whose additive structure is the multiplication of formal series. The multiplication of this ring will be denoted $f \circ g$. It is entirely determined by the condition that the coefficients $c_n = c_n(f \circ g)$ of $f \circ g$ are given by universal polynomials with integral coefficients in the coefficients $a_i = c_i(f)$ and $b_i = c_i(g)$ of f and g, respectively:

$$c_n = P_n(a_1, \ldots, a_n; b_1, \ldots, b_n), \tag{1.6}$$

that it be distributive with respect to the "addition" fg, and be given for linear factors by

$$(1 + at) \circ (1 + bt) = 1 + abt. \tag{1.7}$$

The P_n are computed by calculations with symmetric functions; P_n is isobaric of weight n in the a_i and isobaric of weight n in the b_i (where a_i and b_i have weight i), and moreover is symmetric in $a = (a_i)$ and $b = (b_i)$.

Similarly, the $\lambda^i(f)$ are determined without ambiguity by the condition that the coefficients

$$c_n(\lambda^i(f)) = c_n(\lambda^i(f))$$

of $\lambda^i(f)$ be calculated by universal polynomials with integral coefficients in the coefficients $c_n(f) = a_n$ of f:

$$c_n = P_{i,n}(a_1, \ldots, a_{in}), \tag{1.8}$$

that the relations (1.2) be satisfied, and that, for $i > 1$, one have

$$\lambda^i(1 + at) = 1 \tag{1.9}$$

(1 is the zero element of the ring $K = 1 + k[[t]]^+$). The $P_{i,n}$ are again computed by calculations with symmetric functions, and one observes that $P_{i,n}$ is an isobaric polynomial of weight in in the a_i. Provided with the preceding operations, K becomes a λ-ring; its zero element is 1 and its unit element is $1 + t$. Also note the relation

$$(1 + at) \circ f = f(at). \tag{1.7 bis}$$

Let now K be an arbitrary λ-ring. We say that K is *special* if the additive homomorphism λ_t from K to $1 + K[[t]]^+$ is a λ-homomorphism. Thus this means that one has the formulas

$$\lambda_t(1) = 1 + t, \quad \lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y), \quad \lambda_t(\lambda^i(x)) = \lambda^i(\lambda_t(x)), \tag{1.10}$$

or, more explicitly,

$$\begin{cases} \lambda^i(1) = 0 & \text{if } i > 1, \\ \lambda^n(xy) = P_n(\lambda^1 x, \ldots, \lambda^n x; \lambda^1 y, \ldots, \lambda^n y), \\ \lambda^n(\lambda^i(x)) = P_{i,n}(\lambda^1 x, \ldots, \lambda^{in} x), \end{cases} \tag{1.11}$$

where the P_n and $P_{i,n}$ are the universal polynomials considered above. In particular the formulas (1.5) follow. One may show that the λ-ring $1 + k[[t]]^+$ constructed from a commutative ring k is special, but we shall not need this fact.

2. Examples

1) Let G be a group and k a commutative ring. Consider the group $K(G)$, the quotient of the free group generated by the classes of representations of G in finite-type k-modules by the subgroup generated by elements of the form

$$(\rho \oplus \rho') - (\rho) - (\rho')$$

(where $\oplus$ denotes the direct sum). Tensor product and exterior powers define on $K(G)$ a λ-ring structure. One may also pass to the quotient by elements of the form

$$(\rho) - (\rho') - (\rho''),$$

where ρ is a representation which is an extension of ρ' by ρ'', trivial as an extension of k-modules. One obtains the group $K_r(G)$ of reduced classes of representations of G, and one checks that the λ-ring law passes to the quotient.

When k is an algebraically closed field of characteristic 0,[3] one can show that $K(G)$, and a fortiori $K_r(G)$, is a special λ-ring. This is no longer true if the characteristic is not 0; however, if k is algebraically closed, $K_r(G)$ is still a special λ-ring. To prove this one is reduced to the case where G is a product $\mathrm{Gl}(m, k) \times \mathrm{Gl}(n, k)$ and where only rational representations of G are considered; our assertion follows from the explicit determination of $K_r(G)$ to be made below (Example 3). Finally, when k has characteristic 0, complete reducibility of rational representations of G shows that $K(G) = K_r(G)$.

Variants of the preceding example: G is an algebraic group defined over k and one considers only rational representations of G defined over k; one may consider the case of a topological group, or of a Lie group, and continuous representations, etc.[4]

2) Let X be an algebraic variety, irreducible or not, defined over the field k. We may consider the quotient group of the free group generated by the classes of vector bundles on X (defined over k) by the subgroup generated by the elements of the form

$$(E \oplus E') - (E) - (E'),$$

[3] It is unnecessary for k to be algebraically closed; cf. VI 3.3 for a still more general result, encompassing all those that will be used in the present report.

[4] One could build other examples of λ-rings; for example, the set of equivalence classes of quadratic forms, the set of representation classes of a Lie algebra, etc. We promise that we shall not need all these examples in order to treat the Riemann–Roch theorem.

on which the λ-ring operations are defined by tensor product and exterior powers. If X is complete, this is a free group generated by the indecomposable vector bundles on X (thanks to Krull–Remak–Schmidt); knowledge of this ring and of its basis is equivalent to the complete classification of vector bundles on X and to knowledge of the elementary tensor operations on these bundles. If k is algebraically closed of characteristic 0, all other tensor operations on classes of vector bundles are then known by means of what follows in 3). One may make a more drastic passage to the quotient by the group generated by the elements

$$(E) - (E') - (E''),$$

where E is a vector bundle extension of E' by E''. One then obtains a λ-ring, denoted $K(\xi(X))$, where $\xi(X)$ denotes the additive category of vector bundles on X defined over k. Using the results of Example 1), one shows that, if k is algebraically closed,[5] the λ-ring $K(\xi(X))$ is special, and that the same is true of the first λ-ring defined here if the characteristic of k is 0, but not in the general case.

3) Determination of $K(G)$ and of $K_r(G)$ in some important cases. Suppose the field k is algebraically closed and that the group G is affine algebraic, connected, and "globally reductive" (the radical of G is isomorphic to $(k^*)^s$), and consider only rational linear representations. In characteristic 0, complete reducibility of rational linear representations of G shows that $K(G) = K_r(G)$ is the free group generated by the classes of irreducible rational representations. If T is a maximal torus of G, then

$$K(T) = K_r(T) = \mathbb{Z}(\widehat{T}),$$

the algebra of the group $\widehat{T}$ of rational characters of T; one has $\widehat{T} \simeq \mathbb{Z}^r$ if r is the rank of G (this is independent of the characteristic). The Weyl group W operates on T, hence on $K(T)$, in a way which depends only on the operations of W on the free group $\widehat{T}$. There is an evident homomorphism of λ-rings, deduced from the injection of T into G:

$$K_r(G) \longrightarrow K_r(T)^W = K(T)^W, \tag{1.13}$$

where $K(T)^W$ denotes the set of fixed points of W in $K(T)$. The theory of linear representations of G[6] then proves the following theorem.

Theorem 1.1. The homomorphism (1.13) is an isomorphism.

Thus one sees that $K_r(G)$, once the "type" of G is known (characterized by W operating on a group $\mathbb{Z}^r$), does not depend on the characteristic, and that it is a special λ-ring. There is a canonical basis of $K_r(G)$ corresponding to the orbits of W in $\widehat{T}$; another basis is obtained as follows. Let $(\rho_i)_{1\le i\le r'}$ be the classes of representations of G corresponding to the fundamental representations of $G/\operatorname{rad} G$, and let $(\sigma_j)_{1\le j\le r''}$ be a basis of the group of rational homomorphisms from G to k^*. Thus $r' + r'' = r$ (the rank of G).

Corollary. The ring $K_r(G)$ is identified with the subring

$$\mathbb{Z}[\rho_1, \ldots, \rho_{r'}; \sigma_1, \ldots, \sigma_{r''}, \sigma_1^{-1}, \ldots, \sigma_{r''}^{-1}]$$

of the field of rational functions in the ρ_i and the σ_j with coefficients in $\mathbb{Q}$.[7]

In particular, $K_r(G)$ has no zero divisors. If $G = \prod_i \mathrm{Gl}(n_i, k)$, the ring $K_r(G)$ is made explicit as follows. Let ρ_i be the representation of G deduced from the identical representation of $\mathrm{Gl}(n_i, k)$. If one then considers the field of rational functions over $\mathbb{Q}$ in the indeterminates

[5] Unnecessary condition; cf. the preceding footnote.

[6] Cf. Chevalley Seminar 1956/58, *Groupes de Lie algébriques* (École Normale Supérieure).

[7] The derived group G' of G must be supposed simply connected for this statement to be correct.

$\lambda^j(\rho_i)$, $1 \leq j \leq n_i$, then $K_r(G)$ is identified with the subring generated by these indeterminates and the inverses of the $\lambda^{n_i}(\rho_i)$. This justifies our assertion that, at least in characteristic 0, the operations λ^i make it possible to reconstruct all tensor operations on vector bundles; this remains true over an algebraically closed field of arbitrary characteristic, provided one takes reduced classes.

Remarks. a) In Example 1), the ring $K(G)$ is augmented, the augmentation being defined by the degree of representations; by passage to the quotient, this augmentation defines an augmentation on $K_r(G)$. Likewise, the λ-rings considered in Example 2) are augmented when X is k-irreducible; in $K(\xi(X))$, the augmentation is defined by the rank of a vector bundle, that is, the dimension of its fibres.

b) When the field k is not algebraically closed, it is plausible that $K(G)$ is a special λ-ring in characteristic 0, and likewise for $K_r(G)$ in all characteristics.

c) In non-zero characteristic, Theorem 1.1 was graciously supplied by Chevalley.

3. The λ-ring defined by a graded ring

Let A be a graded commutative ring with unit. To simplify, assume that the degrees of A are positive and that $A^0 = \mathbb{Z}$. Denote by $\widehat{A}$ the product of the A^i (this is a commutative ring with unit), and by $\widehat{A}^+$ the kernel of the evident augmentation $\widehat{A} \to A^0 = \mathbb{Z}$. Then $1 + \widehat{A}^+$ is a subgroup of the multiplicative group of invertible elements of $\widehat{A}$. Consider the abelian group

$$\widetilde{A} = \mathbb{Z} \times (1 + \widehat{A}^+), \tag{1.14}$$

whose elements will be denoted $[n, x]$, with $n \in \mathbb{Z}$ and

$$x = 1 + \sum_{i \geq 1} x^i \in 1 + \widehat{A}^+ \qquad (x^i \in A^i).$$

We write this group additively, thus:

$$[n, x] + [n', x'] = [n + n', xx']. \tag{1.15}$$

The zero element of this group is $[0, 1]$. We shall introduce on it an augmented λ-ring structure, compatible with its additive structure, the augmentation being $\varepsilon : [n, x] \mapsto n$. The product $[m, x][n, y]$ is completely characterized by the fact that it is written

$$[mn, x^n y^m (x * y)],$$

where $x * y$ is an element of $1 + \widehat{A}^+$ whose components are expressed as universal polynomials with integral coefficients in the x^i and the y^i:

$$(x * y)^i = Q_i(x^1, \ldots, x^i; y^1, \ldots, y^i) \qquad (i \geq 1), \tag{1.16}$$

the polynomial Q_i being isobaric of weight i. Moreover, $x * y$ must be distributive with respect to the "addition" xy and, for "linear" factors, must reduce to

$$[1, 1 + x^1][1, 1 + y^1] = [1, 1 + x^1 + y^1]. \tag{1.17}$$

The polynomials Q_i are evaluated by calculations with elementary symmetric functions. This is the calculation of the Chern classes of a tensor product of vector bundles by means of the Chern classes x^i and y^i of the factors and of their respective dimensions m and n. From (1.17) one deduces

$$(1 + x^1) * (1 + y^1) = \frac{1 + x^1 + y^1}{(1 + x^1)(1 + y^1)}, \tag{1.17 bis}$$

and an arbitrary product $x * y$ is computed term by term by decomposing formally

$$x = \prod_{i=1}^{n}(1+\alpha_i), \qquad y = \prod_{i=1}^{n}(1+\beta_i),$$

where the α_i and β_i have degree 1.

In this way $\widetilde{A}$ becomes a commutative augmented ring with unit $[1,1]$; it may be interpreted as the ring obtained by adjoining a unit to the ring $1+\widehat{A}^{+}$. Moreover, on $\widetilde{A}$ one defines a filtration by putting $\widetilde{A}^0 = \widetilde{A}$ and denoting by $\widetilde{A}^n$ the set of the $[0,x]$ with $x^i = 0$ for $1 \leq i < n$. This filtration is compatible with the product, for

$$\left(1+\sum_{i\geq m} x^i\right) * \left(1+\sum_{j\geq n} y^j\right) = 1 - \frac{(m+n-1)!}{(m-1)!(n-1)!}x^m y^n + \cdots, \tag{1.18}$$

the omitted terms having degree $> m+n$. This formula shows that the graded ring $G(\widetilde{A})$ associated with $\widetilde{A}$ is identified with A from the additive point of view, but not from the multiplicative point of view. One can only say that the map η' from A to $G(\widetilde{A})$ defined by

$$\eta'(x^i) = (-1)^{i-1}(i-1)!\, x^i \qquad (x^i \in A^i) \tag{1.19}$$

is a ring homomorphism.

On $\widetilde{A}$ introduce operations λ^i satisfying the axioms of λ-rings (Section 1, formulas (1.2)), well determined by the condition that the components of $\lambda^i[0,x]$ be deduced from the components x^i of x by universal polynomials with integral coefficients:

$$\lambda^i[0,x] = [0, \lambda^i x], \qquad (\lambda^i x)^{(n)} = Q_{i,n}(x^1, \ldots, x^n), \tag{1.20}$$

where $Q_{i,n}$ is isobaric of weight n, and such that

$$\lambda^i[1, 1+x^1] = 0 \qquad \text{for } i > 1. \tag{1.21}$$

The polynomials $Q_{i,n}$ are evaluated by a calculation with elementary symmetric functions: it is the calculation of the Chern classes of the exterior powers of a vector bundle by means of the Chern classes and the rank of that bundle. One then observes that $\widetilde{A}$ is a special augmented λ-ring and that the $\widetilde{A}^i$ are stable under the operations λ^j.

The augmented λ-ring $\widetilde{A}$ may be considered as a functor in A. Consequently, the principal automorphism

$$x = \sum_{i\geq 0} x^i \longmapsto \check{x} = \sum_{i\geq 0} (-1)^i x^i$$

of A defines an automorphism of $\widetilde{A}$, denoted by the same symbol $\check{}$.

Proposition 1.2. Let $x \in \widetilde{A}$ be of the form

$$x = [n, 1 + x^1 + \cdots + x^n]$$

(so $x^i = 0$ for $i > n$). Then $\lambda^i(x) = 0$ for $i > n$, one has $\lambda^n(x) = [1, 1+x^1]$, and finally

$$\sum_{i=0}^{n} (-1)^i \lambda^i(x) = [0, 1-(n-1)!x^n + \cdots]. \tag{1.22}$$

From this proposition, and from the fact that the $\widetilde{A}^i$ are stable under the operations λ^j, it follows that formula (1.22) is valid whenever $\varepsilon(x) = n$, whether or not $x^i = 0$ for $i > n$.

Formula (1.22) admits the following generalizations. Let K be any λ-ring, and let n be arbitrary. For $x \in K$, put

$$\gamma^n(x) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x+n) = \lambda^n(x+n-1) \tag{1.23}$$

(the identity of the last two members follows from an immediate induction). In particular $\gamma^0(x) = 1$ and $\gamma^1(x) = x$. Next put

$$\gamma_t(x) = \sum_{n \geq 0} \gamma^n(x) t^n \in K[[t]]. \tag{1.24}$$

One deduces the formula

$$\gamma_t(x) = \lambda_{t/(1-t)}(x), \tag{1.25}$$

which is equivalent to

$$\lambda_s(x) = \gamma_{s/(1+s)}(x). \tag{1.25 bis}$$

This proves that the λ^i are expressed in terms of the γ^i, and that

$$\gamma_t(x+y) = \gamma_t(x)\gamma_t(y).$$

Consequently the maps γ^i define a new λ-ring structure on K. Formula (1.22) then shows that, for $K = \widetilde{A}$ and every $x \in \widetilde{A}$, one has

$$\gamma^n(x - \varepsilon(x)) = [0, 1 + (-1)^{n-1}(n-1)! x^n + \cdots], \tag{1.22 bis}$$

and hence the component x^n is obtained, up to the factor $(-1)^{n-1}(n-1)!$, by reducing modulo $\widetilde{A}^{n+1}$ the element $\gamma^n(x - \varepsilon(x))$ of $\widetilde{A}^n$.

The Chern homomorphism. For every graded ring A satisfying the preceding conditions, we shall define a homomorphism of augmented rings

$$\mathrm{ch} : \widetilde{A} \longrightarrow \widehat{A} \otimes \mathbb{Q}. \tag{1.26}$$

This homomorphism is defined without ambiguity by the condition that it be an additive homomorphism, send 1 to 1, and be such that the components of $\mathrm{ch}(x)$ in degree > 0 are given by universal isobaric polynomials with rational coefficients in the components of x, and that

$$\mathrm{ch}([1, 1 + x^1]) = \exp x^1 = \sum_{n \geq 0} (x^1)^n / n!. \tag{1.27}$$

One immediately obtains the explicit expression of ch. Let η be the additive endomorphism of $\widehat{A} \otimes \mathbb{Q}$ defined, for x^i of degree i, by

$$\eta(x^i) = \frac{1}{(-1)^{i-1}(i-1)!} x^i. \tag{1.28}$$

Then

$$\mathrm{ch}\left([n, 1 + \sum_{i \geq 1} x^i]\right) = n + \eta\left(\log\left(1 + \sum_{i \geq 1} x^i\right)\right). \tag{1.29}$$

Since this formula can be inverted, at least if A has only finitely many homogeneous components, one sees that in this case one has an isomorphism ch from the ring $\widetilde{A} \otimes \mathbb{Q}$ onto the ring $\widehat{A} \otimes \mathbb{Q}$, which completely elucidates the structure of $\widetilde{A} \otimes \mathbb{Q}$.

Recall that, according to Hirzebruch [1], a formal series $f \in \mathbb{Q}[[t]]$ defines by universal polynomial formulas an additive homomorphism

$$1 + \widehat{A}^+ \longrightarrow 1 + (\widehat{A} \otimes \mathbb{Q})^+,$$

denoted $\mathfrak{C}_f$. When $f(t) = t/(1-\exp(-t))$, we write simply $\mathfrak{C}_f = \mathfrak{C}$ (the initial of Todd). We extend the definition of $\mathfrak{C}(x)$ to $x \in \widetilde{A}$. This said, we have the following result.

Proposition 1.3. Let $N \in \widetilde{A}$ be of the form

$$N = [q, 1 + N^1 + \cdots + N^q]$$

(so $N^i = 0$ for $i > q$). If one puts

$$\lambda_{-1}(N) = \sum_{i \geq 0} (-1)^i \lambda^i(N) = \sum_{i=0}^{n} (-1)^i \lambda^i(N),$$

then

$$\mathrm{ch}(\lambda_{-1}(N)) = (-1)^q N^q \mathfrak{C}(\check{N})^{-1}, \tag{1.30}$$

which is equivalent to

$$\mathrm{ch}(\lambda_{-1}(\check{N})) = N^q \mathfrak{C}(N)^{-1}. \tag{1.30 bis}$$

4. The operations $\lambda^p(N, x)$

Let A be the ring of formal series, with coefficients in $\mathbb{Z}$, in two infinite sequences of indeterminates denoted $\lambda^i N$ and $\lambda^i x$ ($i \geq 1$). Put $\lambda^1 N = N$ and $\lambda^1 x = x$, and introduce on A the unique special λ-ring structure for which

$$\lambda^i(N) = \lambda^i N, \qquad \lambda^i(x) = \lambda^i x,$$

and for which the operations λ^i are continuous. If one restricted oneself to polynomials in the indeterminates $\lambda^i N$ and $\lambda^i x$, one would have the λ-ring freely generated by N and x; our ring A is its completion. One may then form

$$\lambda_{-1}(N) = \sum_{i \geq 0} (-1)^i \lambda^i N, \tag{1.31}$$

which is a formal series with constant term 1, hence invertible. Thus an element $\lambda^p(N, x)$ of A can be defined without ambiguity by putting

$$\lambda^p(N, x) \lambda_{-1}(N) = \lambda^p(x \lambda_{-1}(N)). \tag{1.32}$$

Theorem 1.4. Let J_q be the closed ideal of A generated by the $\lambda^i N$ with $i > q$. Identify A/J_q with the ring of formal series in the $\lambda^i x$ and the $\lambda^j N$ ($1 \leq j \leq q$). Then $\lambda^p(N, x)$ reduced modulo J_q is a polynomial, isobaric of weight p with respect to the variables $\lambda^i x$.

Let N and F be two vector spaces of finite dimensions q, q' over an algebraically closed field k of characteristic 0. Let $N^{(p)}$ be the kernel of the map

$$(a_1, \ldots, a_p) \longmapsto a_1 + \cdots + a_p$$

from N^p to N. We make the symmetric group $\mathfrak{S}_p$ operate on N^p, hence on $N^{(p)}$, hence on $\Lambda(N^{(p)})$, and then on

$$\Lambda(N^{(p)}) \otimes \bigotimes^p F.$$

Consider this vector space as a graded module over $\mathfrak{S}_p \times \mathrm{Gl}(N) \times \mathrm{Gl}(F)$, and take its "alternating part", which is a graded module over $\mathrm{Gl}(N) \times \mathrm{Gl}(F) = G$. Then

$$\lambda^p(N, F) = E_G\left(\left(\Lambda(N^{(p)}) \otimes \bigotimes^p F\right)^{\mathrm{alt}}\right). \tag{1.33}$$

N.B. If M is a complex of G-modules, $E_G(M)$ denotes the alternating sum in $K(G)$ of the classes of the components of M, that is, the Euler–Poincaré characteristic of M. In formula (1.33), the first member $\lambda^p(N,F)$ denotes the element of the ring $K(G)$ obtained by substituting, for the variables λ^i_N and λ^i_F, the classes in $K(G)$ of the G-modules $\lambda^i(N)$ and $\lambda^i(F)$, respectively.

Proof. Expanding the second member of (1.32) by the second formula (1.11), one sees that the first assertion is proved if one proves the analogous assertion obtained by setting $x = 1$, i.e. if one proves that in the λ-ring $\mathbb{Z}[\lambda^1 N, \dots, \lambda^q N]$, where $\lambda^i N = 0$ for $i > q$, the element $\lambda^p(\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$. A priori, the quotient

$$\lambda^p(N,1) = \lambda^p(\lambda_{-1}(N))/\lambda_{-1}(N)$$

is a formal series contained in the quotient field K of the ring of formal series in the $\lambda^i N$ for $1 \le i \le q$. On the other hand, also denote by N a vector space of dimension q over an algebraically closed field k of characteristic 0, and consider the element

$$\mathscr{L}^p(N) = E_G((\Lambda(N^{(p)}))^{\mathrm{alt}}) \in K(G),$$

where now $G = \mathrm{Gl}(N)$. We know (Section 2, Theorem 1.3) that $K(G)$ is identified with the subring of K generated by the λ^i_N for $1 \le i \le q$ and by the inverse of λ^q_N. Moreover, in this ring one verifies easily that

$$\mathscr{L}^p(N)\lambda_{-1}(N) = \lambda^p(\lambda_{-1}(N)).$$

Since we are in a large field K and $\lambda_{-1}(N) \neq 0$, it follows that $\mathscr{L}^p(N) = \lambda^p(N,1)$. Both members lie in the intersection of $K(G)$ with the ring of formal series in the λ^i_N, hence are polynomials in the λ^i_N.

To prove (1.33) in the general case, one verifies that the second member satisfies in $K(G)$ the formula analogous to (1.32), that is, that its product by $\lambda_{-1}(N)$ is $\lambda^p(F\lambda_{-1}(N))$. Likewise, the product of $\lambda^p(N,F) \in K(G)$ by $\lambda_{-1}(N)$ is $\lambda^p(F\lambda_{-1}(N))$, as follows from the fact that $K(G)$ is special. Since $K(G)$ is integral (cf. Section 2), and $\lambda_{-1}(N) \neq 0$ (same reference), formula (1.33) is proved. QED.

Remarks. a) When the dimension q' of F is at least p, formula (1.33) characterizes the polynomial $\lambda^p(N,x)$ in the λ^i_N $(1 \le i \le q)$ and the λ^j_x $(1 \le j \le p)$.

b) It is plausible that (1.33) remains valid if the field k is not algebraically closed.[8] One would have to resolve this question if one wanted to prove the Riemann–Roch theorem over a non-algebraically-closed field of characteristic 0 by the method to be explained. The passage from an $\mathfrak{S}_p$-module to its alternating part has reasonable meaning only when the base field has characteristic 0.

Let now K be any special λ-ring and let N be an element of K such that $\lambda^i N = 0$ for all sufficiently large i. Then for every $x \in K$ and every integer $p \ge 1$ one can consider the elements $\lambda^p(N,x)$, which are expressed by universal polynomials with integral coefficients in the λ^i_N and the λ^j_x. One has $\lambda^1(N,x) = x$ and formula (1.32) in K. To express the properties of the operations $x \mapsto \lambda^p(N,x)$, it is convenient to introduce a new λ-ring, denoted K_N. As a commutative ring, K_N is obtained by adjoining a unit to the ring whose underlying additive group is K and whose multiplication is given by

$$x_N y = xy\mu \qquad (\mu = \lambda_{-1}(N)). \tag{1.34}$$

The map λ_t from K_N to $K_N[[t]]$ is defined by

$$\lambda_t(n,x) = (1+t)^n \sum_{p \ge 0} \lambda^p(N,x)t^p, \tag{1.35}$$

[8] This is in fact proved by the same method.

where $\lambda^0(N,x) = (1,0)$ (the unit element of K_N). I say that, with these operations, K_N is a special λ-ring. Thus one has to verify the formulas

$$\lambda_t(X+Y) = \lambda_t(X)\lambda_t(Y), \qquad \lambda_t(XY) = \lambda_t(X)\lambda_t(Y), \qquad \lambda_t(\lambda^i(X)) = \lambda^i(\lambda_t(X))$$

for $X, Y \in K_N$. One may restrict oneself to verifying these formulas when X and Y lie in K, and one checks that it suffices to verify them when K is the free special λ-ring generated by X, Y and N subject to the relations $\lambda^i N = 0$ for $i > q$. But the group homomorphism

$$K_N \longrightarrow K, \tag{1.36}$$

which sends unit to unit and on K reduces to the homomorphism $x \mapsto x\mu$, is a homomorphism compatible with the ring structures and the maps λ^i, as one verifies immediately. In the present case, its restriction to K is injective. Since K is a special λ-ring, it follows easily that the above relations are indeed verified for $X, Y \in K$, completing the proof of our assertion.

According to formula (1.23), it is natural, for $x \in K$ and $n \geq 1$, to put

$$\gamma^n(N,x) = \sum_{i=0}^{n-1} \binom{n-1}{i} \lambda^i(N,x), \tag{1.37}$$

in other words, $\gamma^n(N,x) = \gamma^n(x_N)$, where x_N denotes x considered as an element of the λ-ring K_N.

Proposition 1.5. Let A be a graded commutative ring with unit, let N be an element of $\widetilde{A}$ of the form

$$N = [q, 1 + N^1 + \cdots + N^q],$$

and let x be an arbitrary element of $\widetilde{A}$:

$$x = [\nu, 1 + \sum_{i \geq 1} x^i].$$

Then, for every integer $j \geq 0$, one has $\gamma^{q+j}(N,x) \in \widetilde{A}^j$, and the component of degree j of $\gamma^{q+j}(N,x)$ is of the form

$$(\gamma^{q+j}(N,x))^{(j)} = (-1)^{j-1}(j-1)!\, G_{q,j}(\nu, x^1, \ldots, x^j; N^1, \ldots, N^j), \tag{1.38}$$

where $G_{q,j}$ is a universal polynomial with integral coefficients characterized by the condition

$$G_{q,j}(\nu, x^1, \ldots, N^j)\,(N^q) = (-1)^q (x * \lambda_{-1}(N))^{(q+j)}. \tag{1.39}$$

Proof. We may suppose that A is the ring of polynomials with integral coefficients in indeterminates x^i and N^j $(1 \leq j \leq q)$. Since, combining (1.32) and (1.37), one has

$$\gamma^n(N,x)\lambda_{-1}(N) = \gamma^n(x\lambda_{-1}(N)), \tag{1.40}$$

and, for $n = q+j$, the second member has filtration $\geq q+j$ (use formula (1.22 bis), taking account of the fact that $x\lambda_{-1}(N)$ has zero augmentation), while $\lambda_{-1}(N)$ has filtration q (formula (1.22)), it follows that $\gamma^n(N,x)$ has filtration j, since by (1.18) the associated graded of A is integral here. Using the preceding formulas, one immediately obtains the indicated form for $(\gamma^{q+j}(N,x))^{(j)}$. QED.

Chapter II. Classes of coherent algebraic sheaves and Chern classes

1. The theory of Chow

Throughout this chapter, a base field k is fixed; for simplicity it will be assumed algebraically closed. A large part of what follows, perhaps all of it, is nevertheless valid without this restriction. The algebraic spaces and the regular maps considered will be defined over k.

An algebraic space is called *quasi-projective* if it is isomorphic to a locally closed subset of a projective space.

Let X be a nonsingular connected quasi-projective variety of dimension n. Denote by $A(X)$, and call the *Chow ring* of X, the graded ring of classes of cycles on X modulo rational equivalence. Thus $A^i(X)$ is formed from the classes of cycles of dimension $n-i$. The fact that $A(X)$ is indeed a graded ring follows from the following theorem of Chow.

Theorem 2.1.[9] Every cycle on X is rationally equivalent to a nonsingular cycle. Given elements of $A^i(X)$ and $A^j(X)$, one can find nonsingular representatives of these elements, say Y^{n-i} and Z^{n-j}, which meet everywhere transversely.

A cycle is called nonsingular if its components are nonsingular subvarieties. Two cycles are said to meet everywhere transversely if every component of one cuts every component of the other transversely, i.e. if the two secant varieties are nonsingular at their intersection points and their tangent varieties there are in general position.

Let f be a morphism, i.e. a regular map, from X to Y (where X and Y are quasi-projective and nonsingular). Then f defines, using the preceding theorem, a homomorphism of graded rings

$$f^* : A(Y) \longrightarrow A(X), \tag{2.1}$$

namely inverse image of cycle classes. Thus $A(X)$ becomes a contravariant functor in X.

Let f be a proper morphism from X^n to Y^m (i.e. in Weil's terminology, the graph of f is complete over Y, cf. [3]). Then f also defines an additive homomorphism

$$f_* : A(X^n) \longrightarrow A(Y^m), \tag{2.2}$$

preserving the dimension of cycles, hence increasing the degree by $m-n$.

Thus, relative to proper morphisms, and ignoring multiplicative structures and gradings, $A(X)$ is a covariant functor in X. Let us record the classical formula

$$f_*(x \cdot f^*(y)) = f_*(x) \cdot y. \tag{2.3}$$

We shall also use Chow's theory of Chern classes. It consists in attaching to every vector bundle E on X (still assumed quasi-projective and nonsingular) Chern classes

$$c^i(E) \in A^i(X) \qquad (i \geq 1),$$

in such a way as to satisfy the following conditions. Put $c^0(E) = 1$ and

$$\widetilde{C}(E) = \left[\operatorname{rank} E, \sum_{i \geq 0} c^i(E)\right] \in \widetilde{A}(X) \tag{2.4}$$

[9] As a result of a misunderstanding on the author's part, the stated theorem is more than is presently known. In fact, one uses only the "moving lemma" of Chow, saying that two cycles can be moved in their classes so that their intersection is not excessive; cf. Chevalley Seminar 1958, *Anneau de Chow et applications*.

(see Chapter I, §3 for the definition of $\widetilde{A}(X)$). The conditions to be verified by a theory of Chern classes are then expressed as follows:

$$\widetilde{C}(E) = \widetilde{C}(E') + \widetilde{C}(E'') \tag{2.5}$$

if the vector bundle E is an extension of E' by E''. If one puts

$$C(E) = \sum_i c^i(E),$$

this condition is expressed by $C(E) = C(E')C(E'')$. Formula (2.5) permits one to extend $\widetilde{C}$ to an additive homomorphism from $K(\xi(X))$ (cf. Chapter I, §2, Example 2) to $\widetilde{A}(X)$:

$$\widetilde{C} : K(\xi(X)) \longrightarrow \widetilde{A}(X), \tag{2.6}$$

and one wants $\widetilde{C}$ to be a λ-homomorphism of λ-rings. This last condition may also be written

$$\widetilde{C}(E \otimes F) = \widetilde{C}(E)\widetilde{C}(F), \qquad \widetilde{C}(\lambda^i E) = \lambda^i \widetilde{C}(E), \tag{2.7}$$

for two vector bundles E and F; of course $\widetilde{C}$ is automatically compatible with augmentations. Let D be a divisor on X and let $L(D)$ be the vector bundle it defines. One wants

$$c^1(L(D)) = \ell(D), \qquad c^i(L(D)) = 0 \quad (i > 1), \tag{2.8}$$

where, for any cycle Z, $\ell(Z)$ denotes its class in $A(X)$. Finally, one imposes that the c^i be functorial, i.e. that if f is a morphism from X to Y and E is a vector bundle on Y, then

$$c^i(f^!(E)) = f^*(c^i(E)), \tag{2.9}$$

where $f^!(E)$ is the inverse image of E on X. One may also write (2.9) in the following form: the notion of inverse image of vector bundles defines a λ-homomorphism of augmented λ-rings

$$f^! : K(Y) \longrightarrow K(X),[10] \tag{2.10}$$

where here one writes $K(X)$ instead of $K(\xi(X))$, and (2.9) means that there is commutativity in the diagram

$$\begin{tikzcd} K(Y) \arrow[r, "f^!"] \arrow[d, "\widetilde{C}_Y"'] & K(X) \arrow[d, "\widetilde{C}_X"] \\ A(Y) \arrow[r, "f^*"'] & A(X). \end{tikzcd}$$

Here the space on which $\widetilde{C}$ is considered has been put as a subscript.

Theorem 2.2. There exists a theory of Chern classes satisfying conditions (2.5), (2.7), (2.8), and (2.9).

We shall also admit the following proposition.

Proposition 2.3. Let Y be a closed subset of X and let U be its complement. If x is an element of $A(X)$ inducing 0 on U, then there exists a representative of x which is a cycle carried by Y.

Corollary. Let $p = \dim X - \dim Y$. Then, if $i < p$, the homomorphism

$$A^i(X) \longrightarrow A^i(U)$$

is bijective. Every element of the kernel of $A^p(X) \to A^p(U)$ is a linear combination of components of dimension $n - p$ of Y; hence it is proportional to $\ell(Y)$ if Y is irreducible.

[10] The notation $f^!$ used here is in conflict with that used in duality theory (cf., for example, R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics, no. 20, Springer, 1966). This is why in this seminar we replace it by the notation f^*.

2. Definition of the Chern classes of coherent algebraic sheaves

Let $\mathcal{C}$ be an abelian category, and let $\mathcal{C}_0$ be a subclass of $\mathcal{C}$. Denote by $K(\mathcal{C}_0)$ the quotient group of the free group generated by the isomorphism classes of elements of $\mathcal{C}_0$ by the subgroup generated by the elements of the form

$$(E) - (E') - (E''),$$

where E is an extension of E' by E'' and $E, E', E'' \in \mathcal{C}_0$. We must suppose that the classes of elements of $\mathcal{C}_0$, for the isomorphism relation, form a set. Denote by $\gamma_{\mathcal{C}_0}(E)$ the class of an object $E \in \mathcal{C}_0$ in $K(\mathcal{C}_0)$, and omit the subscript $\mathcal{C}_0$ when there is no danger of confusion. The group $K(\mathcal{C}_0)$ is the solution of a universal problem relative to functions

$$E \longmapsto f(E)$$

defined on $\mathcal{C}_0$, with values in an arbitrary abelian group A, which are additive in the sense that $f(E) = f(E') + f(E'')$ whenever E is an extension of E' by E''.

Let F be an additive functor from $\mathcal{C}$ to $\mathcal{C}'$ and let $\mathcal{C}'_0$ be a subclass of $\mathcal{C}'$ satisfying the same condition as $\mathcal{C}_0$, such that for every $A \in \mathcal{C}_0$ the $R^qF(A)$ are in $\mathcal{C}'_0$ and are zero for q sufficiently large. We suppose that injective resolutions exist in $\mathcal{C}$, so that the derived functors R^qF of F exist. Then F defines a homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}'_0)$, still denoted F, by the formula

$$F(\gamma_{\mathcal{C}_0}(E)) = \sum_{i\geq 0} (-1)^i \gamma_{\mathcal{C}'_0}(R^iF(E)). \tag{2.11}$$

More generally, if one has a cohomological functor (F^i) from $\mathcal{C}$ to $\mathcal{C}'$ such that for every $E \in \mathcal{C}_0$ the $F^i(E)$ are in $\mathcal{C}'_0$ and are zero except for a finite number of i, one obtains a homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}'_0)$. These remarks extend to multifunctors; moreover, the classical spectral sequences give transitivity properties which we shall not make explicit.

In particular, if F is an exact functor, formula (2.11) simplifies to

$$F(\gamma_{\mathcal{C}_0}(E)) = \gamma_{\mathcal{C}'_0}(F(E)). \tag{2.11 bis}$$

This applies in particular if $\mathcal{C} = \mathcal{C}'$, if F is the identity functor, and if $\mathcal{C}_0 \subset \mathcal{C}'_0$; one has a canonical homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}'_0)$.

Theorem 2.3. Let $\mathcal{C}$ be an abelian category and let $\mathcal{C}_0 \subset \mathcal{C}_1$ be two subclasses such that the isomorphism classes of objects of $\mathcal{C}_1$ form a set. Suppose that the following conditions are satisfied:

(i) For every exact sequence $0 \to A' \to A \to A'' \to 0$ in $\mathcal{C}$ such that A and A'' are in $\mathcal{C}_0$ (respectively $\mathcal{C}_1$), one has $A' \in \mathcal{C}_0$ (respectively $A' \in \mathcal{C}_1$).

(ii) Let $u : A \to B$ and $v : C \to B$, with $A, B, C \in \mathcal{C}_1$ and u surjective. Then there exist $L \in \mathcal{C}_0$ and homomorphisms $u' : L \to C$ and $v' : L \to A$ such that u' is surjective and $vu' = uv'$.

(iii) For every $A \in \mathcal{C}_1$ there exists an integer $d = d(A)$ such that for every left resolution L of A by objects of $\mathcal{C}_0$ one has $Z_d(L) \in \mathcal{C}_0$, where Z_d denotes the cycles in dimension d.

Under these conditions, the canonical homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}_1)$ is an isomorphism.

Principle of the proof. Let $A \in \mathcal{C}_1$. By (ii) and (iii) there exists a finite resolution L of A by objects of $\mathcal{C}_0$; put

$$\ell(A) = \sum_{i\geq 0} (-1)^i \gamma_{\mathcal{C}_0}(L_i).$$

One shows that the element of $K(\mathcal{C}_0)$ thus constructed is independent of the resolution L chosen. If one has a second resolution L', one "caps" L and L' by a third resolution L'', with surjective homomorphisms from L'' to L and L'; this is possible by (i) and (ii). One is then reduced to proving that $E_{\mathcal{C}_0}(L) = E_{\mathcal{C}_0}(L'')$. But the difference of the two is $E_{\mathcal{C}_0}(N)$, where N is the kernel of the surjective homomorphism from L'' to L; this kernel is in $\mathcal{C}_0$ by (i). Since N is an acyclic complex in all dimensions, $E_{\mathcal{C}_0}(N) = 0$, whence the conclusion. Here $E_{\mathcal{C}_0}(L)$, for a finite complex with values in $\mathcal{C}_0$, denotes the Euler–Poincaré characteristic

$$\sum_i (-1)^i \gamma_{\mathcal{C}_0}(L_i).$$

One proves analogously that the function $\ell : \mathcal{C}_1 \to K(\mathcal{C}_0)$ thus constructed is additive; hence it defines a homomorphism from $K(\mathcal{C}_1)$ to $K(\mathcal{C}_0)$, and it is immediate that this homomorphism is inverse to the canonical homomorphism from $K(\mathcal{C}_0)$ to $K(\mathcal{C}_1)$. QED.

Remark. To verify (ii), it is enough to verify that $\mathcal{C}_0$ is contained in a subclass $\mathcal{C}_2$ of $\mathcal{C}$ satisfying the two conditions

(ii a) every $A \in \mathcal{C}_1$ is isomorphic to a quotient of an $L \in \mathcal{C}_2$;

(ii b) if $u : A \to B$, with $A, B \in \mathcal{C}_2$, then $\operatorname{Ker} u \in \mathcal{C}_1$.

Corollary. Let X be a quasi-projective algebraic space. Let $\mathscr{F}(X)$ be the category of coherent algebraic sheaves on X, let $\xi(X)$ be the category of coherent locally free algebraic sheaves, equivalently vector bundles on X, and let $\mathscr{F}_0(X)$ be the category of coherent algebraic sheaves F such that, for every $x \in X$, the stalk F_x is an $\mathcal{O}_x$-module of finite cohomological dimension. Thus $\xi(X) \subset \mathscr{F}_0(X) \subset \mathscr{F}(X)$. Then the canonical homomorphism

$$K(\xi(X)) \longrightarrow K(\mathscr{F}_0(X)) \tag{2.12}$$

is an isomorphism.

We verify the conditions of Theorem 2.3. Conditions (i) and (iii) follow from the definition of cohomological dimension and from the fact that a finite-type projective module over a noetherian local ring $\mathcal{O}_x$ is free. To verify (ii), we shall prove that $\mathcal{C} = \mathscr{F}(X)$ satisfies conditions (ii a) and (ii b) of the remark. The second is trivial; therefore it remains to prove the following result.

Proposition 2.4 (Serre). Let X be a quasi-projective algebraic space. Then every coherent algebraic sheaf on X is isomorphic to the quotient of a locally free algebraic sheaf.

This proposition is true if X is a closed subset of projective space by [2], since, for n large enough, $F(n)$ is generated by its sections, which means that F is a quotient of $\mathcal{O}(-n)^k$. In the general case, X is an open subset of a closed subset of a projective space, and it suffices to use the following result, due to Cartier and Serre in particular cases.

Proposition 2.5. Let F be a coherent algebraic sheaf on an open subset U of an algebraic space X. Then F is isomorphic to the restriction of a coherent algebraic sheaf on X.

By applying Zorn's lemma to the opens containing U on which one can extend F, one is reduced to the case where X is affine. Let $\overline{F}$ be the algebraic sheaf on X whose sections on an open V are the sections of F on $U \cap V$. One checks easily that $\overline{F}$ is quasi-coherent, i.e. defined on every affine open V—hence here on all of X—by a module, not necessarily of finite type, over the coordinate ring of that open. This is immediate if U itself is affine, since $\overline{F}$ is then defined on the open V by $\Gamma(U \cap V, F)$ considered as a module over $\Gamma(V, \mathcal{O}_X)$. In the general case, U is a finite union of affine opens U_i, hence $\overline{F}$ is the kernel of the evident homomorphism of quasi-coherent sheaves

$$\prod_i \overline{F}|_{U_i} \longrightarrow \prod_{i,j} \overline{F}|_{U_i \cap U_j},$$

and is therefore itself quasi-coherent. Since X is affine, $\overline{F}$ is generated by its sections. Since $\overline{F}|_U = F$ is coherent, there exists a finite number of sections of $\overline{F}$ which generate F on U. These sections generate a coherent subsheaf of $\overline{F}$ whose restriction to U is isomorphic to F. QED.

Now suppose that X is nonsingular and quasi-projective. By the theorem on syzygies, one has, with the notation of the corollary to Theorem 2.3,

$$\mathscr{F}_0(X) = \mathscr{F}(X).$$

In this case, therefore,

$$K(\xi(X)) = K(\mathscr{F}(X)). \tag{2.13}$$

One may express this isomorphism as follows.

Corollary 2 to Theorem 2.3. Let X be a nonsingular quasi-projective variety. For every abelian group A, there is a one-to-one correspondence between additive functions f of vector bundles on X with values in A, and additive functions g of coherent algebraic sheaves on X with values in A. To every f there corresponds the unique g such that

$$f(E) = g(\mathcal{O}_X(E)) \tag{2.14}$$

for every vector bundle E on X, where $\mathcal{O}_X(E)$ denotes the sheaf of germs of regular sections of E on X.

To compute $g(F)$ for an arbitrary coherent algebraic sheaf, one considers a finite resolution of F by sheaves of the form $\mathcal{O}_X(E_i)$, and one then has

$$g(F) = \sum_i (-1)^i f(E_i). \tag{2.14 bis}$$

In particular, if X is connected, in §1 we considered an additive function $\widetilde{C}(E)$ of the vector bundle E, with values in the group $A = \widetilde{A}(X)$. We deduce from it an additive function of coherent sheaves, say $\widetilde{C}(F)$, and, by passing to components, functions

$$F \longmapsto c^i(F) \in A^i(X).$$

The $c^i(F)$ are called the *Chern classes of the coherent sheaf* F. Theoretically they are computed from the Chern classes of vector bundles by (2.14 bis), with the understanding that in the second member one must pass to multiplicative notation. As for the component of $\widetilde{C}(F)$ in $\mathbb{Z}$, it is the rank of the sheaf F, corresponding to the rank of a module over an integral domain: if X is affine, to which one can always reduce, F is defined by a finite-type module over the coordinate ring of X, and one takes the rank of this module.

3. Functorial generalities on $K(X)$

If X is an algebraic space, put, for brevity,

$$K(X) = K(\mathscr{F}(X)).$$

When X is nonsingular and quasi-projective, identify this group with $K(\xi(X))$ by the corollary 2 to Theorem 2.3. Since $K(\xi(X))$ is not only an abelian group but a λ-ring, it follows by transport of structure that $K(X)$ is also a λ-ring, and is therefore provided with a multiplicative structure and operations λ^i. It is important to define these structures directly, without passing through vector bundles; this will be done below.

If E is a vector bundle on X, denote by $\gamma(E)$ its class in $K(X)$. If F is a coherent algebraic sheaf on X, $\gamma(F)$ denotes in the same way its class in $K(X)$. If Y is an irreducible subvariety of

X, put $\gamma(Y) = \gamma(\mathcal{O}_Y)$, where $\mathcal{O}_Y$ denotes the sheaf of local rings of Y, considered as a coherent algebraic sheaf on X. By linearity, one defines the symbol $\gamma(Y)$ when Y is any cycle on X. When there is risk of confusion, one writes γ_X for γ.

Let X be an arbitrary algebraic space. We introduce on $K(X)$ a filtration, denoting by $K(X)^i$ the subgroup of $K(X)$ generated by the $\gamma(F)$ with

$$\dim \operatorname{Supp} F \leq n - i,$$

where n is the dimension of X. From Serre's dévissage lemma, in the form given in [3], follows the following result.

Proposition 2.6. The group $K(X)^i$ is generated by the $\gamma(Y)$, where Y runs through the subvarieties of dimension $n - i$ of X. If X is irreducible, then $\operatorname{Gr}^0(K(X)) = \mathbb{Z}$.

The latter isomorphism is given by the homomorphism from $K(X)$ to $\mathbb{Z}$ defined with the aid of the notion of rank of a coherent algebraic sheaf.

Now suppose that X is nonsingular. Following the developments at the beginning of §2, define a multiplication in $K(X)$ by putting

$$\gamma(F)\gamma(G) = \sum_{i \geq 0} (-1)^i \gamma(\operatorname{Tor}_i(F, G)), \tag{2.15}$$

the Tor_i being taken, of course, with respect to the local rings of X. The exact sequence of the Tor_i shows that formula (2.15) indeed defines a biadditive map from $K(X) \times K(X)$ to $K(X)$; associativity follows from the spectral sequence of multiple Tors. One also verifies in this way that one can compute the product of several factors by the formula

$$\gamma(F_1) \cdots \gamma(F_n) = \sum_{i \geq 0} (-1)^i \gamma(\operatorname{Tor}_i(F_1, \ldots, F_n)), \tag{2.16}$$

where, in the second member, the Tor_i are simultaneous Tors, computed by simultaneous projective resolution of the arguments F_i.

If in (2.15) one supposes that G is locally free, one finds

$$\gamma(F)\gamma(G) = \gamma(F \otimes G), \tag{2.15 bis}$$

which shows in particular that $\gamma(\mathcal{O}_X)$ is the unit element of $K(X)$, and that the canonical homomorphism from $K(\xi(X))$ to $K(X)$ is a homomorphism of rings.

Formula (2.15) is evidently suggested by Serre's "Tor formula" in intersection theory. This formula proves the second part of the following proposition, the first part following immediately from (2.15) and from a local Tor calculation.

Proposition 2.7. Let X be a nonsingular variety, and let Y and Z be two cycles in X. If Y and Z meet everywhere transversely, then

$$\gamma(Y)\gamma(Z) = \gamma(Y \cdot Z). \tag{2.17}$$

If Y and Z are homogeneous and of respective dimensions $n - i$ and $n - j$, and if they meet without an excessive component, then

$$\gamma_X(Y)\gamma_X(Z) = \gamma_X(Y \cdot Z) \mod K(X)^{i+j+1}. \tag{2.18}$$

One must be careful: equality does not in general hold in (2.18). We shall see in §4 that, if X is nonsingular and quasi-projective, the ring structure of $K(X)$ is compatible with its filtration, and that the associated graded ring $\mathrm{Gr}(K(X))$ is identified with a quotient of $A(X)$ by a subgroup of torsion, perhaps always zero.[11]

I do not know how to define directly the $\lambda^i(F)$ in terms of sheaves except when the field k has characteristic 0; this is the reason that has prevented us from proving the Riemann–Roch theorem when k is not of characteristic 0. Let F be a coherent algebraic sheaf on X on which the group $\mathfrak{S}_p$ of permutations of p letters operates. Denote by F^{alt} the set of $x \in F$ such that

$$s \cdot x = \varepsilon(s)x$$

for every $s \in \mathfrak{S}_p$, where $\varepsilon(s)$ is the signature of s. It is evidently a coherent algebraic subsheaf of F. This said, one has:

Theorem 2.8. Let X be a nonsingular quasi-projective variety over a field k of characteristic 0. Let F be a coherent algebraic sheaf on X. Then

$$\lambda^p(\gamma(F)) = \sum_{i\geq 0}(-1)^i\gamma\left((\mathrm{Tor}_i(F,F,\dots,F))^{\mathrm{alt}}\right). \tag{2.19}$$

Here it is clear that the group $\mathfrak{S}_p$ operates on the sheaf $\mathrm{Tor}_i(F,\dots,F)$.

Principle of the proof. One considers a finite resolution L of F by locally free sheaves L_i, and uses the following general lemma.

Lemma 2.9. Let $L = (L_i)$ be a graded locally free coherent algebraic sheaf with only finitely many nonzero components. In $K(X)$ one has the formula

$$E\left((\otimes^p L)^{\mathrm{alt}}\right) = \lambda^p(E(L)). \tag{2.20}$$

In this formula, $\otimes^p L$ is considered as a sheaf on which the group $\mathfrak{S}_p$ acts, taking the gradings into account; the gradings therefore introduce signs in the usual way, cf. Cartan–Eilenberg. Moreover, if M is a graded coherent algebraic sheaf on X, put

$$E(M) = \sum_i (-1)^i\gamma(M_i).$$

To prove formula (2.20), one is reduced to proving the analogous formula in $K(G)$, where G is an algebraic group, for instance a product of groups $\mathrm{Gl}(n,k)$, and where L is a finite-dimensional G-module over k (cf. Chapter I, §2, Example 2, for the definition of $K(G)$). In that case, this formula has already been used in Theorem 1.4 to show that the second member of (1.33) satisfies the same functional equation as the first member; the verification is elementary. QED.

Let f be a morphism from X to Y, where Y is an algebraic space. The notion of inverse image of vector bundle defines a homomorphism

$$f^! : K(\xi(Y)) \longrightarrow K(\xi(X))$$

of λ-rings. Identifying vector bundles with locally free coherent algebraic sheaves, the notion of inverse image of vector bundle corresponds to the notion of inverse image of algebraic sheaf, already used in [3, no. 8]. When Y is nonsingular, one defines more generally a homomorphism of groups

$$f^! : K(Y) \longrightarrow K(X) \tag{2.21}$$

[11] No: for a counterexample see this seminar, XIV 4.7.

by a formula involving an alternating sum of Tors, inspired by the developments at the beginning of §2, and which the reader will spell out. One then has commutativity in the following diagram:

$$\begin{tikzcd} K(\xi(Y)) \arrow[r, "f^!"] \arrow[d] & K(\xi(X)) \arrow[d] \\ K(Y) \arrow[r, "f^!"'] & K(X), \end{tikzcd}$$

which shows in particular, when X and Y are both nonsingular and quasi-projective, that the two definitions of $f^!$ coincide under the identifications of $K(X)$ with $K(\xi(X))$ and of $K(Y)$ with $K(\xi(Y))$.

Another case where one can define a homomorphism (2.21), without Y necessarily nonsingular, is the case where the functor $F \mapsto f^!(F)$ from $\mathscr{F}(Y)$ to $\mathscr{F}(X)$ is exact, i.e. where the local rings of X are flat modules over the local rings of Y. It is enough then to put

$$f^!(\gamma_Y(F)) = \gamma_X(f^!(F)).$$

This case occurs in particular when X is a locally trivial fibre space over Y; to see this, one is reduced to the case of a product $X = Y \times T$. Note that one also verifies that in this case one has $f^!(\mathcal{O}_Z) = \mathcal{O}_{f^{-1}(Z)}$ if Z is an irreducible subvariety of Y, whence

$$f^!(\gamma_Y(Z)) = \gamma_X(f^{-1}(Z)) \tag{2.22}$$

if X is flat over Y. Without this latter condition, formula (2.22) becomes false in general, but one has the following analogue of Proposition 2.7.

Proposition 2.8. Let $f : X \to Y$ be a morphism of nonsingular varieties, and let Z be a cycle on Y. If Z is everywhere transverse to f, formula (2.22) is exact. If Z is homogeneous of codimension i and if $f^{-1}(Z)$ has no excessive component, then

$$f^!(\gamma_Y(Z)) = \gamma_X(f^{-1}(Z)) \mod K(X)^{i+1}. \tag{2.23}$$

The proof is analogous to that of Proposition 2.7.

Finally, let X and Y again be arbitrary algebraic spaces, and let f be a proper morphism from X to Y. For every coherent algebraic sheaf F on X, the direct image sheaf $f_*(F)$ is coherent algebraic, and the same holds more generally for the sheaves $R^i f_*(F)$ (cf. [3]). Recall that, by definition, $R^i f_*(F)$ is the sheaf on Y associated with the presheaf which sends an open U to the group $H^i(f^{-1}(U), F)$; in particular

$$R^0 f_*(F) = f_*(F), \qquad R^q f_*(F) = 0 \quad (q > \dim X), \tag{2.24}$$

and moreover

$$\Gamma(U, R^q f_*(F)) = H^q(f^{-1}(U), F) \tag{2.25}$$

if the open U is affine. According to §2, one defines a homomorphism from $K(X)$ to $K(Y)$, still denoted f_*, by the formula

$$f_*(\gamma(F)) = \sum_{q=0}^{\infty} (-1)^q \gamma(R^q f_*(F)). \tag{2.26}$$

Thus $K(X)$ becomes a covariant functor in X relative to proper morphisms. The relation $(fg)_* = f_* g_*$, just like the relation $(fg)^! = g^! f^!$ which we forgot to mention in its place, is a particular case of the transitivity relations alluded to in §2 and which follow from the spectral sequences of composed functors.

Let us note the following formula, valid for Y nonsingular and particularly easy to verify when Y is also quasi-projective by taking for y the class of a vector bundle on Y:

$$f_*(x \cdot f^!(y)) = f_*(x) \cdot y \qquad (x \in K(X),\ y \in K(Y)). \tag{2.27}$$

In the particular case where the inverse images of the points of Y by f are finite and contained in an affine open,[12] one checks that the functor $F \mapsto f_*(F)$ is exact and that $R^q f_*(F) = 0$ for $q > 0$. In this case, therefore, $f_*(\gamma_X(F)) = \gamma_Y(f_*(F))$. In particular, if X is an algebraic subspace of Y and if i is the canonical injection of X into Y, then

$$i_*(\gamma_X(F)) = \gamma_Y(F), \tag{2.28}$$

where, in the second member, F is considered as a sheaf on Y which is zero outside X.

4. Some technical results

Let X be an algebraic space. We have seen in §3 that the projection p from $X \times k$ to X defines a homomorphism

$$p^! : K(X) \longrightarrow K(X \times k). \tag{2.29}$$

Proposition 2.9. The homomorphism (2.29) is surjective.

We argue by induction on $n = \dim X$. The assertion is trivial if $n < 0$; suppose $n \geq 0$. By Proposition 2.6, it is enough to verify that for every irreducible closed subset Y of $X \times k$, $\gamma(Y)$ is in the image of $p^!$. It is enough to continue the reasoning by supposing X irreducible. If $p(Y)$ is not dense in X, our assertion follows from the induction hypothesis. Otherwise, either $Y = X \times k$ and the assertion is trivial, or Y is a hypersurface in $X \times k$. Let U be an affine open of X formed of nonsingular points. If D is the divisor on $U \times k$ induced by Y, one knows by commutative algebra that it is linearly equivalent to a divisor of the form $D' \times k$, where D' is a divisor on U. Then

$$\gamma_{U\times k}(D) - \gamma_{U\times k}(D' \times k)$$

belongs to $K(U \times k)^2$ by the corollary to Proposition 2.10, and by what has already been proved this element is of the form $p^!(x')$ with $x' \in K(U)$. Thus

$$\gamma_{U\times k}(D) = p^!(x' + \gamma_U(D')),$$

but x' is the restriction of an element y of $K(X)$ by Proposition 2.10, and consequently the restriction of $\gamma_{X\times k}(D) - p^!(y)$ to $U \times k$ is zero. By Proposition 2.10, $\gamma_{X\times k}(D) - p^!(y)$ comes from an element of

$$K(X \times k - U \times k) = K(X' \times k), \qquad X' = X - U.$$

The proof is finished by applying the induction hypothesis again. QED.

Corollary. Let X be an algebraic space and let p be the projection from $X \times k^n$ to X. Then the homomorphism

$$p^! : K(X) \longrightarrow K(X \times k^n)$$

is surjective; it is even bijective if X is nonsingular.

[12] This latter condition is evidently a consequence of the first, something the author still did not know in 1957!

The first assertion is by induction on n. For the second, put $f(x) = (x, 0)$, whence $pf = 1$ and $f^! p^! = 1$, which proves that $p^!$ is injective.

Proposition 2.10. Let X be an algebraic space, let U be an open subset of X, and let Y be its complement in X. Then the sequence of natural homomorphisms

$$K(Y) \longrightarrow K(X) \longrightarrow K(U) \longrightarrow 0 \tag{2.30}$$

is exact.

N.B. Compare with Proposition 2.3.

The fact that $K(X) \to K(U)$ is surjective follows at once from Proposition 2.5, or also from Proposition 2.6. It remains to prove that if $x \in K(X)$ has zero restriction in $K(U)$, then it is in the image of $K(Y)$. For this, return to the explicit definition of $K(X)$ and $K(U)$. There are two coherent algebraic sheaves F and G on X with

$$x = \gamma_X(F) - \gamma_X(G)$$

and

$$\gamma_U(F) - \gamma_U(G) = 0.$$

This latter relation means that one can find coherent sheaves P' and Q' on U, provided with filtrations whose factors are pairwise isomorphic on U, and such that one has an isomorphism on U

$$P = F + Q' \longrightarrow Q = G + P'.$$

By Proposition 2.5, the sheaves P' and Q' are restrictions to U of coherent sheaves on X, denoted by the same letters; in the same way, their filtrations come from filtrations on the corresponding sheaves on X, by Lemma 2.11 below. One then has

$$\gamma_X(F) - \gamma_X(G) = (\gamma_X(P) - \gamma_X(Q)) + (\gamma_X(P') - \gamma_X(Q')).$$

Introducing the factors P_i and Q_i of the filtrations of P' and Q' on X, the last term is the sum of terms $\gamma_X(P_i) - \gamma_X(Q_i)$, where $P_i|_U \simeq Q_i|_U$. We are thus reduced to proving the following particular case: if R and S are coherent sheaves on X such that $R|_U \simeq S|_U$, then $\gamma_X(R) - \gamma_X(S)$ is in the image of $K(Y)$. Let T be the graph of the given isomorphism from $R|_U$ to $S|_U$; it is a coherent subsheaf of $(R \times S)|_U$, hence by Lemma 2.11 it is the restriction to U of a coherent subsheaf, again denoted T, of $R \times S$. Then

$$\gamma_X(R) - \gamma_X(S) = (\gamma_X(R) - \gamma_X(T)) - (\gamma_X(S) - \gamma_X(T)),$$

and it is enough to prove, for instance, that the first term of the second member is in the image of $K(Y)$. But the projection $u : T \to R$ is an isomorphism over U, hence

$$\gamma_X(R) - \gamma_X(T) = \gamma_X(\operatorname{Coker} u) - \gamma_X(\operatorname{Ker} u),$$

and $\operatorname{Ker} u$ and $\operatorname{Coker} u$ have support in Y.

It remains, then, to prove the following lemma.

Lemma 2.11. Let X be an algebraic space, let F be a coherent algebraic sheaf on X, let U be an open subset of X, and let G be a coherent subsheaf of $F|_U$. Then G is the restriction to U of a coherent subsheaf of F.

Let $\overline{G}$ be the subsheaf of F whose sections on an open V of X are the sections of F on V whose restriction to $U \cap V$ is a section of G. Everything amounts to proving that this sheaf $\overline{G}$ is coherent. For this one may suppose X and U affine, because U is a union of affine opens U_i

and $\overline{G}$ is then the intersection of the analogous sheaves $\overline{G|U_i}$. Then the coordinate ring of U is a ring of fractions A_S of the coordinate ring A of X, and one immediately verifies that if F is defined by the A-module M, then $F|_U$ is defined by the A_S-module

$$M_S = A_S \otimes_A M.$$

The subsheaf G of $F|_U$ is then defined by a submodule N' of M_S, and one sees that $\overline{G}$ is precisely defined by the submodule N of M, inverse image of N' under the canonical homomorphism from M to M_S. QED.

Corollary 1 to Proposition 2.10. Let $x \in K(X)$. Then $x \in K(X)^i$ if and only if there exists a closed subset Y of X, of dimension $\leq n-i$, such that the restriction of x to $X-Y$ is zero.

Corollary 2. Suppose X is connected of dimension n, with no singularities in dimension $n-1$. If D and D' are two linearly equivalent divisors on X, then

$$\gamma(D) = \gamma(D') \mod K(X)^2.$$

By Corollary 1, one may suppose X nonsingular. The corollary then follows from the general formula

$$\gamma(D) = 1 - \gamma(L(-D)) \mod K(X)^2, \tag{2.31}$$

where $L(-D)$ denotes the rank-one vector bundle defined by the divisor $-D$. To verify (2.31), one writes it in the form

$$\gamma(L(-D)) = 1 - \gamma(D) \mod K(X)^2, \tag{2.31 bis}$$

and notes that the two members are homomorphisms from the group of divisors to the multiplicative group of invertible elements of the ring $K(X)/K(X)^2$. Here $K(X)^1/K(X)^2$ has square zero, which is very easily verified on the explicit formula (2.15), using Corollary 1. Thus one is reduced to the case where D is an irreducible hypersurface. More generally, if D is a hypersurface whose components are disjoint, one has the equality

$$\gamma(D) = 1 - \gamma(L(-D)), \tag{2.32}$$

as follows from the exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X(L(-D)) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

Theorem 2.12. Let X be a nonsingular quasi-projective variety, and let Z and Z' be two homogeneous cycles of codimension p on X which are linearly equivalent. Then

$$\gamma_X(Z) = \gamma_X(Z') \mod K(X)^{p+1}. \tag{2.33}$$

By hypothesis, one can find a cycle Y of dimension $n-p+1$ on $X \times k$ (n being the dimension of X), and two points $a, a' \in k$, such that

$$Z = Y \cdot (X \times a), \qquad Z' = Y \cdot (X \times a').$$

By formula (2.23), one then has

$$\gamma_X(Z) = i_a^!(\gamma_{X\times k}(Y)) \mod K(X)^{p+1},$$

and an analogous formula for $\gamma_X(Z')$. It is therefore enough to prove that $i_a^! = i_{a'}^!$, which follows at once from Proposition 2.9. QED.

Corollary 1. The filtration of $K(X)$ is compatible with its ring structure, i.e. one has

$$K(X)^i K(X)^j \subset K(X)^{i+j}$$

for all integers $i, j \geq 0$. Moreover there is a natural surjective homomorphism of graded rings

$$\varphi : A(X) \longrightarrow \mathrm{Gr}(K(X)), \tag{2.34}$$

where the second member denotes the graded ring associated with $K(X)$, provided with the filtration $(K(X)^i)_{i \geq 0}$, defined by the condition

$$\varphi(\ell(Z^{n-p})) = \gamma(Z^{n-p}) \mod K(X)^{p+1}.$$

This corollary follows immediately from Theorem 2.12, Propositions 2.6 and 2.7, and Chow's theorem 2.1, by induction on the integer $i + j$.

Corollary 2. Let X and Y be two nonsingular quasi-projective varieties and let f be a morphism from X to Y. Then the homomorphism $f^!$ from $K(Y)$ to $K(X)$ is compatible with the filtrations, and one has a commutative diagram

$$\begin{tikzcd}
A(Y) \arrow[r, "\varphi_Y"] \arrow[d, "f^*"'] & \mathrm{Gr}(K(Y)) \arrow[d, "\mathrm{Gr}(f^!)"] \\
A(X) \arrow[r, "\varphi_X"'] & \mathrm{Gr}(K(X)).
\end{tikzcd}$$

Using the graph of f, decompose f as the product of an immersion of X into $X \times Y$ and of a projection of $X \times Y$ onto Y. For the latter the corollary is trivial, and for an immersion one proceeds as for Corollary 1, using Proposition 2.8 in place of Proposition 2.7.

Proposition 2.13 ("cellular decomposition"). Let P be an algebraic space, irreducible or not, provided with an increasing family

$$P^0 \subset P^1 \subset \cdots \subset P^n = P$$

of closed subsets, such that for every i with $0 \leq i \leq n$, the connected components of $P^i - P^{i-1}$ are algebraic spaces $V_{i,\alpha}$ isomorphic to k^i; put $P^{-1} = \varnothing$. Then:

a) For every algebraic space X, the natural map from $K(X) \otimes K(P)$ to $K(X \times P)$, deduced from the tensor product of sheaves on X and P, is surjective.

b) The $\gamma_P(V_{i,\alpha})$ form a system of generators of the group $K(P)$.

The proof is by induction on n, using the corollary to Proposition 2.9 and Proposition 2.10. N.B. Unfortunately, I do not know whether the $\gamma_P(V_{i,\alpha})$ form a basis of $K(P)$ or whether the homomorphism considered in a) is bijective. We shall not need to know this below.

Theorem 2.14. Let x_i $(1 \leq i \leq q)$ and E_j $(1 \leq j \leq r)$ be letters, and let n_j $(1 \leq j \leq r)$ be positive integers. Consider the symbols $\lambda^k(x_i)$ with $k \geq 1$ and $\lambda^k(E_j)$ with $1 \leq k \leq n_j$ as indeterminates, and let R be a polynomial with integral coefficients in these indeterminates. Suppose that, for every graded ring A, when one substitutes in R elements ξ_i for the x_i and elements

$$\eta_j = [n_j, 1 + \eta_j^{(1)} + \cdots + \eta_j^{(n_j)}]$$

of $\widetilde{A}$ for the E_j, subject to the restrictions $\varepsilon(\xi_i) = 0$ and $\eta_j^{(k)} = 0$ for $k > n_j$, one has

$$R((\lambda^k(\xi_i)), (\lambda^k(\eta_j))) \in \widetilde{A}^p.$$

Then, if X is an irreducible nonsingular quasi-projective variety, for arbitrary elements x'_i of $K(X)^1$ and vector bundles E'_j of dimension n_j on X ($1 \leq i \leq q$, $1 \leq j \leq r$), one has

$$R((\lambda^k(x'_i)), (\lambda^k(\gamma(E'_j)))) \in K(X)^p. \tag{*}$$

Proof. By §2, the x'_i are themselves differences of elements of the form $\gamma(E)$, with E a vector bundle. On the other hand, assuming X locally closed in a projective space P, it follows from §2 and Proposition 2.5 that, for every $F \in \mathscr{F}(X)$, the sheaf $F(n)$ is generated by its sections for n sufficiently large. If F is defined by a vector bundle E, this proves that $E(n)$ is the inverse image of the canonical vector bundle on a Grassmannian variety G by a morphism from X to G. Considering the corresponding morphism from X to $P \times G$, one sees that F is the inverse image of a vector bundle on $P \times G$. Applying this to all vector bundles which occur, and using Corollary 2 to Theorem 2.12, we are reduced to the case where X is a product of Grassmannians. In this case the conclusion will follow from the following lemma.

Lemma 2.15. Suppose that X is a product of Grassmannians. Then one can find a λ-homomorphism from $K(X)$ to a torsion-free λ-ring of the form $\widetilde{A}$, where A is a graded ring, compatible with filtrations, and such that the corresponding homomorphism $\mathrm{Gr}(K(X)) \to \mathrm{Gr}(A)$ is injective.

To prove the lemma, we shall use Chow's theory of Chern classes. If one could prove the lemma, or Theorem 2.14 directly, one would deduce a purely "sheaf-theoretic" theory of Chern classes; cf. §5.

We shall consider the homomorphism $\widetilde{C}$ from $K(X)$ to $\widetilde{A}(X)$ and put $A = A(X)$. It remains to prove that the homomorphism $\psi = \mathrm{Gr}(\widetilde{C})$ from $\mathrm{Gr}(K(X))$ to $\mathrm{Gr}(A(X)) \simeq A(X)$ is injective. The fact that $\widetilde{C}$ is compatible with filtrations is true for every nonsingular quasi-projective variety and follows easily from the corollary to Proposition 2.3. On the other hand, since X is a product of Grassmannians G_j, there are vector bundles E_j on X, inverse images of the canonical bundles on the factors. We shall use the following known facts:

a) On X, linear equivalence of cycles is equivalent to numerical equivalence, and $A(X)$ is a group of finite rank.

b) The $c^i(E_j)$ generate the ring $A(X)$.

From a) it follows that

$$\varphi : A(X) \longrightarrow \mathrm{Gr}(K(X))$$

is injective, and hence bijective, by the following lemma.

Lemma 2.15. Let X^n be a nonsingular projective variety and let Z^{n-p} be a cycle on X such that $\varphi(Z) \in \mathrm{Gr}^p(K(X))$ is zero. Then Z^{n-p} is numerically equivalent to zero.

To prove this lemma, it suffices to use the additive function

$$\chi(F) = \sum_{i \geq 0} (-1)^i \dim H^i(X, F)$$

and the fact that this function is equal to 1 for $F = \mathcal{O}_{(x)}$ for every $x \in X$.

Thus, in the case under consideration, $A(X)$ and $\mathrm{Gr}(K(X))$ have the same finite rank in every degree, and $\mathrm{Gr}(K(X))$ is torsion-free, since this is true of $A(X)$. From assertion b) above it follows, in a formal way, that the map $\widetilde{C}$ from $K(X)$ to $\widetilde{A}(X)$ is surjective modulo finite groups. Since $\mathrm{Gr}(K(X))$ and $A(X)$ have the same finite rank in every degree, it then follows easily that the homomorphism $\mathrm{Gr}(\widetilde{C})$ from $\mathrm{Gr}(K(X))$ to $\mathrm{Gr}(\widetilde{A}(X))$ is bijective modulo finite groups in every degree; one uses induction on the degree. Since $\mathrm{Gr}(K(X))$ is torsion-free, this homomorphism $\mathrm{Gr}(\widetilde{C})$ is therefore injective. QED.

N.B. The homomorphism $\widetilde{C}$ is not surjective in general, even if X is a projective space of dimension 2.

5. Sheaf-theoretic definition of Chern classes. Application to the study of injection morphisms

Theorem 2.16. Let X be a nonsingular quasi-projective variety. Then, for every $x \in K(X)$ and every integer $i \geq 0$, the element $\gamma^i(x - \varepsilon(x))$ is of filtration $\geq i$. Let $c^i(x) \in \mathrm{Gr}^i(K(X))$ be the class of this element modulo $K(X)^{i+1}$. If one puts

$$\widetilde{c}(x) = \left[\varepsilon(x), 1 + \sum_{i\geq 0} c^i(x)\right] \in \mathrm{Gr}\widetilde{(K(X)}),$$

then $\widetilde{c}$ satisfies conditions analogous to those of Theorem 2.2.

The first assertion follows from Theorem 2.14, as does the fact that $x \mapsto \widetilde{c}(x)$ is a homomorphism of λ-rings; formula (1.25) implies $\gamma_t(x+y) = \gamma_t(x)\gamma_t(y)$, which already shows that $\widetilde{c}$ is an additive homomorphism. The functoriality of $\widetilde{c}$ is verified immediately, and (2.8) follows from (2.31). QED.

Corollary 1. One has

$$c^i(x) = \varphi(c^i(x)).$$

This follows from a uniqueness theorem for Chern classes, proved as in Hirzebruch by passage to flag varieties. One only has to show that, if Y is a flag variety associated with a vector bundle on X, then $\mathrm{Gr}(K(X)) \to \mathrm{Gr}(K(Y))$ is injective; this is verified by the explicit determination of $K(Y)$, with all its structures, from $K(X)$. I shall not give the details here.

Corollary 2. Suppose that for every element $Z \in A(X)$ there exists a morphism f from X to a product X' of Grassmannians such that Z is in the image of f^*. Then the canonical homomorphisms

$$\varphi : A(X) \longrightarrow \mathrm{Gr}(K(X)), \qquad \psi : \mathrm{Gr}(K(X)) \longrightarrow A(X)$$

satisfy, in dimension i, the relations

$$\psi\varphi = (-1)^{i-1}(i-1)!\,1, \qquad \varphi\psi = (-1)^{i-1}(i-1)!\,1. \tag{2.35}$$

Since φ is surjective, it is clearly enough to prove the first formula, and for this one may suppose that X is a product of Grassmannians. Since then φ is injective, it is enough to prove the second formula (2.35), which by Corollary 1 is written

$$\gamma^i(x - \varepsilon(x)) = (-1)^{i-1}(i-1)!\,x \mod K(X)^{i+1}$$

for every $x \in K(X)^i$. Since ψ is injective, it is enough to prove the analogous relation for $\widetilde{C}(x) \in A(X)^i$, and this is then a consequence of (1.22 bis).

Remark. Corollary 2 makes plausible the validity of (2.35) in all cases; there should exist a direct proof of it. In every case one proves easily the second formula (2.35) modulo the torsion subgroup of $\mathrm{Gr}(K(X))$, and in characteristic 0 this same formula is a consequence of the Riemann–Roch theorem. The interest of (2.35), when true, lies in the fact that it shows that φ is injective modulo torsion groups, perhaps always zero. Thus one does not lose much by defining Chern classes as elements of $\mathrm{Gr}(K(X))$. If one tensors with $\mathbb{Q}$, as will be done later for the Riemann–Roch theorem, the two definitions of Chern classes would then be identical.

Let X still be nonsingular and quasi-projective. Let in addition Y be a nonsingular subvariety of X, and let i be the injection of Y into X. For i_* and $i^!$ one then has the following formulas:

(i) $i^!$ is a homomorphism of λ-rings.

(ii) $i_*(x \cdot i^!(y)) = i_*(x) \cdot y$, whence in particular

$$i_*i^!(y) = \xi y, \qquad \text{with } \xi = i_*i^!(1) = \gamma_X(N).$$

(iii) $i^!i_*(x) = x \cdot \lambda_{-1}(N)$, whence in particular

$$i^!(\xi) = i^!i_*(1) = \lambda_{-1}(N).$$

(iv) $i_*(x)i_*(x') = i_*(xx'\lambda_{-1}(N))$.

Here, by definition, N is the class in $K(Y)$ of the conormal bundle of Y in X. Formulas (i) and (ii) are true for arbitrary morphisms, not necessarily injections, and (iii) and (iv) are verified from the definition (2.15) and the analogous formula for inverse images, taking x and x' to be classes of vector bundles and using the fact that

$$\mathrm{Tor}(\mathcal{O}_Y, \mathcal{O}_Y) = \Lambda(N). \tag{2.36}$$

Formula (iv) expresses that i_* is a homomorphism of rings if one introduces on $K(Y)$ a new multiplicative structure, by the definition

$$x \cdot_N x' = xx'\lambda_{-1}(N)$$

(cf. Chapter I, §4). It is completed by the formula

(v) $i_*(\lambda^p(N, x)) = \lambda^p(i_*(x))$ for $p \geq 1$,

which means that i_* defines a λ-homomorphism from $K(Y)_N$ into $K(X)$. This formula (v), certainly true in all cases, is for the moment proved only in characteristic 0, because it follows from the conjunction of Theorems 1.4 and 2.8.

There are analogous formulas for i_* and i^*:

(i bis) i^* is a homomorphism of graded rings.

(ii bis) $i_*(x \cdot i^*(y)) = i_*(x)y$, whence in particular

$$i_*i^*(y) = \xi' y, \qquad \text{with } \xi' = i_*(1).$$

(iii bis) $i^*i_*(x) = x \cdot (-1)^q N^q$, whence in particular

$$i^*(\xi') = (-1)^q N^q.$$

(iv bis) $i_*(x)i_*(x') = i_*(xx'(-1)^q N^q)$.

(v bis) i_* increases degrees by q units.

In these formulas, q is the codimension of Y in X, and one puts $N^i = c^i(N)$. Only formulas (iii bis) and (iv bis) require proof. I do not know whether they are well known, or even whether they are true, and I confine myself to noting that they follow by reduction from formulas (ii) and (iv), provided one works in $\mathrm{Gr}(K(Y))$ and $\mathrm{Gr}(K(X))$ and not in $A(Y)$ and $A(X)$. One then uses the fact that i_* and i^* come by passage to the quotient from i_* and $i^!$, noting that i_* shifts the filtration by q units.

All universal formulas relating i_* and $i^!$, respectively i_* and i^*, are formal consequences of the preceding formulas, as one can convince oneself without great difficulty.

Formula (v) gives

$$i_*(\gamma^p(N,x)) = \gamma^p(i_*(x)).$$

The conjunction of Theorem 2.14 and Proposition 1.5 shows that $\gamma^p(N,x)$, for $p = q + j$, is of filtration $\geq j$. If one denotes by $G_{q,j}(N,x)$ its class in $\mathrm{Gr}^j(K(Y))$, Theorem 2.16 shows that

$$i_*(G_{q,j}(N,x)) = c^p(i_*(x)).$$

Finally, Proposition 1.5 joined with Theorem 2.14 permits one to express $G_{q,j}(N,x)$ as a polynomial with integral coefficients in the $c^i(N)$, the $c^i(x)$, and $\varepsilon(x)$. One deduces the following theorem.

Theorem 2.17. Suppose the field k has characteristic 0, and let $i : Y \to X$ be an injection morphism of nonsingular quasi-projective varieties. Then one has formulas (i) to (v), and moreover, for every integer $j \geq 0$, the formula

$$c^{q+j}(i_*(x)) = i_*\left(\left(\frac{\widetilde{c}(x) * \lambda_{-1}(\widetilde{c}(N))}{(-1)^q N^q}\right)^{(j)}\right). \tag{2.37}$$

The second member of formula (2.37) is an abbreviated way of denoting the polynomial in the components of $\widetilde{c}(x)$ and $\widetilde{c}(N)$ described in Proposition 1.5.

Corollary. Under the preceding conditions one has

$$\mathrm{ch}(\widetilde{c}(i_*(x))) = i_*\left(\mathrm{ch}(\widetilde{c}(x))\,\mathfrak{C}(\check{N})^{-1}\right), \tag{2.38}$$

where ch is the Chern homomorphism defined by formula (1.26).

It is enough to use (2.37) written in the form

$$\widetilde{c}(i_*(x)) = 1 + i_*\left(\frac{\widetilde{c}(x) * \lambda_{-1}(\widetilde{c}(N)) - 1}{(-1)^q N^q}\right), \tag{2.37 bis}$$

the explicit formula (1.29), formula (iv bis), and finally (1.30).

One may write (2.38) in the form

$$\mathrm{ch}_X(F) = i_*\left(\mathrm{ch}_Y(F)\,\mathfrak{C}(T_{X/Y})^{-1}\right), \tag{2.38 bis}$$

where F is a coherent algebraic sheaf on Y (considered in the first member as a sheaf on X which is zero outside Y), where ch_X and ch_Y are the Chern homomorphisms taken respectively on X and Y, and where $T_{X/Y}$ is the normal bundle of Y in X.

Remarks. Contrary to (2.38), formula (2.37) is a "formula without denominators". It has been proved, even in characteristic 0, only by placing oneself in $\mathrm{Gr}(K(X))$ and not in $A(X)$, which may not be the same thing. It is certainly true in every characteristic and in $A(X)$. Let us note that one verifies at once that the images of the two members by i^* are the same; hence, applying i_*, the product of their difference by N^q is zero. This suggests a completely different method of proof, by trying to obtain Y as a transverse inverse image of a subvariety Y' in a variety X' such that, in dimension $\leq n = \dim X$, the class of Y' in $A(X')$ is not a divisor of 0. Unfortunately one cannot hope for X' to be a Grassmannian or something close to it.

Note that, if torsion elements are neglected, formula (2.38) is equivalent to (2.37).

6. The Riemann–Roch theorem

Its statement is as follows: let f be a proper morphism from a non-singular quasi-projective variety Y into a variety X of the same kind. Then one has

$$f_*(\mathrm{ch}_Y(x)\,\mathfrak{C}(Y)) = \mathrm{ch}_X(f_!(x))\,\mathfrak{C}(X). \tag{2.39}$$

Both sides are in $A(X) \otimes \mathbb{Q}$, and $\mathfrak{C}(X) = \mathfrak{C}(T_X)$.

This formula makes it possible to compute, up to torsion, the Chern classes of $f_!(x)$ from the Chern classes of x, those of T_X and T_Y, and the homomorphism f_*. Or, equivalently, it makes it possible to determine the homomorphism

$$f_! : K(Y) \otimes \mathbb{Q} \longrightarrow K(X) \otimes \mathbb{Q}$$

when these groups are identified, by means of the homomorphisms ch_Y and ch_X, respectively with $A(Y) \otimes \mathbb{Q}$ and $A(X) \otimes \mathbb{Q}$ (provided that formula (2.35) is true), from f_* and the Chern classes of T_X and T_Y.

I can prove formula (2.39) only in characteristic 0, and as an equality in $\mathrm{G}(K(X)) \otimes \mathbb{Q}$. For this, since Y is locally closed in a projective space P, f is decomposed as the product of an injection of Y into $P \times X$ and a projection morphism from $P \times X$ to X, and (2.39) is proved for each of these two morphisms.

For an injection morphism, one sees immediately that formulas (2.38) and (2.39) are equivalent. The characteristic 0 hypothesis enters only through Theorem 2.8, which permits one to express the classes $\lambda^i(\gamma(F))$, for a coherent algebraic sheaf F, by local cohomological invariants of F.

For a projection morphism $f : P \times X \to X$, Proposition 2.13 permits us to suppose that x is of the form $y \otimes z$, with $y \in K(P)$ and $z \in K(X)$. Then the Künneth formula gives

$$f_!(x) = \chi(y)\, z,$$

where, for a coherent algebraic sheaf F on P, one puts

$$\chi(F) = \sum_{i \geq 0} (-1)^i \dim H^i(P, F).$$

One is therefore at once reduced to proving the formula

$$\chi(F) = \pi_r(\mathrm{ch}_p(F)\,\mathfrak{C}(P)), \tag{2.40}$$

where π_r denotes the component of degree $r = \dim P$. By Proposition 2.13, one may suppose that $F = \mathcal{O}_Q$, where Q is a linear subvariety of P. Then $\chi(F) = 1$, since this is the arithmetic genus of Q; it remains only to see that the second member of (2.40) is 1.

Indeed, if V is a hyperplane of P and if one puts

$$\xi = \gamma_P(\mathcal{O}_V), \qquad L = \ell(V),$$

then

$$1 - \xi = \gamma_P(E(-L)),$$

where $E(-L)$ is the vector bundle defined by $-L$. Hence

$$C_p(1-\xi) = [1, 1-L],$$

and consequently

$$\mathrm{ch}_p(1-\xi) = e^{-L}, \qquad \mathrm{ch}_p(\xi) = 1 - e^{-L},$$

and therefore

$$\mathrm{ch}_p(\xi^p) = (1-e^{-L})^p.$$

If p is the dimension of Q in P, one has $\xi^p = \gamma_P(Q)$; hence

$$\mathrm{ch}_p(F)\,\mathfrak{C}(P) = \big(L/(1-e^{-L})\big)^{r+1}(1-e^{-L})^p.$$

It remains to verify that the coefficient of L^r in this formal series in L is 1. This follows from the characteristic property of the formal series $L/(1-e^{-L})$, namely that the coefficient of L^q in $\big(L/(1-e^{-L})\big)^{q+1}$ is 1 (put $q = r+1-p$).

Bibliographical references

[1] Hirzebruch: *Neue topologische Methoden in der algebraischen Geometrie.*

[2] Serre: Faisceaux algébriques cohérents, *Ann. of Math.*

[3] Grothendieck: Sur les faisceaux algébriques cohérents, in Séminaire Cartan 1956/57.

Proof of the Riemann–Roch theorem

(Grothendieck, 1 November 1957)

Reminder of notation

These are the notations of the "Report on Riemann–Roch," cited as RRR below.

All varieties considered are non-singular and quasi-projective, that is, isomorphic to a locally closed subvariety of a projective space. The base field is algebraically closed, of arbitrary characteristic.

If X is such a variety, $K(X)$ denotes the abelian group generated by the classes of coherent sheaves on X, modulo the identification coming from an extension into a sum; the alternating sum of Tor makes $K(X)$ a ring. If F is a sheaf, respectively E a vector bundle, respectively Y a subvariety of X, one denotes by $\gamma_X(F)$, respectively $\gamma_X(E)$, respectively $\gamma_X(Y)$, the element of $K(X)$ corresponding to F, respectively to E, respectively to $\mathcal{O}_Y$. In fact, one often writes simply F, E instead of $\gamma_X(F)$.

Let $A(X)$ be the graded ring of classes of cycles on X for rational equivalence (cf. Chow); $A^i(X)$ is formed by the cycles of codimension i. If E is a vector bundle, $C(E) = \sum C^i(E)$, with $C^i(E) \in A^i(X)$, is its Chern class (defined as inverse image of Schubert cycles). By linearity one deduces the classes $C^i(y)$, for all $y \in K(X)$. If

$$C(y) = \prod(1+a_i),$$

one sets, following Hirzebruch,

$$\mathrm{ch}(y) = \sum e^{a_i}, \qquad T(y) = \prod \frac{a_i}{1-e^{-a_i}}.$$

These are elements of $A(X) \otimes \mathbb{Q}$. When y is the tangent bundle of X, one writes $T(X)$ instead of $T(y)$.

If G is a vector bundle, $\lambda^p G$ denotes its p-th exterior power and

$$\lambda_{-1}(G) = \sum (-1)^p \lambda^p(G);$$

this is an element of $K(X)$.

If f is a proper regular map from a variety Y into a variety X, one defines homomorphisms

$$f^* : A(X) \longrightarrow A(Y), \qquad f_* : A(Y) \longrightarrow A(X).$$

The first is homogeneous of degree 0 and multiplicative. The second is homogeneous of degree $\dim X - \dim Y$. One also defines homomorphisms

$$f^* : K(X) \longrightarrow K(Y), \qquad f_! : K(Y) \longrightarrow K(X).$$

If F is a coherent sheaf on X, $f^*(F)$ is, by definition, the alternating sum of the $\mathrm{Tor}_p^{\mathcal{O}_X}(\mathcal{O}_Y, F)$. For vector bundles, this is simply the inverse-image operation.

If G is a coherent sheaf on Y, $f_*(G)$ is coherent on X. If $R^p f_*$ denotes the p-th derived functor of the functor f_* so defined, one has, by definition,

$$f_!(G) = \sum (-1)^p R^p f_*(G).$$

Riemann–Roch theorem

Let $f : Y \to X$ be a proper morphism (regular map) from a non-singular quasi-projective variety Y into another such variety X, and let $y \in K(Y)$. Then

$$f_*(\mathrm{ch}(y) T(Y)) = \mathrm{ch}(f_!(y)) T(X) \quad \text{in } A(X) \otimes \mathbb{Q}. \tag{1}$$

Proof (abridged).

Lemma 1. Consider proper morphisms of varieties

$$Z \xrightarrow{g} Y \xrightarrow{f} X.$$

If the theorem R.R. is true for f and for g, it is true for fg. If it is true for fg and for g, it is true for f. If f_* is injective, and if the theorem R.R. is true for fg and for f, it is true for g.

This is trivial. In fact, when one supposes R.R. true for fg and g, one must still suppose that $g_!$ is surjective in order to be able to assert that R.R. is true for f.

Since a proper morphism is composed of an injection and a projection $X \times P \to X$, where P is a projective space, one is reduced to treating these two cases separately. The second, as was seen above, is reduced to the standard calculation, by means of Proposition 2.13 ("cellular decomposition") of RRR. Thus suppose $Y \subset X$, with injection morphism i. Formula (1) becomes

$$\mathrm{ch}(i_!(y)) = i_*(\mathrm{ch}(y) T(E)^{-1}), \tag{2}$$

where $E = T_{X/Y}$ is the normal bundle of Y in X.

Lemma 2. Suppose that there exists on X a vector bundle G of rank $p = \mathrm{codim}_X Y$, such that

$$\gamma_X(Y) = i_*(1) = \lambda_{-1}(G), \qquad i^*(G) = E, \qquad (-1)^p C^p(G) = Y. \tag{3}$$

If then

$$y = i^*(x), \qquad x \in K(X), \tag{4}$$

formula (2) is valid.

The proof is immediate from the fact that ch is a homomorphism of rings, and that

$$\mathrm{ch}(\lambda_{-1}(G)) = (-1)^p C^p(G)\, T(G)^{-1}. \tag{5}$$

Corollary. R.R. is true if Y is a hypersurface and if y is of the form $i^*(x)$, with $x \in K(X)$.

Lemma 3. Suppose that $X = Y \times P$, where P is a projective space, and that i is the injection $u \mapsto (u, v_0)$, $v_0 \in P$. Then i_* is injective and R.R. is true.

This is pure standard calculation.

Suppose again that X, Y, y are arbitrary, always with $Y \subset X$. Let X' be the variety obtained by blowing up X along Y, and let Y' be the hypersurface inverse image of Y. One has a diagram of maps

$$\begin{tikzcd} Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\ Y \arrow[r, "i"'] & X. \end{tikzcd}$$

The inverse image of E in Y' admits a subbundle of rank 1, whose dual is denoted L. Moreover L^{-1} is also the restriction to Y' of the line bundle $L(Y')$ defined on X' by the divisor Y'. Put $F = E/L^{-1}$. Elementary computations give the following lemma.

Lemma 4. One has the formulas, where $p = \mathrm{codim}_X\, Y$:

$$f^* i_!(y) = j_*(\lambda_{-1}(\check{F})y), \tag{6}$$
$$\lambda_{-1}(\check{F}) \equiv \gamma^{p-1}(E)\lambda^p(\check{E}) \pmod{1 - L}, \tag{7}$$
$$f_*(1) = 1, \quad \text{hence } f_* f^* = 1, \tag{8}$$
$$f^* i_*(\alpha) \equiv j_*(\alpha C^{p-1}(F)) \pmod{\mathrm{Ker}\, f_*} \quad \text{for all } \alpha \in A(Y). \tag{9}$$

To simplify notation, $K(Y)$ has been made to operate on $K(Y')$, and $A(Y)$ on $A(Y')$, by the homomorphisms g^*. In formula (7), it seems that $\gamma^{p-1}(E)$ denotes $\lambda^{p-1}(E - 2)$, that is,

$$\sum_{i+j=p-1} (\cdots)\lambda^i(E);$$

the congruence takes place in $K(Y')$.

Formula (6) follows from the equality

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y(v), \mathcal{O}_Y(v')) = \lambda^i \check{F}(v') \quad (v' \in Y',\ v = g(v')). \tag{6 bis}$$

Formula (7) follows from the formulas

$$g_*(L^{-i}) = 0 \quad \text{for } 0 < i \leq p - 1, \qquad g_*(1) = 1, \tag{10}$$

themselves consequences of Künneth and of the cohomology of projective spaces.

Formula (8) is trivial. Finally, (9) follows from $g_*(C^{p-1}(F)) = 1$, which is deduced, for example, from the immediate formulas

$$g_*(\xi^i) = 0 \quad \text{if } 0 \leq i < p - 1, \qquad g_*(\xi^{p-1}) = 1, \tag{10 bis}$$

with $\xi = C^1(L) \in A^1(Y')$. In fact, (9) is very likely an identity, and not merely a congruence. The unnecessarily complicated formula (7) may be replaced, for example, by VII 4.1.

From formula (7) it follows at once that $\lambda_{-1}(\check{F}) \in j^*(K(X'))$, provided that

$$y\,\gamma^{p-1}(E)\lambda^p(\check{E}) \in i^*(K(X)). \tag{11}$$

Suppose this condition (11) satisfied. By the corollary to Lemma 2, R.R. is true for $(Y', X', y\lambda_{-1}(\check{F}))$. On the other hand, to prove (2), formula (8) reduces the matter to proving that

$$f^*\operatorname{ch}(i_!(y)) \equiv f^*\big(i_*(\operatorname{ch}(y)T(E)^{-1})\big) \pmod{\operatorname{Ker} f_*}. \tag{12}$$

This last formula is verified immediately, using (6), (9), R.R. for $(Y', X', y\lambda_{-1}(\check{F}))$, and the formula

$$\operatorname{ch}(\lambda_{-1}(\check{F})) = C^{p-1}(F)T(F)^{-1} \tag{13}$$

(cf. formula (5)).

Thus R.R. is proved every time $y\,\gamma^{p-1}(E) = 0$, in particular every time $p-1 > \dim Y$, since $\gamma^{p-1}(E)$ is of filtration $\geq p-1$ in $K(Y)$, filtered by the codimension of supports; that is, every time

$$\dim Y < \frac{\dim X - 1}{2}.$$

One is reduced to this case by combining Lemma 1 and Lemma 3, taking for P a projective space of sufficiently large dimension, for example $r > \dim X + 1$. This proves the assertion.

Remark. We have used the fact that $\gamma^{p-1}(E)$ was of filtration $\geq p-1$. One can prove this fact directly, without passing through Grassmannians as in RRR, 2.16, by studying the projective bundle associated to E. This method more generally gives Theorem 2.14 of RRR.

Extract from a letter of Grothendieck

"Morally," the new method rests on the determination of $K(X)$ and $A(X)$ when X is either the projective bundle associated to a vector bundle, or a variety obtained by blowing up a non-singular subvariety. In this second case I have not made the complete determination; a small equality is still missing. With the notation of the short note, one must prove formula (9), but as an equality:

$$f^*(i_*(\alpha)) = j_*(\alpha C^{p-1}(F)), \qquad \alpha \in A(Y). \tag{9}$$

One observes that the two members have the same image by f_*, and the same image by j^*, so that the question is equivalent to

$$\operatorname{Ker} f_* \cap \operatorname{Ker} j^* = 0. \tag{9 bis}$$

In any case, (9) is true after reducing to $G(K(X'))$ (which destroys a little torsion, at most), for it then follows from (6); there is therefore hardly any doubt that it is true.

I also point out that equality (9) follows very easily from (6), without passing through the troublesome (6 bis).

P.C.C.J-P.S.

Exposé I. Generalities on finiteness conditions in derived categories

by L. Illusie

0. Introduction

The object of Exposés I to IV is to develop, with the degree of generality appropriate in this Seminar, the formalism of finiteness conditions in derived categories of ringed topoi.

As was said in Exposé 0, the need to define, on arbitrary schemes, "Grothendieck groups" possessing good variance properties forced one to generalize and to soften the finiteness notions used up to now.

A classical notion such as that of a coherent sheaf on a ringed space $(X, \mathcal{O}_X)$ makes sense only when $\mathcal{O}_X$ is coherent. One might think of replacing it by the notion of finite presentation. But this has the inconvenience that the kernel of an epimorphism of modules of finite presentation is no longer, in general, of finite presentation. A satisfactory notion is reached by observing that, when $\mathcal{O}_X$ is coherent, a coherent sheaf F is not only of finite presentation but of n-finite presentation for every $n \in \mathbb{N}$, where n-finite presentation means that locally there exists an exact sequence

$$L_n \longrightarrow L_{n-1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow F \longrightarrow 0,$$

with the L_i free of finite type. When $\mathcal{O}_X$ is no longer assumed coherent, say that an $\mathcal{O}_X$-module F is *pseudo-coherent* if it satisfies the preceding condition, that is, if it is of n-finite presentation for every n. Then the notion of pseudo-coherence has, with respect to short exact sequences, the same stability property as coherence: if two terms of a short exact sequence of $\mathcal{O}_X$-modules are pseudo-coherent, then so is the third.

The preceding notion is easily generalized to complexes of sheaves. A complex F of $\mathcal{O}_X$-modules is said to be *pseudo-coherent* if it is n-pseudo-coherent for every $n \in \mathbb{Z}$, where n-pseudo-coherent means that there locally exists a quasi-isomorphism $L \to F$, with L a complex bounded above and with components free of finite type in degrees $> n$. For a complex F reduced to degree 0, this is the same as saying that F is pseudo-coherent as a complex or as a module.

The notion of pseudo-coherent complex has excellent stability properties, described in Exposé I, §§1 and 2. First of all it is stable under isomorphism in the derived category $\mathrm{D}(X)$ of the category of $\mathcal{O}_X$-modules; better still, if two objects of a distinguished triangle of $\mathrm{D}(X)$ are pseudo-coherent, then so is the third. In other words, the full subcategory $\mathrm{D}(X)_{\mathrm{coh}}$ of $\mathrm{D}(X)$ formed by pseudo-coherent complexes is a triangulated subcategory. Moreover, pseudo-coherence is preserved under inverse image and derived tensor product. In the case where $\mathcal{O}_X$ is coherent, to say that a complex F is pseudo-coherent simply means that F has locally bounded-above cohomology and that all the $H^i(F)$ are coherent.

If $\mathcal{O}_X$ is a sheaf of regular local rings, every coherent $\mathcal{O}_X$-module F admits locally a finite resolution to the left by finite-type free modules, that is, there exists locally an exact sequence

$$0 \longrightarrow L_n \longrightarrow \cdots \longrightarrow L_0 \longrightarrow F \longrightarrow 0,$$

where the L_i are free of finite type. It is therefore natural, when $\mathcal{O}_X$ is an arbitrary sheaf of local rings, to study the modules which have the preceding property. Such modules are called *perfect.* More generally, a complex F of $\mathcal{O}_X$-modules is said to be *perfect* if there exists locally a quasi-isomorphism $L \to F$, where L is a bounded complex with components free of finite type. It is shown in Exposé I, §3, that the full subcategory $\mathrm{D}(X)_{\mathrm{parf}}$ of $\mathrm{D}(X)$ formed by perfect complexes is, like $\mathrm{D}(X)_{\mathrm{coh}}$, a triangulated subcategory, stable under inverse image and derived tensor product. One obviously has

$$\mathrm{D}(X)_{\mathrm{parf}} \subset \mathrm{D}(X)_{\mathrm{coh}},$$

but the inclusion is strict in general. Pseudo-coherence can be redefined "by passage to the limit" from perfection: a complex F is pseudo-coherent if and only if, for every $n \in \mathbb{Z}$, F can be locally approximated to order n by a perfect complex, meaning that there exists locally a distinguished triangle

$$\begin{tikzcd} L \arrow[rr] & & F \arrow[dl] \\ & R \arrow[ul] & \end{tikzcd}$$

with L perfect and R acyclic in degrees $> n$. Conversely, perfection is recovered from pseudo-coherence by means of an additional regularity condition: a complex is perfect if and only if it is pseudo-coherent and locally of finite Tor-dimension (Exposé I, §5). Thus one recovers the fact that, if the local rings are regular, every coherent sheaf is perfect, and more generally every pseudo-coherent complex with locally bounded cohomology is perfect.

Pseudo-coherence and perfection are the two fundamental finiteness notions with which we shall work in this Seminar. Sections 1 to 5 of Exposé I are devoted to their definition and to establishing their elementary stability properties. For this purpose, only two or three local properties of the category of finite-type free modules, relative to the category of all modules, are used. It is therefore convenient and useful, notably in Exposé II, to axiomatize the situation by introducing a notion of pseudo-coherence (respectively of perfection) in a category fibered over a site, relative to a fibered subcategory satisfying suitable conditions.

Sections 6 to 8 of Exposé I generalize to perfect complexes a certain number of notions well known for finite-type locally free sheaves: rank, duality, and the trace of an endomorphism.

In Exposé II, one examines the problem of the existence of "global resolutions": under what conditions, on a ringed topos $(X, \mathcal{O}_X)$, is a pseudo-coherent (respectively perfect) complex globally isomorphic in $\mathrm{D}(X)$ to a complex of locally free modules of finite type? The importance of this question lies essentially in the fact that at present one does not know how to generalize to perfect complexes certain usual constructions on locally free modules, such as exterior and symmetric powers or Chern classes. That is why it is useful to have workable sufficient criteria for the existence of global resolutions, which reduce certain questions about pseudo-coherent (respectively perfect) complexes to analogous questions about locally free sheaves of finite type. Such criteria are given in Exposé II.

In particular, one shows that global resolutions exist in the following cases: X is the Zariski topos of an affine scheme, or more generally of a quasi-compact scheme possessing an ample invertible sheaf, or even an ample family of invertible sheaves in the sense of Kleiman, for example a regular scheme; or again X is the topos of sheaves of sets on a compact topological space and $\mathcal{O}_X$ is the sheaf of continuous functions with complex values. By way of illustration, and to show the flexibility of the notions introduced, the appendix gives a "purely sheaf-theoretic" definition of the index of a family of elliptic operators.

Exposé III studies the stability of finiteness conditions under derived direct image. To obtain reasonable statements, one must relativize the notions of pseudo-coherence and perfection. We place ourselves here in the framework of ordinary schemes, which is sufficient for the Seminar, but there is no doubt that one will sooner or later need to develop an analogous theory for relative analytic schemes or spaces. Let $p : X \to S$ be an S-scheme locally of finite type, and let F be a complex of $\mathcal{O}_X$-modules. One says that F is pseudo-coherent (respectively perfect) relative to p if one can locally extend X, by a closed immersion, into a smooth S-scheme X', so that F, extended by zero to X', is pseudo-coherent (respectively perfect). The notion of pseudo-coherence relative to p is especially interesting when S is not locally noetherian; in the opposite case it coincides with the ordinary notion of pseudo-coherence. Moreover, as expected, to say that F is perfect relative to p is equivalent to saying that F is pseudo-coherent relative to p and locally of finite Tor-dimension relative to p (i.e. relative to the sheaf of rings $p^{-1}(\mathcal{O}_S)$).

The central theorem of Exposé III is the following finiteness theorem, stated below in a slightly more precise form: if $f : X \to Y$ is a proper morphism of S-schemes locally of finite type, the functor Rf_* transforms complexes pseudo-coherent relative to S into complexes pseudo-coherent relative to S. This theorem is in fact a conjecture, but it is nevertheless proved in the two particular cases S locally noetherian and f projective. It seems likely, unfortunately, that the extension to the general case is of the same order of difficulty as the analogous theorem in analytic geometry, Grauert's theorem. Combining the finiteness theorem with an essentially trivial formula called the projection formula, one obtains workable criteria for the stability of relative perfection under direct image. As a corollary one recovers Grauert's continuity and semicontinuity theorems (EGA III 7.6).

Exposé IV translates the results of Exposés I to III into the language of "Grothendieck groups." On a ringed topos $(X, \mathcal{O}_X)$, write $K^*(X)$ (respectively $K_\bullet(X)$) for the Grothendieck group of the category of perfect complexes and complexes of finite Tor-dimension (respectively pseudo-coherent complexes with bounded homology). The group $K^*(X)$ is a ring, and $K_\bullet(X)$ is a module over $K^*(X)$. The functor K^* is contravariant, as suggested above, while $K_\bullet$ is covariant for proper pseudo-coherent morphisms of schemes, or of analytic spaces, subject to the availability of the finiteness theorem. There are also unusual variants: covariance of K^*, contravariance of $K_\bullet$, for morphisms satisfying suitable regularity hypotheses. Finally, the globalization criteria of Exposé II make it possible to link these groups with the "Grothendieck groups" defined naïvely from coherent sheaves or finite-type locally free sheaves.

1. Preliminary definitions

1.1. Fibered categories with additive fibers (respectively abelian, respectively triangulated fibers)

1.1.1. Let S be a category, and let $\mathcal{C}$ be an S-category fibered in categories with additive (respectively triangulated) fibers (SGA 1 VI 6.1). We shall say that $\mathcal{C}$ is an *additive* (respectively *triangulated*) S-category, or that $\mathcal{C}$ is additive (respectively triangulated) over S, if, for every arrow $f : X \to Y$ of S, the inverse-image functor

$$f^* : \mathcal{C}_Y \longrightarrow \mathcal{C}_X$$

(determined up to unique isomorphism) is additive (respectively exact). We shall say that $\mathcal{C}$ is an *abelian* S-category if $\mathcal{C}$ is an additive S-category with abelian fibers, and that $\mathcal{C}$ is a *flat abelian* S-category if, moreover, the inverse-image functors are exact.

1.1.2. Let $\mathcal{C}$ be a triangulated S-category. One defines, for example with the aid of a cleavage of $\mathcal{C}$ (SGA 1 VI 7.1), a fibered S-category $\mathrm{Tr}(\mathcal{C})$ whose fiber over each $X \in \mathrm{ob}(S)$ is the category $\mathrm{Tr}(\mathcal{C}_X)$ of distinguished triangles of $\mathcal{C}_X$.

1.1.3. Let $\mathcal{C}$ and $\mathcal{C}'$ be additive (respectively triangulated) S-categories, and let $F : \mathcal{C} \to \mathcal{C}'$ be a functor. We say that F is an additive (respectively exact) S-functor if F is a Cartesian S-functor that is additive (respectively exact) on each fiber.

1.1.3 (continued). If $\mathcal{C}$ and $\mathcal{C}'$ are triangulated S-categories and if F is an exact S-functor, then F defines a Cartesian S-functor

$$\mathrm{Tr}(\mathcal{C}) \longrightarrow \mathrm{Tr}(\mathcal{C}').$$

1.1.4. Let $\mathcal{C}$ be a triangulated S-category, and let $\mathcal{C}'$ be a full fibered sub-S-category of $\mathcal{C}$. We say that $\mathcal{C}'$ is a *triangulated sub-S-category* of $\mathcal{C}$ if, for every $X \in \operatorname{ob} S$, $\mathcal{C}'_X$ is a triangulated subcategory of $\mathcal{C}_X$ [V], Chap. I, §1, 2.3.

1.1.5. Let $\mathcal{C}$ be an additive S-category. One defines (SGA 4 XVII 2.2, 2.4) an additive (respectively triangulated) S-category

$$\mathrm{C}(\mathcal{C}) \qquad \text{respectively} \qquad \mathrm{K}(\mathcal{C}),$$

whose fiber over $X \in \operatorname{ob} S$ is the category $\mathrm{C}(\mathcal{C}_X)$ of complexes with differential of degree $+1$ (respectively the category $\mathrm{K}(\mathcal{C}_X)$ of complexes of $\mathcal{C}_X$ "up to homotopy"). If $\mathcal{C}$ is a flat abelian S-category, one defines, loc. cit., a triangulated S-category $\mathrm{D}(\mathcal{C})$, whose fiber over $X \in \operatorname{ob} S$ is the derived category $\mathrm{D}(\mathcal{C}_X)$; if $f : X \to Y$ is an arrow of S, we shall write

$$f^* : \mathrm{D}(\mathcal{C}_Y) \longrightarrow \mathrm{D}(\mathcal{C}_X)$$

for the inverse-image functor, obtained from

$$f^* : \mathrm{K}(\mathcal{C}_Y) \longrightarrow \mathrm{K}(\mathcal{C}_X)$$

by passage to the quotient. One defines full triangulated sub-S-categories

$$\mathrm{K}^*(\mathcal{C}) \qquad \text{respectively} \qquad \mathrm{D}^*(\mathcal{C}) \quad (* = +, -, b)$$

by the usual degree conditions.

1.2. Terminology

Let S be a site, let $\mathcal{C}$ be a flat abelian S-category (1.1), and let $\mathcal{C}_0$ be a strictly full fibered sub-S-category.

Let $F \in \operatorname{ob} \mathcal{C}_X$, for $X \in \operatorname{ob} S$. The set of arrows $f : U \to X$ in S/X such that there exists an epimorphism

$$E \longrightarrow f^* F, \qquad E \in \operatorname{ob} \mathcal{C}_{0,U},$$

is a sieve on X. If this sieve is covering, we shall say that F is of *$\mathcal{C}_0$-finite type* (or simply of finite type when no confusion is to be feared).

We shall say that $\mathcal{C}_0$ is *locally stable under kernels of epimorphisms* if the following condition is satisfied: for every $X \in \operatorname{ob} S$ and every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$, with $E, F \in \operatorname{ob} \mathcal{C}_{0,X}$, the kernel of u is locally in $\mathcal{C}_0$; in other words the set of $f : U \to X$ in S/X such that

$$f^* \operatorname{Ker}(u) \in \operatorname{ob} \mathcal{C}_{0,U}$$

is a refinement of X.

By the standard process of decomposing a long exact sequence into short exact sequences, one sees that the preceding condition is equivalent to the following apparently stronger one: for every $X \in \operatorname{ob} S$ and every exact sequence in $\mathcal{C}_X$

$$0 \longrightarrow E^0 \longrightarrow E^1 \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0 \tag{*}$$

with $E^i \in \operatorname{ob} \mathcal{C}_{0,X}$ for $1 \leq i \leq n$, the object E^0 is locally in $\mathcal{C}_0$.

We shall say that $\mathcal{C}_0$ is *locally quasi-stable under kernels of epimorphisms* if, for every exact sequence such as $(*)$, E^0 is of $\mathcal{C}_0$-finite type.

On the other hand, we shall say that $\mathcal{C}_0$ is *locally liftable in* $\mathcal{C}$ if, for every $X \in \operatorname{ob} S$, every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$ with $F \in \operatorname{ob} \mathcal{C}_{0,X}$ locally admits a section. Equivalently, for $\mathcal{C}_0$ to be locally liftable in $\mathcal{C}$, it is necessary and sufficient that, for every diagram of $\mathcal{C}_X$, $X \in \operatorname{ob} S$,

$$\begin{tikzcd} & H \arrow[d] \\ E \arrow[r, "u"'] & F \end{tikzcd}$$

where u is an epimorphism and $H \in \operatorname{ob} \mathcal{C}_0$, the set of U over X for which there exists a commutative diagram

$$\begin{tikzcd} & H_U \arrow[d] \\ E_U \arrow[ur] \arrow[r, "u_U"'] & F_U \end{tikzcd}$$

is a refinement of X.

We shall say that $\mathcal{C}_0$ is *locally quasi-liftable in* $\mathcal{C}$ if the following weaker condition is satisfied: for every $X \in \operatorname{ob} S$, and every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$, with $F \in \operatorname{ob} \mathcal{C}_{0,X}$, the set of arrows $f : U \to X$ of S/X for which there exists

$$v : F' \longrightarrow f^*E, \qquad F' \in \operatorname{ob} \mathcal{C}_{0,U},$$

such that $f^*(u)v$ is an epimorphism, is a refinement of X. This condition is equivalent to the following: for every diagram in $\mathcal{C}_X$

$$\begin{tikzcd} & H \arrow[d] \\ E \arrow[r, "u"'] & F, \end{tikzcd}$$

where u is an epimorphism and $H \in \operatorname{ob} \mathcal{C}_0$, the set of U over X such that there exists a commutative diagram

$$\begin{tikzcd} G \arrow[r, "v"] \arrow[d] & H_U \arrow[d] \\ E_U \arrow[r, "u_U"'] & F_U \end{tikzcd}$$

with $G \in \operatorname{ob} \mathcal{C}_0$ and v an epimorphism, is a refinement of X.

If S is a chaotic site, for example the punctual site (one object, one morphism), we shall omit the adverb "locally" in the preceding definitions.

1.3. Examples

a) Let S be the punctual site, let A be a ring, and let $\mathcal{C}$ be the category of left A-modules. The full subcategory $\mathcal{C}_0$ of $\mathcal{C}$ formed by the projective A-modules of finite type is stable under kernels of epimorphisms and is liftable in $\mathcal{C}$; an A-module is of $\mathcal{C}_0$-finite type if and only if it is of finite type in the ordinary sense. The full subcategory $\mathcal{C}_1$ of $\mathcal{C}$ formed by the A-modules of finite type is quasi-liftable in $\mathcal{C}$, but is not liftable in general; if A is noetherian, $\mathcal{C}_1$ is stable under kernels of epimorphisms. The full subcategory $\mathcal{C}_0'$ of $\mathcal{C}$ formed by the free A-modules of finite type is liftable in $\mathcal{C}$ and quasi-stable, though not stable in general, under kernels of epimorphisms.

b) Let $(S, \mathcal{O}_S)$ be a space ringed by local rings, let S be the site of open subsets of S, and let $\mathcal{C}$ be the fibered S-category, even split (SGA 1 VI 9), defined by the functor

$$U \longmapsto \{\text{category of left } \mathcal{O}_U\text{-modules}\}.$$

Then $\mathcal{C}$ is a flat abelian S-category (1.1). Let $\mathcal{C}_0$ be the strictly full additive sub-S-category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } \mathcal{O}_U\text{-modules}\}.$$

An $\mathcal{O}_S$-module M is locally of $\mathcal{C}_0$-finite type if and only if M is of finite type in the ordinary sense (EGA 0_{I}, 5.2). From EGA 0_{I}, 5.2.7 and 5.4.9, it follows that $\mathcal{C}_0$ is locally stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$.

c) Let G be a group, let S be the topos of G-sets (SGA 4 V 3.11), and let k be a field, regarded as the constant ring on S. Let $\mathcal{C}$ be the split S-category defined by

$$U \longmapsto \{\text{category of left } k_U\text{-modules}\};$$

then $\mathcal{C}$ is a flat abelian S-category. Let $\mathcal{C}_0$ be the strictly full additive sub-S-category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } k_U\text{-modules}\}.$$

Since localization in S is equivalent to forgetting the operations of G, it is clear that a G-k-module M is locally of $\mathcal{C}_0$-finite type if and only if M is finite-dimensional as a vector space over k, and that $\mathcal{C}_0$ is locally stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$.

The reader is left to look at the "Galois" variant of c), where G is a profinite group and S is the topos of sets on which G operates continuously.

In fact, examples b) and c) are particular cases of the following example.

d) Let $(S, \mathcal{O}_S)$ be a ringed topos (cf. SGA 4 IV 2.10), and let $\mathcal{C}$ be the split S-category defined by

$$U \longmapsto \{\text{category of left } \mathcal{O}_U\text{-modules}\};$$

then $\mathcal{C}$ is a flat abelian S-category. Let $\mathcal{C}_0$ be the strictly full additive sub-S-category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } \mathcal{O}_U\text{-modules}\}.$$

An $\mathcal{O}_S$-module locally of $\mathcal{C}_0$-finite type will simply be called *of finite type*. The category $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$, as follows from:

Lemma 1.3.1. Let P be an $\mathcal{O}_S$-module. The following conditions are equivalent:

(i) P is of finite type and the functor $\mathrm{Hom}(P, \cdot)$ is exact;

(ii) P is of finite type and every epimorphism $M \to P$ locally admits a section;

(iii) P is locally a direct factor of a free module of finite type.

The proof is immediate and is left as an exercise to the reader.

1.4. An "extension" lemma (cf. EGA 0_{III}, 11.9.1)

Proposition 1.4.1. Let S be a site, let $\mathcal{C}$ be a flat abelian S-category (1.1), and let $\mathcal{C}_0$ be a strictly full additive sub-S-category of $\mathcal{C}$. Assume that $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$ (1.2). Let $S \in \operatorname{ob} S$, let

$$u : E \longrightarrow F$$

be a morphism, of degree 0, of complexes of $\mathcal{C}_S$ (differentials of degree +1), let G be the cone of u (cf. [V], Chap. I, §1, 2.1), let $n \in \mathbb{Z}$, and assume that $\mathrm{H}^n(G)$ is of $\mathcal{C}_0$-finite type (1.2). Then the set of $X \in \operatorname{ob}(S/S)$ satisfying property (P) below is a refinement of S.

(P): there exist an object $M \in \mathcal{C}_{0,X}$ and arrows of $\mathcal{C}_X$

$$r : M \longrightarrow Z^{n+1}(E_X) = \operatorname{Ker}\left(E_X^{n+1} \xrightarrow{d^{n+1}} E_X^{n+2}\right), \qquad s : M \longrightarrow F_X^n$$

such that the following diagram of $\mathcal{C}_X$

$$\begin{array}{ccccccc}
\cdots \to & E_X^{n-1} & \xrightarrow{\binom{d_E^{n-1}}{0}} & E_X^n \oplus M & \xrightarrow{(d_E^n, r)} & E_X^{n+1} & \to \cdots \\
& \Big\downarrow u_X^{n-1} & & \Big\downarrow (u_X^n, s) & & \Big\downarrow u_X^{n+1} & \\
\cdots \to & F_X^{n-1} & \xrightarrow[d_F^{n-1}]{} & F_X^n & \xrightarrow[d_F^n]{} & F_X^{n+1} & \to \cdots
\end{array}$$

is a morphism of complexes $u' : E' \to F_X$, that is,

$$d_F^n s = u_X^{n+1} r,$$

and such that

(i) $\mathrm{H}^{n+1}(u')$ is a monomorphism, with

$$\Im \mathrm{H}^{n+1}(u') = \Im \mathrm{H}^{n+1}(u_X);$$

(ii) $\mathrm{H}^n(u')$ is an epimorphism.

In pictorial terms, one may say that, after modifying E and u locally in degree n only by adding, in degree n, an object of $\mathcal{C}_0$, one can locally make $\mathrm{H}^{n+1}(u)$ injective, without changing its image, and $\mathrm{H}^n(u)$ surjective.

Proof. The procedure is as follows: first one shows that r and s may be chosen locally so as to make $\mathrm{H}^{n+1}(u)$ injective; one verifies that the finiteness hypothesis on $\mathrm{H}^n(G)$ is preserved; a new local correction in degree n then makes it possible to obtain $\mathrm{H}^n(u)$ surjective.

The exact cohomology sequence

$$\cdots \longrightarrow \mathrm{H}^n(G) \longrightarrow \mathrm{H}^{n+1}(E) \longrightarrow \mathrm{H}^{n+1}(F) \longrightarrow \cdots$$

shows that the kernel of $\mathrm{H}^{n+1}(u)$ is of $\mathcal{C}_0$-finite type. In other words, the set of X over S for which there is an exact sequence

$$M \longrightarrow \mathrm{H}^{n+1}(E_X) \xrightarrow{\mathrm{H}^{n+1}(u)} \mathrm{H}^{n+1}(F_X), \qquad M \in \operatorname{ob} \mathcal{C}_{0,X},$$

is a refinement of S. Since $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$, one may moreover impose that the arrow

$$M \longrightarrow \mathrm{H}^{n+1}(E_X)$$

factor through $Z^{n+1}(E_X)$:

$$M \xrightarrow{r} Z^{n+1}(E_X) \longrightarrow \mathrm{H}^{n+1}(E_X).$$

The composite

$$M \xrightarrow{u_X^{n+1} r} Z^{n+1}(F_X) \longrightarrow \mathrm{H}^{n+1}(F_X)$$

is zero by construction, so $u_X^{n+1} r$ factors through

$$B^{n+1}(F_X) = \Im\left(F_X^n \xrightarrow{d_F^n} F_X^{n+1}\right).$$

Applying quasi-liftability once more, one may impose that $u_X^{n+1} r$ factor through F_X^n:

$$M \xrightarrow{s} F_X^n \longrightarrow B^{n+1}(F_X) \longrightarrow Z^{n+1}(F_X).$$

Then u', defined as in (1.4.1), is plainly a morphism of complexes satisfying (i).

Let G' be the cone of $u' : E' \to F_X$. We show that $\mathrm{H}^n(G')$ is of $\mathcal{C}_0$-finite type. By construction there is an exact sequence

$$0 \longrightarrow E_X \xrightarrow{v} E' \longrightarrow M[-n] \longrightarrow 0,$$

with $u'v = u_X$. To compare G_X and G', one may, for instance, form the octahedron in $\mathrm{K}(\mathcal{C}_X)$:

$$\begin{tikzcd}
 & & G_X \arrow[dr, "(*)"'] \arrow[ddll, bend left, "+1"'] & & \\
 & M[-n] \arrow[ur] \arrow[dl, bend left, "+1"'] & & G' \arrow[ll, bend left, "+1"'] \arrow[dl, bend right, "+1"] & \\
E_X \arrow[rr, "v"'] & & E' \arrow[ul] \arrow[rr, "u'"'] & & F_X \arrow[ul] \arrow[uull, bend right]
\end{tikzcd}$$

The exact cohomology sequence of the distinguished triangle marked by the octahedron shows that

$$\mathrm{H}^n(G_X) \longrightarrow \mathrm{H}^n(G')$$

is an epimorphism; hence $\mathrm{H}^n(G')$ is indeed of $\mathcal{C}_0$-finite type.

The preceding correction therefore reduces us to the case where $\mathrm{H}^{n+1}(u)$ is a monomorphism. We suppose this from now on. The exact cohomology sequence of $E \to F \to G$ then gives an exact sequence

$$\mathrm{H}^n(E) \xrightarrow{\mathrm{H}^n(u)} \mathrm{H}^n(F) \longrightarrow \mathrm{H}^n(G) \longrightarrow 0.$$

Since $\mathrm{H}^n(G)$ is of $\mathcal{C}_0$-finite type, the set of X over S for which there exists an epimorphism

$$M \longrightarrow \mathrm{H}^n(G_X), \qquad M \in \operatorname{ob} \mathcal{C}_{0,X},$$

is, by definition, a refinement of S. Using the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, one may impose that this epimorphism factor through $Z^n(F_X)$:

$$M \xrightarrow{s} Z^n(F_X) \longrightarrow \mathrm{H}^n(F_X) \longrightarrow \mathrm{H}^n(G_X).$$

It is clear that u', defined as in (1.4.1) with $r = 0$, is a morphism of complexes and that

$$\mathrm{H}^n(u') : \mathrm{H}^n(E_X) \oplus M \longrightarrow \mathrm{H}^n(F_X)$$

is an epimorphism. This completes the proof of (1.4.1).

Lemma 1.4.2. Let $\mathcal{C}$ be an abelian category, let $u : E \to F$ be a morphism of complexes of $\mathcal{C}$ (respectively an arrow of $\mathrm{D}(\mathcal{C})$), and let G be the cone of u. For a given $n \in \mathbb{Z}$, the following conditions are equivalent:

(i) $\mathrm{H}^i(G) = 0$ for $i \geq n$;

(ii) $\mathrm{H}^i(u)$ is an isomorphism for $i > n$ and an epimorphism for $i = n$.

The proof is immediate from the exact cohomology sequence.

Definition 1.4.3. We shall say that u is an *n-quasi-isomorphism* (respectively an *n-isomorphism*) if u satisfies the equivalent conditions of (1.4.2).

That being so, one deduces at once from (1.4.1) the following:

Corollary 1.4.4. Let $S, S, \mathcal{C}, \mathcal{C}_0$ be as in (1.4.1). Let $u : E \to F$ be an $(n+1)$-quasi-isomorphism of complexes of $\mathcal{C}_S$, with $n \in \mathbb{Z}$ fixed, such that $\mathrm{H}^n(G)$ is of $\mathcal{C}_0$-finite type. Then the set of X over S for which there exist (M, r, s) as in (1.4.1) such that u' is an n-quasi-isomorphism is a refinement of S.

In pictorial terms, one may, using the finiteness condition on $\mathrm{H}^n(G)$, locally "gain one notch," that is, locally modify u in degree n by adding to E^n an object of $\mathcal{C}_0$ in such a way as to make u an n-quasi-isomorphism.

Corollary 1.4.5. Let $\mathcal{C}$ be an abelian category, let $n \in \mathbb{Z}$, and let $u : E \to F$ be an n-quasi-isomorphism of complexes of $\mathcal{C}$. There exists a commutative diagram in $\mathrm{C}(\mathcal{C})$

$$\begin{tikzcd} E \arrow[r, "v"] \arrow[d, "u"'] & E' \arrow[dl, "u'"] \\ F & \end{tikzcd}$$

where u' is a quasi-isomorphism and v is a monomorphism inducing the identity in degrees $\geq n$, that is,

$$E'^i = E^i, \qquad v^i = \mathrm{Id}(E^i) \qquad \text{for } i \geq n.$$

Proof. Applying (1.4.4) in the case where S is the punctual site and $\mathcal{C}_0 = \mathcal{C}$, one obtains a commutative triangle in $\mathrm{C}(\mathcal{C})$

$$\begin{tikzcd} E \arrow[r, "v_{n-1}"] \arrow[d, "u"'] & E'_{n-1} \arrow[dl, "u'_{n-1}"] \\ F & \end{tikzcd}$$

where u'_{n-1} is an $(n-1)$-quasi-isomorphism and v_{n-1} is a monomorphism inducing the identity in degrees $\geq n$. Iterating the process, one finds, for $i \leq n-1$, a sequence of commutative triangles in $\mathrm{C}(\mathcal{C})$

$$\begin{tikzcd} E'_i \arrow[r, "v_{i-1}"] \arrow[d, "u'_i"'] & E'_{i-1} \arrow[dl, "u'_{i-1}"] \\ F & \end{tikzcd}$$

where u'_i is an i-quasi-isomorphism and v_{i-1} is a monomorphism inducing the identity in degrees $\geq i$. The corollary follows by passing to the limit. More precisely, writing d'_i for the differential of E'_i, define E' by

$$E'^i = E^i, \quad d'^i = d^i \quad (i \geq n),$$

and

$$E'^i = (E'_i)^i, \quad d'^i = (d'_i)^i \quad (i < n),$$

and then define v and u' by

$$v^j = (v_i)^j, \qquad u'^j = (u'_i)^j \qquad (i \leq j).$$

One immediately verifies that E', v, and u', so defined, answer the question.

In abbreviated form: one can extend an n-quasi-isomorphism into a quasi-isomorphism "without changing anything given in degree $> n$."

2. Pseudo-coherent complexes

2.0. In this section, $\mathcal{S}$ denotes a site, S an object of $\mathcal{S}$, $\mathcal{C}$ a flat abelian $\mathcal{S}$-category (1.1), and $\mathcal{C}_0$ a strictly full additive sub-$\mathcal{S}$-category of $\mathcal{C}$. We assume that $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms (1.2) and locally quasi-liftable in $\mathcal{C}$ (1.2).

Definition 2.1. Let $X \in \operatorname{ob} \mathcal{S}$, let E be a complex of $\mathcal{C}_X$, and let $n \in \mathbb{Z}$. We shall say that E is *strictly $\mathcal{C}_0$-n-pseudo-coherent* (respectively $\mathcal{C}_0$-pseudo-coherent, respectively $\mathcal{C}_0$-perfect) if $E^i = 0$ for i sufficiently large and $E^i \in \operatorname{ob} \mathcal{C}_{0,S}$ for $i \geq n$ (respectively for every i, respectively for every i and $E^i = 0$ for almost all i).

When no confusion is to be feared, we shall sometimes say "strictly n-pseudo-coherent" (respectively, and so on) instead of "strictly $\mathcal{C}_0$-n-pseudo-coherent".

Lemma 2.1.1. Let $E \in \operatorname{ob} \mathrm{C}(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. If E is strictly n-pseudo-coherent and if $\mathrm{H}^i(E) = 0$ for $i > n$, then $\mathrm{H}^n(E)$ is of finite type.

Proof. The sequence

$$0 \longrightarrow Z^n(E) \longrightarrow E^n \longrightarrow E^{n+1} \longrightarrow \cdots \longrightarrow 0$$

is exact, and $E^i \in \operatorname{ob} \mathcal{C}_0$ for $i \geq n$. Since $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms, it follows that $Z^n(E)$ is of finite type, hence so is $\mathrm{H}^n(E)$, which is a quotient of it.

Proposition 2.2. Let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. Then:

a) If F is strictly n-pseudo-coherent (relative to $\mathcal{C}_0$) and if

$$u : E \longrightarrow F$$

is a quasi-isomorphism, then the set of X over S such that there exists a quasi-isomorphism

$$v : F' \longrightarrow E_X$$

with F' strictly n-pseudo-coherent is a refinement of S.

b) The following conditions are equivalent:

(i) the set of X over S such that there exists a quasi-isomorphism

$$u : E \longrightarrow F_X$$

of $\mathrm{C}(\mathcal{C}_X)$, with E strictly n-pseudo-coherent, is a refinement of S;

(i′) the same condition as (i), but with u an isomorphism of $\mathrm{D}(\mathcal{C}_X)$;

(ii) the set of X over S such that there exists an n-quasi-isomorphism

$$u : E \longrightarrow F_X$$

of $\mathrm{C}(\mathcal{C}_X)$, with E strictly perfect, is a refinement of S;

(ii′) the same condition as (ii), but with u an n-isomorphism of $\mathrm{D}(\mathcal{C}_X)$.

Proof. For a), there exists, for i sufficiently large, an i-quasi-isomorphism

$$v_i : F'_i \longrightarrow E$$

with F'_i strictly n-pseudo-coherent (take $F'_i = 0$). We show that, if $i > n$, there exists, after refining S, an $(i-1)$-quasi-isomorphism

$$v_{i-1} : F'_{i-1} \longrightarrow E$$

with F'_{i-1} strictly n-pseudo-coherent. By (1.4.4), it is enough to show that $\mathrm{H}^{i-1}(M')$, where M' is the cone of v_i, is of $\mathcal{C}_0$-finite type. Now if M'' denotes the cone of uv_i, the hypothesis that u is a quasi-isomorphism and (1.4.2) imply that

$$\mathrm{H}^j(M') \xrightarrow{\sim} \mathrm{H}^j(M'') \qquad \text{for all } j,$$

and

$$\mathrm{H}^j(M') = 0 \qquad \text{for } j \geq i.$$

The cone formula

$$M''^j = F_i'^{\,j+1} \oplus F^j$$

shows, on the other hand, that M'' is strictly $(i-1)$-pseudo-coherent. Hence, by (2.1.1),

$$\mathrm{H}^{i-1}(M'') = \mathrm{H}^{i-1}(M')$$

is of finite type, whence the assertion. It follows, by iteration and after refining S, that there is an n-quasi-isomorphism

$$v_n : F'_n \longrightarrow E$$

where F'_n is strictly n-pseudo-coherent. But, by (1.4.5), one can extend v_n into a quasi-isomorphism

$$v : F' \longrightarrow E$$

without changing anything in degrees $\geq n$; in particular F' is strictly n-pseudo-coherent. This proves a).

For b), the equivalence of (i) and (i′) follows at once from a) and from the definition of isomorphisms of $\mathrm{D}^-(\mathcal{C}_X)$. If $E \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$ is strictly n-pseudo-coherent, there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

where E' is strictly perfect and $E''^i = 0$ for $i \geq n$. Consequently $E' \to E$ is an n-quasi-isomorphism, which gives the implication (i) ⇒ (ii). The implication (ii) ⇒ (ii′) is trivial, so it remains to prove (ii′) ⇒ (i). Let

$$u : E \longrightarrow F$$

be an n-isomorphism of $\mathrm{D}^-(\mathcal{C}_S)$, with E strictly perfect. Then u is represented by a diagram in $\mathrm{C}^-(\mathcal{C}_S)$

$$\begin{tikzcd} & E' \arrow[dl, "s"'] \arrow[dr, "u'"] & \\ E & & F, \end{tikzcd}$$

where s is a quasi-isomorphism and u' an n-quasi-isomorphism. By a), after refining S, there exists a quasi-isomorphism

$$t : E'' \longrightarrow E',$$

with E'' strictly n-pseudo-coherent. Applying (1.4.5) to $u't : E'' \to F$, we extend it to a quasi-isomorphism

$$E''' \longrightarrow F$$

with E''' strictly n-pseudo-coherent. This proves (ii$'$) $\Rightarrow$ (i) and completes the proof of (2.2).

Definition 2.3. Under the hypotheses of (2.2), we shall say that $F \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$ is n-$\mathcal{C}_0$-pseudo-coherent if F satisfies the equivalent conditions b) of (2.2). We shall say that F is $\mathcal{C}_0$-pseudo-coherent if F is n-$\mathcal{C}_0$-pseudo-coherent for every n.

When this cannot cause confusion, we shall sometimes allow ourselves to say n-pseudo-coherent, pseudo-coherent, instead of n-$\mathcal{C}_0$-pseudo-coherent, $\mathcal{C}_0$-pseudo-coherent.

Remark 2.3.1. Let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ be a pseudo-coherent complex. The set of X over S such that there exists a quasi-isomorphism

$$E \longrightarrow F_X$$

with E strictly pseudo-coherent (2.1) is not in general a refinement of S. However, we shall see later (2.7 a) and Exposé II) important cases in which this is so.

The proof of (2.2 a) gives the following result.

Proposition 2.4. Let $n, i \in \mathbb{Z}$, and let

$$u : E \longrightarrow F$$

be an i-quasi-isomorphism of $\mathrm{C}(\mathcal{C}_S)$, with F n-pseudo-coherent and E strictly n-pseudo-coherent. Then, after refining S, there exists a commutative triangle in $\mathrm{C}(\mathcal{C}_S)$

$$\begin{tikzcd} E \arrow[r, "v"] \arrow[d, "u"'] & E' \arrow[dl, "u'"] \\ F & \end{tikzcd}$$

where E' is strictly n-pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degree $\geq i$.

Proposition 2.5. **a)** If $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ is n-pseudo-coherent (respectively pseudo-coherent), then $F[m]$ is $(n-m)$-pseudo-coherent (respectively pseudo-coherent).

b) Let

$$\begin{tikzcd} E \arrow[rr] & & F \arrow[dl] \\ & G \arrow[ul, "+1"] & \end{tikzcd} \tag{*}$$

be a distinguished triangle of $\mathrm{D}(\mathcal{C}_S)$. If E and G are n-pseudo-coherent (respectively pseudo-coherent), then F is n-pseudo-coherent (respectively pseudo-coherent).

Proof. a) is trivial from the definitions.

b) It is enough, of course, to prove the non-parenthesized assertion. The triangle $(*)$ can also be written

$$\begin{tikzcd} G[-1] \arrow[rr] & & E \arrow[dl] \\ & F \arrow[ul, "+1"] & \end{tikzcd}$$

By a), $G[-1]$ is $(n+1)$-pseudo-coherent. Thus we are reduced to proving that, if E is $(n+1)$-pseudo-coherent and F is n-pseudo-coherent, then G is n-pseudo-coherent. After localizing, we may assume E (respectively F) strictly $(n+1)$-pseudo-coherent (respectively n-pseudo-coherent). The arrow

$$E \xrightarrow{u} F$$

of $\mathrm{D}^-(\mathcal{C}_S)$ is represented by a pair of arrows of $\mathrm{C}^-(\mathcal{C}_S)$

$$\begin{tikzcd} & E' \arrow[dl, "s"'] \arrow[dr, "u'"] & \\ E & & F, \end{tikzcd}$$

where s is a quasi-isomorphism. By (2.2 a)), after localizing, there exists a quasi-isomorphism

$$t : E'' \longrightarrow E',$$

where E'' is strictly $(n+1)$-pseudo-coherent. We are therefore reduced to the case where u is defined by a true arrow of $\mathrm{C}^-(\mathcal{C}_S)$. The assertion then follows from the definitions and from the formula

$$C(u)^i = E^{i+1} \oplus F^i,$$

where $C(u)$ denotes the cone of u.

Scholium 2.6. By rotation of triangles, one sees that pseudo-coherence properties for two of the objects of the triangle $(*)$ imply another such property for the third. The various cases may be summarized in the table below; the lower-case letters denote degrees of pseudo-coherence, and in each row the conclusion is underlined:

$$\begin{array}{ccc} E & F & G \\ n+1 & n & \underline{n} \\ n & \underline{n} & n \\ \underline{n} & n & n-1. \end{array}$$

We shall write

$$\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-}n\mathrm{coh}} \quad \text{respectively} \quad \mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$$

for the strictly full fibered subcategory of $\mathrm{D}(\mathcal{C})$ whose fiber over each $X \in \mathrm{ob}\, S$ is formed by the objects of $\mathrm{D}(\mathcal{C}_X)$ that are n-pseudo-coherent (respectively pseudo-coherent) relative to $\mathcal{C}_0$. By (2.5), the category

$$\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$$

is a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$ (1.1).

Proposition 2.7. Suppose that S is the punctual site, so that

$$\mathcal{C} = \mathcal{C}_S, \qquad \mathcal{C}_0 = \mathcal{C}_{0,S}.$$

Then:

a) If $F \in \mathrm{ob}\, \mathrm{D}(\mathcal{C})$ is pseudo-coherent, there exists a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly pseudo-coherent (2.1).

b) Let $\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0)$ denote the full triangulated subcategory of $\mathrm{K}^-(\mathcal{C}_0)$ formed by acyclic complexes. The canonical functor

$$\mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

is fully faithful and has essential image $\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$.

Proof. a) Let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$, and suppose constructed an i-quasi-isomorphism

$$u_i : E_i \longrightarrow F$$

of $\mathrm{C}(\mathcal{C})$, with E_i strictly pseudo-coherent; this is possible for i sufficiently large, taking $E_i = 0$. Let G_i be the cone of u_i. It follows from (1.4.2) and (2.1.1) that $\mathrm{H}^{i-1}(G_i)$ is of $\mathcal{C}_0$-finite type. Hence, by (1.4.4), there exists a commutative triangle in $\mathrm{C}(\mathcal{C})$

$$\begin{tikzcd} E_i \arrow[r, "v_{i-1}"] \arrow[d, "u_i"'] & E_{i-1} \arrow[dl, "u_{i-1}"] \\ F & \end{tikzcd}$$

where E_{i-1} is strictly pseudo-coherent, u_{i-1} an $(i-1)$-quasi-isomorphism, and v_{i-1} a monomorphism inducing the identity in degrees $\geq i$. Passing to the limit (cf. the proof of (1.4.5)) gives a quasi-isomorphism

$$u : E \longrightarrow F$$

with E strictly pseudo-coherent.

b) Apply the criterion of $[V]$, Chap. I, §2, 4.2 b)(i), together with a).

We note that the proof of a) gives the following more precise result, which generalizes (1.4.5).

Proposition 2.7.1. Under the hypotheses of (2.7), let $u : E \to F$ be an n-quasi-isomorphism of $\mathrm{C}(\mathcal{C})$, with F pseudo-coherent and E strictly pseudo-coherent. Then there exists a commutative triangle in $\mathrm{C}(\mathcal{C})$

$$\begin{tikzcd} E \arrow[r, "v"] \arrow[d, "u"'] & E' \arrow[dl, "u'"] \\ F & \end{tikzcd}$$

where E' is strictly pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degree $\geq n$.

Definition 2.8. Let $n \in \mathbb{N}$ and $F \in \operatorname{ob} \mathcal{C}_S$. We shall say that F is of n-$\mathcal{C}_0$-finite presentation if the set of X over S such that there exists an exact sequence

$$E^{-n} \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F_X \longrightarrow 0, \qquad E^i \in \operatorname{ob} \mathcal{C}_{0,X} \quad (-n \leq i \leq 0),$$

is a refinement of S. We shall say that F is of ∞-$\mathcal{C}_0$-finite presentation if F is of n-$\mathcal{C}_0$-finite presentation for every $n \in \mathbb{N}$.

Of course, we shall sometimes simply say n-finite presentation when there is no possible ambiguity about $\mathcal{C}_0$. We shall say “finite presentation” for “1-finite presentation,” and “finite type” for “0-finite presentation,” the latter notion coinciding with the one already introduced in (1.2).

Proposition 2.9. Let $n \in \mathbb{N}$ and $F \in \operatorname{ob} \mathcal{C}_S$. The following conditions are equivalent:

(i) F is of n-finite presentation (respectively of ∞-finite presentation);

(ii) F is $(-n)$-pseudo-coherent (respectively pseudo-coherent).

In the case where S is the punctual site, F is pseudo-coherent if and only if there exists an exact sequence

$$\cdots \longrightarrow E^i \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F \longrightarrow 0, \qquad E^i \in \operatorname{ob} \mathcal{C}_0 \quad \text{for all } i.$$

Proof. For the equivalence (i) $\Leftrightarrow$ (ii), it is plainly enough to treat the non-parenthesized case. The implication (i) $\Rightarrow$ (ii) is immediate from the definitions, and (ii) $\Rightarrow$ (i) follows at once from

(2.4): the arrow $0 \to F$ is a 1-quasi-isomorphism, and from this one deduces, after localizing, a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly $(-n)$-pseudo-coherent and $E^i = 0$ for $i \geq 1$, that is, an n-finite presentation of F. The second assertion is proved in the same way, using (2.7.1).

Proposition 2.10. a) Let $m, n \in \mathbb{Z}$, with $n \leq m$. If $F \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$ is n-pseudo-coherent and acyclic in degrees $\geq m$, then there exists locally a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly n-pseudo-coherent and $E^i = 0$ for $i \geq m$. If S is punctual, and if F is pseudo-coherent and acyclic in degrees $\geq m$, there exists a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly pseudo-coherent and $E^i = 0$ for $i \geq m$.

b) Let $F \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$ be acyclic in degrees $> n$. For F to be n-pseudo-coherent, it is necessary and sufficient that $\mathrm{H}^n(F)$ be of finite type.

Proof. a) Since F is acyclic in degrees $\geq m$, the arrow $0 \to F$ is an m-quasi-isomorphism. The assertion follows from (2.4) and (2.7.1).

b) There exists a distinguished triangle in $\mathrm{D}^-(\mathcal{C}_S)$:

$$\begin{tikzcd} F \arrow[rr] & & F' \arrow[dl] \\ & \mathrm{H}^n(F)[-n] \arrow[ul, "+1"] & \end{tikzcd}$$

(to see this, begin by truncating F, replacing F^n by $Z^n(F)$ and F^i by 0 for $i > n$). The exact cohomology sequence implies

$$\mathrm{H}^i(F') = 0 \quad \text{for } i \geq n-1,$$

so F' is $(n-1)$-pseudo-coherent. We conclude by (2.5) and (2.6).

Exposé I, continued: pseudo-coherent complexes and perfect complexes

Proposition 2.11. Let $n \in \mathbb{Z}$.

a) Let $F \in \operatorname{ob} \mathrm{C}^-(\mathcal{C}_S)$. If, for every $i \geq n$ (respectively for every i), F^i is of finite $(i-n)$-presentation (respectively of finite ∞-presentation), then F is n-pseudo-coherent (respectively pseudo-coherent).

b) Let $F \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$. If, for every $i \geq n$ (respectively for every i), $\mathrm{H}^i(F)$ is of finite $(i-n)$-presentation (respectively of finite ∞-presentation), then F is pseudo-coherent.

Proof. It is enough to treat the non-parenthesized assertions.

a) We argue by induction on N such that $F^i \neq 0$ only for $i \leq n+N$. If $N = 0$, the assertion follows at once from (2.10 b). Suppose $N \geq 1$. We have an exact sequence

$$0 \longrightarrow F^{n+N}[-n-N] \longrightarrow F \longrightarrow F' \longrightarrow 0.$$

By the induction hypothesis, F' is n-pseudo-coherent; on the other hand $F^{n+N}[-n-N]$ is n-pseudo-coherent by hypothesis, using (2.9 and 2.5 a). Hence, by (2.5 b), F is n-pseudo-coherent.

b) We argue by induction on N such that $\mathrm{H}^i(F) = 0$ for $i > n+N$. If $N = 0$, the assertion follows from (2.10 b). Suppose $N \geq 1$. As in the proof of (2.10 b), we have a distinguished triangle in $\mathrm{D}^-(\mathcal{C}_S)$

$$F' \longrightarrow F \longrightarrow \mathrm{H}^{n+N}(F)[-n-N] \longrightarrow F'[1].$$

more plainly, with cone $\mathrm{H}^{n+N}(F)[-n-N]$. The exact cohomology sequence implies

$$\mathrm{H}^i(F') = 0 \quad \text{for } i > n+N-1,$$

that is, for $i > (n-1)+(N-1)$, and

$$\mathrm{H}^i(F') \xrightarrow{\sim} \mathrm{H}^{i+1}(F) \quad \text{for } i < n+N-1.$$

By the induction hypothesis, applied with $n-1$ in place of n, F' is therefore $(n-1)$-pseudo-coherent. On the other hand, $\mathrm{H}^{n+N}(F)[-n-N]$ is n-pseudo-coherent by (2.9 and 2.5 a). Hence, by (2.6), F is n-pseudo-coherent.

Remark 2.11.1. If F is pseudo-coherent, the F^i (respectively the $\mathrm{H}^i(F)$) are not in general pseudo-coherent, nor even of finite type. Nevertheless, in the following paragraph, in the excursions on the notions of coherence, we shall see important examples where pseudo-coherence of F implies that of the $\mathrm{H}^i(F)$.

Proposition 2.12. Let $F', F'' \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$, and let $F = F' \oplus F''$. For F to be n-pseudo-coherent (respectively pseudo-coherent), it is necessary and sufficient that F' and F'' be n-pseudo-coherent (respectively pseudo-coherent).

Proof. It is enough to prove the non-parenthesized assertion. Sufficiency is obvious; let us prove necessity. By induction we may suppose that it has already been proved that F' and F'' are $(n+1)$-pseudo-coherent. After localizing, we may even suppose that F' and F'' are strictly $(n+1)$-pseudo-coherent and that $F = F' \oplus F''$ in $\mathrm{C}^-(\mathcal{C}_S)$. Let $F'_{>n}$ (respectively $F'_{\leq n}$) denote the complex obtained from F' by replacing F'^i by 0 for $i \leq n$ (respectively for $i > n$), and define $F''_{>n}$ and $F''_{\leq n}$ similarly. We have an exact sequence

$$0 \longrightarrow F'_{>n} \oplus F''_{>n} \longrightarrow F \longrightarrow F'_{\leq n} \oplus F''_{\leq n} \longrightarrow 0.$$

The complex $F'_{>n} \oplus F''_{>n}$ is strictly perfect; since F is n-pseudo-coherent, (2.6) implies that $F'_{\leq n} \oplus F''_{\leq n}$ is n-pseudo-coherent. This means, by (2.10 b), that

$$\mathrm{H}^n(F'_{\leq n} \oplus F''_{\leq n})$$

is of finite type. But

$$\mathrm{H}^n(F'_{\leq n} \oplus F''_{\leq n}) = \mathrm{H}^n(F'_{\leq n}) \oplus \mathrm{H}^n(F''_{\leq n}),$$

so $\mathrm{H}^n(F'_{\leq n})$ (respectively $\mathrm{H}^n(F''_{\leq n})$) is of finite type, and again by (2.10 b) $F'_{\leq n}$ (respectively $F''_{\leq n}$) is n-pseudo-coherent. Applying (2.5) to the exact sequence

$$0 \longrightarrow F'_{>n} \longrightarrow F' \longrightarrow F'_{\leq n} \longrightarrow 0$$

(and similarly for F''), we deduce that F' and F'' are n-pseudo-coherent.

Proposition 2.13. Let $(S', \mathcal{C}', \mathcal{C}'_0)$ be a triple satisfying the same hypotheses (2.0) as $(S, \mathcal{C}, \mathcal{C}_0)$. Let $f : S' \to S$ be a morphism of sites, corresponding to an underlying functor $f^{-1} : S \to S'$, and let

$$u : \mathrm{D}^*(\mathcal{C}) \longrightarrow \mathrm{D}^*(\mathcal{C}') \qquad (* = +, -, b)$$

be a Cartesian functor above f, exact on each fiber. Suppose that there exist $a, b \in \mathbb{Z}$ such that, for every $S \in \operatorname{ob} S$, the following conditions hold:

(i) if $E \in \operatorname{ob} \mathcal{C}_{0,S}$, then $u(E)$ is a-$\mathcal{C}'_0$-pseudo-coherent (respectively $\mathcal{C}'_0$-pseudo-coherent);

(ii) if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is acyclic in degrees > 0, then $u(F)$ is acyclic in degrees $> b$.

Then, if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is m-pseudo-coherent relative to $\mathcal{C}_0$ and acyclic in degrees $> n$, the object

$$u(F) \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}'_{S'}), \qquad S' = f^{-1}(S),$$

is $\sup(m+b, n+a)$-pseudo-coherent relative to $\mathcal{C}'_0$ (respectively $(m+b)$-pseudo-coherent), and is acyclic in degrees $> n+b$. In particular, in the parenthesized case, if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is pseudo-coherent relative to $\mathcal{C}_0$, then $u(F)$ is pseudo-coherent relative to $\mathcal{C}'_0$.

Proof. We may restrict to the non-parenthesized case. First suppose that $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}_S)$ is strictly perfect (2.1), with $F^i = 0$ for $i > n$. Then $u(F)$ is $(n+a)$-$\mathcal{C}'_0$-pseudo-coherent. We argue by induction on the number N of indices i such that $F^i \neq 0$. If $N = 1$, then $F = F^p[-p]$ for some $p \leq n$, so $u(F)$ is $(p+a)$-$\mathcal{C}'_0$-pseudo-coherent, hence a fortiori $(n+a)$-pseudo-coherent. If $N > 1$, there is an exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0,$$

where F' and F'' are strictly perfect of length $< N-1$, and where $F'^i = F''^i = 0$ for $i > n$. Applying (2.5 b) to the distinguished triangle

$$u(F') \longrightarrow u(F) \longrightarrow u(F'') \longrightarrow u(F')[1].$$

and using the induction hypothesis, we get that $u(F)$ is $(n+a)$-pseudo-coherent.

Now pass to the general case: F is m-$\mathcal{C}_0$-pseudo-coherent and acyclic in degrees $> n$. After localizing, we may, by (2.10 a), suppose F strictly m-pseudo-coherent and such that $F^i = 0$ for $i > n$. There is then an exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0,$$

where F'' is zero in degrees $\geq m$, and F' is strictly perfect and zero in degrees $> n$. In the distinguished triangle

$$u(F') \longrightarrow u(F) \longrightarrow u(F'') \longrightarrow u(F')[1].$$

the object $u(F'')$ is acyclic in degrees $> m+b$, hence $(m+b)$-pseudo-coherent, while $u(F')$ is $(n+a)$-pseudo-coherent by what was just proved. We conclude by (2.5 b).

Corollary 2.14. Let $\mathcal{C}_1$ be a strictly full additive sub-S-category of $\mathcal{C}$ such that

$$\mathcal{C}_0 \subset \mathcal{C}_1 \subset \mathcal{C}.$$

Assume that every object of $\mathcal{C}_1$ is $\mathcal{C}_0$-pseudo-coherent. Then:

(i) $\mathcal{C}_1$ is locally quasi-liftable in $\mathcal{C}$ (1.2), and locally quasi-stable under kernels of epimorphisms (1.2);

(ii) for $S \in \operatorname{ob} S$ and $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$, in order that F be n-pseudo-coherent (respectively pseudo-coherent) relative to $\mathcal{C}_1$, it is necessary and sufficient that it be so relative to $\mathcal{C}_0$.

Proof. For (i), let $E \to F$ be an epimorphism in $\mathcal{C}_S$, with $F \in \operatorname{ob} \mathcal{C}_{1,S}$. Since F is $\mathcal{C}_0$-pseudo-coherent, hence a fortiori of $\mathcal{C}_0$-finite type, there exists, after localizing, an epimorphism $E' \to F$ with $E' \in \operatorname{ob} \mathcal{C}_{0,S}$. Using the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, we may suppose, after localizing again, that this epimorphism factors through E; this gives the quasi-liftability of $\mathcal{C}_1$. On the other hand, if $E \in \operatorname{ob} \mathcal{C}_S$ admits a finite right resolution by objects of $\mathcal{C}_1$, then E is $\mathcal{C}_0$-pseudo-coherent by (2.11 a), hence a fortiori of $\mathcal{C}_1$-finite type.

For (ii), it is enough to prove necessity. For this one may apply, at choice, (2.11 a) or (2.13), with $S' = S$, $\mathcal{C}' = \mathcal{C}$, $f = \mathrm{Id}$, $u = \mathrm{Id}$, and with $\mathcal{C}'_0$ and $\mathcal{C}_0$ replaced respectively by $\mathcal{C}_0$ and $\mathcal{C}_1$.

Scholium 2.14.1. There is therefore a largest strictly full additive sub-S-category of $\mathcal{C}$ giving the same notion of n-pseudo-coherence (respectively pseudo-coherence) as $\mathcal{C}_0$, namely the category formed by the $\mathcal{C}_0$-pseudo-coherent objects. This latter category is locally stable under kernels of epimorphisms.

Example 2.15. Let (S, A) be a ringed topos, and let $\mathcal{C}, \mathcal{C}_0$ be the S-categories considered in (1.3 d). The fibered category $\mathrm{D}(\mathcal{C})$ will often be denoted by $\mathrm{D}(S, A)$, or $\mathrm{D}_A(S)$, when one wants to suggest that the modules are left A-modules, or even simply $\mathrm{D}(S)$ if there is no ambiguity about A. We denote by S the final object of S, and put $\mathrm{D}(\mathcal{C}_S) = \mathrm{D}_A(S)$ (or $\mathrm{D}(S)$). An object of $\mathrm{D}(S)$ will be said to be n-pseudo-coherent (respectively pseudo-coherent) if it is n-$\mathcal{C}_0$-pseudo-coherent (respectively $\mathcal{C}_0$-pseudo-coherent), and similarly for the strict notions. The category fibered in $\mathrm{D}(\mathcal{C})$, denoted $\mathrm{D}(\mathcal{C})_{n-\mathcal{C}_0-\mathrm{coh}}$ (respectively $\mathrm{D}(\mathcal{C})_{\mathcal{C}_0-\mathrm{coh}}$) in (2.6), will be denoted $\mathrm{D}_A(S)_{n-\mathrm{coh}}$ (respectively $\mathrm{D}_A(S)_{\mathrm{coh}}$). One observes that, by (2.14), the same notion of pseudo-coherence is obtained if $\mathcal{C}_0$ is replaced by the S-category $\mathcal{C}_1$ defined by

$U \longmapsto$ the category of left A_U-modules locally direct summands of free finite-type modules.

If an A-module M is a direct summand of a free finite-type A-module, it does not follow in general that M is locally free of finite type. One does, however, have the following result.

Lemma 2.15.1. Suppose that S satisfies one of the following two hypotheses:

(i) S has enough points (SGA 4 IV 6.4), and for every point s of S, the left A_s-modules which are projective of finite type are free; this last condition is satisfied, for example, if $A_s/\operatorname{rad}(A_s)$ is a field or if $\operatorname{rad}(A_s)$ is nilpotent;

(ii) (S, A) is a "locally ringed" topos in the sense of [1], which means that A is commutative and that, for every $X \in \operatorname{ob} S$ and every $f \in \Gamma(X, A)$, one has

$$X = X_f \cup X_{1-f},$$

where X_f (respectively X_{1-f}) denotes the largest subobject of X on which f (respectively $1 - f$) is invertible.

Then every direct summand of a free finite-type left A-module is locally free of finite type.

Proof. Suppose first that (i) holds. We note that, if E and F are two A-modules and E is of finite presentation, the canonical arrow

$$\operatorname{Hom}(E, F)_s \longrightarrow \operatorname{Hom}(E_s, F_s)$$

is an isomorphism for every point s of S. It follows that, if E and F are both of finite presentation and if $E_s \xrightarrow{\sim} F_s$ at a point s of S, then there is an s-pointed object U of S such that $E|U \xrightarrow{\sim} F|U$. The conclusion follows at once.

Suppose next that (ii) holds. Let $n \in \mathbb{N}$, and let $\mathbb{D}^n$ denote the functor on the category of locally ringed topoi belonging to a fixed universe, defined by

$$T \longmapsto \Gamma(T, \mathcal{O}_T)^n.$$

It is known [1] that the functor $\operatorname{End}_{\mathbb{D}}(\mathbb{D}^n)$ is represented by

$$M_n = \operatorname{Spec} \mathbb{Z}[T_{ij}] \quad (1 \le i \le n, 1 \le j \le n),$$

endowed with the endomorphism U_n of $(\mathcal{O}_{M_n})^n$ defined by the matrix (T_{ij}). Now let

$$p : A^n \longrightarrow A^n$$

be a projector. We show that $p(A^n)$ is locally free of finite type. From what has just been said, there exists a morphism

$$s : S \longrightarrow M_n$$

of locally ringed topoi such that $s^*(u_n) = p$. We may interpret s as a point of M_n with values in $\Gamma(S, A)$, that is, as a matrix of order n with coefficients in $\Gamma(S, A)$. Since this matrix is a projector, s factors through the closed subscheme

$$i : P_n \hookrightarrow M_n$$

corresponding to projectors; say $S \xrightarrow{s'} P_n \xrightarrow{i} M_n$. Hence

$$p = s'^* i^*(u_n),$$

and we are done, because, by the assertion under hypothesis (i), the image of $i^*(u_n)$ is a locally free finite-type sheaf.

Corollary 2.16. Let

$$f : (S', A') \longrightarrow (S, A)$$

be a morphism of ringed topoi, defined by a functor $f^{-1} : S \to S'$ and a morphism of sheaves of rings $f^{-1}(A) \to A'$. Let $\mathrm{D}({}_{A'}S'_A)$ denote the derived category of A'-A-bimodules, left over A' and right over A, and let

$$\otimes^{\mathrm{L}}_A : \mathrm{D}^-({}_{A'}S'_A) \times_{S'} \mathrm{D}^-({}_A S) \longrightarrow \mathrm{D}^-({}_{A'}S')$$

be the derived functor of the tensor product

$$(E', E) \longmapsto E' \otimes_A f^{-1}(E).$$

Let $X \in \operatorname{ob} S$, put $X' = u(X)$, and let $E' \in \operatorname{ob} \mathrm{D}^-({}_{A'}X'_A)$, $E \in \operatorname{ob} \mathrm{D}^-({}_A X)$. If E' is m'-pseudo-coherent (as an object of $\mathrm{D}^-({}_A S')$) and acyclic in degrees $> n'$, and if E is m-pseudo-coherent and acyclic in degrees $> n$, then

$$E' \otimes^{\mathrm{L}}_A E$$

is $\sup(m + n', m' + n)$-pseudo-coherent and acyclic in degrees $> n + n'$.

Proof. After replacing S by S/X, we may suppose $X = S$, and apply (2.13) with

$$u = E' \otimes^{\mathrm{L}}_A : \mathrm{D}^-({}_A S) \longrightarrow \mathrm{D}^-({}_{A'}S').$$

Let $U \in \operatorname{ob} S$, let $U' = f^{-1}(U)$, and let E be a complex of left A_U-modules. If E is acyclic in degrees $> n$, then $E' \otimes^{\mathrm{L}}_A E$ is acyclic in degrees $> n + n'$, by the definition of $\otimes^{\mathrm{L}}_A$. On the other hand, if $E = (A_U)^p$, then

$$E' \otimes^{\mathrm{L}}_A E = (E'|U')^p,$$

which is m'-pseudo-coherent by hypothesis and (2.12). The conditions of (2.13) are therefore satisfied, with $a = m'$ and $b = n'$, whence the conclusion.

In particular:

Corollary 2.16.1. Under the hypotheses of (2.16), one has:

a) The functor

$$\mathrm{L}f^* : \mathrm{D}^-({}_A S) \longrightarrow \mathrm{D}^-({}_{A'}S')$$

(which is none other than the functor $A' \otimes^{\mathrm{L}}_A$) induces a functor

$$\mathrm{L}f^* : \mathrm{D}^-({}_AS)_{\mathrm{coh}} \longrightarrow \mathrm{D}^-({}_{A'}S')_{\mathrm{coh}}$$

(with the notation of (2.6)).

b) If A is commutative, the functor

$$\otimes^{\mathrm{L}}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\otimes^{\mathrm{L}}_A : \mathrm{D}^-(S)_{\mathrm{coh}} \times_S \mathrm{D}^-(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^-(S)_{\mathrm{coh}}.$$

Exercise 2.16.2. Let S be a topos, and let $A \to A'$ be a morphism of rings of S. Assume one of the following hypotheses:

(i) $A \to A'$ is surjective, with nilpotent ideal as kernel;

(ii) the functor $A' \otimes_A -$, on the category of left A-modules, is exact and faithful.

Show that $E \in \operatorname{ob} \mathrm{D}^-({}_AS)$ is n-pseudo-coherent (respectively pseudo-coherent) if and only if

$$A' \otimes_A E \in \operatorname{ob} \mathrm{D}^-({}_{A'}S)$$

is n-pseudo-coherent (respectively pseudo-coherent).

Hints: first show that a left A-module L is of finite type if and only if $A' \otimes_A L$ is of finite type (cf. Bourbaki, Commutative Algebra, Chap. I, §3, Prop. 11 and Chap. II, §3, Prop. 4), and then reduce to this case by truncation, using (2.5) and (2.10 b).

3. Link with the classical notion of coherence

3.0. The hypotheses and notation are those of (2.0), except that $\mathcal{C}_0$ is no longer assumed locally quasi-stable under kernels of epimorphisms.

Definition 3.1. Let $F \in \operatorname{ob} \mathcal{C}_S$. We shall say that F is *precoherent* relative to $\mathcal{C}_0$ if, for every U over S and every morphism

$$u : E \longrightarrow F|U,$$

where $E \in \operatorname{ob} \mathcal{C}_{0,U}$, $\operatorname{Ker}(u)$ is of $\mathcal{C}_0$-finite type (1.2). We shall say that F is *coherent* relative to $\mathcal{C}_0$ if F is precoherent and of finite type.

Technical note. We introduce this notion here only as a technical intermediary; the only notion to retain is that of coherence.

Remark 3.2. Let $F \in \operatorname{ob} \mathcal{C}_S$. It is clear from the definitions that, if F is precoherent (respectively of finite type), then every subobject (respectively quotient) of F is precoherent (respectively of finite type).

Lemma 3.3.

a) Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H$$

be an exact sequence of $\mathcal{C}_S$. If G is of finite type and H is precoherent, then F is of finite type.

b) Let

$$F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be an exact sequence of $\mathcal{C}_S$. If F is of finite type and G is precoherent, then H is precoherent.

c) Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be an exact sequence of $\mathcal{C}_S$. If F and H are precoherent (respectively of finite type), then so is G.

Proof. a) After localizing, there is an epimorphism $G' \to G$ with $G' \in \operatorname{ob}\mathcal{C}_0$. Hence there is a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & F' \arrow[r] \arrow[d] & G' \arrow[r] \arrow[d] & H \arrow[d, "\mathrm{Id}"] \\
0 \arrow[r] & F \arrow[r] & G \arrow[r] & H.
\end{tikzcd}$$

Since H is precoherent, F' is of finite type; but $F' \to F$ is an epimorphism, so F is of finite type.

b) Let U be over S, and let $H' \xrightarrow{f} H|U$ be a morphism of $\mathcal{C}_U$, with $H' \in \operatorname{ob}\mathcal{C}_{0,U}$. After a change of notation, we may suppose $U = S$. By the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, there exists, after localizing, a commutative square

$$\begin{tikzcd}
G' \arrow[r, "p"] \arrow[d] & H' \arrow[d, "f"] \\
G \arrow[r] & H,
\end{tikzcd}$$

where the horizontal arrows are epimorphisms and $G' \in \operatorname{ob}\mathcal{C}_0$. Since $\operatorname{Ker}(f)$ is a quotient of $\operatorname{Ker}(fp)$, it is enough to show that $\operatorname{Ker}(fp)$ is of finite type; hence we may suppose $H' = G'$. The arrows $F \to G$ and $H' \to G$ define

$$g : F \oplus H' \longrightarrow G.$$

Putting $K = \operatorname{Ker}(G \to H)$, we have a commutative diagram, with exact rows and with q an epimorphism,

$$\begin{tikzcd}
0 \arrow[r] & F \arrow[r] \arrow[d, "q"] & F \oplus H' \arrow[r] \arrow[d, "g"] & H' \arrow[r] \arrow[d, "f"] & 0 \\
0 \arrow[r] & K \arrow[r] & G \arrow[r] & H \arrow[r] & 0.
\end{tikzcd}$$

The snake diagram shows that $\operatorname{Ker}(g) \to \operatorname{Ker}(f)$ is an epimorphism. But, since $F \oplus H'$ is of finite type and G is precoherent, $\operatorname{Ker}(g)$ is of finite type by a); hence we are done.

c) First prove the non-parenthesized assertion. Up to a change of notation, it is a matter of showing that, if $g : G' \to G$ is a morphism of $\mathcal{C}_S$, with $G' \in \operatorname{ob}\mathcal{C}_0$, then $\operatorname{Ker}(g)$ is of finite type. Let H' be the image of the composed arrow $G' \to H$. We have a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & F' \arrow[r] \arrow[d, "f"] & G' \arrow[r] \arrow[d, "g"] & H' \arrow[r] \arrow[d, "h"] & 0 \\
0 \arrow[r] & F \arrow[r] & G \arrow[r] & H \arrow[r] & 0.
\end{tikzcd}$$

Since h is a monomorphism, the snake diagram gives an isomorphism $\operatorname{Ker}(f) \xrightarrow{\sim} \operatorname{Ker}(g)$. But F' is of finite type by a), hence $\operatorname{Ker}(f)$ is of finite type, again by a). This proves the non-parenthesized assertion. For the parenthesized assertion, strictly speaking one cannot invoke (2.5 b), since $\mathcal{C}_0$ is not assumed locally quasi-stable under kernels of epimorphisms. After localizing, there are epimorphisms

$$F' \xrightarrow{u} F, \qquad H' \xrightarrow{v} H,$$

with $F', H' \in \operatorname{ob} \mathcal{C}_0$. After localizing further, we may suppose that v factors through G, whence a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & F' \arrow[r] \arrow[d, "u"] & F' \oplus H' \arrow[r] \arrow[d, "w"] & H' \arrow[r] \arrow[d, "v"] & 0 \\
0 \arrow[r] & F \arrow[r] & G \arrow[r] & H \arrow[r] & 0.
\end{tikzcd}$$

Since u and v are epimorphisms, so is w, and this proves the assertion. The proof of (3.3) is complete.

It follows immediately that:

Corollary 3.4. The full (respectively strictly full) subcategory of $\mathcal{C}_S$ formed by coherent objects is stable under finite projective (respectively inductive) limits. In particular, the kernel, cokernel, and image of an arrow between two coherent objects are coherent. Let

$$E^0 \longrightarrow E^1 \longrightarrow E^2 \longrightarrow E^3 \longrightarrow E^4$$

be an exact sequence of $\mathcal{C}_S$. If E^0, E^1, E^3, E^4 are coherent, then E^2 is coherent as well.

Corollary 3.5. Suppose that the objects of $\mathcal{C}_0$ are $\mathcal{C}_0$-coherent (which in particular implies that $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms, that is, that we are under the conditions of (2.0)).

a) For $F \in \operatorname{ob} \mathcal{C}_S$, the following conditions are equivalent:

(i) F is of finite presentation;

(ii) F is coherent;

(iii) F is of finite ∞-presentation (2.8).

b) Let $F \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$, and let $n \in \mathbb{Z}$. Then

$$F \text{ is } n\text{-pseudo-coherent} \iff \begin{array}{l} \mathrm{H}^i(F) \text{ is coherent for } i \geq n+1, \\ \text{and } \mathrm{H}^n(F) \text{ is of finite type.} \end{array}$$

In particular, for F to be pseudo-coherent it is necessary and sufficient that $\mathrm{H}^n(F)$ be coherent for every n.

Proof. Assertion a) and the implication $\Rightarrow$ in b) follow trivially from (3.4). The implication $\Leftarrow$ in b) follows at once from a) and (2.11 b).

Examples 3.6. Let (S, A) be a ringed topos, and let $\mathcal{C}, \mathcal{C}_0$ be the S-categories of (1.3 d), also as in (2.15). In order that the objects of $\mathcal{C}_0$ be coherent, it is necessary and sufficient that A be coherent, that is, precoherent. Here are examples where this is the case:

- A is constant with value a left noetherian ring (SGA 4 IX 2.1);
- S is the topos of sheaves of sets, for the Zariski topology, on a locally noetherian scheme S, and $A = \mathcal{O}_S$;
- S is the topos of sheaves of sets on a complex analytic space S, and $A = \mathcal{O}_S$.

On the other hand, the sheaf of continuous complex-valued functions on a topological space is not coherent in general.

Of course, if (S, A) is defined by a ringed space, the notion of coherence considered here coincides with the usual notion of Serre (EGA 0_I, 5.3.1).

Proposition 3.7. a) Let
$$f : (S', A') \longrightarrow (S, A)$$
be a morphism of ringed topoi. Assume that f is flat on the right, that is, that the functor $f^* = A' \otimes_A -$ is exact. There is, for $F \in \operatorname{ob} \mathrm{D}^-_A(S)$ and $G \in \operatorname{ob} \mathrm{D}^+_A(S)$, a functorial canonical arrow
$$f^*\mathrm{RHom}(F, G) \longrightarrow \mathrm{RHom}(f^*(F), f^*(G)).$$
This arrow is an isomorphism when F is pseudo-coherent.

b) Suppose A is commutative and coherent, and let $F \in \operatorname{ob} \mathrm{D}^-(S)$, $G \in \operatorname{ob} \mathrm{D}^+(S)$. If $\mathrm{H}^i(F)$ and $\mathrm{H}^i(G)$ are coherent for all i, then the same is true of $\mathrm{Ext}^i(F, G)$.

Proof. a) To define the arrow (cf. SGA 4 XVII), one may suppose that G has injective components, and one takes the composite arrow
$$f^* \mathrm{Hom}(F, G) \longrightarrow \mathrm{Hom}(f^*F, f^*G) \longrightarrow \mathrm{RHom}(f^*F, f^*G).$$
Let us show that it is an isomorphism for F pseudo-coherent. For brevity write T (respectively T') for the functor $f^*\mathrm{RHom}(\cdot, G)$ (respectively $\mathrm{RHom}(f^*\cdot, f^*G)$), and let $a \in \mathbb{Z}$ be such that $\mathrm{H}^i(G) = 0$ for $i < a$. It is clear that one has $TA \xrightarrow{\sim} T'A$, hence by dévissage $TF \xrightarrow{\sim} T'F$ for F strictly perfect (2.1). On the other hand, if F is acyclic in degrees $> n$, then TF and $T'F$ are acyclic in degrees $< a - n$. To verify that $TF \xrightarrow{\sim} T'F$ for F pseudo-coherent, the question being local, one may suppose F strictly n-pseudo-coherent; that is, there is a distinguished triangle
$$F' \longrightarrow F \longrightarrow F'' \longrightarrow F'[1].$$
with F' strictly perfect and F'' acyclic in degrees $> n - 1$. From this one obtains a morphism of distinguished triangles
$$\begin{array}{ccccc} TF'' & \to & TF & \to & TF' \\ \downarrow & & \downarrow & & \downarrow \\ T'F'' & \to & T'F & \to & T'F'. \end{array}$$
Now $TF' \xrightarrow{\sim} T'F'$, since F' is strictly perfect, and TF'' and $T'F''$ are acyclic in degrees $\leq a - n$. Writing the induced morphism on the exact cohomology sequences of the preceding triangles, one deduces that $TF \to T'F$ induces an isomorphism
$$\mathrm{H}^i(TF) \xrightarrow{\sim} \mathrm{H}^i(T'F)$$
for $i \leq a - n$. Since n may be chosen arbitrarily negative, this proves the assertion.

b) The proof is left as an exercise to the reader; see, for example, [H] II 3.3.

Remark 3.7.1. We shall see below (7.1.2) another case of compatibility of RHom with $\mathrm{L}f^*$.

Corollary 3.7.2. (cf. EGA 0_I, 5.2.7.) Let (S, A) be a ringed topos, let s be a point of S (SGA 4 VIII 7.8), and let
$$E, F \in \operatorname{ob} \mathrm{D}^b_{A,\mathrm{coh}}(S).$$
If $E_s \xrightarrow{\sim} F_s$, then there exists an s-pointed object U of S such that
$$E|U \xrightarrow{\sim} F|U.$$

Proof. Let $\alpha : E_s \to F_s$ and $\beta : F_s \to E_s$ be reciprocal isomorphisms. By (2.22 a), one has
$$\mathrm{Hom}(E, F)_s \xrightarrow{\sim} \mathrm{Hom}(E_s, F_s), \qquad \mathrm{Hom}(F, E)_s \xrightarrow{\sim} \mathrm{Hom}(F_s, E_s).$$

There are therefore a neighbourhood U of s and a section u (respectively v) of $\mathrm{Hom}(E,F)$ (respectively of $\mathrm{Hom}(F,E)$) over U such that $u_s = \alpha$ (respectively $v_s = \beta$). The morphisms $vu : E|U \to E|U$ and $uv : F|U \to F|U$ induce the identity at s; hence on a suitable V over U, as required.

4. Perfect complexes

4.0. In this section, S denotes a site, S an object of S, $\mathcal{C}$ a flat abelian S-category, and $\mathcal{C}_0$ a strictly full additive sub-S-category of $\mathcal{C}$ (1.1). Unless otherwise mentioned, we suppose that $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$ and locally stable under kernels of epimorphisms (1.2).

Lemma 4.1. Let

$$u : E \longrightarrow F$$

be an arrow of $\mathrm{C}(\mathcal{C}_S)$. If E is strictly $\mathcal{C}_0$-perfect (2.1) and F is acyclic, then u is locally homotopic to zero.

Proof. One has to construct locally a homotopy k such that

$$u = dk + kd.$$

Since E is strictly perfect, there are $a, b \in \mathbb{Z}$, with $a \leq b$, such that $E^i = 0$ for $i \notin [a,b]$. Thus one may take $k^i = 0$ for $i \notin [a,b]$. We construct the k^i, for $i > a$, by descending induction on i. Suppose $k^i : E^i \to F^{i-1}$ has been constructed so that

$$u^i = k^{i+1} d_E^i + d_F^{i-1} k^i$$

for $i > n > a$. We show that, after localizing, there exists

$$k^{n-1} : E^{n-1} \longrightarrow F^{n-2}$$

such that

$$u^{n-1} = k^n d_E^{n-1} + d_F^{n-2} k^{n-1}.$$

By the induction hypothesis, the arrow

$$v^{n-1} = u^{n-1} - k^n d_E^{n-1} : E^{n-1} \longrightarrow F^{n-1}$$

factors through $Z^{n-1}(F)$. But $Z^{n-1}(F) = B^{n-1}(F)$, since F is acyclic. Since $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$, it follows, after localizing, that there exists $k^{n-1} : E^{n-1} \to F^{n-2}$ such that $v^{n-1} = d_F^{n-2} k^{n-1}$, as required.

Corollary 4.2. Let

$$\begin{array}{ccc} P & & \\ \big\downarrow{\scriptstyle v} & & \\ F & \xleftarrow[f]{} & E \end{array}$$

be a diagram of $\mathrm{C}(\mathcal{C}_S)$, where P is strictly $\mathcal{C}_0$-perfect and f is a quasi-isomorphism. The set of X over S for which there exists an arrow

$$u : P_X \longrightarrow E_X$$

of $\mathrm{C}(\mathcal{C}_X)$ such that the diagram

$$\begin{tikzcd} & P_X \arrow[d, "v_X"] \\ E_X \arrow[ur, leftarrow, "u"] \arrow[r, "f_X"'] & F_X \end{tikzcd}$$

is commutative in $\mathrm{K}(\mathcal{C}_X)$, is a refinement of S.

Proof. Let G be the cone of f. There is an exact sequence

$$\longrightarrow \mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,E) \longrightarrow \mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,F) \longrightarrow \mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,G) \longrightarrow .$$

Since G is acyclic, the image of v in $\mathrm{Hom}_{\mathrm{K}(\mathcal{C}_S)}(P,G)$ is locally zero by (4.1). Therefore, after localizing, there exists $u : P \to E$ such that $v = fu$ in $\mathrm{K}(\mathcal{C}_S)$, as required.

Corollary 4.3. Let $f : E \to F$ be a quasi-isomorphism of $\mathrm{C}(\mathcal{C}_S)$, with F strictly perfect relative to $\mathcal{C}_0$. The set of X over S for which there exists a quasi-isomorphism

$$g : F_X \longrightarrow E_X$$

such that $f_X g$ is homotopic to Id_{F_X}, is a refinement of S.

Proof. This is a special case of (4.2).

Scholium 4.4. Let $F \in \mathrm{ob}\,\mathrm{K}(\mathcal{C}_S)$. One may define, for example with the aid of a cleavage of $\mathcal{C}$ (SGA 1 VI 7.1), a category $(\mathrm{Qis}\,/F)$ fibered over S/S, whose fiber over X over S is the category $(\mathrm{Qis}\,/F_X)^\circ$, opposite to the category of quasi-isomorphisms with target F_X. One may then express (4.3) by saying that, if F is strictly perfect, the arrow Id_F is locally cofinal in $(\mathrm{Qis}\,/F)$.

Notation 4.5. Let $\mathcal{M}$ be a fibered S-category. Let $X \in \mathrm{ob}\,S$, and let $E, F \in \mathrm{ob}\,\mathcal{M}_X$. We shall denote by

$$\underline{\mathrm{Hom}}_{\mathcal{M}}(E,F)$$

the sheaf on S/X associated to the presheaf of S-morphisms from E to F in $\mathcal{M}$. For the definition of this presheaf, see [G] I 2.6. One must be careful that, even if $\mathcal{C}$ is a stack ([G]), the corresponding presheaf for $\mathcal{M} = \mathrm{K}(\mathcal{C})$ or $\mathrm{D}(\mathcal{C})$ is not in general a sheaf.

Corollary 4.6. Let $E, F \in \mathrm{ob}\,\mathrm{C}(\mathcal{C}_S)$, with E strictly perfect. The canonical arrow

$$\underline{\mathrm{Hom}}_{\mathrm{K}(\mathcal{C})}(E,F) \longrightarrow \underline{\mathrm{Hom}}_{\mathrm{D}(\mathcal{C})}(E,F)$$

is an isomorphism.

Proof. This is an immediate consequence of (4.4) and of the formula

$$\underline{\mathrm{Hom}}_{\mathrm{D}(\mathcal{C}_X)}(E_X,F_X) = \lim_{(\mathrm{Qis}\,/E_X)^\circ} \underline{\mathrm{Hom}}_{\mathrm{K}(\mathcal{C}_X)}(\,\cdot\,,F_X)$$

for X over S.

Definition 4.7. Let $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C}_S)$, and let $a, b \in \mathbb{Z}$, with $a \le b$. We shall say that F has $\mathcal{C}_0$-perfect amplitude contained in the interval $[a,b]$, and we shall write

$$\mathcal{C}_0\text{-}\,\mathrm{Parf}\text{-}\mathrm{amp}(F) \subset [a,b],$$

if the set of X over S for which there exists a quasi-isomorphism

$$F' \longrightarrow F$$

in $\mathrm{C}(\mathcal{C}_X)$, with F' strictly $\mathcal{C}_0$-perfect (2.1) and such that $F'^i = 0$ for $i \notin [a,b]$, is a refinement of S. By (4.3), this condition is equivalent to the one obtained by replacing "quasi-isomorphism in $\mathrm{C}(\mathcal{C}_X)$" by "isomorphism in $\mathrm{D}(\mathcal{C}_X)$." We shall say that F has finite $\mathcal{C}_0$-perfect amplitude if there are integers a and b such that $\mathcal{C}_0$-Parf-amp$(F) \subset [a,b]$. We shall say that F is $\mathcal{C}_0$-perfect if F is locally of finite $\mathcal{C}_0$-perfect amplitude.

Example 4.8. Let (S,A) be a ringed topos, let $\mathcal{C}$ be the S-category defined by

$$U \longmapsto \text{the category of left } A_U\text{-modules},$$

and let $\mathcal{C}_0$ be the sub-S-category defined by

$U \longmapsto$ the category of left A_U-modules which are direct summands of free finite-type modules.

The pair $(\mathcal{C}, \mathcal{C}_0)$ satisfies the hypotheses (4.0) by (1.3.1). An object F of $\mathrm{D}_A(S)$ (cf. (2.15)) will be said to have perfect amplitude contained in $[a,b]$ (respectively finite perfect amplitude, respectively to be perfect) if F has $\mathcal{C}_0$-perfect amplitude contained in $[a,b]$ (respectively the corresponding property). When the direct summands of free finite-type modules are locally free (cf. (2.15.1)), to say that F is perfect means that F is locally isomorphic, in $\mathrm{D}(S)$, to a bounded complex whose components are free of finite type.

Notation 4.9. We shall denote by $\mathrm{D}(\mathcal{C})_{\mathcal{C}_0\text{-}\mathrm{Parf}}$, or $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$, the strictly full sub-$S$-category of $\mathrm{D}(\mathcal{C})$ formed by the $\mathcal{C}_0$-perfect objects. In the case of Example (4.8), we shall write $\mathrm{D}_A(S)_{\mathrm{Parf}}$, or $\mathrm{Parf}_A(S)$, or simply $\mathrm{Parf}(S)$ if no ambiguity about A is possible.

Proposition 4.10. The category $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$ (1.1.4). The canonical functor

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$$

(respectively

$$\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0)) \longrightarrow \mathrm{Tr}(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0))$$

(1.1.3)) induces an S-equivalence on the associated stacks ([G] II 2.2.4), which means:

(i) for every $X \in \operatorname{ob} S$ and all $E, F \in \operatorname{ob} \mathrm{K}^b(\mathcal{C}_{0,X})$ (respectively $\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_{0,X}))$), the canonical arrow

$$\mathrm{Hom}_{\mathrm{K}^b(\mathcal{C}_0)}(E,F) \longrightarrow \mathrm{Hom}_{\mathrm{D}(\mathcal{C})}(E,F)$$

(respectively

$$\mathrm{Hom}_{\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))}(E,F) \longrightarrow \mathrm{Hom}_{\mathrm{Tr}(\mathrm{D}(\mathcal{C}))}(E,F)$$

) is an isomorphism; and

(ii) every object of $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ (respectively of $\mathrm{Tr}(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0))$) is locally isomorphic in $\mathrm{D}(\mathcal{C})$ (respectively in $\mathrm{Tr}(\mathrm{D}(\mathcal{C}))$) to an object of $\mathrm{K}^b(\mathcal{C}_0)$ (respectively of $\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))$).

Proof. The non-parenthesized assertion (i) is a special case of (4.6), and the non-parenthesized assertion (ii) is true by the definition (4.7) and (4.9). It is clear, moreover, that $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is stable under translation of degrees. To prove that $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$, it remains to show that, if

$$E \xrightarrow{u} F \longrightarrow G \longrightarrow E[1]. \tag{*}$$

is a distinguished triangle of $\mathrm{D}(\mathcal{C})$ with E and F perfect, then G is perfect. After localizing, we may suppose that E and F are objects of $\mathrm{K}^b(\mathcal{C}_0)$. After localizing further, we may suppose, by the non-parenthesized assertion (i), that u is an arrow of $\mathrm{K}^b(\mathcal{C}_0)$; then G is perfect, since it is isomorphic to the cone of u, which is strictly perfect. Hence $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ is indeed a triangulated sub-S-category of $\mathrm{D}(\mathcal{C})$. The preceding argument also proves the parenthesized assertion (ii). It remains to prove the parenthesized assertion (i). Let $X \in \operatorname{ob} S$, and let

$$E = (E_1 \xrightarrow{a_1} E_2 \xrightarrow{a_2} E_3 \xrightarrow{a_3} E_1[+1]), \quad F = (F_1 \xrightarrow{b_1} F_2 \xrightarrow{b_2} F_3 \xrightarrow{b_3} F_1[+1])$$

be distinguished triangles of $\mathrm{K}^b(\mathcal{C}_{0,X})$. We must show that

$$\mathrm{Hom}_{\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))}(E,F) \longrightarrow \mathrm{Hom}_{\mathrm{Tr}(\mathrm{D}(\mathcal{C}))}(E,F) \tag{**}$$

is an isomorphism. By the non-parenthesized assertion (i), we already know that $(**)$ is injective. Let $(u_i) : E \to F$ be a morphism of triangles in $\mathrm{D}(\mathcal{C})$. After localizing, the non-parenthesized assertion (i) permits us to suppose that u_i, for $1 \leq i \leq 3$, is an arrow of $\mathrm{K}^b(\mathcal{C}_0)$. The arrow

$$u_{i+1} a_i - b_i u_i,$$

for $1 \leq i \leq 3$ (with the convention $u_4 = u_1$), is zero in $\mathrm{D}(\mathcal{C})$, hence locally zero in $\mathrm{K}^b(\mathcal{C}_0)$ by the non-parenthesized assertion (i). In other words, (u_i) locally defines a morphism of triangles in $\mathrm{K}^b(\mathcal{C}_0)$. This proves that $(**)$ is surjective and completes the proof of (4.10).

Remark 4.10.1. In (4.10), one may replace $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ by

$$\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-}\mathrm{Parf}} = \mathrm{D}^*(\mathcal{C}) \cap \mathrm{Parf}(\mathcal{C}, \mathcal{C}_0) \qquad (* = - \text{ or } b),$$

and nothing changes in the proof.

Complement 4.11. The triangulated structure of $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$ can be made more precise in terms of perfect amplitude (4.7). If $E \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})$ has perfect amplitude contained in $[a, b]$, then $E[n]$ has perfect amplitude contained in $[a - n, b - n]$.

Let

$$E \longrightarrow F \longrightarrow G \longrightarrow E[1].$$

be a distinguished triangle of $\mathrm{D}(\mathcal{C})$. One may draw up a table analogous to (2.6), where the entries indicate perfect amplitudes and each row corresponds to an implication whose target is underlined:

$$\begin{array}{ccc} E & F & G \\ [a+1, b+1] & [a, b] & \underline{[a, b]} \\ [a, b] & \underline{[a, b]} & [a, b] \\ \underline{[a, b]} & [a, b] & [a-1, b-1]. \end{array}$$

From (4.10) one immediately deduces:

Corollary 4.12. Suppose that S is a punctual site. For $F \in \mathrm{ob}\,\mathrm{D}(\mathcal{C})$ to have perfect amplitude relative to $\mathcal{C}_0$ contained in $[a, b]$, it is necessary and sufficient that there exist a quasi-isomorphism $F' \to F$, with F' strictly perfect and such that $F'^i = 0$ for $i \notin [a, b]$. The canonical functors

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$$

and

$$\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0)) \longrightarrow \mathrm{Tr}(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0))$$

are equivalences of categories.

Remark 4.12.1. When S is a punctual site, the objects of $\mathcal{C}_0$ are projective. When $\mathcal{C}_0$ contains all the projective objects of $\mathcal{C}$, one says "projective amplitude" instead of "$\mathcal{C}_0$-perfect amplitude."

Exposé I, continued: perfect amplitude, finite Tor-dimension, and rank

Remark 4.12.2. Let $\mathcal{C}$ be an abelian category (which may be regarded as fibered over a punctual site), and let $\mathcal{C}_0$ be a strictly full additive subcategory. Suppose that $\mathcal{C}_0$ is stable under kernels of epimorphisms and is quasi-liftable in $\mathcal{C}$. Then an acyclic object of $\mathrm{K}^b(\mathcal{C}_0)$ is not in general zero; hence there is no hope that the functor

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

should be faithful. However, if $\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0)$ denotes the thick subcategory of $\mathrm{K}^b(\mathcal{C}_0)$ formed by the objects which are acyclic in $\mathcal{C}$, the canonical functor

$$\mathrm{K}^b(\mathcal{C}_0)/\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

is fully faithful.

Indications: one already knows from (2.7) that the functor

$$\mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}^-(\mathcal{C})$$

is fully faithful. Show, for example with the aid of the criterion of [V], 4.2 b)(ii), that the functor

$$\mathrm{K}^b(\mathcal{C}_0)/\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0)$$

is fully faithful.

Lemma 4.13. Let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$, and let $a, b, c \in \mathbb{Z}$ be such that $a \leq b$ and $a \leq c$. If F is of perfect amplitude contained in $[a, c]$ (4.7), and if $\mathrm{H}^i(F) = 0$ for $i > b$, then F is of perfect amplitude contained in $[a, d]$, where $d = \inf(b, c)$.

Proof. We may suppose $b \leq c$, and F strictly perfect with $F^i = 0$ for $i \notin [a, c]$. Then F is isomorphic to its truncation F' defined by

$$F'^i = F^i \quad (i < b), \qquad F'^b = Z^b(F), \qquad F'^i = 0 \quad (i > b).$$

But $Z^b(F)$ admits a finite right resolution by objects of $\mathcal{C}_0$, hence is locally an object of $\mathcal{C}_0$, whence the assertion.

Corollary 4.14. Let $F \in \operatorname{ob} \mathcal{C}_S$ and $n \in \mathbb{N}$. The following conditions are equivalent:

(i) $\operatorname{perf.amp}(F) \subset [-n, 0]$;

(ii) F locally admits a left resolution of length n by objects of $\mathcal{C}_0$, that is, the set of objects X over S such that there exists an exact sequence

$$0 \longrightarrow L^{-n} \longrightarrow L^{-n+1} \longrightarrow \cdots \longrightarrow L^0 \longrightarrow F_X \longrightarrow 0,$$

with $L^i \in \operatorname{ob} \mathcal{C}_{0,X}$ for all i, is a refinement of S.

Proof. This is an immediate consequence of (4.7) and (4.13).

Definition 4.14.1. We shall say that F is of perfect dimension $\leq n$, and shall write $\operatorname{perf.dim}(F) \leq n$, if F satisfies the equivalent conditions of (4.14). We shall call the perfect dimension of F the smallest integer n such that $\operatorname{perf.dim}(F) \leq n$; if there is no such integer, we shall say that F is of infinite perfect dimension.

When S is the punctual site and $\mathcal{C}_0$ is the full subcategory of $\mathcal{C}$ formed by the projective objects, the perfect dimension is just the usual projective dimension.

Proposition 4.15. Let $a, b, n \in \mathbb{Z}$, with $a \leq b$ and $n \geq 0$, and let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C})$ be such that $F^i = 0$ (respectively $\mathrm{H}^i(F) = 0$) for $i \notin [a, b]$. If, for every $i \in [a, b]$, F^i (respectively $\mathrm{H}^i(F)$) is of perfect dimension $\leq n + (i - a)$, then F is of perfect amplitude contained in $[a - n, b]$. Consequently, if for every $i \in [a, b]$, F^i (respectively $\mathrm{H}^i(F)$) is perfect, then F is perfect.

Proof. Proceed by induction on b (with a and n fixed), by truncating F and applying (4.11), cf. the proof of (2.11).

Lemma 4.16. Let $a, b \in \mathbb{Z}$, $a \leq b$, and let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C})$ be such that $F^i = 0$ for $i \notin [a, b]$. Suppose that $\operatorname{perf.amp}(F) \subset [a, b]$, and that $F^i \in \operatorname{ob} \mathcal{C}_0$ for $i > a$. Then F^a is locally an object of $\mathcal{C}_0$.

Proof. There is an exact sequence, obtained by truncating F,

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F^a[-a] \longrightarrow 0,$$

with $\operatorname{perf.amp}(F') \subset [a+1, b+1]$. From (4.11) and (4.13) one deduces that $\operatorname{perf.dim}(F^a) = 0$, i.e. that F^a is locally an object of $\mathcal{C}_0$.

Proposition 4.17. Let $F', F'' \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$, and let $a, b \in \mathbb{Z}$, with $a \leq b$. Consider the conditions:

(i) $\operatorname{perf.amp}(F') \subset [a, b]$ and $\operatorname{perf.amp}(F'') \subset [a, b]$;

(ii) $\operatorname{perf.amp}(F' \oplus F'') \subset [a, b]$.

Then (i) implies (ii). If $\mathcal{C}_0$ is locally stable under direct factors, i.e. if the relation $E' \oplus E'' \in \operatorname{ob} \mathcal{C}_0$, for $E', E'' \in \operatorname{ob} \mathcal{C}_X$ and $X \in \operatorname{ob} S$, implies that locally E' and E'' are objects of $\mathcal{C}_0$, then (i) and (ii) are equivalent.

Proof. The implication (i) $\Rightarrow$ (ii) is a particular case of (4.11). Let us prove the converse. Put $F = F' \oplus F''$. Since F is pseudo-coherent, F' and F'' are also pseudo-coherent (2.12). On the other hand, the relation $\mathrm{H}^i(F) = 0$ for $i \notin [a, b]$ implies $\mathrm{H}^i(F') = \mathrm{H}^i(F'') = 0$ for $i \notin [a, b]$. By (2.10 a), we may therefore suppose, after localizing, that F'^i and F''^i are in $\operatorname{ob} \mathcal{C}_0$ for $i \geq a+1$, and that $F'^i = F''^i = 0$ for $i > b$. Then, by truncating, we get $F'^i = F''^i = 0$ for $i < a$. Since $\operatorname{perf.amp}(F) \subset [a, b]$, it follows from (4.16) that F^a is locally an object of $\mathcal{C}_0$. But $F^a = F'^a \oplus F''^a$, hence F'^a and F''^a are locally objects of $\mathcal{C}_0$, since $\mathcal{C}_0$ is locally stable under direct factors.

Proposition 4.18. Let $(S', \mathcal{C}', \mathcal{C}'_0)$ be a triple satisfying the same hypotheses (4.0) as $(S, \mathcal{C}, \mathcal{C}_0)$. Let $f : S' \to S$ be a morphism of sites, defined by $f^{-1} : S \to S'$, and let

$$u : \mathrm{D}^*(\mathcal{C}) \longrightarrow \mathrm{D}^*(\mathcal{C}') \qquad (* = -, +, b)$$

be a functor over f (cf. (2.13)). If u sends objects of $\mathcal{C}_0$ to $\mathcal{C}'_0$-perfect complexes, then u defines a functor

$$\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}} \longrightarrow \mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}.$$

Proof. Let $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}}$. We show that $u(F) \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}$. Since the question is local, one may suppose F strictly perfect. It remains only to use dévissage, taking into account the fact that $\mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}$ is a triangulated subcategory of $\mathrm{D}^*(\mathcal{C}')$ (4.10).

Remark 4.18.1. Suppose that there exist $m, n \in \mathbb{Z}$, $m \leq n$, such that $F \in \operatorname{ob} \mathcal{C}_0$ implies

$$\mathcal{C}'_0\text{-}\operatorname{perf.amp}(u(F)) \subset [m, n].$$

Then, if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C})$, one has

$$\mathcal{C}_0\text{-}\operatorname{perf.amp}(F) \subset [a, b] \quad \Rightarrow \quad \mathcal{C}'_0\text{-}\operatorname{perf.amp}(u(F)) \subset [a+m, b+n].$$

The proof is left as an exercise to the reader (use (4.11)).

Corollary 4.19. Under the hypotheses of (2.16), if $E' \in \operatorname{ob} \mathrm{D}^-(A, S')$ is perfect as an object of $\mathrm{D}^-(A, S')$, and $E \in \operatorname{ob} \mathrm{D}^-(A, S)$ is perfect, then

$$E' \overset{\mathrm{L}}{\otimes}_A E$$

is perfect.

Proof. We may suppose that $E' \in \operatorname{ob} \mathrm{D}^-(A, S'_A)$, where S' is the final object of S', and apply (4.18) to

$$u = E' \overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S').$$

In particular:

Corollary 4.19.1. a) The functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S')$$

induces a functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^-(A, S')_{\mathrm{parf}}.$$

b) If A is commutative, the functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S)_{\mathrm{parf}} \times_S \mathrm{D}^-(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^-(S)_{\mathrm{parf}}.$$

Remark 4.19.2. One can refine (4.19) in terms of perfect amplitude:

$$\operatorname{perf.amp}(E) \subset [a, b], \qquad \operatorname{perf.amp}(E') \subset [a', b'] \quad \Rightarrow \quad \operatorname{perf.amp}(E' \overset{\mathrm{L}}{\otimes}_A E) \subset [a + a', b + b'].$$

In particular,

$$\operatorname{perf.amp}(E) \subset [a, b] \quad \Rightarrow \quad \operatorname{perf.amp}(\mathrm{L}f^*(E)) \subset [a, b].$$

5. Finite Tor-dimension and perfection

5.0. The reader will surely have asked the question: what is missing from a pseudo-coherent complex E in order that it be perfect? Is it enough, for example, that E have bounded cohomology? The answer is no. Indeed, take for $\mathcal{C}$ the category of modules over a commutative noetherian ring A, and for $\mathcal{C}_0$ the full subcategory of projective A-modules of finite type. Any A-module of finite type E is pseudo-coherent and has bounded cohomology, but E is perfect only if, in addition, it is of finite projective dimension (or, equivalently, of finite Tor-dimension). This example suggests that the difference between a pseudo-coherent complex and a perfect complex lies in a question of cohomological dimension. This is what we shall now examine in the case of the category of modules of a ringed topos (cf. (3.3)). We shall need a few preliminary sorties on the notion of finite Tor-dimension. We shall be brief, since this question already appears in the literature: cf. [H] II 4 and SGA 4 XVII 4.1.9.

Proposition 5.1. Let (S, A) be a ringed topos, $X \in \operatorname{ob} S$, $F \in \operatorname{ob} \mathrm{D}^-(A, X)$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. The following conditions are equivalent:

(i) F is isomorphic, in $\mathrm{D}^-(A, X)$, to a complex F' such that F'^i is flat for every i, and zero for $i \notin [m, n]$;

(ii) for every right A_X-module E, one has

$$\mathrm{H}^i(E \overset{\mathrm{L}}{\otimes}_A F) = 0 \quad \text{for } i \notin [m, n].$$

Proof. The implication (i) $\Rightarrow$ (ii) is trivial; the implication (ii) $\Rightarrow$ (i) follows, by truncation, from the following lemma.

Lemma 5.1.1. (cf. (4.16)). Let $F \in \operatorname{ob} \mathrm{D}^-(A, X)$. Suppose that F satisfies hypothesis (ii) of (5.1), that $F^i = 0$ for $i \notin [m, n]$, and that F^i is flat for $i \neq m$. Then F^m is flat.

Proof. There is a distinguished triangle

$$\begin{tikzcd} F' \arrow[rr] & & F \arrow[dl] \\ & F^m[-m] \arrow[ul, "+1"] & \end{tikzcd}$$

where F' has flat components and is zero outside the interval $[m+1, n]$ (suppose $m < n$). Apply the functor $E \overset{\mathrm{L}}{\otimes}_A \cdot$, for an arbitrary right A_X-module E, and write the exact cohomology sequence.

Definition 5.2. An object F of $\mathrm{D}^-(A, S)$ satisfying the equivalent conditions of (5.1) will be said to be of flat amplitude contained in the interval $[m, n]$, and we shall write

$$\operatorname{tor.amp}(F) \subset [m, n].$$

We shall say that F is of finite flat amplitude, or of finite Tor-dimension, if there exist integers m and n such that the preceding condition is satisfied.

5.3. Let $X \in \operatorname{ob} S$. We shall write $\mathrm{D}^-(A, X)_{\mathrm{torf}}$ for the full subcategory of $\mathrm{D}^-(A, X)$ formed by the objects of finite Tor-dimension. By criterion (ii) of (5.1), this is a triangulated subcategory of $\mathrm{D}^-(A, X)$. As X varies, the categories $\mathrm{D}^-(A, X)_{\mathrm{torf}}$ form a triangulated sub-S-category (1.1.4) of $\mathrm{D}^-(A, S)$, which we shall denote by $\mathrm{D}^-(S)_{A\text{-}\mathrm{torf}}$. The functor $\overset{\mathrm{L}}{\otimes}_A$ induces an exact S-functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^b(S_A) \times_S \mathrm{D}(A/S)_{\mathrm{torf}} \longrightarrow \mathrm{D}^b(S_{\mathbb{Z}}).$$

5.4. In order that a left A-module F be of flat amplitude contained in $[-n, 0]$, $n \in \mathbb{N}$, it is necessary and sufficient that F be of Tor-dimension $\leq n$ in the ordinary sense, that is, that F admit a left resolution of length n by flat A-modules.

5.5. Let $X \in \operatorname{ob} S$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. In order that $F \in \operatorname{ob} \mathrm{D}^-(A, X)$ be of flat amplitude contained in $[m, n]$, it is necessary and sufficient that there exist a covering family $(X_i \to X)$ such that $F|X_i$ is of flat amplitude contained in $[m, n]$. In particular, if S/X is of finite type (SGA 4 VI 1.7), and if F is locally of finite Tor-dimension, then F is of finite Tor-dimension.

Proposition 5.6. Under the hypotheses of (2.16), if E' is of flat amplitude contained in $[m', n']$ as an object of $\mathrm{D}^-(A, S')$, and if E is of flat amplitude contained in $[m, n]$, then

$$E' \overset{\mathrm{L}}{\otimes}_A E$$

is of flat amplitude contained in $[m + m', n + n']$.

Proof. This is immediate: use form (5.1 (i)) of the notion of finite Tor-dimension and the fact that, if E is a flat left A-module, then $f^{-1}(E)$ is a flat left $f^{-1}(A)$-module (SGA 4 IV, App.).

In particular:

Corollary 5.6.1. a) The functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S')$$

induces a functor

$$\mathrm{L}f^* : \mathrm{D}(A, S)_{\mathrm{torf}} \longrightarrow \mathrm{D}(A, S')_{\mathrm{torf}}.$$

b) If A is commutative, the functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}(S)_{\mathrm{torf}} \times_S \mathrm{D}(S)_{\mathrm{torf}} \longrightarrow \mathrm{D}(S)_{\mathrm{torf}}.$$

Remark 5.6.2. Let $E \in \operatorname{ob} \mathrm{D}^-(A, S)$, where S is the final object of S, and let $m, n \in \mathbb{Z}$, with $m \leq n$. If E is of flat amplitude contained in $[m, n]$, then so is E_s, for every point s of S. The converse is true if S has enough points; use form (5.1(ii)) of the notion of flat amplitude and the fact that $\overset{\mathrm{L}}{\otimes}_A$ commutes with passage to the fibers.

Exercises 5.7. a) Suppose that A is commutative and that (S', A') is faithfully flat over (S, A) (cf. (2.16.2)). In order that $E \in \operatorname{ob} \mathrm{D}^-(A, S)$[13] be of flat amplitude contained in $[m, n]$, it is necessary and sufficient that $f^*(E)$ have that property.

b) Let $F \in \operatorname{ob} \mathrm{D}^+(A, S)$[13]. Show the equivalence of the following conditions:

(i) F is isomorphic, in $\mathrm{D}^+(A, S)$, to a complex F' such that F'^i is injective for every i and zero for $i \notin [m, n]$;

(ii) for every left A_S-module E, one has $\operatorname{Ext}^i(E, F) = 0$ for $i \notin [m, n]$.

If F satisfies the preceding conditions, one says that F is of injective amplitude contained in $[m, n]$. The full subcategory of $\mathrm{D}^+(A, S)$ formed by the objects of finite injective amplitude is a triangulated subcategory, denoted $\mathrm{D}^+(A, S)_{\mathrm{injf}}$. Show that, if A is commutative, the functor

$$\mathrm{R}\operatorname{Hom} : \mathrm{D}(S)^\circ \times \mathrm{D}^+(S) \longrightarrow \mathrm{D}(S)$$

induces a functor

$$\mathrm{R}\operatorname{Hom} : \mathrm{D}(S)^\circ_{\mathrm{torf}} \times \mathrm{D}(S)_{\mathrm{injf}} \longrightarrow \mathrm{D}(S)_{\mathrm{injf}}.$$

We can now answer the question of (5.0) and state the main result of this section.

Theorem 5.8. Let (S, A) be a ringed topos, $E \in \operatorname{ob} \mathrm{D}(A, S)$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. The following conditions are equivalent:

(i) $\operatorname{perf.amp}(E) \subset [m, n]$ (4.8);

(ii) E is $(m-1)$-pseudo-coherent (2.15) and $\operatorname{tor.amp}(E) \subset [m, n]$ (5.2).

Since "perfect" plainly implies "pseudo-coherent", one deduces:

Corollary 5.8.1. In order that E be perfect (4.8), it is necessary and sufficient that E be pseudo-coherent (2.15) and locally of finite Tor-dimension (5.2).

For the proof, we shall need a few lemmas.

Lemma 5.8.2. Let B be a ring of S. If E is a left A-module, G a right B-module, and F an A-B-bimodule (left over A, right over B), there is a canonical functorial homomorphism

$$\operatorname{Hom}_B(F, G) \otimes_A E \longrightarrow \operatorname{Hom}_B(\operatorname{Hom}_A(E, F), G). \tag{5.8.2.1}$$

[13] Here S denotes the final object of the topos S.

It is an isomorphism for G injective and E of finite presentation.

Proof. First define the arrow. By the adjunction formula dear to Cartan, it amounts to defining a homomorphism of B-modules:

$$\mathrm{Hom}_B(F,G) \otimes_A E \otimes_{\mathbb{Z}_S} \mathrm{Hom}_A(E,F) \longrightarrow G.^{14} \tag{*}$$

There is a canonical homomorphism of A-B-bimodules

$$E \otimes_{\mathbb{Z}_S} \mathrm{Hom}_A(E,F) \longrightarrow F \tag{**}$$

coming, by adjunction, from the identity arrow of $\mathrm{Hom}_A(E,F)$. Similarly, there is a canonical homomorphism of B-modules

$$\mathrm{Hom}_B(F,G) \otimes_A F \longrightarrow G. \tag{***}$$

We then define (*) as the composite of (**) and (***). To see that the arrow (5.8.2.1) is an isomorphism when G is injective and E is of finite presentation, it is enough to note that, if G is injective, the source and the target of the arrow are right-exact functors in E, and that the arrow induces the identity for $E = A$.

Here is now a particular case of (5.8).

Lemma 5.8.3. Let E be a left A-module. The following conditions are equivalent:

(i) E is locally a direct factor of a free module of finite type;

(ii) E is flat and of finite presentation.

Proof (after Bourbaki, Alg. comm., Chap. I, §2, Exer. 15). It is clear that (i) implies (ii). Let us prove (ii) $\Rightarrow$ (i). By (1.3.1), it is enough to show that the functor $\mathrm{Hom}_A(E,\cdot)$ is exact. Let

$$F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence of left A-modules. We must show that the sequence

$$\mathrm{Hom}_A(E,F) \longrightarrow \mathrm{Hom}_A(E,F'') \longrightarrow 0$$

is exact. For this it is enough to prove that, for every injective $\mathbb{Z}_S$-module I, the sequence obtained by applying the functor $\mathrm{Hom}_{\mathbb{Z}_S}(\cdot, I)$ is exact. But there is a commutative diagram, where the vertical arrows are the canonical arrows of (3.17.1):

$$\begin{array}{ccccc}
0 & \to & \mathrm{Hom}_{\mathbb{Z}_S}(F'',I) \otimes_A E & \to & \mathrm{Hom}_{\mathbb{Z}_S}(F,I) \otimes_A E \\
& & \downarrow & & \downarrow \\
0 & \to & \mathrm{Hom}_{\mathbb{Z}_S}(\mathrm{Hom}_A(E,F''),I) & \to & \mathrm{Hom}_{\mathbb{Z}_S}(\mathrm{Hom}_A(E,F),I).
\end{array}$$

The vertical arrows are isomorphisms by (5.8.2). The top row is exact because E is flat. Hence the bottom row is exact, q.e.d.

Proof of (5.8). The implication (i) $\Rightarrow$ (ii) follows trivially from the definitions. Let us prove (ii) $\Rightarrow$ (i). Since the question is local, one may suppose E strictly $(m-1)$-pseudo-coherent. Since, on the other hand, E is acyclic in degrees $< m$, the truncation arrow

$$\begin{array}{ccccccc}
\cdots & \to & E^{m-1} & \to & E^m & \to & E^{m+1} \to \cdots \\
& & \downarrow & & \downarrow & & \downarrow \\
\cdots & \to & 0 & \to & E^m/B^m & \to & E^{m+1} \to \cdots
\end{array}$$

[14] We write S for the final object of the topos S, and $\mathbb{Z}_S$ for the constant sheaf with value $\mathbb{Z}$.

is a quasi-isomorphism. Let E' be the truncated complex. By construction, E' is of flat amplitude contained in $[m,n]$, and E'^{i} is free of finite type for $i > m$. Therefore, by (5.1.1), E'^{m} is flat. On the other hand there is a distinguished triangle

$$\begin{tikzcd} & E'^{m}[-m] \arrow[dl, "+1"'] & \\ E'' \arrow[rr] & & E' \arrow[ul] \end{tikzcd}$$

where E'' is strictly perfect. By (2.6), $E'^{m}[-m]$ is $(m-1)$-pseudo-coherent; i.e. by (2.9), E'^{m} is of finite presentation. We conclude from (5.8.3) that E'^{m} is locally a direct factor of a free module of finite type, hence that $\operatorname{perf.amp}(E) \subset [m,n]$.

Remarks 5.8.4. a) Using (2.5), one recovers the fact that perfect complexes form a triangulated subcategory of $\mathrm{D}(A,S)$.

b) A perfect complex, even with bounded cohomology, is not necessarily of finite Tor-dimension. It is so, however, if S is of finite type (5.5). Write $\mathrm{D}(A,S)_{\mathrm{parf}}$ for the full sub-S-category of $\mathrm{D}^b(A,S)$ formed by perfect complexes of finite Tor-dimension. It follows at once from (2.16.1 b) and (5.3) that, if A is commutative, the functor $\overset{\mathrm{L}}{\otimes}_A$ induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}(S)_{\mathrm{parf}} \times_S \mathrm{D}^b(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}}.$$

This formula is the starting point for the theory of intersections on schemes. Note, on the other hand, that the derived tensor product of two pseudo-coherent complexes with bounded cohomology is not, in general, of bounded cohomology.

Corollary 5.9. Suppose that $S = (\mathrm{Ens})$ (so that A "is" an ordinary ring) and that every left A-module is of finite Tor-dimension; this is the case, for example, if A is of finite global cohomological dimension, cf. (MULT VI 2.6) and (EGA 0_{IV} 17.2.8). Then

$$\text{(i)} \quad \mathrm{D}^b(A,S) = \mathrm{D}(A,S)_{\mathrm{torf}}, \qquad \text{and} \qquad \text{(ii)} \quad \mathrm{D}^b(A,S)_{\mathrm{coh}} = \mathrm{D}(A,S)_{\mathrm{parf}}.$$

In other words, in order that $E \in \operatorname{ob}\mathrm{D}(A,S)$ be of finite Tor-dimension (respectively finite Tor-dimension and perfect), it is necessary and sufficient that E have bounded cohomology (respectively bounded and pseudo-coherent cohomology).

Proof. By (5.8), it is enough to prove (i). Let $E \in \operatorname{ob}\mathrm{D}^b(A,S)$. We may suppose, after extending E by 0, that $E \in \operatorname{ob}\mathrm{D}^b(A,S)$, where S is the final object of S. By induction on the number of non-zero cohomology objects of E, one is reduced, by dévissage, to the case where E is an ordinary A-module; then one wins.

More generally, one may state:

Corollary 5.10. Let $X \in \operatorname{ob} S$. Suppose that S/X has enough points (SGA 4 IV 6.4.1) and that, for every point s of S/X, every left A_s-module is of finite Tor-dimension. Then

$$\mathrm{D}^b(A,X)_{\mathrm{coh}} = \mathrm{D}^b(A,X)_{\mathrm{parf}}.$$

Equivalently, in order that $E \in \operatorname{ob}\mathrm{D}^b(A,X)$ be pseudo-coherent, it is necessary and sufficient that E be perfect.

Proof. Let $E \in \operatorname{ob}\mathrm{D}^b(A,X)_{\mathrm{coh}}$ and let s be a point of S/X. One must show that E is perfect in a neighbourhood of s, that is, that there exists an s-pointed object U of S over X such that $E|U$ is perfect. Now, since the functor s^* is exact, (2.16.1 a) gives $E_s \in \operatorname{ob}\mathrm{D}^b(A_s)_{\mathrm{coh}}$. Hence, by (5.9), E_s is perfect. The assertion is therefore a consequence of (3.7.2) and of the following lemma.

Lemma 5.10.1. Let s be a point of S, and let $F \in \operatorname{ob} \mathrm{D}(A_s)_{\mathrm{parf}}$. There exist an s-pointed object U of S and $F' \in \operatorname{ob} \mathrm{D}(A_U)_{\mathrm{parf}}$ such that $F'_s = F$.

Proof. This is evident if F is a free A_s-module of finite type. Suppose now that F is projective of finite type, hence the image of a projector

$$p : L \longrightarrow L,$$

where L is free of finite type. There is then a neighbourhood U of s, a free A_U-module L', and a homomorphism $p' : L' \to L'$ such that $p'_s = p$. Since $p_s^2 = p_s$, we may suppose, after shrinking U, that $p'^2 = p'$. Thus $F' = \operatorname{Im}(p')$ is a direct factor of L' and satisfies $F'_s = F$. The case where F is strictly perfect, that is, a bounded complex made up of projective modules of finite type, follows immediately.

Examples 5.11. Corollary (5.10) applies in particular when (S, A) is defined by a locally ringed space in regular local rings, for example a regular scheme or a smooth complex analytic space over $\mathbb{C}$.

6. Rank of a perfect complex

6.0. Let $(X, \mathcal{O}_X)$ be a ringed space in local rings. It is well known that one can attach, to every locally free sheaf of finite type L on an open set U of X, a function $\mathrm{rg}_U(L)$, locally constant on U and with integral values, called the rank of L. The family of functions $\mathrm{rg}_U(L)$ is characterized by the following properties:

(i) if $V \subset U$ is an inclusion of open sets, then $\mathrm{rg}_V(L|V) = \mathrm{rg}_U(L)|V$;

(ii) for every exact sequence of locally free $\mathcal{O}_U$-modules of finite type

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0,$$

one has $\mathrm{rg}_U(L) = \mathrm{rg}_U(L') + \mathrm{rg}_U(L'')$;

(iii) $\mathrm{rg}_X(\mathcal{O}_X) = 1$.

We propose in this section to extend the notion of rank to perfect complexes.

Definition 6.1 (cf. SGA 5 VIII 2). Let $\mathcal{C}$ be an additive (respectively triangulated) category, and let F be an abelian group. A function

$$f : \mathrm{ob}(\mathcal{C}) \longrightarrow F$$

is said to be an additive function from $\mathcal{C}$ to F if, for all $L, L', L'' \in \mathrm{ob}(\mathcal{C})$ such that $L \simeq L' \oplus L''$ (respectively such that there is a distinguished triangle

$$\begin{tikzcd} & L'' \arrow[dl, dashed] & \\ L' \arrow[rr] & & L \arrow[ul] \end{tikzcd}$$

), one has

$$f(L) = f(L') + f(L'').$$

The additive functions from $\mathcal{C}$ to F form an abelian group, denoted $\mathrm{Add}(\mathcal{C}, F)$.

6.2. Let $\mathcal{C}$ be a triangulated category, and let f be an additive function from $\mathcal{C}$ to an abelian group F. Then f induces an additive function on the underlying additive category $\mathcal{C}^{\mathrm{ad}}$ of $\mathcal{C}$; hence an inclusion

$$\mathrm{Add}(\mathcal{C}, F) \subset \mathrm{Add}(\mathcal{C}^{\mathrm{ad}}, F).$$

In particular, $f(0) = 0$ and $f(L) = f(M)$ if $L \simeq M$. On the other hand,

$$f(L[n]) = (-1)^n f(L)$$

for all $L \in \mathrm{ob}(\mathcal{C})$ and $n \in \mathbb{Z}$.

6.3. Let $\mathcal{C}$ be an additive (respectively triangulated) category. The group $\mathrm{Add}(\mathcal{C}, F)$ depends functorially on F. The functor $\mathrm{Add}(\mathcal{C}, \cdot)$ is representable (cf. SGA 5 VIII 2) by an abelian group, denoted $k(\mathcal{C})$[15], endowed with an additive application

$$\mathrm{cl}_{\mathcal{C}} : \mathrm{ob}(\mathcal{C}) \longrightarrow k(\mathcal{C}).$$

If $\mathcal{C}$ is additive, one takes for $k(\mathcal{C})$ the symmetrized monoid of isomorphism classes of objects of $\mathcal{C}$. If $\mathcal{C}$ is triangulated, one takes for $k(\mathcal{C})$ the quotient of the free abelian group generated by $\mathrm{ob}(\mathcal{C})$ by the relations identifying L with $L' + L''$ whenever there is a distinguished triangle

$$L' \longrightarrow L \longrightarrow L'' \longrightarrow L'[1].$$

The group $k(\mathcal{C})$ depends functorially on $\mathcal{C}$, with respect to additive (respectively exact) functors. If $u, u' : \mathcal{C} \to \mathcal{D}$ are isomorphic additive (respectively exact) functors, then $k(u) = k(u')$.

If $\mathcal{C}$ is a triangulated category, one has a canonical epimorphism

$$k(\mathcal{C}^{\mathrm{ad}}) \longrightarrow k(\mathcal{C})$$

(cf. (6.2)).

Lemma 6.4. Let $\mathcal{C}$ be an additive category. The canonical functor

$$\mathcal{C} \longrightarrow \mathrm{K}^b(\mathcal{C})$$

induces an isomorphism, functorial in $\mathcal{C}$,

$$k(\mathcal{C}) \xrightarrow{\sim} k(\mathrm{K}^b(\mathcal{C})).$$

Proof. This is an immediate consequence of the following formula, which follows trivially from (6.2):

$$f(L) = \sum_i (-1)^i f(L^i) \tag{6.4.1}$$

for $L \in \mathrm{ob}\,\mathrm{K}^b(\mathcal{C})$ and f an additive function on $\mathrm{K}^b(\mathcal{C})$.

Definition 6.5. Let S be a category, $\mathcal{C}$ an additive (respectively triangulated) S-category (1.1.1), and F an abelian presheaf on S. A family of functions

$$f_X : \mathrm{ob}(\mathcal{C}_X) \longrightarrow F(X), \qquad X \in \mathrm{ob}\,S,$$

is said to be an additive S-function from $\mathcal{C}$ to F if the following conditions are satisfied:

(i) for every $X \in \mathrm{ob}\,S$, f_X is an additive function (6.1);

[15] We shall not use here the usual notation $K(\mathcal{C})$, since $K(\mathcal{C})$ already denotes, when $\mathcal{C}$ is additive, the category of complexes of $\mathcal{C}$ “up to homotopy.”

(ii) for every arrow $u : X \to Y$ of S, the following diagram is commutative:

$$\begin{tikzcd} \operatorname{ob}\mathcal{C}_Y \arrow[r, "f_Y"] \arrow[d, "u^*"'] & F(Y) \arrow[d, "F(u)"] \\ \operatorname{ob}\mathcal{C}_X \arrow[r, "f_X"'] & F(X). \end{tikzcd}$$

The additive S-functions from $\mathcal{C}$ to F form an abelian group, denoted $\mathrm{Add}_S(\mathcal{C}, F)$.

Remarks 6.6. a) Let $\mathcal{C}^{\mathrm{is}}$ denote the fibered S-category whose fiber over $X \in \operatorname{ob} S$ is the category with the same objects as $\mathcal{C}_X$ but whose arrows are the isomorphisms of $\mathcal{C}_X$. Let $\underline{F}$ denote the fibered S-category whose fiber over $X \in \operatorname{ob} S$ is the discrete category (i.e. the only arrows are identities) $F(X)$, the inverse-image functor by $u \in \mathrm{Fl}(S)$ being $F(u)$. Then an additive S-function from $\mathcal{C}$ to F is simply a Cartesian S-functor from $\mathcal{C}^{\mathrm{is}}$ to $\underline{F}$, inducing on each fiber an additive function in the sense of (6.1). It will be convenient for us to interpret additivity as follows. Suppose, for definiteness, that $\mathcal{C}$ is triangulated. The category

$$\mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$$

(respectively $\mathrm{Cart}_S(\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F})$ (1.1.2)) of Cartesian S-functors from $\mathcal{C}^{\mathrm{is}}$ (respectively $\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}$) to $\underline{F}$ is the discrete category associated to an abelian group, and there is a homomorphism

$$\rho : \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F}) \longrightarrow \mathrm{Cart}_S(\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F}),$$

defined by

$$\rho(f)(T) = f(L) - f(L') - f(L'')$$

for $f \in \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$ and $T = (L' \to L \to L'' \to L'[1]) \in \operatorname{ob}\mathrm{Tr}(\mathcal{C})$. To say that $f \in \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$ is additive means that $\rho(f) = 0$, or equivalently that one has a functorial canonical exact sequence

$$0 \longrightarrow \mathrm{Add}_S(\mathcal{C}, F) \longrightarrow \mathrm{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F}) \xrightarrow{\rho} \mathrm{Cart}_S(\mathrm{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F}). \tag{6.6.1}$$

b) Let $u : X \to Y$ be an arrow of S. The inverse-image functors by u, being all isomorphic to one another, define by (6.3) a unique homomorphism

$$k(Y) \longrightarrow k(X).$$

Consequently, $X \mapsto k(X)$ defines an abelian presheaf on S, denoted $k(\mathcal{C})$. An additive S-function from $\mathcal{C}$ to F is then simply interpreted as a homomorphism of presheaves

$$k(\mathcal{C}) \longrightarrow F,$$

or in other words

$$\mathrm{Add}_S(\mathcal{C}, F) = \mathrm{Hom}(k(\mathcal{C}), F).$$

The presheaf $k(\mathcal{C})$ of course depends functorially on $\mathcal{C}$ with respect to additive (respectively exact) S-functors.

Proposition 6.7. Let S be a site, $\mathcal{C}$ a flat abelian S-category, and $\mathcal{C}_0$ a strictly full additive sub-S-category of $\mathcal{C}$ satisfying the hypotheses (4.0). Then the morphism of abelian presheaves

$$k(\mathcal{C}_0) \longrightarrow k(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)) \qquad \text{(cf. (4.9))}$$

induced by the inclusion of $\mathcal{C}_0$ in $\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0)$, defines an isomorphism on the associated sheaves. In other words, for every abelian sheaf F on S, the restriction homomorphism

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C}, \mathcal{C}_0), F) \longrightarrow \mathrm{Add}_S(\mathcal{C}_0, F)$$

is an isomorphism.

Proof. Let F be an abelian sheaf on S. We show that the arrow

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \longrightarrow \mathrm{Add}_S(\mathcal{C}_0,F)$$

is an isomorphism. It factors as

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \xrightarrow{a} \mathrm{Add}_S(\mathrm{K}^b(\mathcal{C}_0),F) \xrightarrow{b} \mathrm{Add}_S(\mathcal{C}_0,F).$$

By (6.4), the arrow b is an isomorphism. We are reduced to proving that a is an isomorphism.

Consider the following commutative diagram, whose rows are exact by (6.6.1), and whose vertical arrows are the restriction arrows:

$$\begin{array}{ccccccc} 0 & \to & \mathrm{Add}_S(\mathrm{Parf}(\mathcal{C}),F) & \to & \mathrm{Cart}_S(\mathrm{Parf}(\mathcal{C})^{\mathrm{is}},\underline{F}) & \to & \mathrm{Cart}_S(\mathrm{Tr}(\mathrm{Parf}(\mathcal{C}))^{\mathrm{is}},\underline{F}) \\ & & \downarrow a & & \downarrow & & \downarrow \\ 0 & \to & \mathrm{Add}_S(\mathrm{K}^b(\mathcal{C}_0),F) & \to & \mathrm{Cart}_S(\mathrm{K}^b(\mathcal{C}_0)^{\mathrm{is}},\underline{F}) & \to & \mathrm{Cart}_S(\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))^{\mathrm{is}},\underline{F}). \end{array}$$

Since F is a sheaf, $\underline{F}$ is a stack (G II 2.2.4); hence, by (4.10), the two right vertical arrows are isomorphisms. The same is therefore true of a, q.e.d.

Remark 6.7.1 (cf. 4.10.1). In (6.7), one may replace $\mathrm{Parf}(\mathcal{C},\mathcal{C}_0)$ by $\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}}$ ($* = -$ or b).

6.8. Let (S,A) be a ringed topos, and let $\mathcal{C}$ and $\mathcal{C}_0$ be the fibered S-categories considered in (4.8). There are homomorphisms of abelian presheaves

$$k(\mathcal{C}_0) \xrightarrow{\alpha} k(\mathrm{D}^-(A,S)_{\mathrm{parf}}) \xrightarrow{\beta} k(\mathrm{Parf}(A,S)),$$

which, by (6.7), induce isomorphisms on the associated sheaves. The homomorphism α is compatible with inverse images: that is, if

$$u : (S',A') \longrightarrow (S,A)$$

is a morphism of ringed topoi, there is a commutative diagram, whose vertical arrows are induced by the functor $\mathrm{L}u^*$ (cf. (4.19.1)):

$$\begin{array}{ccc} k(\mathcal{C}_0) & \longrightarrow & k(\mathrm{D}^-(A,S)_{\mathrm{parf}}) \\ \downarrow & & \downarrow \\ k(\mathcal{C}'_0) & \longrightarrow & k(\mathrm{D}^-(A',S')_{\mathrm{parf}}), \end{array}$$

where $\mathcal{C}'_0$ is defined analogously to $\mathcal{C}_0$. If A is commutative, the derived tensor product endows $k(\mathcal{C}_0)$ and $k(\mathrm{D}^-(A,S)_{\mathrm{parf}})$ with structures of presheaves of rings (cf. (4.19.1)), and α is a homomorphism of presheaves of rings.

6.9. In the situation of (6.8), suppose that (S,A) is locally ringed (cf. (2.15.1)). Then, for every $X \in \mathrm{ob}\, S$, $\mathcal{C}_{0,X}$ is the category of locally free A_X-modules of finite type. Let $\mathcal{C}_1$ be the fibered S-category defined by

$$X \longmapsto \text{the category of free } A_X\text{-modules of finite type.}$$

It is clear that $\mathcal{C}_1$ satisfies the hypotheses (4.0). An object of $\mathrm{D}(S)$ is perfect if and only if it is perfect relative to $\mathcal{C}_1$. There are homomorphisms of abelian presheaves

$$k(\mathcal{C}_1) \longrightarrow k(\mathcal{C}_0) \longrightarrow k(\mathrm{D}^-(S)_{\mathrm{parf}}) \longrightarrow k(\mathrm{Parf}(S))$$

(the two on the left being homomorphisms of presheaves of rings), which, by (6.7), induce isomorphisms on the associated sheaves. We shall show that the associated sheaves are in fact canonically isomorphic to the constant sheaf $\mathbb{Z}_S$.

Lemma 6.10. Let (S, A) be a locally ringed topos, let U be an object of S distinct from the empty object, and let $p, q \in \mathbb{N}$. If

$$A_U^p \simeq A_U^q,$$

then $p = q$.

Proof. By an argument similar to that given in the proof of (2.15.1), one is reduced to the case where (S, A) is the locally ringed topos defined by a prescheme; in this case the assertion is trivial: take a point in U.

Definition 6.11. The rank, denoted rg, is the additive S-function from $\mathcal{C}_1$ to $\mathbb{Z}_S$ defined by

$$\mathrm{rg}(L) = p \quad \text{if } L \simeq A_X^p, \qquad X \neq \varnothing,$$

$$\mathrm{rg}(L) = 0 \quad \text{if } L \in \operatorname{ob} \mathcal{C}_{1,X}, \qquad X = \varnothing.$$

By (6.7), the rank function extends uniquely to an additive S-function from $\mathrm{Parf}(S)$ to $\mathbb{Z}_S$, which we shall again call rank and denote by rg.

Proposition 6.12. a) For $L, M \in \operatorname{ob} \mathrm{D}^-(X)_{\mathrm{parf}}$, with $X \in \operatorname{ob} S$, one has

$$\mathrm{rg}(L \overset{\mathrm{L}}{\otimes} M) = \mathrm{rg}(L)\ \mathrm{rg}(M).$$

b) Let

$$u : (S', A') \longrightarrow (S, A)$$

be a morphism of locally ringed topoi. For $M \in \operatorname{ob} \mathrm{D}^-(S)_{\mathrm{parf}}$, one has

$$\mathrm{rg}(\mathrm{L}u^*(M)) = u^*(\mathrm{rg}(M)).$$

In other words, the rank homomorphism

$$\mathrm{rg} : k(\mathrm{D}^-(S)_{\mathrm{parf}}) \longrightarrow \mathbb{Z}_S$$

is a homomorphism of presheaves of rings, compatible with inverse images.

Proof. It is enough to show that the homomorphism induced by rank on the sheaf associated to $k(\mathrm{D}^-(S)_{\mathrm{Parf}})$ is a homomorphism of sheaves of rings compatible with inverse images. By (6.9), one may replace $k(\mathrm{D}^-(S)_{\mathrm{Parf}})$ by $k(\mathcal{C}_1)$. We are thus reduced to verifying a) and b) for free modules of finite type, which is trivial.

6.13. With (S, A) still denoting a locally ringed topos, let $X \in \operatorname{ob} S$ and $n \in H^0(X, \mathbb{Z}_S)$. We define an object of $\mathrm{Parf}(X)$, denoted A_X^n, as follows. If n is constant of value an integer $n \geq 0$ (respectively $n < 0$), put $A_X^n = A_X^{\oplus n}$ (respectively $A_X^n = A_X^{-n}[1]$). In the general case, decompose X into disjoint pieces X_i on which $n|X_i$ is constant, and define A_X^n as the unique object of $\mathrm{Parf}(X)$ inducing $A_{X_i}^{n|X_i}$ on each X_i. More generally, for $L \in \operatorname{ob} \mathrm{Parf}(X)$, one may define L^n by putting $L^n = A_X^n \otimes_A L$.

The map

$$i_X : H^0(X, \mathbb{Z}_S) \longrightarrow k(\mathrm{Parf}(X)), \qquad i_X(n) = \mathrm{cl}(A_X^n),$$

is a homomorphism of rings, with $i_X(1) = 1$, and defines, as X varies, a homomorphism of presheaves of rings

$$i : \mathbb{Z}_S \longrightarrow k(\mathrm{Parf}(S)).$$

Proposition 6.14. Under the hypotheses of (6.13), one has

$$\mathrm{rg} \circ i = \mathrm{Id}(\mathbb{Z}_S).$$

In other words, $k(\mathrm{Parf}(S))$, considered as a presheaf of $\mathbb{Z}_S$-algebras by means of i, is augmented toward $\mathbb{Z}_S$ by the rank. Moreover, the rank

$$\mathrm{rg} : k(\mathrm{Parf}(S)) \longrightarrow \mathbb{Z}_S$$

defines an isomorphism on the associated sheaves.

Proof. The first assertion follows trivially from the relations

$$\mathrm{cl}(L[1]) = -\,\mathrm{cl}(L), \qquad \mathrm{cl}(L^n) = n\,\mathrm{cl}(L), \qquad L \in \mathrm{ob}\,\mathrm{Parf}(S).$$

Let us prove the second. First observe that i factors in the evident way through $k(\mathcal{C}_1)$. Since $\mathrm{rg} \circ i$ is the identity, we are reduced, by (6.9), to proving that the homomorphism

$$i \circ \mathrm{rg} : k(\mathcal{C}_1) \longrightarrow k(\mathcal{C}_1)$$

induces an isomorphism on the associated sheaves; equivalently, for $L \in \mathrm{ob}\,\mathcal{C}_1$, that $i(\mathrm{rg}(L))$ is locally isomorphic to L. This is immediate.

One deduces, by (6.6 b):

Corollary 6.15. Under the hypotheses of (6.13), for every abelian sheaf F on S, the group of S-additive functions from $\mathrm{Parf}(S)$ to F is canonically identified with $H^0(S, F)$, where S is the final object of S. The rank function corresponds, for $F = \mathbb{Z}_S$, to the unit section of $\mathbb{Z}_S$.

7. Duality of perfect complexes

7.0. In this section, (S, A) denotes a topos ringed in commutative rings, and S also denotes the final object of S. We propose to generalize to perfect complexes the duality notions that are well known for locally free sheaves of finite type.

Proposition 7.1. Let $E \in \mathrm{ob}\,\mathrm{D}^-(S)_{\mathrm{Parf}}$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$.

a) If F has locally bounded cohomology, then so does $\mathrm{RHom}(E, F)$. If F is pseudo-coherent (respectively locally of finite Tor-dimension, respectively perfect), then the same is true of $\mathrm{RHom}(E, F)$.

b) Suppose, moreover, that E is of finite Tor-dimension, a condition satisfied if S is of finite type. Then, if F has bounded cohomology (respectively finite Tor-dimension), the same is true of $\mathrm{RHom}(E, F)$. More precisely, let a, b, m, n be integers such that

$$\mathrm{tor.\,amp}(E) \subset [a, b]$$

(or, equivalently by (5.8), $\mathrm{parf.\,amp}(E) \subset [a, b]$), and such that $\mathrm{H}^i(F) = 0$ for $i \notin [m, n]$ (respectively $\mathrm{tor.\,amp}(F) \subset [m, n]$). Then

$$\mathrm{H}^i(\mathrm{RHom}(E, F)) = 0 \quad (i \notin [m - b, n - a]),$$

respectively

$$\mathrm{tor.\,amp}(\mathrm{RHom}(E, F)) \subset [m - b, n - a].$$

Proof. Assertion a) is local, so we may suppose E strictly perfect, and then conclude by devissage on E, reducing to $E = A$. For assertion b), we may suppose E strictly perfect with $E^i = 0$ for

$i \notin [a,b]$, and $F^i = 0$ for $i \notin [m,n]$ (respectively F^i flat for every i and $F^i = 0$ for $i \notin [m,n]$). It is then enough to apply the following lemma.

Lemma 7.1.1. Let $E \in \operatorname{ob} \mathrm{D}^-(S)$ and $F \in \operatorname{ob} \mathrm{D}^+(S)$. If E is strictly pseudo-coherent, that is, in the sense of (2.1), with components direct factors of free finite-type modules, then the canonical arrow of $\mathrm{D}^+(S)$

$$\mathrm{Hom}^\bullet(E,F) \longrightarrow \mathrm{RHom}(E,F)$$

is an isomorphism.

Proof. The spectral sequence

$$E_2^{p0} = \mathrm{H}^p(\mathrm{Hom}^\bullet(E,F)) \Longrightarrow \mathrm{Ext}^p(E,F)$$

where $\mathrm{Ext}^q(E^i,F^j) = 0$ for every $q > 0$ by (1.3.1), degenerates. One obtains isomorphisms

$$E_2^{p0} = \mathrm{H}^p(\mathrm{Hom}^\bullet(E,F)) \xrightarrow{\sim} \mathrm{Ext}^p(E,F) = \mathrm{H}^p(\mathrm{RHom}(E,F)),$$

which prove the assertion.

Proposition 7.1.2. Let

$$f : (S', A') \longrightarrow (S, A)$$

be a morphism of ringed topoi. Let $E \in \operatorname{ob} \mathrm{D}^-(S)$ and $F \in \operatorname{ob} \mathrm{D}^b(S)$ be such that $\mathrm{RHom}(E,F) \in \operatorname{ob} \mathrm{D}^b(S)$ and $\mathrm{L}f^*(F) \in \operatorname{ob} \mathrm{D}^b(S')$. There is a functorial canonical arrow

$$\mathrm{L}f^*\mathrm{RHom}(E,F) \longrightarrow \mathrm{RHom}(\mathrm{L}f^*E, \mathrm{L}f^*F).$$

This is an isomorphism if E is perfect.

Proof. By the adjunction of $\mathrm{L}f^*$ and $\mathrm{R}f_*$, it is enough to define an arrow

$$\mathrm{RHom}(E,F) \longrightarrow \mathrm{R}f_*\mathrm{RHom}(\mathrm{L}f^*E, \mathrm{L}f^*F).$$

For this we use the adjunction arrow $F \to \mathrm{R}f_*\mathrm{L}f^*F$ and the duality isomorphism

$$\mathrm{R}f_*\mathrm{RHom}(\mathrm{L}f^*E, \mathrm{L}f^*F) \xrightarrow{\sim} \mathrm{RHom}(E, \mathrm{R}f_*\mathrm{L}f^*F).$$

To see that the arrow of (7.1.2) is an isomorphism for E perfect, one reduces trivially, by localization and devissage, to the case $E = A$, where it is a tautology.

Proposition 7.2. There exists (cf. SGA 5 I 1.6), for $E \in \operatorname{ob} \mathrm{D}(S)$ and $F \in \operatorname{ob} \mathrm{D}^+(S)$, a functorial canonical arrow

$$E \longrightarrow \mathrm{RHom}(\mathrm{RHom}(E,F),F).$$

This is an isomorphism if $E \in \operatorname{ob} \mathrm{D}(S)_{\mathrm{Parf}}$ and $F = A$.

Proof. Define the arrow by resolving F injectively and using the canonical arrow in $\mathrm{K}(S)$

$$E \longrightarrow \mathrm{Hom}^\bullet(\mathrm{Hom}^\bullet(E,F),F).$$

To see that the arrow in question is an isomorphism if E is perfect and $F = A$, one reduces, by localization and devissage, to $E = A$, where it is evident.

Definition 7.3. Let $E \in \operatorname{ob} \mathrm{D}(S)$. We shall put

$$\mathrm{RHom}(E,A) = E^\vee,$$

and we shall say that $E^\vee$ is the dual of E.

By (7.1) and (7.2), if $E \in \operatorname{ob} \mathrm{D}(S)$ is perfect (respectively perfect and of finite Tor-dimension), then so is $E^\vee$, and the canonical arrow $E \to E^{\vee\vee}$ is an isomorphism. In other words, the functor

$\mathrm{RHom}(\,\cdot\,, A)$ defines an anti-auto-equivalence of the category $\mathrm{D}(S)_{\mathrm{Parf}}$ (respectively $\mathrm{D}(S)_{\mathrm{parf}}$, cf. (5.8.4 b)) with itself.

Lemma 7.4 (Cartan isomorphism) (cf. ([H] II 5.15)). For $E \in \mathrm{ob}\,\mathrm{D}(S)$, $F \in \mathrm{ob}\,\mathrm{D}^-(S)$, and $G \in \mathrm{ob}\,\mathrm{D}^+(S)$, there are functorial canonical isomorphisms

$$\begin{aligned}\mathrm{RHom}(E \overset{\mathrm{L}}{\otimes} F, G) &\xrightarrow{\sim} \mathrm{RHom}(E, \mathrm{RHom}(F, G)),\\ \mathrm{RHom}(E \otimes F, G) &\xrightarrow{\sim} \mathrm{RHom}(E, \mathrm{RHom}(F, G)),\\ \mathrm{Hom}(E \otimes F, G) &\xrightarrow{\sim} \mathrm{Hom}(E, \mathrm{RHom}(F, G)).\end{aligned}$$

Proof. This is standard: resolve G injectively and F flatly.

Lemma 7.5. For $E \in \mathrm{ob}\,\mathrm{D}^-(S)$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, there is a functorial canonical homomorphism

$$E \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(E, F) \longrightarrow F.$$

Proposition 7.6 (cf. ([H] II 5.13)). Let $E \in \mathrm{ob}\,\mathrm{D}(S)$, $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, and $G \in \mathrm{ob}\,\mathrm{D}^+(S)$. Under either of the following hypotheses:

(i) G is of finite Tor-dimension;

(ii) $E \in \mathrm{ob}\,\mathrm{D}^-(S)$, F is of finite Tor-dimension, and $G \in \mathrm{ob}\,\mathrm{D}^b(S)$,

there is a functorial canonical homomorphism

$$\mathrm{RHom}(E, F) \overset{\mathrm{L}}{\otimes} G \longrightarrow \mathrm{RHom}(E, F \overset{\mathrm{L}}{\otimes} G).$$

This is an isomorphism when E is perfect.

Proof. Let us define the arrow. Under hypothesis (i), it is enough to resolve F injectively and G flatly. Under hypothesis (ii), by (7.4) it is equivalent to define an arrow

$$E \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(E, F) \overset{\mathrm{L}}{\otimes} G \longrightarrow F \overset{\mathrm{L}}{\otimes} G.$$

Apply the functor $- \otimes^{\mathrm{L}} G$ to the arrow of (7.5).

To see that the arrow is an isomorphism when E is perfect, use the standard method (cf. the proof of (7.2)): reduce, by localization and devissage, to the case $E = A$.

Corollary 7.7. There exists, for $E \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$ and $F \in \mathrm{ob}\,\mathrm{D}^+(S)$, a functorial isomorphism

$$E^\vee \overset{\mathrm{L}}{\otimes} F \xrightarrow{\sim} \mathrm{RHom}(E, F),$$

with the notation of (7.3).

Proof. Apply (7.6), with F and G replaced respectively by A and F, and use the fact (7.1) that $E^\vee \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$.

Corollary 7.8. Let $E, F \in \mathrm{ob}\,\mathrm{D}(S)_{\mathrm{Parf}}$. There is a natural arrow

$$\mathrm{RHom}(E, F) \overset{\mathrm{L}}{\otimes} \mathrm{RHom}(F, E) \longrightarrow A,$$

which is a “perfect duality,” that is, modulo (7.4), it identifies each of the complexes $\mathrm{RHom}(E, F)$ and $\mathrm{RHom}(F, E)$ with the dual of the other, in the sense of (7.3).

Proof. By (7.5), one has natural pairings $E^\vee \otimes^{\mathrm{L}} E \to A$ and $F^\vee \otimes^{\mathrm{L}} F \to A$. Hence, by tensor product, one obtains a natural pairing

$$E^\vee \overset{\mathrm{L}}{\otimes} F \overset{\mathrm{L}}{\otimes} F^\vee \overset{\mathrm{L}}{\otimes} E \longrightarrow A,$$

which, in view of (7.7), gives the announced pairing on the RHom's. Let us show that it is a perfect duality. There are canonical isomorphisms

$$\begin{aligned} \mathrm{RHom}(F^\vee \overset{\mathrm{L}}{\otimes} E, A) &\xrightarrow{\sim} \mathrm{RHom}(F^\vee, E^\vee) && (7.4)\\ &\xrightarrow{\sim} F^{\vee\vee} \overset{\mathrm{L}}{\otimes} E^\vee && (7.6)\\ &\xrightarrow{\sim} F \overset{\mathrm{L}}{\otimes} E^\vee && (7.2). \end{aligned}$$

Thus, up to a compatibility verification, the result follows. One could also localize and devissage to reduce to $E = F = A$.

Exercise 7.9. Let $E, F \in \operatorname{ob} \mathrm{D}(S)_{\mathrm{Parf}}$. Show that there is a canonical isomorphism

$$E^\vee \overset{\mathrm{L}}{\otimes} F^\vee \xrightarrow{\sim} (E \overset{\mathrm{L}}{\otimes} F)^\vee.$$

8. Traces and cup-products

8.0. In this section we generalize to perfect complexes the notion of the trace of an endomorphism of a projective module of finite type (see SGA 5 III for applications). The hypotheses and notation are those of (7.0).

8.1. Let $E \in \operatorname{ob} \mathrm{D}(S)_{\mathrm{parf}}$ (5.8.4 b). There is a natural arrow

$$\tau : \mathrm{RHom}(E, E) \longrightarrow A,$$

obtained by composing the inverse of the isomorphism (7.7)

$$\mathrm{RHom}(E, E) \xrightarrow{\sim} E^\vee \overset{\mathrm{L}}{\otimes} E$$

with the canonical arrow (7.5) $E^\vee \otimes^{\mathrm{L}} E \to A$.

Let $U \in \operatorname{ob} S$. Applying to τ the functor $H^0(U, \cdot)$, and using the fact that $H^0(U, \mathrm{RHom}(E, E))$ identifies with $\mathrm{Hom}_{\mathrm{D}(U)}(E|U, E|U)$, one obtains a homomorphism of $H^0(U, A)$-modules

$$\mathrm{Tr}_{E|U} : \mathrm{Hom}_{\mathrm{D}(U)}(E|U, E|U) \longrightarrow H^0(U, A),$$

which we shall call the trace homomorphism. The finite Tor-dimension hypothesis on E is in fact superfluous for the definition of $\mathrm{Tr}_{E|U}$; see (8.5).

Before going further, we give some properties of the trace that follow immediately from the definition.

8.1.1. If $E = A$, then

$$\mathrm{Tr}_{A|U} : \mathrm{Hom}(A|U, A|U) \longrightarrow H^0(U, A)$$

is just the well-known isomorphism. If E is a direct factor of a free module of finite type, an endomorphism of E over U is locally defined by

$$e \longmapsto \sum_i \langle e, x_i^\vee \rangle x_i,$$

where x_i (respectively $x_i^\vee$) is a local section of E (respectively $E^\vee$); the corresponding trace is $\sum_i \langle x_i, x_i^\vee \rangle$. It follows easily that, if E and F are direct factors of free modules of finite type, and if

$$u : (E+F)|U \longrightarrow (E+F)|U$$

is defined by a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$\operatorname{Tr}_{(E+F)|U}(u) = \operatorname{Tr}_{E|U}(a) + \operatorname{Tr}_{F|U}(d).$$

8.1.2. Let E be a strictly perfect complex, let $U \in \operatorname{ob} S$, and let

$$u : E|U \longrightarrow E|U$$

be a homomorphism of complexes, with components $u^i : E^i|U \to E^i|U$. Then

$$\operatorname{Tr}_{E|U}(u) = \sum_i (-1)^i \operatorname{Tr}_{E^i|U}(u^i).$$

Indeed, the question being local, one may suppose that u is defined by a section of $H^0(E^\vee \otimes E)$, and even that this section is of the form

$$\sum_{i,j} x_{ij}^\vee \otimes x_{ij},$$

where x_{ij} (respectively $x_{ij}^\vee$) is a section of E^i (respectively $E^{i\vee}$). Making explicit the arrow (7.5) $E^\vee \otimes^{\mathrm{L}} E \to A$, one finds

$$\operatorname{Tr}_{E|U}(u) = \sum_{i,j} (-1)^i \langle x_{ij}, x_{ij}^\vee \rangle = \sum_i (-1)^i \operatorname{Tr}_{E^i|U}(u^i).$$

8.2. Let $E, F \in \operatorname{ob} \mathrm{D}(S)_{\mathrm{Parf}}$. We have seen in (7.8) that there is a canonical homomorphism

$$\chi : \operatorname{RHom}(E,F) \overset{\mathrm{L}}{\otimes} \operatorname{RHom}(F,E) \longrightarrow A.$$

Let $U \in \operatorname{ob} S$, and let

$$u \in H^0(U, \operatorname{RHom}(E,F)), \qquad v \in H^0(U, \operatorname{RHom}(F,E)).$$

One may interpret u (respectively v) as a homomorphism, in $\mathrm{D}(U)$, from $A|U$ to $\operatorname{RHom}(E,F)|U$ (respectively $\operatorname{RHom}(F,E)|U$), and form

$$u \overset{\mathrm{L}}{\otimes} v : A|U \longrightarrow (\operatorname{RHom}(E,F) \overset{\mathrm{L}}{\otimes} \operatorname{RHom}(F,E))|U.$$

The map $(u,v) \mapsto \langle u, v \rangle = \chi \circ (u \otimes^{\mathrm{L}} v)$ defines an $H^0(U,A)$-bilinear pairing

$$\operatorname{Hom}_{\mathrm{D}(U)}(E|U, F|U) \otimes_{H^0(U,A)} \operatorname{Hom}_{\mathrm{D}(U)}(F|U, E|U) \longrightarrow H^0(U,A),$$

which we shall call the cup-product. The same comment as for the definition of $\operatorname{Tr}_{E|U}$ applies here.

The relation between trace and cup-product is expressed in the following proposition.

Proposition 8.3. In the situation of (8.2), for $u \in \operatorname{Hom}(E|U, F|U)$ and $v \in \operatorname{Hom}(F|U, E|U)$, one has

$$\operatorname{Tr}_{E|U}(vu) = \operatorname{Tr}_{F|U}(uv) = \langle u, v \rangle.$$

In particular, if $E = F$, then

$$\mathrm{Tr}_{E|U}(u) = \langle u, \mathrm{Id}\rangle = \langle \mathrm{Id}, u\rangle.$$

Proof. For simplicity we suppose $U = S$. We therefore have two arrows

$$u : A \longrightarrow E^\vee \overset{\mathrm{L}}{\otimes} F, \qquad v : A \longrightarrow F^\vee \overset{\mathrm{L}}{\otimes} E,$$

and hence

$$u \overset{\mathrm{L}}{\otimes} v : A \longrightarrow E^\vee \overset{\mathrm{L}}{\otimes} F \overset{\mathrm{L}}{\otimes} F^\vee \overset{\mathrm{L}}{\otimes} E.$$

On the other hand, we have seen that $vu : A \to E^\vee \otimes^{\mathrm{L}} E$ and $uv : A \to F^\vee \otimes^{\mathrm{L}} F$. The proposition follows from the commutativity of the following diagrams, whose verification is purely mechanical:

$$\begin{tikzcd} A \arrow[r, "u \otimes^{\mathrm{L}} v"] \arrow[dr, "vu"'] & E^\vee \otimes^{\mathrm{L}} F \otimes^{\mathrm{L}} F^\vee \otimes^{\mathrm{L}} E \arrow[d, "\mathrm{Id} \otimes \tau_F"] \\ & E^\vee \otimes^{\mathrm{L}} E \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "u \otimes^{\mathrm{L}} v"] \arrow[dr, "uv"'] & E^\vee \otimes^{\mathrm{L}} F \otimes^{\mathrm{L}} F^\vee \otimes^{\mathrm{L}} E \arrow[d, "\tau_E \otimes \mathrm{Id}"] \\ & F^\vee \otimes^{\mathrm{L}} F \end{tikzcd}$$

$$\begin{tikzcd} E^\vee \otimes^{\mathrm{L}} F \otimes^{\mathrm{L}} F^\vee \otimes^{\mathrm{L}} E \arrow[r, "\mathrm{Id} \otimes \tau_E"] \arrow[d, "\mathrm{Id} \otimes \tau_F"'] & F^\vee \otimes^{\mathrm{L}} F \arrow[d, "\tau_F"] \\ E^\vee \otimes^{\mathrm{L}} E \arrow[r, "\tau_E"'] & A. \end{tikzcd}$$

Corollary 8.4. Let E, F, U be as in (8.2), let $u \in \mathrm{Hom}(E|U, E|U)$, and let $s : E|U \to F|U$ be an isomorphism. Then $sus^{-1} \in \mathrm{Hom}(F|U, F|U)$, and

$$\mathrm{Tr}_{F|U}(sus^{-1}) = \mathrm{Tr}_{E|U}(u).$$

Proof. Immediate.

Remark 8.5. If $\mathcal{C}$ is a fibered S-category, denote by $\mathcal{E}(\mathcal{C})$ the fibered S-category whose fiber over $U \in \mathrm{ob}\, S$ is the category whose objects are pairs (L, u), with $L \in \mathrm{ob}\, \mathcal{C}_U$ and $u \in \mathrm{End}_S(L)$, and whose arrows from (L, u) to (M, v) are the arrows $f : L \to M$ such that $fu = vf$. By (8.4) one may regard the trace homomorphism as an S-functor from $\mathcal{E}(\mathrm{D}(S)_{\mathrm{Parf}})^{\mathrm{is}}$ to $\mathcal{A}$, where the exponent is means that only isomorphisms are considered, and where $\mathcal{A}$ is the S-category with discrete fibers defined by A (cf. (6.6 a)). Since every perfect complex is locally of finite Tor-dimension, the forgetful S-functor

$$\mathcal{E}(\mathrm{D}(S)_{\mathrm{Parf}})^{\mathrm{is}} \longrightarrow \mathcal{E}(\mathrm{D}(S)_{\mathrm{parf}})^{\mathrm{is}}$$

induces an isomorphism on the associated stacks. Consequently, the trace homomorphism extends uniquely to an S-functor from $\mathcal{E}(\mathrm{D}(S)_{\mathrm{Parf}})^{\mathrm{is}}$ to $\mathcal{A}$.

For $E \in \mathrm{ob}\,\mathrm{Parf}(U)$, $U \in \mathrm{ob}\, S$, we again write

$$\mathrm{Tr}_E : \mathrm{Hom}(E, E) \longrightarrow H^0(U, A)$$

for the corresponding homomorphism, which is an $H^0(U, A)$-module homomorphism. One sees in the same way that the definition of the cup-product (8.2), as well as assertions (8.3) and (8.4), generalize to perfect complexes, not necessarily of finite Tor-dimension.

Proposition 8.6. Let $U \in \operatorname{ob} S$, let $E, F \in \operatorname{Parf}(U)$, and let $u \in \operatorname{Hom}(E, E)$, $v \in \operatorname{Hom}(F, F)$; hence $u \otimes^{\mathrm{L}} v \in \operatorname{Hom}(E \otimes^{\mathrm{L}} F, E \otimes^{\mathrm{L}} F)$. Then

$$\operatorname{Tr}_{E \otimes^{\mathrm{L}} F}(u \overset{\mathrm{L}}{\otimes} v) = \operatorname{Tr}_E(u)\ \operatorname{Tr}_F(v).$$

Proof. The question being local, by (8.4) and (4.10) one may suppose that u and v are arrows of $K^b(\mathcal{C}_0)$. Using (8.1.2), one is reduced to the case where u and v are arrows of $\mathcal{C}_0$. One finishes by (8.1.1).

Bibliography

[H] Hartshorne, R. *Residues and Duality*, Lecture Notes no. 20, Springer, 1966.

[V] Verdier, J.-L. *Catégories dérivées*, I.H.E.S., 1963.

[G] Giraud, J. *Cohomologie non abélienne*, to appear in North-Holland Publishing Company.

SGA 4 *Cohomologie étale des schémas*, by M. Artin, A. Grothendieck, and J.-L. Verdier, to appear in North-Holland Publishing Company.

SGA 5 *Cohomologie ℓ-adique et fonctions L*, by A. Grothendieck, to appear in North-Holland Publishing Company.

[1] Hakim, M. *Topos annelés et schémas relatifs*, thesis (in preparation).

[2] Verdier, J.-L. *Les catégories dérivées des catégories abéliennes*, to appear in North-Holland Publishing Company.

[MUL] Cartan, H. and Eilenberg, S. *Homological Algebra*, Princeton, 1956.

Exposé II. Existence of global resolutions

by L. Illusie

1. General globalization criteria

1.0

In this section we are given a site S, a flat abelian S-category $\mathcal{C}$ (I 1.1), and a strictly full additive sub-S-category $\mathcal{C}_0$ (I 1.1). We assume that $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$ (I 1.2) and locally quasi-stable under kernels of epimorphisms (I 1.2).

Proposition 1.1. Let $S \in \operatorname{ob}(S)$, and suppose that:

(i) $\mathcal{C}_{0S}$ is quasi-liftable in $\mathcal{C}_S$ (I 1.2), where $\mathcal{C}_S$ is regarded as a category fibered over a punctual site;

(ii) every object of $\mathcal{C}_S$ which is of finite $\mathcal{C}_0$-type (I 1.2), respectively of finite $\mathcal{C}_0$-presentation (I 2.8), is of finite $\mathcal{C}_{0S}$-type.

Then:

a) $\mathcal{C}_{0S}$ is quasi-stable under kernels of epimorphisms (I 1.2), where $\mathcal{C}_{0S}$ is regarded as a category fibered over a punctual site.

b) Let $E \in \operatorname{ob} \mathrm{D}^-(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. If E is n-$\mathcal{C}_0$-pseudo-coherent (I 2.2), then E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. In other words,

$$\mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_0\text{-coh}} = \mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_{0S}\text{-coh}},$$

and, respectively,

$$\mathrm{D}^-(\mathcal{C}_S)_{n-\mathcal{C}_0\text{-coh}} \subset \mathrm{D}^-(\mathcal{C}_S)_{(n+1)-\mathcal{C}_{0S}\text{-coh}} \subset \mathrm{D}^-(\mathcal{C}_S)_{(n+1)-\mathcal{C}_0\text{-coh}},$$

inclusions of strictly full subcategories of $\mathrm{D}(\mathcal{C}_S)$, the inclusion on the right being trivial.

c) For an object $E \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ to be pseudo-coherent relative to $\mathcal{C}_0$ (I 2.3), it is necessary and sufficient that it be pseudo-coherent relative to $\mathcal{C}_{0S}$; equivalently,

$$\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}} = \mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_{0S}\text{-coh}}.$$

Moreover, the canonical functor

$$\mathrm{K}^-(\mathcal{C}_{0S})/\mathrm{K}^{-,\delta}(\mathcal{C}_{0S}) \longrightarrow \mathrm{D}(\mathcal{C}_S),$$

where $\mathrm{K}^{-,\delta}(\mathcal{C}_{0S})$ denotes the thick subcategory of $\mathrm{K}^-(\mathcal{C}_{0S})$ formed by acyclic complexes, is fully faithful and has as essential image $\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}}$. If $\mathcal{C}_{0S}$ is liftable in $\mathcal{C}_S$ (I 1.2), that is, if the objects of $\mathcal{C}_{0S}$ are projective, then $\mathrm{K}^{-,\delta}(\mathcal{C}_{0S})$ is reduced to zero objects, and consequently the functor

$$\mathrm{K}^-(\mathcal{C}_{0S}) \longrightarrow \mathrm{D}(\mathcal{C}_S)$$

is fully faithful and has essential image $\mathrm{D}^-(\mathcal{C}_S)_{\mathcal{C}_0\text{-coh}}$.

Proof. a) Let

$$0 \longrightarrow L \longrightarrow L^0 \longrightarrow \cdots \longrightarrow L^m \longrightarrow 0$$

be an exact sequence in $\mathcal{C}_S$, with $L^i \in \operatorname{ob} \mathcal{C}_{0S}$ for all i. It is enough to prove that L is of finite $\mathcal{C}_{0S}$-type. By (I 2.2), L is $\mathcal{C}_0$-pseudo-coherent, hence, by (I 2.9), of finite $\mathcal{C}_0$-presentation; the assertion follows from (ii).

b) We prove by descending induction on i that E is i-$\mathcal{C}_{0S}$-pseudo-coherent for $i \geq n$, respectively for $i \geq n+1$. One may already suppose that E is strictly $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent, respectively strictly $(n+2)$-$\mathcal{C}_{0S}$-pseudo-coherent (I 2.1); the point is then to prove that E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. Thus there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

where E' is strictly $\mathcal{C}_{0S}$-perfect (I 2.1) and $E''^i = 0$ for $i > n$, respectively for $i > n+1$. Replacing E by E'', one may therefore suppose (I 2.6) that $E^i = 0$ for $i > n$, respectively for $i > n+1$. Then, by (I 2.10 a)), $H^n(E)$, respectively $H^{n+1}(E)$, is of finite $\mathcal{C}_0$-type, respectively of finite $\mathcal{C}_0$-presentation. Hence, by (ii), $H^n(E)$, respectively $H^{n+1}(E)$, is of finite $\mathcal{C}_{0S}$-type. Applying (I 2.10 b)) again, one obtains that E is n-$\mathcal{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\mathcal{C}_{0S}$-pseudo-coherent. The hypothesis (i) and part a) have been used implicitly in the references to (I 2.6 and 2.10).

c) The first assertion follows from b), the second from (I 2.7), and the last is evident.

Proposition 1.2. Suppose that $\mathcal{C}_0$ is locally liftable in $\mathcal{C}$ (I 1.2) and locally stable under kernels of epimorphisms (I 1.2), and make the hypotheses (1.1 (i) and (ii), respectively). Let $\mathcal{C}_{1S}$ denote the strictly full subcategory of $\mathcal{C}_S$ formed by the objects which are locally in $\mathcal{C}_0$. Then:

a) $\mathcal{C}_{1S}$ is stable under kernels of epimorphisms (I 1.2) and is quasi-liftable in $\mathcal{C}_S$ (I 1.2), as a category fibered over a punctual site.

b) Let $E \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$ and let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have $\mathcal{C}_0$-perfect amplitude contained in $[m, n]$, it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(\mathcal{C}_S)$, to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \operatorname{ob} \mathcal{C}_{0S}$ for $i > m$, and $F^m \in \operatorname{ob} \mathcal{C}_{1S}$.

c) The canonical functor
$$\mathrm{K}^b(\mathcal{C}_{1S})/\mathrm{K}^{b,\delta}(\mathcal{C}_{1S}) \longrightarrow \mathrm{D}(\mathcal{C}_S)$$
is fully faithful and has as essential image the full subcategory of $\mathrm{D}(\mathcal{C}_S)$ formed by complexes of finite $\mathcal{C}_0$-perfect amplitude.

d) If $\mathcal{C}_{1S}$ is not merely quasi-liftable but liftable in $\mathcal{C}_S$, one may replace, in b), the expression "E is isomorphic to F in $\mathrm{D}(\mathcal{C}_S)$" by "there exists a quasi-isomorphism $F \to E$"; and, in c), $\mathrm{K}^{b,\delta}(\mathcal{C}_{1S})$ is reduced to zero objects.

Proof. a) It is clear that $\mathcal{C}_{1S}$ is stable under kernels of epimorphisms. On the other hand, the objects of $\mathcal{C}_{1S}$ are $\mathcal{C}_0$-pseudo-coherent, hence $\mathcal{C}_{0S}$-pseudo-coherent by (1.1 c)); the second assertion follows from (I 2.14), taking S to be the punctual site.

b) It suffices to prove necessity. By (1.1 c)), E is already $\mathcal{C}_{0S}$-pseudo-coherent. Since E is acyclic in degree $> n$, one may, after replacing E by an isomorphic complex in $\mathrm{D}(\mathcal{C}_S)$, suppose (I 2.10) that E is strictly $(m+1)$-$\mathcal{C}_{0S}$-pseudo-coherent and that $E^i = 0$ for $i > n$. Since E is acyclic in degree $< m$, one may also replace E by its truncation
$$0 \longrightarrow E^m/B^m \longrightarrow E^{m+1} \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0,$$
so that $E^i = 0$ for $i < m$. Then, by (I 4.16), E^m is locally an object of $\mathcal{C}_0$, and therefore an object of $\mathcal{C}_{1S}$.

c) In view of (1.2 a)), the assertion follows from (1.2 b)) and (I 4.12.2). d) The fact that $\mathrm{K}^{b,\delta}(\mathcal{C}_{1S})$ is reduced to zero objects is evident. The other assertion follows from (I 4.6), applied over the punctual site.

2. Application to certain categories of modules

2.0. Notation

Let (S, A) be a ringed topos. For $U \in \operatorname{ob} S$, write $\mathrm{Mod}(U)$ for the category of left A_U-modules, and $\mathrm{Lib}(U)$, respectively $\mathrm{Loclib}(U)$, for the full subcategory of $\mathrm{Mod}(U)$ formed by free A_U-modules, respectively modules locally direct factors of free modules, of finite type.

2.1. Lifting lemmas

Lemma 2.1.1 (cf. EGA 0_{I}, 5.2.3). Let (S, A) be a ringed topos, and let $S \in \operatorname{ob} S$. Suppose that S is quasi-compact, that is, every covering family $(S_i \to S)$ is dominated by a finite covering family.

a) Let F be a left A_S-module which is the filtered inductive limit, over a filtered category Λ, of A_S-modules F_λ, and let
$$u = \varinjlim u_\lambda : F \longrightarrow G$$
be an epimorphism, where G is an A_S-module of finite type. Then there exists $\lambda \in \Lambda$ such that u_λ is an epimorphism.

b) Let F be a left A_S-module generated by its sections, that is, such that there exists an epimorphism $A_S^{(I)} \to F$ for a suitable set of indices I. If F is of finite type, then there exists an epimorphism $A_S^p \to F$ for some $p \in \mathbb{N}$.

The proof of a) is analogous to the proof of the cited result and is left to the reader; b) follows immediately from a).

Lemma 2.1.2. Let (S, A) be a ringed topos, let $\mathcal{C}$ be the S-category fibered in left A-modules, let $\mathcal{C}_0$ be the fibered subcategory of free A-modules of finite type (I 1.3 d)), let $S \in \operatorname{ob} S$, and let $\mathcal{C}'_S$ be a strictly full additive subcategory of $\mathcal{C}_S$ containing $\mathcal{C}_{0S}$ and stable under kernels, cokernels, and extensions. Suppose that

$$H^1(S, F) = 0 \quad \text{for every } F \in \operatorname{ob} \mathcal{C}'_S.$$

Then:

a) Every exact sequence in $\mathcal{C}_S$

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0,$$

such that $\operatorname{Hom}(G, E) \in \operatorname{ob} \mathcal{C}'_S$ and such that G is locally a direct factor of a free module of finite type, is split. The category $\mathcal{C}_{0S}$ is liftable in $\mathcal{C}'_S$ (I 2.2).

b) Let $\operatorname{Mod}(A(S))$ denote the category of $A(S)$-modules. The functor

$$\Gamma_S : \mathcal{C}'_S \longrightarrow \operatorname{Mod}(A(S))$$

is exact. Moreover:

(i) if every object F of $\mathcal{C}'_S$ admits a finite global presentation

$$A_S^q \longrightarrow A_S^p \longrightarrow F \longrightarrow 0,$$

then Γ_S is fully faithful and has as essential image the full subcategory of $\operatorname{Mod}(A(S))$ formed by $A(S)$-modules of finite presentation;

(ii) if $\mathcal{C}'_S$ contains the free A_S-modules, if every object F of $\mathcal{C}'_S$ admits a global presentation

$$A_S^{(J)} \longrightarrow A_S^{(I)} \longrightarrow F \longrightarrow 0,$$

and if Γ_S commutes with filtered inductive limits, then Γ_S is an equivalence of categories.

Proof. a) Since G is locally a direct factor of a free module of finite type, the spectral sequence

$$E_2^{pq} = H^p(S, \mathcal{E}xt^q(G, E)) \Longrightarrow \operatorname{Ext}^{p+q}(G, E)$$

degenerates (I 1.3.1) and gives an isomorphism

$$H^1(S, \operatorname{Hom}(G, E)) \simeq \operatorname{Ext}^1(G, E),$$

whence the first assertion. The second is a particular case, since for $G \in \operatorname{ob} \mathcal{C}_{0S}$ and $E \in \operatorname{ob} \mathcal{C}'_S$ one has $\operatorname{Hom}(G, E) \in \operatorname{ob} \mathcal{C}'_S$.

b) The first assertion is trivial. Let us prove, in case (i) or (ii), the full faithfulness of $\Gamma_S : \mathcal{C}'_S \to \operatorname{Mod}(A(S))$. Let $F, G \in \operatorname{ob} \mathcal{C}'_S$ and let

$$A_S^{(J)} \longrightarrow A_S^{(I)} \longrightarrow F \longrightarrow 0 \tag{$*$}$$

be a global presentation of F, with I and J finite in case (i). Applying Γ_S, which is exact on $\mathcal{C}'_S$ and, in case (ii), commutes with filtered inductive limits, one obtains an exact sequence

$$A(S)^{(J)} \longrightarrow A(S)^{(I)} \longrightarrow \Gamma_S(F) \longrightarrow 0. \tag{$**$}$$

Applying the left exact functors $\mathrm{Hom}_{\mathcal{C}_S}(-, G)$ and $\mathrm{Hom}_{A(S)}(-, \Gamma_S(G))$ to $(*)$ and $(**)$ gives a commutative diagram with exact rows:

$$\begin{array}{ccccccc}
0 \longrightarrow & \mathrm{Hom}_{\mathcal{C}_S}(F, G) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}_S}(A_S^{(I)}, G) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}_S}(A_S^{(J)}, G) \\
& \downarrow & & \downarrow\wr & & \downarrow\wr \\
0 \longrightarrow & \mathrm{Hom}_{A(S)}(\Gamma_S F, \Gamma_S G) & \longrightarrow & \mathrm{Hom}_{A(S)}(A(S)^{(I)}, \Gamma_S G) & \longrightarrow & \mathrm{Hom}_{A(S)}(A(S)^{(J)}, \Gamma_S G).
\end{array}$$

The two vertical arrows on the right are visibly isomorphisms, hence so is the arrow on the left. The description of the essential image in cases (i) and (ii) is immediate and is left to the reader.

Lemma 2.1.3. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, and let $\varphi : \mathcal{A} \to \mathcal{B}$ be an exact functor.

a) For $M \in \mathrm{ob}\,\mathcal{A}$, the following conditions are equivalent:

  (i) for every $L \in \mathrm{ob}\,\mathcal{A}$, the canonical map induced by φ

$$\mathrm{Ext}^n_{\mathcal{A}}(L, M) \longrightarrow \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$$

  is an isomorphism for all n;

  (ii) for every $L \in \mathrm{ob}\,\mathcal{A}$, the canonical map

$$\mathrm{Hom}_{\mathcal{A}}(L, M) \longrightarrow \mathrm{Hom}_{\mathcal{B}}(\varphi L, \varphi M)$$

  is an isomorphism, and for every $n \geq 1$ and every $x \in \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$ there exists an epimorphism $L' \to L$ in $\mathcal{A}$ which kills x.

b) The following conditions are equivalent:

  (i) for all $L, M \in \mathrm{ob}\,\mathcal{A}$ and all $n \in \mathbb{Z}$, the canonical map

$$\mathrm{Ext}^n_{\mathcal{A}}(L, M) \longrightarrow \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M)$$

  is an isomorphism;

  (ii) the canonical functor induced by φ

$$\mathrm{D}^b(\mathcal{A}) \longrightarrow \mathrm{D}^b(\mathcal{B})$$

  is fully faithful and has as essential image the full subcategory of $\mathrm{D}^b(\mathcal{B})$ formed by complexes F such that $H^i(F)$ is in the essential image of φ for every i.

Proof. For (i)⇒(ii) in a), it is enough to show that, if $n \geq 1$, $L \in \mathrm{ob}\,\mathcal{A}$, and $y \in \mathrm{Ext}^n_{\mathcal{A}}(L, M)$, then there exists an epimorphism $L' \to L$ killing y. Using Yoneda's interpretation ([18], III 3.2), y is the class of an exact sequence

$$0 \longrightarrow M \longrightarrow E_{n-1} \longrightarrow \cdots \longrightarrow E_0 \longrightarrow L \longrightarrow 0,$$

and the epimorphism $E_0 \to L$ kills y ([18], III 3.2.5 and 3.2.8). Conversely, put

$$F^n(L) = \mathrm{Ext}^n_{\mathcal{A}}(L, M), \qquad G^n(L) = \mathrm{Ext}^n_{\mathcal{B}}(\varphi L, \varphi M).$$

The effaceability hypothesis says, for $n \geq 1$ and $L \in \operatorname{ob} \mathcal{A}$, that

$$G^n(L) = \varinjlim \operatorname{Coker} igl(G^{n-1}(L_0) \to G^{n-1}(L_1)),$$

the limit being taken over the exact sequences $L_1 \to L_0 \to L \to 0$. As above, one has the same formula for $F^n(L)$. Since $F^0 \to G^0$ is an isomorphism, induction on n shows that the morphism $F^n \to G^n$ induced by φ is an isomorphism for every n. The equivalence in b) follows by an immediate dévissage, left to the reader.

Remarks 2.1.3.1. There is of course a dual version of a), which we leave to the reader to state. On the other hand, if $\mathcal{A}$ and $\mathcal{B}$ have enough injectives, the conditions (i) and (ii) of b) are equivalent to:

(iii) for $L \in \operatorname{ob} \mathrm{D}^-(\mathcal{A})$ and $M \in \operatorname{ob} \mathrm{D}^+(\mathcal{A})$, the canonical map

$$\operatorname{RHom}(L, M) \longrightarrow \operatorname{RHom}(\varphi L, \varphi M)$$

is an isomorphism,

as is seen at once by a spectral sequence argument.

2.2. Noetherian schemes and divisorial schemes

Notation 2.2.1. Let S be a scheme. We write $\operatorname{Qcoh}(S)$, respectively $\operatorname{Coh}(S)$, for the abelian category of quasi-coherent, respectively coherent, $\mathcal{O}_S$-modules. Let $[m, n]$ be an interval of $\mathbb{Z}$. We shall say that an object E of $\mathrm{D}(\operatorname{Qcoh}(S))$ is n-pseudo-coherent, respectively pseudo-coherent, respectively of perfect amplitude contained in $[m, n]$, respectively perfect, if it is so relative to the category fibered over the Zariski site of S defined by

$$U \longmapsto \operatorname{Loclib}(U)$$

((I 2.3), (I 4.7), and (2.0)). We denote by $\mathrm{D}^-(\operatorname{Qcoh}(S))_{\mathrm{coh}}$, respectively $\mathrm{D}(\operatorname{Qcoh}(S))_{\mathrm{parf}}$, the full triangulated subcategory of $\mathrm{D}(\operatorname{Qcoh}(S))$ formed by pseudo-coherent, respectively perfect, complexes.

Proposition 2.2.2. Let S be a noetherian scheme. The canonical functor

$$\mathrm{D}^-(\operatorname{Coh}(S)) \longrightarrow \mathrm{D}(\operatorname{Qcoh}(S))$$

is fully faithful and has as essential image $\mathrm{D}^-(\operatorname{Qcoh}(S))_{\mathrm{coh}}$.

Proof. Apply (1.1), taking for the site the Zariski site of S, and for $\mathcal{C}$, respectively $\mathcal{C}_0$, the fibered categories $U \mapsto \operatorname{Qcoh}(U)$, respectively $U \mapsto \operatorname{Coh}(U)$. Since $\mathcal{O}_S$ is coherent, the objects of $\mathcal{C}_0$ are pseudo-coherent (I 3.5); hence an object of $\mathrm{D}(\operatorname{Qcoh}(U))$ is pseudo-coherent, in the sense of (2.2.1), if and only if it is pseudo-coherent relative to $\mathcal{C}_0$ (I 2.14.1). By (1.1 c)), it remains to verify conditions (i) and (ii) of (1.1). For (ii) this is trivial, since a quasi-coherent module which is a quotient of a coherent module is coherent, and coherence is local. For (i), one uses (2.1.1 a)) and the fact that every quasi-coherent $\mathcal{O}_S$-module is a filtered inductive limit of coherent $\mathcal{O}_S$-modules (EGA I 9.4.9).

Granting (3.5) and (3.7), which are independent of the present section, one easily deduces the following.

Corollary 2.2.2.1. Let S be a noetherian scheme, respectively a noetherian scheme of finite dimension. The canonical functor

$$\mathrm{D}^b(\operatorname{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

respectively

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}^b(S)_{\mathrm{coh}}$, respectively $\mathrm{D}^-(S)_{\mathrm{coh}}$ (I 2.15).

Proposition 2.2.3 (cf. EGA II 4.5.2 and 4.5.5). Let S be a quasi-compact and quasi-separated scheme, and let $(L_i)_{i\in I}$ be a family of invertible sheaves on S. For every $\mathcal{O}_S$-module F and every pair $(i,n) \in I \times \mathbb{Z}$, let $F_{(i,n)}$ denote the sub-$\mathcal{O}_S$-module of $F \otimes L_i^{\otimes n}$ generated by the sections of $F \otimes L_i^{\otimes n}$ over S. The following conditions are equivalent:

(i) the open sets S_f, for $f \in \Gamma(S, L_i^{\otimes n})$, $i \in I$ and $n \in \mathbb{N}$, form a base for the topology of S;

(ii) those of the S_f which are affine cover S;

(iii) for every quasi-coherent $\mathcal{O}_S$-module F of finite type, there exists a finite family (i_α, n_α) with $n_\alpha \geq 0$ and an epimorphism

$$\bigoplus_\alpha L_{i_\alpha}^{\otimes(-n_\alpha)} \longrightarrow F;$$

(iv) for every $F \in \operatorname{ob}\mathrm{Qcoh}(S)$, one has

$$F = \sum_{(i,n)\in I\times\mathbb{N}} F_{(i,n)} \otimes L_i^{\otimes(-n)},$$

a sum, not necessarily direct, of sub-sheaves of F;

(v) same condition as (iv), but with F a quasi-coherent ideal of $\mathcal{O}_S$.

Proof. This is a simple paraphrase of the cited passages; we recall the argument briefly for the reader's convenience. For (i)⇒(ii), choose for $x \in S$ an affine open U containing x. An open S_f with $x \in S_f \subset U$, where $f \in \Gamma(S, L_i^{\otimes n})$ and $n \geq 0$, is necessarily affine, by the following lemma.

Lemma 2.2.3.1 (EGA II 5.5.8). Let X be a scheme, L an invertible $\mathcal{O}_X$-module, and $f \in \Gamma(X, L)$. The canonical immersion $X_f \to X$ is affine.

Proof. This is trivial: the question is local on X, so one may suppose X affine and $L \simeq \mathcal{O}_X$.

For (ii)⇒(iii), choose a finite covering of S by affine opens $S_{f_{ij}}$, with $f_{ij} \in \Gamma(S, L_i^{\otimes m_{ij}})$. Replacing the f_{ij} by suitable powers, one may suppose that the m_{ij} with fixed i are all equal to an integer m_i. If F is quasi-coherent of finite type, then, for each i, j, there is an integer $q_i \geq 0$ such that every section of F over $S_{f_{ij}}$ extends to a section of $F \otimes L_i^{\otimes q_i m_i}$ over S. Thus $F \otimes L_i^{\otimes q_i m_i}$ is generated over $S_{f_{ij}}$ by finitely many global sections. Proceeding as in the proof of EGA II 4.5.5, one obtains, for each i, an integer $n_i \geq 0$ such that $F \otimes L_i^{\otimes n_i}$ is generated over the union of the corresponding $S_{f_{ij}}$ by finitely many sections over S. Hence there are integers $r_i \geq 0$ and a morphism

$$L_i^{-n_i r_i} \longrightarrow F$$

which induces an epimorphism over that union; the assertion follows.

For (iii)⇒(iv), since every quasi-coherent sheaf on S is the inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9 and IV 1.7.7), it is enough to treat the case where F is of finite type. There exist morphisms

$$u_i : L_i^{-n_i r_i} \longrightarrow F$$

whose sum is an epimorphism. But

$$\mathrm{Im}(u_i) = \mathrm{Im}\big(u_i \otimes \mathrm{Id}(L_i^{n_i})\big) \otimes L_i^{-n_i},$$

and $\mathrm{Im}(u_i \otimes \mathrm{Id}(L_i^{n_i}))$ is a subsheaf of $F_{(i,n_i)}$, which gives the assertion. The implication (iv)⇒(v) is trivial. Finally, for (v)⇒(i), let $x \in S$, let U be an open containing x, and let F be a quasi-coherent ideal defining $S - U$. By (v), there are $(i, n) \in I \times \mathbb{N}$ and a section f of $F \otimes L_i^{\otimes n}$ over S with $f(x) \neq 0$. Then $x \in S_f \subset U$, proving (i).

Definition 2.2.4. Let S be a quasi-compact and quasi-separated scheme, and let $(L_i)_{i \in I}$ be a family of invertible sheaves on S. We say that the L_i form an *ample family* if the equivalent conditions of (2.2.3) hold. In particular, a family consisting of a single sheaf L is ample if and only if L is ample in the usual sense (EGA II 4.5.3).

Definition 2.2.5. A scheme S is called *divisorial* if S is quasi-compact and quasi-separated and if there exists an ample family (2.2.4) of invertible $\mathcal{O}_S$-modules.

Lemma 2.2.6. Let X be a locally noetherian scheme. Let U be an open of X such that the inclusion $U \to X$ is affine, for example U affine and X separated. Then $X - U$ is purely of codimension 0 or 1; that is, at every maximal point x of $X - U$ one has $\dim \mathcal{O}_{X,x} = 0$ or 1. It is purely of codimension 1 if U contains the maximal points of X.

Proof. Put $Y = X - U$. Let x be a maximal point of Y and make the base change $X' = \operatorname{Spec} \mathcal{O}_{X,x} \to X$; one obtains a cartesian diagram

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X. \end{array}$$

Let $Y' = X' - U'$. Since x is a maximal point of Y, Y' is just the closed point of X'. Since X' is affine, U' is an affine open of X'. We show that the dimension d of X' is 0 or 1. Indeed, one knows (SGA 2 V 5.7) that $H^d_{Y'}(X', \mathcal{O}_{X'}) \neq 0$. But if $d \geq 2$, then

$$H^d_{Y'}(X', \mathcal{O}_{X'}) = H^{d-1}(U', \mathcal{O}_{X'}),$$

and the latter group is zero because U' is affine and $d - 1 \geq 1$, a contradiction.

Remark 2.2.6.1. Without the separation hypothesis on X, (2.2.6) would be false for an affine open U of X, as shown by the example of the affine plane with the origin doubled (EGA I 5.5.11).

Proposition 2.2.7. Let S be a noetherian separated locally factorial scheme, that is, a scheme whose local rings are factorial. Then S is divisorial (2.2.5). In particular (EGA IV 21.11.1):

Corollary 2.2.7.1. Every noetherian regular separated scheme is divisorial.

Proof of 2.2.7. One may suppose that S is integral. Let $(S_i)_{i \in I}$ be a finite covering of S by nonempty affine opens. By (2.2.6), $S - S_i$ is purely of codimension 1; hence, by EGA IV 21.7.4, it is the support of a positive divisor. In other words, for each $i \in I$, there is an invertible sheaf L_i and a section $f_i \in \Gamma(S, L_i)$ such that $S_i = S_{f_i}$. The family (L_i) is then ample by (2.2.4), using (2.2.3 (ii)).

Remark 2.2.7.2. As in (2.2.6), the separation hypothesis on S is essential. Let S again be the affine plane with the origin doubled. Recall that S is obtained by gluing two copies of the affine plane $\mathbb{A}^2_k$ along the identity on the complement U of the origin O; hence there is a canonical projection

$$p : S \longrightarrow \mathbb{A}^2_k$$

which is an isomorphism over U, while $p^{-1}(O)$ consists of two points O_1 and O_2. An invertible sheaf on S is obtained by gluing two invertible sheaves on $\mathbb{A}^2_k$ by an isomorphism over U. Since every invertible sheaf on $\mathbb{A}^2_k$ is trivial, an invertible sheaf on S is defined by an element

$g \in \Gamma(U, \mathcal{O}_U^*)$. But g is necessarily constant, since the restriction $\Gamma(\mathbb{A}_k^2, \mathcal{O}_{\mathbb{A}^2}) \to \Gamma(U, \mathcal{O}_U)$ is an isomorphism, the origin being of depth 2 in the plane. Thus every invertible sheaf on S is trivial. It is immediate, on the other hand, that p induces an isomorphism

$$\Gamma(\mathbb{A}_k^2, \mathcal{O}_{\mathbb{A}^2}) \xrightarrow{\sim} \Gamma(S, \mathcal{O}_S).$$

Consequently every global section of $\mathcal{O}_S$ which vanishes at O_1 also vanishes at O_2, and the opens S_f, for $f \in \Gamma(S, \mathcal{O}_S)$, do not form a base for the topology of S.

Theorem 2.2.8. Let S be a divisorial scheme (2.2.5), and let $(L_i)_{i \in I}$ be an ample family of invertible $\mathcal{O}_S$-modules. Let $L(S)$ be the strictly full additive subcategory of $\mathrm{Qcoh}(S)$ generated by the $L_i^{\otimes(-n)}$, for $(i, n) \in I \times \mathbb{N}$. Let $E \in \operatorname{ob} \mathrm{D}(\mathrm{Qcoh}(S))$.

a) The categories $L(S)$ and $\mathrm{Loclib}(S)$ are quasi-liftable in $\mathrm{Qcoh}(S)$ (I 1.2) and quasi-stable under kernels of epimorphisms (I 1.2).

b) Let $n \in \mathbb{Z}$. For E to be n-pseudo-coherent, respectively pseudo-coherent, in the sense of (2.2.1), it is necessary and sufficient that E be so relative to $L(S)$ or to $\mathrm{Loclib}(S)$ (I 2.3). The canonical functors

$$\mathrm{K}^-(L(S))/\mathrm{K}^{-,\delta}(L(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S)),$$

$$\mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

are fully faithful and have as essential image $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{coh}}$.

c) Let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have perfect amplitude contained in $[m, n]$ (2.2.1), it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(\mathrm{Qcoh}(S))$, to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \operatorname{ob} L(S)$ for $i > m$, and F^m is locally free of finite type. The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has as essential image $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$.

Proof. Let $F \to G$ be an epimorphism in $\mathrm{Qcoh}(S)$, with G of finite type. Since F is the inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9), there exists, by (2.1.1 a)), a quasi-coherent subsheaf F' of finite type in F such that $F' \to F \to G$ is an epimorphism. By (2.2.3 (iii)), there is an epimorphism $E \to F'$, with $E \in \operatorname{ob} L(S)$; hence $E \to F' \to F \to G$ is an epimorphism. This proves that the category of quasi-coherent $\mathcal{O}_S$-modules of finite type is quasi-liftable in $\mathrm{Qcoh}(S)$, and a fortiori the same is true for $\mathrm{Loclib}(S)$ and for $L(S)$.

For the second assertion of a), apply (1.1) on the Zariski site of S, with $\mathcal{C}$ the fibered category $U \mapsto \mathrm{Qcoh}(U)$ and $\mathcal{C}_0$ the fibered category $U \mapsto L(U)$. Since every object of L is pseudo-coherent, the notions of n-pseudo-coherence and pseudo-coherence in the fibered category $\mathrm{D}(\mathrm{Qcoh})$ coincide with the same notions relative to L (I 2.14). Condition (ii) of (1.1) is satisfied by (2.2.3 (iii)), and thus (1.1 a)) gives the part of a) concerning $L(S)$. The assertion for $\mathrm{Loclib}(S)$ is proved similarly. Part b) follows immediately from (1.1 b), c)), taking into account that, if $E \in \operatorname{ob} \mathrm{D}(\mathrm{Qcoh}(S))$ is n-pseudo-coherent, then E lies in $\mathrm{D}^-(\mathrm{Qcoh}(S))$, since S is quasi-compact. An object of the fibered category Qcoh is locally in L if and only if it is locally free of finite type; hence L is locally liftable in Qcoh and locally stable under kernels of epimorphisms. Thus c) follows from (1.2), with $\mathcal{C} = \mathrm{Qcoh}$, $\mathcal{C}_0 = L$, and $\mathcal{C}_1 = \mathrm{Loclib}(S)$.

Proposition 2.2.9. The hypotheses and notation are those of (2.2.8), but assume moreover that S is separated or noetherian.

a) The canonical functors

$$\mathrm{K}^-(L(S))/\mathrm{K}^{-,\delta}(L(S)) \longrightarrow \mathrm{D}(S), \qquad \mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

are fully faithful and have as essential image $\mathrm{D}^-(S)_{\mathrm{coh}}$. Moreover, if S is of finite cohomological dimension in the sense of (3.7), for example if the underlying space of S is noetherian of finite Krull dimension, then the functors

$$\mathrm{K}(L(S))/\mathrm{K}^{\delta}(L(S)) \longrightarrow \mathrm{K}(\mathrm{Loclib}(S))/\mathrm{K}^{\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^-(S)_{\mathrm{coh}}$$

are equivalences of categories.

b) The canonical functor

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\delta}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}(S)_{\mathrm{parf}}$ (I 4.9). For $E \in \mathrm{ob}\,\mathrm{D}(S)$ to have perfect amplitude contained in $[m,n]$ (I 4.8), it is necessary and sufficient that E be isomorphic, in $\mathrm{D}(S)$, to a complex F such that $F^i = 0$ for $i \notin [m,n]$, $F^i \in \mathrm{ob}\,L(S)$ for $i > m$, and F^m is locally free of finite type.

Proof. In view of (3.5) and (3.7), this is an immediate corollary of (2.2.8). The only point to check, and it follows trivially from (3.5), respectively (3.7), is that for an object E of $\mathrm{D}^-(\mathrm{Qcoh}(S))$, respectively of $\mathrm{D}(\mathrm{Qcoh}(S))$ when S has finite cohomological dimension in the sense of (3.7), the notions of pseudo-coherence and perfection of (2.2.1) coincide with the usual notions (I 2.15 and 4.9) after regarding E as an object of $\mathrm{D}(S)$.

Remark 2.2.10. a) If $S = \mathrm{Spec}\,A$ is affine, the family consisting of the single sheaf $\mathcal{O}_S$ is ample, and for it $L(S) = \mathrm{Lib}(S)$. It is also clear that the categories $\mathrm{K}^{-,\delta}(\mathrm{Lib}(S))$ and $\mathrm{K}^{-,\delta}(\mathrm{Loclib}(S))$ are reduced to zero objects. It follows from (2.2.9) that, writing $\mathrm{Mod}(A)$ for the category of A-modules and $\mathrm{D}(A) = \mathrm{D}(\mathrm{Mod}(A))$, the functor $R\Gamma_S : \mathrm{D}(S) \to \mathrm{D}(A)$ induces fully faithful functors

$$\mathrm{D}^-(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}(A), \qquad \mathrm{D}(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}(A),$$

whose essential images are respectively the triangulated subcategories $\mathrm{D}^-(A)_{\mathrm{coh}}$ and $\mathrm{D}(A)_{\mathrm{parf}}$, formed by complexes with bounded-above cohomology and pseudo-coherent, respectively perfect, relative to the subcategory $\mathrm{Lib}(A)$, respectively $\mathrm{Loclib}(A)$, of free, respectively projective, modules of finite type. There are commutative diagrams of equivalences of categories

$$\begin{tikzcd}
\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \arrow[r] \arrow[d, "\Gamma_S"'] & \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \arrow[r] \arrow[d, "\Gamma_S"'] & \mathrm{D}^b(S)_{\mathrm{coh}} \arrow[d, "R\Gamma_S"] \\
\mathrm{K}^{-,b}(\mathrm{Lib}(A)) \arrow[r] & \mathrm{K}^{-,b}(\mathrm{Loclib}(A)) \arrow[r] & \mathrm{D}^b(A)_{\mathrm{coh}},
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{K}^b(\mathrm{Loclib}(S)) \arrow[r] \arrow[d, "\Gamma_S"'] & \mathrm{D}(S)_{\mathrm{parf}} \arrow[d, "R\Gamma_S"] \\
\mathrm{K}^b(\mathrm{Loclib}(A)) \arrow[r] & \mathrm{D}(A)_{\mathrm{parf}}.
\end{tikzcd}$$

In the upper diagram, one may replace b by $-$ when S has finite cohomological dimension in the sense of (3.7).

b) For an arbitrary scheme S, it follows from a) that a complex of $\mathcal{O}_S$-modules which is pseudo-coherent and has bounded cohomology is locally isomorphic, in $\mathrm{D}(S)$, to a strictly pseudo-coherent complex, which was by no means evident a priori (cf. I 2.3.1).

2.3. Soft sheaves

2.3.1. Recall a few definitions and results from [T.F.]. Let X be a paracompact space. A sheaf of sets F on X is called *soft* if, for every closed subset Y of X, every section of F over Y extends to a section of F over X. Soft abelian sheaves are acyclic: if F is soft, then $H^i(X, F) = 0$ for $i > 0$ (T.F. II 4.4.3).

Let A be a sheaf of rings on X. If F is a soft A-module, then F is generated by its sections (cf. 2.1.1): indeed, the homomorphism $A_X^{(I)} \to F$ defined by $I = H^0(X, F)$ is an epimorphism. Moreover, if A is soft, every A-module is soft (T.F. II 3.7.1). Examples of soft sheaves of rings include the sheaf of continuous real or complex-valued functions on a paracompact space and the sheaf of C^∞ functions on a C^∞ manifold.

Proposition 2.3.2. Let S be a compact space equipped with a soft sheaf of rings A. Write $\mathrm{Mod}(S)$ for the category of left A-modules and $\mathrm{D}(S) = \mathrm{D}(\mathrm{Mod}(S))$.

a) The category $\mathrm{Lib}(S)$, respectively $\mathrm{Loclib}(S)$ (2.0), is liftable in $\mathrm{Mod}(S)$ and quasi-stable, respectively stable, under kernels of epimorphisms (I 1.2). This assertion holds more generally for S paracompact.

b) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$ and $n \in \mathbb{Z}$. For E to be n-pseudo-coherent, respectively pseudo-coherent, it is necessary and sufficient that it be so relative to $\mathrm{Lib}(S)$ or to $\mathrm{Loclib}(S)$. The canonical functors
$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{D}(S), \qquad \mathrm{K}^-(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$
are fully faithful and have as essential image $\mathrm{D}^-(S)_{\mathrm{coh}}$ (I 2.15).

c) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$ and let $[m, n]$ be an interval of $\mathbb{Z}$. For E to have perfect amplitude contained in $[m, n]$ (I 4.8), it is necessary and sufficient that there exist a quasi-isomorphism $F \to E$, where F is a complex such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$, and $F^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$. The canonical functor
$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)$$
is fully faithful and has as essential image $\mathrm{D}(S)_{\mathrm{parf}}$ (I 4.9).

d) Let $\mathrm{Mod}(A(S))$ be the category of left $A(S)$-modules and $\mathrm{D}(A(S))$ the corresponding derived category. The functor
$$H^0(S, -) : \mathrm{Mod}(S) \longrightarrow \mathrm{Mod}(A(S))$$
is exact and is an equivalence of categories. Moreover it defines equivalences of categories
$$\mathrm{D}(S) \xrightarrow{\sim} \mathrm{D}(A(S)), \qquad \mathrm{D}^-(S)_{\mathrm{coh}} \xrightarrow{\sim} \mathrm{D}^-(A(S))_{\mathrm{coh}}, \qquad \mathrm{D}(S)_{\mathrm{parf}} \xrightarrow{\sim} \mathrm{D}(A(S))_{\mathrm{parf}},$$
where pseudo-coherence, respectively perfection, in $\mathrm{D}(A(S))$ is understood relative to the category of free, respectively projective, modules of finite type. It also gives an equivalence from the full subcategory of $\mathrm{D}(S)$ formed by n-pseudo-coherent objects, respectively by objects of perfect amplitude contained in $[m, n]$, onto the analogous full subcategory of $\mathrm{D}(A(S))$.

Proof. The first assertion of a) follows from (2.1.2 a)), applied with S the topos of sheaves of sets on S and $\mathcal{C}'_S = \mathrm{Mod}(S)$, using (2.3.1). The second follows from (1.1 a)), respectively (1.2 a)), with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$, respectively Loclib; the condition (ii) of (1.1) is verified by (2.3.1) and (2.1.1 b)). Part b) follows from (1.1 b), c)) with the same $\mathcal{C}$ and $\mathcal{C}_0$. Part c) follows from (1.2 b), c), d)), with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$. The first assertion of d) follows, using (2.3.1), from (2.1.2 b) (ii)) with $\mathcal{C}'_S = \mathcal{C}_S = \mathrm{Mod}(S)$. The other assertions in d) follow from it, together with b) and c), and from the fact that Γ_S carries free modules, respectively direct factors of free modules, of finite type to free, respectively projective, modules of finite type.

2.4. Stein spaces

Definition 2.4.1. Let S be a locally ringed topos (I 2.15.1 (ii)), with final object also denoted S. We say that S is *Stein* if the structural sheaf $\mathcal{O}_S$ is coherent (I 3.1) and if the following properties hold:

Theorem A: every coherent $\mathcal{O}_S$-module (I 3.1) is generated by its sections;

Theorem B: for every coherent $\mathcal{O}_S$-module F, one has $H^i(S, F) = 0$ for $i > 0$.

Remark 2.4.1.1. If S has enough closed points s such that the ideal $\mathfrak{m}_s$ defined by s is coherent, then the relation $H^1(S, F) = 0$ for every coherent sheaf F implies Theorem A. Indeed, let F be a coherent $\mathcal{O}_S$-module and let s be a closed point of S. There is an exact sequence

$$0 \longrightarrow \mathfrak{m}_s F \longrightarrow F \longrightarrow F/\mathfrak{m}_s F \longrightarrow 0,$$

where $F/\mathfrak{m}_s F$ is a coherent sheaf concentrated at s. Applying $H^0(S, -)$ and writing the cohomology exact sequence, one finds that

$$H^0(S, F) \longrightarrow H^0(S, F/\mathfrak{m}_s F)$$

is an epimorphism. Since

$$H^0(S, F/\mathfrak{m}_s F) = F_s/\mathfrak{m}_s F_s,$$

where $\mathfrak{m}_s$ is the maximal ideal of $\mathcal{O}_s$, Nakayama's lemma gives the conclusion. The editor does not know whether the preceding implication is true in general when one assumes only that $\mathcal{O}_S$ is a coherent sheaf of local rings. The editor also does not know whether the relation $H^1(S, F) = 0$ for every coherent sheaf F implies Theorem B.

Examples 2.4.2. A complex analytic Stein space, a compact convex subset of $\mathbb{C}^n$ equipped with the sheaf induced by the sheaf of holomorphic functions, the spectrum of a noetherian ring, and an affine rigid analytic space define Stein topoi.

Lemma 2.4.3. Let S be a Stein topos (2.4.1). Suppose that the final object S is quasi-compact (2.1.1), and let $\mathrm{Coh}(S)$ be the abelian category of coherent $\mathcal{O}_S$-modules, that is, modules of finite presentation. Then:

a) $\mathrm{Lib}(S)$, respectively $\mathrm{Loclib}(S)$, is liftable in $\mathrm{Coh}(S)$ and quasi-stable, respectively stable, under kernels of epimorphisms (I 1.2).

b) The canonical functors

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^-(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^-(\mathrm{Coh}(S))$$

are equivalences of categories.

c) Put $A = \Gamma(S, \mathcal{O}_S)$, and write $\mathrm{Mod}(A)$ for the category of A-modules and $\mathrm{D}(A)$ for the corresponding derived category. The functor

$$\Gamma_S : \mathrm{Coh}(S) \longrightarrow \mathrm{Mod}(A)$$

is exact, fully faithful, and has as essential image the strictly full additive subcategory of $\mathrm{Mod}(A)$ formed by A-modules of finite presentation. It induces a fully faithful functor

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(A),$$

whose essential image is the full subcategory $\mathrm{D}^-(A)_{\mathrm{coh}}$ formed by complexes pseudo-coherent relative to the subcategory of free modules of finite type.

Here "enough closed points" means enough fiber functors (SGA 4 IV 6.4.1) of the form $i^* : S \to s$, with i_* the inclusion of a closed subtopos of S; and $\mathfrak{m}_s$ denotes the sections of $\mathcal{O}_S$ whose fiber at s belongs to the maximal ideal of $\mathcal{O}_s$.

Proof. For the first assertion of a), apply (2.1.2 a)) with $\mathcal{C}'_S = \mathrm{Coh}(S)$; the hypothesis of (2.1.2) is verified by Theorem B, since if E is coherent and G is locally free of finite type, then $\mathrm{Hom}(G, E)$ is coherent. Thus $\mathrm{Loclib}(S)$, and a fortiori $\mathrm{Lib}(S)$, is liftable in $\mathrm{Coh}(S)$. The second assertion of a) follows from (1.1 a)), respectively (1.2 a)), applied with $\mathcal{C}_0 = \mathrm{Lib}$, respectively Loclib; condition (ii) of (1.1) is verified by Theorem A and (2.1.1 b)). Part b) follows from (1.1 c)), applied successively with $\mathcal{C}_0 = \mathrm{Lib}$ and $\mathcal{C}_0 = \mathrm{Loclib}$. The first assertion of c) follows from (2.1.2 b)(i)), applied with $\mathcal{C}'_S = \mathrm{Coh}(S)$. To prove that $\Gamma_S : \mathrm{D}^-(\mathrm{Coh}(S)) \to \mathrm{D}(A)$ is fully faithful, it is enough, by b), to prove that the composite

$$\mathrm{K}^-(\mathrm{Lib}(S)) \longrightarrow \mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(A)$$

is so. But this composite factors as

$$\mathrm{K}^-(\mathrm{Lib}(S)) \xrightarrow{\Gamma_S} \mathrm{K}^-(\mathrm{Lib}(A)) \longrightarrow \mathrm{D}(A),$$

where $\mathrm{Lib}(A)$ denotes the category of free A-modules of finite type; the first arrow is an equivalence of categories, and the second is fully faithful by (I 2.7 b)), applied with $\mathcal{C} = \mathrm{Mod}(A)$ and $\mathcal{C}_0 = \mathrm{Lib}(A)$, since $\mathrm{K}^{-,\delta}(\mathrm{Lib}(A)) = 0$. This proves the last assertion of c).

Lemma 2.4.4. Under the hypotheses of (2.4.3), the canonical functor

$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image $\mathrm{D}^b(S)_{\mathrm{coh}}$ (I 2.15).

Proof. Apply (2.1.3) with $\mathcal{A} = \mathrm{Coh}(S)$, $\mathcal{B} = \mathrm{Mod}(S)$, and $\varphi : \mathcal{A} \to \mathcal{B}$ the inclusion functor. Since this functor is fully faithful, it is enough, by (2.1.3 a)(ii)), to check that for fixed $M \in \mathrm{ob}\,\mathrm{Coh}(S)$ and $n > 0$ the functor

$$L \longmapsto \mathrm{Ext}^n_{\mathrm{Mod}(S)}(L, M)$$

is coeffaceable on $L \in \mathrm{ob}\,\mathrm{Coh}(S)$. By Theorem A and (2.1.1 b)), every coherent $\mathcal{O}_S$-module is a quotient of a free module of finite type; and if L is free of finite type, then $\mathrm{Ext}^n_{\mathrm{Mod}(S)}(L, M) = 0$ by Theorem B. The assertion follows. Hence the functor $\mathrm{D}^b(\mathrm{Coh}(S)) \to \mathrm{D}(S)$ is fully faithful and has as essential image the full subcategory of $\mathrm{D}(S)$ formed by complexes with bounded coherent cohomology, which is precisely $\mathrm{D}^b(S)_{\mathrm{coh}}$ since $\mathcal{O}_S$ is coherent (I 3.5).

Proposition 2.4.5. With the hypotheses and notation of (2.4.3), one has:

a) The canonical functors

$$\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}}$$

are equivalences of categories. Moreover, the functor $R\Gamma_S : \mathrm{D}^b(S) \to \mathrm{D}^b(A)$ induces a fully faithful functor

$$R\Gamma_S : \mathrm{D}^b(S)_{\mathrm{coh}} \longrightarrow \mathrm{D}^b(A),$$

whose essential image is $\mathrm{D}^b(A)_{\mathrm{coh}}$, the triangulated subcategory of $\mathrm{D}(A)$ formed by complexes with bounded cohomology and pseudo-coherent relative to the category of free modules of finite type.

b) Let $E \in \mathrm{ob}\,\mathrm{D}(S)$, and let $[m, n]$ be an interval of $\mathbb{Z}$. If E has perfect amplitude contained in $[m, n]$ (I 4.5), then E is isomorphic to a complex F such that $F^i = 0$ for $i \notin [m, n]$, $F^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$, and $F^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$.

c) The canonical functor
$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S)_{\mathrm{parf}}$$
is an equivalence of categories (I 4.9). Moreover, $R\Gamma_S : \mathrm{D}^b(S) \to \mathrm{D}^b(A)$ induces a fully faithful functor
$$R\Gamma_S : \mathrm{D}(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^b(A),$$
whose essential image is $\mathrm{D}^b(A)_{\mathrm{parf}}$, the triangulated subcategory of $\mathrm{D}(A)$ formed by complexes perfect relative to the category of projective modules of finite type.

Proof. a) The functors
$$\mathrm{K}^{-,b}(\mathrm{Lib}(S)) \longrightarrow \mathrm{K}^{-,b}(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(A)_{\mathrm{coh}}$$
are equivalences of categories by (2.4.3 b), c). Since the fully faithful functor $\Gamma_S : \mathrm{D}^b(\mathrm{Coh}(S)) \to \mathrm{D}(A)$ factors as
$$\mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(S)_{\mathrm{coh}} \xrightarrow{R\Gamma_S} \mathrm{D}(A),$$
the conclusion follows from (2.4.4).

b) Suppose $\mathrm{parf.amp}(E) \subset [m,n]$. In particular E is pseudo-coherent; by a), after replacing E by an isomorphic object, we may assume that $E^i \in \mathrm{ob}\,\mathrm{Lib}(S)$ for $i > m$ and $E^i = 0$ for $i > n$. Truncating, we may also suppose that $E^i = 0$ for $i < m$. By (I 4.16), applied with $\mathcal{C} = \mathrm{Mod}$ and $\mathcal{C}_0 = \mathrm{Lib}$, one then has $E^m \in \mathrm{ob}\,\mathrm{Loclib}(S)$.

c) The hypotheses (1.1 (i)) and, respectively, (1.1 (ii)) are verified for $\mathcal{C} = \mathrm{Coh}(S)$ and $\mathcal{C}_0 = \mathrm{Lib}$, by (2.4.3 a)) and by (2.1.1 b)) together with Theorem A. Hence (1.2 c), d)) shows that the functor
$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}^b(\mathrm{Coh}(S))$$
is fully faithful. Therefore, by (2.4.4), the same is true for
$$\mathrm{K}^b(\mathrm{Loclib}(S)) \longrightarrow \mathrm{D}(S),$$
and its essential image is $\mathrm{D}(S)_{\mathrm{parf}}$ by b). By an argument analogous to that used in the proof of (2.4.3 c)), one sees that the functor $\mathrm{K}^b(\mathrm{Loclib}(S)) \to \mathrm{D}(S)$ induced by Γ_S is fully faithful and has essential image $\mathrm{D}^b(A)_{\mathrm{parf}}$; one finishes the proof as in a).

Remark 2.4.6. The editor does not know whether in (2.4.4) one may replace D^b by D^-, and consequently whether (2.4.5 a)) remains true when one replaces $\mathrm{K}^{-,b}$ by K^- and D^b by D^-.

We finish with an interesting complement to (2.4.5). First we give a definition.

Definition 2.4.7. A *complex pseudo-analytic space* is a ringed space locally isomorphic to a ringed space of the form $(Y, \mathcal{O}_X|Y)$, where X is a complex analytic space and Y is a closed subset of X.

The structural sheaf of a complex pseudo-analytic space is therefore coherent. We say that a complex pseudo-analytic space is Stein if the corresponding topos is Stein (2.4.1).

Exposé II. Global resolutions (continued)

Proposition 2.4.8. Let S be a compact Stein complex pseudo-analytic space (2.4.7). Suppose that, for every open U of S and every reduced closed subspace X of U, the normalization of X is locally connected. This condition is verified, for example, when S is locally defined by real analytic equations or inequalities in a space $\mathbb{R}^n$. Then
$$A = \Gamma(S, \mathcal{O}_S)$$

is a noetherian ring.

Proof. It is enough to show that, if M is an A-module of finite presentation, then the set of submodules of M, or equivalently the set of submodules of finite type of M, satisfies the maximal condition. Indeed, if N is a submodule of finite type of M, then M/N is of finite presentation by I 2.6 and 2.9, so that N is itself of finite presentation by the equivalence (2.4.3 c). Using again the equivalence (2.4.3 c) between the category of coherent $\mathcal{O}_S$-modules and the category of A-modules of finite presentation, together with the compactness of S, one is reduced to proving the following lemma.

Lemma 2.4.8.1. Let S be a complex pseudo-analytic space satisfying the local connectedness condition of (2.4.8). Let F be a coherent $\mathcal{O}_S$-module, let (E_n) be an increasing sequence of coherent sub-$\mathcal{O}_S$-modules of F, and let s be a point of S. Then there exists an open neighbourhood U of s such that the sequence $(E_n|U)$ is stationary.

Proof. Since the ring $\mathcal{O}_{S,s}$ is noetherian, there exists an integer n_0 such that $(E_n)_s = (E_{n_0})_s$ for $n \geq n_0$. Replacing the sequence (E_n) by the sequence (E_n/E_{n_0}), one may therefore suppose that $(E_n)_s = 0$ for every n. On the other hand $F_s = M$ is a finite $\mathcal{O}_{S,s}$-module; hence it has a finite filtration

$$M = M_0 \supset M_1 \supset \cdots \supset M_q = 0$$

such that

$$M_i/M_{i+1} \simeq \mathcal{O}_s/\mathfrak{p}_i,$$

where $\mathfrak{p}_i$ is a prime ideal of $\mathcal{O}_s$. Since the category of finite-type $\mathcal{O}_s$-modules is equivalent to the category of germs at s of coherent $\mathcal{O}_S$-modules, after shrinking around s one may suppose that the M_i, respectively the $\mathfrak{p}_i$, are the fibres at s of coherent sub-$\mathcal{O}_S$-modules F_i, respectively coherent ideals $\mathfrak{p}_i$, of F, respectively of $\mathcal{O}_S$, and that one has

$$F = F_0 \supset F_1 \supset \cdots \supset F_q = 0, \qquad F_i/F_{i+1} \simeq \mathcal{O}_S/\mathfrak{p}_i.$$

To prove that the sequence (E_n) is stationary in a neighbourhood of s is therefore equivalent to proving the same assertion for the induced sequences on the graded pieces of this filtration. Thus one may suppose that $F = \mathcal{O}_S$ and that $\mathcal{O}_{S,s}$ is an integral domain; the E_n then form an increasing sequence of ideals of $\mathcal{O}_S$ with $(E_n)_s = 0$.

Since $\mathcal{O}_{S,s}$ is an integral domain, hence in particular reduced, one may moreover suppose, by Oka's theorem, after localizing at s, that S is reduced. We have to show that there is an open neighbourhood U of s such that $E_n|U = 0$ for n large enough. Consider, to do this, the normalization $\widetilde{S}$ of S. It is the analytic spectrum of a finite $\mathcal{O}_S$-algebra $\widetilde{\mathcal{O}}_S$ whose fibre at each $t \in S$ is the integral closure of $\mathcal{O}_{S,t}$. Since $E_n \subset E_n\widetilde{\mathcal{O}}_S$, it is enough to prove that $E_n\widetilde{\mathcal{O}}_S = 0$ for n large. This reduces us to the case where S is normal.

In that case, since $(E_n)_s = 0$, the closed immersion

$$i_n : S_n \longrightarrow S$$

defined by E_n is étale at s. But, S being normal, the set of points at which i_n is étale is both open and closed by the lemma below. Hence its image in S is simultaneously open and closed. The restriction of E_n to the connected component of s in S is therefore zero; this component is open because S is locally connected. The lemma is proved.

Lemma 2.4.8.2 (cf. EGA IV 18.10.3). Let Y be a normal pseudo-analytic space, and let $f : X \to Y$ be a non-ramified morphism. The set of points of X at which f is étale is both open and closed.

Proof. It is immediate that the set of points at which f is étale is open. It remains to show that the set of points at which f is not étale is open. Since a non-ramified morphism is locally an immersion, one may suppose that f is a closed immersion defined by a coherent ideal I. Let x be a point of X such that $I_x \neq 0$. Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_{Y,x}$ containing I_x. One has

$$\dim(\mathcal{O}_{Y,x}/\mathfrak{p}) > 0,$$

for otherwise $\mathfrak{p} = 0$, since $\mathcal{O}_{Y,x}$ is integral. Thus Y has positive codimension at x, hence positive codimension in a neighbourhood of x; the assertion follows.

3. Complements on quasi-coherent sheaves on schemes

3.0. In this section we fix a quasi-compact and quasi-separated scheme S. The canonical, fully faithful inclusion functor

$$\iota : \mathrm{Qcoh}(S) \longrightarrow \mathrm{Mod}(S)$$

extends to a functor, again denoted ι, from $\mathrm{D}(\mathrm{Qcoh}(S))$ to $\mathrm{D}(S) = \mathrm{D}(\mathrm{Mod}(S))$, which we shall study.

Lemma 3.1. The category $\mathrm{Qcoh}(S)$ is an abelian category satisfying AB5, and the quasi-coherent sheaves of finite type form a small family of generators. In particular $\mathrm{Qcoh}(S)$ has enough injectives.

Proof. The assertion about generators follows at once from the fact that every quasi-coherent sheaf on S is the filtered inductive limit of its quasi-coherent subsheaves of finite type (EGA I 9.4.9 and IV 1.7.7). The other assertions are evident.

Remark 3.1.1. The editor does not know whether $\mathrm{Qcoh}(S)$ still has enough injectives when one assumes only that S is quasi-separated.

Lemma 3.2. The functor

$$\iota : \mathrm{Qcoh}(S) \longrightarrow \mathrm{Mod}(S)$$

has a right adjoint

$$Q : \mathrm{Mod}(S) \longrightarrow \mathrm{Qcoh}(S),$$

called the coherer.

Proof. If S is affine, the existence of Q is immediate: one takes $Q = \widetilde{\Gamma_S}$ (EGA I 1.3.7). Let now $f : U \to S$ be an affine open of S. It is clear that, for $E \in \mathrm{ob}\,\mathrm{Qcoh}(S)$ and $F \in \mathrm{ob}\,\mathrm{Mod}(U)$, one has a canonical isomorphism

$$\mathrm{Hom}(E, f_*F) \simeq \mathrm{Hom}(E, f_*Q(F)). \tag{1}$$

Let $(f_i : U_i \to S)$ be a finite covering of S by affine opens, and, for each pair (i, j), let $(U_{ijk} \to U_i \cap U_j)_k$ be a finite covering of $U_i \cap U_j$ by affine opens, which is possible since S is quasi-separated. For $F \in \mathrm{ob}\,\mathrm{Mod}(S)$ one has an exact sequence

$$0 \longrightarrow F \longrightarrow \prod_i f_{i*}(F|U_i) \longrightarrow \prod_{i,j,k} f_{ijk*}(F|U_{ijk}), \tag{2}$$

where f_{ijk} denotes the inclusion of U_{ijk} into S. If $g : V \to U$ is an inclusion of affine opens of S, there is an evident natural arrow

$$Q(F|U) \longrightarrow g_*Q(F|V).$$

Consequently (2) gives an arrow

$$\prod_i f_{i*}Q(F|U_i) \longrightarrow \prod_{i,j,k} f_{ijk*}Q(F|U_{ijk}), \tag{3}$$

whose kernel we denote by QF. If $E \in \operatorname{ob} \mathrm{Qcoh}(S)$, applying the functor $\mathrm{Hom}(E, \cdot)$ to (2), and using (1), one obtains a functorial isomorphism in E,

$$\mathrm{Hom}(E, F) \simeq \mathrm{Hom}(E, QF).$$

This proves the existence of Q; its uniqueness is evident, since it is an adjoint.

Lemma 3.3.

a) The coherer is an additive functor and transforms injectives into injectives.

b) Let
$$RQ : \mathrm{D}^+(S) \longrightarrow \mathrm{D}^+(\mathrm{Qcoh}(S))$$
be the right derived functor of the coherer. For $E \in \operatorname{ob} \mathrm{D}(\mathrm{Qcoh}(S))$ and $F \in \operatorname{ob} \mathrm{D}^+(S)$ there are canonical functorial isomorphisms
$$R\,\mathrm{Hom}_{\mathrm{Mod}}(E, F) \simeq R\,\mathrm{Hom}_{\mathrm{Qcoh}}(E, RQ(F)), \tag{i}$$
$$\mathrm{Hom}_{\mathrm{Mod}}(E, F) \simeq \mathrm{Hom}_{\mathrm{Qcoh}}(E, RQ(F)). \tag{ii}$$

Proof. Assertion a) is an immediate consequence of the definition. For b), it is enough to establish the isomorphism (i), since (ii) follows by passing to cohomology. For this purpose one may suppose that F has injective components. Then QF has injective components by a), and the isomorphism (i) follows directly from the definition of Q.

Lemma 3.4. Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. For $E \in \operatorname{ob} \mathrm{Mod}(X)$ there is a canonical functorial isomorphism

$$Q_Y f_*(E) \simeq f_* Q_X(E).$$

Proof. Since f is quasi-compact and quasi-separated (EGA IV 1.2.3 and 1.2.4), f_* transforms quasi-coherent sheaves into quasi-coherent sheaves (EGA III 1.4.10 and IV 1.7.21). The functors $Q_Y f_*$ and $f_* Q_X$ from $\mathrm{Mod}(X)$ to $\mathrm{Qcoh}(Y)$ are therefore both right adjoints to the functor

$$f^* : \mathrm{Qcoh}(Y) \longrightarrow \mathrm{Mod}(X),$$

and hence are canonically isomorphic.

Proposition 3.5. If S is noetherian, or quasi-compact and separated, the functor

$$\iota : \mathrm{D}^+(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}^+(S)$$

is fully faithful and has as essential image the full subcategory $\mathrm{D}^+(S)_{\mathrm{qcoh}}$ of $\mathrm{D}^+(S)$ formed by complexes with quasi-coherent cohomology.

Proof. One may restrict to proving the assertion in the case where S is separated, since the noetherian case is already known. It is plainly equivalent to prove the following two assertions.

(3.5.1) For $E \in \operatorname{ob} \mathrm{D}^+(\mathrm{Qcoh}(S))$, the adjunction arrow

$$E \longrightarrow RQ(E)$$

is an isomorphism.

(3.5.2) For $E \in \operatorname{ob} \mathrm{D}^+(S)_{\mathrm{qcoh}}$, the adjunction arrow

$$RQ(E) \longrightarrow E$$

is an isomorphism.

Proof of (3.5.1). By an immediate dévissage one is reduced to proving that quasi-coherent sheaves are acyclic for the coherer, that is, for $E \in \operatorname{ob}\operatorname{Qcoh}(S)$, one has $R^iQ(E) = 0$ for $i > 0$, and that, moreover, the adjunction arrow $E \to QE$ is an isomorphism. If S is affine, this follows from the formula $Q = \widetilde{\Gamma_S}$ used in the proof of (3.2). In the general case, cover S by a finite number of affine opens U_i. Since S is separated, the inclusions

$$f_{i_0\ldots i_p} : U_{i_0} \cap \cdots \cap U_{i_p} \longrightarrow S$$

are affine morphisms. Let $E \in \operatorname{ob}\operatorname{Qcoh}(S)$, and consider the resolution of E defined by this covering,

$$0 \longrightarrow E \longrightarrow \prod_i f_{i*}f_i^*E \longrightarrow \cdots \longrightarrow \prod_{i_0\ldots i_p} f_{i_0\ldots i_p*}f_{i_0\ldots i_p}^*E \longrightarrow \cdots . \tag{*}$$

It gives a spectral sequence

$$E_1^{pq} = \prod_{i_0\ldots i_p} R^qQ(f_{i_0\ldots i_p*}f_{i_0\ldots i_p}^*E) \Longrightarrow R^{p+q}Q(E). \tag{**}$$

Since $f_{i_0\ldots i_p}$ is affine, (3.4) gives, for every q,

$$R^qQ(f_{i_0\ldots i_p*}f_{i_0\ldots i_p}^*E) \simeq f_{i_0\ldots i_p*}R^qQ(f_{i_0\ldots i_p}^*E). \tag{***}$$

In particular $E_1^{pq} = 0$ for $q > 0$, by the affine case already treated. From (**) one deduces isomorphisms

$$R^nQ(E) \simeq H^n(QE') \quad (n \geq 0),$$

where E' is the resolution of E defined by the exact sequence (*). But by (***) one has $QE^n \simeq E^n$ for $n \geq 0$; hence finally

$$R^nQ(E) = 0 \quad (n > 0),$$

and $Q(E) \simeq E$. This proves (3.5.1).

Proof of (3.5.2). The spectral sequence

$$R^pQ(H^q(E)) \Longrightarrow R^{p+q}Q(E)$$

reduces us immediately to the case where $E \in \operatorname{ob}\operatorname{Qcoh}(S)$. The composite

$$E \longrightarrow RQE \longrightarrow E$$

of the adjunction arrows is the identity, and the assertion follows from (3.5.1). This completes the proof of (3.5).

Remark 3.6. The conclusion of (3.5) would be false if one omitted the hypothesis "separated or noetherian", as is shown by a neat counterexample of J.-L. Verdier; see Appendix I to the present exposé.

Proposition 3.7. Under the hypothesis of (3.5), suppose moreover that there is an integer N such that, for every $F \in \operatorname{ob}\operatorname{Mod}(S)$ and every quasi-compact open U of S, one has

$$H^i(U, F) = 0 \quad \text{for } i > N.$$

Then:

a) The coherer has finite cohomological dimension.

b) The functor

$$\iota : \mathrm{D}(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}(S)$$

is fully faithful and has as essential image the full subcategory $\mathrm{D}(S)_{\mathrm{qcoh}}$ of $\mathrm{D}(S)$ formed by complexes with quasi-coherent cohomology.

Proof of a). If S is affine, the formula $Q = \widetilde{\Gamma_S}$ implies that Q is of cohomological dimension $\leq N$. We pass to the general case. Let us show that there is an integer a such that $R^qQ(E) = 0$ for every $E \in \mathrm{ob}\,\mathrm{Mod}(S)$ and every $q > a$. The hypothesis implies that every $E \in \mathrm{ob}\,\mathrm{Mod}(S)$ admits a finite right resolution $E^{\cdot}$ of length N such that the E^i are acyclic, that is, $H^q(U, E^i) = 0$ for every open U of S and every $q > 0$. Thus it is enough to prove that there is an integer b such that

$$R^qQ(E) = 0$$

for every acyclic $\mathcal{O}_S$-module E and every $q > b$; one may then take $a = b + N$.

Let (U_i) be a finite covering of S by N' affine opens. Let E be an acyclic $\mathcal{O}_S$-module and let $C^{\cdot}(E)$ be the resolution of E defined by the covering (U_i). There are two cases.

If S is separated, the components of $C^{\cdot}(E)$ are sums of terms of the form f_*f^*E, where $f : U \to S$ is the inclusion of an affine open, hence an affine morphism. One has

$$RQ(Q_X f_*)f^*E \simeq RQ\, Rf_*f^*E \simeq RQ\, f_*f^*E,$$

since f^*E is acyclic. On the other hand, by (3.4),

$$R(Q_X f_*)f^*E \simeq R(f_*Q_X)f^*E \simeq f_*RQf^*E,$$

since f is affine. Hence

$$R^nQ(f_*f^*E) \simeq f_*R^nQ(f^*E) \quad (n \geq 0).$$

By the affine case,

$$R^nQ(f_*f^*E) = 0 \quad (n > N).$$

Since $C^{\cdot}(E)$ has length N', it follows that

$$R^nQ(E) = 0 \quad (n > N + N').$$

If S is noetherian, the components of $C^{\cdot}(E)$ are sums of terms of the same form f_*f^*E, where now $f : U \to S$ is the inclusion of a quasi-compact and separated open. As in the first case, one has

$$R(Q_X f_*)f^*E \simeq RQ\, f_*f^*E.$$

Since U is noetherian, $\mathrm{Qcoh}(U)$ has enough injective objects in $\mathrm{Mod}(U)$, and hence the right derived functor of $f_* : \mathrm{Qcoh}(U) \to \mathrm{Qcoh}(S)$ is induced by $Rf_* : \mathrm{D}^+(U) \to \mathrm{D}^+(S)$ when $\mathrm{D}^+(\mathrm{Qcoh}(U))$ and $\mathrm{D}^+(\mathrm{Qcoh}(S))$ are identified with full subcategories of $\mathrm{D}^+(U)$ and $\mathrm{D}^+(S)$ by (3.5). Thus

$$R(f_*Q_X)f^*E \simeq Rf_*RQf^*E.$$

One deduces a spectral sequence

$$E_2^{pq} = R^pf_*R^qQ(f^*E) \Longrightarrow R^{p+q}(f_*Q)f^*E.$$

By the first case there is an integer $n(U)$ depending only on U such that Q is of cohomological dimension $< n(U)$ on $\mathrm{Mod}(U)$; hence $E_2^{pq} = 0$ for $q > n(U)$. Also $E_2^{pq} = 0$ for $p > N$, by the cohomological-dimension hypothesis on S. Thus

$$R^n(f_*Q)f^*E = 0 \quad (n > N + N''),$$

where N'' is an integer greater than all the $n(U)$, with U running over the intersections $U_{i_0} \cap \cdots \cap U_{i_p}$. Taking account of the preceding displayed isomorphism, one obtains

$$R^n Q(f_* f^* E) = 0 \quad (n > N + N''),$$

and hence finally

$$R^n Q(E) = 0 \quad (n > N + N' + N'').$$

This proves a).

Proof of b). The coherer, being of finite cohomological dimension by a), admits a right derived functor

$$RQ : \mathrm{D}(S) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

whose restriction to $\mathrm{D}^+(S)$ coincides with the functor RQ defined above. For $E \in \operatorname{ob} \mathrm{D}(\mathrm{Qcoh}(S))$ there is a canonical functorial isomorphism

$$E \xrightarrow{\sim} RQ(E), \tag{1}$$

since, by (3.5.1), the E^i are acyclic for Q and satisfy $E^i \xrightarrow{\sim} Q(E^i)$. On the other hand, for $E \in \operatorname{ob} \mathrm{D}(S)$ one has a canonical functorial homomorphism

$$RQ(E) \longrightarrow E, \tag{2}$$

which, in the case where the E^i are Q-acyclic, comes from the ordinary adjunction homomorphism between ι and Q. It is checked directly that these homomorphisms make

$$RQ : \mathrm{D}(S) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

a right adjoint of

$$\iota : \mathrm{D}(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}(S).$$

The fact that (1) is an isomorphism implies that ι is fully faithful. Finally, using the spectral sequence

$$R^p Q(H^q(E)) \Longrightarrow R^{p+q} Q(E),$$

which converges since Q has finite cohomological dimension, one sees that (2) is an isomorphism for $E \in \operatorname{ob} \mathrm{D}(S)_{\mathrm{qcoh}}$; hence ι has image $\mathrm{D}(S)_{\mathrm{qcoh}}$. This proves b) and completes the proof of (3.7).

Appendix I. A counterexample of Verdier

0. Introduction

If S is a scheme, write $\mathrm{Mod}(S)$, respectively $\mathrm{Qcoh}(S)$, for the category of $\mathcal{O}_S$-modules, respectively of quasi-coherent $\mathcal{O}_S$-modules. The purpose of this appendix is to draw attention to certain pathological properties of the category $\mathrm{D}(\mathrm{Qcoh}(S))$, and in particular to prove the following facts.

0.1. Let X be an affine scheme with ring A, and let I be an injective A-module. We know that, if A is noetherian, the sheaf $\widetilde{I}$ is an injective $\mathcal{O}_X$-module. If A is not noetherian, $\widetilde{I}$ is not necessarily injective, nor even flasque, even if the underlying space of X is noetherian of finite Krull dimension. Moreover, the restriction of $\widetilde{I}$ to an open U of X is not, in general, an injective object of $\mathrm{Qcoh}(U)$.

0.2. Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. The functor

$$Rf_* : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(Y)$$

does not in general induce, on $\mathrm{D}^+(\mathrm{Qcoh}(X))$ and $\mathrm{D}^+(\mathrm{Qcoh}(Y))$, the right derived functor of

$$f_* : \mathrm{Qcoh}(X) \longrightarrow \mathrm{Mod}(Y).$$

More precisely, an injective object of $\mathrm{Qcoh}(X)$ is not necessarily acyclic for $f_* : \mathrm{Mod}(X) \to \mathrm{Mod}(Y)$.

0.3. Let S be a quasi-compact and quasi-separated scheme. An object of $\mathrm{Qcoh}(S)$, even an injective object, is not necessarily acyclic for the coherer (II 3.2). The functor

$$\iota : \mathrm{D}^b(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}^b(S)$$

is not in general fully faithful.

1. The counterexamples

We first recall the following lemma (SGA 2 II 9).

Lemma 1.1. Let A be a ring, $X = \mathrm{Spec}(A)$, let $\mathbf{f} = (f_\alpha)$ be a finite system of elements of A, and let Y be the closed subset defined by $\mathbf{f}(x) = 0$. For $i > 0$, the following conditions are equivalent:

(i) for every injective A-module I, one has $H^i_Y(X, \widetilde{I}) = 0$;

(ii) the pro-object $n \mapsto H_i(\mathbf{f}^n, A)$ is zero, that is, for every n there exists $n' \geq n$ such that

$$H_i(\mathbf{f}^{n'}, A) \longrightarrow H_i(\mathbf{f}^n, A)$$

is zero.

1.2. We shall use (1.1) to prove (0.1). Let k be a field,

$$B = k[x, y], \qquad \mathbf{f} = (x, y), \qquad M = \bigoplus_{n>0} B/(\mathbf{f}^n),$$

and let A be the B-algebra $D_B(M)$ (EGA 0_{IV} 18.2.3). One has trivially

$$H^A_2(\mathbf{f}^n, A) \simeq H^B_2(\mathbf{f}^n, A).$$

On the other hand,

$$H^B_2(\mathbf{f}^n, A) \simeq H^B_2(\mathbf{f}^n, B) \oplus H^B_2(\mathbf{f}^n, M),$$

and

$$H^B_2(\mathbf{f}^n, B) = 0,$$

since $\mathbf{f}$ is a regular sequence. Thus

$$H^B_2(\mathbf{f}^n, A) \simeq H^B_2(\mathbf{f}^n, M).$$

Finally,

$$H^B_2(\mathbf{f}^n, M) \simeq H^0_B(\mathbf{f}^n, M),$$

where $H^0_B(\mathbf{f}^n, M)$ is canonically identified with the submodule $(\mathbf{f}^n, M)$ of M annihilated by $\mathbf{f}^n$. Hence

$$H^A_2(\mathbf{f}^n, A) \simeq (\mathbf{f}^n, M). \tag{*}$$

Since the transition morphism

$$H_2^A(\mathbf{f}^{n'}, A) \longrightarrow H_2^A(\mathbf{f}^n, A)$$

is induced by multiplication by $\mathbf{f}^{\,n'-n}$, one checks immediately from (*) that the pro-object $n \mapsto H_2^A(\mathbf{f}^n, A)$ is not zero. Hence, by (1.1), there is an injective A-module I such that

$$H_Y^2(X, \widetilde{I}) \neq 0,$$

and therefore $\widetilde{I}$ is not flasque. Notice also that the projection $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is a homeomorphism, so that the underlying space of X is noetherian of dimension 2. If U denotes the complement of Y in X, then by SGA 2 II 4,

$$H_Y^2(X, I) \simeq H^1(U, I) \neq 0.$$

But, since U is separated, the functor $\mathrm{D}^+(\mathrm{Qcoh}(U)) \to \mathrm{D}^+(U)$ is fully faithful by II 3.5, so $\widetilde{I}|U$ is not injective in $\mathrm{Qcoh}(U)$. This proves assertion (0.1).

Let Z be the quasi-compact and quasi-separated scheme obtained by gluing two copies of X along U; let j_1, j_2 be the two canonical open immersions of X into Z. Thus one has a Cartesian square

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[d, "i"'] & X \arrow[d, "j_1"] \\ X \arrow[r, "j_2"'] & Z. \end{tikzcd}$$

One has

$$j_2^* R^1 j_{1*}(\widetilde{I}) \simeq R^1 i_*(i^* \widetilde{I}),$$

and hence

$$H^0(X, R^1 j_{1*}(\widetilde{I})) \simeq H^1(U, \widetilde{I}) \neq 0,$$

so that $R^1 j_{1*}(\widetilde{I}) \neq 0$, which proves (0.2).

On the other hand, by II 3.4 and 3.5 one has

$$RQ\, Rj_{1*}(\widetilde{I}) \simeq j_{1*}(\widetilde{I}).$$

It follows from this a spectral sequence whose exact sequence in low degrees gives the isomorphism

$$R^1 j_{1*}(\widetilde{I}) \simeq R^2 Q(j_{1*}\widetilde{I}).$$

Since the sheaf $j_{1*}\widetilde{I}$ is injective in $\mathrm{Qcoh}(Z)$, the first part of (0.3) is proved.

Finally, let $E \in \operatorname{ob}\mathrm{Qcoh}(Z)$. We have the duality isomorphism

$$R\operatorname{Hom}_{\mathrm{Mod}}(E, Rj_{1*}\widetilde{I}) \simeq R\operatorname{Hom}_{\mathrm{Mod}}(j_1^* E, \widetilde{I}),$$

but, since X is affine, II 3.5 gives

$$R\operatorname{Hom}_{\mathrm{Mod}}(j_1^* E, \widetilde{I}) \simeq R\operatorname{Hom}_{\mathrm{Qcoh}}(j_1^* E, I) \simeq \operatorname{Hom}(j_1^* E, I).$$

Consequently

$$R\operatorname{Hom}_{\mathrm{Mod}}(E, Rj_{1*}\widetilde{I}) \simeq \operatorname{Hom}(j_1^* E, I).$$

A spectral sequence then gives, in low degree, the isomorphism

$$\operatorname{Ext}^2_{\mathrm{Mod}}(E, j_{1*}\widetilde{I}) \simeq \operatorname{Hom}(E, R^1 j_{1*}\widetilde{I}).$$

Thus the identity morphism of $E = R^1 j_{1*}\widetilde{I}$ supplies a non-zero element of

$$\operatorname{Ext}^2_{\mathrm{Mod}}(E, j_{1*}\widetilde{I}).$$

It follows that the functor

$$\mathrm{D}^b(\mathrm{Qcoh}(Z)) \longrightarrow \mathrm{D}^b(Z)$$

is not fully faithful. This completes the proof of (0.3).

Appendix II. Definition of the analytic index of a relative elliptic complex

In this appendix we propose to show that the notion of a perfect complex leads to a simple and natural definition of the "analytic index" of a continuous family of elliptic differential operators, or more generally of a relative elliptic complex. For the initiated reader, let us say that the definition given here has the advantage, compared with the classical definitions (see for example [2] and [17]), of not requiring any deformation of the elliptic complex under consideration.

Sections 1 and 2 consist of reminders on the classical notion of a "mixed structure". [16] We give, of course, only the minimum of definitions and results required in order to state and prove, in Section 3, the central theorem we have in view. It says, more precisely, that the direct image, by the projection (assumed proper) onto the base (assumed locally compact), of a relative elliptic complex is a perfect complex. When the base S is compact, the definition of the analytic index as an element of $K(S)$, the Grothendieck group of complex vector bundles of finite rank on S, then follows from the globalization sorite of II 2.3.2.

Throughout what follows, if S is a topological space, $\mathcal{O}_S$ will denote the sheaf of continuous real-valued functions on S.

1. The sorite of mixed structures

In this section a topological space S is fixed.

1.1. Definition of mixed varieties

Let V be an open subset of $\mathbb{R}^n$. For $r \in \mathbb{N}$ one writes $C^r(V)$ for the Fréchet space of real-valued functions of class C^r on V. Put

$$C^\infty(V) = \varprojlim_r C^r(V).$$

Let U be an open subset of S. For $r \in \mathbb{N} \cup \{\infty\}$, denote by ${}^r\mathcal{O}_{U\times V}$ the sheaf of rings on $U \times V$ defined by

$$\Gamma(U' \times V', {}^r\mathcal{O}_{U\times V}) = C(U', C^r(V')),$$

where U' (respectively V') is an open subset of U (respectively of V), and $C(U', C^r(V'))$ denotes the space of continuous maps from U' into $C^r(V')$. There is a canonical injection

$$\mathrm{pr}_1^{-1}(\mathcal{O}_U) \longrightarrow {}^r\mathcal{O}_{U\times V}$$

which makes $(U \times V, {}^r\mathcal{O}_{U\times V})$ into a ringed space over U.

A ringed space X over S is said to be a *mixed variety of class C^r* ($r \in \mathbb{N} \cup \{\infty\}$) if X is locally isomorphic, as a ringed space over S, to a space of the type just described. If S is reduced to one point, a mixed variety of class C^r over S is just a variety of class C^r in the usual sense.

Let $f : X \to S$ be a mixed variety of class C^r. For $s \in S$, the fiber $X_s = f^{-1}(s)$, endowed with the sheaf $i_s^{-1}(\mathcal{O}_X)/I_s$, where $i_s : X_s \to X$ is the injection and I_s is the ideal of sections which vanish on X_s, is a variety of class C^r.

[16] The informed reader is advised to skip them.

1.2. Relative differential operators

It is easy, on the model of algebraic geometry (EGA IV 16, SGA 3 VII), to develop an infinitesimal study of mixed varieties. We shall only review rapidly the notions we shall need.

1.2.1. Let $f : X \to S$ be a mixed variety of class C^∞. For each $n \in \mathbb{N}$ one defines the sheaf $P^n_{X/S}$ of principal parts of order n on X. In two different ways, corresponding to the two projections pr_1 and pr_2 of $X \times_S X$ onto X, it is an $\mathcal{O}_X$-algebra augmented toward $\mathcal{O}_X$, and is locally free of finite type as an $\mathcal{O}_X$-module. When the $\mathcal{O}_X$-module structure on $P^n_{X/S}$ is not specified, it will be understood to be the one coming from pr_1. One has the canonical identifications

$$P^0_{X/S} = \mathcal{O}_X, \qquad P^1_{X/S}/\mathcal{O}_X = \Omega^1_{X/S}.$$

The sheaf $\Omega^1_{X/S}$ is the sheaf of relative differentials of X over S, or relative cotangent sheaf. Its dual is denoted $T_{X/S}$ and called the relative tangent sheaf. The sheaf $T_{X/S}$, respectively $\Omega^1_{X/S}$, is the sheaf of sections of a vector bundle of finite rank on X, called the relative tangent, respectively cotangent, bundle; these bundles are mixed varieties over X.

The $f^{-1}(\mathcal{O}_S)$-linear homomorphism, corresponding to the $\mathcal{O}_X$-algebra structure defined by pr_2,

$$d^n_{X/S} : \mathcal{O}_X \longrightarrow P^n_{X/S},$$

which associates to a section of $\mathcal{O}_X$ its principal part of order n, is called the universal differential operator of order n on X relative to S.

1.2.2. More generally, if E is a locally free $\mathcal{O}_X$-module of finite type, one may consider the sheaf of principal parts of order n of E,

$$P^n_{X/S}(E) = P^n_{X/S} \otimes_{\mathcal{O}_X} E,$$

where the tensor product is defined by means of the second $\mathcal{O}_X$-module structure on $P^n_{X/S}$. One has a universal differential operator

$$d^n_{X/S}(E) : E \longrightarrow P^n_{X/S}(E).$$

Let E, F be locally free $\mathcal{O}_X$-modules of finite type. By definition, a differential operator from E to F of order $\leq n$ is an $f^{-1}(\mathcal{O}_S)$-linear homomorphism from E to F which factors through $d^n_{X/S}(E)$ as an $\mathcal{O}_X$-linear homomorphism $P^n_{X/S}(E) \to F$; the factorization is then unique. In other words, $d^n_{X/S}(E)$ gives a bijection

$$\mathrm{Hom}_{\mathcal{O}_X}(P^n_{X/S}(E), F) \xrightarrow{\sim} \mathrm{Diff}^n(E, F), \tag{1.2.2.1}$$

where $\mathrm{Diff}^n(E, F)$ denotes the set of differential operators from E to F of order $\leq n$.

1.2.3. We shall also have to consider differential operators with complex coefficients. Let $\mathcal{O}_X \otimes \mathbb{C}$ (respectively $\mathcal{O}_S \otimes \mathbb{C}$) be the complexification of $\mathcal{O}_X$ (respectively $\mathcal{O}_S$). Let E, F be locally free $(\mathcal{O}_X \otimes \mathbb{C})$-modules of finite type. By a differential operator from E to F of order $\leq n$ one means an $f^{-1}(\mathcal{O}_S \otimes \mathbb{C})$-linear homomorphism from E to F which factors through $d^n_{X/S}(E)$ as an $(\mathcal{O}_X \otimes \mathbb{C})$-linear homomorphism $P^n_{X/S}(E) \to F$. If $\mathrm{Diff}^n_{\mathbb{C}}(E, F)$ denotes the set of such operators, $d^n_{X/S}(E)$ therefore defines a bijection

$$\mathrm{Hom}_{\mathcal{O}_X \otimes \mathbb{C}}(P^n_{X/S}(E), F) \xrightarrow{\sim} \mathrm{Diff}^n_{\mathbb{C}}(E, F). \tag{1.2.3.1}$$

1.2.4. For $n \geq 1$ there is a canonical exact sequence

$$0 \longrightarrow S^n(\Omega^1_{X/S}) \longrightarrow P^n_{X/S} \longrightarrow P^{n-1}_{X/S} \longrightarrow 0.$$

Hence, tensoring over $\mathcal{O}_X$ with a locally free $\mathcal{O}_X$-module, respectively $(\mathcal{O}_X \otimes \mathbb{C})$-module, E of finite type, one obtains an exact sequence

$$0 \longrightarrow S^n(\Omega^1_{X/S}) \otimes E \longrightarrow P^n_{X/S}(E) \longrightarrow P^{n-1}_{X/S}(E) \longrightarrow 0. \tag{1.2.4.1}$$

Here $S^n(\Omega^1_{X/S})$ denotes the n-th symmetric power.

Let F be a locally free $\mathcal{O}_X$-module, respectively $(\mathcal{O}_X \otimes \mathbb{C})$-module, of finite type. Applying $\mathrm{Hom}(\ ,F)$, respectively $\mathrm{Hom}_{\mathcal{O}_X \otimes \mathbb{C}}(\ ,F)$, to (1.2.4.1), and using (1.2.2.1) and (1.2.3.1), one obtains an exact sequence

$$0 \longrightarrow \mathrm{Diff}^{n-1}(E,F) \longrightarrow \mathrm{Diff}^n(E,F) \xrightarrow{\sigma_n} \mathrm{Hom}_{\mathcal{O}_X}\big(S^n(\Omega^1_{X/S}) \otimes E, F\big), \tag{1.2.4.2}$$

and analogously in the complex case. If X is paracompact and $\mathcal{O}_X$ is soft, the sequence may be completed on the right by adding a zero. The homomorphism σ_n is called the *symbol*. It associates to a differential operator from E to F of order $\leq n$ a homogeneous polynomial map of degree n from $T^*_{X/S}$ into $\mathrm{Hom}(E,F)$, or equivalently a homomorphism $p^*E \to p^*F$, where p is the projection of the relative cotangent bundle onto X, satisfying the usual homogeneity condition.

Let E, F, G be locally free $\mathcal{O}_X$-modules of finite type, $u \in \mathrm{Diff}^n(E,F)$ and $v \in \mathrm{Diff}^m(F,G)$. Then $vu \in \mathrm{Diff}^{m+n}(E,G)$ and one checks at once the relation

$$\sigma_{m+n}(vu) = \sigma_m(v)\sigma_n(u). \tag{1.2.4.3}$$

The same relation holds for differential operators with complex coefficients. In particular, if the composite of two differential operators is zero, then the composite of their symbols is zero.

1.2.5. Let $E^\bullet$ be a bounded complex of differential operators with complex coefficients on X relative to S, and let $d^i : E^i \to E^{i+1}$ be of order n_i. By (1.2.4.3),

$$\sigma_{n_{i+1}}(d^{i+1})\sigma_{n_i}(d^i) = 0.$$

The complex $E^\bullet$ is called an *elliptic complex of order* (n_i) if the symbol complex on the relative cotangent bundle, pulled back to that bundle, is acyclic outside the zero section.

When S is reduced to one point, this notion of elliptic complex on X coincides with the usual one, studied in detail in [4] and [16].

1.2.6. Let $s \in S$, and let i_s be the inclusion of the fiber X_s in X, considered as a monomorphism of ringed spaces (1.1). For $n \in \mathbb{N}$ one has a canonical isomorphism

$$i_s^* P^n_{X/S} \simeq P^n_{X_s},$$

and hence a restriction homomorphism

$$i_s^* : \mathrm{Diff}^n(E,F) \longrightarrow \mathrm{Diff}^n(i_s^*E, i_s^*F),$$

for E, F locally free $\mathcal{O}_X$-modules of finite type, and similarly for operators with complex coefficients. If $E^\bullet$ is a bounded complex of differential operators with complex coefficients on X, then $E^\bullet$ is elliptic if and only if, for every $s \in S$, $i_s^*(E^\bullet)$ is an elliptic complex. The verification is left to the reader.

2. Rigidity lemmas

2.0. It is “well known” that compact differentiable varieties are rigid, that is, they admit no continuous deformation other than the trivial one. Since this result will be used essentially in

Section 3, and references seem rather scarce, it is perhaps useful to give precise statements and a few proofs.

A topological space S is fixed. The mixed varieties under consideration are assumed to be of class C^r, with $r \gg 1$; possibly $r = \infty$.

2.1. Notation. Let X, Y be mixed S-varieties. We denote by $\mathrm{Hom}(X, Y)$ the set of S-morphisms from X to Y, and by $\mathcal{H}om(X, Y)$ the sheaf on X of germs of S-morphisms from X to Y, defined by

$$\mathcal{H}om(X, Y)(U) = \mathrm{Hom}(U, Y)$$

for U open in X.

Let $s \in S$, and write i_s for the inclusion of X_s in X. There is a canonical induction homomorphism

$$i_s^{-1}\mathcal{H}om(X, Y) \longrightarrow \mathcal{H}om(X_s, Y_s) \tag{2.1.1}$$

which is an epimorphism, as is immediate. In the case $Y = S \times \mathbb{R}$, the sheaf $\mathcal{H}om(X, Y)$ is canonically identified with $\mathcal{O}_X$, and the epimorphism (2.1.1) with the epimorphism

$$i_s^{-1}(\mathcal{O}_X) \longrightarrow i_s^*(\mathcal{O}_X)$$

considered in (1.1).

Proposition 2.2. Let X, Y be mixed S-varieties, and let $s \in S$. Suppose that the fiber X_s is paracompact and that s has a fundamental system of paracompact neighborhoods. Let A be a closed subset of X_s, and let

$$f \in H^0(A, \mathcal{H}om(X_s, Y_s)).$$

The subsheaf $\mathcal{F}$ of $i_s^{-1}\mathcal{H}om(X, Y)$ consisting of the germs which extend f, that is, whose image by (2.1.1) coincides with f over A, is soft.

Proof. Let X' be an open neighborhood in X which is isomorphic, as a mixed variety, to $S' \times U$, where S' is a paracompact neighborhood of s and U is a relatively compact open subset of $\mathbb{R}^n$, and such that the image by f of $X'_s \cap A$ is contained in an open subset of Y isomorphic to $S' \times V$, where V is a convex open subset of $\mathbb{R}^q$ containing the origin. By T.F. II 3.4.1 it is enough to show that every section of $\mathcal{F}$ over a closed subset of X_s contained in X'_s extends to X'_s. Since $X'_s = \{s\} \times U$ is relatively compact, a closed subset of X_s contained in X'_s is compact. Identifying X' with $S' \times U$ and Y' with $S' \times V$, putting $A' = A \cap X'_s$, and using T.F. II 3.3.1, the assertion is reduced to the following elementary lemma.[17]

Lemma 2.2.1. Let U be an open subset of $\mathbb{R}^n$, and let V be a convex open subset of $\mathbb{R}^q$ containing 0. Let A_1, A_2 be closed subsets of $\{s\} \times U$. Let U_1, respectively U_2, be an open subset of $S \times U$, respectively of $\{s\} \times U$, containing A_1, respectively A_2. Let $f_1 : U_1 \to V$ be a morphism and let $f_2 : U_2 \to S \times V$ be an S-morphism which, on $U_1 \cap (\{s\} \times U)$, defines the same germ as f_1 near A_1. Then there exists an open subset U' of $S \times U$ containing $\{s\} \times U$ and an S-morphism

$$F : U' \longrightarrow S \times V$$

such that the germ of F along $\{s\} \times U$ extends the germ defined by f_1, and the morphism induced by F on $\{s\} \times U$ defines the same germ as f_2 near A_2.

Proof. Replacing f_1 by af_1, where a is a C^∞ function on U equal to 1 near A_1 and with support in U_1, one may suppose $U_1 = U$. Let b be a C^∞ function on U, equal to 1 on A_2 and with support in $U_2 \cap (\{s\} \times U)$. Then

$$U' = U_1 \cup (S \times (U - \mathrm{Supp}\, b))$$

[17] Communicated to the writer by A. Douady. A closed subset of a paracompact space has a fundamental system of paracompact neighborhoods.

is an open subset of $S \times U$ containing $\{s\} \times U$. The morphism $F : U' \to S \times V$ defined by

$$F(t, x) = (t,\, b(x) \operatorname{pr}_2 f_2(x) + (1 - b(x)) f_1(t, x))$$

visibly satisfies the required conditions.

Corollary 2.3. Under the hypotheses of (2.2), the sheaves $i_s^{-1}\mathcal{H}om(X, Y)$ and $\mathcal{H}om(X_s, Y_s)$ are soft. In particular, the sheaves $i_s^{-1}(\mathcal{O}_X)$ and $\mathcal{O}_{X_s}$ are soft. For every closed subset A of X_s, the map

$$H^0(X_s, i_s^{-1}\mathcal{H}om(X, Y)) \longrightarrow H^0(A, \mathcal{H}om(X_s, Y_s))$$

deduced from (2.1.1) is surjective.

Proof. The softness of $i_s^{-1}\mathcal{H}om(X, Y)$ follows from (2.2) applied with $A = \varnothing$. Taking $S = \{s\}$, one obtains the softness of $\mathcal{H}om(X_s, Y_s)$. The last assertion follows from (2.2).[18]

Corollary 2.4. Under the hypotheses of (2.2), let E be an $\mathcal{O}_X$-module. For every closed subset A of X_s, the canonical homomorphism

$$H^0(X_s, i_s^{-1}E) \longrightarrow H^0(A, i_s^* E)$$

deduced from the epimorphism

$$i_s^{-1}E \longrightarrow i_s^* E = i_s^{-1}E \otimes_{i_s^{-1}\mathcal{O}_X} \mathcal{O}_{X_s}$$

is an epimorphism.

Proof. Since $i_s^{-1}\mathcal{O}_X$ is soft by (2.3), the functor $H^0(A,\ )$ is exact on the category of $i_s^{-1}\mathcal{O}_X$-modules (T.F. II 3.7.1, 3.4.2, 4.4.3). Hence

$$H^0(A, i_s^{-1}E) \longrightarrow H^0(A, i_s^* E)$$

is an epimorphism. Since $i_s^{-1}E$ is soft, the homomorphism

$$H^0(X_s, i_s^{-1}E) \longrightarrow H^0(A, i_s^{-1}E)$$

is an epimorphism, and this proves the assertion.

Corollary 2.5. Let $p : X \to S$ be a mixed variety and let $s \in S$. Suppose that s has a fundamental system of paracompact neighborhoods S' such that the $p^{-1}(S')$ form a fundamental system of paracompact neighborhoods of X_s. This condition is satisfied, for example, if S is locally compact and p is a proper map of topological spaces. Then, for every $\mathcal{O}_X$-module E,

$$(R^i p_* E)_s = 0 \qquad (i > 0).$$

Proof. By T.F. II 4.17.1,

$$(R^i p_* E)_s = H^i(X_s, i_s^{-1}E).$$

But $i_s^{-1}\mathcal{O}_X$ is soft by (2.3), hence $H^i(X_s, i_s^{-1}E) = 0$ for $i > 0$.

Corollary 2.6. Let X, Y be mixed S-varieties, and let $s \in S$. Suppose that X satisfies the hypotheses of (2.5). For every morphism $f : X_s \to Y_s$, there exists an open neighborhood S' of s and a morphism

$$F : X|S' \longrightarrow Y|S'$$

such that $F_s = f$.

[18]For $\mathcal{O}_{X_s}$ this is very well known. The writer does not know whether $\mathcal{O}_X$ is soft when the hypotheses of (2.2) are satisfied for every $s \in S$ and X is paracompact. This would trivially imply, under those hypotheses, the softness of $i_s^{-1}(\mathcal{O}_X)$ as well as (2.5) below.

Proof. This is an immediate consequence of (2.3), with $A = X_s$, and of the fact that the hypotheses on X give

$$H^0(X_s, i_s^{-1}\mathcal{H}om(X,Y)) = \lim_{S' \ni s} \mathrm{Hom}(X|S', Y|S')$$

for open S' containing s (T.F. II 8.5.5).

Proposition 2.7. Let X, Y be mixed S-varieties, let $f : X \to Y$ be a morphism, let $s \in S$, and let $x \in X_s$. If the tangent linear map of

$$f_s : X_s \longrightarrow Y_s$$

is invertible at x, then f is a local isomorphism at x; that is, there is an open neighborhood U of x in X and an open subset V of Y such that f induces an isomorphism $U \simeq V$.

Proof. This is an easy consequence of the implicit function theorem. The details are left to the reader.

Corollary 2.8. Let X, Y be mixed S-varieties and let $s \in S$. Suppose that s has a fundamental system of neighborhoods S' such that the $X|S'$, respectively $Y|S'$, form a fundamental system of neighborhoods of X_s, respectively Y_s; this is the case, for example, if S is locally compact and X, respectively Y, is proper over S. Let $f : X \to Y$ be an S-morphism such that

$$f_s : X_s \longrightarrow Y_s$$

is an isomorphism. Then there exists a neighborhood S' of s such that

$$f|X|S' : X|S' \longrightarrow Y|S'$$

is an isomorphism.

Proof. By (2.7), there exists an open neighborhood U of X_s in X such that $f|U$ is étale, that is, a local isomorphism. We may suppose that $U = X|S'$ for a neighborhood S' of s; after shrinking S, we may therefore assume that f is étale. Then the fiber product $Z = X \times_Y X$ is naturally a mixed S-variety, étale over X for each projection, and the diagonal morphism $\Delta : X \to Z$ is an open immersion. Moreover the sets $Z|S'$ form a fundamental system of neighborhoods of Z_s. Since

$$\Delta_s : X_s \longrightarrow Z_s$$

is an isomorphism, it follows, after shrinking S, that Δ is an isomorphism. Thus f is an open immersion, and after shrinking S again it is an isomorphism.

Corollary 2.9. Let X be a mixed S-variety and let $s \in S$. Suppose that the hypothesis of (2.5) is satisfied. Then there exists a neighborhood S' of s in S such that $X|S'$ is S'-isomorphic to $S' \times X_s$.

Proof. Apply (2.6) and (2.8).

Corollary 2.10. Let $s \in S$ and let V be a paracompact variety. Put $X = S \times V$, and let i_s be the injection of V into X defined by s. Suppose that the hypothesis of (2.5) is satisfied. Let E be a locally free $\mathcal{O}_X$-module, respectively $\mathcal{O}_X \otimes \mathbb{C}$-module, of finite type. Then there exists a neighborhood S' of s in S such that $E|X|S'$ is isomorphic to $\mathrm{pr}_2^*(i_s^*E)$, where $\mathrm{pr}_2 : S' \times V \to V$ is the second projection.

Proof. Put $F = \mathrm{pr}_2^*(i_s^*E)$. By (2.4), applied to $\mathcal{H}om(E,F)$, the induction homomorphism

$$\mathrm{Hom}(i_s^{-1}E, i_s^{-1}F) \longrightarrow \mathrm{Hom}(i_s^*E, i_s^*F)$$

is an epimorphism. On the other hand, by T.F. II 3.3.1,

$$\mathrm{Hom}(i_s^{-1}E, i_s^{-1}F) = \varinjlim_{S' \ni s} \mathrm{Hom}(E|X|S', F|X|S').$$

Thus the canonical isomorphism $i_s^* E \simeq i_s^* F$ is induced, after shrinking S, by a homomorphism $f : E \to F$. Since f_s is an isomorphism, f is an isomorphism near X_s by Nakayama, hence over $X|S'$ for a suitable neighborhood S' of s.

3. The finiteness theorem

3.0. Conventions and notation

3.0.1. Let X and Y be topological spaces. We write $C(X, Y)$ for the set of continuous maps from X to Y, and $\mathcal{C}(X, Y)$ for the sheaf on X defined by $U \mapsto C(U, Y)$.

3.0.2. Let E, F be Banach spaces, real or complex. We write $L(E, F)$ for the Banach space of continuous linear maps from E to F.

3.0.3. The mixed varieties occurring in this section are assumed to be of class C^∞. If X is a mixed S-variety, we shall often identify a locally free $\mathcal{O}_X$-module, respectively $\mathcal{O}_X \otimes \mathbb{C}$-module, of finite type with the corresponding real, respectively complex, vector bundle.

3.0.4. Let X be a compact ordinary variety, and let E be a real or complex vector bundle of finite rank on X. Write $C^\infty(X, E)$ for the Fréchet space of C^∞ sections of E on X. For $r \in \mathbb{Z}$, write $C_r(X, E)$ for the Hilbert space of sections of E which are r times differentiable in L^2 ([16] IX). For $s \leq r$, $C_r(X, E)$ maps to $C_s(X, E)$ by a continuous linear injective map with dense image, and

$$C^\infty(X, E) = \varprojlim_r C_r(X, E).$$

3.0.5. Let S be a topological space. We shall need the notion of a Hilbert bundle over S, for which we refer the reader to [12] or [14]. There is an additive functor from the category of real, respectively complex, Hilbert bundles over S to the category of $\mathcal{O}_S$-, respectively $\mathcal{O}_S \otimes \mathbb{C}$-, modules: it sends a Hilbert bundle E to the sheaf of continuous sections of E.

3.1. Auxiliary technical results

Lemma 3.1.1. Let X, Y be topological spaces, with X locally compact. Let $\mathcal{C}'(X, Y)$ be the presheaf, on the basis of relatively compact open subsets of X, defined by

$$\mathcal{C}'(X, Y)(U) = C(\overline{U}, Y).$$

The canonical restriction morphism

$$\mathcal{C}'(X, Y) \longrightarrow \mathcal{C}(X, Y)$$

induces an isomorphism on the associated sheaves.

Proof. Immediate.

Lemma 3.1.2. Let E, F be locally convex topological vector spaces over $\mathbb{C}$, and let $u : E \to F$ be a continuous linear map, respectively a continuous linear map with dense image. Let X be compact. Endow $C(X, E)$ and $C(X, F)$ with the topology of uniform convergence. Then the induced linear map

$$u_X : C(X, E) \longrightarrow C(X, F)$$

is continuous, respectively continuous with dense image.

Proof. It is clear that u_X is continuous. On the other hand, by Bourbaki, *Intégration*, chapter II, §2, lemmas 1 and 2, the subspace $C(X,\mathbb{C})\otimes E$ of $C(X,E)$, formed by the continuous maps from X to E whose values lie in a finite-dimensional vector subspace, is dense; and similarly $C(X,\mathbb{C})\otimes F$ is dense in $C(X,F)$. If u has dense image, then $u_X(C(X,E))$ is dense in $C(X,\mathbb{C})\otimes F$, and taking closures gives the assertion.

3.1.3. Let V be a compact variety, let E and F be complex vector bundles of finite rank on V, and let $n\in\mathbb{N}$. The space $\mathrm{Diff}^n_{\mathbb{C}}(E,F)$, identified by (1.2.3.1) with

$$\mathrm{Hom}(P^n(E),F)\simeq C^\infty(V,\mathrm{Hom}(P^n(E),F)),$$

is naturally a Fréchet space. Let $u\in\mathrm{Diff}^n_{\mathbb{C}}(E,F)$. One checks easily that the linear map $C^\infty(V,E)\to C^\infty(V,F)$ defined by u is continuous. If $L(C^\infty(V,E),C^\infty(V,F))$ denotes the space of continuous linear maps, one therefore has a linear map

$$\delta:\mathrm{Diff}^n_{\mathbb{C}}(E,F)\longrightarrow L(C^\infty(V,E),C^\infty(V,F)).$$

Moreover, for $r\in\mathbb{Z}$, u acts by extension of scalars on the C_r sections of E and defines a linear map

$$\delta_r:\mathrm{Diff}^n_{\mathbb{C}}(E,F)\longrightarrow L(C_r(V,E),C_{r-n}(V,F)).$$

The maps δ_r are continuous, and for $u\in\mathrm{Diff}^n_{\mathbb{C}}(E,F)$, $\delta(u)$ is the inverse-limit element of the projective system of the $\delta_r(u)$. These assertions follow easily from [16] IX: one reduces immediately to $n=0$; the statement on the C_r follows directly from the definitions, while the statement for C^∞ follows, after localizing by a partition of unity, from the formula for successive derivatives of a product. The analogous real statements are also valid.

Lemma 3.1.4. Let S be a topological space, let V be a compact variety, let E,F be complex vector bundles of finite rank on V, and let $n\in\mathbb{N}$. Put $X=S\times V$. For

$$u\in\mathrm{Diff}^n_{\mathbb{C}}(\mathrm{pr}_2^*E,\mathrm{pr}_2^*F),$$

the map $s\mapsto i_s^*u$ from S into $\mathrm{Diff}^n_{\mathbb{C}}(E,F)$ is continuous. The linear map

$$\mathrm{Diff}^n_{\mathbb{C}}(\mathrm{pr}_2^*E,\mathrm{pr}_2^*F)\longrightarrow C(S,\mathrm{Diff}^n_{\mathbb{C}}(E,F))$$

so defined is an isomorphism.

Proof. By (1.2.3.1) one may restrict to the case $n=0$. Putting $G=\mathrm{Hom}(E,F)$, one is reduced to showing that for $u\in\Gamma(X,\mathrm{pr}_2^*G)$, the map $s\mapsto u_s=i_s^*u$ from S into $C^\infty(V,G)$ is continuous and that the resulting map

$$\Gamma(X,\mathrm{pr}_2^*G)\longrightarrow C(S,C^\infty(V,G))$$

is an isomorphism. This follows immediately from the definition of mixed varieties.

3.1.5. Under the hypotheses of (3.1.4), let

$$u\in\mathrm{Diff}^n_{\mathbb{C}}(\mathrm{pr}_2^*E,\mathrm{pr}_2^*F).$$

Identifying u, by (3.1.4), with a continuous map from S into $\mathrm{Diff}^n(E,F)$, one sees that, for U open in S, the morphism

$$\mathrm{pr}_{1*}(u)(U):C^\infty(U,E)\longrightarrow C^\infty(U,F)$$

is canonically identified with the linear map

$$C(U,C^\infty(V,E))\longrightarrow C(U,C^\infty(V,F))$$

deduced from
$$\delta u : U \longrightarrow L(C^\infty(V,E), C^\infty(V,F)).$$
Suppose S locally compact. By (3.1.1), the morphism of sheaves
$$\operatorname{pr}_{1*}(u) : \operatorname{pr}_{1*}\operatorname{pr}_2^* E \longrightarrow \operatorname{pr}_{1*}\operatorname{pr}_2^* F$$
is the morphism associated to the morphism of presheaves
$$\mathcal{C}'(S, C^\infty(V,E)) \longrightarrow \mathcal{C}'(S, C^\infty(V,F))$$
defined by δu. On the other hand, for $r \in \mathbb{Z}$,
$$\delta_r u : S \longrightarrow L(C_r(V,E), C_{r-n}(V,F))$$
defines a morphism of presheaves
$$u_r : \mathcal{C}'(S, C_r(V,E)) \longrightarrow \mathcal{C}'(S, C_{r-n}(V,F)).$$
By (3.1.3), the u_r form a projective system whose limit is the previous morphism. It will be convenient to interpret $\delta_r u$ as a morphism of trivial Hilbert bundles
$$S \times C_r(V,E) \longrightarrow S \times C_{r-n}(V,F),$$
inducing on sheaves of sections the sheaf morphism associated to u_r.

3.2. Definition of the index

Theorem 3.2.1. Let S be a locally compact space, let $p : X \to S$ be a proper mixed variety, and let $E^\bullet$ be an elliptic complex of order (n_i) of complex vector bundles on X (1.2.5). Then the complex $Rp_*(E^\bullet)$, which by (2.5) is isomorphic to $p_*(E^\bullet)$ in the derived category of $\mathcal{O}_S \otimes \mathbb{C}$-modules, is perfect.

Proof. The question being local on S, one may suppose by (2.9) that $X = S \times V$, where V is a compact variety, and that $p = \operatorname{pr}_1$. Moreover, by (2.10), one may suppose that for all i,
$$E^i = \operatorname{pr}_2^*(F^i),$$
where F^i is a complex vector bundle of finite rank on V. By (3.1.5), $p_*(E^\bullet)$ is associated to the complex of presheaves
$$\mathcal{C}'(S, C^\infty(V, F^\bullet)),$$
where the differentials are induced by the given differential operators. Put, for each $i \in \mathbb{Z}$,
$$m_i = \sum_{j<i} n_j,$$
with the convention $n_i = 0$ if $E^i = 0$. For each $r \in \mathbb{Z}$ one has, again by (3.1.5), a complex of presheaves
$$\mathcal{C}'(S, C_{r-m_i}(V, F^i)),$$
which we denote by $\mathcal{C}'(S, C_r(V, F^\bullet))$. These complexes form a projective system, and
$$\mathcal{C}'(S, C^\infty(V, F^\bullet)) = \lim_r \mathcal{C}'(S, C_r(V, F^\bullet)). \tag{3.2.1.1}$$

Finally, denote by $F_r^\bullet$ the complex of trivial Hilbert bundles over S defined by the corresponding $\delta_r(d^i)$, and by $\mathcal{F}_r^\bullet$ the complex of sheaves of sections; by (3.1.5), it is just the complex of sheaves associated to $\mathcal{C}'(S, C_r(V, F^\bullet))$. To prove the theorem it is enough to prove the following assertions:

(a) For $r \in \mathbb{Z}$, the canonical morphism

$$\mathcal{C}'(S, C^\infty(V, F^\bullet)) \longrightarrow \mathcal{C}'(S, C_r(V, F^\bullet))$$

is a quasi-isomorphism of complexes of presheaves, that is, it induces isomorphisms on the presheaves of cohomology.

(b) For $r \in \mathbb{Z}$, the complex $\mathcal{F}_r^\bullet$ is perfect.

Proof of (b). The ellipticity of $E^\bullet$ implies ([16] XI, theorem 6) that, for every $s \in S$, the differential of $(F_r^\bullet)_s$ has closed image and the spaces $H^i((F_r^\bullet)_s)$ are finite-dimensional vector spaces. In the terminology of [11], $F_r^\bullet$ is a direct quasi-acyclic complex of Hilbert bundles. Hence, by [12] 2.3, $F_r^\bullet$ is locally homotopically equivalent to a bounded complex of vector bundles of finite rank. Since the sheaf-of-sections functor is additive (3.0.5), $\mathcal{F}_r^\bullet$ is locally homotopically equivalent to a bounded complex of locally free sheaves of finite type; hence it is perfect.

For the proof of (a) we shall need the following intermediate assertion:

(a') For $p \gg q$, the canonical morphism

$$F_p^\bullet \longrightarrow F_q^\bullet$$

is a quasi-isomorphism.

Proof of (a'). Let $M_{pq}^\bullet$ be the complex of Hilbert bundles which is the cone of the morphism $F_p^\bullet \to F_q^\bullet$. The ellipticity of $E^\bullet$ implies that, for every $s \in S$, the morphism $(F_p^\bullet)_s \to (F_q^\bullet)_s$ is a quasi-isomorphism ([16] XI, corollary 2 and theorem 6). In other words, $(M_{pq}^\bullet)_s$ is a direct acyclic complex. By [11] 2.3, $M_{pq}^\bullet$ is locally homotopically trivial; hence its sheaf complex, the cone of $\mathcal{F}_p^\bullet \to \mathcal{F}_q^\bullet$, is acyclic.

Proof of (a). It is enough to show that, for every relatively compact open subset U of S, the morphism

$$C^\infty(U, C^\infty(V, F^\bullet)) \longrightarrow C^\infty(U, C_r(V, F^\bullet))$$

is a quasi-isomorphism. Replacing S by $\overline{U}$ and $E^\bullet$ by the induced complex over $\overline{U}$, one is reduced to proving that

$$C(S, C^\infty(V, F^\bullet)) \longrightarrow C(S, C_r(V, F^\bullet))$$

is a quasi-isomorphism, with S compact. By (3.2.1.1),

$$C(S, C^\infty(V, F^\bullet)) = \lim_i C(S, C_i(V, F^\bullet)).$$

For each i, $C(S, C_i(V, F^j))$ is a Banach space for the topology of uniform convergence, and the transition morphisms

$$C(S, C_i(V, F^j)) \longrightarrow C(S, C_{i-1}(V, F^j))$$

are continuous linear maps with dense image by (3.0.4) and (3.1.2). On the other hand, applying to the quasi-isomorphisms (a') the functor "sections", which is exact since $\mathcal{O}_S$ is soft, one sees that the transition morphisms are quasi-isomorphisms. Assertion (a) follows from EGA 0_{I} 13.2.3, topological variant. This completes the proof of (3.2.1).

3.2.2. If S is a topological space, we denote by $\mathrm{Parf}(S)$ the triangulated category of perfect complexes of $(\mathcal{O}_S \otimes \mathbb{C})$-modules, and by $K^0(S)$ the corresponding Grothendieck group.[19]

[19]See I 6.3. For a general review of K-functors, see Exposé IV.

Let $S, X, E^\bullet$ be as in (3.2.1). The *analytic index* of $E^\bullet$ is the element

$$\mathrm{ind}_{\mathrm{an}}(E^\bullet) = [Rp_*(E^\bullet)] \in K^0(S).$$

If S is compact, $\mathrm{ind}_{\mathrm{an}}(E^\bullet)$ canonically defines an element in the usual group $K(S)$ of topologists. Indeed, in general, for every topological space S, let $\mathrm{Loclib}(S)$ be the additive category of locally free $(\mathcal{O}_S \otimes \mathbb{C})$-modules of finite type, and let $K(S)$ be the Grothendieck group of the triangulated category

$$K^b(\mathrm{Loclib}(S))/K^b_{\mathrm{ac}}(\mathrm{Loclib}(S)).$$

This category is in fact identified with $K^b(\mathrm{Loclib}(S))$ when S is paracompact (II 2.2.2(a)), so in this case, by I 6.4, $K(S)$ is canonically isomorphic to the Grothendieck group of the additive category $\mathrm{Loclib}(S)$. There is a natural homomorphism

$$K(S) \longrightarrow K^0(S).$$

By II 2.3.2 this is an isomorphism for compact S, proving the assertion. More generally, $\mathrm{ind}_{\mathrm{an}}(E^\bullet)$ defines an element in the completed group

$$\widehat{K}(S) = \lim_{A} K(A),$$

the projective limit being taken over the compact subsets A of S.

3.2.3. The writer does not know whether the finiteness theorem (3.2.1) remains valid if, instead of assuming S locally compact, one merely assumes that each point of S has a fundamental system of paracompact neighborhoods. Moreover, the local compactness hypothesis on S does not seem very natural. The space of "universal local parameters" for elliptic complexes of fixed order is in fact a subspace of infinite dimension in a Fréchet space. More precisely, let V be a compact variety, let $F = (F^i)$ be a sequence of complex vector bundles of finite rank on V, zero except for finitely many indices, and let $n = (n_i)$ be a sequence of integers ≥ 0. Let $\mathrm{Comp.\,diff}_n(F)$ be the space of complexes of differential operators of base F and order n, that is, the closed subspace of

$$\prod_i \mathrm{Diff}^{n_i}(F^i, F^{i+1})$$

defined by the equations $d^{i+1}d^i = 0$. This product is a Fréchet space. The subset $\mathrm{Ell}_n(F)$ formed by the elliptic complexes of order n is open. It is this space, in general not locally compact, which plays the role of a space of universal local parameters.

3.2.4. Nevertheless, even if the generalization of (3.2.1) just mentioned is impossible, one can always associate to $E^\bullet$, under the sole hypothesis that the fibration p is proper and locally trivial, an element of $K^0(S)$ deserving the name analytic index of $E^\bullet$. Indeed, suppose, for simplicity, that we are in the local situation where $X = S \times V$ with V compact, and $E^\bullet$ is given by a continuous map from S into $\mathrm{Ell}_n(F)$ (3.2.3). Assertions (a') and (b) in the proof of (3.2.1) are in fact valid without any hypothesis on S; it is therefore reasonable to put, with the notation of that proof,

$$\mathrm{ind}_{\mathrm{an}}(E^\bullet) = [\mathcal{F}^\bullet_r] \in K^0(S),$$

this element being independent of r.

3.2.5. Suppose S paracompact. Let H be a complex Hilbert space of infinite Hilbert dimension, and let $I(H)$ be the subspace of $L(H, H)$ formed by the Fredholm operators. The complexes $F^\bullet_r$ define [12] a canonical element in the group $[S, I(H)]$ of homotopy classes of continuous maps from S to $I(H)$. But the writer does not know, for non-compact S (even when S is locally compact), what the relations are between $[S, I(H)]$ and $K^0(S)$: is there a natural arrow from one to the other? are they isomorphic? A fortiori, the relation between the two indices thus defined remains unclear.

For a generalization, under the hypotheses of (3.2.1), of the Atiyah–Singer index theorem, we refer to the recent work of Shih [17], or to work in preparation of M. F. Atiyah and I. M. Singer.[20]

Bibliography

[1] Artin, M., Grothendieck, A., and Verdier, J.-L., *Cohomologie étale des schémas*, to appear with North-Holland Publishing Company (cited as SGA 4).

[2] Atiyah, M., and Segal, G., secret papers.

[3] Cartan, H., *Funzioni e varietà complesse. Faisceaux analytiques cohérents*, course at the Centro Internazionale Matematico Estivo, 1964, Rome.

[4] Cartan, H., and Schwartz, L., *Théorème d'Atiyah–Singer*, Seminar 1963–64, Institut Henri Poincaré, Paris.

[5] Frisch, J., Fonctions analytiques sur un ensemble semi-analytique, note in C.R.A.S. Paris, t. 260, pp. 2974–2976 (1965).

[6] Godement, R., *Théorie des faisceaux*, Hermann (1958), cited as T.F.

[7] Grothendieck, A., *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux*, Séminaire de Géométrie Algébrique de l'I.H.E.S. 1962, to appear with North-Holland Publishing Company, cited as SGA 2.

[8] Grothendieck, A., *Fondements de la géométrie analytique*, Séminaire Cartan 1959–60, Institut Henri Poincaré, Paris.

[9] Hartshorne, R., *Residues and Duality*, Lecture Notes in Mathematics 20, Springer, cited as [H].

[10] Houzel, C., *Géométrie analytique locale*, Séminaire Cartan 1959–60, Institut Henri Poincaré, Paris.

[11] Illusie, L., *Complexes de fibrés et définitions du foncteur K*, Séminaire Shih Weishu, I.H.E.S. 1964–65.

[12] Illusie, L., Complexes quasi-acycliques directs de fibrés banachiques, note in C.R.A.S. Paris, t. 260, pp. 6499–6502.

[13] Kiehl, R., Theorem A und Theorem B in der nichtarchimedischen Funktionentheorie, *Inventiones Mathematicae* 2, pp. 256–273 (1967).

[14] Lang, S., *Introduction to differentiable manifolds.*

[15] Łojasiewicz, S., *Annali della Scuola Normale Superiore di Pisa*, Serie III, 18, fasc. IV, 1964.

[16] Palais, R., lectures at the Princeton Seminar 1963–64 on the index theorem.

[17] Shih, W., *Fiber cobordism and the index of a family of elliptic differential operators.*

[18] Verdier, J.-L., *Les catégories dérivées des catégories abéliennes*, to appear with North-Holland Publishing Company.

[20] For the general case, use the notion of "weak Hilbert bundle", or continuous field of Hilbert spaces, of [12].

Exposé III. Relative finiteness conditions

by L. Illusie

1. Relative pseudo-coherence

Theorem 1.1. Let $n \in \mathbb{Z}$. For each locally finite type morphism of schemes

$$f : X \longrightarrow Y$$

there exists one and only one full subcategory $D(f)_{n\text{-}\mathrm{coh}}$ of $D(X)$ such that the following conditions are satisfied:

(i) If $E \in \operatorname{ob} D(f)_{n\text{-}\mathrm{coh}}$, then $i^*(E) \in \operatorname{ob} D(fi)_{n\text{-}\mathrm{coh}}$ for every open immersion $i : U \to X$.

(ii) If $E \in \operatorname{ob} D(X)$, and if there exists an open covering $(i_\alpha : U_\alpha \to X)_{\alpha \in A}$ such that $i_\alpha^*(E) \in \operatorname{ob} D(fi_\alpha)_{n\text{-}\mathrm{coh}}$ for every $\alpha \in A$, then $E \in \operatorname{ob} D(f)_{n\text{-}\mathrm{coh}}$.

(iii) If

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "h"'] & Y \arrow[d, "g"] \\ & Z \end{tikzcd}$$

is a commutative triangle of locally finite type morphisms, where f is a closed immersion and g is smooth, then one has

$$E \in \operatorname{ob} D(h)_{n\text{-}\mathrm{coh}} \iff f_*(E) \in D(Y)_{n\text{-}\mathrm{coh}}.$$

Here $D(X)$ denotes the derived category of $\mathcal{O}_X$-modules, and $D(Y)_{n\text{-}\mathrm{coh}}$ denotes the full subcategory of $D(Y)$ formed by the n-pseudo-coherent objects.

Proof. Uniqueness is evident. Let indeed $f : X \to Y$ be a locally finite type morphism. Choose an open covering $(i_\alpha : X_\alpha \to X)_{\alpha \in A}$ such that, for every α, fi_α factors as

$$X_\alpha \xrightarrow{i'_\alpha} Y_\alpha \xrightarrow{f'_\alpha} Y,$$

where i'_α is a closed immersion and f'_α is smooth. It follows at once from (i), (ii), (iii) that the objects of $D(f)_{n\text{-}\mathrm{coh}}$ are precisely the objects $E \in D(X)$ such that

$$i'_{\alpha *} i_\alpha^*(E) \in \operatorname{ob} D(Y_\alpha)_{n\text{-}\mathrm{coh}};$$

therefore $D(f)_{n\text{-}\mathrm{coh}}$ is well determined, since it is a full subcategory of $D(X)$. To prove existence we shall need several lemmas.

Lemma 1.1.1. Let $i : X \to Y$ be a closed immersion of schemes such that $i_*(\mathcal{O}_X)$ is pseudo-coherent, and let $E \in \operatorname{ob} D(X)$. In order that E be n-pseudo-coherent, it is necessary and sufficient that $i_*(E)$ be so.

Proof. The necessity follows from the general stability criterion for pseudo-coherence under application of a functor. Conversely, suppose that $i_*(E)$ is n-pseudo-coherent, and prove that the same is true of E. Since $i_*(E)$ has locally bounded-above cohomology, the same is true of E. The question being local, one may suppose that E is m-pseudo-coherent for m sufficiently large. We prove by descending induction on m, with E assumed m-pseudo-coherent for some

$m \geq n$. Suppose already proved that E is $(n+1)$-pseudo-coherent. After localization, there is a distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1]$$

where E' is perfect and $H^i(E'') = 0$ for $i > n$. By Exposé I, Proposition 2.6, it remains to show that E'' is n-pseudo-coherent. Applying i_* to the preceding triangle, we get a distinguished triangle

$$i_*(E') \longrightarrow i_*(E) \longrightarrow i_*(E'') \longrightarrow i_*(E')[1]$$

and, by the first part of the proof, $i_*(E')$ is pseudo-coherent. Hence, again by Exposé I, Proposition 2.6, $i_*(E'')$ is n-pseudo-coherent. Since $i_*(E'')$ is zero in degrees $> n$, this means, by Exposé I, Proposition 2.10 b), that $H^n(i_*(E''))$ is of finite type. But

$$H^n(i_*(E'')) = i_* H^n(E''),$$

so the assertion is reduced to showing, by Exposé I, Proposition 2.10 b), that if M is an $\mathcal{O}_X$-module such that $i_*(M)$ is of finite type, then M is of finite type. This is immediate and independent of the hypothesis on $i_*(\mathcal{O}_X)$.

Remark 1.1.2. The preceding lemma applies in particular to the case where i is a regular immersion. Indeed, by means of the Koszul complex, $i_*(\mathcal{O}_X)$ has locally a finite left resolution by finite-type free $\mathcal{O}_Y$-modules; hence $i_*(\mathcal{O}_X)$ is perfect, and a fortiori pseudo-coherent.

Lemma 1.1.3. Let

$$\begin{tikzcd} X' \arrow[r, "f"] \arrow[d, "i'"'] & X \arrow[d, "i"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

be a Cartesian square of schemes, where i is a closed immersion, and let E be a complex of $\mathcal{O}_X$-modules. Then the canonical base-change arrow

$$g^* i_*(E) \longrightarrow i'_* f^*(E)$$

is an isomorphism.

If $E' = f^*(E)$, if g is flat and $i_*(E)$ is n-pseudo-coherent, then $i'_*(E')$ is n-pseudo-coherent. Conversely, if g is faithfully flat and $i'_*(E')$ is n-pseudo-coherent, then $i_*(E)$ is n-pseudo-coherent.

Proof. The assertion is verified trivially on the fibers.

Lemma 1.1.4. Let

$$\begin{tikzcd} X' & X \arrow[l, "i'"'] \arrow[r, "i''"] & X'' \\ & Y \arrow[ul, "f'"] \arrow[ur, "f''"'] & \end{tikzcd}$$

be a commutative diagram of schemes, where i' (respectively i'') is a closed immersion and f' (respectively f'') is smooth, and let $E \in \operatorname{ob} D(X)$. Then

$$i'^*(E) \text{ is } n\text{-pseudo-coherent} \iff i''^*(E) \text{ is } n\text{-pseudo-coherent.}$$

Proof. Following a device due to Lichtenbaum, send X diagonally into $X' \times_Y X''$. One is at once reduced to proving the assertion in the case where one of the morphisms f', f'' is the identity. Changing notation, one has a commutative triangle

$$\begin{tikzcd} & Y \arrow[d, "f"] \\ X \arrow[ur, "j"] \arrow[r, "i"'] & Z \end{tikzcd}$$

where i and j are closed immersions and f is smooth, and one must show that $i_*(E)$ is n-pseudo-coherent if and only if $j_*(E)$ is n-pseudo-coherent. The question being local on Y and on Z in a neighborhood of a point of X, one may suppose that X, Y, Z are affine and that there is a factorization

$$\begin{tikzcd} Y \arrow[r, "i'"] \arrow[d, "f"'] & Z[T_1, \dots, T_r] \arrow[dl] \\ Z & \end{tikzcd}$$

where i' is a closed immersion and the oblique arrow is the canonical projection. Then, by Exposé VII, 1.10, i' is a regular immersion; hence by Lemma 1.1.1 an object F of $D(Y)$ is n-pseudo-coherent if and only if $i'_*(F)$ is so.

Thus one may suppose that $Y = Z[T_1, \dots, T_r]$, f being the canonical projection. The preceding diagram then corresponds to a diagram of rings

$$A \xleftarrow{\ v\ } C \xrightarrow{\ u\ } B = C[T_1, \dots, T_r]$$

where u and v are epimorphisms and $c : C \to B$ is the canonical inclusion. For $1 \leq i \leq r$, put $u(T_i) = x_i$, and let $z_i \in C$ be such that $v(z_i) = x_i$. The retraction

$$\sigma : B \longrightarrow C, \qquad \sigma(T_i) = z_i,$$

is defined by $c(T_i) = z_i$, is such that $v\sigma = u$. Geometrically, it corresponds to a section s of f such that $i = sj$. By Exposé VII, 1.10, s is a regular immersion; the assertion therefore follows from Lemma 1.1.1.

End of the proof of Theorem 1.1. Let $f : X \to Y$ be a locally finite type morphism. There exists an open covering (X_α) of X such that each X_α embeds by a closed immersion into a scheme Y_α smooth over Y. Put $\coprod X_\alpha = X'$, $\coprod Y_\alpha = Y'$. We have a commutative diagram

$$\begin{tikzcd} X' \arrow[r, "i'"] \arrow[d, "p'"'] & Y' \arrow[dl, "f'"] \\ X \arrow[d, "f"'] & \\ Y & \end{tikzcd}$$

where p' is the morphism which is the sum of a surjective family of open immersions (we shall say *special*, for short), i' is a closed immersion, and f' is smooth. We define $D(f)_{n\text{-}\mathrm{coh}}$ as the full subcategory of $D(X)$ formed by the objects E such that $i'_* p'^*(E)$ is n-pseudo-coherent. First we show that this definition makes sense, that is, that $D(f)_{n\text{-}\mathrm{coh}}$ is independent of the choice of i', p', f'. Let

$$\begin{tikzcd} X'' \arrow[r, "i''"] \arrow[d, "p''"'] & Y'' \arrow[dl, "f''"] \\ X \arrow[d, "f"'] & \\ Y & \end{tikzcd}$$

be an analogous commutative diagram, with p'' special, i'' a closed immersion, and f'' smooth.

From the two diagrams one deduces a commutative diagram

$$\begin{tikzcd}
X' \times_X X'' \arrow[r] \arrow[d] & Y' \times_Y Y'' \arrow[d] \\
X' \arrow[r, "i'"] \arrow[d] & Y' \arrow[d, "f'"] \\
X'' \arrow[r, "i''"'] \arrow[d] & Y'' \arrow[d, "f''"'] \\
X \arrow[r, "f"'] & Y
\end{tikzcd}$$

where the horizontal arrows, except for f, are closed immersions, and the arrows of the left (respectively right) Cartesian square are special morphisms (respectively smooth morphisms). Thus one is reduced to proving the following assertion.

$$\begin{tikzcd}
X_1 \arrow[r, "i_1"] \arrow[d, "p"'] & Y_1 \arrow[d, "q"] \\
X_2 \arrow[r, "i_2"'] & Y_2
\end{tikzcd}$$

Let this be a commutative square of schemes, where p is special, q is smooth, i_1 and i_2 are closed immersions, and let $E \in \operatorname{ob} D(X_2)$. Then $i_{2*}(E)$ is n-pseudo-coherent if and only if $i_{1*}p^*(E)$ is so.

Proof of the assertion. By definition, p is the sum of a family of opens U_α of X_2 covering X_2. Each U_α is induced by an open V_α of Y_2, and the V_α cover an open neighborhood X_2' of X_2 in Y_2. There is therefore a Cartesian commutative diagram

$$\begin{tikzcd}
X_1 \arrow[r, "i_1'"] \arrow[d, "p"'] & X_1' \arrow[d, "p'"] & \\
X_2 \arrow[r, "i_2'"'] & X_2' \arrow[r, "j"'] & Y_2
\end{tikzcd}$$

where i_1' and i_2' are closed immersions, p and p' are special morphisms, and j is an open immersion. The condition that $i_{2*}(E)$ be n-pseudo-coherent is plainly equivalent to the condition that $i'_{2*}(E)$ be n-pseudo-coherent. On the other hand, by Lemma 1.1.3, the condition that $i'_{2*}(E)$ be n-pseudo-coherent is equivalent to the condition that $i'_{1*}p^*(E)$ be n-pseudo-coherent. Applying Lemma 1.1.4 to the factorizations $(jp')i_1'$ and qi_1 of i_2p, and to $p^*(E) \in \operatorname{ob} D(X_1)$, one finally concludes that $i'_{2*}(E)$ is n-pseudo-coherent if and only if $i_{1*}p^*(E)$ is n-pseudo-coherent, as required.

We have thus shown that the category $D(f)_{n\text{-}\mathrm{coh}}$ defined above by means of a diagram as above depends in fact only on f. Moreover it is trivial from the definition that it satisfies (i) to (iii). This proves Theorem 1.1.

Definition 1.2. Let $f : X \to Y$ be a locally finite type morphism, let $E \in \operatorname{ob} D(X)$, and let $n \in \mathbb{Z}$. We say that E is n-pseudo-coherent relative to f if $E \in \operatorname{ob} D(f)_{n\text{-}\mathrm{coh}}$ (cf. 1.1). We say that E is pseudo-coherent, or $(-\infty)$-pseudo-coherent, relative to f if E is n-pseudo-coherent relative to f for every $n \in \mathbb{Z}$. We write $D(f)_{\mathrm{coh}}$ for the full subcategory of $D(X)$ formed by the objects pseudo-coherent relative to f.

We shall say that f is n-pseudo-coherent (respectively pseudo-coherent) if $\mathcal{O}_X$ is n-pseudo-coherent (respectively pseudo-coherent) relative to f. When this cannot lead to confusion, we shall sometimes say "n-pseudo-coherent (respectively pseudo-coherent) relative to Y" instead of "relative to f," and we shall also write

$$D(X)_{Y,n\text{-}\mathrm{coh}} \quad (\text{respectively } D(X)_{Y,\mathrm{coh}})$$

instead of

$$D(f)_{n\text{-}\mathrm{coh}} \quad (\text{respectively } D(f)_{\mathrm{coh}}).$$

Finally, for $* = +, -, b$, put

$$D^*(f)_{n\text{-}\mathrm{coh}} = D^*(X) \cap D(f)_{n\text{-}\mathrm{coh}}, \qquad D^*(f)_{\mathrm{coh}} = D^*(X) \cap D(f)_{\mathrm{coh}}.$$

Remarks 1.2.1.

a) The categories $D(f)_{n\text{-}\mathrm{coh}}$, $D(f)_{\mathrm{coh}}$ are not in general contained in $D^-(X)$. This is the case, however, if X is quasi-compact, since an object of $D(X)$ with locally bounded-above cohomology then has bounded-above cohomology.

b) The objects of $D(f)_{\mathrm{coh}}$ have quasi-coherent cohomology. Indeed, if $E \in \operatorname{ob} D(f)_{\mathrm{coh}}$, the question being local, one may suppose that f is a closed immersion; then $f_*(E)$ is pseudo-coherent, hence has quasi-coherent cohomology, and one uses the fact that an $\mathcal{O}_X$-module M is quasi-coherent if and only if $f_*(M)$ is.

c) When it is a matter of an $\mathcal{O}_X$-module, we shall use the terminology "of finite presentation relative to f" in preference to "(-1)-pseudo-coherent relative to f." For an $\mathcal{O}_X$-module E to be of finite presentation relative to f, it is necessary and sufficient, by Exposé I, Proposition 2.9, that there exist locally a closed immersion $i : X \to X'$, with X' smooth over Y, such that $i_*(E)$ is of finite presentation in the ordinary sense. In particular, to say that f is (-1)-pseudo-coherent is equivalent to saying that f is locally of finite presentation in the sense of EGA IV 1.4.

Examples 1.3. A smooth morphism and a regular immersion are pseudo-coherent morphisms. The same is true of an arbitrary locally finite type morphism $f : X \to Y$ if Y is locally noetherian.

Proposition 1.4. Let $f : X \to Y$ be a locally finite type morphism and let

$$E \longrightarrow F \longrightarrow G \longrightarrow E[1]$$

be a distinguished triangle of $D(X)$. The following table, where $n \in \mathbb{Z} \cup \{-\infty\}$ indicates a degree of pseudo-coherence relative to f, has the property that each line represents an implication whose conclusion is underlined:

$$\begin{array}{ccc} E & F & G \\ n+1 & n & \underline{n} \\ n & \underline{n} & n \\ \underline{n} & n & n-1. \end{array}$$

In particular, $D(f)_{\mathrm{coh}}$ is a triangulated subcategory of $D(X)$.

Proof. By the properties (i) to (iii) of Theorem 1.1, one may suppose that f is a closed immersion. We then have a distinguished triangle

$$f_*(E) \longrightarrow f_*(F) \longrightarrow f_*(G) \longrightarrow f_*(E)[1]$$

and the assertion follows from Exposé I, Proposition 2.6.

Definition 1.5. Let X and Y be two S-schemes. We shall say that X and Y are *Tor-independent over S* if

$$\operatorname{Tor}_i^S(\mathcal{O}_X, \mathcal{O}_Y) = 0 \quad \text{for } i \neq 0,$$

where the Tor_i^S are the local hyper-Tor sheaves defined in EGA III 6.5. This is a condition of local nature. If X, Y, S are affine, with respective rings B, C, A, it means that the canonical map

$$B \otimes_A^{\mathbf{L}} C \longrightarrow B \otimes_A C$$

is an isomorphism. It is therefore satisfied in particular if X or Y is flat over S.

Lemma 1.5.1. Let

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \uparrow & & \uparrow & & \uparrow \\ Y & \longleftarrow & Y' & \longleftarrow & Y'' \end{array}$$

be Cartesian squares of schemes. If X and Y' are Tor-independent over Y, and if X' and Y'' are Tor-independent over Y', then X and Y'' are Tor-independent over Y.

Proof. This is an easy exercise on associativity of the tensor product.

Notation 1.6. Let X and Y be two S-schemes, $E \in \operatorname{ob} D^-(X)$, $F \in \operatorname{ob} D^-(Y)$. We put

$$E \otimes_S^{\mathbf{L}} F = L\,\mathrm{pr}_1^*(E) \otimes^{\mathbf{L}} L\,\mathrm{pr}_2^*(F),$$

where $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections from $X \times_S Y$.

Lemma 1.7. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be a closed S-immersion. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Then, for $E_i \in \operatorname{ob} D^-(X_i)_{\mathrm{qcoh}}$, there is a canonical isomorphism

$$f_{1*}(E_1) \otimes_S^{\mathbf{L}} f_{2*}(E_2) \xrightarrow{\sim} (f_1 \times_S f_2)_*(E_1 \otimes_S^{\mathbf{L}} E_2).$$

Proof. Put $f_1 \times_S f_2 = f$. One easily defines, by adjunction, a canonical arrow

$$f_{1*}(E_1) \otimes_S^{\mathbf{L}} f_{2*}(E_2) \longrightarrow f_*(E_1 \otimes_S^{\mathbf{L}} E_2).$$

We show that it is an isomorphism. The question being local, one may suppose S, X_i, Y_i affine with respective rings C, A_i, B_i. By dévissage, one is reduced to $E_i = \mathcal{O}_{X_i}$. One must then verify that

$$A_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} A_2 \longrightarrow A$$

is an isomorphism, where $A = A_1 \otimes_C A_2$, $B = B_1 \otimes_C B_2$. In view of the Tor-independence hypotheses, this follows at once from associativity of the tensor product.

Proposition 1.8. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be a locally finite type S-morphism. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Let $[a_i, b_i]$ be intervals of $\mathbb{Z}$, and let

$$E_i \in \operatorname{ob} D^-(X_i)_{\mathrm{qcoh}}$$

be a complex which is a_i-pseudo-coherent relative to f_i and acyclic in degree $> b_i$. Then

$$E_1 \otimes_S^{\mathbf{L}} E_2$$

is a-pseudo-coherent relative to $f = f_1 \times_S f_2$ and acyclic in degree $> b$, where

$$a = \sup(a_1 + b_2, a_2 + b_1), \qquad b = b_1 + b_2.$$

Proof. The question being local on $X_1 \times_S X_2$, one may suppose that f_i factors as

$$X_i \xrightarrow{i_i} Y_i' \xrightarrow{p_i} Y_i,$$

where i_i is a closed immersion and p_i a smooth morphism. Put $f' = f_1' \times_S f_2'$. We must show that $f'_*(E_1 \otimes^{\mathbf{L}}_S E_2)$ is a-pseudo-coherent and acyclic in degree $> b$. By Lemma 1.5.1, Y_1' and Y_2' are Tor-independent over S. After changing notation, one may suppose that $f_i = f_i'$, in other words that the f_i are closed immersions. By hypothesis, $f_{i*}(E_i)$ is a_i-pseudo-coherent and acyclic in degree $> b_i$; hence by Exposé I, Proposition 2.16,

$$f_{1*}(E_1) \otimes^{\mathbf{L}}_S f_{2*}(E_2)$$

is a-pseudo-coherent and acyclic in degree $> b$. One concludes by Lemma 1.7.

Corollary 1.9. Under the hypotheses of Proposition 1.8, the functor

$$\otimes^{\mathbf{L}}_S : D^-(X_1) \times D^-(X_2) \longrightarrow D^-(X)$$

induces a functor

$$\otimes^{\mathbf{L}}_S : D^-(f_1)_{\mathrm{coh}} \times D^-(f_2)_{\mathrm{coh}} \longrightarrow D^-(f)_{\mathrm{coh}}.$$

In particular, if f_1 and f_2 are pseudo-coherent, then so is f.

Corollary 1.10. Let

$$\begin{tikzcd} X' \arrow[r, "g"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "h"'] & Y \end{tikzcd}$$

be a Cartesian square, with f locally of finite type. Then:

a) Suppose that X and Y' are Tor-independent over Y. For $n \in \mathbb{Z} \cup \{-\infty\}$, the functor $Lg^* : D^-(X) \to D^-(X')$ induces a functor

$$Lg^* : D^-(f)_{n\text{-}\mathrm{coh}} \longrightarrow D^-(f')_{n\text{-}\mathrm{coh}}.$$

In particular, if f is n-pseudo-coherent, then so is f'.

b) Suppose that h is faithfully flat. Let $n \in \mathbb{Z} \cup \{-\infty\}$, and let $E \in \operatorname{ob} D^-(X)$ be a complex with quasi-coherent cohomology. In order that E be n-pseudo-coherent relative to f, it is necessary and sufficient that $g^*(E)$ be n-pseudo-coherent relative to f'. In particular, in order that f be n-pseudo-coherent, it is necessary and sufficient that f' be so.

Proof. Taking Remark 1.2.1 b) into account, all assertions of Corollaries 1.9 and 1.10 follow trivially from Proposition 1.8, except the necessity in 1.10 b). For this one may, the question being local on X, suppose that f factors as a closed immersion followed by a smooth morphism. One is then reduced by Lemma 1.1.3 to the case where f (respectively f') is the identity of X (respectively X'). The assertion then follows, by a standard dévissage, from the fact that a quasi-coherent sheaf L on X is of finite type if and only if g^*L is of finite type.

Proposition 1.11. Let

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "h"'] & Y \arrow[d, "g"] \\ & Z \end{tikzcd}$$

be a commutative triangle of locally finite type morphisms. Let $E \in \operatorname{ob} D(X)$, $F \in \operatorname{ob} D(Y)$, and let $[a, b]$, $[c, d]$ be intervals of $\mathbb{Z}$.

a) If E is a-pseudo-coherent relative to f and acyclic in degree $> b$, and if F is c-pseudo-coherent relative to g and acyclic in degree $> d$, then

$$E \otimes^{\mathbf{L}}_{\mathcal{O}_Y} F$$

is m-pseudo-coherent relative to h and acyclic in degree $> n$, where

$$m = \sup(a + d, b + c), \qquad n = b + d.$$

b) If E is a-pseudo-coherent relative to h and acyclic in degree $> b$, and if g is $(a + 1 - b)$-pseudo-coherent, then E is a-pseudo-coherent relative to f.

Proof. The question being local on X, one may suppose that there is a Cartesian commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & E^r_Y & \xrightarrow{j} & E^{r+s}_Z \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ Y & \xrightarrow{k} & E^s_Z & \longrightarrow & Z \end{array}$$

where i, j, k are closed immersions and p, q are the canonical projections of affine spaces. If F is c-pseudo-coherent relative to Z, then $k_*(F)$ is c-pseudo-coherent by Theorem 1.1 (iii), and $j_*p^*(F)$ is c-pseudo-coherent by Lemma 1.1.3. Similarly, if g is x-pseudo-coherent, then $j_*(\mathcal{O}_{E^r_Y})$ is x-pseudo-coherent. By the definition of relative pseudo-coherence, one is thus reduced to proving a) and b) when f and g are closed immersions.

Proof of a). By Theorem 1.1 (iii), one has to show the following. Suppose $f_*(E)$ is a-pseudo-coherent and acyclic in degree $> b$, and $g_*(F)$ is c-pseudo-coherent and acyclic in degree $> d$. Then

$$h_*(E \otimes^{\mathbf{L}}_Y F)$$

is m-pseudo-coherent and acyclic in degree $> n$. There is a canonical isomorphism

$$f_*(E) \otimes^{\mathbf{L}}_Y F \xrightarrow{\sim} f_*(E \otimes^{\mathbf{L}}_Y F),$$

a special case of the isomorphism used in the proof of Proposition 1.8. Since the functor $g_*(- \otimes^{\mathbf{L}} F)$ takes finite-type locally free sheaves to c-pseudo-coherent complexes, and complexes acyclic in degree > 0 to complexes acyclic in degree $> d$, the assertion follows from Exposé I, Proposition 2.13.

Proof of b). One must prove the following assertion: suppose $f_*(E)$ is a-pseudo-coherent and acyclic in degree $> b$, and $g_*(\mathcal{O}_Y)$ is $(a+1-b)$-pseudo-coherent. Then $f_*(E)$ is a-pseudo-coherent. Since the assertion concerns $f_*(E)$, one may suppose $X = Y$ and $f = \mathrm{Id}_X$. One then proceeds as in the proof of Lemma 1.1.1, proving by descending induction that E is pseudo-coherent in degree a. Suppose already proved that E is $(a + 1)$-pseudo-coherent. By Exposé I, Proposition 2.10, after localizing there is a distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1]$$

where E'^i is locally free for $i \in [a + 1, b]$, $E'^i = 0$ for $i \notin [a + 1, b]$, and $E''^i = 0$ for $i > a$. The hypothesis that $g_*(\mathcal{O}_Y)$ is $(a + 1 - b)$-pseudo-coherent implies, by Exposé I, Proposition 2.11 a), that $g_*(E')$ is $(a + 1)$-pseudo-coherent. Since $g_*(E)$ is a-pseudo-coherent, it follows from Exposé I, Proposition 2.6, that $g_*(E'')$ is a-pseudo-coherent. We conclude by Exposé I, Proposition 2.10 b), and by the fact that an $\mathcal{O}_Y$-module M is of finite type if and only if $g_*(M)$ is. Corollaries 1.12–1.14 follow immediately.

Corollary 1.12. In the situation of Proposition 1.11, if g is pseudo-coherent, then, for $r \in \mathbb{Z} \cup \{-\infty\}$,

$$D(f)_{r\text{-}\mathrm{coh}} = D(h)_{r\text{-}\mathrm{coh}}.$$

In particular, taking $X = Y$, the notions of r-pseudo-coherence over Y, in the absolute sense or relative to g, coincide.

Corollary 1.13. In the situation of Proposition 1.11, if f is pseudo-coherent, the functor

$$\mathcal{O}_X \otimes^{\mathbf{L}}_Y - = Lf^* : D^-(Y) \longrightarrow D^-(X)$$

induces, for $r \in \mathbb{Z} \cup \{-\infty\}$, a functor

$$Lf^* : D^-(g)_{r\text{-}\mathrm{coh}} \longrightarrow D^-(h)_{r\text{-}\mathrm{coh}}.$$

Corollary 1.14. In the situation of Proposition 1.11, if f is r-pseudo-coherent and g is s-pseudo-coherent, with $r, s \in \mathbb{Z} \cup \{-\infty\}$, then h is $\sup(r,s)$-pseudo-coherent. If h is r-pseudo-coherent and g is $(r+1)$-pseudo-coherent, then f is r-pseudo-coherent.

Remark 1.14.1. In particular, for $r = s = -1$, one recovers the assertions (ii) and (v) of EGA IV 1.4.3.

2. The finiteness theorem

Conjecture 2.1. Let S be a scheme, let $f : X \to Y$ be a proper S-morphism of schemes locally of finite type over S, and let $E \in \operatorname{ob} D^+(X)$. If E is pseudo-coherent relative to S, then the same is true of $\mathbf{R}f_*(E)$.

Theorem 2.2. The preceding conjecture is true in the following cases:

(i) S is locally noetherian; (ii) f is projective.

Proof in case (i). Since S is locally noetherian, the projection of X (respectively Y) onto S is a pseudo-coherent morphism, and the notions of pseudo-coherence over X (respectively Y) and relative to S coincide (Corollary 1.12). On the other hand, since X (respectively Y) is locally noetherian, hence $\mathcal{O}_X$ (respectively $\mathcal{O}_Y$) coherent, an object M of $D(X)$ (respectively $D(Y)$) is pseudo-coherent if and only if its $H^i(M)$ are coherent and M is locally in D^- (Exposé I, 3.5). The theorem therefore follows from:

Theorem 2.2.1. Let $f : X \to Y$ be a proper morphism of locally noetherian schemes, and let $E \in \operatorname{ob} D^+(X)$. If the $H^i(E)$ are coherent, then the same is true of $R^i f_*(E)$. If moreover E is locally in D^-, then the same is true of $\mathbf{R}f_*(E)$.

Proof. The first assertion follows from the spectral sequence

$$R^p f_*(H^q(E)) \Longrightarrow R^{p+q} f_*(E)$$

and from the finiteness theorem EGA III 3.2.1. For the second, one may suppose Y affine. Then X is quasi-compact and separated; suppose it covered by r affine opens. One knows (EGA III 1.4.12) that under these conditions, for a quasi-coherent $\mathcal{O}_X$-module M, one has $R^i f_*(M) = 0$ for every $i > r$. If E is locally in D^-, then $E \in D^-$, since X is quasi-compact; say $H^i(E) = 0$ for $i > N$. The preceding spectral sequence then gives $R^i f_*(E) = 0$ for $i > N + r$, which proves the assertion.

Proof in case (ii). The question being local on Y, one may suppose S and Y affine. There is then a Cartesian commutative diagram

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "f"'] & \mathbb{P}^r_Y \arrow[r, "j"] \arrow[d, "p"] & \mathbb{P}^r_{Y'} \arrow[d, "p'"] \\ Y \arrow[r] & Y' \arrow[r, "g"'] & S \end{tikzcd}$$

where i, j, j' are closed immersions, g' is smooth and affine, and p, p' are projections of projective spaces. Suppose that E is pseudo-coherent relative to S. By Theorem 1.1 (iii), this means that $(ji)_*(E)$ is pseudo-coherent. We must show that $\mathbf{R}f_*(E)$ is pseudo-coherent relative to S, that is, by Theorem 1.1 (iii), that

$$j'_*\mathbf{R}f_*(E)$$

is pseudo-coherent. But

$$j'_*\mathbf{R}f_*(E) \simeq \mathbf{R}p'_*((ji)_*(E)),$$

so the assertion is a consequence of the following more precise result.

Theorem 2.2.2. Let Y be an affine scheme, let

$$f : X = \mathbb{P}^r_Y \longrightarrow Y$$

be the projection of a projective space of finite type, and let $E \in \operatorname{ob} D^b(X)_{\mathrm{coh}}$. Then $\mathbf{R}f_*(E)$ is pseudo-coherent. Moreover, if E is acyclic in degree $> a$, then $\mathbf{R}f_*(E)$ is acyclic in degree $> a + r$, and there exists an integer N such that, for all $n \geq N$,

$$\mathbf{R}f_*(E(n)), \qquad E(n) = E \otimes \mathcal{O}_X(n),$$

is acyclic in degree $> a$.

Proof. Since Y is affine, X is quasi-compact and separated. By Exposé II, 2.2.9 a), E is isomorphic, in $D(X)$, to a complex F bounded above such that, for every i, $F^i \in \operatorname{ob} L(X)$, that is, F^i is a finite direct sum of sheaves of the type $\mathcal{O}_X(-n)$, $n \geq 0$. If E is acyclic in degree $> a$, F may be chosen so that $F^i = 0$ for $i > a$; this follows formally from the preceding assertion, taking account of Exposé I, 2.10 a), and Exposé II, 2.2.8 a). Replacing E by F, one may suppose that $E^i \in \operatorname{ob} L(X)$ for every i, and that $E^i = 0$ for $i > a$. Since X is covered by $r + 1$ affine opens, EGA III 1.4.10 and the hypercohomology spectral sequence give

$$R^i f_*(E) = 0 \qquad \text{for } i > a + r.$$

Let $t \leq a$ be an integer and put

$$E' = (0 \to E^t \to \cdots \to E^a \to 0).$$

There is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

where E'' satisfies the same conditions as E, but with a replaced by $t - 1$. Applying $\mathbf{R}f_*$, one obtains a distinguished triangle

$$\mathbf{R}f_*(E') \longrightarrow \mathbf{R}f_*(E) \longrightarrow \mathbf{R}f_*(E'') \longrightarrow \mathbf{R}f_*(E')[1]$$

Since t can be chosen arbitrarily negative, it is enough, for the first assertion, to prove that $\mathbf{R}f_*(E')$ is $(t+r)$-pseudo-coherent. But $\mathbf{R}f_*(E'')$ is acyclic in degree $\geq t + r$, hence $(t+r)$-pseudo-coherent. Thus one wins by Exposé I, Proposition 2.6, if one proves that $\mathbf{R}f_*(E')$ is perfect.

For this one is reduced by dévissage to the case where E' is concentrated in degree 0 and equal to $\mathcal{O}_X(-n)$, $n > 0$. Then, by EGA III 2.1.16,

$$\mathbf{R}f_*(\mathcal{O}_X(-n)) \simeq R^r f_*(\mathcal{O}_X(-n))[-r].$$

Therefore $\mathbf{R}f_*(\mathcal{O}_X(-n))$ is perfect, and this ends the proof of the first assertion of Theorem 2.2.2. We have already seen that $\mathbf{R}f_*(E)$ is acyclic in degree $> a + r$. It remains to prove the last assertion. For $t < -r + a$, $\mathbf{R}f_*(E'')$ is acyclic in degree $> a$. Taking account of the preceding triangle, one is reduced to doing the proof for E' instead of for E. Since E' is bounded above and has components in $L(X)$, there is an integer N such that, for all $n \geq N$, the components of $E'(n)$ are finite direct sums of sheaves of type $\mathcal{O}_X(p)$ with $p \geq 0$. Then, by EGA III 2.1.16 and the hypercohomology spectral sequence, $\mathbf{R}f_*(E'(n))$ is acyclic in degree $> a$ for all $n \geq N$, which completes the proof of Theorem 2.2.2 and hence of Theorem 2.2.

We also have the following classical complement to 2.2 (ii), which will be given for later reference.

Corollary 2.3 (EGA III 2.2.1). Let S be a quasi-compact scheme, $f : X \to Y$ a projective S-morphism of finite type schemes over S, and let

$$L = i^*\mathcal{O}_P(1)$$

be the invertible sheaf defined by an extension $i : X \to P = P(F)$, where F is a quasi-coherent $\mathcal{O}_Y$-module of finite type. For $M \in \operatorname{ob} D(X)$ and $n \in \mathbb{Z}$, put

$$M(n) = M \otimes L^{\otimes n}.$$

If $E \in \operatorname{ob} D^+(X)$ is pseudo-coherent relative to S and acyclic in degree $> a$, then there exists an integer N such that, for all $n \geq N$, $\mathbf{R}f_*(E(n))$ has the same properties.

Proof. One reduces trivially, as in the proof of Theorem 2.2 (ii), to the situation of Theorem 2.2.2: $Y = S$, $f : X = \mathbb{P}^r_Y \to Y$ the canonical projection, $L = \mathcal{O}_X(1)$ the canonical sheaf, $E \in \operatorname{ob} D^b(X)$ a pseudo-coherent complex acyclic in degree $> a$. The conclusion then follows from Theorem 2.2.2.

Corollary 2.4. Let S be a quasi-compact scheme and let $f : X \to Y$ be a proper S-morphism of schemes of finite type over S. If S is noetherian, or if f is projective, the functor

$$\mathbf{R}f_* : D^+(X) \longrightarrow D^+(Y)$$

induces a functor

$$\mathbf{R}f_* : D^b(X)_{S\text{-}\mathrm{coh}} \longrightarrow D^b(Y)_{S\text{-}\mathrm{coh}}.$$

Proof. Consequence of Theorem 2.2 and of the fact that there is then an integer N such that, for every quasi-coherent $\mathcal{O}_X$-module E, $R^i f_*(E) = 0$ for all $i > N$.

Corollary 2.5. Let $f : X \to Y$ be a proper and pseudo-coherent morphism, and let $E \in \operatorname{ob} D^+(X)$. Suppose Y locally noetherian or f projective. If E is pseudo-coherent in the absolute sense, then the same is true of $\mathbf{R}f_*(E)$. If Y is quasi-compact, the functor $\mathbf{R}f_*$ induces a functor

$$\mathbf{R}f_* : D^b(X)_{\mathrm{coh}} \longrightarrow D^b(Y)_{\mathrm{coh}}.$$

Proof. Consequence of Theorem 2.2, Corollary 2.4, and Corollary 1.12.

Scholie 2.6. To prove Conjecture 2.1 in general, new techniques will surely have to be introduced. One may hope that they will one day make it possible to prove the analogous conjecture in analytic geometry, for a proper morphism of analytic spaces over a suitably topologized locally ringed topos in $\mathbb{C}$-algebras. For the present this conjecture has been proved only over a complex analytic space, by Grauert's theorem.[21]

[21]Added in April 1971: see a forthcoming article of R. Kiehl and J.-L. Verdier, where the finiteness theorem is proved by different, simpler techniques and under more general hypotheses.

3. Relative finite Tor-dimension

Definition 3.1. Let

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

be a morphism of ringed topoi, let $E \in \operatorname{ob} D^-(X)$, and let $[m, n]$ be an interval of $\mathbb{Z}$. We say that E is of flat amplitude relative to f (or to Y, if there is no possible confusion) contained in $[m, n]$, and write

$$\operatorname{tor.amp}_f(E) \subset [m, n] \qquad (\text{or } \operatorname{tor.amp}_Y(E) \subset [m, n]),$$

if E, considered as a complex of $f^{-1}(\mathcal{O}_Y)$-modules, is of flat amplitude contained in $[m, n]$ in the sense of Exposé I, 5.2. We say that f is of flat amplitude contained in $[-a, 0]$, or of Tor-dimension $\leq a$, if $\mathcal{O}_X$ is of flat amplitude relative to f contained in $[-a, 0]$. We say that E is of finite flat amplitude relative to f if there exists an interval $[m, n]$ such that $\operatorname{tor.amp}_f(E) \subset [m, n]$.

Notation 3.2. We write

$$D(f)_{\text{torf}}, \qquad D(X)_{f\text{-torf}}, \qquad D(X)_{Y\text{-torf}}$$

if no confusion is possible, for the full subcategory of $D^-(X)$ formed by complexes of finite flat amplitude relative to f. By Exposé I, 5.3, this is a triangulated subcategory of $D^b(X)$.

Proposition 3.3. With the notation of Definition 3.1, suppose that X has enough points (SGA 4 XVII IV 6.4). The following conditions are then equivalent:

(i) $\operatorname{tor.amp}_f(E) \subset [m, n]$.

(ii) For every point x of X, one has

$$\operatorname{tor.amp}_y(E_x) \subset [m, n],$$

where $y = f(x)$, ringed by $\mathcal{O}_{Y,y}$.

(iii) For every object V of Y and every $\mathcal{O}_V$-module M, one has

$$H^i\left(E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} M\right) = 0 \quad \text{for } i \notin [m, n].$$

Equivalently, for every $\mathcal{O}_V$-module M,

$$E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} M \simeq E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} L_! f^*(M).$$

(iv) The same condition as (iii), but with M the cokernel of a homomorphism

$$\mathcal{O}_V^q \longrightarrow \mathcal{O}_V^p, \qquad p, q \in \mathbb{N}.$$

Proof. The equivalence (i)$\Longleftrightarrow$(ii) follows from Exposé I, 5.6.2, and the implications (ii)$\Rightarrow$(iii)$\Rightarrow$(iv) are trivial. We prove (iv)$\Rightarrow$(ii). Let x be a point of X, $y = f(x)$. One has to show that

$$H^i(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} M) = 0$$

for every $\mathcal{O}_{Y,y}$-module M and every $i \notin [m, n]$. Since every $\mathcal{O}_{Y,y}$-module is an inductive filtered limit of modules of finite presentation, one may restrict to M of finite presentation. If M is a finitely presented $\mathcal{O}_{Y,y}$-module, the same argument as for ordinary ringed spaces (EGA I, 5.3.9) shows that there exists a y-pointed object V of Y and an $\mathcal{O}_V$-module P, cokernel of a homomorphism

$$\mathcal{O}_V^q \longrightarrow \mathcal{O}_V^p, \qquad p, q \in \mathbb{N},$$

such that $P_y \simeq M$. One then has

$$(E|_{f^{-1}(V)} \otimes^{\mathbf{L}}_{\mathcal{O}_V} P)_x \simeq E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} M,$$

by the commutation of tensor product with passage to fibers, and hypothesis (iv) implies the desired vanishing.

Remark 3.3.1. The editor does not know whether the equivalences (i)$\Longleftrightarrow$(iii)$\Longleftrightarrow$(iv) are true without the hypothesis that X has enough points.

Corollary 3.4. Let

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "h"'] & Y \arrow[d, "g"] \\ & Z \end{tikzcd}$$

be a commutative triangle of ringed topoi, where X is assumed to have enough points. Let $E \in \operatorname{ob} D(X)$, $F \in \operatorname{ob} D(Y)$, and let $[a,b]$, $[c,d]$ be intervals of $\mathbb{Z}$. Then

$$\operatorname{tor.amp}_f(E) \subset [a,b] \quad \text{and} \quad \operatorname{tor.amp}_g(F) \subset [c,d]$$

imply

$$\operatorname{tor.amp}_h(E \otimes^{\mathbf{L}}_Y F) \subset [a+c, b+d].$$

Proof. By Proposition 3.3, one must show that the relation on the left implies

$$H^i\Big(((E \otimes^{\mathbf{L}}_Y F)|_{h^{-1}(Z')}) \otimes^{\mathbf{L}}_{\mathcal{O}_{Z'}} M\Big) = 0$$

for every $i \notin [a+c, b+d]$ and every $\mathcal{O}_{Z'}$-module M, where Z' is an object of Z. Put $Y' = g^{-1}(Z')$, $X' = f^{-1}(Y') = h^{-1}(Z')$. Then

$$((E \otimes^{\mathbf{L}}_Y F)|_{X'}) \otimes^{\mathbf{L}}_{Z'} M \simeq (E|_{X'}) \otimes^{\mathbf{L}}_{Y'} ((F|_{Y'}) \otimes^{\mathbf{L}}_{Z'} M),$$

and the assertion follows at once from the hypotheses.

In particular:

Corollary 3.4.1. Under the hypotheses of Corollary 3.4:

a) If g is of Tor-dimension $\leq n$, then

$$\operatorname{tor.amp}_f(E) \subset [a,b] \Longrightarrow \operatorname{tor.amp}_h(E) \subset [a-n, b].$$

b) If f is of Tor-dimension $\leq m$, then

$$\operatorname{tor.amp}_g(F) \subset [c,d] \Longrightarrow \operatorname{tor.amp}_h(Lf^*(F)) \subset [c-m, d].$$

Remark 3.4.2. Let $f : X \to Y$ be a morphism of finite Tor-dimension and $E \in \operatorname{ob} D(X)$. By Corollary 3.4.1 a), if E is of finite flat amplitude in the absolute sense, then E is of finite flat amplitude relative to f. But the converse is false, even if f is flat, as is shown by the example of a singularity of a scheme over a field.

Proposition 3.5. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be an S-morphism, with $E_i \in \operatorname{ob} D^-(X_i)$, and let $[a_i, b_i]$ be an interval of $\mathbb{Z}$. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Put

$$a = a_1 + a_2, \qquad b = b_1 + b_2, \qquad f = f_1 \times_S f_2 : X \to Y.$$

Then:

a) If $\operatorname{tor.amp}_{f_1}(E_1) \subset [a_1, b_1]$, and if E_2 is acyclic outside $[a_2, b_2]$, then

$$E_1 \otimes_S^{\mathbf{L}} E_2$$

is acyclic outside $[a, b]$.

b) If $\operatorname{tor.amp}_{f_i}(E_i) \subset [a_i, b_i]$ for $i = 1, 2$, then

$$\operatorname{tor.amp}_f(E_1 \otimes_S^{\mathbf{L}} E_2) \subset [a, b].$$

Proof. By Proposition 3.3, it is enough to verify the assertion on the fibers. Let s be a point of S, x a point of X above s, $y = f(x)$, and let x_i (respectively y_i) be the canonical projection of x (respectively y) onto X_i (respectively Y_i). Put

$$(E_i)_{x_i} = M_i, \quad \mathcal{O}_{X_i,x_i} = A_i, \quad \mathcal{O}_{Y_i,y_i} = B_i, \quad \mathcal{O}_{S,s} = C,$$

and

$$\mathcal{O}_{X,x} = A = A_1 \otimes_C A_2, \qquad \mathcal{O}_{Y,y} = B = B_1 \otimes_C B_2.$$

Then

$$(E_1 \otimes_S^{\mathbf{L}} E_2)_x \simeq M_1 \otimes_{A_1}^{\mathbf{L}} A \otimes_{A_2}^{\mathbf{L}} M_2.$$

The Tor-independence hypotheses imply on the other hand that

$$A \simeq A_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} A_2.$$

Consequently, the second member of the displayed isomorphism is canonically isomorphic, in the derived category $D(B)$ of B-modules, to

$$M_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} M_2.$$

The relations $\operatorname{tor.amp}_{f_i}(E_i) \subset [a_i, b_i]$, $i = 1, 2$, imply

$$\operatorname{tor.amp}_{B_i}(M_i) \subset [a_i, b_i],$$

hence, by Exposé I, 5.5,

$$\operatorname{tor.amp}_B(M_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} M_2) \subset [a, b],$$

which proves b). If, on the other hand, $\operatorname{tor.amp}_{f_1}(E_1) \subset [a_1, b_1]$, M_1 is isomorphic to a complex of flat B_1-modules, zero outside $[a_1, b_1]$, and if E_2, hence M_2, is acyclic outside $[a_2, b_2]$, then

$$M_1 \otimes_{B_1}^{\mathbf{L}} B \otimes_{B_2}^{\mathbf{L}} M_2$$

is acyclic outside $[a, b]$. This proves a) and completes the proof.

Corollary 3.5.1. Under the hypotheses of Proposition 3.5, if f_1 and f_2 are of finite Tor-dimension, then so is f.

Corollary 3.5.2. Let

$$\begin{tikzcd} X' \arrow[r, "g"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "h"'] & Y \end{tikzcd}$$

be a Cartesian square of schemes, where X and Y' are assumed Tor-independent over Y, and let $E \in \operatorname{ob} D(X)$. Then one has

$$\operatorname{tor.amp}_f(E) \subset [a, b] \Longrightarrow \operatorname{tor.amp}_{f'}(Lg^*(E)) \subset [a, b],$$

and consequently the functor $Lg^* : D^-(X) \to D^-(X')$ induces a functor

$$Lg^* : D(f)_{\text{torf}} \longrightarrow D(f')_{\text{torf}}.$$

Proposition 3.6. Let

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "f"'] & X' \arrow[dl, "f'"] \\ Y & \end{tikzcd}$$

be a commutative triangle of schemes, where i is a closed immersion and f' is smooth, and let $E \in \operatorname{ob} D(X)$. For E to be locally of finite flat amplitude relative to f, it is necessary and sufficient that $i_*(E)$ be locally of finite flat amplitude in the absolute sense. More precisely, if f' is of relative dimension $\leq d$, then

$$\begin{aligned} &\text{(i)} \quad \operatorname{tor.amp}(i_*E) \subset [a,b] \Longrightarrow \operatorname{tor.amp}_f(E) \subset [a,b], \\ &\text{(ii)} \quad \operatorname{tor.amp}_f(E) \subset [a,b] \Longrightarrow \operatorname{tor.amp}(i_*E) \subset [a-d,b]. \end{aligned}$$

Proof. If $[m,n]$ is an interval in $\mathbb{Z}$, one has evidently

$$\operatorname{tor.amp}(E) \subset [m,n] \Longleftrightarrow \operatorname{tor.amp}(i_*E) \subset [m,n]$$

in the reduction in which one tests by modules after the closed immersion. Thus it remains to treat the case $X = X'$, $i = \mathrm{Id}_X$, $f = f'$. Assertion (i) is immediate from (3.4.1 a). For (ii), consider the diagonal section

$$s : X \longrightarrow X \times_Y X$$

of the two projections. If f is smooth of relative dimension d, then, by VII 1.10, s is a regular immersion of codimension $\leq d$, hence of Tor-dimension $\leq d$ (as is seen from the Koszul complex). By (3.5.2),

$$\operatorname{tor.amp}_f(E) \subset [a,b] \Longleftrightarrow \operatorname{tor.amp}_{\mathrm{pr}_2}(\mathrm{pr}_1^* E) \subset [a,b].$$

By (3.4.1 b) it follows that

$$\operatorname{tor.amp}(s^* \mathrm{pr}_1^* E) \subset [a-d,b],$$

and the implication follows because $s^* \mathrm{pr}_1^* \simeq \mathrm{Id}$.

Remark 3.6.1. The statement (3.6) would be false if one supposed only that f' were flat; see (3.4.2).

Proposition 3.7 (projection formula). Let S be a quasi-compact scheme, let $f : X \to Y$ be a quasi-compact and quasi-separated S-morphism of quasi-compact schemes over S, let $E \in \operatorname{ob} D^-(X)_{\text{qcoh}}$, and let $M \in \operatorname{ob} D^b(S)_{\text{qcoh}}$. If M has finite flat amplitude, or if E has finite flat amplitude relative to S, there is a functorial canonical isomorphism in $D(Y)_{\text{qcoh}}$

$$Rf_*(E) \otimes^{\mathbf{L}}_S M \xrightarrow{\sim} Rf_*(E \otimes^{\mathbf{L}}_S M).$$

Proof. First, by EGA III 1.4.12 and IV 1.7.21, there is an integer N such that $R^i f_*(F) = 0$ for $i > N$ and for every quasi-coherent $\mathcal{O}_X$-module F. Hence Rf_* sends $D^-(X)_{\text{qcoh}}$ into $D^-(Y)_{\text{qcoh}}$. The adjunction arrow

$$Lf^* Rf_*(E) \longrightarrow E$$

yields, after tensoring with M, an arrow

$$Lf^*(Rf_*(E) \otimes^{\mathbf{L}}_S M) \longrightarrow E \otimes^{\mathbf{L}}_S M,$$

and therefore, by adjunction, the canonical arrow

$$Rf_*(E) \otimes^{\mathbf{L}}_S M \longrightarrow Rf_*(E \otimes^{\mathbf{L}}_S M). \tag{*}$$

Both sides have quasi-coherent cohomology. The assertion is local on Y, so assume S and Y affine. If M has finite flat amplitude, devissage and the way-out property of the two functors in $(*)$ reduce successively to M an ordinary quasi-coherent module, then to M free, then to M free of finite type, and finally to $M = \mathcal{O}_S$, where $(*)$ is tautological. If instead E has finite flat amplitude relative to S, devissage reduces to the case $M = P$, with P a flat A-module and $S = \operatorname{Spec} A$. Put $A' = D_A(P)$ (EGA 0_{IV}, 18.2.3) and $S' = \operatorname{Spec} A'$. Then the assertion is the flat base-change assertion that

$$q^* Rf_*(E) \longrightarrow Rf'_* p^*(E)$$

is an isomorphism for the base change to S', which is EGA III 1.4.15 together with IV 1.7.21.

Corollary 3.7.1. Let S be a scheme, let $f : X \to Y$ be a quasi-compact and quasi-separated S-morphism of quasi-compact schemes over S, and let $E \in \operatorname{ob} D^-(X)_{\mathrm{qcoh}}$. If E is locally of finite flat amplitude relative to S, then the same is true of $Rf_*(E)$. More precisely, if S is quasi-compact and if N is an integer such that $R^i f_*(F) = 0$ for $i > N$ and every quasi-coherent $\mathcal{O}_X$-module F, then

$$\operatorname{tor.amp}_S(E) \subset [a,b] \Longrightarrow \operatorname{tor.amp}_S(Rf_*E) \subset [a, b+N].$$

Proof. By (3.3 (1) (iv)), and using that Rf_* commutes with restriction to an open of S, it suffices to prove that the hypothesis implies

$$H^i(Rf_*E \otimes^{\mathbf{L}}_S M) = 0$$

for $i \notin [a, b+N]$ and every quasi-coherent $\mathcal{O}_S$-module M. But the hypothesis says that $H^i(E \otimes^{\mathbf{L}}_S M) = 0$ for $i \notin [a,b]$, and (3.7) gives the result.

Corollary 3.7.2. Let $f : X \to Y$ be quasi-compact, quasi-separated, and locally of finite Tor-dimension, and let $E \in \operatorname{ob} D^-(X)_{\mathrm{qcoh}}$. If E is locally of finite flat amplitude in the absolute sense, then so is $Rf_*(E)$.

Proof. Combine (3.4.1 a) and (3.7.1).

4. Relative perfection

Definition 4.1. Let $f : X \to Y$ be a morphism locally of finite type of schemes, let $E \in \operatorname{ob} D(X)$, and let $[a,b]$ be an interval of $\mathbb{Z}$. We say that E is of perfect amplitude relative to f (or to Y, if there is no ambiguity), contained in $[a,b]$, and write

$$\operatorname{parf.amp}_f(E) \subset [a,b]$$

if E, relative to f, is pseudo-coherent (1.2) and of flat amplitude contained in $[a,b]$ (3.1). We say that f is of perfect amplitude contained in $[-n, 0]$, or of perfect dimension $\leq n$, if $\mathcal{O}_X$ has perfect amplitude contained in $[-n,0]$. We say that E is of finite perfect amplitude relative to f if such an interval $[a,b]$ exists. We say that E is perfect relative to f if it is locally of finite perfect amplitude relative to f. We say that f is perfect if $\mathcal{O}_X$ is locally of finite perfect amplitude relative to f.

Example 4.1.1. A regular immersion of codimension d (VII 1.4) is of perfect dimension $\leq d$, as follows from the Koszul complex. A morphism locally a complete intersection, i.e. locally the composite of a regular immersion and a smooth morphism, is a perfect morphism.

Notation 4.2. Let $f : X \to Y$ be locally of finite type. We write

$$D(f)_{\text{parf}}, \qquad D(f)_{\text{parf}}, \qquad D(f)^{-}_{\text{parf}}$$

when no confusion is possible, for the full subcategories of $D(X)$ formed by complexes perfect, respectively of finite perfect amplitude, relative to f. By (I 5.3) and (1.4), these are triangulated subcategories of $D(X)$. If X is quasi-compact, the two finite-amplitude/perfect notations coincide in the evident bounded sense.

Proposition 4.3 (cf. I 5.8). Let $f : X \to Y$ be locally of finite type, let $E \in \operatorname{ob} D(X)$, let $[a, b]$ be an interval of $\mathbb{Z}$, and let $x \in X$, $y = f(x)$. The following conditions are equivalent:

i) $\operatorname{parf.amp}_f(E) \subset [a, b]$ in a neighbourhood of x;

ii) in a neighbourhood of x, E is $(a-1)$-pseudo-coherent relative to f, is acyclic outside $[a, b]$, and

$$\operatorname{tor.amp}_{\mathcal{O}_{Y,y}}(E_x) \subset [a, b].$$

Proof. The implication i) $\Rightarrow$ ii) is immediate. For the converse, the question is local around x, and by the definitions one reduces to the case where f is smooth. Then E is $(a-1)$-pseudo-coherent near x; after localizing, one may assume that E is strictly $(a-1)$-pseudo-coherent and zero in degrees $> b$. After truncating, one may represent it by a complex with E^i free of finite type for $i > a$, E^a of finite presentation, and zero outside $[a, b]$. By (I 5.1.1), the term in degree a modulo boundaries is flat relative to f near x. Since f is smooth, one embeds locally into an affine space over Y and descends finite presentation and flatness from a noetherian approximation; pseudo-coherence follows from (1.10 a), and the desired perfect-amplitude assertion follows.

Corollary 4.3.1. Let $f : X \to Y$ be locally of finite type and let E be an $\mathcal{O}_X$-module. One has $\operatorname{parf.amp}_f(E) \subset [0, 0]$ if and only if E is flat and of finite presentation relative to f (1.2.1 c). In particular, f is of perfect dimension zero if and only if f is flat and locally of finite presentation.

Corollary 4.3.2. Let $f : X \to Y$ be locally of finite type, let E be a complex of $\mathcal{O}_X$-modules flat and of finite presentation relative to f, and let $[a, b]$ be an interval. If $E^i = 0$ for $i \notin [a, b]$, then $\operatorname{parf.amp}_f(E) \subset [a, b]$. If E is bounded above, then E is pseudo-coherent relative to f.

Proof. This follows from (4.3.1) and from I 4.15 and I 4.8.

Proposition 4.4. Let

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "f"'] & X' \arrow[dl, "g"] \\ Y & \end{tikzcd}$$

be a commutative triangle of schemes, where i is a closed immersion and g is smooth, and let $E \in \operatorname{ob} D(X)$. Then E is perfect relative to f if and only if $i_*(E)$ is perfect in the absolute sense. More precisely, if g has relative dimension $\leq d$, then

$$\operatorname{parf.amp}(i_*E) \subset [a, b] \Longrightarrow \operatorname{parf.amp}_f(E) \subset [a, b],$$
$$\operatorname{parf.amp}_f(E) \subset [a, b] \Longrightarrow \operatorname{parf.amp}(i_*E) \subset [a-d, b].$$

Proof. This is an immediate consequence of (3.6).

Proposition 4.5. Let

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "h"'] & Y \arrow[d, "g"] \\ & S \end{tikzcd}$$

be a commutative triangle of morphisms locally of finite type, let $E \in \operatorname{ob} D(X)$, $F \in \operatorname{ob} D(Y)$, and let $[a,b]$, $[c,d]$ be intervals of $\mathbb{Z}$. Then

$$\operatorname{parf.amp}_f(E) \subset [a,b], \quad \operatorname{parf.amp}_g(F) \subset [c,d] \Longrightarrow \operatorname{parf.amp}_h(E \otimes^{\mathbf{L}}_Y F) \subset [a+c, b+d].$$

In particular, if E is perfect relative to f and F is perfect relative to g, then $E \otimes^{\mathbf{L}}_Y F$ is perfect relative to h.

Proof. Combine (1.1) and (3.4).

Corollary 4.5.1. Under the hypotheses of (4.5):

a) if g is of perfect dimension $\leq n$, then

$$\operatorname{parf.amp}_f(E) \subset [a,b] \Longrightarrow \operatorname{parf.amp}_h(E) \subset [a-n, b];$$

therefore, if g is perfect, E perfect relative to f implies E perfect relative to h;

b) if f is of perfect dimension $\leq m$, then

$$\operatorname{parf.amp}_g(F) \subset [c,d] \Longrightarrow \operatorname{parf.amp}_h(Lf^*F) \subset [c-m, d];$$

therefore, if f is perfect, F perfect relative to g implies Lf^*F perfect relative to h.

Remark 4.5.2. With the preceding notation, if g is perfect, E perfect relative to h does not necessarily imply that E is perfect relative to f; the usual example of a singular point of a scheme over a field already shows this. The following criterion is available.

Proposition 4.6 (criterion of perfection by fibers). Let

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "h"'] & Y \arrow[d, "g"] \\ & S \end{tikzcd}$$

be a commutative triangle of morphisms of schemes. Suppose that f and h are locally of finite type, and that g is locally of finite presentation and flat. Let $x \in X$, $y = f(x)$, $s = h(x) = g(y)$, and let $E \in \operatorname{ob} D(X)$. Let $[a,b]$ be an interval of $\mathbb{Z}$, and let $f_s : X_s \to Y_s$ be the fiber of f over s, i.e. the morphism obtained from f by the base change $\operatorname{Spec} k(s) \to S$. Then the following conditions are equivalent:

i) $\operatorname{parf.amp}_h(E) \subset [a,b]$ near x, and

$$\operatorname{tor.amp}_{\mathcal{O}_{Y_s,y}}(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y_s,y}) \subset [a,b];$$

ii) $\operatorname{parf.amp}_f(E) \subset [a,b]$ near x.

Proof. The implication ii) $\Rightarrow$ i) follows because g has perfect dimension zero by (4.3.1), hence (4.5.1 a) gives $\operatorname{parf.amp}_h(E) \subset [a,b]$ near x; the asserted Tor-amplitude condition follows from (4.3). Conversely, the question is local around x. We may factor f as a closed immersion followed by a smooth morphism. Replacing E by its direct image under the immersion and replacing the smooth factor accordingly does not change the conditions; hence one may suppose f smooth. Since g is perfect of dimension zero and h is perfect by hypothesis, E is pseudo-coherent. After localizing, use (II 2.2.10 b) and (I 2.10 a) to suppose that E is strictly pseudo-coherent, zero in degrees $> b$, and quasi-isomorphic to the truncation

$$t : E \longrightarrow (0 \to E^a/B^a \to E^{a+1} \to \cdots \to E^b \to 0).$$

It remains to prove that E^a/B^a is f-flat near x. Let $i : X_s \to X$ be the canonical map. One has

$$E_x \otimes^{\mathbf{L}}_{mathcalO_{Y,y}} \mathcal{O}_{Y_s,y} \simeq i^*(E)_x.$$

Using (3.5.2), the Tor-amplitude hypothesis on the fiber shows that the truncated complex $i^*(t)$ is a quasi-isomorphism, so $i^*(E^a/B^a)_x$ is flat over $\mathcal{O}_{Y_s,y}$. On the other hand, the Tor-amplitude of E relative to h implies that E^a/B^a is h-flat. The usual criterion of flatness by fibers (EGA IV 11.3.10) then gives that E^a/B^a is f-flat near x.

Corollary 4.6.1. Let $f : X \to S$ be locally of finite presentation and flat, let $x \in X$, $s = f(x)$, let $E \in \operatorname{ob} D(X)$, and let $[a, b]$ be an interval. If $i_s : X_s = X \times_S \operatorname{Spec} k(s) \to X$ is the canonical projection and $E_s = Li_s^* E$, then

$$\operatorname{parf.amp}_S(E) \subset [a, b] \text{ near } x, \quad \operatorname{tor.amp}(E_s) \subset [a, b]$$

if and only if

$$\operatorname{parf.amp}(E) \subset [a, b] \text{ near } x.$$

Consequently E is perfect if and only if E is perfect relative to f and E_s is perfect for every $s \in S$.

Proof. Apply (4.6) with $X = Y$ and $f = \mathrm{Id}_X$.

Remark 4.6.2. From (4.6.1) and I 5.10 it follows that, if f is locally of finite presentation, flat, and has regular schemes as fibers, then E perfect relative to f is equivalent to E perfect. This equivalence was already obtained in (4.4) when f is smooth.

Proposition 4.7. Let S be a scheme, and for $i = 1, 2$ let

$$f_i : X_i \longrightarrow Y_i$$

be an S-morphism locally of finite type, with $E_i \in \operatorname{ob} D(X_i)$ and intervals $[a_i, b_i]$. Suppose that X_1 and X_2, and also Y_1 and Y_2, are Tor-independent over S. Put $a = a_1 + a_2$, $b = b_1 + b_2$, and $f = f_1 \times_S f_2$. Then:

a) if $\operatorname{parf.amp}_{f_1}(E_1) \subset [a_1, b_1]$, and if E_2 is pseudo-coherent relative to f_2 and acyclic outside $[a_2, b_2]$, then $E_1 \otimes^{\mathbf{L}}_S E_2$ is pseudo-coherent relative to f and acyclic outside $[a, b]$;

b) if $\operatorname{parf.amp}_{f_i}(E_i) \subset [a_i, b_i]$ for $i = 1, 2$, then

$$\operatorname{parf.amp}_f(E_1 \otimes^{\mathbf{L}}_S E_2) \subset [a, b].$$

Proof. Combine (1.9) and (3.5).

Corollary 4.7.1. Under the hypotheses of (4.7), if f_i has perfect dimension $\leq n_i$ for $i = 1, 2$, then f has perfect dimension $\leq n_1 + n_2$. Consequently, if f_1 and f_2 are perfect, then so is f.

Corollary 4.7.2. In a Cartesian square

$$\begin{tikzcd} X' \arrow[r, "g"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "h"'] & Y, \end{tikzcd}$$

where f is locally of finite type and X and Y' are Tor-independent over Y, one has

$$\operatorname{parf.amp}_f(E) \subset [a, b] \Longrightarrow \operatorname{parf.amp}_{f'}(Lg^* E) \subset [a, b].$$

Consequently $Lg^* : D^-(X) \to D^-(X')$ sends $D(f)_{\mathrm{parf}}$, respectively $D(f)_{\mathrm{parf}}$, into the corresponding categories for f'.

Proposition 4.8. Let S be a scheme, let $f : X \to Y$ be a proper S-morphism of S-schemes locally of finite type, and let $E \in \operatorname{ob} D^-(X)$. Under the proviso that (2.1) is valid for f (hence at least if S is locally noetherian or if f is projective):

a) if E is perfect relative to S, then so is $Rf_*(E)$;

b) if S is quasi-compact, X and Y are of finite type over S, $[a, b]$ is an interval, and N is such that $R^i f_*(M) = 0$ for $i > N$ and every quasi-coherent $\mathcal{O}_X$-module M, then

$$\operatorname{parf.amp}_S(E) \subset [a, b] \Longrightarrow \operatorname{parf.amp}_S(Rf_*E) \subset [a, b + N].$$

Consequently Rf_* sends perfect complexes relative to S to perfect complexes relative to S.

Proof. Assertion a) follows from b), and b) is obtained by combining (2.1) and (3.7).

Corollary 4.8.1. Let $f : X \to Y$ be proper and perfect, and let $E \in \operatorname{ob} D^-(X)$. Under the proviso that (2.1) holds for f, if E is perfect then $Rf_*(E)$ is perfect. If moreover Y is quasi-compact, Rf_* sends $D(X)_{\mathrm{parf}}$ to $D(Y)_{\mathrm{parf}}$.

Proof. If E is perfect, then E is perfect relative to f by (4.5.1 a).

Proposition 4.9. Let S be noetherian, let X and Y be S-schemes of finite type, let $f : X \to Y$ be a compactifiable S-morphism, i.e. the composite of an open immersion and a proper morphism, and let $E \in \operatorname{ob} D(X)$, $F \in \operatorname{ob} D^-(Y)$. If E is perfect relative to f and F is perfect relative to S, then

$$\operatorname{RHom}(E, f^! F)$$

is perfect relative to S.

Proof. The question is local on X. One may suppose that f factors as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y,$$

where i is a closed immersion and f' is smooth. If E is perfect relative to f, then it is perfect relative to i by (4.4); if F is perfect relative to S, then $f'^! F$ is perfect relative to S by (4.5.1 b), since $f'^! F$ is locally isomorphic to $f'^* F$ up to translation. As $f^! = i^! f'^!$, it remains to prove the assertion when f is a closed immersion. Then $f_* E$ is perfect and, by global duality ([H] III 6.7),

$$f_* \operatorname{RHom}(E, f^! F) \simeq \operatorname{RHom}(f_* E, F).$$

Since $f_* E$ is perfect, (I 7.7) gives

$$\operatorname{RHom}(f_* E, F) \simeq (f_* E)^\vee \otimes^{\mathbf{L}} F,$$

and the result follows from (4.5) with the identity morphism on Y.

Corollary 4.9.1. Under the hypotheses of (4.8), if f is perfect, then $f^!$ sends $D(Y)_{\mathrm{parf}}$ into $D(X)_{\mathrm{parf}}$.

Proof. Take $E = \mathcal{O}_X$ in (4.9).

Corollary 4.9.2. Let S be noetherian and let $f : X \to S$ be compactifiable. Put $K_X = f^!(\mathcal{O}_S)$, and $D_X(\cdot) = \operatorname{RHom}(\cdot, K_X)$. For $E \in \operatorname{ob} D(f)_{\mathrm{parf}}$, one has $D_X(E) \in \operatorname{ob} D(f)_{\mathrm{parf}}$, and the canonical arrow

$$E \longrightarrow D_X D_X(E)$$

is an isomorphism.

Proof. For the first assertion, apply (4.9) with $Y = S$ and $F = \mathcal{O}_S$. For the second, one reduces easily to the case where f is a closed immersion. It is enough then to apply f_* to the canonical arrow. By global duality one has $f_* D_X = D_S f_*$, and one is reduced to ordinary biduality for perfect complexes (I 7.2).

5. Applications: exchange and semi-continuity theorems

Lemma 5.1. Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos, let $\mathrm{Mod}(X)$ be the category of $\mathcal{O}_X$-modules, and write $D(X) = D(\mathrm{Mod}(X))$. Let $E \in \mathrm{ob}\, D(X)$ and let $p \in \mathbb{Z}$. Consider the following conditions:

a) the functor $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ from $\mathrm{Mod}(X)$ to $\mathrm{Mod}(X)$ is right exact;

a$'$ the canonical map
$$H^p(E) \otimes M \longrightarrow H^p(E \otimes^{\mathbf{L}} M)$$
is an isomorphism for all $M \in \mathrm{ob}\,\mathrm{Mod}(X)$;

b the functor $M \mapsto H^{p+1}(E \otimes^{\mathbf{L}} M)$ is left exact;

b$'$ the canonical map defined below
$$H^{p+1}(E \otimes^{\mathbf{L}} M) \longrightarrow \mathrm{Hom}(H^{-p-1}(E^\vee), M)$$
is an isomorphism for all $M \in \mathrm{ob}\,\mathrm{Mod}(X)$.

Then
$$\mathrm{a} \Longleftrightarrow \mathrm{a}' \Longleftrightarrow \mathrm{b} \Longleftarrow \mathrm{b}'.$$
If E is pseudo-coherent, then a), a'), b), and b') are equivalent.

The map in b') is obtained as follows. The canonical evaluation arrow $E^\vee \otimes^{\mathbf{L}} E \to \mathcal{O}_X$, tensorized with M, gives an arrow
$$E \otimes^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(E^\vee, M).$$
On H^{p+1}, and then by the canonical map
$$\mathrm{Ext}^{p+1}(E^\vee, M) o\, \mathrm{Hom}(H^{-p-1}(E^\vee), M),$$
one obtains the arrow b').

Proof. The implications $\mathrm{a}' \Rightarrow \mathrm{a} \Leftrightarrow \mathrm{b} \Leftarrow \mathrm{b}'$ are formal. To prove $\mathrm{a} \Rightarrow \mathrm{a}'$, present M by a sequence $M^0 \to M^1 \to M \to 0$ in which M^0 and M^1 are direct sums of sheaves of the form $\mathcal{O}_{U,X}$, with U an object of X. Right exactness and compatibility with direct sums reduce the assertion to $M = \mathcal{O}_{U,X}$. Then both sides are identified with extension by zero from U, and exactness of that extension gives the result.

Assume now that E is pseudo-coherent and prove $\mathrm{b} \Rightarrow \mathrm{b}'$. It is enough to show that the preceding comparison arrow
$$E \otimes^{\mathbf{L}} M \longrightarrow \mathrm{RHom}(E^\vee, M)$$
is an isomorphism when M is injective. Let $T(F) = F \otimes^{\mathbf{L}} M$ and $T'(F) = \mathrm{RHom}(F^\vee, M)$. Both functors have the same boundedness behaviour under truncation; the arrow is local and is compatible with restriction to opens, and an injective remains injective after such a restriction. After localizing and truncating, one may suppose E strictly perfect. A devissage then reduces to $E = \mathcal{O}_X$, where the assertion is immediate.

Remarks 5.2.

i) Conditions a), a'), b), and b') are local on the topos: if (X_α) is an open covering of the final object and the condition holds for each X_α and $E|X_\alpha$, then it holds on X.

ii) If x is a point of X, write a_x, a'_x, b_x, etc. for the corresponding conditions for E_x and $\mathrm{Mod}(\mathcal{O}_{X,x})$. If X has enough points, condition a), respectively a'), b), is equivalent to the corresponding stalkwise condition for every x; if E is pseudo-coherent, this is also equivalent to b') stalkwise. Moreover, a') may be tested on quotients of modules of the form $\mathcal{O}_{U,x}^{(I)} \to \mathcal{O}_{U,x}$.

iii) The criterion EGA III 7.4.2 extends word for word: if E is represented by flat terms, condition a) says that $\mathrm{Coker}(E^p \to E^{p+1})$ is flat, hence locally a direct factor of a free module of finite type when E^p and E^{p+1} are of finite presentation. In particular, for E perfect, condition a) for degree p is equivalent to the corresponding condition for $E^\vee$ in degree $-p-1$.

iv) Let X be a quasi-separated scheme and let $E \in \mathrm{ob}\, D(X)$ have locally bounded quasi-coherent cohomology. Let a_{qcoh}, a'_{qcoh}, etc. be the analogous conditions with $\mathrm{Mod}(X)$ replaced by $\mathrm{Qcoh}(X)$. Then

$$a \Rightarrow a_{\mathrm{qcoh}}, \qquad a' \Rightarrow a'_{\mathrm{qcoh}}, \qquad b \Rightarrow b_{\mathrm{qcoh}},$$

and these conditions are all equivalent when E is pseudo-coherent. This follows by reducing to affine opens, using II 3.5 and II 2.2.10 a), and the fact that quasi-coherent sheaves extend from affine opens on a quasi-separated scheme.

Proposition 5.3 (exchange property; EGA III 7.7.5 II). Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes, let $E \in \mathrm{ob}\, D^-(X)$ have quasi-coherent cohomology and finite flat amplitude relative to f, and let $p \in \mathbb{Z}$. Consider:

a) $M \mapsto R^p f_*(E \otimes_Y^{\mathbf{L}} M)$, from $\mathrm{Qcoh}(Y)$ to $\mathrm{Qcoh}(Y)$, is right exact;

a′ the canonical arrow
$$R^p f_*(E) \otimes M \longrightarrow R^p f_*(E \otimes_Y^{\mathbf{L}} M)$$
is an isomorphism for every $M \in \mathrm{ob}\,\mathrm{Qcoh}(Y)$;

b $M \mapsto R^{p+1} f_*(E \otimes_Y^{\mathbf{L}} M)$ is left exact;

b′ the canonical arrow

$$R^{p+1} f_*(E \otimes_Y^{\mathbf{L}} M) \longrightarrow \mathrm{Hom}(H^{-p-1}((Rf_*E)^\vee), M)$$

is an isomorphism for every $M \in \mathrm{ob}\,\mathrm{Qcoh}(Y)$;

c for every base change $Y' \to Y$, the canonical arrow

$$R^p f_*(E) \otimes_Y \mathcal{O}_{Y'} \longrightarrow H^p(Rf_*E \otimes_Y^{\mathbf{L}} \mathcal{O}_{Y'})$$

is an isomorphism.

Then
$$a \Longleftrightarrow a' \Longleftrightarrow c \Longleftrightarrow b \Longleftarrow b'.$$

If, moreover, f is proper, E is pseudo-coherent relative to f (hence perfect relative to f), and (2.1) is valid for f, for instance if f is projective or Y is noetherian, then the preceding conditions are all equivalent.

Proof. The implications among a), a'), b), and b') follow at once from (3.7), (5.1), and (5.2 iv); the implication $b \Rightarrow b'$ in the proper pseudo-coherent case follows from (2.1). It remains to compare a') and c). This is supplied by the following lemma.

Lemma 5.3.1. Let X be a quasi-separated scheme, and let $E \in \operatorname{ob} D^-(X)$ have locally bounded quasi-coherent cohomology. Then condition (5.1 a) is equivalent to:

c) for every $X' \to X$, the canonical map

$$H^p(E) \otimes_X \mathcal{O}_{X'} \longrightarrow H^p(E \otimes^{\mathbf{L}}_X \mathcal{O}_{X'})$$

is an isomorphism.

If E is pseudo-coherent, condition (5.1 a) also implies that, for every $X' \to X$, the map

$$H^{-p-1}(E^\vee) \otimes_X \mathcal{O}_{X'} \longrightarrow H^{-p-1}((E \otimes^{\mathbf{L}}_X \mathcal{O}_{X'})^\vee) \tag{5.3.1.1}$$

is an isomorphism.

Proof of the lemma. For $a \Rightarrow c$, localize on X', and assume $X = \operatorname{Spec} A$, $X' = \operatorname{Spec} A'$, and $E = \widetilde{F}$. By the affine case the hypothesis gives

$$H^p(F) \otimes_A A' \longrightarrow H^p(F \otimes^{\mathbf{L}}_A A')$$

an isomorphism, which is exactly c). Conversely, by (5.2 iv) it suffices to prove $c \Rightarrow a'_{\mathrm{qcoh}}$. Again localizing, write $X = \operatorname{Spec} A$, $E = \widetilde{F}$, and let $M = \widetilde{N}$ be quasi-coherent. Taking $A' = D_A(N)$ and applying c) to $\operatorname{Spec} A' \to X$ gives the required map

$$H^p(F) \otimes_A N \longrightarrow H^p(F \otimes^{\mathbf{L}}_A N)$$

as an isomorphism.

For the last assertion, the question is local on X'. By (II 2.2.10 b) and truncation, one may suppose E strictly perfect. Then $(E \otimes^{\mathbf{L}}_X \mathcal{O}_{X'})^\vee$ identifies canonically with $E^\vee \otimes^{\mathbf{L}}_X \mathcal{O}_{X'}$, and (5.3.1.1) is the canonical map

$$H^{-p-1}(E^\vee) \otimes_X \mathcal{O}_{X'} \longrightarrow H^{-p-1}(E^\vee \otimes^{\mathbf{L}}_X \mathcal{O}_{X'}). \tag{*}$$

If E satisfies (5.1 a) for p, then by (5.2 iii) $E^\vee$ satisfies the same condition for $-p-1$, and $(*)$ is an isomorphism by the first part of the lemma. The lemma, and therefore (5.3), follow.

Remark 5.3.2. Proposition (5.3) applies in particular when E is a bounded complex of quasi-coherent flat $\mathcal{O}_X$-modules of finite presentation relative to f: by (4.3.2), such a complex has finite perfect amplitude relative to f.

Definition 5.4. Let X be a topos, let $\operatorname{Pt}(X)$ be its set of points (SGA 4 IV 6.1), let M be an ordered set, and let $\varphi : \operatorname{Pt}(X) \to M$ be a function. We say that φ is upper semi-continuous at x (respectively lower semi-continuous, respectively locally constant) if there exists an x-pointed object U of X such that for every point y of U, one has $\varphi(y) \leq \varphi(x)$ (respectively $\varphi(y) \geq \varphi(x)$, respectively $\varphi(y) = \varphi(x)$). We say that φ is upper semi-continuous (respectively lower semi-continuous, respectively locally constant) if it is so at every point.

Examples 5.4.1.

a) If $\underline{M}$ is the constant sheaf with value M and $s \in H^0(X, \underline{M})$, then $x \mapsto x^*s$ is locally constant. In particular, if X is locally ringed and E is a perfect complex on X, the rank of E (I 6.11) defines a locally constant function on $\operatorname{Pt}(X)$ with values in $\mathbb{Z}$.

b) If X is locally ringed and P is an $\mathcal{O}_X$-module of finite type, then for $x \in \mathrm{Pt}(X)$, $P_x \otimes_{\mathcal{O}_{X,x}} k(x)$ is a finite-dimensional vector space over $k(x)$, and

$$x \longmapsto \dim_{k(x)}(P_x \otimes k(x))$$

is upper semi-continuous. This is the usual Nakayama argument: an epimorphism $\mathcal{O}_{X,x}^n \to P_x$ lifts over a pointed neighbourhood.

Lemma 5.5. Let $(X, \mathcal{O}_X)$ be a locally ringed topos, let $E \in \mathrm{ob}\, D(X)$, and let $p \in \mathbb{Z}$. Then:

i) If E is perfect, the vector spaces

$$H^i(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} k(x))$$

are finite-dimensional over $k(x)$, zero except for finitely many i, and the function

$$x \longmapsto \sum_i (-1)^i \dim_{k(x)} H^i(E_x \otimes^{\mathbf{L}} k(x))$$

is locally constant on $\mathrm{Pt}(X)$.

ii) If E is $(p-1)$-pseudo-coherent, then

$$H^p(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} k(x))$$

is finite-dimensional over $k(x)$, and its dimension is upper semi-continuous on $\mathrm{Pt}(X)$. If moreover $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ is exact on $\mathrm{Mod}(X)$, then $H^p(E)$ is locally free of finite type and the preceding function is locally constant.

iii) If E is p-pseudo-coherent (respectively $(p-1)$-pseudo-coherent), and if for a point x the functor

$$M \longmapsto H^p(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} M)$$

is right exact (respectively left exact) on $\mathrm{Mod}(\mathcal{O}_{X,x})$, then there exists an x-pointed object U such that

$$M \longmapsto H^p(E|_U \otimes^{\mathbf{L}} M)$$

is right exact (respectively left exact) on $\mathrm{Mod}(U)$.

Proof. For i), by I 4.19.1 the complex $E_x \otimes^{\mathbf{L}} k(x)$ is a perfect complex of $k(x)$-modules; hence its cohomology spaces have finite rank and vanish except for finitely many degrees. The identity

$$\mathrm{rank}(E_x \otimes^{\mathbf{L}} k(x)) = \sum_i (-1)^i \dim_{k(x)} H^i(E_x \otimes^{\mathbf{L}} k(x))$$

follows by devissage from the additivity of rank. The assertion follows from the compatibility of the rank with inverse images (I 6.12).

For ii), after truncation and translation, one may assume for the first assertion that E is strictly perfect, $p = 0$, and $E^i = 0$ for $i \notin [-1, 1]$. Put $E(x) = E_x \otimes^{\mathbf{L}} k(x)$. Then

$$0 = \mathrm{rank}(E(x)) = \dim H^{-1}(E(x)) - \dim H^0(E(x)) + \dim H^1(E(x))$$

up to the corresponding alternating-rank convention; hence upper semi-continuity of $\dim H^0(E(x))$ follows from that of the neighbouring cohomology dimensions and from local constancy of the rank of the terms. For H^1 this is immediate from (5.4.1 b). For H^{-1}, replacing E by the truncated complex $(0 \to E^{-1} \to E^0 \to 0)$ reduces to the same argument.

For the second assertion in ii), after truncating one may suppose E strictly perfect and $E^i = 0$ for $i < p - 1$. Then $Z'^{p+1}(E) = \operatorname{Coker}(E^p \to E^{p+1})$ is of finite presentation and one has an exact sequence

$$0 \to H^p(E) \to Z'^p(E) \to E^{p+1} \to Z'^{p+1}(E) \to 0.$$

Exactness of $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ implies, by (5.2 iii), that $Z'^p(E)$ and $Z'^{p+1}(E)$ are flat, hence locally free of finite type by I 5.8.3. The exact sequence then shows that $H^p(E)$ is locally free of finite type, and the function $x \mapsto \dim_{k(x)} H^p(E)_x$ is locally constant.

For iii), one may assume that E is strictly perfect and that $E^i = 0$ for $i < p$ (respectively for $i < p - 1$). Saying that the functor

$$M \longmapsto H^p(E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} M)$$

is right exact (respectively left exact) on $\operatorname{Mod}(\mathcal{O}_{X,x})$ is equivalent, by (5.2 iii), to saying that $Z'^{p+1}(E_x)$ (respectively $Z'^p(E_x)$) is flat, hence free of finite type since it is of finite presentation. Since

$$Z'^{p+1}(E_x) \simeq Z'^{p+1}(E)_x \qquad \text{(respectively } Z'^p(E_x) \simeq Z'^p(E)_x\text{)},$$

the usual argument (EGA 0_I, 5.2.7) gives an x-pointed object U of X on which $Z'^{p+1}(E|_U)$ (respectively $Z'^p(E|_U)$) is locally free of finite type. This proves the exactness assertion.

Corollary 5.6 (EGA III 7.6.9). Let X be a scheme, let $p \in \mathbb{Z}$, and let $E \in \operatorname{ob} D^-(X)$ be $(p-1)$-pseudo-coherent. Then:

i) For $x \in X$, $H^p(E \otimes^{\mathbf{L}}_X k(x))$ has finite rank over $k(x)$, and

$$x \longmapsto \dim_{k(x)} H^p(E \otimes^{\mathbf{L}}_X k(x))$$

is constructible and upper semi-continuous on X.

ii) Suppose X is quasi-separated and E has locally bounded quasi-coherent cohomology. If $M \mapsto H^p(E \otimes^{\mathbf{L}} M)$ is exact on the category of quasi-coherent $\mathcal{O}_X$-modules, then the preceding function is locally constant and $H^p(E)$ is locally free of finite type. Conversely, if X is locally noetherian and reduced and that function is locally constant, then the same functor is exact on quasi-coherent modules.

Proof. If X is integral with generic point x, the functor $M \mapsto H^p(E_x \otimes M)$ is exact on $\mathcal{O}_{X,x}$-modules, since $\mathcal{O}_{X,x} = k(x)$. By (5.5 iii) there is a non-empty open neighbourhood U of x on which the corresponding functor is exact, and by (5.5 ii) the rank function is locally constant on U. This gives constructibility. The other assertions follow from (5.5 ii, iii) and (5.2 iv), except for the converse in ii), for which one reduces locally and by truncation to the strictly perfect case and applies EGA III 7.6.2 and 7.6.5 to the functor $T_M = H^p(E \otimes^{\mathbf{L}} M)$.

Proposition 5.7 (EGA III 7.7.5 and 7.9.4). Let $f : X \to Y$ be a proper morphism of quasi-compact schemes, let $E \in \operatorname{ob} D(X)_{\mathrm{parf}}$, and let $p \in \mathbb{Z}$. Assuming (2.1) is valid for f, for instance if f is projective or if Y is noetherian, the following hold:

i) For $y \in Y$, the vector spaces $H^i(Rf_*E \otimes^{\mathbf{L}}_Y k(y))$ are finite-dimensional over $k(y)$, zero except for finitely many i, and their alternating rank is locally constant on Y.

ii) The function

$$y \longmapsto \dim_{k(y)} H^p(Rf_*E \otimes^{\mathbf{L}}_Y k(y))$$

is constructible and upper semi-continuous on Y.

iii) Suppose Y is quasi-separated. If $M \mapsto R^p f_*(E \otimes^{\mathbf{L}}_Y M)$ is exact on the category of quasi-coherent $\mathcal{O}_Y$-modules, then $R^p f_*(E)$ is locally free of finite type and the function in ii) is locally constant. Conversely, if Y is noetherian and reduced and the function in ii) is locally constant, then the same functor is exact on quasi-coherent modules.

Proof. By (4.8), $Rf_*(E)$ is a perfect complex. Assertion i), respectively ii), follows from (5.5 i), respectively (5.6 i). If M is quasi-coherent on Y, the canonical arrow

$$Rf_*(E) \otimes^{\mathbf{L}}_Y M \xrightarrow{\sim} Rf_*(E \otimes^{\mathbf{L}}_Y M)$$

is an isomorphism by (3.7), and iii) follows from (5.6 ii).

Remarks 5.8.

a) In the situation of (5.7), let $i_y : X_y \to X$ be the inclusion of the fiber over a point $y \in Y$. One may show, under the usual flatness hypotheses, the Künneth formula

$$H^i(Rf_*E \otimes^{\mathbf{L}}_Y k(y)) \simeq H^i(X_y, E_y),$$

where $E_y = Li_y^* E$, or $E_y = i_y^* E$ in the flat case.

b) One can develop for complex analytic spaces a theory analogous to that of this section: exchange, semi-continuity, and continuity properties. Its essential inputs are Grauert's finiteness theorem for proper morphisms and the analytic analogue of the projection formula. More precisely, for a morphism $f : X \to Y$ of analytic spaces, with X of bounded dimension, and for $E \in \operatorname{ob} D(X)$, $M \in \operatorname{ob} D(Y)_{\text{coh}}$, the canonical arrow

$$Rf_*(E) \otimes^{\mathbf{L}}_Y M \longrightarrow Rf_*(E \otimes^{\mathbf{L}}_Y M)$$

is an isomorphism. Unlike Grauert's theorem, this assertion is essentially formal; it follows by the same devissage as in the proof of (3.7), and requires no hypothesis on E beyond belonging to the appropriate derived category.

Bibliography

[1] Grauert, H., *Ein Theorem der Analytischen Garbentheorie*, Publ. IHES, no. 5.

[2] Hartshorne, R., *Residues and Duality*, Lecture Notes in Mathematics, no. 20, Springer, 1966.

[3] Godement, R., *Théorie des faisceaux*, Hermann, 1958.

Exposé IV. Grothendieck groups of ringed topoi

by L. Illusie

1. Reminders and generalities on Grothendieck groups

1.1. Let C be a triangulated category. Recall that a map f from $\operatorname{ob} C$ to an abelian group G is called *additive* if

$$f(E) = f(E') + f(E'')$$

for every distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1].$$

The additive maps from $\operatorname{ob} C$ to G form an abelian group, depending functorially on G. The functor so obtained is represented by an abelian group $k(C)$ and a universal additive map

$$\mathrm{cl}_C : \operatorname{ob} C \longrightarrow k(C),$$

which we shall write simply as cl when no confusion is possible.

The group $k(C)$ depends functorially on C with respect to exact functors. If C, C' are triangulated categories, two isomorphic exact functors from C to C' induce the same homomorphism $k(C) \to k(C')$; in particular, if $u : C \to C'$ is an exact equivalence of categories, then $k(u) : k(C) \to k(C')$ is an isomorphism.

Lemma 1.2. Let C be a triangulated category and let $L \in \operatorname{ob} C$.

a) For every $n \in \mathbb{Z}$,
$$\mathrm{cl}(L[n]) = (-1)^n \, \mathrm{cl}(L).$$
If L', L'' are objects of C such that $L \simeq L' \oplus L''$, then
$$\mathrm{cl}(L) = \mathrm{cl}(L') + \mathrm{cl}(L'').$$

b) Suppose that there exists $n \in \mathbb{Z}$ such that
$$\coprod_{i \geq 0} L[2ni]$$
is representable. Then $\mathrm{cl}(L) = 0$.

Proof. Assertion a) follows directly from the definitions. For b), it is enough to observe that

$$\coprod_{i \geq 0} L[2ni] \simeq L \oplus \left(\coprod_{i \geq 0} L[2ni] \right) [2n],$$

and to apply a).

Lemma 1.3. Let C be a triangulated category and let A be a thick subcategory of C ([V] I 2.1). The inclusion and quotient functors define an exact sequence

$$k(A) \longrightarrow k(C) \longrightarrow k(C/A) \longrightarrow 0.$$

Proof. See SGA 5 VIII 3.1.

Lemma 1.4. Let A be an additive category, respectively an abelian category, and let f be an additive function on $\operatorname{ob} K^b(A)$, respectively on $\operatorname{ob} D^b(A)$. For $E \in \operatorname{ob} K^b(A)$, respectively $E \in \operatorname{ob} D^b(A)$, one has

$$f(E) = \sum_i (-1)^i f(E^i),$$

respectively

$$f(E) = \sum_i (-1)^i f(H^i(E)).$$

Proof. Left as an exercise to the reader.

1.5. Let C be an abelian category and let A be a full subcategory of C, stable under finite sums. A map f from $\operatorname{ob} A$ to an abelian group G is called *additive* if

$$\sum_i (-1)^i f(E^i) = 0$$

for every acyclic bounded complex E of C whose components belong to A. When A is stable under kernels of epimorphisms (I 1.2), this amounts to saying that

$$f(L) = f(L') + f(L'')$$

for every exact sequence

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0$$

in C whose terms belong to A. The additive maps from $\operatorname{ob} A$ to G form an abelian group depending functorially on G. Denote by $K^{b,\emptyset}(A)$ the thick subcategory of $K^b(A)$ consisting of acyclic complexes. It follows easily from (1.3) and (1.4) that the map

$$\mathrm{cl}_A : \operatorname{ob} A \longrightarrow k\big(K^b(A)/K^{b,\emptyset}(A)\big) \tag{1.5.1}$$

induced by $\mathrm{cl}_{K^b(A)/K^{b,\emptyset}(A)}$ is a universal additive map, i.e. that the pair

$$\Big(k\big(K^b(A)/K^{b,\emptyset}(A)\big), \mathrm{cl}_A\Big)$$

represents the functor which associates to each abelian group G the abelian group of additive maps from $\operatorname{ob} A$ to G. We put

$$k\big(K^b(A)/K^{b,\emptyset}(A)\big) = k(A), \tag{1.5.2}$$

and write $k(A, C)$ instead of $k(A)$ when it is necessary to avoid ambiguity about the ambient abelian category.[22] In particular, for $A = C$, one has

$$k(D^b(C)) = k(C). \tag{1.5.3}$$

Lemma 1.6. Let C be an abelian category and let A be a thick abelian subcategory, i.e. a full subcategory stable under kernels, cokernels, and extensions. Let $D_A(C)$ be the full subcategory of $D(C)$ formed by the objects E such that $H^i(E) \in \operatorname{ob} A$ for every i. Then $D_A(C)$ is a triangulated subcategory ([V] I 2.3) of $D(C)$, and the canonical functor

$$D^b(A) \longrightarrow D^b_A(C) = D^b(C) \cap D_A(C)$$

induces an isomorphism

$$k(A) \xrightarrow{\sim} k(D^b_A(C)),$$

where $k(A)$ is defined by (1.5.2).

Proof. The first assertion is an immediate consequence of the long exact cohomology sequence. For the second, consider the map

$$f : \operatorname{ob} D^b_A(C) \longrightarrow k(A)$$

defined by

$$f(E) = \sum_i (-1)^i \, \mathrm{cl}_A(H^i(E)).$$

[22] The notation $k(A)$ had been introduced in I 6.3 for the group $k(K^b(A))$, which is not in general isomorphic to the group defined by (1.5.2). We therefore change convention here: in this exposé the notation $k(A)$ will mean only the group defined by (1.5.2).

By the long exact cohomology sequence, f is additive in the sense of (1.1), hence defines a homomorphism

$$\varphi : k(D^b_A(C)) \longrightarrow k(A).$$

It follows at once from (1.4) that φ is inverse to the canonical homomorphism $k(A) \to k(D^b_A(C))$.

Remark 1.6.1. In the situation of (1.6), the canonical functor

$$D^b(A) \longrightarrow D^b_A(C)$$

is not in general an equivalence of categories. This is shown by the example where S is a scheme, C is the category of $\mathcal{O}_S$-modules, and A is the category of quasi-coherent $\mathcal{O}_S$-modules (II App. I 0.3). See, however, the criterion (II 2.1.3).

Definition 1.7. Let C be a triangulated category, respectively an abelian category, and let C' be a subset of $\operatorname{ob} C$. We shall say that C' is an *exact subset* if, for every distinguished triangle

$$E' \longrightarrow E \longrightarrow E'' \longrightarrow E'[1],$$

respectively for every exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0,$$

such that two of the terms E', E, E'' belong to C', the third also belongs to C'. We shall say that C' is *dense* in $\operatorname{ob} C$ if $\operatorname{ob} C$ is the smallest exact subset of $\operatorname{ob} C$ containing C'.

Lemma 1.8. Let C be a triangulated category, respectively an abelian category, and let C' be a dense subset of $\operatorname{ob} C$. Then the elements $\operatorname{cl}_C(E)$, for $E \in C'$, generate $k(C)$.

Proof. Put $C' = C_0$, and for $i \geq 1$ define C_i recursively as the set of objects of C occurring in a distinguished triangle, respectively a short exact sequence, two of whose terms belong to C_{i-1}. The C_i form an increasing filtration of $\operatorname{ob} C$, which is exhaustive by hypothesis. For $i \geq 0$, let $k(C)_i$ be the subgroup of $k(C)$ generated by the elements $\operatorname{cl}_C(E)$, $E \in C_i$. It is immediate from the definitions that $k(C)_i = k(C)_{i+1}$ for every $i \geq 0$, hence $k(C)_i = k(C)_0$. If $x \in k(C)$, one can write $x = \operatorname{cl}(E') - \operatorname{cl}(E'')$ for $E', E'' \in \operatorname{ob} C$ (in the non-respectable case one may even arrange $E'' = 0$). Since the filtration (C_i) is exhaustive, there exists $i \geq 0$ such that E' and E'' belong to C_i; hence $x \in k(C)_i = k(C)_0$, as required.

Exercise 1.9. Let C be an abelian category and A a full subcategory stable under finite sums, subobjects, and quotients, such that $\operatorname{ob} A$ is dense in $\operatorname{ob} C$. Then the homomorphism

$$k(A, C) \longrightarrow k(C)$$

defined by the additive function $E \mapsto \operatorname{cl}_C(E)$ on $\operatorname{ob} A$ (cf. (1.5)) is an isomorphism.

Hint. Put $A = A_0$, and define A_i, for $i \geq 1$, as the full subcategory of C consisting of objects of C which are extensions of two objects of A_{i-1}. Prove that A_i satisfies the same hypotheses as A, and then prove that the canonical homomorphism

$$\varinjlim_i k(A_i, C) \longrightarrow k(C)$$

is an isomorphism. The desired conclusion follows at once.

2. The functors $K_\bullet$ and K' of a ringed topos

2.1. Let (X, A) be a ringed topos. Recall the following notation:

$$D^b(A_X)_{\text{coh}}$$

is the category of pseudo-coherent complexes with bounded cohomology of left A-modules (I 2.15);

$$D(A_X)_{\mathrm{parf}}$$

is the category of complexes of left A-modules of finite perfect amplitude (I 5.8.4);

$$\mathrm{Loclib}(A_X)$$

is the category of left A-modules locally direct factors of finite free modules (II 2.0);

$$\mathrm{Coh}(A_X)$$

is the category of coherent left A-modules, relative to the category of finite-type free A-modules (I 3.1).

The last two categories are full subcategories of the category $\mathrm{Mod}(A_X)$ of left A-modules, and the first two are full triangulated subcategories of

$$D(A_X) = D(\mathrm{Mod}(A_X)).$$

Definition 2.2. Let (X, A) be a ringed topos. Unless otherwise stated, we shall put

$$K_\bullet(A_X) = k(D^b(A_X)_{\mathrm{coh}}), \qquad K'(A_X) = k(D(A_X)_{\mathrm{parf}}),$$

where $k(\ )$ is defined in (1.1), and

$$K_\bullet(A_X)_{\mathrm{naif}} = k(\mathrm{Coh}(A_X)), \qquad K'(A_X)_{\mathrm{naif}} = k(\mathrm{Loclib}(A_X)),$$

where $k(\ )$ is defined by (1.5.2), the ambient abelian category being $\mathrm{Mod}(A_X)$.[23] We say that $K_\bullet(A_X)$, respectively $K'(A_X)$, is the Grothendieck group of pseudo-coherent, respectively perfect, A-module complexes on X, and that $K_\bullet(A_X)_{\mathrm{naif}}$, respectively $K'(A_X)_{\mathrm{naif}}$, is the Grothendieck group of coherent, respectively locally free, left A-modules.

Remarks 2.3.

a) One can define variants of $K_\bullet(\ )$ and $K'(\ )$ by putting

$$K_\bullet^{\mathrm{loc}}(A_X) = k(D^{\mathrm{loc}b}(A_X)_{\mathrm{coh}}),$$

where "loc b" means "with locally bounded cohomology", and

$$K'_{\mathrm{loc}}(A_X) = k(D(A_X)_{\mathrm{parf}}).$$

These groups coincide with those defined in (2.2) when the final object of X is quasi-compact, but in general they lend themselves less well to computations in intersection theory.

b) The group $K_\bullet(A_X)_{\mathrm{naif}}$ has practically no interest unless A is coherent as a left A-module; otherwise there is a strong risk that very few coherent sheaves exist.

c) Considering the k of triangulated categories which are "too large" is of no interest. Thus, from (1.2 b), one immediately deduces the relations

$$k(D(A_X)) = k(D^-(A_X)) = k(D^+(A_X)) = k(D^b(A_X)) = k(D^-(A_X)_{\mathrm{coh}}) = 0,$$

for the last equality observing that $L \in \mathrm{ob}\, D^-(A_X)_{\mathrm{coh}}$ implies

$$\coprod_{i\geq 0} L[2i] \in \mathrm{ob}\, D^-(A_X)_{\mathrm{coh}}.$$

[23] We shall allow ourselves to omit the index A when no confusion can result. We shall sometimes also drop the index "naive", but will then say so explicitly.

2.4. Case of a coherent sheaf of rings. Let (X, A) be a ringed topos, with A assumed coherent as a left A-module (I 3.1 and 3.6). Every coherent left A-module is a pseudo-coherent complex (I 3.5), and one has a fully faithful functor

$$\mathrm{Coh}(A_X) \longrightarrow D^b(A_X)_{\mathrm{coh}}. \tag{2.4.1}$$

Moreover $\mathrm{Coh}(A_X)$ is a thick subcategory of $\mathrm{Mod}(A_X)$ (I 3.4), and, by (I 3.5), $D^b(A_X)_{\mathrm{coh}}$ is the full subcategory of $D^b(A_X)$ formed by the objects L such that $H^i(L) \in \mathrm{ob}\,\mathrm{Coh}(A_X)$ for all i. Consequently, it follows from (1.6) that the homomorphism

$$K_\bullet(A_X)_{\mathrm{naif}} \longrightarrow K_\bullet(A_X) \tag{2.4.2}$$

defined by (2.4.1) is an isomorphism.

2.5. The homomorphism $K' \to K_\bullet$. Let (X, A) be a ringed topos. The inclusion functor

$$D(A_X)_{\mathrm{parf}} \longrightarrow D^b(A_X)_{\mathrm{coh}}$$

defines a homomorphism

$$K'(A_X) \longrightarrow K_\bullet(A_X). \tag{2.5.1}$$

It is not in general injective or surjective. However, assume the following hypotheses:

(i) the final object of X is quasi-compact;

(ii) X has enough points, and for every point x of X, every left A_x-module has finite Tor-dimension.

Then, by (I 5.10),

$$D(A_X)_{\mathrm{parf}} = D^b(A_X)_{\mathrm{coh}},$$

and (2.5.1) is an isomorphism. The hypotheses (i) and (ii) are satisfied, for example, by a compact space ringed by regular local rings.

2.6. Pairings between K' and K' in the non-commutative case. Let

$$f : (X', A') \longrightarrow (X, A)$$

be a morphism of ringed topoi, and denote by $\mathrm{Mod}(A', X', A)$ the category of bimodules which are left A'-modules and right $f^{-1}(A)$-modules. Let

$$\mathrm{Loclib}^b_A(A', X', A)$$

be the full subcategory formed by the objects which belong, as objects over A', to $\mathrm{Loclib}(A', X')$. By analogy with (2.2), put

$$K'_{\mathrm{naif}}(A', X', A)_{A'} = k(\mathrm{Loclib}^b_A(A', X', A)).$$

Tensor product

$$\otimes_A : \mathrm{Mod}(A', X', A) \times \mathrm{Mod}(A_X) \longrightarrow \mathrm{Mod}(A', X')$$

sends

$$(E', E) \longmapsto E' \otimes_{f^{-1}(A)} f^{-1}(E),$$

and induces a functor

$$\mathrm{Loclib}^b_A(A', X', A) \times \mathrm{Loclib}(A_X) \longrightarrow \mathrm{Loclib}(A', X'). \tag{2.6.1}$$

The map $(E', E) \mapsto \mathrm{cl}(E' \otimes_A E)$ deduced from (2.6.1) is biadditive in the sense of (1.5), and therefore defines a homomorphism

$$K'_{\mathrm{naif}}(A', X', A)_{A'} \otimes K'_{\mathrm{naif}}(A_X) \longrightarrow K'_{\mathrm{naif}}(A', X'). \tag{2.6.2}$$

Let $D^-(A', X', A)$ be the derived category of $\mathrm{Mod}(A', X', A)$, and let

$$D^-(A', X', A)_{A'\text{-}\mathrm{parf}}$$

be the full subcategory formed by the complexes of finite perfect amplitude relative to A', i.e. as objects of $D^-(A', X')$. Put, as in (2.2),

$$K'(A', X', A)_{A'} = k(D^-(A', X', A)_{A'\text{-}\mathrm{parf}}).$$

The inclusion functors define evident homomorphisms

$$K'_{\mathrm{naif}}(A', X', A)_{A'} \longrightarrow K'(A', X', A)_{A'}, \tag{2.6.3.1}$$
$$K'_{\mathrm{naif}}(A_X) \longrightarrow K'(A_X), \tag{2.6.3.2}$$
$$K'_{\mathrm{naif}}(A', X') \longrightarrow K'(A', X'). \tag{2.6.3.3}$$

On the other hand, one knows (I 4.18, 4.19) that the derived tensor functor

$$\otimes^{\mathbf{L}}_A : D^-(A', X', A) \times D^-(A_X) \longrightarrow D^-(A', X')$$

induces an exact functor

$$D^-(A', X', A)_{A'\text{-}\mathrm{parf}} \times D(A_X)_{\mathrm{parf}} \longrightarrow D(A', X')_{\mathrm{parf}}. \tag{2.6.4}$$

The latter defines a homomorphism

$$K'(A', X', A)_{A'} \otimes K'(A_X) \longrightarrow K'(A', X'). \tag{2.6.5}$$

Since (2.6.1) is induced by (2.6.4), the homomorphisms (2.6.2) and (2.6.5) are compatible with the homomorphisms (2.6.3.1), (2.6.3.2), and (2.6.3.3), in other words one has a commutative diagram

$$\begin{tikzcd} K'_{\mathrm{naif}}(A', X', A)_{A'} \otimes K'_{\mathrm{naif}}(A_X) \arrow[r] \arrow[d] & K'_{\mathrm{naif}}(A', X') \arrow[d] \\ K'(A', X', A)_{A'} \otimes K'(A_X) \arrow[r] & K'(A', X'), \end{tikzcd} \tag{2.6.6}$$

where the upper and lower horizontal arrows are respectively (2.6.2) and (2.6.5), and the vertical arrows are defined by (2.6.3.1), (2.6.3.2), and (2.6.3.3).

2.7. Contravariant law and ring structure on K'. We spell out some important particular cases of the pairings (2.6), although their direct study is immediate.

a) Let

$$f : (X', A') \longrightarrow (X, A)$$

be a morphism of ringed topoi. The functor

$$Lf^* : D^-(A_X) \longrightarrow D^-(A', X')$$

sends $D(A_X)_{\mathrm{parf}}$ to $D(A', X')_{\mathrm{parf}}$ (I 4.19.1), and likewise sends $\mathrm{Loclib}(A_X)$ to $\mathrm{Loclib}(A', X')$. Thus it defines a homomorphism

$$f^* : K'(A_X) \longrightarrow K'(A', X') \tag{2.7.1}$$

and, respectively,

$$f^* : K'(A_X)_{\mathrm{naif}} \longrightarrow K'(A', X')_{\mathrm{naif}}. \tag{2.7.2}$$

This can also be interpreted as multiplication by $\mathrm{cl}(A')$ in (2.6.2), respectively (2.6.5). One has a commutative diagram, a special case of (2.6.6),

$$\begin{array}{ccc} K'(A_X)_{\mathrm{naif}} & \xrightarrow{f^*} & K'(A', X')_{\mathrm{naif}} \\ \downarrow & & \downarrow \\ K'(A_X) & \xrightarrow{f^*} & K'(A', X'), \end{array}$$

where the left and right vertical arrows are respectively (2.6.3.2) and (2.6.3.3).

Let $g : (X, A) \to (X'', A'')$ be a morphism of ringed topoi. The transitivity of inverse-image functors implies that the homomorphisms of type (2.7.1), respectively (2.7.2), satisfy

$$(gf)^* = f^* g^*. \tag{2.7.3}$$

Thus $K'(A_X)$, respectively $K'(A_X)_{\mathrm{naif}}$, is a contravariant functor of (X, A).

b) Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos. Thanks to the associativity and commutativity of tensor product, (2.6.5), respectively (2.6.2), with $A = \mathcal{O}_X$, $(X', A') = (X, A)$, and $f = \mathrm{Id}$, endows $K'(X)$, respectively $K'(X)_{\mathrm{naif}}$, with the structure of a commutative ring with unit

$$1 = \mathrm{cl}(\mathcal{O}_X).$$

The canonical homomorphism

$$K'(X)_{\mathrm{naif}} \longrightarrow K'(X)$$

is a homomorphism of unital rings.

If $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of commutatively ringed topoi, compatibility of tensor product with inverse image shows that

$$f^* : K'(Y) \longrightarrow K'(X),$$

respectively

$$f^* : K'(Y)_{\mathrm{naif}} \longrightarrow K'(X)_{\mathrm{naif}},$$

is a homomorphism of unital rings.

2.8. Augmentation of K' toward $H^0(\ ,\mathbb{Z})$. Let $(X, \mathcal{O}_X)$ be a locally ringed topos (I 2.15.1). Recall that one has defined (I 6.11 to 6.14) a homomorphism of unital rings

$$i : H^0(X, \mathbb{Z}) \longrightarrow K'(X)_{\mathrm{naif}},$$

thereby endowing $K'(X)_{\mathrm{naif}}$, and then $K'(X)$ through the canonical homomorphism $K'(X)_{\mathrm{naif}} \to K'(X)$, with the structure of a unital algebra over $H^0(X, \mathbb{Z})$. Moreover, $K'(X)_{\mathrm{naif}}$, respectively $K'(X)$, is augmented toward $H^0(X, \mathbb{Z})$ by the rank, and the canonical homomorphism $K'(X)_{\mathrm{naif}} \to K'(X)$ is a homomorphism of augmented algebras.

If $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of locally ringed topoi, the homomorphisms

$$f^* : K'(Y) \longrightarrow K'(X), \qquad f^* : K'(Y)_{\mathrm{naif}} \longrightarrow K'(X)_{\mathrm{naif}}$$

are compatible (I 6.12, 6.13) with the preceding augmented-algebra structures; this is expressed by the commutativity of the diagram, where rg denotes the rank,

$$\begin{array}{ccccccc} H^0(Y,\mathbb{Z}) & \xrightarrow{i} & K'(Y)_{\mathrm{naif}} & \to & K'(Y) & \xrightarrow{\mathrm{rg}} & H^0(Y,\mathbb{Z}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H^0(X,\mathbb{Z}) & \xrightarrow{i} & K'(X)_{\mathrm{naif}} & \to & K'(X) & \xrightarrow{\mathrm{rg}} & H^0(X,\mathbb{Z}). \end{array}$$

Later in the Seminar (V 3.9.1 and VI) one will see that $K'(X)_{\mathrm{naif}}$ can be endowed with a structure richer than that of an algebra augmented toward $H^0(X,\mathbb{Z})$, namely with a structure of a λ-algebra augmented toward $H^0(X,\mathbb{Z})$, viewed as a binomial ring (V 2.8), this structure being compatible with inverse-image homomorphisms. Deligne has recently proved that $K'(X)$ can be endowed with an analogous structure, at least when X is of characteristic $p \geq 0$, and that the canonical homomorphism

$$K'(X)_{\mathrm{naif}} \longrightarrow K'(X)$$

is a homomorphism of augmented λ-algebras, thereby answering affirmatively a conjecture of Grothendieck (XIV 1).

2.9. Comparison of K' and K'_{naif}. Let (X, A) be a ringed topos. There is in general no reason for the canonical homomorphism

$$K'(X)_{\mathrm{naif}} \longrightarrow K'(X)$$

to be injective or surjective. However, the "global resolution existence criteria" of Exposé II show that it is an isomorphism in the following cases:

(i) X is the Zariski topos of a separated or noetherian divisorial scheme, and $A = \mathcal{O}_X$ (II 2.2.9);

(ii) X is the topos defined by a compact space and A is a soft sheaf (II 2.3.2);

(iii) X is a Stein topos whose final object is quasi-compact and $A = \mathcal{O}_X$ (II 2.4.5).

2.10. Pairing between K' and $K_\bullet$. Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos. One knows (I 5.8.4 b) that there is a pairing

$$\otimes^{\mathbf{L}}_{\mathcal{O}_X} : D(X)_{\mathrm{parf}} \times D^b(X)_{\mathrm{coh}} \longrightarrow D^b(X)_{\mathrm{coh}}.$$

It defines a homomorphism

$$K'(X) \otimes K_\bullet(X) \longrightarrow K_\bullet(X). \tag{2.10.1}$$

Thanks to associativity of tensor product, (2.10.1) endows $K_\bullet(X)$ with the structure of a $K'(X)$-module. More generally, if

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

is a morphism of commutatively ringed topoi, $K_\bullet(X)$ may be regarded as a $K'(Y)$-module through the homomorphism

$$f^* : K'(Y) \longrightarrow K'(X).$$

2.11. Covariant law on $K_\bullet$ and projection formula. Let (X, A) be a ringed topos. In contrast with $K'(A_X)$, the group $K_\bullet(A_X)$ has no natural variance with respect to (X, A). Nevertheless, in algebraic geometry and analytic geometry, $K_\bullet(X)$ tends to be covariant in X for proper morphisms. We now make this precise.

2.11.1. Let $f : X \to Y$ be a proper and pseudo-coherent morphism (III 1.2) of quasi-compact schemes. Then, subject to the validity of the finiteness conjecture for f (III 2.1), hence in particular if Y is noetherian (in which case the pseudo-coherence hypothesis on f is automatically verified (III 1.3)) or if f is projective, the functor Rf_* induces, by (III 2.6), a functor

$$Rf_* : D^b(X)_{\mathrm{coh}} \longrightarrow D^b(Y)_{\mathrm{coh}}.$$

It therefore defines a homomorphism

$$f_* : K_\bullet(X) \longrightarrow K_\bullet(Y). \tag{2.11.1.1}$$

This homomorphism is $K'(Y)$-linear (cf. (2.10)); in other words, for $y \in K'(Y)$ and $x \in K_\bullet(X)$, one has the *projection formula*

$$f_*(yx) = yf_*(x). \tag{2.11.1.2}$$

Indeed, if $x = \mathrm{cl}(E)$ for $E \in \mathrm{ob}\, D^b(X)_{\mathrm{coh}}$ and $y = \mathrm{cl}(M)$ for $M \in \mathrm{ob}\, D(Y)_{\mathrm{parf}}$, then by (III 3.7) one has

$$Rf_*(E) \otimes^{\mathbf{L}}_Y M \simeq Rf_*(E \otimes^{\mathbf{L}}_Y M).$$

Finally, if $g : Y \to Z$ is a proper morphism satisfying the same hypotheses as f, and for which the finiteness conjecture (III 2.1) is true, the transitivity of direct-image functors gives, for the homomorphisms (2.11.1.1), the formula

$$g_* f_* = (gf)_*. \tag{2.11.1.3}$$

2.11.2. Let $f : X \to Y$ be a proper morphism of finite-dimensional analytic spaces. Then, by Grauert's finiteness theorem [2], Rf_* sends $D^b(X)_{\mathrm{coh}}$ into $D^b(Y)_{\mathrm{coh}}$, and hence defines a homomorphism

$$f_* : K_\bullet(X) \longrightarrow K_\bullet(Y). \tag{2.11.2.1}$$

Like its algebraic analogue (2.11.1.1), this homomorphism is $K'(Y)$-linear; equivalently, formula (2.11.1.2) is valid, as follows from the projection formula at the level of derived categories (III 5.9 b). One also has the transitivity formula (2.11.1.3).

2.11.3. Outside the algebraic or analytic framework, this covariance of $K_\bullet$ for proper morphisms disappears. For instance, let

$$f : X \to Y$$

be a proper morphism of topological spaces ringed by the sheaves of complex continuous functions. Then, if $E \in \mathrm{ob}\, D^b(X)_{\mathrm{coh}}$, in general $Rf_*(E)$ need not belong to $D^b(Y)_{\mathrm{coh}}$, as is already evident for $E = \mathcal{O}_X$ and Y reduced to a point. In fact, the definition (2.2) of $K_\bullet$ is especially adapted to the needs of algebraic and analytic geometry, and a good definition of $K_\bullet$ for a topological space remains to be found. It seems, moreover, that one will have to leave the framework of $\mathcal{O}_X$-linear complexes, and define, for a morphism $f : X \to Y$ of topological spaces, or of mixed varieties (II App. II), a suitable class of complexes of topological vector-sheaves on X, meant to contain, when f is the structural morphism of a mixed variety, the elliptic complexes on X relative to Y. This is suggested by the finiteness theorem (II App. II), as well as by the analogous theorem, where Y is assumed reduced to a point, proved by Grothendieck in [3].

2.12. Unusual variances of K' and $K_\bullet$. Let

$$f : (X, A) \longrightarrow (Y, B)$$

be a morphism of finite Tor-dimension of ringed topoi (III 3.1). The functor

$$Lf^* : D^-(B_Y) \longrightarrow D^-(A_X)$$

sends $D^b(B_Y)_{\mathrm{coh}}$ into $D^b(A_X)_{\mathrm{coh}}$ (I 2.16.1), and therefore defines a homomorphism

$$f^* : K_\bullet(B_Y) \longrightarrow K_\bullet(A_X). \tag{2.12.1}$$

If A and B are commutative, commutation of inverse image with tensor product shows that (2.12.1) is compatible with the module structures of $K_\bullet(Y)$ and $K_\bullet(X)$ over $K'(Y)$ and $K'(X)$ respectively (2.10). In other words, for $a \in K'(Y)$ and $y \in K_\bullet(Y)$, one has

$$f^*(ay) = af^*(y). \tag{2.12.2}$$

If $g : (Y, B) \to (Z, C)$ is a morphism of finite Tor-dimension of ringed topoi, transitivity of inverse-image functors implies that the homomorphisms (2.12.1) satisfy

$$f^*g^* = (gf)^*. \tag{2.12.1.1}$$

Now suppose that we are in the situation of (2.11.1), respectively (2.11.2), with f of finite Tor-dimension. Then by (III 4.8.1 and 5.9 b), the functor Rf_* sends $D(X)_{\mathrm{parf}}$ into $D(Y)_{\mathrm{parf}}$, and hence defines a homomorphism

$$f_* : K'(X) \longrightarrow K'(Y). \tag{2.12.3}$$

This is not generally a ring homomorphism. Nevertheless, for $y \in K_\bullet(Y)$, respectively $K'(Y)$, and $x \in K'(X)$, one has

$$f_*(f^*(y)x) = yf_*(x), \tag{2.12.4}$$

where f^* is the homomorphism (2.12.1), respectively (2.7.1). The proof is analogous to that of (2.11.1.2). If $g : Y \to Z$ is a morphism satisfying the same hypotheses as f, the homomorphisms (2.12.3) satisfy the transitivity formula

$$g_*f_* = (gf)_*. \tag{2.12.3.1}$$

The homomorphism (2.12.3) is one of the essential ingredients of the Riemann–Roch theorem, both in algebraic geometry and in analytic geometry, for the indications in the analytic case see Exposé 0, 5.

Suppose that X and Y are separated and divisorial schemes (II 2.2.5), for example schemes possessing ample invertible sheaves. Then (2.9 (i)) identifies $K'(X)$, respectively $K'(Y)$, with $K'(X)_{\mathrm{naif}}$, respectively $K'(Y)_{\mathrm{naif}}$, and (2.12.3) therefore defines a homomorphism

$$f_* : K'(X)_{\mathrm{naif}} \longrightarrow K'(Y)_{\mathrm{naif}}. \tag{2.12.3'}$$

As was already explained in Exposé 0, 3.1, the definition of (2.12.3') was far from evident a priori. In practice, it is precisely one of the main motivations of this Seminar, from Exposé I through Exposé IV.

3. Complements on Grothendieck groups of schemes

The results of this section will not be used in Exposés V to VIII.

Proposition 3.1.0. Let

$$\begin{tikzcd} X' \arrow[r, "g"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "h"'] & Y \end{tikzcd}$$

be a cartesian square of schemes, where Y is quasi-compact and f is quasi-compact and quasi-separated. Let $E \in \operatorname{ob} D^b(X)$ be a complex with quasi-coherent cohomology. Suppose that X and Y' are Tor-independent over Y (III 1.5). Suppose also either that h has finite Tor-dimension, or that E has finite flat amplitude relative to f. Under these conditions, the base-change arrow

$$Lh^* \, Rf_* E \longrightarrow Rf'_* \, Lg^* E$$

is defined and is an isomorphism.

Proof. To define the arrow, one uses the trivial duality formula (SGA 4 XVII 2.3) for h, which says that there is a canonical functorial isomorphism

$$\operatorname{Hom}(Lh^* L, M) \xrightarrow{\sim} \operatorname{Hom}(L, Rh_* M)$$

for $L \in \operatorname{ob} D^-(Y)$ and $M \in \operatorname{ob} D^+(Y')$. Using (III 3.5.2) and (III 3.7.1), one easily verifies that one may take $L = Rf_* E$ and $M = Rf'_* Lg^* E$, so that applying this formula reduces the problem to defining an arrow

$$Rf_* E \longrightarrow Rf_* Rg_* Lg^* E.$$

For this arrow we take the image under Rf_* of the adjunction map

$$E \longrightarrow Rg_* Lg^* E,$$

which comes from the trivial duality formula for g. The base-change arrow is therefore defined.

It remains to show that it is an isomorphism. For this, one may first suppose that Y and Y' are affine. Then, by the Leray spectral sequence of an open covering of X, one is reduced to the case where X and X' are affine. One thus has a cartesian square of rings

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A', \end{array}$$

and one may suppose that E is defined by a bounded-above complex of B-modules. The Tor-independence of A' and B over A, written

$$B \otimes^{\mathbf{L}}_A A' \xrightarrow{\sim} B',$$

then implies, in $D(A')$,

$$E \otimes^{\mathbf{L}}_A A' \xrightarrow{\sim} E \otimes^{\mathbf{L}}_B B',$$

which proves the proposition.

Translating (3.1.0) into terms of Grothendieck groups gives:

Proposition 3.1.1. In the cartesian square of schemes

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[h]{} & Y, \end{array}$$

suppose that Y and Y' are quasi-compact, that X and Y' are Tor-independent over Y (III 1.5), that f is proper and pseudo-coherent, and that the finiteness conjecture (III 2.1) is valid for f and f', for example if Y and Y' are noetherian, or if f is projective. Then, if f is perfect, respectively if h has finite Tor-dimension, for $x \in K'(X)$, respectively $x \in K_\bullet(X)$, one has

$$h^* f_*(x) = f'_* g^*(x),$$

where h^*, g^* are defined by (2.7.1), respectively (2.12.1), and f_*, f'_* by (2.12.3), respectively (2.11.1.1). Notice that by (III 4.7.1) f' is perfect if f is, and by (III 3.5.1) g has finite Tor-dimension if h does.

3.2. Passage to the limit on K'_{naif} and $K_\bullet$. Let $(X_i, f_{ij})_{i \in I}$ be a filtered projective system of quasi-compact and quasi-separated schemes whose transition morphisms are affine (cf. EGA IV 8.2.2). Put

$$X = \varprojlim_i X_i,$$

and let $f_i : X \to X_i$ denote the canonical morphism.

3.2.1. The homomorphisms (2.7.2)

$$f_i^* : K'(X_i)_{\text{naif}} \longrightarrow K'(X)_{\text{naif}}$$

define, by the transitivity formula (2.7.3), a homomorphism

$$\varinjlim_i K'(X_i)_{\text{naif}} \longrightarrow K'(X)_{\text{naif}}. \tag{3.2.1.1}$$

I say that this homomorphism is an isomorphism. Indeed, let E be a locally free $\mathcal{O}_X$-module of finite type. By EGA IV 8.5.2 and 8.5.5, there exists an index i and a locally free $\mathcal{O}_{X_i}$-module of finite type E_i such that

$$E \simeq f_i^*(E_i).$$

By EGA IV 8.5.2.5, the image of $\mathrm{cl}(E_i)$ in $\varinjlim_i K'(X_i)_{\text{naif}}$ does not depend on the choice of E_i; denote it by $u(E)$. By EGA IV 8.5.2 and 8.5.6, the map

$$u : \operatorname{ob} \operatorname{Loclib}(X) \longrightarrow \varinjlim_i K'(X_i)_{\text{naif}}$$

so defined is additive in the sense of (1.5). It therefore defines a homomorphism

$$K'(X)_{\text{naif}} \longrightarrow \varinjlim_i K'(X_i)_{\text{naif}},$$

and one verifies immediately, using EGA IV 8.5.2, that it is inverse to (3.2.1.1). This proves the assertion.

3.2.2. Likewise one defines a canonical homomorphism

$$\varinjlim_i K'(X_i) \longrightarrow K'(X). \tag{3.2.2.1}$$

Using Deligne's localization technique (SGA 4 XVII), which implies that the category of perfect complexes on X is equivalent to the category

$$\varinjlim_i \operatorname{Parf}(X_i)$$

of the analogous categories of perfect complexes on the X_i, one proves again that (3.2.2.1) is an isomorphism.

3.2.3. Suppose that the X_i, and X, are noetherian, and that the morphisms f_i are flat (then so are the f_{ij}). Then the homomorphisms (2.12.1)

$$f_i^* : K_\bullet(X_i) \longrightarrow K_\bullet(X)$$

define an isomorphism

$$\varinjlim_i K_\bullet(X_i) \xrightarrow{\sim} K_\bullet(X).$$

The proof is analogous to that of (3.2.1), using the fact (2.4) that $K_\bullet(X) \simeq K_\bullet(X)_{\text{naif}}$ is computed in terms of coherent sheaves on X.

Moreover, the module structures of the $K_\bullet(X_i)$ over the $K'(X_i)_{\text{naif}}$ are compatible with the inverse images f_{ij}^* by (2.7.1) and (2.12.2). Passing to the limit, one finds that $\varinjlim_i K_\bullet(X_i)$ is a module over $\varinjlim_i K'(X_i)_{\text{naif}}$; this is compatible, through the isomorphisms (3.2.1.1) and (3.2.2.1), with the module structure of $K_\bullet(X)$ over $K'(X)_{\text{naif}}$. In other words, one has a commutative diagram

$$\begin{tikzcd} \varinjlim K'(X_i)_{\text{naif}} \otimes \varinjlim K_\bullet(X_i) \arrow[r] \arrow[d, "\sim"'] & \varinjlim K_\bullet(X_i) \arrow[d, "\sim"] \\ K'(X)_{\text{naif}} \otimes K_\bullet(X) \arrow[r] & K_\bullet(X). \end{tikzcd}$$

3.3. Relative Grothendieck groups.

3.3.1. Let

$$f : X \longrightarrow Y$$

be a morphism of schemes locally of finite type. Put

$$K'(f) = k(D(f)_{\text{parf}}), \qquad K_\bullet(f) = k(D^b(f)_{\text{coh}}),$$

where $D^b(f)_{\text{coh}}$ denotes the category of complexes which are pseudo-coherent relative to f and have bounded cohomology (III 1.2), and $D(f)_{\text{parf}}$ the category of complexes of finite perfect amplitude relative to f (III 4.2). When there is no possible ambiguity, we shall sometimes write $K'(X/Y)$, respectively $K_\bullet(X/Y)$, instead of $K'(f)$, respectively $K_\bullet(f)$.

The groups $K'(f)$ and $K_\bullet(f)$ do not possess the rich structure of their absolute analogues $K'(X)$ and $K_\bullet(X)$, for tensor product does not in general define homomorphisms

$$K'(f) \otimes K'(f) \longrightarrow K'(f), \qquad K'(f) \otimes K_\bullet(f) \longrightarrow K_\bullet(f).$$

On the other hand, they satisfy interesting variance properties, which we now review.

3.3.2. Let

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "h"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

be a commutative triangle of morphisms locally of finite type of schemes.

3.3.2.1. If g is pseudo-coherent, one has

$$K_\bullet(f) = K_\bullet(h)$$

(III 1.12). If g is smooth and X quasi-compact, one has

$$K'(f) = K'(h)$$

(III 4.4.1).

3.3.2.2. By (III 1.11 and 4.5), the functor $\otimes^{\mathbf{L}}_Y$ defines pairings

$$D(f)_{\text{parf}} \times D^b(g)_{\text{coh}} \longrightarrow D^b(h)_{\text{coh}},$$

$$D(f)_{\text{parf}} \times D(g)_{\text{parf}} \longrightarrow D(h)_{\text{parf}},$$

which induce homomorphisms

$$K'(f) \otimes K_\bullet(g) \longrightarrow K_\bullet(h),$$

$$K'(f) \otimes K'(g) \longrightarrow K'(h).$$

In particular, taking $X = Y$ and $f = \mathrm{Id}$, one deduces that $K_\bullet(g)$ is a $K'(Y)$-module. On the other hand, if f is perfect, product by

$$\mathrm{cl}(\mathcal{O}_X) \in K'(f)$$

defines $K'(Y)$-linear homomorphisms

$$f^* : K_\bullet(Y/S) \longrightarrow K_\bullet(X/S),$$

and

$$f^* : K'(Y/S) \longrightarrow K'(X/S).$$

3.3.2.3. Suppose S quasi-compact, X and Y of finite type over S, and f proper. Then, subject to the validity of the finiteness conjecture (III 2.1), in particular if Y is noetherian or if f is projective, the functor Rf_* sends $D^b(h)_{\mathrm{coh}}$ into $D^b(g)_{\mathrm{coh}}$, and therefore defines a homomorphism

$$f_* : K_\bullet(X/S) \longrightarrow K_\bullet(Y/S).$$

If in addition f is perfect, Rf_* sends $D(h)_{\mathrm{parf}}$ into $D(g)_{\mathrm{parf}}$ (III 4.8), and defines a homomorphism

$$f_* : K'(X/S) \longrightarrow K'(Y/S).$$

From the projection formula (III 3.7) one deduces

$$f_*(ax) = af_*(x),$$

for f proper, $a \in K'(S)$, $x \in K_\bullet(X)$, respectively for f proper and perfect, $a \in K_\bullet(S)$, $x \in K'(X/S)$.

3.3.2.4. Suppose S noetherian, X and Y of finite type over S, and f compactifiable and perfect. Then by (III 4.9.1), the functor $f^!$ sends $D(g)_{\mathrm{parf}}$ into $D(h)_{\mathrm{parf}}$, and defines a homomorphism

$$f^! : K'(Y/S) \longrightarrow K'(X/S).$$

3.3.3. Under the hypotheses of (III 4.7), the functor $\otimes^{\mathbf{L}}_S$ defines pairings

$$D(f_1)_{\mathrm{parf}} \times D^b(f_2)_{\mathrm{coh}} \longrightarrow D^b(f)_{\mathrm{coh}},$$

$$D(f_1)_{\mathrm{parf}} \times D(f_2)_{\mathrm{parf}} \longrightarrow D(f)_{\mathrm{parf}},$$

which induce homomorphisms

$$K'(f_1) \otimes K_\bullet(f_2) \longrightarrow K_\bullet(f),$$

$$K'(f_1) \otimes K'(f_2) \longrightarrow K'(f).$$

Of course, for $X_i = Y_i = S$, one recovers the $K'(S)$-module structure on $K_\bullet(S)$.

If X is flat over S, one deduces, for $n \geq 1$, homomorphisms

$$K'(X/S)^{\otimes n} \longrightarrow K'((X/S)^n),$$

which should, in a sense, replace the absent ring structure on $K'(X/S)$.

Bibliography

[1] Deligne, P., papiers secrets.

[2] Grauert, H., *Ein Theorem der Analytischen Garbentheorie*, Publ. Math. IHES, no. 5 (1960).

[3] Grothendieck, A., *Théorèmes de finitude pour la cohomologie des faisceaux*, Bull. Soc. Math. France, 84, 1956, pp. 1–7.

[V] Verdier, J.-L., *Catégories dérivées*, IHES (1964), miméographié.

Exposé V. Generalities on λ-rings

by P. Berthelot

The theory of λ-rings is useful whenever one uses rings such as the ring of classes of vector bundles on a ringed site, the representation ring of a group, and so on, for which the exterior-power operations define maps λ^i. We shall use it abundantly later in the Seminar. It is therefore necessary first to develop the general formalism. This exposé is accordingly entirely independent of the rest of the Seminar, and may be useful to users of λ-rings outside algebraic geometry.

The first paragraph essentially gives the theory of Hirzebruch's "multiplicative sequences of polynomials" (cf. [2]). Besides its generality, the functorial viewpoint adopted here has the advantage of considerably reducing the calculations. The following paragraphs will show how, to every augmented λ-ring, one can associate a "Chern ring" which permits the development of a theory of Chern classes. This is the theory which will allow us to state the Riemann–Roch theorem in the general form of Exposé VIII.

1. Universal polynomials

1.1. Let K be a commutative ring with unit, fixed throughout this paragraph. Let $\mathcal{C}$ denote the category of filtered commutative K-algebras with unit, whose filtration is decreasing and such that the group of elements of filtration zero is the whole algebra. We define on $\mathcal{C}$ a functor with values in the category of abelian groups as follows. If $A \in \mathrm{Ob}(\mathcal{C})$, associate to A the set of formal series $\psi \in A[[t]]$ satisfying the following conditions: if

$$\psi = a_0 + a_1 t + \cdots + a_n t^n + \cdots ,$$

then

$$a_0 = 1, \qquad \forall i \geq 1, \quad a_i \in \mathrm{Fil}^i A. \tag{1.1.1}$$

This set will be denoted $\widehat{G}_K(A)$, or $\widehat{G}(A)$ when no confusion is possible. It is clearly an abelian group for multiplication of formal series, and this defines a functor

$$\widehat{G}_K : K\text{-Alg.fil.} \longrightarrow \mathrm{Ab}.$$

If A has the gross filtration, then $\widehat{G}(A)$ is the group of formal series with augmentation 1, which will also be denoted

$$1 + A[[t]]^+.$$

Lemma 1.2. The functor $\widehat{G}_K$ is representable.

Let $(X_i)_{i\in\mathbb{N},\ i\geq 1}$ be a family of indeterminates. Assign to X_i the weight i, and filter the ring $K[X_i]_{i\geq 1}$ by weights with respect to the X_i. One then obtains a bijection

$$\widehat{G}_K(A) \xrightarrow{\sim} \operatorname{Hom}_{K\text{-Alg.fil.}}(K[X_i]_{i\geq 1}, A)$$

by associating to every series

$$\psi = 1 + a_1 t + \cdots + a_n t^n + \cdots \in \widehat{G}_K(A)$$

the morphism f_ψ determined by $f_\psi(X_i) = a_i$. This homomorphism is compatible with the filtrations, and one thus obtains an isomorphism of functors.

Put

$$\varphi = 1 + X_1 t + \cdots + X_n t^n + \cdots .$$

For every filtered K-algebra A and every series $\psi \in \widehat{G}(A)$, there exists a unique homomorphism

$$f_\psi \in K[X_i]_{i\geq 1} \longrightarrow A$$

such that

$$\widehat{G}(f_\psi)(\varphi) = \psi.$$

1.3. For every $A \in \mathrm{Ob}(\mathcal{C})$ we define a filtration on $\widehat{G}(A)$ as follows: a series

$$\psi = 1 + a_1 t + \cdots$$

is said to be of filtration $\geq k$ if $a_i = 0$ for $1 \leq i < k$. Let $F^k\widehat{G}(A)$ be the subgroup so obtained. This is a functor on $\mathcal{C}$, and from now on we shall regard $\widehat{G}$ as a functor with values in the category Ab fil of filtered abelian groups by means of this filtration, with $F^0\widehat{G} = \widehat{G}$.

Let H be a functor on $\mathcal{C}$ with values in Ab fil. We shall suppose that H satisfies the following two conditions:

i) $\forall A \in \mathrm{Ob}(\mathcal{C})$, $H(A)$ is separated and complete,

ii) if $A \in \mathrm{Ob}(\mathcal{C})$ is a graded K-algebra, with the filtration associated to its grading, and if a finite group G acts on A by graded morphisms, then $H(A^G) \simeq H(A)^G$.

Here $H(A)^G$ denotes the invariants under G.

In fact we shall need these conditions only when A is a polynomial ring and G is a finite product of symmetric groups acting by permutation of the indeterminates.

Examples.

1) $H(A) = \widehat{G}(A)$,
2) $H(A) = 1 + A[[t]]^+$.

The latter functor will be denoted $\widehat{G}_g$, with g standing for "gross". Finally,

3) $H(A) = \widehat{G}_a(A)$,

where $\widehat{G}_a(A)$ denotes the group, for addition, of formal series

$$a = a_0 + a_1 t + \cdots$$

such that $a_0 = 0$ and $a_i \in \mathrm{Fil}^i A$ for $i \geq 1$, endowed with the filtration analogous to that on $\widehat{G}$.

1.4. For every n, denote by f_n the endomorphism of $K[X_1, \dots, X_n]$ defined by

$$f_n(X_i) = X_i \quad (i < n), \qquad f_n(X_n) = 0.$$

It factors through the inclusion of $K[X_1, \dots, X_{n-1}]$ in $K[X_1, \dots, X_n]$. Hence the homomorphism

$$H(K[X_1, \dots, X_{n-1}]) \longrightarrow H(K[X_1, \dots, X_n])$$

is injective. This will allow us to identify the first group with a subgroup of the second; similarly, we shall regard $H(K[X_1, \dots, X_n])$ as contained in $H(K[X_i]_{i \geq 1})$.

One should note that f_1 is none other than the augmentation of $K[X]$.

1.5. Embed $K[X_1, \dots, X_n]$ in $K[T_1, \dots, T_n]$ by sending X_i to the i-th elementary symmetric function in the T_j. Let the symmetric group Sym_n act by permutation of the T_j. Using the theorem on symmetric polynomials and (1.3.ii), one sees that the induced homomorphism

$$H(K[X_1, \dots, X_n]) \longrightarrow H(K[T_1, \dots, T_n])$$

is injective, which again permits us to identify the former with a subgroup of the latter:

$$H(K[X_1, \dots, X_n]) \simeq H(K[T_1, \dots, T_n])^{\mathrm{Sym}_n} \subset H(K[T_1, \dots, T_n]).$$

Let

$$u_i^n : K[X] \longrightarrow K[T_1, \dots, T_n]$$

be the homomorphism defined by $u_i^n(X) = T_i$, and put

$$u_H^n = \sum_{i=1}^{n} H(u_i^n) : H(K[X]) \longrightarrow H(K[T_1, \dots, T_n]).$$

The symmetric group permutes the morphisms $H(u_i^n)$, and so leaves u_H^n invariant; thus u_H^n takes its values in

$$H(K[T_1, \dots, T_n])^{\mathrm{Sym}_n} \simeq H(K[X_1, \dots, X_n])$$

by (1.3.ii). Notice, finally, that if $H = \widehat{G}$, then

$$u_H^n(1 + Xt) = \prod_{i=1}^{n}(1 + T_i t) = 1 + X_1 t + \cdots + X_n t^n.$$

Theorem 1.6. The map from $\mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G}, H)$ to $H(K[X])$ which, to a morphism of functors, associates its value on the element $1 + Xt$ of $\widehat{G}(K[X])$, is injective. Its image is the subgroup of $H(K[X])$ formed by the elements θ such that, with the notation of 1.4 for the f_n,

i) $H(f_1)(\theta) = 0$,

ii) $\forall n > 1, \quad H(f_n)\big(u_H^n(\theta)\big) \equiv u_H^n(\theta) \pmod{F^n H(K[X_1, \dots, X_n])}$.

First prove injectivity. By Lemma 1.2 it is enough to see that a morphism of functors ξ which is zero on $1 + Xt$ is zero on

$$\varphi = 1 + X_1 t + \cdots + X_n t^n + \cdots .$$

The commutative diagram

$$\begin{tikzcd} \widehat{G}(K[X]) \arrow[r, "u^n_{\widehat{G}}"] \arrow[d, "\xi"'] & \widehat{G}(K[X_1, \dots, X_n]) \arrow[d, "\xi"] \\ H(K[X]) \arrow[r, "u^n_H"'] & H(K[X_1, \dots, X_n]) \end{tikzcd} \tag{1.6.1}$$

shows that, for every n,

$$\xi(1 + X_1 t + \cdots + X_n t^n) = 0.$$

But

$$\varphi \equiv 1 + X_1 t + \cdots + X_n t^n \pmod{F^{n+1}\widehat{G}},$$

hence

$$\xi(\varphi) \equiv \xi(1 + X_1 t + \cdots + X_n t^n) \pmod{F^{n+1}H};$$

therefore $\xi(\varphi) \equiv 0 \pmod{F^{n+1}H}$ for every n. Since $H(K[X_i]_{i\geq 1})$ is separated and complete, $\xi(\varphi) = 0$.

Now let ξ be any element of $\mathrm{Hom}(\widehat{G}, H)$, and set

$$\theta = \xi(1 + Xt).$$

We show that θ satisfies conditions i) and ii). First,

$$H(f_1)(\theta) = H(f_1)\xi(1 + Xt) = \xi\, \widehat{G}(f_1)(1 + Xt) = \xi(1) = 0.$$

Second,

$$u^n_H(\theta) = \xi(1 + \cdots + X_n t^n)$$

by (1.6.1), while

$$1 + \cdots + X_n t^n \equiv 1 + \cdots + X_{n-1} t^{n-1} \pmod{F^n\widehat{G}},$$

and

$$1 + \cdots + X_{n-1} t^{n-1} = \widehat{G}(f_n)(1 + \cdots + X_n t^n).$$

Since $H \circ \widehat{G}(f_n) = H(f_n) \circ \xi$, and since ξ is a morphism of filtered functors, condition ii) follows.

It remains to show that if $\theta \in H(K[X])$ satisfies i) and ii), then there exists $\xi \in \mathrm{Hom}(\widehat{G}, H)$ with

$$\theta = \xi(1 + Xt).$$

First determine the value of ξ on the element $1 + \cdots + X_n t^n$ of $\widehat{G}(K[X_1, \ldots, X_n])$. In view of (1.6.1), it must be, by definition,

$$\xi(1 + \cdots + X_n t^n) = \sum_{i=1}^{n} \xi(1 + T_i t),$$

where the $\xi(1 + T_i t)$ are known by functoriality from the morphisms u^n_i.

One then glues the $\xi(1 + \cdots + X_n t^n)$, for variable n, by means of the commutative diagram

$$\begin{tikzcd} K[X_1, \ldots, X_n] \arrow[r, "f_n"] \arrow[d, "\sigma_n"'] & K[X_1, \ldots, X_n] \arrow[d, "\sigma'_n"] \\ K[T_1, \ldots, T_n] \arrow[r, "f'_n"'] & K[T_1, \ldots, T_n], \end{tikzcd} \tag{1.6.2}$$

where f'_n is determined by $f'_n(T_i) = T_i$ if $i < n$, and $f'_n(T_n) = 0$; where σ_n sends X_i to the i-th elementary symmetric function in $T_1, \ldots, T_n$; and where σ'_n sends X_i to the i-th elementary symmetric function in $T_1, \ldots, T_{n-1}$ for $i < n$, and sends X_n to 0.

Indeed one obtains from this

$$\begin{aligned} H(f_n)(\xi(1 + \cdots + X_n t^n)) &= H(f'_n)\left(\sum_{i=1}^{n} \xi(1 + T_i t)\right) \\ &= \sum_{i=1}^{n-1} \xi(1 + T_i t) = \xi(1 + \cdots + X_{n-1} t^{n-1}), \end{aligned}$$

inside $H(K[T_1, \ldots, T_{n-1}])$ by the identifications of 1.4. By 1.6.ii the elements $\xi(1 + \cdots + X_n t^n)$ define an element ψ of $H(K[X_i]_{i\geq 1})$, since this is a separated complete group. Put

$$\xi(\varphi) = \psi.$$

Finally, if $A \in \mathrm{Ob}(\mathcal{C})$ and $x \in \widehat{G}(A)$, define

$$\xi_A(x) = H(f_x)(\psi),$$

where $f_x : K[X_i]_{i\geq 1} \to A$ is the homomorphism defined in 1.2. Since ξ has already been defined on the elements $1 + \cdots + X_n t^n = \varphi_n$, one must verify that the two definitions give the same result, i.e.

$$H(f_{\varphi_n})(\psi) = \xi(1 + \cdots + X_n t^n).$$

But for every p one has

$$\psi \equiv \xi(1 + \cdots + X_p t^p) \pmod{F^{p+1} H(K[X_i]_{i\geq 1})}.$$

Thus, for $p \geq n$,

$$H(f_{\varphi_n})(\psi) \equiv H(f_{\varphi_n})(\xi(1 + \cdots + X_p t^p)) \pmod{F^{p+1} H(K[X_i]_{i\geq 1})}.$$

Using the identifications of 1.4 and the gluing of the $\xi(\varphi_m)$ for variable m, one obtains

$$H(f_{\varphi_n})(\xi(\varphi_p)) = \xi(\varphi_n).$$

Hence finally

$$\forall p \geq n, \quad H(f_{\varphi_n})(\psi) \equiv \xi(1 + \cdots + X_n t^n) \pmod{F^p H(K[X_1, \ldots, X_n])}.$$

Since H is separated, the required equality follows.

We have thus defined a system of maps

$$\xi_A : \widehat{G}(A) \longrightarrow H(A).$$

To verify that they constitute a morphism of functors with values in groups, one must prove first that, for every morphism $h : A \to A'$, the square

$$\begin{tikzcd}
\widehat{G}(A) \arrow[r, "\widehat{G}(h)"] \arrow[d, "\xi_A"'] & \widehat{G}(A') \arrow[d, "\xi_{A'}"] \\
H(A) \arrow[r, "H(h)"'] & H(A')
\end{tikzcd}$$

is commutative. If $x \in \widehat{G}(A)$, then $h \circ f_x = f_{\widehat{G}(h)(x)}$, whence

$$H(h)(\xi_A(x)) = H(h)H(f_x)(\psi) = H(f_{\widehat{G}(h)(x)})(\psi) = \xi_{A'}(\widehat{G}(h)(x)).$$

It remains only to show that the morphism of functors so constructed is compatible with the structures of filtered groups.

Suppose $x \in F^n\widehat{G}(A)$. To say that x is of filtration $> n$ is to say that $f_x : K[X_i]_{i\geq 1} \to A$ factors through the endomorphism g_n of $K[X_i]_{i\geq 1}$ defined by

$$g_n(X_i) = 0 \quad (1 \leq i \leq n-1), \qquad g_n(X_i) = X_i \quad (i \geq n).$$

To show that ξ is compatible with filtrations, it is enough to show that

$$H(g_n)(\xi(\varphi)) \in F^n H.$$

But by construction

$$\xi(\varphi) \equiv \xi(1 + \cdots + X_{n-1}t^{n-1}) \pmod{F^n H}.$$

Therefore

$$H(g_n)(\xi(\varphi)) = H(g_n)(\xi(1 + \cdots + X_{n-1}t^{n-1})) \equiv 0 \pmod{F^n H}.$$

To verify the additivity of ξ, it suffices to do this for the elements of finite degree of $\widehat{G}(A)$, and then pass to the limit. Let X_1, X_2 be two elements of degree n of $\widehat{G}(A)$, and let X be their product. Consider the commutative diagram

$$\begin{tikzcd} K[X_1, \ldots, X_{2n}] \arrow[r, "f_X"] \arrow[d] & A \\ (K[T_1, \ldots, T_{2n}])^{\mathrm{Sym}_n \times \mathrm{Sym}_n} \arrow[ur, "f'_X"'] & \end{tikzcd}$$

where the first factor of $\mathrm{Sym}_n \times \mathrm{Sym}_n$ permutes $T_1, \ldots, T_n$ and the second permutes $T_{n+1}, \ldots, T_{2n}$. The ring $K[X_1, \ldots, X_{2n}]$ is embedded into the ring of invariants by sending X_j to the j-th elementary symmetric function of the T_i; f' sends the i-th elementary symmetric function of $T_1, \ldots, T_n$, respectively of $T_{n+1}, \ldots, T_{2n}$, to the i-th coefficient of X_1, respectively X_2. Then

$$\xi(X) = H(f'_X)\left(\xi\left(\prod_{i=1}^{2n}(1 + T_i t)\right)\right),$$

$$\xi(X_1) = H(f'_X)\left(\xi\left(\prod_{i=1}^{n}(1 + T_i t)\right)\right),$$

and similarly for X_2, using $i = n+1, \ldots, 2n$. Since $H(f'_X)$ is a homomorphism of groups, we are reduced to the case

$$X = \prod_{i=1}^{2n}(1 + T_i t),$$

and in this case the additivity of ξ follows from the definition of the $\xi(1 + X_1 t + \cdots + X_k t^k)$.

1.7. One can generalize the preceding considerations to the case of a product of functors $\widehat{G}$. The functor $(\widehat{G})^k$ is representable by the ring

$$K[X_i^{(j)}]_{j=1,\ldots,k;\ i\geq 1},$$

where $X_i^{(j)}$ has weight i. We denote by $\varphi^{(j)}$ the series

$$1 + X_1^{(j)}t + \cdots + X_n^{(j)}t^n + \cdots .$$

The element $(\varphi^{(j)})_j$ of

$$\left(\widehat{G}(K[X_i^{(j)}])\right)^k$$

has the universal property analogous to that of φ (cf. 1.2).

One can introduce the morphisms $f_n^{(j)}$ and $u_H^{n,(j)}$ as above, by repeating the construction for each family $(X_i^{(j)})_{i\geq 1}$ separately. One obtains a result analogous to Theorem 1.6 by considering morphisms of functors from $(\widehat{G})^k$ to H which are $\mathbb{Z}$-multilinear and compatible with the filtrations in each of the k arguments separately. They are determined by their value on

$$(1+X^{(j)}t)_{j=1,\dots,k} \in \big(\widehat{G}(K[X^{(j)}]_j)\big)^k.$$

An element

$$\theta \in H(K[X^{(j)}]_j)$$

determines such a morphism of functors if and only if it satisfies

$$\begin{aligned} &\text{i)} \quad \forall j, \quad H(f_1^{(j)})(\theta) = 0, \\ &\text{ii)} \quad \forall n, j, \quad H(f_n^{(j)})(u_H^{n,(j)}(\theta)) \equiv u_H^{n,(j)}(\theta) \pmod{F^n H}. \end{aligned}$$

The reader will easily convince himself that, apart from notational complications, it is enough to copy the proof of 1.6.

1.8. Let $\mathcal{C}_0$ be the category of K-algebras, and let

$$i : \mathcal{C}_0 \longrightarrow \mathcal{C}$$

be the functor which assigns to a K-algebra A the K-algebra A endowed with the gross filtration. Let

$$g : \mathcal{C} \longrightarrow \mathcal{C}_0$$

be the forgetful functor which forgets the filtration. These functors satisfy the following relations:

$$g \circ i = \mathrm{Id}_{\mathcal{C}_0}, \qquad \widehat{G} \circ i \circ g = \widehat{G}_g, \qquad \widehat{G}_g \circ i \circ g = \widehat{G}_g$$

(cf. 1.3, example 2).

Lemma 1.9. Let $H : \mathcal{C}_0 \to \mathrm{Ab\,fil}$ be a functor on $\mathcal{C}_0$. Then the canonical homomorphism

$$\mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G}, H \circ g) \longrightarrow \mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G} \circ i, H)$$

is an isomorphism.

To see that it is injective, one uses the sequence of canonical homomorphisms

$$\mathrm{Hom}(\widehat{G}, H \circ g) \to \mathrm{Hom}(\widehat{G} \circ i, H) \to \mathrm{Hom}(\widehat{G}_g, H \circ g) \to \mathrm{Hom}(\widehat{G}, H \circ g),$$

the last arrow coming from the fact that $\widehat{G}$ is a subfunctor of $\widehat{G}_g$. To verify that the composite is the identity, observe that a morphism of functors

$$\xi : \widehat{G} \longrightarrow H \circ g$$

is determined by its values on objects of $\mathcal{C}$ whose filtration is gross. Indeed, if $A \in \mathrm{Ob}(\mathcal{C})$, and if A_g is A endowed with the gross filtration, the canonical morphism $A \to A_g$ defines a commutative diagram

$$\begin{array}{ccc} \widehat{G}(A) & \xrightarrow{\ \xi_A\ } & (H \circ g)(A) \\ \downarrow & & \downarrow \\ \widehat{G}(A_g) & \xrightarrow[\ \xi_{A_g}\ ]{} & (H \circ g)(A_g), \end{array}$$

which shows that ξ_A is the restriction of ξ_{A_g} to $\widehat{G}(A)$.

Next consider the composite homomorphism

$$\mathrm{Hom}(\widehat{G} \circ i, H) \longrightarrow \mathrm{Hom}(\widehat{G}_g, H \circ g) \longrightarrow \mathrm{Hom}(\widehat{G}, H \circ g) \longrightarrow \mathrm{Hom}(\widehat{G} \circ i, H).$$

It is clear that this is the identity, whence the asserted result.

Theorem 1.10. Let H be a functor on the category $\mathcal{C}_0$ of K-algebras, with values in the category of separated complete filtered abelian groups, and commuting with invariants under a finite group. Then the map from

$$\mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G} \circ i, H)$$

to $H(K[X])$ which, to a morphism of functors, associates its value on the element $1 + Xt$ of $\widehat{G} \circ i(K[X])$, is injective. Its image is the subgroup of $H(K[X])$ formed by the elements θ such that, with notation analogous to that of 1.4 and 1.5,

i) $H(f_1)(\theta) = 0$,

ii) $\forall n > 1, \quad H(f_n)(u_H^n(\theta)) \equiv u_H^n(\theta) \pmod{F^n H(K[X_1, \dots, X_n])}$.

The theorem follows immediately from Theorem 1.6 and Lemma 1.9, since $H \circ g(K[X])$ does not depend on the chosen filtration on $K[X]$.

The analogous theorem obtained by replacing $\widehat{G}$ by $(\widehat{G})^k$ is also true (cf. 1.7).

When there is no risk of confusion, we write the functor $\widehat{G} \circ i$ simply as $\widehat{G}$. This is legitimate because i identifies $\mathcal{C}_0$ with a subcategory of $\mathcal{C}$.

1.11. Let $\mathcal{C}_g$ denote the category of graded K-algebras, and let

$$j : \mathcal{C}_g \longrightarrow \mathcal{C}$$

be the functor which assigns to a graded K-algebra B the K-algebra B endowed with the filtration canonically defined by its grading. The associated graded functor

$$\mathrm{Gr} : \mathcal{C} \longrightarrow \mathcal{C}_g$$

will be denoted Gr. There is an isomorphism of functors

$$\varepsilon : \mathrm{Id}_{\mathcal{C}_g} \xrightarrow{\sim} \mathrm{Gr} \circ j.$$

If $B \in \mathrm{Ob}(\mathcal{C}_g)$, denote by

$$1 + \widehat{B}^+$$

the multiplicative group of elements of

$$\prod_{i=0}^{\infty} B^i$$

having augmentation 1. With the filtration analogous to that on $\widehat{G}$, one obtains a functor

$$\mathcal{C}_g \longrightarrow \mathrm{Ab\,fil}.$$

Furthermore, for every $A \in \mathrm{Ob}(\mathcal{C})$, one defines a homomorphism of filtered groups

$$\widehat{G}(A) \longrightarrow 1 + \widehat{\mathrm{Gr}\,A}^+$$

by sending the series $1 + \cdots + a_n t^n + \cdots$ to the series $1 + \cdots + \bar{a}_n + \cdots$, where $\bar{a}_n$ is the class of a_n in $\mathrm{Gr}^n A$. Its kernel is the group $\widehat{G}'(A)$ of formal series such that

$$a_0 = 1, \qquad \forall i \geq 1, \quad a_i \in \mathrm{Fil}^{i+1} A.$$

Thus one has a functorial isomorphism in A:

$$\widehat{G}(A)/\widehat{G}'(A) \xrightarrow{\sim} 1 + \widehat{\mathrm{Gr}\,A}^+.$$

Lemma 1.12. Let $H : \mathcal{C}_g \to \mathrm{Ab\,fil}$ be a functor on $\mathcal{C}_g$. Then the canonical homomorphism

$$\rho : \mathrm{Hom}_{\mathrm{Ab\,fil}}(1 + \widehat{\cdot}^{+}, H) \longrightarrow \mathrm{Hom}_{\mathrm{Ab\,fil}}(\widehat{G}, H \circ \mathrm{Gr})$$

is an isomorphism.

The homomorphism ρ is defined as the composite

$$\mathrm{Hom}(1 + \widehat{\cdot}^{+}, H) \longrightarrow \mathrm{Hom}(1 + \widehat{\mathrm{Gr}(\,\cdot\,)}^{+}, H \circ \mathrm{Gr}) \longrightarrow \mathrm{Hom}(\widehat{G}, H \circ \mathrm{Gr}),$$

the last arrow being defined by the homomorphism

$$\widehat{G} \longrightarrow 1 + \widehat{\mathrm{Gr}(\,\cdot\,)}^{+}.$$

We construct an inverse morphism

$$\rho' : \mathrm{Hom}(\widehat{G}, H \circ \mathrm{Gr}) \longrightarrow \mathrm{Hom}(1 + \widehat{\cdot}^{+}, H).$$

For this, we show that every morphism of functors from $\widehat{G}$ to $H \circ \mathrm{Gr}$ factors through the canonical morphism

$$\widehat{G} \longrightarrow 1 + \widehat{\mathrm{Gr}(\,\cdot\,)}^{+}.$$

Equivalently, it suffices to show that it vanishes on $\widehat{G}'$, by 1.11. Let $A \in \mathrm{Ob}(\mathcal{C})$ and let $x \in \widehat{G}'(A)$. Define a homomorphism of filtered K-algebras

$$K[X_i]_{i \geq 1} \longrightarrow A$$

by sending X_1 to 0 and X_i to the $(i-1)$-st coefficient of x for $i > 1$. Using this homomorphism, it is enough to verify that the morphism of functors considered vanishes on the element

$$1 + X_2 t + \cdots + X_{n+1} t^n + \cdots .$$

Use the endomorphism f of $K[X_i]_{i \geq 1}$ defined by

$$f(X_i) = X_{i+1} \quad \text{for all } i,$$

and the commutative diagram

$$\begin{array}{ccc} \widehat{G}(K[X_i]) & \longrightarrow & (H \circ \mathrm{Gr})(K[X_i]) \\ {\scriptstyle \widehat{G}(f)}\big\downarrow & & \big\downarrow{\scriptstyle H(\mathrm{Gr}(f))} \\ \widehat{G}(K[X_i]) & \longrightarrow & (H \circ \mathrm{Gr})(K[X_i]). \end{array}$$

Now $\mathrm{Gr}(f) = \mathrm{Gr}(f_1)$, where f_1 is the augmentation homomorphism of $K[X_i]_{i \geq 1}$. Since the latter annihilates φ, the analogous diagram written for f_1 shows that $H(\mathrm{Gr}(f))$ annihilates the image of φ in $H \circ \mathrm{Gr}(K[X_i]_{i \geq 1})$, and the result follows.

Passing to the quotient, one associates to every morphism of functors

$$\widehat{G} \longrightarrow H \circ \mathrm{Gr}$$

a morphism of functors

$$1 + \widehat{\mathrm{Gr}}^{+} \longrightarrow H \circ \mathrm{Gr}\,.$$

By composing with j and using the isomorphism ε, one obtains a morphism

$$1 + \widehat{\cdot}^{+} \longrightarrow H,$$

which defines the desired homomorphism ρ'. It is immediate to verify that ρ and ρ' are inverses of one another.

Theorem 1.13. Let H be a functor on $\mathcal{C}_g$, with values in the category of separated complete filtered abelian groups, and commuting with invariants under a finite group. Then the map from

$$\mathrm{Hom}_{\mathrm{Ab\,fil}}(1+\widehat{\cdot}^{+}, H)$$

to $H(K[X])$, which to a morphism of functors associates its value on the element $1+X$ of $1+\widehat{K[X]}^{+}$, is injective. Its image is the subgroup of $H(K[X])$ formed by the elements θ such that, with notation as in 1.4 and 1.5,

$$\begin{array}{ll} \text{i)} & H(f_1)(\theta)=0, \\ \text{ii)} & \forall n>1, \quad H(f_n)(u_H^n(\theta)) \equiv u_H^n(\theta) \pmod{F^n H(K[X_1,\dots,X_n])}. \end{array}$$

This is immediate from 1.6 and 1.12. There is an analogous statement replacing $\widehat{G}$ by $(\widehat{G})^k$.

Remark 1.14. If H is a functor on $\mathcal{C}_g$, with values in abelian groups, commuting with invariants under a finite group, and satisfying the condition

$$\forall B\in \mathrm{Ob}(\mathcal{C}_g), \qquad H(B)\simeq \varprojlim_n H(B/B^{\geq n}), \qquad B^{\geq n}=\prod_{i\geq n} B^i,$$

then condition ii) of 1.13 is always satisfied if one takes for the filtration on H the filtration by the kernels of

$$H(B)\longrightarrow H(B/B^{\geq n}),$$

which is separated and complete. Indeed, if p_n is the canonical homomorphism

$$K[X_1,\dots,X_n]\longrightarrow K[X_1,\dots,X_n]/K[X_1,\dots,X_n]^{\geq n},$$

then

$$p_n\circ f_n=p_n,$$

and hence

$$H(p_n)[H(f_n)\circ u_H^n(\theta)-u_H^n(\theta)]=0.$$

This says precisely that

$$H(f_n)\circ u_H^n(\theta)-u_H^n(\theta)\in F^n H(K[X_1,\dots,X_n]).$$

Thus, in order to give a morphism of functors from $1+\widehat{\cdot}^{+}$ to H, it is enough to give an element of $H(K[X])$ satisfying only condition i) of 1.13.

1.15. Let $H:\mathcal{C}\to \mathrm{Ab\,fil}$ (respectively $\mathcal{C}_0\to \mathrm{Ab\,fil}$, $\mathcal{C}_g\to \mathrm{Ab\,fil}$) be a functor satisfying the conditions of 1.3 (respectively 1.10, 1.13), and let

$$\xi:(\widehat{G})^k\to H$$

(respectively $(\widehat{G}\circ i)^k\to H$, $(1+\widehat{\cdot}^{+})^k\to H$) be a morphism of functors. We have seen in 1.7 that ξ is determined by its value on the element

$$(\varphi^{(j)})_{j=1,\dots,k}$$

of

$$\left(\widehat{G}(K[X_i^{(j)}]_{j=1,\dots,k;\ i\geq 1})\right)^k$$

(respectively similarly). Suppose H is one of the functors

$$\widehat{G}, \quad \widehat{G}_g, \quad \widehat{G}_a, \quad 1 + \widehat{\mathrm{Gr}}^{+}, \quad \text{etc.}$$

Then the element $\xi((\varphi^{(j)}))$ is a formal series with coefficients in

$$K[X_i^{(j)}]_{i \geq 1,\ j=1,\ldots,k}.$$

The n-th coefficient of this series is called the n-th universal polynomial of ξ. The knowledge of these polynomials is equivalent to the knowledge of ξ.

According to the proof of 1.6, these polynomials are themselves determined by the element

$$\xi(1 + X^{(1)}t, \ldots, 1 + X^{(k)}t).$$

The calculation is performed by the method of the proof: one computes the elements

$$\xi(1 + X_1^{(1)}t + \cdots + X_n^{(1)}t^n, \ldots, 1 + X_1^{(k)}t + \cdots + X_n^{(k)}t^n),$$

by decomposing

$$1 + X_1^{(1)}t + \cdots + X_n^{(1)}t^n = \prod_{j=1}^{n} (1 + T_j^{(1)}t)$$

and by using the additivity of ξ. The polynomials thus obtained are symmetric with respect to the $T_j^{(i)}$, for fixed i and variable j, and can be expressed as functions of the $X_j^{(i)}$. Gluing by means of the filtration of H, one thus determines the universal polynomials of ξ. If

$$(\psi^{(1)}, \ldots, \psi^{(k)}) \in (\widehat{G}(A))^k,$$

the calculation of $\xi(\psi^{(1)}, \ldots, \psi^{(k)})$ is made by substituting the coefficients of the $\psi^{(i)}$ for the $X_j^{(i)}$ in the universal polynomials.

Suppose that there exists an integer m such that the coefficient of t^n in

$$\theta = \xi(1 + X^{(1)}t, \ldots, 1 + X^{(k)}t)$$

is a polynomial whose isobaric components have weights between n and m. Then the k-th universal polynomial of ξ has isobaric components whose weights range between k and km. This follows immediately from the calculation just described and from the theorem on symmetric functions.

In particular, if $H = \widehat{G}$, then the n-th universal polynomial of ξ has no component of weight $< n$. Now consider the canonical morphism

$$H \longrightarrow 1 + \widehat{\mathrm{Gr}}^{+}.$$

Composing it with ξ gives a morphism

$$(\widehat{G})^k \longrightarrow 1 + \widehat{\mathrm{Gr}}^{+},$$

which, by the proof of 1.12, factors through

$$(1 + \widehat{\mathrm{Gr}}^{+})^k$$

and defines a morphism

$$\bar{\xi} : (1 + \widehat{\ }^{+})^k \longrightarrow 1 + \widehat{\ }^{+}.$$

One sees at once that the n-th universal polynomial of $\bar{\xi}$ is the same as that of

$$(\widehat{G})^k \longrightarrow 1 + \widehat{\mathrm{Gr}}^+,$$

that is, the isobaric component of weight n of the n-th universal polynomial of ξ.

The same remarks apply to the case $H = \widehat{G}_a$. Then one may define a morphism of functors

$$H \longrightarrow \widehat{\mathrm{Gr}}^+$$

analogously to 1.11, by putting

$$\widehat{B}^+ = \prod_{i\geq 1} B^i$$

for a graded algebra B; one repeats the same reasoning and reaches the conclusions of the preceding paragraph.

1.16. We finish with an example. Suppose $K = \mathbb{Q}$. Define an endomorphism of the functor $1 + \widehat{\cdot}^+$, denoted Todd and called the Todd character, by putting

$$\mathrm{Todd}(1+X) = \frac{X}{1-e^{-X}}.$$

The conditions i) and ii) of Theorem 1.13 are immediately verified, which shows that an endomorphism of $1 + \widehat{\cdot}^+$ has indeed been defined.

2. Definition of λ-rings; examples

In this paragraph the base ring of paragraph 1 will be $\mathbb{Z}$, except in 2.9. We place ourselves in the category $\mathcal{C}_0$ of 1.8, and write $\widehat{G}$ for the restriction of $\widehat{G}$ to $\mathcal{C}_0$, as in 1.10.

Definition 2.1. A *pre-λ-ring* is a commutative ring K with unit, endowed with a homomorphism of abelian groups

$$\lambda_t : K \longrightarrow 1 + K[[t]]^+$$

such that, for every $x \in K$,

$$\lambda_t(x) = 1 + xt + \cdots .$$

We shall write

$$\lambda_t(x) = \sum_{i\geq 0} \lambda^i(x) t^i.$$

Thus giving a pre-λ-ring structure on a ring K is equivalent to giving a family of maps

$$\lambda^i : K \longrightarrow K \qquad (i \in \mathbb{N})$$

such that

$$\begin{array}{ll} \text{i)} & \forall x \in K, \quad \lambda^0(x) = 1, \\ \text{ii)} & \forall x \in K, \quad \lambda^1(x) = x, \\ \text{iii)} & \forall x, y \in K,\ \forall n \in \mathbb{N}, \quad \lambda^n(x+y) = \displaystyle\sum_{p+q=n} \lambda^p(x)\lambda^q(y). \end{array}$$

If K, K' are two pre-λ-rings and $f : K \to K'$ is a ring homomorphism, we say that f is a homomorphism of pre-λ-rings, or a λ-homomorphism, if the diagram

$$\begin{tikzcd} K \arrow[r, "\lambda_t"] \arrow[d, "f"'] & 1 + K[[t]]^+ \arrow[d, "\widehat{G}(f)"] \\ K' \arrow[r, "\lambda_t"'] & 1 + K'[[t]]^+ \end{tikzcd}$$

is commutative. This is equivalent to the relations

$$\forall x \in K,\ \forall i \in \mathbb{N}, \qquad \lambda^i(f(x)) = f(\lambda^i(x)).$$

Examples 2.2.

a) One can give all pre-λ-ring structures on $\mathbb{Z}$. It is enough to give $\lambda_t(1)$, which is then a formal series with integral coefficients subject to the sole condition that it begin with $1+t$.

b) Let (T, A) be a commutatively ringed topos, and let K be the "Grothendieck group" of A-modules locally isomorphic to a direct factor of a free A-module of finite type. We know that K can be given a ring structure by the tensor product (Exposé IV), and we now endow it with a pre-λ-ring structure as follows. For every A-module E locally a direct factor of a free module of finite type, let $\lambda^i(E)$ denote the class in K of the i-th exterior power of E, which is again locally a direct factor of a free module of finite type. This defines a function from the category of such A-modules to the abelian group $1 + K[[t]]^+$. It remains to show that this function is additive, which will imply that it factors through K and hence gives the desired structure.

Lemma 2.2.1. Let

$$0 \longrightarrow E' \xrightarrow{u} E \xrightarrow{v} E'' \longrightarrow 0$$

be an exact sequence of A-modules which is locally split, for example with E'' locally a direct factor of a free module of finite type. There exists on $\Lambda^n(E)$ a finite filtration and isomorphisms

$$\Lambda^{n-p}(E') \otimes_A \Lambda^p(E'') \xrightarrow{\sim} \mathrm{Gr}^p\, \Lambda^n(E).$$

Take for $F^p\Lambda^n(E)$ the image of the canonical morphism

$$\Lambda^{n-p}(E') \otimes_A \Lambda^p(E) \longrightarrow \Lambda^n(E) \quad (0 \le p \le n).$$

If $p > n$, put $F^p\Lambda^n(E) = \Lambda^n(E)$, and if $p < 0$, put $F^p\Lambda^n(E) = 0$. This does indeed define a finite filtration on $\Lambda^n(E)$.

Define a homomorphism

$$f_p : \Lambda^{n-p}(E') \otimes_A \Lambda^p(E'') \longrightarrow F^p\Lambda^n(E)/F^{p-1}\Lambda^n(E)$$

by

$$f_p(x'_1 \wedge \cdots \wedge x'_{n-p} \otimes x''_1 \wedge \cdots \wedge x''_p) = \mathrm{cl}\big(u(x'_1) \wedge \cdots \wedge u(x'_{n-p}) \wedge x_1 \wedge \cdots \wedge x_p\big),$$

where the x_i are lifts in E of the x''_i in E''. This visibly does not depend on the choice of the x_i. If the exact sequence is locally split, then f_p is an isomorphism, since locally

$$\Lambda^n(E) \simeq \bigoplus_{p+q=n} \Lambda^p(E') \otimes_A \Lambda^q(E'').$$

When E', E, E'' are locally direct factors of free modules of finite type, the $\mathrm{Gr}^p\, \Lambda^n(E)$ are locally direct factors of free modules of finite type, and hence so are the $F^p\Lambda^n(E)$. Thus one may write in K

$$\lambda^n(E) = \sum_{p=0}^{n} \mathrm{cl}(\mathrm{Gr}^p\, \Lambda^n(E)) = \sum_{p=0}^{n} \lambda^{n-p}(E')\lambda^p(E''),$$

which gives the required additivity.

Let (T, A) and (T', A') be two commutatively ringed topoi, and let

$$f : (T, A) \longrightarrow (T', A')$$

be a morphism of ringed topoi. We know (Exposé IV, 2.7) that f defines a morphism of rings on the corresponding K's, namely

$$f^* : K^*_{\text{naif}}(T', A') \longrightarrow K^*_{\text{naif}}(T, A).$$

Since inverse image commutes with exterior powers, the corresponding morphism f^* on K is a λ-homomorphism for the structure just defined.

We point out two special cases:

- If X is a scheme, then $K^*_{\text{naif}}(X)$ is thus endowed with a pre-λ-ring structure. This is the case which will be used later in the Seminar, and it may be regarded as the model for the general theory of λ-rings.
- If A is a ring, $X = \operatorname{Spec} A$, and G is a group acting trivially on A, then, taking for (T, A) the topos of sheaves of sets on X on which G acts, the ring K thus obtained is the representation ring of G in the projective A-modules of finite type.

2.3. Let K be a commutative ring with unit. We propose to endow the abelian group

$$\widehat{G}(K) = 1 + K[[t]]^+$$

with a pre-λ-ring structure, functorially in K. To this end, we apply the results of 1.10 to the functor $\widehat{G}$.

First define multiplication by giving a $\mathbb{Z}$-bilinear morphism of functors

$$\widehat{G} \times \widehat{G} \longrightarrow \widehat{G}$$

through the element $1 + XYt$ of $\widehat{G}(\mathbb{Z}[X, Y])$. This element satisfies 1.7.i; to see that it satisfies 1.7.ii, one must show that in

$$\prod_{i=1}^{n} (1 + T_i Y t),$$

when expressed by means of the elementary symmetric functions X_j in the T_i, the variable X_n does not occur in the coefficients of t^k for $k < n$. This is evident by degree considerations.

The law so defined will be denoted by the sign $\circ$. Thus, by definition,

$$(1 + Xt) \circ (1 + Yt) = 1 + XYt. \tag{2.3.1}$$

It is distributive by construction, and it is associative and commutative by the diagrams

$$\begin{array}{ccc} \widehat{G} \times \widehat{G} \times \widehat{G} & \to & \widehat{G} \times \widehat{G} \\ \downarrow & & \downarrow \\ \widehat{G} \times \widehat{G} & \longrightarrow & \widehat{G} \end{array} \qquad \begin{array}{ccc} \widehat{G} \times \widehat{G} & \to & \widehat{G} \times \widehat{G} \\ \downarrow & & \downarrow \\ \widehat{G} & \xrightarrow[\mathrm{Id}]{} & \widehat{G}, \end{array}$$

where the meaning of the arrows is evident, and where it is enough to verify commutativity on the elements $1 + Xt, 1 + Yt, 1 + Zt$, respectively $1 + Xt, 1 + Xt$, of $\widehat{G}(\mathbb{Z}[X, Y, Z])$, respectively $\widehat{G}(\mathbb{Z}[X, Y])$, by 1.10; this verification is immediate.

By 1.8, the product under this law of two series is calculated by substituting their coefficients into the universal polynomials of the law. The n-th universal polynomial will be denoted P_n. It is a polynomial in

$$\mathbb{Z}[X_i, Y_j]_{1\le i,j\le n}$$

which is isobaric of weight n with respect to each of the two groups of variables (see 1.15), and symmetric with respect to these two groups. For example,

$$P_2(X_1, X_2; Y_1, Y_2) = X_2Y_1^2 + Y_2X_1^2 - 2X_2Y_2.$$

One also has the following relation: if $K \in \mathrm{Ob}(\mathcal{C}_0)$, $\psi \in \widehat{G}(K)$, and $a \in K$, then

$$\psi(t) \circ (1 + at) = \psi(at). \tag{2.3.2}$$

As usual, by the standard functorial argument this reduces to the case $K = \mathbb{Z}[X, Y]$, $\psi = 1 + Xt$, $a = Y$, where it is nothing other than (2.3.1).

Applying (2.3.2) to $\widehat{G}(K) = 1 + K[[t]]^+$ with $a = 1$, one sees that $1 + t$ is the unit for this multiplication. To avoid confusion, we sometimes denote it by 1, or by $1_\circ$.

The functor $\widehat{G}$ can now be considered as a functor from $\mathcal{C}_0$ to the category of commutative rings with unit, and hence can be composed with itself. To avoid confusion, choose different indeterminates for each of the two factors in the product of $\widehat{G}$ with itself; we indicate the indeterminate by an exponent when necessary:

$$\widehat{G}^t, \quad \widehat{G}^u, \quad \text{etc.}$$

For $A \in \mathrm{Ob}(\mathcal{C}_0)$ one has

$$\widehat{G}^u \circ \widehat{G}^t(A) = 1 + \{1 + A[[t]]^+\}[[u]]^+.$$

To define the operations λ^i on $1 + K[[t]]^+$, we are therefore going to give a morphism of functors

$$\widehat{G} \longrightarrow \widehat{G}^u \circ \widehat{G}^t.$$

For this, one must first define a filtration of the functor $\widehat{G}^u \circ \widehat{G}^t$ which permits the gluing of 1.6.

First note that $F^k\widehat{G}$ is an ideal for the law $\circ$. For this follows, for the product

$$\left(1 + \sum_{i\ge k} X_i t^i\right) \circ (1 + Yt),$$

by reducing by functoriality to the case $A = \mathbb{Z}[X_i, Y]_{i\ge 1}$, and using (2.3.2). Therefore $F^n(\widehat{G}^u \circ \widehat{G}^t)$ is defined to be the set of series of the form

$$1 + \psi_1 u + \cdots + \psi_i u^i + \cdots, \qquad \psi_i \in F^{[n/i]}\widehat{G},$$

where $[n/i]$ denotes the integral part of n/i.

The element defining the morphism λ_u is

$$1 + \{1 + Xt\}u.$$

The reader will verify as an exercise that the conditions of 1.6 are indeed fulfilled; in fact the filtration was chosen ad hoc.

Again, the operations λ^i are calculated by substitution from

$$\lambda_u(1 + \cdots + X_n t^n + \cdots),$$

and so are determined by universal polynomials. Denote by

$$Q_{i,j}(X_1, \dots, X_{ij})$$

the coefficient of t^j in

$$\lambda^i(1 + \cdots + X_n t^n + \cdots).$$

It is an isobaric polynomial of weight ij with respect to the variables $X_1, \dots, X_{ij}$ (by a verification analogous to that of 1.15). It is calculated by symmetric functions from the defining element

$$\lambda_u(1 + Xt) = 1 + \{1 + Xt\}u. \tag{2.3.3}$$

For example,

$$Q_{2,2}(X_1, X_2, X_3, X_4) = X_1X_3 - X_4.$$

Finally, we indicate the origin of formulas (2.3.1) and (2.3.3). Consider the example 2.2.b, where the operations λ^i are defined by exterior powers of A-modules. We want

$$\lambda_t : K \longrightarrow \widehat{G}(K)$$

to be a λ-homomorphism. Thus on $\widehat{G}$ one must define laws such that, for all A-modules E, F,

$$\lambda_t(EF) = \lambda_t(E) \circ \lambda_t(F), \qquad \lambda_u(\lambda_t(E)) = \widehat{G}^u_{\lambda_t}(\lambda_t(\lambda_u(E))).$$

The relations (2.3.1) and (2.3.3) are precisely the preceding relations applied to the case where E and F are invertible modules.

We shall see in Exposé VI to what extent the desired property is verified.

Definition 2.4. A pre-λ-ring K will be called a *λ-ring* if the homomorphism of abelian groups

$$\lambda_t : K \longrightarrow 1 + K[[t]]^+$$

is a λ-homomorphism for the pre-λ-ring structure on the second member defined in 2.3. Thus the following relations hold for all $x, y \in K$:

$$\lambda_t(1) = 1 + t, \qquad \lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y), \qquad \lambda_u(\lambda_t(x)) = \widehat{G}(\lambda_t)(\lambda_u(x)). \tag{2.4.1}$$

In terms of the maps λ^i, the last two relations are equivalent, for $x, y \in K$ and $i, j \in \mathbb{N}$, to

$$\lambda^i(xy) = P_i(\lambda^1(x), \dots, \lambda^i(x); \lambda^1(y), \dots, \lambda^i(y)), \tag{2.4.2}$$

$$\lambda^j(\lambda^i(x)) = Q_{i,j}(\lambda^1(x), \dots, \lambda^{ij}(x)). \tag{2.4.3}$$

We note that here we depart from the usual terminology (cf. [1]), where what we call a pre-λ-ring was called a λ-ring, and our λ-rings were called "special" λ-rings. The reason is that only "special" λ-rings are useful in practice.

Examples 2.5.

a) We have determined all pre-λ-ring structures on $\mathbb{Z}$. It follows from (2.4.1) that only the structure defined by

$$\lambda_t(1) = 1 + t$$

is capable of being a λ-ring structure. One then has

$$\lambda_t(n) = (1 + t)^n,$$

hence

$$\forall n, n' \in \mathbb{Z}, \quad \lambda_t(n) \circ \lambda_t(n') = (1+t)^n \circ (1+t)^{n'} = (1+t)^{nn'} = \lambda_t(nn'),$$

and

$$\lambda_u((1+t)^n) = (\lambda_u(1+t))^n = (1+\{1+t\}u)^n = \widehat{G}(\lambda_t)((1+u)^n) = \widehat{G}(\lambda_t)(\lambda_u(n)).$$

Thus this is indeed a λ-ring structure. If $\binom{X}{k}$ denotes the polynomial with rational coefficients

$$\frac{1}{k!}X(X-1)\cdots(X-k+1),$$

then

$$\lambda^i(n) = \binom{n}{i}. \tag{2.5.1}$$

b) For every ring K, the pre-λ-ring $1+K[[t]]^+$ is a λ-ring. First,

$$\lambda_u(1) = \lambda_u(1+t) = 1+\{1+t\}u = 1+1\cdot u.$$

The other relations are verified by the commutativity of the diagrams

$$\begin{tikzcd}
\widehat{G}^t \times \widehat{G}^t \arrow[r, "\lambda_u \times \lambda_u"] \arrow[d, "\circ"'] & (\widehat{G}^u \circ \widehat{G}^t) \times (\widehat{G}^u \circ \widehat{G}^t) \arrow[d, "\circ"] \\
\widehat{G}^t \arrow[r, "\lambda_u"'] & \widehat{G}^u \circ \widehat{G}^t,
\end{tikzcd}
\qquad
\begin{tikzcd}
\widehat{G}^t \arrow[r, "\lambda_v"] \arrow[d, "\lambda_u"'] & \widehat{G}^v \circ \widehat{G}^t \arrow[d, "\widehat{G}^v(\lambda_u)"] \\
\widehat{G}^u \circ \widehat{G}^t \arrow[r, "\lambda_v"'] & \widehat{G}^v \circ \widehat{G}^u \circ \widehat{G}^t,
\end{tikzcd}$$

which are checked by 1.10 on the elements $1+Xt$ and $1+Yt$. For the first,

$$\begin{aligned}
\lambda_u(1+Xt) \circ \lambda_u(1+Yt) &= (1+\{Xt\}u) \circ (1+\{1+Yt\}u) \\
&= 1+\{(1+Xt)\circ(1+Yt)\}u \\
&= 1+\{1+XYt\}u \\
&= \lambda_u((1+Xt)\circ(1+Yt)).
\end{aligned}$$

For the second,

$$\lambda_v(\lambda_u(1+Xt)) = 1+\{1+\{1+Xt\}u\}v = \widehat{G}^v(\lambda_u)\lambda_v(1+Xt).$$

c) The geometric examples of 2.2 will be treated later.

2.6. One can generalize Example 2.5.a as follows. Let K be a commutative ring with unit such that:

i) K is torsion-free,

ii) $\forall x \in K,\ \forall n \in \mathbb{N}, \quad \binom{x}{n} \in K$.

Then, for all $x \in K$ and all $n \in \mathbb{N}$, put

$$\lambda^n(x) = \binom{x}{n},$$

that is,

$$\lambda_t(x) = \sum_{n=0}^{\infty} \binom{x}{n} t^n = (1+t)^x = \exp(x \operatorname{Log}(1+t)), \tag{2.6.1}$$

where the last identity is considered in $K \otimes_{\mathbb{Z}} \mathbb{Q}[[t]]$. By multiplicativity of the exponential, this defines a pre-λ-ring structure on K.

To see that it is a λ-ring structure, one must prove the relations (2.4.2), and it is enough to prove them in $K \otimes \mathbb{Q}$, by i). One is then reduced by functoriality to the case $K = \mathbb{Q}[X, Y]$ and to the elements X, Y. In view of the definition of the λ^i, the two sides of the equalities (2.4.2) are polynomials in X, Y, with rational coefficients. It is therefore enough, in order to prove their equality, to show that they take the same value for all integral values of X, Y. But the relations obtained by substituting integral values in X, Y are precisely those expressing that the pre-λ-ring $\mathbb{Z}$ of 2.5.a is a λ-ring. They are therefore satisfied.

Definition 2.7. A *binomial ring* is a ring satisfying the conditions 2.6.i and ii), endowed with the λ-ring structure defined by (2.6.1).

Example 2.8. Let X be a site, K a binomial ring, and $\mathcal{K}$ the sheaf associated to the constant presheaf with value K on X. Then $H^0(X, \mathcal{K})$ is a binomial ring. Indeed, the conditions 2.6 are preserved under passage to an inductive limit. In particular, $H^0(X, \mathbb{Z})$ is a binomial ring.

2.9. Let K be a binomial ring and A a K-algebra. For every series

$$\varphi \in \widehat{G}(A)$$

and every $a \in K$, one can define a series φ^a in the usual way. Writing $\varphi = 1 + \psi$, put

$$\varphi^a = (1+\psi)^a = \sum_{n=0}^{\infty} \binom{a}{n} \psi^n \;(= \exp(a \operatorname{Log} \varphi))\,, \tag{2.9.1}$$

where the last equality can be written only modulo torsion. These formulas are obtained by substitution from the analogous formulas for $\varphi = 1 + t$. Therefore the following relations, the classical properties of the exponential, remain valid for $a, b \in K$ and $\varphi, \psi \in \widehat{G}(A)$:

$$\varphi^a \varphi^b = \varphi^{a+b}, \qquad \varphi^{ab} = (\varphi^a)^b, \qquad \varphi^a \psi^a = (\varphi\psi)^a. \tag{2.9.2}$$

Continuation of 2.9

We still suppose that K is binomial. Multiplication, for the law $\circ$, by $\lambda_t(a)$, where $a \in K$, defines a morphism of functors

$$\widehat{G}_K \longrightarrow \widehat{G}_K$$

which is characterized by its value on the element $1 + Xt$ of $G(K[X])$. Thus

$$\lambda_t(a) \circ (1 + Xt) = \lambda_{Xt}(a),$$

and hence

$$\lambda_t(a) \circ (1 + Xt) = (1 + Xt)^a = \exp(a \operatorname{Log}(1 + Xt)) = \sum_{n=0}^{\infty} \binom{a}{n} X^n t^n. \tag{2.9.3}$$

It follows that

$$\lambda_t(a) \circ \varphi = \varphi^a \quad (= \exp(a \operatorname{Log} \varphi)). \tag{2.9.4}$$

Indeed, the two members are additive functors which coincide on $1 + Xt$. From (2.9.3) and 1.15 one obtains that the universal polynomial of degree n defining multiplication by $\lambda_t(a)$ is isobaric of weight n. We denote it by $R_{n,a}$.

Finally, if A is a K-λ-algebra and $b \in A$, then

$$\lambda_t(ab) = (\lambda_t(b))^a. \tag{2.9.5}$$

This follows from $\lambda_t(ab) = \lambda_t(a) \circ \lambda_t(b)$ and from (2.9.4).

3. The operations γ

We still work over the category $\mathscr{C}_0$.

3.1. Let K be a λ-ring. We shall define from the operations λ^i new operations denoted γ^i. They are the operations which are adapted to Chern classes. If E is a vector bundle of rank d, the top Chern class of E is represented, in K-theoretic language, by the alternating sum of the exterior powers. To obtain the n-th class, one first replaces E by $E \oplus \mathbf{1}^{n-d}$, so that the same top-class recipe applies. This motivates the formula below.

Proposition 3.2. Let K be a λ-ring.

i) For every $x \in K$ and every $n \in \mathbb{N}$,

$$\lambda^n(x+n-1) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x+n).$$

ii) If

$$\gamma^n(x) = \lambda^n(x+n-1), \qquad \gamma_t(x) = \sum_{n\geq 0} \gamma^n(x) t^n,$$

then

$$\gamma_t(x) = \lambda_{t/(1-t)}(x), \qquad \text{hence} \qquad \lambda_t(x) = \gamma_{t/(1+t)}(x).$$

Thus knowledge of the operations γ is equivalent to knowledge of the operations λ.

Proof. For i),

$$\lambda^n(x+n-1) = \sum_{i=0}^{n} \lambda^i(x+n)\lambda^{n-i}(-1).$$

Since $\lambda_t(1) = 1+t$, one has $\lambda_t(-1) = 1/(1+t)$, and hence $\lambda^k(-1) = (-1)^k$, giving the desired relation. For ii),

$$\lambda_{t/(1-t)}(x) = \sum_{n\geq 0} \lambda^n(x) t^n (1-t)^{-n},$$

and

$$t^n(1-t)^{-n} = \sum_{p\geq n} \binom{p-1}{p-n} t^p.$$

The coefficient of t^p is therefore

$$\sum_{n=0}^{p} \lambda^n(x) \binom{p-1}{p-n} = \sum_{n=0}^{p} \lambda^n(x)\lambda^{p-n}(p-1) = \lambda^p(x+p-1).$$

3.3. If $f : K \to K'$ is a homomorphism of λ-rings, then f is a λ-homomorphism if and only if, for every $i \in \mathbb{N}$,

$$f(\gamma^i(x)) = \gamma^i(f(x)),$$

or equivalently

$$\gamma_t(f(x)) = \widehat{G}(f)(\gamma_t(x)).$$

3.4. From 3.2 one obtains

$$\gamma^0(x) = 1, \qquad \gamma^1(x) = x, \qquad \gamma^n(x+y) = \sum_{p+q=n} \gamma^p(x)\gamma^q(y). \tag{3.4.1}$$

Thus γ_t defines a new pre-λ-ring structure. It is not, in general, a λ-ring structure. Let ρ_t denote the automorphism of the functor $\widehat{G}$ induced by the change of variable

$$t \longmapsto \frac{t}{1-t}.$$

Then $\gamma_t = \rho_t\lambda_t$. Transporting the λ-ring functor structure of $\widehat{G}$ by ρ_t gives a second structure on $\widehat{G}$, for which $\gamma_t : K \to \widehat{G}(K)$ is a λ-homomorphism whenever K is a λ-ring.

3.5. The multiplication in this transported structure is the composite

$$\widehat{G} \times \widehat{G} \xrightarrow{\rho_t^{-1} \times \rho_t^{-1}} \widehat{G} \times \widehat{G} \xrightarrow{\circ} \widehat{G} \xrightarrow{\rho_t} \widehat{G}.$$

It will be denoted ∇. It is determined by

$$\begin{aligned}
(1+Xt)\nabla(1+Yt) &= \rho_t\Big(\rho_t^{-1}(1+Xt) \circ \rho_t^{-1}(1+Yt)\Big) \\
&= \rho_t\left(\frac{1+(X+1)t}{1+t} \circ \frac{1+(Y+1)t}{1+t}\right) \\
&= \rho_t\left(\frac{(1+((X+1)(Y+1))t)(1+t)}{(1+(X+1)t)(1+(Y+1)t)}\right) \\
&= \frac{1+(1+X+Y+XY)t}{(1+Xt)(1+Yt)}.
\end{aligned} \tag{3.5.1}$$

Thus, for $x, y \in K$,

$$\gamma_t(xy) = \gamma_t(x)\nabla\gamma_t(y). \tag{3.5.2}$$

The universal polynomial of degree n for this law will be denoted P'_n. Its isobaric components have weights between n and $2n$ in each of the two groups of variables, and

$$\gamma^n(xy) = P'_n(\gamma^1(x), \ldots, \gamma^n(x), \gamma^1(y), \ldots, \gamma^n(y)). \tag{3.5.3}$$

The unit for ∇ is $\gamma_t(1) = 1/(1-t)$.

3.6. The operations λ' of the transported structure are defined by

$$\widehat{G}$$

$$\begin{tikzcd}
1+K[[t]]^+ \arrow[r, "\lambda_u"] \arrow[d, "\rho_t"'] & 1+\{1+K[[t]]^+\}[[u]]^+ \arrow[d, "\widehat{G}(\rho_t)"] \\
1+K[[t]]^+ \arrow[r, "\lambda'_u"'] & 1+\{1+K[[t]]^+\}[[u]]^+.
\end{tikzcd} \tag{3.6.1}$$

If γ' denotes the γ-operations attached to these operations λ', then

$$\gamma'_u(1+Xt) = \frac{1}{1-u} + \{1+Xt\}u. \tag{3.6.2}$$

Let $Q'_{i,j}$ be the coefficient of t^j in $\gamma'^i(1+X_1t+X_2t^2+\cdots)$. It is a polynomial in $\mathbb{Z}[X_1, \ldots, X_{ij}]$, whose isobaric components have weights from j to ij. For a λ-ring K and $x \in K$,

$$\gamma_t(\lambda^i(x)) = \lambda'^i(\gamma_t(x)), \qquad \gamma_t(\gamma^i(x)) = \gamma'^i(\gamma_t(x)), \tag{3.6.3}$$

and therefore

$$\gamma^j(\gamma^i(x)) = Q'_{i,j}(\gamma^1(x), \ldots, \gamma^{ij}(x)). \tag{3.6.4}$$

The transported pre-λ-ring structure on $\widehat{G}$ is itself a λ-ring structure.

3.7. We denote by $\widehat{G}_\circ$ the functor $\widehat{G}$ with the structure of §2, and by $\widehat{G}_\nabla$ the same functor with the structure of §3. Giving a λ-ring structure on a ring A is equivalent either to giving a ring homomorphism $A \to \widehat{G}_\circ(A)$ satisfying the usual diagram for λ_t, or to giving a ring homomorphism $A \to \widehat{G}_\nabla(A)$ satisfying the corresponding diagram for γ_t. Explicitly, the two diagrams are

$$\begin{tikzcd} A \arrow[r, "\lambda_u"] \arrow[d, "\lambda_t"'] & \widehat{G}^u_\circ(A) \arrow[d, "\widehat{G}^u_\circ(\lambda_t)"] \\ \widehat{G}^t_\circ(A) \arrow[r, "\lambda_u"'] & \widehat{G}^u_\circ(\widehat{G}^t_\circ(A)) \end{tikzcd} \tag{3.7.1}$$

and

$$\begin{tikzcd} A \arrow[r, "\gamma_u"] \arrow[d, "\gamma_t"'] & \widehat{G}^u_\nabla(A) \arrow[d, "\widehat{G}^u_\nabla(\gamma_t)"] \\ \widehat{G}^t_\nabla(A) \arrow[r, "\gamma'_u"'] & \widehat{G}^u_\nabla(\widehat{G}^t_\nabla(A)). \end{tikzcd} \tag{3.7.2}$$

3.8. Let K be a binomial ring. From (2.6.1), for $x \in K$,

$$\gamma_t(x) = \left(\frac{1}{1-t}\right)^x = \exp(-x\operatorname{Log}(1-t)). \tag{3.8.1}$$

For $a \in K$, multiplication by $\gamma_t(a)$ for the law ∇ is characterized by its value on $1 + Xt$:

$$\gamma_t(a)\nabla(1+Xt) = (1+Xt)^a = \exp(a\operatorname{Log}(1+Xt)) = \sum_{n=0}^{\infty}\binom{a}{n}X^n t^n. \tag{3.8.2}$$

Thus the universal polynomial of degree n defining this multiplication is again $R_{n,a}$.

Definition 3.9. Let K be a λ-ring. An *augmented K-λ-algebra* is a K-λ-algebra A, together with a λ-homomorphism of K-algebras

$$\epsilon : A \longrightarrow K.$$

Morphisms are λ-homomorphisms compatible with the augmentations.

Example 3.9.1. Let (X, A) be a locally ringed topos. The Grothendieck group $K^\circ(X)$ of locally free finite-type A-modules has a natural pre-λ-ring structure by exterior powers. There is a natural λ-homomorphism from $H^0(X, \mathbb{Z})$ to $K^\circ(X)$: a section a of the constant sheaf $\mathbb{Z}$ partitions the final object into opens U_n on which $a = n$, and is sent to the class of the locally free module which is A^n on U_n for $n \geq 0$, minus the analogous class on the negative pieces. The rank map is the augmentation. This is the augmented structure used below.

3.10. Let A be an augmented K-λ-algebra. For $n > 0$, define $\operatorname{Fil}^n A$ to be the ideal generated by products

$$\gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k), \qquad i_1 + \cdots + i_k \geq n, \qquad \epsilon(x_i) = 0. \tag{3.10.1}$$

If A is generated as an abelian group by (x_α), then $\operatorname{Fil}^n A$ is generated over K by

$$\gamma^{i_1}(x_{\alpha_1} - \epsilon(x_{\alpha_1}))\cdots\gamma^{i_k}(x_{\alpha_k} - \epsilon(x_{\alpha_k})), \qquad i_1 + \cdots + i_k \geq n. \tag{3.10.2}$$

The filtration is decreasing and multiplicative, and

$$\operatorname{Fil}^0 A = A, \qquad \operatorname{Fil}^1 A = \operatorname{Ker}\epsilon.$$

It is functorial for homomorphisms of augmented K-λ-algebras.

Proposition 3.11. Let A be a λ-ring and let $y_1, \dots, y_n$ be elements of A such that $\gamma^j(y_i) = 0$ for $j \geq 2$. Put $x = \sum_i y_i$. Then $\gamma^k(x-n) = 0$ for $k > n$, and, for $1 \leq k \leq n$, $\gamma^k(x-n)$ is the k-th elementary symmetric function of the $y_i - 1$. This follows from

$$\gamma_t(x-n) = \prod_{i=1}^{n} \gamma_t(y_i - 1).$$

4. λ-rings generated by generators and relations

4.1. Let K be a λ-ring and let $(X_i)_{i\in I}$ be indeterminates. Put

$$A = K[X_i, {}^nX_i]_{i\in I,\ n\geq 2}.$$

Define

$$\lambda_t(X_i) = 1 + X_i t + \sum_{n\geq 2} {}^nX_i t^n, \qquad \lambda_t({}^nX_i) = \lambda^n(\lambda_t(X_i)). \tag{4.1.1}$$

This gives a λ-ring structure on A extending that of K.

Proposition 4.2. For every K-λ-algebra B and every family $(b_i)_{i\in I}$ in B, there exists a unique K-λ-homomorphism $A \to B$ carrying X_i to b_i. Necessarily it carries ${}^nX_i$ to $\lambda^n(b_i)$, and this condition is sufficient.

4.3. The same construction may be made using γ's. Let

$$A' = K[X_i, {}^{\gamma}_{n}X_i]_{i\in I,\ n\geq 2},$$

and define

$$\gamma_t(X_i) = 1 + X_i t + \sum_{n\geq 2} {}^{\gamma}_{n}X_i t^n, \qquad \gamma_t({}^{\gamma}_{n}X_i) = \gamma'^n(\gamma_t(X_i)). \tag{4.3.1}$$

This gives a λ-ring satisfying the same universal property, hence canonically isomorphic to A.

Definition 4.4. The preceding λ-ring is the *free λ-ring* over K generated by the family (X_i).

Definition 4.5. An ideal J of a λ-ring A is a *λ-ideal* if $\lambda^n(x) \in J$ for every $x \in J$ and $n > 1$. Kernels of λ-homomorphisms are λ-ideals; conversely, if J is a λ-ideal, A/J receives a unique λ-ring structure. The intersection of λ-ideals is a λ-ideal. Thus every family of elements has a smallest containing λ-ideal, generated by the elements and all their iterated λ^n's.

Definition 4.6. The quotient of the free λ-ring by the λ-ideal generated by prescribed relations (P_i) is called the λ-ring generated by the generators (X_i) subject to the relations (P_i). Equivalently, one may formulate the same construction in terms of γ-operations.

Proposition 4.7. This quotient has the expected universal property: every family (b_i) in a K-λ-algebra satisfying the relations (P_i) determines a unique K-λ-homomorphism from the quotient to that algebra.

Proposition 4.8. Let $n \geq 0$. In the free λ-ring over K on one generator X, the ideal generated by the $\lambda^k X$, $k > n$, is a λ-ideal. Indeed, after quotienting, one has only to check that if $\varphi \in \widehat{G}(C)$ has no terms of degree $> n$, then $\lambda^k(\varphi) = 0$ for $k > n$. By the splitting principle, it is enough to write

$$\varphi = \prod_{i=1}^{n} (1 + T_i t),$$

and then

$$\lambda_u(\varphi) = \prod_{i=1}^{n} (1 + \{1 + T_i t\} u),$$

which has degree n in u.

Corollary 4.9. The λ-ring generated over K by one element X, subject to $\lambda^k(X) = 0$ for $k > n$, is

$$K[X, \lambda^2 X, \dots, \lambda^n X].$$

More generally, for a family (X_i) with bounds (n_i), the quotient subject to $\lambda^k(X_i) = 0$ for $k > n_i$ is the polynomial ring generated by $X_i, \lambda^2 X_i,$
$ldots, \lambda^{n_i} X_i$.

4.10. Let K be binomial and let A be an augmented K-λ-algebra. Put

$$B = A[X, \gamma^2 X, \gamma^3 X, \dots],$$

the free A-λ-algebra on X, written in γ-coordinates. Extend the augmentation by $\epsilon(X) = 0$, hence $\epsilon(\gamma^i X) = 0$ for $i > 0$. Let M_k be the set of monomials in the variables $\gamma^j X$, counted with weight j, of total weight k, and set $M_0 = \{1\}$.

Lemma 4.11. If $x \in \operatorname{Fil}^p A$, then every coefficient of $\gamma^n(x + X)$ which is homogeneous of weight k in the $\gamma^j X$ belongs to $\operatorname{Fil}^{p+n-k} A$.

Proposition 4.12. For every $k \geq 0$,

$$\operatorname{Fil}^k B = \operatorname{Fil}^k A + \operatorname{Fil}^{k-1} A\, M_1 + \cdots + \operatorname{Fil}^1 A\, M_{k-1} + \operatorname{Fil}^0 A\, M_k + B\, M_{k+1}. \tag{4.12.1}$$

The inclusion from right to left follows from Lemma 4.11. The reverse inclusion is obtained by writing the generators of $\operatorname{Fil}^k B$ as products of γ-operations and reducing, term by term, to the displayed weights.

Remarks 4.13. The proof gives the same assertion for a free λ-algebra on any finite or infinite family of generators, with the evident multi-weight. It is also unchanged when one uses the λ-coordinates, since λ and γ coordinates differ by triangular universal formulas.

Corollary 4.14. Passing to the associated graded ring, one has

$$\operatorname{Gr}^k B \simeq \operatorname{Gr}^k A \oplus \operatorname{Gr}^{k-1} A\, M_1 \oplus \cdots \oplus \operatorname{Gr}^0 A\, M_k. \tag{4.14.1}$$

Corollary 4.15. If $B' = K[X, \gamma^2 X, \dots, \gamma^n X]$ is the K-λ-algebra generated by one element satisfying $\gamma^k(X) = 0$ for $k > n$, then the λ-ring filtration of B' is exactly the weight filtration in the variables $\gamma^j X$.

5. The elements $\lambda^p(N, x)$

This section will be used only in Exposé VII, §4, in the study of regular immersions; it may be omitted on first reading.

5.1. Let $d \geq 1$. Let A be the $\mathbb{Z}$-λ-algebra generated by N and x, subject to the relations $\lambda^i(N) = 0$ for $i > d$. Thus

$$A = \mathbb{Z}[N, \dots, \lambda^d N; x, \dots, \lambda^j x, \dots].$$

Put

$$\lambda_{-1}(N) = \sum_{i=0}^{d} (-1)^i \lambda^i(N). \tag{5.1.1}$$

Lemma 5.2. For every $p \geq 1$, the element $\lambda^p(x\lambda_{-1}(N))$ is divisible by $\lambda_{-1}(N)$.

It is enough, by isobaricity, to prove this for $\lambda^p(\lambda_{-1}(N))$. Let B be the λ-ring generated by $N_1, \ldots, N_d$, with $\lambda^i(N_j) = 0$ for $i \geq 2$. The map $N \mapsto \sum_i N_i$ identifies A with the symmetric polynomials in the N_i, and

$$\lambda_{-1}(N) = \prod_{i=1}^{d}(1 - N_i).$$

If some $N_i = 1$, then $\lambda_t(\lambda_{-1}(N)) = \lambda_t(0) = 1$, so every positive coefficient vanishes. Thus each $\lambda^p(\lambda_{-1}(N))$ is divisible by every $1 - N_i$, hence by their product.

Definition 5.3. Since A is integral, define $\lambda^p(N, x)$ by

$$\lambda^p(N, x)\lambda_{-1}(N) = \lambda^p(x\lambda_{-1}(N)), \qquad p \geq 1. \tag{5.3.1}$$

Thus $\lambda^p(N, x)$ is a polynomial in the $\lambda^i(x)$ and $\lambda^j(N)$, isobaric of weight p in the $\lambda^i(x)$. If a λ-ring B contains an element N' with $\lambda^k(N') = 0$ for $k > d$, then $\lambda^p(N', x')$ is defined by substitution; it is independent of the chosen bound d.

5.4. In the λ-ring generated over $\mathbb{Z}$ by N, x, y, subject to $\lambda^k(N) = 0$ for $k > d$, formula (5.3.1) gives

$$\begin{aligned}
\lambda^p(N, x + y)\lambda_{-1}(N) &= \lambda^p\big((x + y)\lambda_{-1}(N)\big) \\
&= \sum_{i+j=p} \lambda^i\big(x\lambda_{-1}(N)\big)\lambda^j\big(y\lambda_{-1}(N)\big) \\
&= \sum_{\substack{i+j=p \\ i,j\neq 0}} \lambda^i(N, x)\lambda^j(N, y)\big(\lambda_{-1}(N)\big)^2 + \big(\lambda^p(N, x) + \lambda^p(N, y)\big)\lambda_{-1}(N).
\end{aligned}$$

Since the ring under consideration is an integral domain, this yields the relation, valid by substitution in every λ-ring,

$$\lambda^p(N, x + y) = \sum_{\substack{i+j=p \\ i,j\neq 0}} \lambda^i(N, x)\lambda^j(N, y)\lambda_{-1}(N) + \lambda^p(N, x) + \lambda^p(N, y). \tag{5.4.1}$$

Now consider the λ-ring generated over $\mathbb{Z}$ by N, N', x, subject to $\lambda^k(N) = 0$ for $k > d$ and $\lambda^k(N') = 0$ for $k > d'$. Then $\lambda^k(N + N') = 0$ for $k > d + d'$, and

$$\begin{aligned}
\lambda^p(N + N', x)\lambda_{-1}(N)\lambda_{-1}(N') &= \lambda^p\big(x\lambda_{-1}(N)\lambda_{-1}(N')\big) \\
&= \lambda^p\big(N', x\lambda_{-1}(N)\big)\lambda_{-1}(N').
\end{aligned}$$

Here we used $\lambda_{-1}(N + N') = \lambda_{-1}(N)\lambda_{-1}(N')$. Cancelling in the integral domain gives the relation, again valid by substitution,

$$\lambda^p(N + N', x)\lambda_{-1}(N) = \lambda^p\big(N', x\lambda_{-1}(N)\big). \tag{5.4.2}$$

5.5. Let A be a λ-ring and let $N \in A$ satisfy $\lambda^k(N) = 0$ for $k > d$. Define on the underlying abelian group of A a new multiplication by

$$x \cdot_N y = xy\lambda_{-1}(N).$$

After adjoining a unit, one obtains a ring $\mathbb{Z} \times A$ with

$$(n, x)(m, y) = (nm, ny + mx + x \cdot_N y).$$

The formulas

$$\lambda_t(n, x) = (1 + t)^n \left(1 + \sum_{p\geq 1} \lambda^p(N, x)t^p\right) \tag{5.5.1}$$

define a pre-λ-ring structure. In fact, when A is a λ-ring, this is a λ-ring structure; it is the one transported from the original structure by multiplication by $\lambda_{-1}(N)$. We denote this augmented λ-ring by $\widehat{A}_N$. Its γ-operations satisfy

$$\gamma^p(N,x)\lambda_{-1}(N) = \gamma^p(x\lambda_{-1}(N)). \tag{5.5.2}$$

5.6. Suppose that, in addition, L is an element such that $\lambda^k(L) = 0$ for $k > 1$. Then $\lambda^p(N+L,x)$ is the coefficient of t^p in the series

$$\frac{\sum_{k\geq 1} \lambda^k(N,x)(1+L+\cdots+L^{k-1})t^k}{1+\sum_{k\geq 1} \lambda^k(N,x)\lambda_{-1}(N)L^k t^k}. \tag{5.6.1}$$

This is obtained by applying (5.4.2) to $N+L$ and using $\lambda_{-1}(L) = 1-L$.

Proposition 5.7. Let K be a binomial ring, A an augmented K-λ-algebra, and $N \in A$ such that $\lambda^k(N) = 0$ for $k > d$. Then, for every $x \in A$ and every $p \geq 1$,

$$\gamma^p(N,x) \in \operatorname{Fil}^{p-d} A. \tag{5.7.1}$$

Indeed, let A_0 be the K-λ-algebra generated by N_0', X_0', subject to $\gamma^k(N_0') = 0$ for $k > d$, and augmented by $\epsilon(N_0') = \epsilon(X_0') = 0$. This is a polynomial algebra over K in the variables $\gamma^i(N_0')$, $1 \leq i \leq d$, and $\gamma^j(X_0')$, $j \geq 1$, and its λ-filtration is the weight filtration, with weight i assigned to $\gamma^i(N_0')$ and $\gamma^i(X_0')$ (4.12 and 4.15). Put

$$N_0 = N_0' + d, \qquad X_0 = X_0' + \epsilon(x).$$

For N_0, X_0, the assertion is equivalent to

$$\gamma^p(N_0, X_0)\gamma^d(N_0') \in \operatorname{Fil}^p A_0.$$

Since $\gamma^d(N_0') = (-1)^d \lambda_{-1}(N_0)$, this becomes

$$(-1)^d \gamma^p(X_0 \gamma^d(N_0')) \in \operatorname{Fil}^p A_0,$$

by (5.5.4). This holds because $\gamma^d(N_0')$ has zero augmentation. By 4.7 there is a K-λ-homomorphism $A_0 \to A$, compatible with augmentations and sending N_0' to $N-d$ and X_0' to $x-\epsilon(x)$, hence sending N_0 to N and X_0 to x. Compatibility with the filtrations gives the result.

6. Chern ring: opening

Throughout this section the base ring is a fixed binomial ring K. We use the functor G of 1.1, and return to the notation of §1: G denotes the functor there defined and G_i its restriction to $\mathscr{C}_0$.

6.1. In 3.5 and 3.6 we defined morphisms of functors

$$G_i \times G_i \xrightarrow{\star} G_i, \qquad G_i^\circ \xrightarrow{\gamma'} G_i^\circ \circ G_S, \qquad G_i \xrightarrow{a} G_i,$$

where the last map is multiplication, for the law ∇, by $\gamma_t(a)$ with $a \in K$. By 1.9 these give morphisms

$$G \times G \xrightarrow{\star} G, \qquad G \xrightarrow{\gamma'} G \circ G_S, \qquad G \xrightarrow{a} G.$$

The isobaricity of the universal polynomials involved implies that the first and third images lie in G, while the second lies in the subgroup of formal series in u whose non-zero terms have the prescribed degrees.

Definition 6.2. The functor defined in 6.1 on the category of graded K-algebras, with values in the category of K-λ-algebras, is called the *Chern ring*; its value on $A \in \mathrm{Ob}(\mathscr{C})$ is denoted $\mathrm{Ch}(A)$, or, if there is a risk of confusion, $\mathrm{Ch}_K(A)$.

From formulas (6.1.1), (6.1.2), (6.1.4), and (3.6.2), one deduces the relations

$$(1, 1+X)(1, 1+Y) = (1, 1+X+Y), \qquad \lambda^i(1, 1+X) = 0 \quad \text{for } i \geq 2, \tag{6.2.1}$$

the relations which served in [1] to define the Chern ring. Conversely, from these formulas one can recover the relations (6.1.1) and (6.1.4).

Let us indicate why the Chern ring is introduced. We place ourselves in the situation of Example 3.9, and write A for the graded ring associated, by the filtration defined in 3.10, with the $K^\bullet$ of a commutatively locally ringed topos; for simplicity suppose that $K^\bullet$ is a λ-ring. The augmentation ring is K. We shall see below that for every element of an augmented λ-ring one can define Chern classes with values in the associated graded ring (6.7). To every locally free sheaf of finite type one can therefore associate an element of $\mathrm{Ch}(A)$, whose component on the factor K is the rank of the sheaf and whose component on the factor $1 + \widehat{A}^+$ is the sum of the Chern classes. Formulas (6.2.1) are then the usual formulas giving the Chern classes and the rank of the tensor product of two invertible sheaves and of the exterior powers of an invertible sheaf; in other words, the map just defined from $K^\bullet$ to $\mathrm{Ch}(A)$ is a λ-homomorphism.

6.3. For every graded K-algebra A, denote by $A_{\mathbb{Q}}$ the K-algebra $A \otimes \mathbb{Q}$, graded in the evident way. By 1.13 one defines a functorial homomorphism

$$1 + \widehat{A}^+ \xrightarrow{\mathrm{ch}} \widehat{A}^+_{\mathbb{Q}}$$

by giving its value on $1 + X$:

$$\mathrm{ch}(1+X) = \exp(X) - 1. \tag{6.3.1}$$

This homomorphism is compatible with the multiplication defined in 6.1 on $1 + \widehat{A}^+$. By 1.13 it is enough to show this for the elements $1+X$, $1+Y$ of the universal ring $1 + K[X,Y]^+$. One has

$$\begin{aligned} \mathrm{ch}\big((1+X) * (1+Y)\big) &= \mathrm{ch}\left(\frac{1+X+Y}{(1+X)(1+Y)}\right) \\ &= \mathrm{ch}(1+X+Y) - \mathrm{ch}(1+X) - \mathrm{ch}(1+Y) \\ &= \exp(X+Y) - \exp(X) - \exp(Y) + 1 \\ &= \mathrm{ch}(1+X)\,\mathrm{ch}(1+Y). \end{aligned}$$

Let η be the endomorphism of $\widehat{A}^+_{\mathbb{Q}}$, functorial in A, defined by

$$\eta\left(\sum_{k \geq 1} a_k t^k\right) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{(k-1)!}\, a_k t^k.$$

Then, for every $\varphi \in 1 + \widehat{A}^+$,

$$\mathrm{ch}(\varphi) = \eta(\operatorname{Log} \varphi). \tag{6.3.2}$$

Indeed, ch and $\eta \circ \operatorname{Log}$ are two functorial homomorphisms of abelian groups, and it is enough for them to agree on $1 + X$, which is evident.

It follows that ch commutes with the operations of K: again it is enough to show this for $1+X$, and for $a \in K$ one has

$$\begin{aligned}\operatorname{ch}((1+X)^a) &= \operatorname{ch}(\exp(a \operatorname{Log}(1+X)))\\ &= \eta(a \operatorname{Log}(1+X))\\ &= a\,\eta(\operatorname{Log}(1+X))\\ &= a \operatorname{ch}(1+X).\end{aligned}$$

Thus one can extend ch in the evident way to a homomorphism of K-algebras

$$K \times (1+\widehat{A}^{+}) \longrightarrow K \oplus \widehat{A}^{+}_{\mathbb{Q}},$$

namely

$$\operatorname{ch}(a, \varphi) = a + \eta(\operatorname{Log} \varphi). \tag{6.3.3}$$

Definition 6.4. The homomorphism

$$\operatorname{ch} : \operatorname{Ch} A \longrightarrow K \oplus \widehat{A}^{+}_{\mathbb{Q}}$$

just defined is called the *Chern character*.

If A is of characteristic zero, that is, is a $\mathbb{Q}$-algebra, the Chern character is an isomorphism. Indeed, if $\psi \in K \oplus \widehat{A}^{+}$, one may write $\psi = a + \psi_1$, where ψ_1 is of positive degree, and then an inverse morphism is defined by

$$\psi \longmapsto (a, \exp(\eta^{-1}(\psi_1))).$$

Proposition 6.5. Let A be a graded K-algebra. On $\operatorname{Ch}(A)$ consider the filtration defined as follows, the degree filtration:

$$(\operatorname{Ch}(A))_0 = \operatorname{Ch}(A), \qquad (\operatorname{Ch}(A))_n = 0 \times (1+\widehat{A}^{\geq n}) \quad (n \geq 1), \tag{6.5.1}$$

where $\widehat{A}^{\geq n} = \prod_{i \geq n} A^i$. The following congruences hold.

i) If $n \geq 1$ and $\varphi \in (\operatorname{Ch} A)_n$, writing $\varphi = 1 + a_n + \cdots$, then

$$\operatorname{ch}(\varphi) = \frac{(-1)^{n-1}}{(n-1)!}\, a_n + \cdots . \tag{6.5.2}$$

ii) If $\varphi \in (\operatorname{Ch} A)_m$, $\psi \in (\operatorname{Ch} A)_n$, and if $\varphi = 1 + a_m + \cdots$, $\psi = 1 + b_n + \cdots$, then

$$\varphi * \psi = 1 - \frac{(m+n-1)!}{(m-1)!(n-1)!}\, a_m b_n + \cdots . \tag{6.5.3}$$

Consequently the degree filtration is a ring filtration.

For (i), the Chern character is by construction a morphism of filtered functors; from the congruence $\varphi \equiv 1 + a_n \pmod{(\operatorname{Ch} A)_{n+1}}$ one obtains

$$\operatorname{ch}(\varphi) = \operatorname{ch}(1+a_n) \pmod{\widehat{A}^{\geq n+1}},$$

and, since $\operatorname{ch}(1+a_n) = \eta(\operatorname{Log}(1+a_n))$, this is

$$\frac{(-1)^{n-1}}{(n-1)!} a_n + \cdots .$$

For (ii), by substitution it is enough to prove (6.5.3) for the torsion-free universal polynomial algebra

$$A = K[X_i, Y_j]_{i,j\in\mathbb{N}}$$

and for $\varphi = 1+\sum_{k\geq m} X_k$, $\psi = 1+\sum_{k\geq n} Y_k$. Since K is torsion-free in the universal case, one can embed this algebra in its tensor product with $\mathbb{Q}$, so that ch is injective. Put $\varphi * \psi = 1 + Z_p + \cdots$, where Z_p is the first non-zero term different from 1. The multiplicativity of ch, together with (6.5.2), gives

$$1 + \frac{(-1)^{p-1}}{(p-1)!} Z_p + \cdots = \left(1 + \frac{(-1)^{m-1}}{(m-1)!} X_m + \cdots\right)\left(1 + \frac{(-1)^{n-1}}{(n-1)!} Y_n + \cdots\right),$$

which gives the stated value of Z_p by comparing the terms of lowest degree.

6.6. The λ-ring $\mathrm{Ch}\, A$ is augmented over K: one defines a λ-homomorphism $\mathrm{Ch}\, A \to K$ by $(a, \varphi) \mapsto a$. It is therefore equipped with the canonical filtration of an augmented λ-ring. We compare it with the filtration just defined.

Proposition 6.6.1. Let $x \in \mathrm{Ch}\, A$ be an element of the form $x = (n, 1 + \cdots + a_n)$. Then:

$$\begin{aligned} &\text{i)} && \lambda^i(x) = 0 \quad \text{for } i > n,\\ &\text{ii)} && \lambda^n(x) = (1, 1 + a_1),\\ &\text{iii)} && \sum_{i=0}^{n} (-1)^i \lambda^i(x) = (0, 1 - (n-1)! a_n + \cdots), \end{aligned}$$

where the unwritten terms have degree $> n$.

By functoriality, it is enough to prove this in the case where $A = K[X_1, \ldots, X_n]$, filtered by weight with respect to the X_i, X_i having weight i, and for the element $(n, 1 + X_1 + \cdots + X_n)$. Embed A in $A' = K[T_1, \ldots, T_n]$ by the symmetric functions. Then

$$(n, 1 + X_1 + \cdots + X_n) = \left(n, \prod_{i=1}^{n} (1 + T_i)\right) = \sum_{i=1}^{n} (1, 1 + T_i).$$

Hence, by (6.2.1),

$$\lambda_t(x) = \prod_{i=1}^{n} (1 + (1, 1 + T_i)t),$$

so that

$$\lambda^k(x) = 0 \quad (k > n), \qquad \lambda^n(x) = \prod_{i=1}^{n} (1, 1 + T_i) = \left(1, 1 + \sum_{i=1}^{n} T_i\right) = (1, 1 + X_1).$$

Finally,

$$\sum_{i=1}^{n} (-1)^i \lambda^i(x) = \lambda_{-1}(x) = \prod_{i=1}^{n} (1 - (1, 1 + T_i)) = (-1)^n \prod_{i=1}^{n} (0, 1 + T_i).$$

By (6.5.3),

$$\prod_{i=1}^{n} (0, 1 + T_i) = \left(0, 1 + (-1)^{n-1}(n-1)! \prod_{i=1}^{n} T_i + \cdots\right),$$

hence

$$\lambda_{-1}(x) = (0, 1 - (n-1)! X_n + \cdots).$$

Corollary 6.6.2. Relation 6.6.1(iii) is true for every element X of augmentation n.

For this we shall use the following lemma.

Lemma 6.6.3. For every n, $(\operatorname{Ch} A)_n$ is a λ-ideal of $\operatorname{Ch} A$.

We may suppose $n > 1$. Let $x = (0, \varphi) \in (\operatorname{Ch} A)_n$, and write

$$\lambda^i(0, \varphi) = (0, 1 + Q^i_{i,1} + \cdots + Q^i_{i,j} + \cdots),$$

where the $Q^i_{i,j}$ are the isobaric components of weight j of the Q^i_i of (3.6). For every j, the polynomial $Q^i_{i,j}$ is a polynomial in the first j coefficients of φ. If $j < n$, these coefficients are zero; hence the $Q^i_{i,j}$ are zero, and so the $\lambda^i(x)$ belong to $(\operatorname{Ch} A)_n$. Thus $(\operatorname{Ch} A)_n$ is a λ-ideal.

Let $x = (n, 1 + a_1 + \cdots + a_n + \cdots)$ be an element of augmentation n in $\operatorname{Ch} A$. Then

$$x \equiv (n, 1 + \cdots + a_n) \pmod{(\operatorname{Ch} A)_{n+1}},$$

and therefore

$$\lambda_{-1}(x) \equiv \lambda_{-1}(n, 1 + \cdots + a_n) = (0, 1 - (n-1)! a_n + \cdots),$$

which proves the corollary.

Corollary 6.6.4. Let ϵ be the augmentation homomorphism of $\operatorname{Ch} A$. Then, for every $x \in \operatorname{Ch} A$, one has

$$\gamma^n(x - \epsilon(x)) = (0, 1 + (-1)^{n-1}(n-1)! a_n + \cdots).$$

By definition,

$$\gamma^n(x - \epsilon(x)) = (-1)^n \sum_{i=0}^{n} (-1)^i \lambda^i(x + n - \epsilon(x)).$$

Since $x + n - \epsilon(x)$ has augmentation n, Corollary 6.6.2 applies to it, giving the result.

Corollary 6.6.5. With the notation of 3.10,

$$\operatorname{Fil}^n \operatorname{Ch} A \subset (\operatorname{Ch} A)_n.$$

This follows from the preceding corollary, which gives $\gamma^n(x - \epsilon(x)) \in (\operatorname{Ch} A)_n$, and from the fact that the two filtrations are ring filtrations and that the elements $\gamma^n(x - \epsilon(x))$, for n and x variable, generate the λ-ring filtration.

Corollary 6.6.7. Let A be a graded K-algebra of characteristic zero, bounded by degree. Then the two filtrations defined on $\operatorname{Ch} A$ are identical.

Suppose that $\operatorname{Fil}^p A = 0$, and let $x = (0, 1 + a_n + \cdots + a_p)$ be an element of $(\operatorname{Ch} A)_n$. Put

$$x' = \left(0, 1 + \frac{(-1)^{n-1}}{(n-1)!} a_n + a_{n+1} + \cdots + a_p\right).$$

Then $\gamma^n(x') = (0, 1 + a_n + \cdots + a'_p)$, since x' has zero augmentation. Hence $x - \gamma^n(x') \in (\operatorname{Ch} A)_{n+1}$. Iterating the operation $p - n$ times, one reaches $(\operatorname{Ch} A)_{p+1} = 0$, and therefore x is written as an element of $\operatorname{Fil}^n \operatorname{Ch} A$. In view of (6.6.5), the two filtrations are identical.

6.7. Now suppose that A is a λ-ring augmented over K, and let ϵ be the augmentation homomorphism. Then A is equipped with its canonical filtration of augmented λ-ring, and one can associate to $\operatorname{Gr} A$ its Chern ring.

If $x \in A$, then $\gamma_t(x - \epsilon(x)) \in \widehat{G}(A)$ by definition of the filtration of an augmented λ-ring. We write $c(x)$ for the image of $\gamma_t(x - \epsilon(x))$ in

$$\widehat{G}(A)/\widehat{G}'(A) = 1 + \widehat{\operatorname{Gr} A}^{+}.$$

It is called the *total Chern class* of x. We write $c^i(x)$, and call it the i-th Chern class of x, for the i-th coefficient of $c(x)$. Thus

$$c^i(x) = \mathrm{cl}_{\mathrm{Gr}^i A}\Big(\gamma^i(x - \epsilon(x))\Big), \qquad c(x) = 1 + c^1(x) + \cdots . \tag{6.7.1}$$

Put

$$\widetilde{c}(x) = (\epsilon(x), c(x)). \tag{6.7.2}$$

This is called the *completed Chern class* of x.

Proposition 6.8. The completed Chern class

$$\widetilde{c} : A \longrightarrow \mathrm{Ch}(\mathrm{Gr}\, A)$$

is a K-λ-homomorphism of augmented K-λ-algebras.

It is clear that the homomorphism so defined is additive. To verify that it is multiplicative, write

$$\begin{aligned}\gamma_t(xy - \epsilon(xy)) &= \gamma_t(\epsilon(y)(x - \epsilon(x)))\gamma_t(\epsilon(x)(y - \epsilon(y)))\gamma_t((x - \epsilon(x))(y - \epsilon(y))) \\ &= \big(\gamma_t(x - \epsilon(x))\big)^{\epsilon(y)}\big(\gamma_t(y - \epsilon(y))\big)^{\epsilon(x)}\big(\gamma_t(x - \epsilon(x)) \circ \gamma_t(y - \epsilon(y))\big),\end{aligned}$$

which gives the result by (6.1.2) after passage to the quotient. If $a \in K$, then $\widetilde{c}(a) = (a, 1)$, so $\widetilde{c}$ is a homomorphism of K-algebras; moreover it is compatible with augmentations.

It remains to see that it is a λ-homomorphism. One must prove the commutativity of the diagram

$$\begin{tikzcd} A \arrow[r, "\gamma_t"] \arrow[d, "\widetilde{c}"'] & \widehat{G}(A) \arrow[d, "\widehat{G}(\widetilde{c})"] \\ \mathrm{Ch}\, A \arrow[r, "\gamma_t"'] & \widehat{G}(\mathrm{Ch}\, A). \end{tikzcd}$$

By additivity, it is enough to prove it for the elements of K, where it is evident, and for the elements of zero augmentation in A, where it follows, by the definition of c and (6.1.4), from diagram (3.7.2).

Proposition 6.9. Let A be an augmented K-λ-algebra and let $z \in \mathrm{Fil}^m A$. Then

$$\gamma^m(z) = (-1)^{m-1}(m-1)!z \pmod{\mathrm{Fil}^{m+1} A},$$

that is,

$$c^m(z) = \mathrm{cl}_{\mathrm{Gr}^m A}\Big((-1)^{m-1}(m-1)!z\Big).$$

One must see that these two elements give the same image in $\mathrm{Gr}^m A$, that is, that the m-th Chern class of z is the image of $(-1)^{m-1}(m-1)!z$.

One may write z as a sum

$$\sum a\, \gamma^{i_1}(x_1) \cdots \gamma^{i_k}(x_k),$$

with $a \in A$, $\epsilon(x_i) = 0$ for all i, and $i_1 + \cdots + i_k \geq m$. We then have

$$\widetilde{c}(z) = \sum \widetilde{c}(a)\, \widetilde{c}(\gamma^{i_1}(x_1)) \cdots \widetilde{c}(\gamma^{i_k}(x_k)).$$

Since $\widetilde{c}$ is a λ-homomorphism, one may write

$$\widetilde{c}(\gamma^i(x_j)) = \gamma^i(\widetilde{c}(x_j)).$$

Taking account of the definition of c and of (6.6.4), one obtains

$$\widetilde{c}(\gamma^{i_j}(x_j)) = \Big(0, 1 + (-1)^{i_j-1}(i_j-1)!\,\gamma^{i_j}(x_j) + \cdots\Big).$$

By (6.5.3), and since $\widetilde{c}(a) = (a, 1)$, one finally gets

$$\begin{aligned}\widetilde{c}(z) &= \sum \Big(0, \Big(1 + (-1)^{i_1+\cdots+i_k-1}(i_1+\cdots+i_k-1)!\,\gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k) + \cdots\Big)^a\Big)\\ &= \left(0, 1 + \sum_{\substack{(x_1,\ldots,x_k)\\ i_1+\cdots+i_k=m}} (-1)^{m-1}(m-1)!a\,\gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k) + \cdots\right)\\ &= (0, 1 + (-1)^{m-1}(m-1)!z + \cdots),\end{aligned}$$

which proves the assertion.

Proposition 6.10. Let K be a binomial ring, R an augmented K-λ-algebra, $N \in R$ an element such that $\lambda^k(N) = 0$ for $k > d$, and $x \in \operatorname{Fil}^i R$. Then

$$\gamma^{d+i}(N, x) - (-1)^{d+i-1}(d+i-1)!x \in \operatorname{Fil}^{i+1} R.$$

One can find elements $z_{\alpha,j} \in R$ and $a_\alpha \in K$ such that

$$x = \sum_\alpha a_\alpha\,\gamma^{i_1}(x_{\alpha,1})\cdots\gamma^{i_k}(x_{\alpha,k}),$$

with $i_1 + \cdots + i_k \geq i$ and $\epsilon(x_{\alpha,j}) = 0$ for every α and every j. Let R_0 be the K-λ-algebra generated by N_0' and $X_{\alpha,j}$, subject to the relations

$$\gamma^k(N_0') = 0 \quad \text{for } k > d.$$

Define an augmentation homomorphism $R_0 \to K$ by $\epsilon(N_0') = \epsilon(X_{\alpha,j}) = 0$. By (4.9.1), R_0 is the ring of polynomials with coefficients in K in the indeterminates $\gamma^k(N_0')$, $1 \leq k \leq d$, and $\gamma^p(X_{\alpha,j})$, $p \geq 1$; the λ-filtration of R_0 is its weight filtration, where $\gamma^k(N_0')$ and $\gamma^k(X_{\alpha,j})$ have weight k (4.12 and 4.15).

Put

$$X_0 = \sum_\alpha a_\alpha\,\gamma^{i_1}(X_{\alpha,1})\cdots\gamma^{i_k}(X_{\alpha,k}).$$

We shall prove 6.10 first for X_0 and $N_0' = N_0 + d$, which do satisfy the hypotheses of 6.10. In view of the preceding remark about the filtration of R_0, it is enough to show that

$$\gamma^{d+i}(N_0, X_0)\gamma^d(N_0') - (-1)^{d+i-1}(d+i-1)!X_0\gamma^d(N_0') \in \operatorname{Fil}^{d+i+1} R_0.$$

But $\lambda_{-1}(N_0) = (-1)^d\gamma^d(N_0')$. Thus this relation becomes

$$(-1)^d\gamma^{d+i}(X_0, (-1)^d\gamma^d(N_0')) - (-1)^{d+i-1}(d+i-1)!X_0\gamma^d(N_0') \in \operatorname{Fil}^{d+i+1} R_0.$$

Putting $z = (-1)^d X_0\gamma^d(N_0')$, we have $z \in \operatorname{Fil}^{d+i} R_0$, and the statement is reduced to

$$\gamma^{d+i}(z) - (-1)^{d+i-1}(d+i-1)!z \in \operatorname{Fil}^{d+i+1} R_0,$$

which is just 6.9.

To deduce 6.10 from this particular case, consider the K-λ-homomorphism $R_0 \to R$ which sends N_0' to $N - d$ and $X_{\alpha,j}$ to $x_{\alpha,j}$, for every α and j. This homomorphism is compatible with the λ-filtrations and sends N_0 to N and X_0 to x. Hence the result follows.

Proposition 6.11. Suppose that K is of characteristic zero, and let $n \geq 0$ be an integer. Then the functor Ch is an equivalence from the category of graded K-algebras A such that $A_0 = K$ and $A_k = 0$ for $k > n$, to the category of augmented K-λ-algebras Λ such that $\mathrm{Fil}^{n+1} \Lambda = 0$. A quasi-inverse functor is given by $\Lambda \mapsto \mathrm{Gr}\,\Lambda$.

Let A be a graded K-algebra such that $A_k = 0$ for $k > n$, equipped with the filtration associated to its grading. Then $(\mathrm{Ch}\,A)_{n+1} = 0$, and hence $\mathrm{Fil}^{n+1}\,\mathrm{Ch}\,A = 0$ by (6.6.5). Conversely, if Λ is an augmented K-λ-algebra such that $\mathrm{Fil}^{n+1} \Lambda = 0$, then $\mathrm{Gr}^0 \Lambda = K$ and $\mathrm{Gr}^k \Lambda = 0$ for $k > n$.

We shall define isomorphisms of functors

$$A \longrightarrow \mathrm{Gr}(\mathrm{Ch}\,A), \qquad \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}\,\Lambda).$$

Since A is of finite grading, the associated gradeds of the λ-ring filtration on $\mathrm{Ch}\,A$ and of the degree filtration are the same by (6.6.7). Define

$$u : A \longrightarrow \mathrm{Gr}_{\deg}(\mathrm{Ch}\,A)$$

to be the identity on A_0, and, for $i \geq 1$ and $a_i \in A_i$,

$$u(a_i) = \mathrm{cl}_{\mathrm{Gr}^i}\left(0, 1 + (-1)^{i-1}(i-1)!a_i\right),$$

then extend by additivity. By relation (6.5.3) this is a homomorphism of graded rings, and it is easy to see that it is compatible with the operations of K. The inverse morphism u^{-1} may be defined as follows: for $i \geq 1$ and $\bar{\varphi} \in \mathrm{Gr}^i(\mathrm{Ch}\,A)$, the i-th coefficient a_i of φ is independent of the lift of $\bar{\varphi}$ in $(\mathrm{Ch}\,A)_i$, and one puts

$$u^{-1}(\bar{\varphi}) = \frac{(-1)^{i-1}}{(i-1)!} a_i.$$

Thus u is an isomorphism.

The isomorphism $\Lambda \to \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$ is defined by the completed Chern class $\widetilde{c}$. To verify that it really defines an isomorphism, it is enough to verify that the homomorphism obtained by passing to associated graded rings is one, the filtrations being discrete. One gets

$$\mathrm{Gr}\,\Lambda \xrightarrow{\mathrm{Gr}(\widetilde{c})} \mathrm{Gr}_\lambda(\mathrm{Ch}(\mathrm{Gr}\,\Lambda)) \simeq \mathrm{Gr}_{\deg}(\mathrm{Ch}(\mathrm{Gr}\,\Lambda)) \xrightarrow{u^{-1}} \mathrm{Gr}\,\Lambda.$$

It remains to show that the composite is the identity. Since the homomorphisms are homomorphisms of graded K-algebras, it is enough, by the definition of the λ-ring filtration of Λ, to show that it is the identity on the $\gamma^i(x)$, where $\epsilon(x) = 0$. One then obtains

$$\mathrm{Gr}(\widetilde{c})(\gamma^i(x)) = \mathrm{cl}_{\mathrm{Gr}^i}(\gamma^i(c(x))).$$

But $\gamma^i(\widetilde{c}(x)) = (0, 1 + (-1)^{i-1}(i-1)!c^i(x) + \cdots)$, by (6.6.4). Therefore

$$\mathrm{cl}_{\mathrm{Gr}^i}(\gamma^i(\widetilde{c}(x))) = u(c^i(x)),$$

which shows that $u^{-1} \circ \mathrm{Gr}(\widetilde{c}) = \mathrm{Id}$.

Corollary 6.12. If K is of characteristic zero, and if Λ is a K-λ-algebra with discrete filtration, then the completed Chern class

$$\widetilde{c} : \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$$

is an isomorphism.

The proof is contained in that of 6.11.

Definition 6.13. Let A be a graded K-algebra and Λ an augmented K-λ-algebra. A *Chern theory for* Λ *with values in* A is a λ-homomorphism

$$\Lambda \longrightarrow \mathrm{Ch}(A)$$

compatible with augmentations.

Thus one defines a functor on the category of graded K-algebras, with values in the category of sets, by associating with a graded K-algebra A the set of Chern theories of Λ with values in A, namely

$$\mathrm{Hom}_\lambda(\Lambda, \mathrm{Ch}(A)).$$

Proposition 6.14. The functor defined in 6.13 is representable. If K is of characteristic zero and Λ has discrete filtration, it is represented by the graded ring associated with Λ.

First prove the first assertion. Consider the graded K-algebra generated by generators $c^i(x)$, with $i \geq 1$, $x \in \mathrm{Fil}^1 \Lambda$, $c^i(x)$ of degree i, and take its quotient by the following relations:

$$\begin{aligned}
&\text{i)} \quad c^k(x+y) = \sum_{i+j=k} c^i(x)c^j(y), \quad k \geq 1,\ x, y \in \mathrm{Fil}^1 \Lambda,\\
&\text{ii)} \quad c^k(xy) = P''_k(c^i(x), c^j(y)) \quad (\text{cf. } 6.1), \quad k \geq 1,\ x, y \in \mathrm{Fil}^1 \Lambda,\\
&\text{iii)} \quad c^k(ax) = R_{k,a}(c^i(x)) \quad (\text{cf. } 6.1), \quad k \geq 1,\ a \in K,\ x \in \mathrm{Fil}^1 \Lambda,\\
&\text{iv)} \quad c^k(\gamma^j(x)) = Q''_{j,k}(c^i(x)) \quad (\text{cf. } 6.1), \quad j, k \geq 1,\ x \in \mathrm{Fil}^1 \Lambda.
\end{aligned}$$

These relations are isobaric, so the quotient is a graded K-algebra Ω. Define a morphism

$$\widetilde{c}_0 : \Lambda \longrightarrow \mathrm{Ch}(\Omega)$$

by

$$x \longmapsto (\epsilon(x), 1 + c^1(x - \epsilon(x)) + \cdots).$$

The imposed relations show that this is a Chern theory for Λ with values in Ω. Conversely, if f is a Chern theory with values in a graded K-algebra A, then there is a unique homomorphism $\Omega \to A$ through which f factors through $\mathrm{Ch}(\Omega)$: it sends, for every $i \geq 1$ and every $x \in \mathrm{Fil}^1 \Lambda$, the element $c^i(x)$ to the i-th coefficient of $f(x)$. This is well defined because f is a λ-homomorphism.

Now suppose that K is of characteristic zero and Λ has discrete filtration. Let n be such that $\mathrm{Fil}^n \Lambda = 0$. Then the graded algebra Ω is zero in degrees $> n$. To show this, it is enough to prove that, for

$$x_1, \ldots, x_p \in \mathrm{Fil}^1 \Lambda \quad \text{and} \quad k_1 + \cdots + k_p \geq n,$$

one has

$$c^{k_1}(x_1) \cdots c^{k_p}(x_p) = 0.$$

By hypothesis,

$$\gamma^{k_1}(x_1) \cdots \gamma^{k_p}(x_p) = 0.$$

Applying the λ-homomorphism $\widetilde{c}_0$, one gets

$$\gamma^{k_1}(\widetilde{c}_0(x_1)) \cdots \gamma^{k_p}(\widetilde{c}_0(x_p)) = (0, 1).$$

By (6.6.4) and (6.5.3), this implies

$$(k_1 + \cdots + k_p)!\, c^{k_1}(x_1) \cdots c^{k_p}(x_p) = 0.$$

Since K is of characteristic zero, the desired result follows.

The universal property of Ω, applied to the canonical homomorphism $\widetilde{c}$, gives a commutative diagram

$$\begin{tikzcd} \Lambda \arrow[r, "\widetilde{c}"] \arrow[d, "\widetilde{c}_0"'] & \mathrm{Ch}(\mathrm{Gr}\,\Lambda) \\ \mathrm{Ch}(\Omega) \arrow[ur] & \end{tikzcd}$$

By Proposition 6.11, the homomorphism $\Omega \to \mathrm{Gr}\,\Lambda$ is an isomorphism if and only if the homomorphism

$$\mathrm{Ch}(\Omega) \longrightarrow \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$$

is one. Since it is visibly surjective, the latter is also surjective; it remains to show that $\mathrm{Ch}(\Omega) \to \mathrm{Ch}(\mathrm{Gr}\,\Lambda)$ is injective. By functoriality of Ch, one obtains the diagram

$$\begin{tikzcd} \Lambda \arrow[r, "\widetilde{c}_0"] \arrow[d, "\widetilde{c}"'] & \mathrm{Ch}(\Omega) \arrow[d, "\widetilde{c}"] \\ \mathrm{Ch}(\mathrm{Gr}\,\Lambda) \arrow[r, "\mathrm{Ch}(\widetilde{c}_0)"'] & \mathrm{Ch}\big(\mathrm{Gr}_\Lambda(\mathrm{Ch}(\Omega))\big). \end{tikzcd}$$

Since Ω has bounded degrees, the right vertical arrow is an isomorphism by (6.12). Using 6.11, one therefore obtains a morphism

$$\mathrm{Gr}\,\Lambda \longrightarrow \Omega$$

whose two composites with $\Omega \to \mathrm{Gr}\,\Lambda$ are the identity, by the universal property of Ω.

7. Appendix: the Adams operations ψ^k

We follow here a presentation due to J.-P. Serre.

7.1. Let K be a pre-λ-ring. For $x \in K$, write

$$\frac{d}{dt}(\lambda_t(x))$$

for the derivative with respect to t of $\lambda_t(x)$. Define elements of K, denoted $\psi^k(x)$, by putting

$$\frac{d}{dt}(\lambda_t(x))/\lambda_t(x) = \sum_{k=1}^{\infty}(-1)^{k-1}\psi^k(x)t^{k-1}. \tag{7.1.1}$$

The maps $\psi^k : K \to K$ are additive. Indeed,

$$\begin{aligned}\frac{d}{dt}(\lambda_t(x+y))/\lambda_t(x+y) &= \frac{d}{dt}(\lambda_t(x)\lambda_t(y))/\lambda_t(x)\lambda_t(y) \\ &= \frac{d}{dt}(\lambda_t(x))/\lambda_t(x) + \frac{d}{dt}(\lambda_t(y))/\lambda_t(y),\end{aligned}$$

and therefore, for every k,

$$\psi^k(x+y) = \psi^k(x) + \psi^k(y).$$

By (7.1.1), the ψ^k are isobaric polynomials of weight k in the λ^i, $i = 1, \ldots, k$. For example,

$$\psi^1(x) = \lambda^1(x) = x, \qquad \psi^2(x) = x^2 - 2\lambda^2(x).$$

7.2. Let K be a ring of characteristic zero. Suppose given additive maps $\psi^k : K \to K$ such that $\psi^1 = \mathrm{Id}_K$. Define a pre-λ-structure on K by

$$\lambda_t(x) = \exp\left(\sum_{k=1}^{\infty}(-1)^{k-1}\psi^k(x)t^k/k\right). \tag{7.2.1}$$

Multiplicativity follows from the additivity of the ψ^k. Thus there is an equivalence between giving a pre-λ-structure on K and giving the corresponding operations ψ^k.

7.3. It is clear from formula (7.1.1) that the ψ^k commute with λ-homomorphisms. If K, K' are pre-λ-rings of characteristic zero, it is necessary and sufficient for a homomorphism $f : K \to K'$ to be a λ-homomorphism that it commute with the ψ^k. This follows from 7.2, since the λ^i are expressed as polynomials in the ψ^k.

Examples 7.4. i) For $K = \mathbb{Z}$, with its λ-ring structure, one has

$$\frac{d}{dt}(\lambda_t(n))/\lambda_t(n) = n(1+t)^{n-1}/(1+t)^n = n\sum_{k=1}^{\infty}(-1)^{k-1}t^{k-1}.$$

Consequently the ψ^k are all equal to the identity map of $\mathbb{Z}$.

ii) Work in the category $\mathscr{C}_0$ of $\mathbb{Z}$-algebras (1.8). By 7.3, the operations ψ^k define additive endomorphisms of the functor G on λ-rings. By 1.10 they are characterized by their value on $1 + Xt$. One has

$$\frac{d}{dt}(\lambda_u(1+Xt))/\lambda_u(1+Xt) = \frac{1+Xt}{1+\{1+Xt\}u} = \sum_{k=1}^{\infty}(-1)^{k-1}\{1+Xt\}^{\circ k}u^{k-1}.$$

Thus

$$\psi^k(1+Xt) = 1 + X^k t. \tag{7.4.1}$$

Proposition 7.5. i) If K is a λ-ring, the operations ψ^k satisfy the identities

$$\psi^k(1) = 1, \qquad \psi^k(xy) = \psi^k(x)\psi^k(y), \qquad \psi^k \circ \psi^{k'} = \psi^{kk'}. \tag{7.5.1}$$

ii) Conversely, if the ψ^k satisfy these identities and if K is of characteristic zero, then K is a λ-ring.

For (i), if K is a λ-ring, $\lambda_t(1) = 1 + t$, hence

$$\frac{d}{dt}(\lambda_t(1))/\lambda_t(1) = \frac{1}{1+t} = \sum_{k=1}^{\infty}(-1)^{k-1}t^{k-1},$$

and therefore $\psi^k(1) = 1$.

To prove the second relation of (7.5.1), consider the diagram

$$\begin{tikzcd}
K \times K \arrow[r, "\lambda_t \times \lambda_t"] \arrow[d] & \widehat{G}(K) \times \widehat{G}(K) \arrow[r, "g^k \times g^k"] \arrow[d, "\circ"] & K \times K \arrow[d] \\
K \arrow[r, "\lambda_t"'] & \widehat{G}(K) \arrow[r, "g^k"'] & K,
\end{tikzcd} \tag{7.5.2}$$

where the vertical arrows are defined by multiplication, and the maps g^k as follows: the logarithmic derivative is a group homomorphism from $G(K)$ to $K[[t]]$, and the map which to a series assigns its $(k-1)$-st coefficient, multiplied by $(-1)^{k-1}$, is an additive homomorphism from $K[[t]]$ to K; g^k is their composite. Clearly $g^k \circ \lambda_t = \psi^k$ by definition (7.1.1).

The second of formulas (7.5.1) therefore expresses the commutativity of the diagram just written. If K is a λ-ring, the left square is commutative, and it is enough to verify the commutativity of the right square. This is a diagram of functor homomorphisms on $\mathscr{C}_0$, with source $\widehat{G} \times \widehat{G}$. By 1.10 it is enough to verify commutativity on $(1+Xt, 1+Yt)$. Then

$$g^k(1+Xt) = X^k, \qquad g^k(1+Yt) = Y^k,$$

and

$$g^k((1+Xt) \circ (1+Yt)) = g^k(1+XYt) = (XY)^k = g^k(1+Xt)g^k(1+Yt),$$

which gives the result.

To prove the last relation, first observe that the diagram

$$\begin{tikzcd} \widehat{G}_t(K) \arrow[r, "\lambda_u"] \arrow[d, "g^k"'] & \widehat{G}_u\widehat{G}_t(K) \arrow[d, "g^k_u"] \\ K \arrow[r, "\lambda_u"'] & \widehat{G}^u(K) \end{tikzcd} \tag{7.5.3}$$

is commutative. As above, it is enough to see this for $1+Xt$, and one has

$$g_u^{k'}\lambda_u(1+Xt) = g_u^{k'}(1+\{1+Xt\}u) = (1+Xt)^{\circ k'} = 1+X^{k'}t,$$

and then

$$g^k(1+X^{k'}t) = X^{kk'} = g^k(1+Xt).$$

Consequently

$$\psi^k \circ \psi^{k'} = g^k \circ g_u^{k'} \circ \lambda_u \circ \lambda_t = g^k \circ \lambda_t \circ \psi^{k'}.$$

Moreover

$$\psi^k \circ \psi^{k'} = g^k \circ \lambda_t \circ \psi^{k'} = \psi^k \circ \lambda_t \circ \psi^{k'}.$$

If K is a λ-ring, the diagram

$$\begin{tikzcd} K \arrow[r, "\lambda_t"] \arrow[d, "\psi^{k'}"'] & \widehat{G}(K) \arrow[d, "\psi^{k'}"] \\ K \arrow[r, "\lambda_t"'] & \widehat{G}(K) \end{tikzcd} \tag{7.5.4}$$

is commutative by 7.3, since λ_t is a λ-homomorphism. The desired relations follow at once.

For (ii), if K is of characteristic zero, the logarithmic derivative is an isomorphism from $\widehat{G}(K)$ onto $K[[t]]$, since one can integrate. Therefore two elements of $\widehat{G}(K)$ are equal if and only if, for every k, they have the same image by g^k.

To show that λ_t is a ring homomorphism, it is enough to show that the left square in (7.5.2) is commutative. By the preceding remark, it is enough to show that, for every k, diagram (7.5.2) is commutative; the right square always is. But this is exactly the relation

$$\psi^k(xy) = \psi^k(x)\psi^k(y).$$

To show that λ_t is a λ-homomorphism, it is enough, by 7.3, to show that diagram (7.5.4) is commutative for every k'. By the preceding remark, this is equivalent to showing, for all k, k', that

$$g^k \circ \psi^{k'} \circ \lambda_t = g^k \circ \lambda_t \circ \psi^{k'},$$

that is, the relations $\psi^{kk'} = \psi^k \circ \psi^{k'}$. Finally, the relation $\lambda_t(1) = 1+t$ is obtained by integration from the relations $\psi^k(1) = 1$.

Bibliography

[1] A. Grothendieck, *La théorie des classes de Chern*, Bull. Soc. math. France, 86, 1958, p. 137 à 154.

[2] F. Hirzebruch, *Neue topologische Methoden in der algebraischen Geometrie.*

Exposé VI. The $K^\bullet$ of a projective bundle: computations and consequences

Pierre Berthelot

1. Computation of $K^\bullet$ of a projective bundle: the case of locally free sheaves of finite type

In this section $K^\bullet(X)$ denotes the Grothendieck group of locally free sheaves of finite type on a scheme X. The aim is to prove the following structure theorem.

Theorem 1.1. Let S be a quasi-compact scheme, let E be a locally free $\mathcal{O}_S$-module of rank $r+1$, and let

$$X = \mathbf{P}(E)$$

be the associated projective bundle. The structural morphism $f : X \to S$ gives $K^\bullet(X)$ the structure of a $K^\bullet(S)$-algebra. If $\xi \in K^\bullet(X)$ denotes the class of $\mathcal{O}_X(1)$, then $K^\bullet(X)$ is a free $K^\bullet(S)$-module with basis

$$1, \xi, \ldots, \xi^r.$$

1.2. Plan of the proof

The proof is divided into three parts.

First, one proves that the powers of ξ generate $K^\bullet(X)$ as a $K^\bullet(S)$-module. For this it is enough to establish the following facts. If F is locally free of finite type on X, one can find a graded $\mathrm{Sym}(E)$-module M, of finite presentation and flat over $\mathcal{O}_S$, such that $F = \mathrm{Proj}_0(M)$. Such an M admits a finite graded resolution

$$0 \longrightarrow L_{r+1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

by graded $\mathrm{Sym}(E)$-modules locally free of finite type. Finally, each L_i has a finite filtration whose quotients are of the form

$$N \otimes_{\mathcal{O}_S} \mathrm{Sym}(E)(n),$$

with N locally free of finite type on S. Applying Proj_0 then gives the required generation by the powers of ξ.

Second, one proves that $1, \xi, \ldots, \xi^{r+1}$ are linearly dependent over $K^\bullet(S)$. This is obtained from the canonical epimorphism $f^*E \to \mathcal{O}_X(1)$.

Third, one proves that $1, \xi, \ldots, \xi^r$ are linearly independent over $K^\bullet(S)$.

Let S be a scheme, let $B = \bigoplus_{n\geq 0} B_n$ be a graded $\mathcal{O}_S$-algebra, and let M be a graded B-module. If N is a graded B-module, $N(n)$ denotes the graded module defined by

$$N(n)_k = N_{n+k}.$$

A graded B-module is said to be free, respectively locally free, respectively locally free of finite type, if locally on S it is a direct sum, respectively locally a direct sum, respectively locally a finite direct sum, of modules $B(n)$. It is said to be of finite presentation if, locally on S, it admits a presentation by free graded modules of finite type.

For every integer n there is a canonical homomorphism of graded B-modules

$$M_n \otimes_{\mathcal{O}_S} B(-n) \longrightarrow M, \tag{1.2.1}$$

defined by $x \otimes a \mapsto ax$. A free graded module of finite type admits a decomposition

$$M \simeq \bigoplus_{n=n_0}^{n_1} B(-n)^{k_n}. \tag{1.2.2}$$

Proposition 1.2. The integers k_n occurring in a decomposition (1.2.2) are independent of the chosen decomposition.

Indeed, after restricting to an open set where $B_0 = \mathcal{O}_S$, one compares the first non-zero homogeneous component. This determines k_{n_0}; passing to the quotient and arguing by induction gives all the k_n.

Proposition 1.3. Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of finite type, let $X = \mathbf{P}(E)$, and let F be an $\mathcal{O}_X$-module of finite presentation which is flat over S. Then there exists an integer n_0 such that, for all $n \geq n_0$,

(i) $R^i f_*(F(n)) = 0 \quad (i \geq 1)$,

(ii) $f_*(F(n))$ is locally free of finite type on S,

(iii) $M = \bigoplus_{n \geq n_0} f_*(F(n))$ is a graded $\operatorname{Sym}(E)$-module of finite presentation.

The statement is local on S. Since S is quasi-compact, one is reduced to the affine case. If S is noetherian, (i) is Serre's theorem, (ii) follows from EGA III, 7.9.10, and (iii) from the usual comparison between a coherent sheaf on a projective bundle and its associated graded module.

In the general affine case one uses passage to the limit. Write $S = \operatorname{Spec}(A)$ as a filtered projective limit of noetherian affine schemes $S_\lambda = \operatorname{Spec}(A_\lambda)$, with affine transition morphisms. The locally free module E, the scheme X, and the sheaf F descend, after increasing λ, to corresponding data $E_\lambda, X_\lambda, F_\lambda$, with F_λ flat over S_λ. The noetherian case applies to F_λ, and the theorem of change of base gives the assertions after pullback to S.

In this reduction one uses the Cartesian square

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & X_0 \arrow[d, "f_0"] \\ S \arrow[r, "g_0"'] & S_0. \end{tikzcd}$$

Here S_0 is noetherian, $X_0 = \mathbf{P}(E_0)$, and $F = g^*(F_0)$. The previous paragraph proves the assertion for F_0, and EGA III, 7.7.5 and 7.8.5 give the required base-change identifications for F.

Corollary 1.4. Under the hypotheses of Proposition 1.3 there exists a graded $\operatorname{Sym}(E)$-module M, of finite presentation and flat over $\mathcal{O}_S$, such that

$$F = \operatorname{Proj}_0(M).$$

Choose an integer n_0 for which the properties of Proposition 1.3 hold. For

$$M = \bigoplus_{n \geq n_0} f_*(F(n)),$$

we have seen that M is of finite presentation; it is flat over S, since the $f_*(F(n))$, for $n \geq n_0$, are locally free on S. Moreover, $F = \mathrm{Proj}_0(M)$ by EGA II 3.4. We now turn to part b) of the proof.

Lemma 1.5.1. Let A be a ring, let B be a positively graded A-algebra with $B_0 = A$, and let M be a graded B-module. Suppose that M, regarded as an ungraded B-module, is projective of finite type. Let $\widetilde{B}$ be the sheaf of algebras on $\mathrm{Spec}(A)$ defined by B, and let $\widetilde{M}$ be the associated graded $\widetilde{B}$-module. Then, locally on $\mathrm{Spec}(A)$, $\widetilde{M}$ is free of finite type over $\widetilde{B}$ as a graded module (1.2).

Indeed, since M is of finite type, its degrees are bounded below. Further, $M \otimes_B A$ is projective of finite type over A, and therefore has only finitely many non-zero homogeneous components, each projective of finite type over A. After localizing on $\mathrm{Spec}(A)$, one may thus suppose that $M \otimes_B A$ is free of finite type over A. Finally, projectivity of M gives

$$\mathrm{Tor}_1^B(M, A) = 0.$$

These properties imply that M is free of finite type over B as a graded module (Bourbaki, *Algèbre*, ch. II, § 11, prop. 7), and hence $\widetilde{M}$ is free of finite type over $\widetilde{B}$, as required.

Lemma 1.5.2. Let A be a ring, let B be an A-algebra of finite presentation, and let M be a B-module of finite presentation. Suppose that B and M are flat over A, and that, for every residue field k of A, the $B \otimes_A k$-module $M \otimes_A k$ has projective dimension at most n (as is the case if $B = A[t_0, \ldots, t_r]$). Then M has a resolution of length n by projective B-modules of finite type:

$$0 \longrightarrow L_n \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0.$$

This is the local criterion of EGA IV, 12.3.2.

Proposition 1.6. Let S be quasi-compact, let E be locally free of rank $r+1$, let $B = \mathrm{Sym}(E)$ with its usual grading, and let M be a graded B-module of finite presentation, flat over S. Then M admits a graded resolution

$$0 \longrightarrow L_{r+1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0,$$

where the L_i are graded B-modules locally free of finite type.

First, M admits a graded epimorphism $L \to M$ with L locally free of finite type: choose an integer N such that M is generated in degrees $\leq N$, and take

$$L = \bigoplus_{i \leq N} M_i \otimes_{\mathcal{O}_S} B(-i),$$

with the morphism (1.2.1). Repeating the construction for successive kernels gives a global graded resolution. After $r+1$ steps, Lemma 1.5.2 shows locally on affine opens that the final kernel is projective over B, and Lemma 1.5.1 then shows that it is locally free as a graded B-module; truncation gives the asserted resolution.

Proposition 1.7. Let S be quasi-compact, let E be locally free of finite type, let $B = \mathrm{Sym}(E)$, and let L be a graded B-module locally free of finite type. Then L admits a finite decreasing filtration by graded submodules $L^{(m)}$ such that

$$\begin{array}{ll} \text{(i)} & L^{(m)} = L \quad \text{for } m \leq m_0, \\ \text{(ii)} & L^{(m)} = 0 \quad \text{for } m \geq m_1, \\ \text{(iii)} & \mathrm{gr}^m L \simeq N_m \otimes_{\mathcal{O}_S} B(m), \end{array}$$

where N_m is locally free of finite type on S.

The question is local on S. Since S is quasi-compact, the degrees of homogeneous bases that occur locally lie between two fixed integers. One argues by induction on the length of this interval. The first step is the canonical map from the lowest non-zero homogeneous component,

$$L_{n_0} \otimes_{\mathcal{O}_S} B(-n_0) \longrightarrow L,$$

which locally identifies the corresponding direct summand. The quotient has local homogeneous bases in higher degrees, and induction applies.

1.8. Generation by the powers of ξ

Let F be locally free of finite type on X. By Corollary 1.4 and Proposition 1.6, F comes from a graded module M admitting a finite graded resolution by L_i's. Applying Proj_0 gives an exact sequence of locally free sheaves on X, and therefore in $K^\bullet(X)$

$$\mathrm{cl}(F) = \sum_i (-1)^i \, \mathrm{cl}(\mathrm{Proj}_0(L_i)).$$

Using the filtration of Proposition 1.7, each term is a sum of classes

$$\mathrm{cl}\big(f^* N_{ij}(j)\big) = f^* \, \mathrm{cl}(N_{ij}) \, \xi^j.$$

Thus $K^\bullet(X)$ is generated as a $K^\bullet(S)$-module by the powers of ξ.

1.9. The relation

For linear dependence, use the canonical epimorphism $f^*E \to \mathcal{O}_X(1)$. Its kernel is locally free of rank r. Hence

$$\lambda^{r+1}(f^*E - \xi) = 0.$$

Since $\lambda_t(-\xi) = 1/(1+\xi t)$, one obtains the relation

$$\sum_{i=0}^{r+1} (-1)^i \, \lambda^i(E) \, \xi^{r+1-i} = 0, \tag{1.9.1}$$

equivalently the characteristic equation of ξ associated with the locally free module E.

1.10–1.11. Linear independence

The proof of the linear-independence assertion is due to J.-P. Jouanolou.

Lemma 1.10. With the hypotheses of Theorem 1.1, let F and G be locally free sheaves of finite type on X, and suppose that they have the same class in $K^\bullet(X)$. Then there is an integer n_0 such that, for all $n \geq n_0$,

(i) $f_*(F(n))$ and $f_*(G(n))$ are locally free of finite type on S,
(ii) $\mathrm{cl}\, f_*(F(n)) = \mathrm{cl}\, f_*(G(n))$ in $K^\bullet(S)$.

An equality of classes in $K^\bullet(X)$ is a finite integral combination of relations coming from short exact sequences of locally free sheaves. Twisting by $\mathcal{O}_X(n)$ and applying f_*, for n large enough all higher direct images vanish and all direct images are locally free of finite type; the same relations then hold in $K^\bullet(S)$.

Suppose that

$$f^*(x_0) + f^*(x_1)\xi + \cdots + f^*(x_r)\xi^r = 0. \tag{1.11.1}$$

Writing each x_i as a difference of classes of locally free sheaves and adding suitable summands, (1.11.1) is converted into an equality $\mathrm{cl}(F) = \mathrm{cl}(G)$ for locally free sheaves of finite type on X. By Lemma 1.10, for all sufficiently large n one has

$$\sum_{i=0}^{r} x_i\, \sigma_{n+i}(E) = 0 \quad \text{in } K^\bullet(S), \tag{1.11.2}$$

where $\sigma_p(E)$ denotes the class of $\mathrm{Sym}^p(E)$, and $\sigma_p(E) = 0$ for $p < 0$.

The Koszul complex of the symmetric algebra gives, for $p \geq 1$,

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E)\, \sigma_{p-i}(E) = 0. \tag{1.11.3}$$

Using this relation, one descends by induction from large n in (1.11.2). It follows that (1.11.2) holds for all $n \geq -r$. Taking $n = -r$ gives $x_r = 0$; repeating the argument gives successively all $x_i = 0$. This proves the linear independence and hence Theorem 1.1.

Remark 1.12. Let

$$\sigma_t(E) = \sum_{i \geq 0} \sigma_i(E) t^i.$$

The relation (1.11.3) is equivalent to

$$\lambda_t(E)\, \sigma_{-t}(E) = 1.$$

It follows in particular that σ_t is additive and extends to a homomorphism

$$K^\bullet(S) \longrightarrow 1 + K^\bullet(S)[[t]]_+.$$

Remark 1.13. Any sequence of $r+1$ consecutive powers of ξ is a basis of $K^\bullet(X)$ over $K^\bullet(S)$. If

$$x = 1 - \xi^{-1}$$

is the class of a hyperplane section, then $1, x, \ldots, x^r$ is also a basis. The dependence relation becomes

$$\sum_{k=0}^{r+1} \lambda^{r+1-k}(\check{E} - r - 1)\, x^k = 0. \tag{1.13.1}$$

For $E = \mathcal{O}_S^{r+1}$, so that $X = \mathbf{P}_S^r$, this reduces to

$$x^{r+1} = 0.$$

Using relative schemes in Hakim's sense, Theorem 1.1 also makes sense when S is a locally ringed topos; if the final object of the topos is quasi-compact, the proof transports without essential change.

2. Computation of $K^\bullet$ of a projective bundle: the case of perfect complexes

The present section will not be used in the rest of the Seminar. In this section the notation $K^\bullet(X)$ denotes the Grothendieck group of perfect complexes on X (IV 2.2). We propose to prove the following theorem, analogous to 1.1:

Theorem 2.1. Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of rank $r+1$, let $X = \mathbf{P}(E)$ be the associated projective bundle; let ξ be the class in $K^\bullet(X)$ of the complex equal to $\mathcal{O}_X(1)$ in degree zero, and zero in the other degrees. Then $K^\bullet(X)$, endowed with its natural $K^\bullet(S)$-algebra structure, is a free $K^\bullet(S)$-module with basis

$$1, \xi, \ldots, \xi^r.$$

2.2. Notation

Fix the notation: B will denote the symmetric algebra of E; write $\mathcal{A}$ for the category of $\mathcal{O}_X$-modules, $\mathcal{G}$ for the category of graded B-modules with graded morphisms as morphisms, $\mathcal{G}_0$ for the full subcategory of $\mathcal{G}$ whose objects are the B-modules locally free of finite type as graded B-modules, and $\mathcal{G}'$ for the category of $\mathcal{O}_S$-modules. For every n, and every complex $L^\bullet$ of $D^+(\mathcal{A})$, we shall write $L^\bullet(n)$ for the complex

$$L^\bullet(n) = L^\bullet \otimes_{\mathcal{O}_X} \mathcal{O}_X(n).$$

2.3. The functor Proj_0

We shall need to apply the functor Proj_0 to arbitrary graded B-modules, and not only to quasi-coherent graded B-modules as in the construction of EGA II 3.2. We shall therefore indicate briefly how this construction can be made in the general case.

2.3.1. First let S be a scheme, let A be a quasi-coherent $\mathcal{O}_S$-algebra, let $Y = \mathrm{Spec}(A)$, and let $f : Y \to S$ be the structural morphism. From the isomorphism

$$A \simeq f_* \mathcal{O}_Y$$

one obtains a canonical homomorphism

$$f^{-1}(A) \longrightarrow \mathcal{O}_Y, \tag{2.3.1.1}$$

where f^{-1} denotes inverse image in the sense of sheaves of sets; in other words, one has a morphism of ringed spaces

$$\bar{f} : (Y, \mathcal{O}_Y) \longrightarrow (S, A). \tag{2.3.1.2}$$

It gives an inverse-image functor $\bar{f}^*$ from the category of A-modules to the category of $\mathcal{O}_Y$-modules, left adjoint to the direct-image functor $\bar{f}_*$ from the category of $\mathcal{O}_Y$-modules to the category of $f_*(\mathcal{O}_Y)$-modules. Thus

$$\bar{f}^*(M) = f^{-1}(M) \otimes_{f^{-1}A} \mathcal{O}_Y \tag{2.3.1.3}$$

for every A-module M.

Since the homomorphism (2.3.1.1) is flat (in each fiber it corresponds to a localization), the functor $\bar{f}^*$ just defined is exact (cf. [4], p. 164). If M is a quasi-coherent A-module, this construction gives the module $\widetilde{M}$ defined in EGA II 1.4.3. Indeed, the isomorphism

$$M \simeq f_*(\widetilde{M}) = \bar{f}_*(\widetilde{M})$$

gives, by adjunction, a homomorphism

$$\bar{f}^*(M) \longrightarrow \widetilde{M}$$

and one checks immediately on each fiber that it is an isomorphism.

2.3.2. Now let B be a quasi-coherent graded $\mathcal{O}_S$-algebra, let M be a graded B-module, and let $X = \operatorname{Proj}(B)$. For every affine open U of S, and every section s of B over U, homogeneous of degree $d > 0$, write $M_{(s),U}$ for the homogeneous component of degree 0 of the module of fractions M_s; thus

$$M_{(s),U} = \big((M|_U) \otimes_{B|_U} (B|_U)_s\big)_0.$$

If V is an affine open of U, then $M_{(s),V}$ is the restriction to V of $M_{(s),U}$.

Consider the base of opens of X formed by the affine opens

$$D_+(U, s) \simeq \operatorname{Spec}\big((B|_U)_{(s)}\big)$$

Write f for the structural morphism $X \to S$, and $f_{U,s}$ for the affine morphism

$$f_{U,s} : D_+(U, s) \longrightarrow U.$$

Put

$$\operatorname{Proj}_0(M)_{U,s} = \bar{f}^*_{U,s}(M_{(s),U}).$$

With the notation of 2.3.1, we shall show that the $\operatorname{Proj}_0(M)_{U,s}$ glue as U and s vary, thereby defining an $\mathcal{O}_X$-module denoted $\operatorname{Proj}_0(M)$. The gluing for variable U and fixed s being evident, one is reduced (cf. EGA II 2.5.2) to showing that, if t is a section of B on U, homogeneous of degree $e > 0$, then the restriction of $\operatorname{Proj}_0(M)_{U,s}$ to $D_+(U, st)$ is canonically isomorphic to $\operatorname{Proj}_0(M)_{U,st}$. Consider

$$\begin{tikzcd} D_+(st) \arrow[r, "i"] \arrow[dr, "f_{U,st}"'] & D_+(s) \arrow[d, "f_{U,s}"] \\ & U \end{tikzcd}$$

It is a matter of showing that the modules

$$i^*\bar{f}^*_{U,s}(M_{(s),U}) \quad \text{and} \quad \bar{f}^*_{U,st}(M_{(st),U})$$

are canonically isomorphic. On the one hand, one has the canonical isomorphisms

$$\begin{aligned} i^*\bar{f}^*_{U,s}(M_{(s),U}) &\simeq i^*\Big(f^{-1}_{U,s}(M_{(s),U}) \otimes_{f^{-1}_{U,s}((B|_U)_{(s)})} \mathcal{O}_{D_+(s)}\Big) \\ &\simeq i^{-1}\Big(f^{-1}_{U,s}(M_{(s),U}) \otimes_{f^{-1}_{U,s}((B|_U)_{(s)})} \mathcal{O}_{D_+(s)}\Big) \otimes_{i^{-1}(\mathcal{O}_{D_+(s)})} \mathcal{O}_{D_+(st)} \\ &\simeq f^{-1}_{U,st}(M_{(s),U}) \otimes_{f^{-1}_{U,st}((B|_U)_{(s)})} \mathcal{O}_{D_+(st)}. \end{aligned}$$

On the other hand,

$$\bar{f}^*_{U,st}(M_{(st),U}) \simeq f^{-1}_{U,st}(M_{(st),U}) \otimes_{f^{-1}_{U,st}((B|_U)_{(st)})} \mathcal{O}_{D_+(st)}.$$

Finally, it follows from EGA II 2.2.2, translated into sheaf language, that one has the canonical isomorphisms

$$M_{(st),U} \simeq \big(M_{(s),U}\big)_{t^d/s^e} \simeq M_{(s),U} \otimes_{(B|_U)_{(s)}} (B|_U)_{(st)},$$

and hence the desired isomorphism.

It is clear that, if M is quasi-coherent, the module $\operatorname{Proj}_0(M)$ thus constructed is indeed the one defined in EGA II.

2.3.3. It follows from 2.3.1 and 2.3.2 that the functor Proj_0 is exact. Returning to the notation of 2.2, by passage to derived categories one therefore obtains a functor, still denoted Proj_0:

$$\operatorname{Proj}_0 : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{A}).$$

It is also clear from the preceding constructions that the functor Proj_0, on the category of graded B-modules, is compatible with tensor product; the same remains true of the functor obtained on the derived categories.

2.4. The functor F_k

For an $\mathcal{O}_X$-module L and an integer k, set

$$F_k(L) = \bigoplus_{n \geq k} f_*(L(n)). \tag{2.4.1}$$

The functor F_k is left exact and therefore has a right derived functor

$$RF_k : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{G}). \tag{2.4.2}$$

If L is locally free of finite type, then for k sufficiently large it is F_k-acyclic, by the vanishing theorem used above in 1.3.

For every k there is a functorial morphism

$$\beta : \mathrm{Proj}_0(F_k(F)) \longrightarrow F, \tag{2.5.1}$$

constructed on each $D_+(U, s)$ by the usual localization formula: a fraction represented by a homogeneous section of sufficiently high degree is sent to the corresponding local section of F. In the quasi-coherent case this is the canonical morphism of EGA. Passing to derived categories gives

$$\mathrm{Proj}_0 \circ RF_k \longrightarrow \mathrm{id}_{D^+(\mathcal{A})}\,. \tag{2.5.2}$$

Proposition 2.6. Let S be quasi-compact, let E be locally free of finite type, let $X = \mathbf{P}(E)$, and let $L^\bullet$ be a perfect complex on X. Then, for k sufficiently large,

(i) $RF_k(L^\bullet)$ is $\mathcal{G}_0$-perfect,

(ii) $\mathrm{Proj}_0\, RF_k(L^\bullet) \longrightarrow L^\bullet$ is an isomorphism in $D^+(\mathcal{A})$.

The assertion is local on S. One may therefore assume S affine. Since $\mathcal{O}_X(1)$ is ample, the perfect complex $L^\bullet$ is locally, and then globally after refining, represented by a bounded complex whose terms are locally free of finite type. For k sufficiently large all those finitely many terms are F_k-acyclic, and the graded modules $F_k(L^i)$ are $\mathcal{G}_0$-perfect. This gives (i), and the morphism (2.5.2) is then an isomorphism term by term, giving (ii).

2.7. Decomposition of a $\mathcal{G}_0$-perfect complex

On $\mathcal{G}$, the functor which associates to a graded B-module its n-th homogeneous component is exact with values in $\mathcal{G}'$; it defines a functor

$$\mathrm{Comp}_n : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{G}'). \tag{2.7.1}$$

The homomorphism (1.2.1) gives a morphism of functors

$$\mathrm{Comp}_n \otimes_{\mathcal{O}_S} B(-n) \longrightarrow \mathrm{id}_{D^+(\mathcal{G})}\,. \tag{2.7.2}$$

Let $M^\bullet$ be a $\mathcal{G}_0$-perfect complex. We shall show that there exists a finite sequence of distinguished triangles

$$\begin{array}{ccc} & P_1^\bullet & \\ \swarrow & & \nwarrow \\ N_1^\bullet & \longrightarrow & M^\bullet, \end{array} \qquad \begin{array}{ccc} & P_2^\bullet & \\ \swarrow & & \nwarrow \\ N_2^\bullet & \longrightarrow & P_1^\bullet, \end{array} \quad \cdots, \quad \begin{array}{ccc} & P_k^\bullet & \\ \swarrow & & \nwarrow \\ N_k^\bullet & \longrightarrow & P_{k-1}^\bullet, \end{array}$$

where each $P_k^\bullet$ at the last stage is acyclic, the terms $N_i^\bullet$ are of the form

$$N_i^\bullet = K_i^\bullet \otimes_{\mathcal{O}_S} B(d_i),$$

and the $K_i^\bullet$ are perfect complexes on S in the ordinary sense.

Indeed, because $M^\bullet$ is $\mathcal{G}_0$-perfect and S is quasi-compact, there exists a finite covering of S and an integer d_0 such that, on every open of the covering, $M^\bullet$ is represented by a bounded complex of locally free B-modules of finite type, with no homogeneous generators in degrees $< d_0$. Taking d_0 minimal, the morphism (2.7.2) extracts the part of the lowest degree. Its cone has all homogeneous generators in strictly higher degrees. Repeating the process, and using the finiteness of the degree interval from Proposition 1.7, one obtains the desired finite sequence of triangles.

2.8. Proof of Theorem 2.1

Let $L^\bullet$ be a perfect complex on X. For k sufficiently large put

$$M^\bullet = RF_k(L^\bullet).$$

By Proposition 2.6, $M^\bullet$ is $\mathcal{G}_0$-perfect and $\mathrm{Proj}_0(M^\bullet) \simeq L^\bullet$. Applying the decomposition of 2.7 and then Proj_0, one obtains in $K^\bullet(X)$

$$\mathrm{cl}(L^\bullet) = \sum_i \mathrm{cl}\big(f^* K_i^\bullet(d_i)\big) = \sum_i f^* \,\mathrm{cl}(K_i^\bullet)\, \xi^{d_i}.$$

Thus the powers of ξ generate $K^\bullet(X)$ as a $K^\bullet(S)$-module.

The dependence relation is the same relation (1.9.1), now interpreted in the Grothendieck group of perfect complexes. For linear independence one uses the $K^\bullet(S)$-linear homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(S).$$

For the structure sheaf and negative twists of the projective bundle one has

$$f_*(1) = 1, \qquad f_*(\xi^n) = 0 \quad (-r \leq n < 0).$$

Multiplying a relation among $1, \xi, \ldots, \xi^r$ by suitable negative powers of ξ and applying f_* successively gives all its coefficients equal to zero. This completes the proof of Theorem 2.1.

This completes the induction, for if $0 \leq s \leq r$ and if $x_j = 0$ for $s < j \leq r$, then from a relation

$$\sum_{i=0}^{s} f^*(x_i)\xi^i = 0$$

after multiplication by ξ^{r-s} one gets

$$\sum_{i=0}^{s} f^*(x_i)\xi^{i+r-s} = 0,$$

and hence $x_s = 0$ by what precedes.

3. Consequences of the structure theorem for the $K^\bullet$ of a projective bundle

Proposition 3.1. Let S be a quasi-compact scheme, let $E_1, \ldots, E_n$ be $\mathcal{O}_S$-modules locally free of finite type, and let $D(E_i)$ be the flag scheme of E_i. Put

$$X = D(E_1) \times_S \cdots \times_S D(E_n).$$

Then the canonical homomorphism

$$K^\bullet(S) \longrightarrow K^\bullet(X)$$

is injective, where $K^\bullet$ is defined either from $\mathcal{O}_S$-modules locally free of finite type or from perfect complexes.

One verifies this immediately by using the universal property of flag schemes. The scheme X can be constructed as follows: one considers a sequence of S-schemes $X_1, \ldots, X_n$, defined by induction by taking X_k to be the flag scheme of the inverse image of E_k on X_{k-1}; then $X = X_n$. Thus, to prove the proposition, it is enough to prove it when $n = 1$, the general case following by induction. We may therefore suppose that X is the flag scheme of an $\mathcal{O}_S$-module E, locally free of finite type.

The scheme X may then be constructed as follows. Put $E = F_0$, $S = X_0$, and define successively schemes $X_1, \ldots, X_n$ and, for each i, an $\mathcal{O}_{X_i}$-module locally free of finite type F_i, by

$$X_i = \mathbb{P}(F_{i-1}), \qquad F_i = \operatorname{Ker}\big(f_i^*(F_{i-1}) \longrightarrow \mathcal{O}_{X_i}(1)\big),$$

where $f_i : X_i \to X_{i-1}$ is the structural morphism. It follows that it is enough to prove the proposition for a projective bundle. This is precisely the content of 1.1, respectively of 2.1, according to which definition of $K^\bullet$ is being used.

Theorem 3.2. Let S be a quasi-compact scheme. The pre-λ-ring $K^\bullet(S)$, defined from $\mathcal{O}_S$-modules locally free of finite type, is a λ-ring.

The homomorphism

$$\lambda_t : K^\bullet(S) \longrightarrow 1 + K^\bullet(S)[[t]]^+$$

being additive, it is enough, in order to verify that it is a λ-homomorphism, to check the universal relations on elements of the form $\mathrm{cl}(E)$, where E is an $\mathcal{O}_S$-module locally free of finite type. Thus one has to show that for every pair E, F of such modules, with classes again denoted E, F, the relations V 2.4.2 are satisfied:

$$\lambda^i(EF) = P_i(\lambda^1(E), \ldots, \lambda^i(E); \lambda^1(F), \ldots, \lambda^i(F)),$$

$$\lambda^j(\lambda^i(E)) = Q_{i,j}(\lambda^1(E), \ldots, \lambda^{ij}(E)).$$

Since S is quasi-compact, it admits a finite covering by disjoint opens such that on each of them E and F have constant rank. Since the $K^\bullet$ of a finite disjoint union is the product of the corresponding $K^\bullet$'s, it is enough to check the relations on the members of this covering; hence we may suppose that E and F have constant ranks on S.

Put then

$$X = D(E) \times_S D(F).$$

By Proposition 3.1 the λ-homomorphism $K^\bullet(S) \to K^\bullet(X)$ is injective. Consequently, in order to prove the relations V 2.4.2, we may work in $K^\bullet(X)$. But in $K^\bullet(X)$ the elements E and F split as sums of classes of invertible sheaves. By additivity it is therefore enough to prove the same relations for elements ξ, η which are classes of invertible sheaves. Written in the form V 2.4.1, the relations are

$$1 + \xi\eta t = (1 + \xi t) \circ (1 + \eta t), \qquad \lambda_u(1 + \xi t) = 1 + \{1 + \xi t\}u.$$

These are just the relations V 2.3.1 and V 2.3.3 defining the λ-ring structure on $1 + K^\bullet(S)[[t]]^+$.

Let us also record the following more general result.

Theorem 3.3. Let (T, A) be a commutatively ringed topos, and let K be the Grothendieck group of A-modules locally free of bounded rank on T. Then K, endowed with its canonical pre-λ-structure, is a λ-ring.

To prove that K is a λ-ring, it is necessary to prove the relations V 2.4.1; by additivity, it is enough to prove them for classes of A-modules E, F locally free of bounded rank. Since the ranks of E and F are bounded, one may decompose T as a finite disjoint sum of opens over which E and F have constant rank. Using the compatibility of K with finite disjoint sums, and writing the relations V 2.4.1 in the form V 2.4.2, we are reduced to the case of modules of constant rank.

Let E be an A-module locally free of finite rank n. There is a torsor P under $\mathrm{GL}(n, A)$ such that

$$E = P \times_{\mathrm{GL}(n,A)} A^n,$$

where A^n is considered as a $\mathrm{GL}(n, A)$-module by the standard representation. One can then define a λ-homomorphism from the representation ring $R_{\mathbb{Z}}(\mathrm{GL}(n))$ to K by associating to a representation ρ of $\mathrm{GL}(n)$ in a free $\mathbb{Z}$-module of rank p the class of

$$P \times_{\mathrm{GL}(n,A)} A^p.$$

Under this homomorphism, the class of E is the image of the standard representation. Thus any polynomial relation among the $\lambda^i(E)$'s follows from the corresponding relation for the standard representation in $R_{\mathbb{Z}}(\mathrm{GL}(n))$.

Similarly, for two locally free modules E, F of ranks n, p, one uses a torsor under $\mathrm{GL}(n, A) \times \mathrm{GL}(p, A)$, with $\mathrm{GL}(n, A)$ acting through the standard representation and $\mathrm{GL}(p, A)$ through the trivial one on the first factor, and conversely for F. We are reduced to proving that, for every group scheme G over $\operatorname{Spec} \mathbb{Z}$ which is a finite product of linear group schemes, the pre-λ-ring $R_{\mathbb{Z}}(G)$ is a λ-ring. This may be obtained by the explicit computation of $R_{\mathbb{Z}}(G)$ in [5]; alternatively, one can reason as in 3.2, using 1.14.

4. Computation of the $K^\bullet$ of a flag bundle

4.1

Let S be a scheme and let E be an $\mathcal{O}_S$-module locally free of rank n. Let

$$\pi = (p_1, \ldots, p_k), \qquad p_i \geq 1, \qquad \sum_{i=1}^{k} p_i = n.$$

The flag functor of type π of E is the functor on the category of S-schemes, with values in sets, which to an S-scheme $f : X \to S$ associates the set of chains of quotients of f^*E

$$(L_{k-1}, \ldots, L_1),$$

where, for each i, L_i is a quotient of L_{i+1}, locally free of rank $p_1 + \cdots + p_i$. This functor is representable by an S-scheme $D_\pi(E)$, endowed with a canonical sequence of quotients of the inverse image of E, locally free of ranks $p_1, p_1 + p_2, \ldots$. If $\pi = (1, n-1)$, then $D_\pi(E) = \mathbb{P}(E)$; if $\pi = (r, n-r)$, then $D_\pi(E) = G_{r,n}(E)$, the Grassmannian of type $(r, n-r)$ of E. If all p_i's are equal to 1, one obtains the usual flag bundle of E, denoted $D(E)$.

Lemma 4.2. Let S be a scheme, let E be an $\mathcal{O}_S$-module locally free of rank n, and let

$$\pi = (p_1, \ldots, p_i, \ldots, p_k), \qquad p_1 + \cdots + p_k = n.$$

Let $p', p'' \geq 1$ be such that $p' + p'' = p_i$, and put

$$\pi' = (p_1, \dots, p_{i-1}, p', p'', p_{i+1}, \dots, p_k), \qquad \pi'' = (p', p'').$$

Let $L_1, \dots, L_{k-1}$ be the canonical quotients of the inverse image of E on $D_\pi(E)$, and let F be the kernel of $L_i \to L_{i-1}$. Then there is a canonical S-isomorphism

$$D_{\pi'}(E) \simeq D_{\pi''}(F).$$

The proof, immediate from the functorial definition of flag schemes, is left to the reader.

Lemma 4.3. Let A be a commutative ring with unit, let $n > 0$, and let $(c_i)_{1 \leq i \leq n}$ be elements of A. Let C be the A-algebra generated by $\xi_1, \dots, \xi_n$, subject to the relations

$$\sigma_i(\xi_1, \dots, \xi_n) = c_i, \qquad i = 1, \dots, n,$$

where σ_i denotes the elementary symmetric functions. Let $p_1, \dots, p_k$ be integers ≥ 1 whose sum is n. For $1 \leq j \leq k$ put

$$\xi^{(j)} = (\xi_1^{(j)}, \dots, \xi_{p_j}^{(j)}), \qquad \xi_i^{(j)} = \xi_{\sum_{\alpha<j} p_\alpha + i}.$$

Put

$$c_i^{(j)} = \sigma_i(\xi_1^{(j)}, \dots, \xi_{p_j}^{(j)}), \qquad 1 \leq j \leq k, \quad 1 \leq i \leq p_j,$$

and let B be the subalgebra of C generated by the $c_i^{(j)}$'s. Then:

i) The elements

$$(\xi^{(1)})^{\alpha(1)} \cdots (\xi^{(k)})^{\alpha(k)}, \qquad 0 \leq \alpha_i^{(j)} \leq p_j - i,$$

form a basis of C, considered as a B-module.

ii) The generators $c_i^{(j)}$ of the A-algebra B satisfy the n isobaric relations deduced from

$$\sum_{i=1}^{n} c_i = \prod_{j=1}^{k} \left(\sum_{i=1}^{p_j} c_i^{(j)} \right),$$

where c_i and $c_i^{(j)}$ are assigned weight i; these n relations generate the ideal of relations among the $c_i^{(j)}$'s.

Corollary 4.4. The elements of C of the form

$$\xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}}, \qquad 0 \leq \alpha_i \leq n - i,$$

form a basis of C as an A-module.

For the proof of the lemma and corollary we refer to [2], Exposé 4, pages 4.19, from which they are borrowed.

4.5

Let S be quasi-compact, let E be an $\mathcal{O}_S$-module locally free of rank n, let $\pi = (p_1, \dots, p_k)$ be a sequence of integers ≥ 1 with sum n, and put $X = D_\pi(E)$. The scheme X is equipped with canonical quotients L_i. Put $L_0 = 0$, $L_k = f^*E$, where $f : X \to S$ is the structural morphism. For $1 \leq i \leq k$, let F_i be the kernel of $L_i \to L_{i-1}$. Let $K^\bullet(X)$ be the Grothendieck group of locally free finite-type sheaves on X, respectively of perfect complexes. Denote by F_j the class of F_j, and by $\lambda_i^{(j)}$ the class of $\lambda^i(F_j)$. In the case where $K^\bullet(X)$ is defined by perfect complexes, one gives the same evident meaning to the series $\lambda_t(E)$ and $\lambda_t(F_j)$, where E is the class of E. Finally,

$$\sum_{j=1}^{k} F_j = E.$$

Theorem 4.6. With the hypotheses and notations of 4.5, the $K^\bullet(S)$-algebra $K^\bullet(X)$ is generated by the $\lambda_i^{(j)}$, and the ideal of relations among the $\lambda_i^{(j)}$'s is generated by the relations deduced from

$$\lambda_t(E) = \prod_{j=1}^{k} \lambda_t(F_j).$$

Equivalently, $K^\bullet(X)$ is the $K^\bullet(S)$-λ-algebra generated by $F_1, \dots, F_k$, subject to the relations

$$\text{(i)} \quad \lambda^p(F_i) = 0 \quad (p > p_i), \qquad \text{(ii)} \quad \sum_i F_i = E.$$

We first treat the case where all p_i's are equal to 1. Then the theorem reads as follows.

Corollary 4.7. Let S be quasi-compact, let E be an $\mathcal{O}_S$-module locally free of rank n, let $X = D(E)$ be the flag bundle of E, and let $\xi_1, \dots, \xi_n$ be the classes of the invertible sheaves $F_1, \dots, F_n$. Then $K^\bullet(X)$ is the $K^\bullet(S)$-algebra generated by the ξ_i's, subject to the relations

$$\sigma_i(\xi_1, \dots, \xi_n) = \lambda^i(E).$$

We argue by induction on n. For $n = 1$ the assertion is trivial. Suppose it known for $n-1$. Consider $\mathbb{P}(E)$, and on $\mathbb{P}(E)$ the exact sequence

$$0 \longrightarrow F \longrightarrow g^*E \longrightarrow \mathcal{O}_{\mathbb{P}(E)}(1) \longrightarrow 0,$$

where g is the structural morphism of $\mathbb{P}(E)$. By Lemma 4.2, X identifies with the usual flag bundle $D(F)$ over $\mathbb{P}(E)$, and the canonical sheaves correspond under this identification. The diagram

$$\begin{tikzcd} D(E) = D(F) \arrow[r, "h"] \arrow[dr, "f"'] & \mathbb{P}(E) \arrow[d, "g"] \\ & S \end{tikzcd}$$

shows that $K^\bullet(S) \to K^\bullet(X)$ factors through $K^\bullet(\mathbb{P}(E))$. By the induction hypothesis applied to $D(F)$, and by Corollary 4.4, $K^\bullet(X)$ has over $K^\bullet(\mathbb{P}(E))$ a basis formed by

$$\xi_2^{\alpha_2} \cdots \xi_n^{\alpha_{n-1}}, \qquad 0 \le \alpha_i \le n - i.$$

Moreover $K^\bullet(\mathbb{P}(E))$ has over $K^\bullet(S)$ the basis $1, \xi_1, \dots, \xi_1^{n-1}$. Hence $K^\bullet(X)$ has over $K^\bullet(S)$ the basis

$$\xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}}, \qquad 0 \le \alpha_i \le n - i.$$

In addition, the additivity of λ_t gives $\prod_i (1 + \xi_i t) = \lambda_t(E)$, so the displayed relations are satisfied. If C is the $K^\bullet(S)$-algebra generated by ξ_i's and subjected to these relations, then there is a surjective homomorphism $C \to K^\bullet(X)$. By Corollary 4.4 this homomorphism sends a basis of C to a basis of $K^\bullet(X)$, hence it is an isomorphism.

For a general flag bundle of type π, the usual flag bundle $D(E)$ is obtained from $D_\pi(E)$ by constructing successively the flag bundles of the F_j's. It follows that $K^\bullet(D(E))$, considered as a module over $K^\bullet(X)$, has a basis formed by the elements appearing in 4.3.i), the $\xi_i^{(j)}$'s being the classes of the sheaves analogous to the F_j's in the splitting of the inverse images of the F_j's on $D(E)$. Thus $K^\bullet(X)$ is a subring of $K^\bullet(D(E))$. Moreover, additivity gives

$$\lambda_i^{(j)} = \sigma_i(\xi_1^{(j)}, \dots, \xi_{p_j}^{(j)}),$$

and therefore $K^\bullet(X)$ contains the sub-$K^\bullet(S)$-algebra of $K^\bullet(D(E))$ generated by the $\lambda_i^{(j)}$'s. If B is this subalgebra, then by 4.3.i) the same family of elements is a basis both for the B-module structure and for the $K^\bullet(X)$-module structure. Hence $K^\bullet(X) = B$, proving generation. The ideal of relations among the $\lambda_i^{(j)}$'s is then given by 4.3.ii), taking into account the structure of $K^\bullet(D(E))$ over $K^\bullet(S)$ determined by 4.7, and the fact that the $\xi_i^{(j)}$'s are precisely the classes involved in the splitting of the inverse image of E on $D(E)$. This proves the first form of 4.6.

For the second form, one observes that the elements F_i of the $K^\bullet(S)$-λ-algebra $K^\bullet(X)$ satisfy (i) and (ii). If A is the $K^\bullet(S)$-λ-algebra generated by elements F'_i satisfying (i) and (ii), there is therefore a $K^\bullet(S)$-λ-homomorphism $A \to K^\bullet(X)$ sending F'_i to F_i. By 4.6 this homomorphism is surjective. Since the λ-ideal generated by (i) and (ii) in $K^\bullet(S)[F_1, \dots, F_k]$ contains the usual ideal generated by the first form of the relations, the homomorphism is also injective. Thus it is an isomorphism.

Corollary 4.8. With the hypotheses and notations of 4.5, put

$$c_i^{(j)} = \gamma^i(F_j - p_j), \qquad 1 \le i \le p_j.$$

Then the $K^\bullet(S)$-algebra $K^\bullet(X)$ is generated by the $c_i^{(j)}$'s, and the ideal of relations among them is generated by the relations deduced from

$$\gamma_t(E - n) = \prod_{j=1}^{k} \gamma_t(F_j - p_j).$$

This is merely another formulation of 4.6. Indeed, the $c_i^{(j)}$'s may be expressed as functions of the $\lambda_i^{(j)}$'s and conversely. Moreover the relation $\lambda_t(E) = \prod \lambda_t(F_j)$ is equivalent to $\gamma_t(E - n) = \prod \gamma_t(F_j - p_j)$.

4.9

Let S be a scheme. Denote by $G_S^{n,r}$ the Grassmannian of type $(n, r-n)$ of $\mathcal{O}_S^r$, with $r \ge n$. For variable r, the $G_S^{n,r}$ form an inductive system, the morphism

$$G_S^{n,r} \longrightarrow G_S^{n,r+1}$$

being defined by considering $\mathcal{O}^r_{G_S^{n,r}}$ as the quotient of $\mathcal{O}^{r+1}_{G_S^{n,r}}$ by the $(r+1)$-st factor. In this system of morphisms the canonical quotients therefore correspond by inverse image. The inductive system so defined will be denoted $G_S^{n,\infty}$.

We define its $K^\bullet$ by the formula

$$K^\bullet(G_S^{n,\infty}) = \varprojlim_r K^\bullet(G_S^{n,r}). \tag{4.9.1}$$

Let F_1, F_2 be the canonical sheaves on $G_S^{n,r}$ (4.5), and F_1, F_2 their classes in $K^\bullet(G_S^{n,r})$. Put

$$c_i = \gamma^i(F_1 - n).$$

When r varies, the c_i correspond under the transition morphisms and consequently define elements, still denoted c_i, of $K^\bullet(G_S^{n,\infty})$.

Proposition 4.10. Let S be a quasi-compact scheme. Then, with the notation of 4.9, one has an isomorphism

$$K^\bullet(G_S^{n,\infty}) \simeq K^\bullet(S)[[c_1, \ldots, c_n]].$$

By 4.8, $K^\bullet(G_S^{n,r})$ is the $K^\bullet(S)$-algebra generated by

$$c_i = \gamma^i(F_1 - n), \quad 1 \le i \le n,$$

and by

$$\gamma^j(F_2 - r + n), \quad 1 \le j \le r - n,$$

subject to the relations deduced from

$$\gamma_t(F_1 - n) \cdot \gamma_t(F_2 - r + n) = \gamma_t(r - r) = 1,$$

or equivalently

$$\gamma_t(F_2 - r + n) = 1/\gamma_t(F_1 - n).$$

This relation makes it possible to express the $\gamma^j(F_2 - r + n)$ in terms of the c_i, and gives a series of isobaric relations among the c_i (where c_i is assumed to have weight i) by writing that the coefficients of $1/\gamma_t(F_1 - n)$ vanish in degree $> r - n$; moreover, these relations generate the ideal of relations among the c_i. We may therefore write

$$K^\bullet(G_S^{n,r}) = K^\bullet(S)[c_1, \ldots, c_n]/I_r,$$

where I_r is a homogeneous ideal of $K^\bullet(S)[c_1, \ldots, c_n]$, of degree $> r - n$. It then follows from the definition (4.9.1) that $K^\bullet(G_S^{n,\infty})$ is the separated completion of $K^\bullet(S)[c_1, \ldots, c_n]$ for the topology defined by the I_r. This topology is finer than the topology defined by weight; we shall show that it is equivalent to it, which will imply the proposition.

Put $A = K^\bullet(S)[c_1, \ldots, c_n]/I_r$, and let C be the A-algebra generated by generators $x_1, \ldots, x_n$ subject to the relations

$$\sigma_j(x_1, \ldots, x_n) = c_j,$$

where the σ_j denote the elementary symmetric functions. By 4.4, the canonical homomorphism $A \to C$ is injective. We obtain

$$\gamma_t(F_1 - n) = 1 + c_1 t + \cdots + c_n t^n = \prod_{i=1}^{n}(1 + x_i t).$$

Since $1/\gamma_t(F_1 - n)$ is a polynomial, so is $1/(1 + x_i t)$, for every i. Hence the x_i are nilpotent, and there exists an integer N such that every polynomial of degree $> N$ in the x_i, with coefficients in $K^\bullet(S)$, is zero. Consequently every polynomial of A of weight $> N$ is zero, i.e. every polynomial of $K^\bullet(S)[c_1, \ldots, c_n]$ of weight $> N$ belongs to I_r. It follows that the two topologies under consideration are equivalent, completing the proof.

5. Applications to projective bundles; study of f_*

5.1

Let S be quasi-compact, let E be locally free of rank $r + 1$, and put $X = \mathbb{P}(E)$, with structural morphism $f : X \to S$. On X one has the canonical exact sequence

$$0 \longrightarrow F \longrightarrow f^*E \longrightarrow \mathcal{O}_X(1) \longrightarrow 0.$$

Let E, F, ξ denote the corresponding classes in $K^\bullet$, and put $x = \xi - 1$.

5.2

For a locally free sheaf G on X, the higher direct images $R^i f_* G$ need not be locally free of finite type on S. Thus the direct image $f_* : K^\bullet(X) \to K^\bullet(S)$ is not defined here by alternating sums of higher direct images. Theorem 1.1 gives a substitute: every $y \in K^\bullet(X)$ is uniquely expressible in the form

$$y = a_0 + a_1\xi + \cdots + a_r\xi^r, \qquad a_i \in K^\bullet(S),$$

and we put

$$f_*(y) = a_0.$$

This additive homomorphism satisfies the projection formula. Moreover,

$$f_*(1) = 1, \qquad f_*(\xi^n) = 0 \quad (-r \leq n < 0), \qquad f_*(\xi^n) = s_n \quad (n \geq 0),$$

where s_n is the n-th complete symmetric function in the roots of E. The last relations follow by induction from the dependence relation and the Koszul relation.

Proposition 5.3. For every $k \geq 0$,

$$\operatorname{Fil}^k K^\bullet(X) = \operatorname{Fil}^k K^\bullet(S) + \operatorname{Fil}^{k-1} K^\bullet(S)x + \cdots + \operatorname{Fil}^{k-r} K^\bullet(S)x^r,$$

as a direct sum of $K^\bullet(S)$-modules, where the filtration is the canonical filtration of augmented λ-rings.

Here x has augmentation zero and

$$\gamma_t(x) = \lambda_{t/(1-t)}(1 - \xi^{-1}) = \frac{1}{1 - xt},$$

so that $\gamma^i(x) = x^i$. The result follows from the computation of the filtration for free λ-rings in V, 4.12 and V, 4.13, together with the relation (1.9.1). Directness follows from the independence of $1, x, \ldots, x^r$.

Corollary 5.4. The filtration induced on $K^\bullet(S)$ by $K^\bullet(X)$ is the original filtration of $K^\bullet(S)$.

Corollary 5.5. If X is a product over S of flag bundles of locally free sheaves, then

$$\operatorname{Gr} K^\bullet(S) \longrightarrow \operatorname{Gr} K^\bullet(X)$$

is injective.

Corollary 5.6. With the notation of the projective bundle above, $\operatorname{Gr} K^\bullet(X)$ is a free $\operatorname{Gr} K^\bullet(S)$-module with basis $1, x, \ldots, x^r$, and

$$\operatorname{Gr}^k K^\bullet(X) = \operatorname{Gr}^k K^\bullet(S) \oplus \operatorname{Gr}^{k-1} K^\bullet(S)x \oplus \cdots \oplus \operatorname{Gr}^{k-r} K^\bullet(S)x^r.$$

The elements $1, x, \ldots, x^{r+1}$ satisfy

$$\sum_{i=0}^{r+1} (-1)^i c_i(E) x^{r+1-i} = 0,$$

where $c_i(E)$ is the i-th Chern class of E.

Corollary 5.7. For a flag bundle as in 4.5, the $\operatorname{Gr} K^\bullet(S)$-algebra $\operatorname{Gr} K^\bullet(X)$ is generated by the classes $\xi_i^{(\alpha)}$ of $\xi_i^{(\alpha)} - 1$, with relations generated by

$$c_t(E) = \prod_{i,\alpha} c_t(\xi_i^{(\alpha)}),$$

where c_t denotes the total Chern class.

Corollary 5.8. For every k,

$$f_*(\mathrm{Fil}^k K^\bullet(X)) \subset \mathrm{Fil}^k K^\bullet(S).$$

This follows immediately from Proposition 5.3 and the projection formula.

Proposition 5.9. With the notation of 5.1, for every $y \in K^\bullet(X)$, one has

$$f_*(\lambda_{-1}(F)y) = 0, \qquad f_*(\gamma^*(F)y) = 0.$$

In particular,

$$f_*(\lambda_{-1}(F)) = 1.$$

If

$$\xi_t : 1 + K^\bullet(X)[[t]]^+ \longrightarrow 1 + K^\bullet(S)[[t]]^+$$

is defined coefficientwise by

$$\xi_t\left(\sum a_i t^i\right) = \sum f_*(a_i)t^i,$$

then, although it is not in general a homomorphism of groups, it satisfies

$$\xi_t(mf^*y) = \xi_t(m)y.$$

Using $E = F + \xi$ and $\lambda_t(E) = \lambda_t(F)(1 + \xi t)$, one obtains the asserted identities.

Proposition 5.10. Let $z \in K^\bullet(X)$. The condition

$$z(1 - \xi) = 0$$

is equivalent to the existence of an element $y \in K^\bullet(S)$ such that

$$\text{(i)} \quad z = y\lambda_{-1}(F), \qquad \text{(ii)} \quad y\lambda_{-1}(E) = 0.$$

Sufficiency follows from

$$\lambda_{-1}(E) = \lambda_{-1}(F)\lambda_{-1}(\xi) = \lambda_{-1}(F)(1 - \xi).$$

Conversely, write $z = \sum_{i=0}^r a_i\xi^i$. Combining the dependence relation among the powers of ξ with $z(1 - \xi) = 0$, and using the independence of $1, \ldots, \xi^r$, expresses z as a multiple of $\lambda_{-1}(F)$. The top relation gives $y\lambda_{-1}(E) = 0$.

Proposition 5.11. Let $z \in \mathrm{Gr}\, K^\bullet(X)$. The condition $zx = 0$ is equivalent to the existence of $y \in \mathrm{Gr}\, K^\bullet(S)$ such that

$$\text{(i)} \quad z = y\, c_{\mathrm{top}}(F), \qquad \text{(ii)} \quad y\, c_{\mathrm{top}}(E) = 0.$$

The proof is the same as that of 5.10, after passing to associated graded rings and using 5.6.

6. Study of the filtration of $K^\bullet(X)$ when X has an ample sheaf

Let X be a scheme possessing an ample sheaf. Then the two definitions of $K^\bullet(X)$, by locally free finite-type sheaves and by perfect complexes, coincide. Let

$$\varepsilon : K^\bullet(X) \longrightarrow H^0(X, \mathbb{Z})$$

be the augmentation.

Proposition 6.1. If X is quasi-compact and has an ample sheaf, then $\mathrm{Fil}^1 K^\bullet(X)$ is a nilideal.

It is enough to show that $E - \varepsilon(E)$ is nilpotent, where E is the class of a locally free sheaf. By decomposing X into finitely many disjoint opens and then passing to the flag bundle of E, one reduces to the case where E is the class of an invertible sheaf. Let L be the class of an ample invertible sheaf. For n sufficiently large, $E^\vee(n)$ is generated by global sections, giving a surjection $(E(-n))^m \to \mathcal{O}_X$ with locally free kernel. Thus in $K^\bullet(X)$

$$mEL^{-n} = 1 + F.$$

The corresponding γ-relation shows that $1 - EL^{-n}$, and therefore $L^n - E$, is nilpotent. Since $E - 1 = (L^n - 1) - (L^n - E)$, and since $L^n - 1$ is also nilpotent for n large, the assertion follows.

Proposition 6.2. Let R be an augmented sub-λ-algebra of $K^\bullet(X)$, generated by the classes of finitely many sheaves. Then the λ-ring filtration of R is discrete and $\mathrm{Fil}^1 R$ is nilpotent. Moreover, R is a finite module over its image in $H^0(X, \mathbb{Z})$.

After reducing to constant ranks, R is generated by finitely many elements $x_i = E_i - \varepsilon(E_i)$ and their γ-operations. Proposition 6.1 shows that all these elements are nilpotent. Hence the augmentation ideal of R is nilpotent; only finitely many monomials in the x_i's and the $\gamma^j(x_i)$'s remain. This gives both discreteness and finiteness.

Corollary 6.3. Let A be a commutative ring. An element y of $K^\bullet(X) \otimes_{\mathbb{Z}} A$ is invertible if and only if its augmentation is invertible. In particular, the class of every nowhere-zero locally free sheaf of finite type becomes invertible in $K^\bullet(X) \otimes_{\mathbb{Z}} A$.

Indeed, the augmentation ideal is nil.

Corollary 6.4. Let X be quasi-compact with an ample sheaf, and let $f : X' \to X$ be proper and perfect. Assume either that f is projective or that X is noetherian, so that the finiteness theorem applies. If there is an element $x' \in K^\bullet(X') \otimes A$ such that the augmentation of $f_*(x')$ is invertible, then

$$f^* : K^\bullet(X) \otimes A \longrightarrow K^\bullet(X') \otimes A$$

is injective. Indeed, by the projection formula,

$$f_*(f^*y \cdot x') = y \cdot f_*(x'),$$

and $f_*(x')$ is invertible by 6.3.

6.5. The codimension filtration

Assume now that X is noetherian of dimension d, and has an ample sheaf. Define a filtration $K^\bullet(X)^i$ as follows: an element $x \in K^\bullet(X)$ lies in $K^\bullet(X)^i$ if, for every closed subscheme $Y \subset X$, it can be represented by a perfect complex $M^\bullet(Y)$ such that

a) $\mathrm{cl}(M^\bullet) = x$;

b) $\mathrm{codim}(\mathrm{Supp}\, H^j(M^\bullet) \cap Y, Y) \geq i + j$, and $H^j(M^\bullet) = 0$ for $j \gg 0$.

These $K^\bullet(X)^i$'s are additive subgroups.

Proposition 6.6. The filtration of 6.5 is a ring filtration, and $K^\bullet(X)^{d+1} = 0$.

If x and x' are represented by perfect complexes $M^\bullet$ and $M'^\bullet$ satisfying the above codimension estimates, then xx' is represented by $M^\bullet \otimes M'^\bullet$. The support of $H^n(M^\bullet \otimes M'^\bullet)$ is contained in the union of the intersections of supports with total degree n, and the codimension estimate follows from EGA 0_{IV}, 14.2.2.3. Finally $K^\bullet(X)^{d+1} = 0$, because a support in codimension greater than d is empty.

Lemma 6.7. Let $X = \operatorname{Spec} A$ be affine noetherian, let $x_1, \dots, x_n$ be finitely many points of X, and let E be invertible. There exists a section $s \in \Gamma(X, E)$ which is a basis of E_{x_i} for every i.

The proof uses closed specializations x_i' with distinct maximal ideals $\mathfrak{m}_i$, the isomorphisms $M/\mathfrak{m}_i M \simeq A/\mathfrak{m}_i$, for $M = \Gamma(X, E)$, and Nakayama's lemma.

Lemma 6.8. Let X be noetherian with an ample sheaf L, let E be invertible, and let $x_1, \dots, x_n$ be points of X. There exists an integer N such that, for every $k \geq 1$, $E(kN)$ has a global section which is a basis at every x_i.

Choose a power of the ample sheaf and a section whose non-vanishing locus is affine and contains all the x_i's; apply Lemma 6.7 there, and extend after multiplying by a further power.

Theorem 6.9. Let X be a noetherian scheme of dimension d with an ample sheaf. Then

$$\operatorname{Fil}^{d+1} K^\bullet(X) = 0,$$

and therefore $(\operatorname{Fil}^1 K^\bullet(X))^{d+1} = 0$.

First, if E and E' are invertible, with classes still denoted E, E', then $E - E' \in K^\bullet(X)^1$. For a closed $Y \subset X$, choose one point in each irreducible component. Lemma 6.8 gives morphisms from suitable negative powers of the ample sheaf to E and E' which are isomorphisms at those points. The two-term complex built from these morphisms has class $E - E'$, and is acyclic at the chosen generic points of Y; hence its cohomology supports have codimension at least 1 in Y.

Now take $x \in \operatorname{Fil}^{d+1} K^\bullet(X)$. It is enough to treat products

$$\gamma^{i_1}(E_1 - F_1) \cdots \gamma^{i_k}(E_k - F_k), \qquad \sum i_j \geq d+1,$$

where the E_i, F_i are represented by locally free sheaves of the same constant rank. Pull back to the product of the flag bundles of these sheaves. The pullback is injective on $K^\bullet$, and each class splits into sums of invertible classes. For invertible classes ξ, η,

$$\gamma_t(\xi - \eta) = \frac{\gamma_t(\xi - 1)}{\gamma_t(\eta - 1)} = \frac{1 + (\xi - 1)t}{1 + (\eta - 1)t},$$

so

$$\gamma^i(\xi - \eta) = (-1)^{i-1}(\xi - \eta)(\eta - 1)^{i-1} \in K^\bullet(X')^i.$$

Thus the pullback of x belongs to $K^\bullet(X')^{d+1}$. Multiplication by a basis element of $K^\bullet(X')$ over $K^\bullet(X)$ of relative codimension r gives an element of $K^\bullet(X')^{d+1+r} = 0$, since $\dim X' = d + r$. The basis property and injectivity then give $x = 0$.

Problem 6.10. Under the hypotheses of Theorem 6.9, is it true that for every k one has

$$K^\bullet(X)^k = \operatorname{Fil}^k K^\bullet(X)?$$

Bibliography for Exposé VI

[1] H. Bass, *K-Theory and stable algebra*, Publications Mathématiques de l'IHÉS, no. 22.

[2] C. Chevalley, Séminaire 1958: *Anneaux de Chow et applications*.

[3] M. Hakim, *Topos localement annelés et schémas relatifs*, thesis in preparation.

[4] R. Hartshorne, *Residues and duality*, Lecture Notes in Mathematics 20, Springer-Verlag.

[5] J.-P. Serre, *Le groupe de Grothendieck des schémas en groupes réductifs*, Publications Mathématiques de l'IHÉS, no. 34.

Exposé VII. Regular immersions and the computation of $K^\bullet$ of a blow-up scheme

by Pierre Berthelot

1. Generalities on regular immersions

The classical notion of regular immersion (EGA IV 16.9.2) proves inadequate in the non-noetherian case. We shall define a weaker notion, one which gives the desired properties, and which is equivalent to the usual definition in the noetherian case.

Definition 1.1. Let A be a commutative ring, let E be a projective A-module of finite type, and let $u : E \to A$ be an A-linear homomorphism. We say that u is *regular* if the Koszul complex $\mathrm{K}(u)$ defined by u is acyclic in dimensions > 0, and hence is a resolution of $B = A/u(E)$.

Recall that the Koszul complex associated with u is the graded A-module $\bigwedge(E)$, with boundary the interior multiplication by u, regarded as an element of the dual $\check{E}$. This boundary is an antiderivation of degree -1, equal to u on E.

Lemma 1.2. Let $u : E \to A$ be regular. Let A' be an A-algebra and let $u' : E' \to A'$ be obtained from u by base change; suppose that u' is regular. Put $J = u(E)$, $J' = JA' = u(E')$. For every $n \geq 0$ one has

$$J^n/J^{n+1} \otimes_A A' \xrightarrow{\sim} J'^n/J'^{n+1}, \tag{1.2.1}$$

$$\mathrm{Tor}_i^A(J^n/J^{n+1}, A') = 0 \quad (i \geq 1), \tag{1.2.2}$$

$$J^n \otimes_A A' \xrightarrow{\sim} J'^n, \tag{1.2.3}$$

$$\mathrm{Tor}_i^A(J^n, A') = 0 \quad (i \geq 1). \tag{1.2.4}$$

First assume that the modules J^n/J^{n+1} are projective over $B = A/J$. Since J is of finite type they are locally free of finite type over B. The assertions are checked pointwise on $\mathrm{Spec}(A)$, so that one may suppose A local and $u(E) \subset \mathrm{rad}(A)$. Then J^n/J^{n+1} is free over B, and (1.2.2) is equivalent to $\mathrm{Tor}_i^A(B, A') = 0$ for $i \geq 1$. This follows by computing with the Koszul resolution $\mathrm{K}(u)$, since after tensoring with A' it becomes $\mathrm{K}(u')$, acyclic in positive degrees.

The exact sequences

$$0 \to J^n/J^{n+1} \to A/J^{n+1} \to A/J^n \to 0, \qquad 0 \to J^n \to A \to A/J^n \to 0$$

then give (1.2.1), (1.2.3), and (1.2.4) by induction on n. It remains to know that J^n/J^{n+1} is projective over B. Locally one reduces to $E = A^d$, with u given by $f_1, \dots, f_d$, and to the universal homomorphism $\mathbb{Z}[T_1, \dots, T_d]^d \to \mathbb{Z}[T_1, \dots, T_d]$. Since the sequence $(T_1, \dots, T_d)$ is regular, the conclusion follows by base change.

Proposition 1.3. Let A be a ring, E a projective A-module of finite type, $u : E \to A$ a regular homomorphism, and $J = u(E)$. Then J is quasi-regular: J is of finite type, J/J^2 is a projective A/J-module of finite type, and the canonical homomorphism

$$\mathrm{Sym}_{A/J}(J/J^2) \longrightarrow \mathrm{gr}_J(A) = \bigoplus_{n \geq 0} J^n/J^{n+1} \tag{1.3.1}$$

is an isomorphism. This is obtained from the preceding proof, again by reduction to the universal polynomial case.

Definition 1.4. Let (X, A) be a commutatively locally ringed site, let E be an A-module locally free of finite type, and let $u : E \to A$ be A-linear. We say that u is regular if $\mathrm{K}(u)$ is acyclic in degrees > 0. If J is an ideal of A, we say that J is regular if, locally on X, it is the image of a surjective regular homomorphism from a free A-module of finite type.

Let $i : Y \to X$ be an immersion of preschemes. If U is an open subset of X containing $i(Y)$, and if i is a closed immersion into U, then i is said to be regular if the ideal defining $i(Y)$ in $\mathcal{O}_U$ is regular. This condition is independent of U, and regularity may be tested in a neighbourhood of every point.

The definition replaces here that of EGA IV 16.9.2, and in the noetherian case the two definitions are equivalent. A closed immersion locally defined by a regular sequence is regular, and a regular immersion is quasi-regular by Proposition 1.3. A minimal system of generators of an ideal J at a point is a family whose images form a basis of $J_x \otimes k(x)$; regularity is equivalently the local acyclicity of the corresponding Koszul complex. If J is of finite type and J/J^2 is locally free, these systems correspond to local bases of J/J^2.

Proposition 1.5. If $i : Y \to X$ is regular, it remains regular after every flat base change. Conversely, i is regular if and only if it is so locally for the faithfully flat quasi-compact topology.

Definition 1.6. If $i : Y \to X$ is defined by an ideal J, the $\mathcal{O}_Y$-module J/J^2 is the conormal sheaf of Y in X, denoted $N_{X/Y}$. If i is regular it is locally free of finite type, and its rank is the codimension. For closed immersions $Z \xrightarrow{j} Y \xrightarrow{i} X$ one has the exact conormal sequence

$$j^* N_{X/Y} \longrightarrow N_{X/Z} \longrightarrow N_{Y/Z} \longrightarrow 0. \tag{1.6.1}$$

Proposition 1.7. Let $i : Y \to X$ and $j : Z \to Y$ be immersions. If i and j are regular, then $i \circ j$ is regular and the conormal sequence, completed by 0 on the left, is exact. If $i \circ j$ and i are regular and if this sequence is exact and locally split, then j is regular. If X is noetherian and if $i \circ j$ and j are regular, then i is regular at the points of Z.

Locally, write $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $Z = \mathrm{Spec}(C)$, with $B = A/I$ and $C = B/K$. Choosing liftings $f_1, \dots, f_n$ of a basis of I/I^2 and $g_1, \dots, g_m$ of a basis of K/K^2, one obtains the diagram

$$\begin{array}{ccc} A_0 = \mathbb{Z}[T_1, \dots, T_{n+m}] & \longrightarrow & A \\ \downarrow & & \downarrow \\ B_0 = \mathbb{Z}[T_1, \dots, T_m] & \longrightarrow & B \\ \downarrow & & \downarrow \\ C_0 = \mathbb{Z} & \longrightarrow & C, \end{array} \tag{1.7.1}$$

Therefore:

$$\begin{aligned} i \text{ regular} &\iff \forall i \geq 1, \quad \mathrm{Tor}_i^{A_0}(B_0, A) = 0, \\ i \circ j \text{ regular} &\iff \forall i \geq 1, \quad \mathrm{Tor}_i^{A_0}(C_0, A) = 0, \\ j \text{ regular} &\iff \forall i \geq 1, \quad \mathrm{Tor}_i^{B_0}(C_0, B) = 0. \end{aligned} \tag{1.7.2}$$

One then uses the transitivity spectral sequence

$$E^2_{pq} = \mathrm{Tor}_p^{B_0}(C_0, \mathrm{Tor}_q^{A_0}(B_0, A)) \Rightarrow \mathrm{Tor}_n^{A_0}(C_0, A). \tag{1.7.3}$$

This gives the first two assertions; the noetherian assertion is that of EGA IV 19.1.5, with the definitions identified in the noetherian case.

Proposition 1.8. Let X be a scheme, E a locally free $\mathcal{O}_X$-module of finite type, $u : E \to \mathcal{O}_X$ a regular homomorphism, and $J = u(E)$. Let Y denote the closed subscheme of X defined by J, let X' be the scheme obtained by blowing up Y, let $P = \mathbf{P}(E)$ be the projective bundle associated with E, and let $p : P \to X$ be the structural morphism. On P consider the canonical exact sequence

$$0 \to F \to p^*E \to \mathcal{O}_P(1) \to 0, \tag{1.8.1}$$

and the composite morphism

$$v : F \longrightarrow p^*E \xrightarrow{p^*(u)} \mathcal{O}_P.$$

Then:

1. the formation of X' commutes with every base change $f : T \to X$ such that $f^*(u)$ is regular;
2. the closed immersion $j : X' \to P$ defined by the surjective homomorphism

$$\mathcal{S}(E) \longrightarrow \bigoplus_{n\geq 0} J^n$$

   defined by u in degree 1 is a regular immersion, its ideal being the image of the regular morphism v. In particular, if $\operatorname{codim}(i)$ is constant, then $\operatorname{codim}(j) = \operatorname{codim}(i) - 1$;
3. the $\mathcal{S}(E)$-module $\bigoplus_{n\geq 0} J^n$ is perfect as a graded $\mathcal{S}(E)$-module, i.e. locally there is a finite graded resolution of $\bigoplus_{n\geq 0} J^n$ by graded $\mathcal{S}(E)$-modules locally free of finite type.

For (i), since $X' = \operatorname{Proj}(\bigoplus_{n\geq 0} J^n)$, the result is an immediate consequence of the relations (1.2.3) and of EGA II.

For (ii), the asserted properties being local on X, one may suppose that $X = \operatorname{Spec}(A)$, and moreover that E is free on X. We first prove the assertion in the standard case

$$A = \mathbb{Z}[T_1, \ldots, T_d],$$

where E is given the basis $e_1, \ldots, e_d$, with $u(e_i) = T_i$. Then

$$P = \operatorname{Proj}(\mathbb{Z}[T_1, \ldots, T_d, X_1, \ldots, X_d]),$$

the grading being taken with respect to the X_i. To show that v is a regular morphism whose image is the ideal of j, it suffices to do so locally on P, hence on the open sets $D_+(X_i)$; let us place ourselves, for instance, on $D_+(X_1)$. Then $p^*(E)$ is defined by the module

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1] \otimes E,$$

and $\mathcal{O}_P(1)$ by

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1],$$

the canonical morphism sending e_i to X_i/X_1 for every i. Thus F has as a basis the elements

$$e_i - e_1 X_i/X_1 \qquad (i > 1).$$

One has

$$v(e_i - e_1 X_i/X_1) = T_i - T_1 X_i/X_1; \qquad T_i' = T_i - T_1 X_i/X_1.$$

Since the sequence of the T_i is regular in

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1],$$

the same is true of the sequence $T_1, T_2', \ldots, T_d'$, which is obtained from it by an invertible matrix, and therefore also of the extracted sequence $T_2', \ldots, T_d'$.

It remains to show that the T_i' generate the ideal of j. In $\bigoplus_{n\geq 0} J^n$, for $m \leq p$, we shall write $x^{(m)}$ for the element x of J^p, considered as being of degree m. Over $D_+(X_1)$,

$$X' = \operatorname{Spec}\left(\left(\bigoplus_{n\geq 0} J^n\right)_{(T_1^{(1)})}\right),$$

and j is defined by the morphism sending X_i to $T_i^{(1)}$. In this morphism the image of T_i' is

$$T_i^{(0)} - T_1^{(0)} T_i^{(1)}/T_1^{(1)} = ((T_iT_1)^{(1)} - (T_1T_i)^{(1)})/T_1^{(1)} = 0.$$

Thus the T_i' belong to the ideal of j. Since every element of

$$\mathbb{Z}[T_1, \ldots, T_d, X_2/X_1, \ldots, X_d/X_1]$$

can be expressed as a polynomial in $T_1, T_2', \ldots, T_d', X_2/X_1, \ldots, X_d/X_1$, it suffices, in order to show that the T_i' generate the ideal of j, to show that no non-zero element of

$$\mathbb{Z}[T_1, X_2/X_1, \ldots, X_d/X_1]$$

belongs to the ideal of j. This amounts to showing that the homomorphism

$$\mathbb{Z}[T_1, X_1, \ldots, X_d] \longrightarrow \bigoplus_{n\geq 0} J^n$$

is injective, since $T_1^{(1)}$ is not a zero divisor in $\bigoplus_{n\geq 0} J^n$. If one grades $\mathbb{Z}[T_1, X_1, \ldots, X_d]$ by assigning degree 0 to T_1 and degree 1 to the X_i, the kernel of the homomorphism in question is a graded ideal. But it is contained in the kernel of the composite homomorphism

$$\mathbb{Z}[T_1, X_1, \ldots, X_d] \longrightarrow \bigoplus_{n\geq 0} J^n \longrightarrow A,$$

the last homomorphism being defined in degree n by the injection of J^n into A. The kernel of the composite homomorphism is the principal ideal generated by $X_1 - T_1$, which, by integrality, contains no non-zero homogeneous element. Therefore the first homomorphism is injective, which completes the proof in this case.

Return to the general case where A is arbitrary, with E still assumed free on X. Put index 0 on the special case just treated. By part (i), the situation under consideration is obtained from the preceding one by the base change $A_0 \to A$, where the homomorphism $A_0 \to A$ is defined by choosing a basis of E, its image $f_1, \ldots, f_d$ in J, and sending T_i to f_i. Put

$$B_0 = \left(\bigoplus_{n\geq 0} J_0^n\right)_{(T_1^{(1)})}, \qquad B = \left(\bigoplus_{n\geq 0} J^n\right)_{(f_1^{(1)})}.$$

Then $B = B_0 \otimes_{A_0} A$, and the Koszul complex defined by $v : F \to \mathcal{O}_P$ is obtained by base change from the Koszul complex defined by v_0. Since the latter is a projective resolution of B_0 on a suitable open set, the homology groups of $K_\bullet(v)$ are, on this open set, the groups

$$\operatorname{Tor}_i^{A_0}(A, B_0).$$

These are the components of degree 0 of

$$\operatorname{Tor}_i^{A_0}\left(A, \left(\bigoplus_{n\geq 0} J_0^n\right)_{T_1^{(1)}}\right) = \left(\bigoplus_{n\geq 0} \operatorname{Tor}_i^{A_0}(A, J_0^n)\right)_{T_1^{(1)}}.$$

But the $\mathrm{Tor}_i^{A_0}(A, J_0^n)$ vanish for $i \geq 1$ by the relations 1.2; hence $K_\bullet(v)$ is a resolution of B, which shows that j is regular.

For (iii), the property being local on X, one may again suppose X affine and E free of finite type. We then resume the notation of (ii). First consider the standard case of (ii). The ring A_0 is noetherian and of finite cohomological dimension, and so is the algebra $\mathcal{S}(E_0)$. One may then construct, step by step, a partial graded resolution of $\bigoplus_{n\geq 0} J_0^n$ by graded free $\mathcal{S}(E_0)$-modules of finite type, of length greater than the cohomological dimension of A_0. The last kernel is then a graded $\mathcal{S}(E_0)$-module, projective of finite type as an ungraded $\mathcal{S}(E_0)$-module, hence locally free of finite type on $\mathrm{Spec}(A_0)$ as a graded $\mathcal{S}(E_0)$-module by VI 1.5.1. In this way one obtains the announced finite resolution, replacing the last module of the partial resolution by the corresponding kernel.

Tensoring the resolution thus constructed by A, one obtains a graded complex of graded $\mathcal{S}(E)$-modules locally free of finite type, and it remains to see that this is a resolution of $\bigoplus_{n\geq 0} J^n$. But the i-th homology module of the resulting complex is a graded $\mathcal{S}(E)$-module whose n-th component is $\mathrm{Tor}_i^{A_0}(A, J_0^n)$, which is zero for $i \geq 1$ by (1.2.4). Consequently, by (1.2.3), one has thus constructed a resolution of the required type.

Proposition 1.9. If $i : Y \to X$ is a regular closed immersion, if X' is the blow-up of Y, and if $f : X' \to X$ is structural, then i and f are perfect morphisms: K resolves $\mathcal{O}_Y$ by finite locally free $\mathcal{O}_X$-modules, and f factors through a regular immersion into a projective bundle followed by a smooth morphism.

Proposition 1.10. If X and Y are smooth over S, every S-immersion from X into Y is regular. In particular, every section of a smooth morphism is a regular immersion.

2. Computations on regular immersions

2.1–2.3. For two A-algebras B, C, the graded module $\bigoplus_{i\geq 0} \mathrm{Tor}_i^A(B, C)$ is given a natural structure of associative graded $B \otimes_A C$-algebra by choosing a projective resolution $P_\bullet \to B$, using the Kunneth morphisms, and extending the augmentation $B \otimes_A B \to B$. If the resolution is an alternating differential graded algebra, the Tor algebra is alternating. Globally this gives, for two X-schemes X', X'', a graded algebra structure on

$$\bigoplus_i \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''}).$$

The multiplication is represented by the commutative square of complexes

$$\begin{array}{ccc} P_i \otimes P_j & \longrightarrow & (P_{i-1} \otimes P_j) \oplus (P_i \otimes P_{j-1}) \\ \downarrow & & \downarrow \\ P_{i+j} & \longrightarrow & P_{i+j-1}. \end{array} \tag{2.2.1}$$

Lemma 2.4. For two X-schemes X', X'' and a flat $\mathcal{O}_{X'}$-module M, one has canonically

$$\mathrm{Tor}_i^{\mathcal{O}_X}(M, \mathcal{O}_{X''}) \simeq M \otimes_{\mathcal{O}_{X'}} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_{X''}). \tag{2.4.1}$$

This is immediate locally from a projective resolution, and then glues.

Proposition 2.5. Let Y and Z be closed subschemes of X, with $T = Y \times_X Z$. Under the local Koszul hypotheses of this proposition, one has

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y, \mathcal{O}_Z), \tag{2.5.1}$$

and

$$\operatorname{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Z) \simeq I\cap J/IJ \simeq \operatorname{Ker}(i^*N_{Y/X}\times j^*N_{Z/X}\to N_{T/X}). \tag{2.5.2}$$

In particular, for a regular closed immersion $Y\to X$,

$$\operatorname{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y) \simeq \bigwedge^i \operatorname{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y), \qquad \operatorname{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y)\simeq N_{Y/X}. \tag{2.5.3}$$

The proof is the standard comparison of Koszul complexes, together with the bicomplex spectral sequence and the preceding multiplicative construction.

2.6–2.7. Assume that X has an ample sheaf, so that the two definitions of $K^\bullet$ coincide. If $i: Y\to X$ is a regular closed immersion, one defines

$$i_*: K^\bullet(Y)\to K^\bullet(X)$$

using the class of the direct image, and one has, for every $x\in K^\bullet(Y)$,

$$i^*i_*(x) = x\lambda_{-1}(N), \qquad \lambda_{-1}(N) = \sum_{r\geq 0}(-1)^r\lambda^r(N), \tag{2.7.1}$$

where N is the conormal sheaf. Indeed, if x is represented by a locally free finite type $\mathcal{O}_Y$-module M, the homology sheaves of the derived inverse image of i_*M are $M\otimes \operatorname{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Y,\mathcal{O}_Y)$, and Proposition 2.5 gives the displayed formula.

Corollary 2.8. Under the hypotheses of 2.7, for $x,y\in K^\bullet(Y)$,

$$i_*(xy\lambda_{-1}(N)) = i_*(x)i_*(y). \tag{2.8.1}$$

This follows from (2.7.1) and the projection formula.

Corollary 2.9. Under the same hypotheses, if $i_*(1)$ is invertible in $K^\bullet(X)$, then i_* is an isomorphism of $K^\bullet(Y)$ onto a direct summand.

3. Computation of $K^\bullet$ of a blow-up scheme

3.1. Let $i: Y\to X$ be a regular closed immersion. Let X' be the blow-up of Y, and let $Y' = Y\times_X X'$. By EGA II 8.1.8, Y' is a closed subscheme of X' defined by an ideal canonically isomorphic to $\mathcal{O}_{X'}(1)$. Thus $j: Y'\to X'$ is a regular closed immersion of codimension 1. Let J be the ideal of Y in X, and J' the ideal of Y' in X'; then

$$J'\simeq J\mathcal{O}_{X'}\simeq \mathcal{O}_{X'}(1).$$

Writing $f: X'\to X$ and $g: Y'\to Y$ for the structural morphisms, one has the commutative diagram

$$\begin{tikzcd} Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\ Y \arrow[r, "i"'] & X. \end{tikzcd} \tag{3.1.1}$$

Moreover,

$$X' = \operatorname{Proj}(S), \qquad S = \bigoplus_{n\geq 0} J^n,$$

and

$$Y' = \operatorname{Proj}(i^*S) = \operatorname{Proj}(\bigoplus_{n\geq 0} J^n/J^{n+1}).$$

Since i is regular, $\bigoplus J^n/J^{n+1} \simeq \mathrm{Sym}_{\mathcal{O}_Y}(J/J^2)$. If $N = J/J^2$, then $Y' = \mathbf{P}(N)$. On Y' there is the exact sequence

$$0 \to F \to g^*N \to \mathcal{O}_{Y'}(1) \to 0. \tag{3.1.2}$$

The sheaf $\mathcal{O}_{Y'}(1) = j^*\mathcal{O}_{X'}(1) = J'/J'^2$ is the conormal sheaf of j; denote it by L.

Lemma 3.2. With the hypotheses and notation of 3.1, one has

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y), \tag{3.2.i}$$

$$\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq F. \tag{3.2.ii}$$

For (i), since $\mathcal{O}_Y$ can locally be resolved by a Koszul complex, the algebra $\bigoplus_i \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y)$ is alternating. Hence there is a unique algebra homomorphism

$$\bigwedge \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \longrightarrow \bigoplus_{i\geq 0} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \tag{3.2.1}$$

extending the injection of degree one; it is enough to verify locally that it is an isomorphism. On an affine open, choose a regular homomorphism $E \to A$ defining J, with basis images $f_1, \ldots, f_d$. On the open of X' where $f_1^{(1)}$ is invertible, change the basis by

$$e_1' = e_1, \qquad e_i' = e_i - (f_i^{(1)}/f_1^{(1)})e_1 \quad (i \neq 1).$$

The Koszul differential then gives local isomorphisms

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq (\mathcal{O}_{X'}/f_1^{(0)}\mathcal{O}_{X'}) \otimes_{\mathcal{O}_{X'}} \bigwedge^i(\mathcal{O}_{X'}^{d-1}) \simeq \bigwedge^i(\mathcal{O}_{Y'}^{d-1}). \tag{3.2.2}$$

In particular, for $i = 1$ one obtains $\mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \mathcal{O}_{Y'}^{d-1}$, and hence

$$\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq \bigwedge^i \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y). \tag{3.2.3}$$

For (ii), tensor the exact sequence $0 \to J \to \mathcal{O}_X \to \mathcal{O}_Y \to 0$ with $\mathcal{O}_{X'}$. Since the image of $J \otimes \mathcal{O}_{X'} \to \mathcal{O}_{X'}$ is $J\mathcal{O}_{X'} = \mathcal{O}_{X'}(1)$, and then tensoring with $\mathcal{O}_{Y'}$, one obtains

$$0 \to \mathrm{Tor}_1^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \to g^*N \to \mathcal{O}_{Y'}(1) \to 0, \tag{3.2.4}$$

which identifies this Tor sheaf with F in (3.1.2).

3.3. Suppose now that X has an ample sheaf. Then the same is true for X', which is projective over X, and for Y and Y'. Moreover i and j are perfect morphisms, g is smooth hence perfect because $Y' = \mathbf{P}(N)$, and f is perfect by 1.9. Since these morphisms are projective, one has additive homomorphisms

$$i_* : K^\bullet(Y) \to K^\bullet(X), \quad j_* : K^\bullet(Y') \to K^\bullet(X'), \quad f_* : K^\bullet(X') \to K^\bullet(X), \quad g_* : K^\bullet(Y') \to K^\bullet(Y).$$

Proposition 3.4. With the hypotheses and notation of 3.1 and 3.3, for every $x \in K^\bullet(Y)$ one has

$$f^*(i_*(x)) = j_*\big(g^*(x)\lambda_{-1}(F)\big). \tag{3.4.1}$$

By additivity, one may assume x represented by a locally free finite type $\mathcal{O}_Y$-module M. The class of $f^*i_*(x)$ is represented by the complex obtained from a projective resolution of i_*M and application of f^*. Its homology sheaves are $\mathrm{Tor}_i^{\mathcal{O}_X}(M, \mathcal{O}_{X'})$. By 2.4 and 3.2,

$$\mathrm{Tor}_i^{\mathcal{O}_X}(M, \mathcal{O}_{X'}) \simeq M \otimes_{\mathcal{O}_Y} \mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_{X'}, \mathcal{O}_Y) \simeq M \otimes_{\mathcal{O}_Y} \bigwedge^i F,$$

and the alternating sum of their classes is exactly the right hand side of (3.4.1).

Lemma 3.5. With the hypotheses and notation of 3.1, one has the following isomorphisms:

i) for all $n \geq 0$ and $i \geq 1$,

$$R^i f_*(\mathcal{O}_{X'}(n)) = 0;$$

ii) for all $n \geq 0$,

$$f_*(\mathcal{O}_{X'}(n)) = J^n.$$

One may suppose X affine, hence quasi-compact. For (i) one uses the following lemma: if $f : X' \to X$ is projective and perfect, with X quasi-compact, and if L' is a perfect complex on X' acyclic in degrees > 0, then there exists n_0 such that, for $n \geq n_0$, $Rf_*(L' \otimes \mathcal{O}_{X'}(n))$ is acyclic in degrees > 0. Applied to $\mathcal{O}_{X'}$, this gives the vanishing for large n. The exact sequence

$$0 \to \mathcal{O}_{X'}(n+1) \to \mathcal{O}_{X'}(n) \to \mathcal{O}_{Y'}(n) \to 0$$

and the fact that $Y' = \mathbf{P}(N)$ then allow descending induction to $n = 0$.

For (ii), the canonical graded homomorphism

$$\bigoplus_{n \geq 0} J^n \longrightarrow \bigoplus_{n \geq 0} f_*(\mathcal{O}_{X'}(n))$$

is shown to be an isomorphism locally. Write $S = \bigoplus J^n$ and use a surjective graded homomorphism $\mathcal{O}_X[T_1, \ldots, T_d] \to S$ given by a set of generators of J. This realizes X' as a regular closed subscheme of $\mathbf{P}^{d-1}_X$. With a finite graded presentation $L'' \to L' \to S \to 0$, one obtains on $\mathbf{P}^{d-1}_X$ a presentation $M'' \to M' \to \mathcal{O}_{X'} \to 0$. For each n there is a commutative diagram

$$\begin{tikzcd}
L''_n \arrow[r] \arrow[d, "\sim"'] & L'_n \arrow[r] \arrow[d, "\sim"'] & J^n \arrow[r] \arrow[d] & 0 \arrow[d] \\
p_*M''(n) \arrow[r] & p_*M'(n) \arrow[r] & p_*\mathcal{O}_{X'}(n) \arrow[r] & 0,
\end{tikzcd}$$

where $p : \mathbf{P}^{d-1}_X \to X$. For n sufficiently large the bottom row is exact; hence the middle vertical arrow is an isomorphism for large n. Descending induction, using

$$0 \to J^{n+1} \to J^n \to J^n/J^{n+1} \to 0$$

and the analogous direct-image exact sequence, completes the proof.

Proposition 3.6. With the hypotheses and notation of 3.1 and 3.3,

$$f_* f^* = \mathrm{Id}_{K^\bullet(X)}\,. \tag{3.6.1}$$

Indeed, by the projection formula, $f_*f^*(x) = xf_*(1)$, and Lemma 3.5 says that $Rf_*\mathcal{O}_{X'}$ has $\mathcal{O}_X$ in degree 0 and no higher cohomology.

Theorem 3.7. Let $i : Y \to X$ be a regular closed immersion, let X' be the blow-up of Y, and let $Y' = X' \times_X Y$, so that one has

$$\begin{tikzcd}
Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\
Y \arrow[r, "i"'] & X.
\end{tikzcd}$$

Assume that X is quasi-compact and has an ample sheaf. Then the sequence

$$0 \to K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \tag{3.7.0}$$

is exact, where

$$u(y) = \big(g^*(y)\lambda_{-1}(F), -i_*(y)\big), \tag{3.7.u}$$

and, for $y' \in K^\bullet(Y')$, $x \in K^\bullet(X)$,

$$v(y', x) = j_*(y') + f^*(x). \tag{3.7.v}$$

Moreover u has a left inverse u', defined by $u'(y', x) = g_*(y')$. If, in addition, X is noetherian and Y regular, then v is surjective and

$$0 \to K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \to 0 \tag{3.7.2}$$

is exact.

First, $u' \circ u = 1$, because $g_*(g^* y \lambda_{-1}(F)) = y\, g_*(\lambda_{-1}(F)) = y$. Also $v \circ u = 0$, by Proposition 3.4. If (y', x) belongs to the kernel of v, then $j_*(y') + f^*(x) = 0$. One must find $y \in K^\bullet(Y)$ such that

$$x = -i_*(y), \qquad y' = g^*(y)\lambda_{-1}(F). \tag{3.7.1}$$

Necessarily $y = u'(y', x) = g_*(y')$. Applying f_* to the relation $f^*(x) = -j_*(y')$, and using Proposition 3.6, gives $x = -i_*(y)$. To prove the second equality in (3.7.1), put $z = y' - g^*(g_* y')\lambda_{-1}(F)$. Then $g_* z = 0$, and the preceding formulas give $j_* z = 0$. Since $\operatorname{Ker} j_* \cap \operatorname{Ker} g_* = 0$, it follows that $z = 0$. This proves exactness at the middle term.

For the final surjectivity assertion, assume X noetherian and Y regular. Then Y' is regular, being a projective bundle over Y, and local rings of X' along Y' are regular because Y' is cut out by one regular element. Therefore coherent modules on X' with support in Y' are perfect. If such a module E has class in $K^\bullet(X')$, filtration by the powers $J'^n E$ shows that its class is in the image of j_*. For arbitrary $x \in K^\bullet(X')$, choose a perfect complex M' representing x, and form the distinguished triangle

$$\begin{tikzcd} & N' \arrow[dl] \arrow[dr] & \\ M' \arrow[rr] & & Lf^* Rf_*(M'). \end{tikzcd}$$

The horizontal arrow is the canonical morphism. It is an isomorphism off Y', so the cohomology sheaves of N' are coherent sheaves supported on Y'. Their class lies in $\operatorname{Im}(j_*)$. Hence

$$x - f^* f_*(x) \in \operatorname{Im}(j_*),$$

and v is surjective.

Remark 3.8

In the case where X is noetherian and Y regular, 3.7 determines the abelian group $K^\bullet(X')$ entirely from the λ-rings $K^\bullet(X)$, $K^\bullet(Y)$, the maps i_*, i^* between these rings, and the element $N \in K^\bullet(Y)$. Indeed, $K^\bullet(Y')$ can be computed from $K^\bullet(Y)$ and N by VI 1.1. We shall see in the following paragraph that the multiplicative structure of $K^\bullet(X')$ is determined by the same data, and so is the λ-structure after tensoring with $\mathbb{Q}$ (this restriction is probably unnecessary; see 4.5).

4. Regular immersions and the filtration of $K^\bullet$

Throughout this paragraph we keep the notation and hypotheses of 3.1.

Lemma 4.1. Suppose that there exists on Y a locally free sheaf N' of finite type such that, if N, N' denote the classes in $K^\bullet(Y)$ of N, N', then $N = N' + 2$. Then the following congruence is verified:

$$\lambda_{-1}(F) \equiv 0 \pmod{1 - L}.$$

Let d be the rank of N, which we may suppose constant; the rank of F is then $d-1$. By definition,

$$\begin{aligned}\lambda_{-1}(F) &= \sum_{i \geq 0} (-1)^i \lambda^i(F) \\ &= (-1)^{d-1}\lambda^{d-1}(F-1) \\ &= (-1)^{d-1}\lambda^{d-1}(F+L-1-L) \\ &= (-1)^{d-1}\lambda^{d-1}(N-2+1-L).\end{aligned}$$

Now

$$\lambda_t(1-L) = \frac{\lambda_t(1)}{\lambda_t(L)} = \frac{1+t}{1+Lt}.$$

Thus, for every $i \geq 1$, $\lambda^i(1-L)$ is divisible by $1-L$. Consequently

$$\begin{aligned}\lambda_{-1}(F) &\equiv (-1)^{d-1}\lambda^{d-1}(N-2) \pmod{1-L} \\ &= (-1)^{d-1}\lambda^{d-1}(N') \pmod{1-L}.\end{aligned}$$

Since N' has rank $d-2$, $\lambda^{d-1}(N') = 0$, proving the result.

4.2

Let $i : Y \to X$ be a regular immersion. Let $r \geq 1$ be an integer and let

$$s : X \longrightarrow \mathbf{P}^r_X$$

be the section defined by a homomorphism $(\mathcal{O}_X)^{r+1} \to \mathcal{O}_X$, after passing to the quotient by the sub-bundle generated by the corresponding sections. If $\mathbf{P}^r_X$ is defined by the graded $\mathcal{O}_X$-algebra $\mathcal{O}_X[T_0, \ldots, T_r]$, the ideal I defining the section s is generated by the sections $T_1, \ldots, T_r$. Therefore I/I^2 is isomorphic to $(\mathcal{O}_X[T_0])^r$, so the ideal corresponding to $\mathcal{O}_{\mathbf{P}^r}$ is $(\mathcal{O}_X)^r$.

Since the section s is evidently a regular immersion, the same is true for $s \circ i$, and one may write the corresponding exact sequence of conormal sheaves (1.7). Thus, if N denotes the conormal sheaf of i, and N_1 the conormal sheaf of $s \circ i$, one obtains the equality

$$N_1 = N + r.$$

It follows that every regular immersion i can be made to satisfy the property of Lemma 4.1 by composing it with a section $X \to \mathbf{P}^2_X$, or with a section $X \to \mathbf{P}^1_X$.

Theorem 4.3. Let $i : Y \to X$ be a regular closed immersion, and let N be the class in $K^\bullet(Y)$ of the conormal sheaf of Y in X. Suppose that X is quasi-compact and has an ample sheaf. Then, with the notation of 3.3, for every $y \in K^\bullet(Y)$ and every $p \geq 1$,

$$i_*(\lambda^p(N, y)) \equiv \lambda^p(i_*(y)) \pmod{\text{torsion}}, \tag{4.3.1}$$

where $\lambda^p(N, y)$ is the element defined in V 5.3.

First suppose that the hypothesis of 4.1 is satisfied; in that case we shall show

$$i_*(\lambda^p(N,y)) = \lambda^p(i_*(y)).$$

Since $f_*f^* = \mathrm{Id}$, this relation is equivalent to

$$f^*(i_*(\lambda^p(N,y))) = f^*(\lambda^p(i_*(y))).$$

Now one has

$$\begin{aligned} f^*(\lambda^p(i_*(y))) &= \lambda^p(f^*(i_*(y))) \\ &= \lambda^p(j_*g^*(y)\lambda_{-1}(F)), \end{aligned} \tag{3.4}$$

whereas

$$f^*(i_*(\lambda^p(N,y))) = j_*(g^*(\lambda^p(N,y))\lambda_{-1}(F)) \tag{3.4}$$

by Proposition 3.4. Since $g^*(N) = F + L$, formula V 5.4.2 gives

$$g^*(\lambda^p(N,y))\lambda_{-1}(F) = \lambda^p(g^*(N), g^*(y))\lambda_{-1}(F) = \lambda^p(L, g^*(y)\lambda_{-1}(F)).$$

Putting $z = g^*(y)\lambda_{-1}(F)$, we are reduced to showing

$$j_*(\lambda^p(L,z)) = \lambda^p(j_*(z)).$$

This is the analogous relation for the immersion j and the element z. By hypothesis, $\lambda_{-1}(F)$ is a multiple of $1 - L$, so one can find z' such that $z = z'(1-L)$. Then

$$z = j_*(u') \quad \text{with } u' = g_*(z')$$

by (2.7), and since $L = j^*(J')$, it remains only to verify

$$j_*(j^*(\lambda^p(J',u'))) = \lambda^p(j_*j^*(u')).$$

The exact sequence

$$0 \longrightarrow J' \longrightarrow \mathcal{O}_{X'} \longrightarrow \mathcal{O}_{Y'} \longrightarrow 0$$

gives

$$j_*j^*(1) = 1 - J'. \tag{4.3.2}$$

The desired relation is therefore, after applying the projection formula,

$$\lambda^p(J',u')(1-J') = \lambda^p(u'(1-J')).$$

Since $1 - J' = \lambda_{-1}(J')$, this is precisely the formula which serves to define $\lambda^p(J',u')$.

We now pass to the general case. Compose i first with a section $X \to \mathbf{P}^1_X$, and then with a section $\mathbf{P}^1_X \to \mathbf{P}^1_{\mathbf{P}^1_X}$, in such a way that the composed immersion satisfies the condition of 4.1 (see 4.2). Proposition 4.3 is then true for the immersion $s \circ i$ which precedes it; it remains to descend to X. We descend first to $\mathbf{P}^1_X$, and then to X, so that the proof is made by applying twice the following lemma, due to J.-P. Jouanolou.

Lemma. Under the conditions of 4.3, let $s : X \to \mathbf{P}^1_X$ be the section defined in 4.2. If Proposition 4.3 is true for $s \circ i$ and for s, then it is true for i.

Suppose 4.3 true for $s \circ i$. Then, for every $p \geq 1$ and every $y \in K^\bullet(Y)$,

$$(s \circ i)_*(\lambda^p(N+1,y)) \equiv \lambda^p((s \circ i)_*(y)) \quad (\text{mod torsion}). \tag{4.3.3}$$

The section s is defined by an ideal isomorphic to $\mathcal{O}_{\mathbf{P}}(-1)$ (4.2). Let ξ be its class in $K^\bullet(\mathbf{P}^1_X)$. Then, for every $x \in K^\bullet(X)$,

$$s_*(x) = p^*(x)(1-\xi),$$

where p is the structural morphism $\mathbf{P}^1_X \to X$. Indeed, since $p \circ s = \mathrm{Id}_X$, one has $x = s^*(p^*(x))$, whence $s_*(x) = s_*(s^*p^*(x)) = p^*(x)s_*(1)$; and $s_*(1) = 1 - \xi$. Thus

$$\begin{aligned}(s \circ i)_*(\lambda^p(N+1, y)) &\equiv \lambda^p(p^*(i_*(y))(1-\xi)) \quad \text{(mod torsion)}\\ &= \lambda^p(\xi, p^*(i_*(y)))(1-\xi) \quad \text{(mod torsion)}.\end{aligned}$$

Now $\lambda^p(\xi, p^*(i_*(y))) - \lambda^p(1, p^*(i_*(y)))$ is divisible by $1-\xi$. To see this, one may work in the universal λ-ring used to define $\lambda^p(\xi, x)$ (V 5). Then $\lambda^p(\xi, x) - \lambda^p(1, x)$ is a polynomial in ξ which vanishes for $\xi = 1$, hence is divisible by $\xi - 1$. By the independence equation for ξ on $K^\bullet(X)$, one has $(1-\xi)^2 = 0$ (VI 1.13.1). Hence

$$\begin{aligned}(s \circ i)_*(\lambda^p(N+1, y)) &\equiv \lambda^p(1, p^*(i_*(y)))(1-\xi) \quad \text{(mod torsion)}\\ &= p^*(\lambda^p(1, i_*(y)))(1-\xi) \quad \text{(mod torsion)}.\end{aligned}$$

Applying p_*, we obtain

$$i_*(\lambda^p(N+1, y)) \equiv \lambda^p(1, i_*(y)) \quad \text{(mod torsion)}, \tag{4.3.4}$$

since $p_*(\xi) = 0$.

By formula V 5.6, applied with $L = 1$, $\lambda^p(N+1, y)$ is the coefficient of t^p in the series

$$\frac{\sum_{k\geq 1} \lambda^k(N, y)t^k}{1 + \sum_{k\geq 1} \lambda^k(N, y)\lambda_{-1}(N)t^k}.$$

Similarly, the same formula applied to $N = 0$ and $L = 1$ shows that $\lambda^p(1, y)$ is the coefficient of t^p in

$$\frac{\sum_{k\geq 1} \lambda^k(y)t^k}{\sum_{k\geq 0} \lambda^k(y)t^k}.$$

The congruences (4.3.4) can therefore be written

$$i_*\left(\frac{\sum_{k\geq 1} k\lambda^k(N, y)t^k}{1 + \sum_{k\geq 1} \lambda^k(N, y)\lambda_{-1}(N)t^k}\right) = \frac{\sum_{k\geq 1} k\lambda^k(i_*(y))t^k}{\sum_{k\geq 0} \lambda^k(i_*(y))t^k} \quad \text{(mod torsion)}.$$

Applying formula 2.8, one obtains

$$\frac{\sum_{k\geq 0} k\, i_*(\lambda^k(N, y))t^k}{1 + \sum_{k\geq 1} i_*(\lambda^k(N, y))t^k} = \frac{\sum_{k\geq 0} k\lambda^k(i_*(y))t^k}{\sum_{k\geq 0} \lambda^k(i_*(y))t^k} \quad \text{(mod torsion)}.$$

If

$$u(t) = 1 + \sum_{k\geq 1} i_*(\lambda^k(N, y))t^k, \qquad v(t) = \sum_{k\geq 0} \lambda^k(i_*(y))t^k,$$

this last relation can also be written

$$\frac{d}{dt}\left(\frac{u(t)}{v(t)}\right) \equiv 0 \quad \text{(mod torsion)}.$$

Since the constant term of $u(t)/v(t)$ is 1, and since the congruences are modulo torsion, one may integrate and write the congruence $u(t) = v(t)$. This gives the announced congruences.

Corollary 4.4. With the notation and hypotheses of 4.3, for every $p \geq 1$ and every $y \in K^\bullet(Y)$, one has

$$i_*(\psi^p(N, y)) \equiv \psi^p(i_*(y)) \quad \text{(mod torsion)}.$$

This follows immediately from 4.3, using the expression of the ψ^p as integral linear combinations of the λ^p.

Remark 4.5. The proof of 4.3 itself suggests that (4.3.1) should be true not merely modulo torsion, but also as an equality. In RRR, the equality was proved in characteristic 0 by means of a geometric characterization of $\lambda^p(N, x)$. An extension of this method to all characteristics would probably be possible if one knew how to define exterior-power operations on perfect complexes (cf. XIV).

Theorem 4.6. Let X be a quasi-compact scheme having an ample sheaf, let $d > 0$ be an integer, and let $i : Y \to X$ be a regular immersion of codimension d. Then, for every k,

$$i_*((\operatorname{Fil}^k K^\bullet(Y)) \otimes \mathbb{Q}) \subset (\operatorname{Fil}^{k+d} K^\bullet(X)) \otimes \mathbb{Q}.$$

First note that $i_*(K^\bullet(Y)) \subset \operatorname{Fil}^1 K^\bullet(X)$. Indeed, $X - Y$ is dense in X; otherwise Y would contain an affine open, and on this open the ideal J defining Y would be a nilideal, hence a locally nilpotent ideal, since it is of finite type. This is impossible, since the J^n/J^{n+1} are non-zero locally free modules. The augmentation in $K^\bullet(X)$ of an element of $i_*(K^\bullet(Y))$ is a locally constant function on X; since it vanishes on $X - Y$, it vanishes on all of X.

For every abelian group A, write $A_\mathbb{Q} = A \otimes \mathbb{Q}$. Let $x \in \operatorname{Fil}^k K^\bullet(Y)_\mathbb{Q}$. Then x can be written in the form

$$x = \sum a\, \gamma^{i_1}(x_1) \cdots \gamma^{i_n}(x_n),$$

with $\epsilon(x_i) = 0$ for all i, $a \in \mathbb{Q}$, and $i_1 + \cdots + i_n \geq k$. The x_i can be expressed as integral linear combinations of the classes E_α of finitely many locally free finite-type $\mathcal{O}_Y$-modules E_α. Let R be the sub-$\mathbb{Q}$-λ-algebra of $K^\bullet(Y)_\mathbb{Q}$ generated by the E_α and by N. Then $x \in \operatorname{Fil}^k R$.

The filtration of the λ-ring R is discrete (VI 6.2); choose m such that $\operatorname{Fil}^m R = 0$. Finally, after decomposing X as a finite sum of disjoint open subsets and proving the assertion on each such open, one may suppose that the E_α have constant rank r_α. The augmentation ring of R is then $\mathbb{Q}$.

We shall show that, for every i,

$$i_*(\operatorname{Fil}^i R) \subset \operatorname{Fil}^{i+d} K^\bullet(X)_\mathbb{Q}.$$

This will imply in particular

$$i_*(x) \in \operatorname{Fil}^{k+d} K^\bullet(X),$$

and so prove the theorem.

The proof is by descending induction on i, the case $i = m$ being trivial. Let $y \in \operatorname{Fil}^i R$, and set, for j,

$$a_j = (-1)^{j-1}(j-1)!.$$

Then, by V 6.10,

$$\gamma^{d+i}(N, y) - a_{d+i} y \in \operatorname{Fil}^{i+1} R.$$

The induction hypothesis can therefore be applied, giving

$$i_*(\operatorname{Fil}^{i+1} R) \subset \operatorname{Fil}^{i+d+1} K^\bullet(X)_\mathbb{Q}.$$

Thus

$$i_*(\gamma^{d+i}(N, y) - a_{d+i} y) \in \operatorname{Fil}^{i+d+1} K^\bullet(X)_\mathbb{Q}.$$

By 4.4 this implies

$$\gamma^{d+i}(i_*(y)) - a_{d+i} i_*(y) \in \operatorname{Fil}^{i+d+1} K^\bullet(X)_\mathbb{Q}.$$

Since $i_*(y)$ has zero augmentation,

$$\gamma^{d+i}(i_*(y)) \in \operatorname{Fil}^{i+d+1} K^\bullet(X).$$

As we are working modulo torsion, this gives

$$i_*(y) \in \operatorname{Fil}^{i+d} K^\bullet(X)_{\mathbb{Q}}.$$

Remark 4.6.1. One sees that, even if 4.3 were true without tensoring by $\mathbb{Q}$, the preceding proof works only modulo torsion. For a counterexample, see XIV 4.6.

4.7

We return to the situation of 3.7, where one had the commutative diagram

$$\begin{tikzcd} Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\ Y \arrow[r, "i"'] & X, \end{tikzcd}$$

where i is a regular immersion, X' is the X-scheme obtained by blowing up Y, and $Y' = X' \times_X Y$. Suppose X noetherian with an ample sheaf, and Y regular. We saw in 3.7 that there is then the exact sequence

$$0 \longrightarrow K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \longrightarrow 0.$$

We shall show that this exact sequence permits one to describe the whole situation formed by the $K^\bullet$ of X, Y, X', Y', from the data of the λ-rings $K^\bullet(X)$, $K^\bullet(Y)$, the maps i^* and i_*, and the element $N \in K^\bullet(Y)$, at least up to torsion, up to one further point.

Let K, K' be two λ-rings, let $i_* : K \to K'$ be an additive homomorphism, let $i^* : K' \to K$ be a λ-homomorphism, let d be an integer, and let $N \in K$ be such that $\lambda^d(N) \neq 0$ for $i > d$ and $\lambda^d(N)$ is invertible. Suppose, moreover, that the following properties hold:

i) i_* is K'-linear (the "projection formula");

ii) $i^* i_*$ is multiplication by $\lambda_{-1}(N)$;

iii) for all $n \geq 1$ and all $y \in K$,
$$i_*(\lambda^n(N, y)) = \lambda^n(i_*(y))$$
(relation (4.3.1)).

We first use VI 1.1 to construct a K-λ-algebra K_1 which plays the role of $K^\bullet(Y')$. Let L be an indeterminate, and let $K[L]$ be the K-λ-algebra generated by L, subject to the relations $\lambda^i(L) = 0$ for $i \geq 2$ (V 4.9). Put

$$P(L) = \lambda^d(N - L) = \sum_{i=0}^{d} (-1)^i \lambda^{d-i}(N) L^i.$$

The ideal generated by P is a λ-ideal. To see this, first observe that the relations $\lambda^i(N) = 0$ for $i > d$ imply that the $\lambda^j(N - L)$, for $j \geq d$, are multiples of $\lambda^d(N - L)$. If one works in $1 + K[(P)][[t]]$, the image $\lambda_t(N - L)$ is a polynomial of degree $< d$, and a universal argument shows that its λ^i for $i \geq d$ are equal to 1 (cf. V 4.8). Since in

$$1 + K[[t]]^+[L]/(L^N)$$

one has $\lambda_t(\lambda^d(N - L)) = \lambda(\lambda_t(N - L))$, it follows that the $\lambda^i(\lambda^d(N - L))$, $i \geq 1$, are multiples of P. By the relations V 2.4.2, the ideal generated by P is therefore a λ-ideal.

Put $K_1 = K/(P)$. The natural homomorphism $K \to K_1$ will be denoted g^*; it is a λ-homomorphism. Moreover K_1 is a free K-module with basis

$$1, L, \ldots, L^{d-1}.$$

Indeed, from

$$\lambda^d(N) = \lambda^{d-1}(N-L)L$$

and the invertibility hypothesis on $\lambda^d(N)$, it follows that L is invertible. Thus K_1 admits as a basis over K the elements $1, L^{-1}, \ldots, L^{-d+1}$. Define an application $g_* : K_1 \to K$ by assigning to

$$y' = y_0 + y_1 L + \cdots + y_{d-1}L^{d-1}$$

the element y_{d-1} of K. It is clear that the application so defined is an additive, K-linear homomorphism, and that $g_* g^* = \mathrm{Id}$.

We next equip $K_1 \times K'$ with a λ-ring structure by putting

$$(y_1', x_1) * (y_2', x_2) = (y_1' y_2'(1-L) + x_1 y_2' + x_2 y_1',\ x_1 x_2), \tag{4.7.1}$$

$$\lambda_*^n(y', x) = \sum_{p=1}^{n} \lambda^{n-p}(x)\lambda^p(L, y') + \lambda^n(x). \tag{4.7.2}$$

In these formulas K_1 is regarded as a K'-algebra by the homomorphism

$$K' \xrightarrow{i^*} K \xrightarrow{g^*} K_1.$$

Next, for $y_1, y_2 \in K$ and $n \geq 1$, put

$$y_1 *_y y_2 = -\, y_1 y_2 \lambda_{-1}(N), \tag{4.7.3}$$

$$\lambda_*^n(y_1) = -\, \lambda^n(N, -y_1), \tag{4.7.4}$$

and define a homomorphism

$$u : K \longrightarrow K_1 \times K'$$

by

$$u(y) = \big(g^*(y)\lambda_{-1}(F), -i_*(y)\big), \qquad F = N - L.$$

Proposition 4.7.5. The operations defined by (4.7.1) and (4.7.2) furnish $K_1 \times K'$ with a λ-ring structure, and u is an injective homomorphism which identifies K, equipped with the operations defined by (4.7.3) and (4.7.4), with a λ-ideal of $K_1 \times K'$.

It is immediate to verify that $K_1 \times K'$ is a λ-ring; for example, one may refer to V 5.5, where an entirely analogous construction is made. The verification of the second part of the proposition is also straightforward from the preceding relations, the relation

$$i_*(y_1 y_2 \lambda_{-1}(N)) = i_*(y_1)\, i_*(y_2),$$

which follows immediately from the hypotheses, the relation $g_*(\lambda_{-1}(F)) = 1$ (VI 5.9), and the relations of V 5.

Put

$$K_1' = (K_1 \times K')/K,$$

and let $v : K_1 \times K' \to K_1'$ be the canonical homomorphism. Regard K_1' as a λ-ring via the quotient structure. Define additive homomorphisms

$$j_* : K_1 \to K_1', \quad j^* : K_1' \to K_1, \quad f_* : K_1' \to K', \quad f^* : K' \to K_1'.$$

The homomorphism j_* is the composite $K_1 \to K_1 \times K' \to K'_1$. Similarly, f^* is the composite $K' \to K_1 \times K' \to K'_1$. Thus

$$v = j_* + f^*.$$

Define next a homomorphism $K_1 \times K' \to K_1$ by assigning to (y', x) the element

$$y'(1 - L) + g^* i^*(x).$$

This homomorphism vanishes on K, and hence defines a homomorphism

$$j^* : K'_1 \longrightarrow K_1.$$

Finally, define a homomorphism $K_1 \times K' \to K'$ by assigning to (y', x) the element

$$i_* g_*(y') + x.$$

This homomorphism vanishes on K, and hence defines a homomorphism

$$f_* : K'_1 \longrightarrow K'.$$

Proposition 4.7.6.

i) j^* and f^* are λ-homomorphisms;

ii) j_* is K'_1-linear and f_* is K'-linear;

iii) $j^* j_*$ is the homomorphism "multiplication by $1 - L$";

iv) for every $y' \in K_1$ and every $p \geq 1$,

$$j_*(\lambda^p(L, y')) = \lambda^p(j_*(y'));$$

v) $f_* f^* = \mathrm{Id}$.

It suffices to let the machine run.

If one returns to the geometric situation recalled above, one sees that $K^\bullet(X')$ is constructed from $K^\bullet(Y)$, $K^\bullet(X)$, the maps i^* and i_*, and the element $N \in K^\bullet(Y)$, by the preceding method. However, the λ-operations on $K^\bullet(X')$ agree with the λ-operations defined above on K'_1 only up to torsion, so long as relation (4.3.1) has been proved only modulo torsion.

Theorem 4.8. Let $i : Y \to X$ be a regular closed immersion of constant codimension d, let X' be the scheme obtained by blowing up Y, and let $Y' = X' \times_X Y$, so that one has the commutative diagram

$$\begin{tikzcd} Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\ Y \arrow[r, "i"'] & X. \end{tikzcd}$$

Assume that X is quasi-compact and has an ample sheaf. Then, for every k, there is an exact sequence

$$0 \longrightarrow \mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \xrightarrow{u^{\mathrm{gr}}} \mathrm{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \times \mathrm{Gr}^k K^\bullet(X)_\mathbb{Q} \xrightarrow{v^{\mathrm{gr}}} \mathrm{Gr}^k K^\bullet(X')_\mathbb{Q},$$

where, for $y \in \mathrm{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q}$,

$$u^{\mathrm{gr}}(y) = (g^{\mathrm{gr}}(y)(-1)^{d-1} c^{d-1}(F), -\, i_{\mathrm{gr}}(y)),$$

and, for $y' \in \mathrm{Gr}^{k-1} K^\bullet(Y')_{\mathbb{Q}}$ and $x \in \mathrm{Gr}^k K^\bullet(X)_{\mathbb{Q}}$,

$$v^{\mathrm{gr}}(y', x) = j_{\mathrm{gr}}(y') + f^{\mathrm{gr}}(x).$$

The notation is that of 3.1 and 3.3, and $c^{d-1}(F)$ denotes the $(d-1)$-st Chern class of F (V 6.7). Moreover, u^{gr} has a left inverse $u'^{\,\mathrm{gr}}$, defined by

$$u'^{\,\mathrm{gr}}(y', x) = g_{\mathrm{gr}}(y').$$

If, in addition, X is noetherian and Y regular, one has the exact sequence

$$0 \longrightarrow \mathrm{Gr}^{k-d} K^\bullet(Y)_{\mathbb{Q}} \xrightarrow{u^{\mathrm{gr}}} \mathrm{Gr}^{k-1} K^\bullet(Y')_{\mathbb{Q}} \times \mathrm{Gr}^k K^\bullet(X)_{\mathbb{Q}} \xrightarrow{v^{\mathrm{gr}}} \mathrm{Gr}^k K^\bullet(X')_{\mathbb{Q}} \longrightarrow 0.$$

Proof. Notice that one could give, in the case where Y is regular, a purely algebraic proof in terms of the theory of λ-rings, using the description 4.7 of the situation. We shall instead compute it from the proof of 3.7.

We know that the inverse-image homomorphisms f^*, g^* are λ-homomorphisms, are compatible with the filtrations of the λ-rings, and therefore define homomorphisms on the associated gradeds:

$$f^{\mathrm{gr}} : \mathrm{Gr}^k K^\bullet(X) \to \mathrm{Gr}^k K^\bullet(X'), \qquad g^{\mathrm{gr}} : \mathrm{Gr}^{k-d} K^\bullet(Y) \to \mathrm{Gr}^{k-d} K^\bullet(Y'),$$

and similarly after tensoring with $\mathbb{Q}$. Moreover, 4.6 shows that, after tensoring with $\mathbb{Q}$, the homomorphisms i_*, j_* pass to the associated gradeds, giving

$$i_{\mathrm{gr}} : \mathrm{Gr}^{k-d} K^\bullet(Y)_{\mathbb{Q}} \to \mathrm{Gr}^k K^\bullet(X)_{\mathbb{Q}}, \qquad j_{\mathrm{gr}} : \mathrm{Gr}^{k-1} K^\bullet(Y')_{\mathbb{Q}} \to \mathrm{Gr}^k K^\bullet(X')_{\mathbb{Q}}.$$

Finally, since

$$\lambda_{-1}(F) = (-1)^{d-1}\gamma^{d-1}(F),$$

the morphisms u^{gr} and v^{gr} are obtained by passage to the quotient from the morphisms u and v of 3.7. Thus $v^{\mathrm{gr}}u^{\mathrm{gr}} = 0$.

By VI 5.8,

$$g_*(\mathrm{Fil}^{k-1} K^\bullet(Y')) \subset \mathrm{Fil}^{k-d} K^\bullet(Y)$$

for every k, since $Y' = \mathbf{P}(N)$ and N has rank d. We therefore obtain a homomorphism

$$g_{\mathrm{gr}} : \mathrm{Gr}^{k-1} K^\bullet(Y') \to \mathrm{Gr}^{k-d} K^\bullet(Y),$$

and an analogous homomorphism after tensoring with $\mathbb{Q}$. The homomorphism $u'^{\,\mathrm{gr}}$ is obtained from u' (3.7) by passage to the associated graded. Since u' is a left inverse of u, $u'^{\,\mathrm{gr}}$ is a left inverse of u^{gr}, proving at the same time the injectivity of u^{gr}.

To show that $\mathrm{Ker}(v^{\mathrm{gr}}) = \Im(u^{\mathrm{gr}})$, suppose that

$$y' \in \mathrm{Fil}^{k-1} K^\bullet(Y')_{\mathbb{Q}}, \qquad x \in \mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}$$

are such that

$$j_*(y') + f^*(x) \in \mathrm{Fil}^{k+1} K^\bullet(X')_{\mathbb{Q}}. \tag{4.6.1}$$

We must show that there exists $y \in \mathrm{Fil}^{k-d} K^\bullet(Y)_{\mathbb{Q}}$ such that

$$\begin{aligned} &\text{i)} \quad x + i_*(y) \in \mathrm{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}}, \\ &\text{ii)} \quad y' - g^*(y)\lambda_{-1}(F) \in \mathrm{Fil}^k K^\bullet(Y')_{\mathbb{Q}}. \end{aligned}$$

Put

$$y = g_*(y').$$

By VI 5.8, $y \in \mathrm{Fil}^{k-d} K^\bullet(Y)$. Since f_* preserves the filtration (using results of Exposé VIII, which is inconvenient here, for 4.8 is not used to prove this fact), it follows from 1.8 ii) that f is a complete-intersection morphism of virtual relative dimension 0 (VIII 4.2). Since f is projective, the result follows from VIII 3.2. Applying f_* to (4.6.1), we get

$$f_*(j_*(y')) + f_* f^*(x) \in \mathrm{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}}.$$

Thus, by 3.6,

$$i_*(y) + x \in \mathrm{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}}.$$

To prove (ii), one must show that

$$y' - g^*(g_* y')\lambda_{-1}(F) \in \mathrm{Fil}^{k} K^\bullet(Y')_{\mathbb{Q}}.$$

Put

$$z = y' - g^*(g_* y')\lambda_{-1}(F).$$

Then

$$\begin{aligned} j_*(z) &= j_*(y') - j_*\big(g^*(g_* y')\lambda_{-1}(F)\big) \\ &= j_*(y') - f^*(i_* g_* y') \qquad \text{by 3.4.} \end{aligned}$$

But relation (i) gives

$$f^*(i_* g_* y') + f^*(x) \in \mathrm{Fil}^{k+1} K^\bullet(X')_{\mathbb{Q}}.$$

Together with (4.6.1), this gives

$$j_*(z) \in \mathrm{Fil}^{k+1} K^\bullet(X')_{\mathbb{Q}}.$$

Consequently

$$j^*(j_*(z)) = z(1-L) \in \mathrm{Fil}^{k+1} K^\bullet(Y')_{\mathbb{Q}}.$$

Let z' be the image of z in $\mathrm{Gr}^{k-1} K^\bullet(Y')_{\mathbb{Q}}$, and let $(1-L)$ denote the class of $1-L$ in $\mathrm{Gr}^1 K^\bullet(Y')_{\mathbb{Q}}$. We know that $z'(1-L) = 0$. We may then apply VI 5.11 (which remains valid after tensoring with $\mathbb{Q}$, since its proof uses only the structure of $K^\bullet(Y')$ as a free $K^\bullet(Y)$-algebra) and find $z'' \in K^\bullet(Y)_{\mathbb{Q}}$ such that

$$z = g^*(z'')\lambda_{-1}(F) \in \mathrm{Fil}^{k} K^\bullet(Y')_{\mathbb{Q}}.$$

Applying g_*, one obtains, since $g_*(z) = 0$,

$$z'' \in \mathrm{Fil}^{k+1-d} K^\bullet(Y)_{\mathbb{Q}},$$

and hence, since $\lambda_{-1}(F) \in \mathrm{Fil}^{d-1} K^\bullet(Y')_{\mathbb{Q}}$,

$$z \in \mathrm{Fil}^{k} K^\bullet(Y')_{\mathbb{Q}},$$

which finishes the proof.

Assume now that X is noetherian and Y regular. Then v is surjective (3.7). We saw in the proof of 3.7 that for every $x' \in K^\bullet(X')$ there exists $y' \in K^\bullet(Y')$ such that

$$x' = f^*(f_* x') + j_*(y').$$

The same property remains true after tensoring with $\mathbb{Q}$. Hence, if $x' \in \mathrm{Fil}^k K^\bullet(X')_{\mathbb{Q}}$, then there exist $(y', x) \in K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}$ such that

$$x' = v(y', x), \qquad x \in \mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

Indeed it suffices to take $x = f_*(x')$.

Filter $K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}$, as in 4.7, by putting

$$\operatorname{Fil}^k(K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}) = \operatorname{Fil}^{k-1} K^\bullet(Y')_{\mathbb{Q}} \times \operatorname{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

It follows immediately from 4.7.1 that this is a ring filtration. To show that for every k the homomorphism

$$\operatorname{Fil}^1(K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}) \longrightarrow \operatorname{Fil}^1 K^\bullet(X')_{\mathbb{Q}}$$

is surjective, it is enough to show that, if $x' = y(x'_1)$ with $\epsilon(x'_1) = 0$, there is $(y', x) \in \operatorname{Fil}^1(K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}})$ such that $x' = v(y', x)$.

By what precedes, there exists $(y'_1, x_1) \in K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}$ such that $x'_1 = v(y'_1, x_1)$ and $x_1 \in \operatorname{Fil}^1 K^\bullet(X)_{\mathbb{Q}}$. By 4.7,

$$x' = v(\gamma_*^k(y'_1, x_1)).$$

But

$$\gamma_*^k(y'_1, x_1) = \sum_{p=1}^{k} \gamma^{k-p}(x_1)\gamma^p(L, y'_1),$$

which is an immediate consequence of (4.7.2). Since x_1 has zero augmentation, $\gamma^k(x_1) \in \operatorname{Fil}^1 K^\bullet(X)_{\mathbb{Q}}$, and by V 5.7

$$\sum_{p=1}^{k} \gamma^{k-p}(x_1)\gamma^p(L, y'_1) \in \operatorname{Fil}^{k-1} K^\bullet(Y')_{\mathbb{Q}}.$$

This proves the assertion.

Remark 4.9. In the case where X is noetherian and Y regular, one can, proceeding as in 4.7, describe the structure of graded ring of $\operatorname{Gr}^* K^\bullet(X')_{\mathbb{Q}}$ in terms of the graded rings $\operatorname{Gr}^* K^\bullet(X)_{\mathbb{Q}}$ and $\operatorname{Gr}^* K^\bullet(Y)_{\mathbb{Q}}$, of the additive homomorphisms i_{gr} and i^{gr} induced by i^* and i_*, and of the element $c(N) \in \operatorname{Gr}^* K^\bullet(Y)$. Similarly, one can describe, from the same data, the homomorphisms $f^{\mathrm{gr}}, f_{\mathrm{gr}}, j^{\mathrm{gr}}, j_{\mathrm{gr}}$ defined from f^*, f_*, j^*, j_* by passage to the associated gradeds.

4.10

Let X be a regular scheme having an ample sheaf. We know (IV) that the Grothendieck groups formed with locally free finite-type $\mathcal{O}_X$-modules, and with coherent $\mathcal{O}_X$-modules, are isomorphic. We shall use this fact to define on $K^\bullet(X)$ a "topological" filtration: take $\operatorname{Fil}^i_{\mathrm{top}} K^\bullet(X)$ to be the subgroup generated by the classes of coherent sheaves whose support has codimension at least i. One shows easily that $\operatorname{Fil}^i_{\mathrm{top}} K^\bullet(X)$ is generated by the classes of the sheaves $\mathcal{O}_Z$, where Z ranges over the closed integral subschemes of X of codimension $\geq i$.

Proposition 4.11. Let X be a regular scheme having an ample sheaf. Then, for every $k \geq 0$,

$$\operatorname{Fil}^k_{\mathrm{top}} K^\bullet(X)_{\mathbb{Q}} \subset \operatorname{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

The reverse inclusion, valid without tensoring by $\mathbb{Q}$, will be proved in X.

By 4.10, it is enough to show that, if Z is an integral closed subscheme of X, of codimension k, then

$$\operatorname{cl}(\mathcal{O}_Z) \in \operatorname{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

To this end we proceed by noetherian induction on the integral closed subschemes of X. Suppose the property already proved for the proper integral closed subschemes of Z, and let us prove it for Z. Let J be the ideal of $\mathcal{O}_X$ defining Z, and let z be the generic point of Z. Since Z is integral, J_z is the maximal ideal of $\mathcal{O}_{X,z}$; and since X is regular, J_z can be generated by an

$\mathcal{O}_{X,z}$-regular sequence. One can therefore find a sequence of sections of J, in a neighbourhood of z, forming an $\mathcal{O}_X$-regular sequence. Consequently there is an open neighbourhood U of z, such that $Z' = X - U$ is contained in Z, and such that $J\mathcal{O}_U$ is a regular ideal of $\mathcal{O}_U$. The immersion $Z \cap U \to U$ is then a regular immersion of codimension

$$\dim(\mathcal{O}_{X,z}) = \operatorname{codim}(Z, X)$$

(EGA IV 5.1.2). Moreover, one has the exact sequence (IV)

$$K^\bullet(Z') \xrightarrow{j_*} K^\bullet(X) \xrightarrow{u^*} K^\bullet(U) \to 0,$$

where j is the immersion $Z' \to X$ and u is the immersion $U \to X$. After tensoring with $\mathbb{Q}$, one obtains

$$\operatorname{cl}(\mathcal{O}_Z) \equiv i_*(1_{\mathcal{O}_{Z\cap U}}),$$

and by 4.6 the image under u^* of the class of $\mathcal{O}_Z$ belongs to $\operatorname{Fil}^k K^\bullet(U)_\mathbb{Q}$. Since u^* is a surjective λ-homomorphism, it induces a surjection

$$\operatorname{Fil}^k K^\bullet(X) \longrightarrow \operatorname{Fil}^k K^\bullet(U).$$

Thus $u^*(1_Z) = u^*(x)$, with $x \in \operatorname{Fil}^k K^\bullet(X)_\mathbb{Q}$. Therefore there exists

$$y \in j_*(K^\bullet(Z'))$$

such that

$$\operatorname{cl}(\mathcal{O}_Z) - y \in \operatorname{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

But $Z' \subset Z$, and since $z \notin Z'$, $\operatorname{codim}(Z', X) > k$. Thus y can be expressed as a combination of classes of sheaves $\mathcal{O}_{Z''_\alpha}$, where the Z''_α are integral closed subschemes of Z, different from Z. By the induction hypothesis, $y \in \operatorname{Fil}^k K^\bullet(X)_\mathbb{Q}$, and this completes the proof.

Bibliography

[1] Cartan–Eilenberg, *Homological Algebra.*

Exposé VIII. The Riemann–Roch theorem

by Pierre Berthelot

1. Complete-intersection morphisms

Definition 1.1. Let $f : X \to Y$ be a morphism of schemes. One says that f is a *complete-intersection morphism* if, for every point x of X, there exist an open neighbourhood U of x and a smooth Y-scheme V such that the restriction of f to U factors as

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & {\scriptstyle f}\searrow & \downarrow{\scriptstyle g} \\ & & Y, \end{array}$$

where i is a regular immersion (VII 1.4). After replacing V by an open subset, one may suppose that i is a regular closed immersion.

In this definition one may replace V by a vector bundle $Y[T_1, \dots, T_n]$. Indeed, since the notion is local, one may suppose that V is affine and that its image in Y is contained in an affine open W of Y. There then exists an integer n and a closed W-immersion

$$V \longrightarrow W[T_1, \dots, T_n],$$

hence a Y-immersion $V \to Y[T_1, \dots, T_n]$, which is regular by VII 1.10. Since the composite of two regular immersions is regular, one obtains a regular Y-immersion

$$U \longrightarrow Y[T_1, \dots, T_n]$$

factorizing f.

Proposition 1.2. Let $f : X \to Y$ be a complete-intersection morphism. Then every Y-immersion $i : X \to Z$ of X in a smooth Y-scheme is regular.

We first prove the following corollary.

Corollary 1.3. Let

$$\begin{tikzcd} & V' \arrow[d, "p"] \\ X \arrow[ur, "j"] \arrow[r, "i"'] & V \end{tikzcd}$$

be a commutative diagram in which i and j are immersions, and p is a smooth morphism. Then i is a regular immersion if and only if j is a regular immersion.

(i) Suppose i regular. Put $X' = V' \times_V X$. The datum of j defines a section j' of X' over X, such that $j = i'j'$:

$$\begin{tikzcd} X' \arrow[r, "i'"] \arrow[d, "p'"'] & V' \arrow[d, "p"] \\ X \arrow[ur, "j"] \arrow[r, "i"'] & V. \end{tikzcd}$$

Now j' is a regular immersion, being a section of a smooth morphism (VII 1.10); i' is a regular immersion because it is obtained from i by the smooth, hence flat, base change (VII 1.5). Therefore j is a regular immersion (VII 1.7).

(ii) Suppose j regular and p étale. Then j' is an open immersion (EGA IV 17.9.3), hence an isomorphism of X onto $j'(X)$; put $U = j'(X)$. Let i'' be the restriction of i' to U, and let U' be an open subset of V' such that $U = U' \cap X'$. Then

$$i'' = j \circ (j')^{-1},$$

so that i'' is a regular immersion of U into U'. But the morphism $U' \to V$, being smooth, is flat and locally of finite presentation, hence open. Consequently it is faithfully flat and quasi-compact from U' onto an open neighbourhood of X in V. It follows from VII 1.5 that i is a regular immersion.

(iii) Suppose j regular, and suppose that V' is a vector bundle over V. Since the property to be proved is local on V, one may suppose V affine, i closed, and $V' = V[T_1, \dots, T_n]$. Put $V = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$. One has the commutative diagram

$$\begin{tikzcd} & A[T_1, \dots, T_n] \arrow[d, "v"] \\ A \arrow[ur, "u"] \arrow[r] & B, \end{tikzcd}$$

where the morphisms u and v are surjective. Choose elements $t_i \in A$, $1 \leq i \leq n$, such that $u(t_i) = v(T_i)$ for every i. Define a homomorphism

$$w : A[T_1, \dots, T_n] \longrightarrow A, \qquad w(T_i) = t_i.$$

This gives a closed immersion $s : V \to V'$ which is a section of p, and such that $j = s \circ i$. Moreover, being a section of a smooth morphism, s is a regular immersion (VII 1.10). Since s and $s \circ i$ are regular, it will follow that i is regular if one proves that the sequence

$$0 \longrightarrow i^*(K/K^2) \longrightarrow J/J^2 \longrightarrow I/I^2 \longrightarrow 0$$

is exact and locally split; here I, J, K denote respectively the ideals defining i, j, s (VII 1.7). It is enough to show that $i^*(K/K^2)$ is a direct factor of J/J^2. But j' is obtained from s by the base change $i' : X' \to V'$. Moreover j' and s are regular immersions (VII 1.10). Thus, by VII 1.2, the ideal defining j' is $i'^*(K)$, and the conormal sheaf of j' is $i'^*(K/K^2)$. The exact sequence of conormal sheaves attached to the decomposition $j = i'j'$ therefore gives in particular a homomorphism

$$J/J^2 \longrightarrow i^*(K/K^2).$$

It is immediate to verify that the composite

$$i^*(K/K^2) \longrightarrow J/J^2 \longrightarrow i^*(K/K^2)$$

is the identity. In affine terms, this corresponds to the decomposition of J as the direct sum of an ideal of A and the ideal generated by the $T_i - t_i$. This gives the required splitting.

(iv) Finally suppose j regular and p smooth. For every $x \in X$, one can find an open neighbourhood W of $i(x)$, an open neighbourhood W' of $j(x)$ lying over W, and a commutative diagram

$$\begin{tikzcd} W' \arrow[r, "q"] \arrow[d, "p"'] & W[T_1, \dots, T_n] \arrow[d, "p_0"] \\ W \arrow[r, equal] & W, \end{tikzcd}$$

where p_0 is the structural morphism, and q is an étalemorphism (EGA IV 17.11.4). One may then suppose $V = W$ and $V' = W'$. It is easy to see that qj is an immersion, by introducing the fiber product $X[T_1, \dots, T_n]$. Applying (ii) and (iii), one deduces that qj is regular, and then that i is regular.

We now prove Proposition 1.2. Since regularity is local on Z, we may work in a neighbourhood of a point $x \in X$. There is then an open neighbourhood U of x in X, a smooth Y-scheme V, and a regular Y-immersion $i' : U \to V$. We have a commutative diagram

$$\begin{tikzcd} U \arrow[r, "i'"] \arrow[d, "i"'] & V \arrow[d] \\ Z \arrow[r] & Y, \end{tikzcd}$$

where the unlabeled vertical arrows are the structural morphisms. From it one deduces a closed immersion

$$i'' : U \longrightarrow Z \times_Y V,$$

and a commutative diagram

$$\begin{tikzcd} & U \arrow[dl, "i"'] \arrow[dr, "i'"] & \\ Z & Z \times_Y V \arrow[l, "p"'] \arrow[r, "p'"] & V. \end{tikzcd}$$

Since i' is regular and p, p' are smooth, the corollary implies that i'' and i are regular.

The following proposition will not be used in the sequel of the Seminar.

Proposition 1.4. Let $f : X \to Y$ be a flat morphism locally of finite presentation. For f to be a complete-intersection morphism in the sense of 1.1, it is necessary and sufficient that it be a complete-intersection morphism in the sense of EGA IV 19.3.6.

Suppose that f is a complete-intersection morphism in the sense of 1.1, and let $x \in X$. Then there are an open neighbourhood U of x and a factorization

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[dr, "f"'] & V \arrow[d, "g"] \\ & Y, \end{tikzcd}$$

where i is a regular immersion and g is smooth. Let $y = f(x)$. Then the immersion $i_y : U_y \to V_y$ is regular at x. Indeed, after possibly restricting V, one may suppose that there is a finite-type free $\mathcal{O}_V$-module E and a regular homomorphism $u : E \to \mathcal{O}_V$ such that the Koszul complex associated to u is a resolution of $\mathcal{O}_U$. Let u_y be the homomorphism induced on the fiber over y, and let $K_\bullet(u_y) = K_\bullet(u)_y$ be the associated Koszul complex. Since g is smooth, hence flat, $K_\bullet(u)$ is a Y-flat resolution of $\mathcal{O}_U$. Consequently the homology groups of $K_\bullet(u)_y$ are the groups

$$\mathrm{Tor}_i^{\mathcal{O}_Y}(\mathcal{O}_U, k(y)),$$

where $k(y)$ is the residue field of $\mathcal{O}_{Y,y}$. Since f is flat, these groups are zero; hence i_y is regular at x. Since g is smooth, the local ring of V_y at x is regular. The local ring of U_y at y is therefore the quotient of a regular ring by a regular ideal, hence a complete-intersection ring (EGA IV 19.3.2). It follows that f is a complete intersection at x in the sense of EGA IV 19.3.6.

Conversely, suppose that f is a complete-intersection morphism in the sense of EGA IV 19.3.6, and let $x \in X$. Let U be an affine neighbourhood of x lying over an affine neighbourhood U' of $y = f(x)$. One can then find a smooth U'-scheme V, and a closed immersion $U \to V$ (for instance by taking for V a vector bundle over U'). We shall show that i is regular. By EGA IV 19.3.7, i is transversally regular at x, since V is smooth over Y, hence V_y is regular at x, and U is flat over Y. It follows from EGA IV 19.2.4, a)3c), that i is regular at x. Therefore f is a complete-intersection morphism in the sense of 1.1.

Proposition 1.5. Let $f : X \to Y$ and $g : Y \to Z$ be two locally complete-intersection morphisms. Then gf is locally complete-intersection.

Let $x \in X$. One can find open neighbourhoods U, V of $x, f(x)$ and regular immersions i, j such that one has factorizations

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[dr, "f"'] & V[T_1, \dots, T_n] \arrow[d] \\ & V \end{tikzcd} \qquad \begin{tikzcd} V \arrow[r, "j"] \arrow[dr, "g"'] & Z[T'_1, \dots, T'_m] \arrow[d] \\ & Z. \end{tikzcd}$$

This diagram can be completed by the commutative square

$$\begin{tikzcd} V[T_1, \dots, T_n] \arrow[r, "j'"] \arrow[d] & Z[T'_1, \dots, T'_m, T_1, \dots, T_n] \arrow[d] \\ V \arrow[r, "j"'] & Z[T'_1, \dots, T'_m], \end{tikzcd}$$

where j' is obtained from j by base change. This base change is flat; hence j' is a regular immersion, and so $j'i$ is a regular immersion of U in a scheme smooth over W, where W is the relevant open of Z. This proves the assertion.

Proposition 1.6. Let $f : X \to Y$ be a morphism and let $g : Y' \to Y$ be a flat base change. If f is complete-intersection, then the morphism

$$f' : X \times_Y Y' \longrightarrow Y'$$

deduced from f is complete-intersection. The converse is true if, moreover, g is quasi-compact and surjective.

One can find an open covering of X by open subsets U_i such that, for every i, the restriction of f to U_i factors as

$$U_i \longrightarrow V_i \longrightarrow Y,$$

where $U_i \to V_i$ is a regular immersion and $V_i \to Y$ is smooth. The $U_i \times_Y Y'$ form an open covering of $X \times_Y Y'$, the immersions

$$U_i \times_Y Y' \longrightarrow V_i \times_Y Y'$$

are regular by VII 1.5, and the morphisms $V_i \times_Y Y' \to Y'$ are smooth, giving the first assertion. For the converse, EGA IV 2.7.1 (iv) first implies that f is locally of finite presentation; this reduces us easily, using 1.2, to the case where f is a closed immersion. In that case the assertion is just VII 1.5.

Proposition 1.7. Let $f : X \to Y$ be a complete-intersection morphism. Then f is a perfect morphism.

Since the property is local on X, one may work in a neighbourhood of a point $x \in X$ and suppose that there is an open subset U containing x and a factorization

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[dr, "f"'] & V \arrow[d, "g"] \\ & Y, \end{tikzcd}$$

where i is a regular immersion and g is smooth. Since g is smooth, it is a perfect morphism (III 4.1). By VII 1.9, i is also a perfect morphism. Since the composite of two perfect morphisms is perfect (III 4.5), the proposition follows.

Proposition 1.8. Let $f : X \to Y$ be a complete-intersection morphism, let x be a point of X, and let

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[dr, "f"'] & V \arrow[d, "g"] \\ & Y \end{tikzcd}$$

be a factorization in which U is an open neighbourhood of x, i is a regular immersion, and g is smooth. Let J be the ideal of $\mathcal{O}_V$ defining U. Then the integer

$$d_x = \operatorname{rank}\big((\Omega^1_{V/Y})_{i(x)}\big) - \operatorname{rank}\big((J/J^2)_x\big)$$

is independent of the chosen factorization of f.

Suppose that the restriction of f to U factors through two smooth Y-schemes V and V', and put $V'' = V \times_Y V'$. Then V'' is smooth over Y, and also over V and V'. Moreover, the immersions

$X \to V$ and $X \to V'$ define an immersion $X \to V''$, which is regular by 1.2. It is therefore enough to compare the integers d_x attached to two factorizations through smooth Y-schemes V and V' such that there exists a commutative diagram over Y

$$\begin{tikzcd} & V' \arrow[d, "p"] \\ X \arrow[ur, "j"] \arrow[r, "i"'] & V, \end{tikzcd}$$

where p is smooth. We may also suppose that $\Omega^1_{V/Y}$, $\Omega^1_{V'/Y}$, and $\Omega^1_{V'/V}$ have constant rank, as do the conormal sheaves of i and j.

From the commutative diagram, all morphisms being smooth,

$$\begin{tikzcd} V' \arrow[r, "p"] \arrow[dr] & V \arrow[d] \\ & Y, \end{tikzcd}$$

one deduces

$$\operatorname{rank}(\Omega^1_{V'/Y}) = \operatorname{rank}(\Omega^1_{V/Y}) + \operatorname{rank}(\Omega^1_{V'/V}). \tag{1.8.1}$$

Let $X' = V' \times_V X$; then the rank of $\Omega^1_{V'/V}$ is equal to that of $\Omega^1_{X'/X}$. Further, j defines a section $j' : X \to X'$, which is a regular immersion of codimension equal to the rank of $\Omega^1_{X'/X}$ (EGA IV 17.10.4). If i' is the immersion obtained from i by the base change $V' \to V$, then $j = i'j'$, and consequently (VII 1.7)

$$\operatorname{rank}(N_{V'/X}) = \operatorname{rank}(N_{V/X}) + \operatorname{rank}(\Omega^1_{V'/V}). \tag{1.8.2}$$

Comparing (1.8.1) and (1.8.2) gives the proposition.

Definition 1.9. The integer d_x of Proposition 1.8 is called the *relative virtual dimension* of X over Y (or of f) at the point x.

The relative virtual dimension of X over Y is therefore a locally constant function on X, with integral values, positive or negative. If f is a smooth morphism of relative dimension n, its relative virtual dimension is n. If f is a regular immersion of codimension n, its relative virtual dimension is $-n$.

Proposition 1.10. Let $f : X \to Y$ and $g : Y \to Z$ be two complete-intersection morphisms, and let $x \in X$. Then

$$\operatorname{dim.rel.virt.}_x(gf) = \operatorname{dim.rel.virt.}_x(f) + \operatorname{dim.rel.virt.}_{f(x)}(g).$$

This follows immediately from 1.9 and from the proof of 1.5.

2. The relative cotangent complex

2.1. We give here a definition of the cotangent complex which, without having the greatest possible generality (see Exposé 0, 4.4), will be sufficient for the needs of the Riemann–Roch theorem. Of course the definition given in Exposé 0 coincides, in the case considered there, with the definition given here.

Let $f : X \to Y$ be a smoothable morphism, that is, a morphism admitting a factorization

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr, "f"'] & V \arrow[d, "g"] \\ & Y, \end{tikzcd}$$

where g is smooth and i is an immersion, which may be supposed closed. Let J be the ideal of X in V. The canonical derivation

$$d : \mathcal{O}_V \longrightarrow \Omega^1_{V/Y}$$

defines a homomorphism

$$J \longrightarrow \mathcal{O}_V \xrightarrow{d} \Omega^1_{V/Y} \longrightarrow \Omega^1_{V/Y} \otimes_{\mathcal{O}_V} \mathcal{O}_X$$

which vanishes on J^2, and hence defines a homomorphism, still denoted d,

$$J/J^2 \xrightarrow{d} i^*(\Omega^1_{V/Y}). \tag{2.1.1}$$

One checks easily that this homomorphism is $\mathcal{O}_X$-linear. One then defines a chain complex $L^{X/Y}_\bullet$ by putting

$$L^{X/Y}_0 = i^*(\Omega^1_{V/Y}), \qquad L^{X/Y}_1 = J/J^2, \qquad L^{X/Y}_i = 0 \quad (i \neq 0, 1),$$

and taking as homomorphism $L^{X/Y}_1 \to L^{X/Y}_0$ the homomorphism (2.1.1).

Proposition 2.2. The complex $L^{X/Y}_\bullet$ is independent, up to canonical isomorphism in the derived category $D(X)$, of the factorization chosen through a smooth Y-scheme.

Let V, V' be two smooth Y-schemes through which f factors. Put $V'' = V \times_Y V'$. Then V'' is smooth over Y, and also over V and V'. Moreover the immersions $X \to V$ and $X \to V'$ define an immersion $X \to V''$ which factors f. It is therefore enough to compare the complexes $L^{X/Y}_\bullet$ attached to two factorizations through smooth Y-schemes V and V' for which there is a commutative diagram over Y

$$\begin{tikzcd} & V' \arrow[d, "p"] \\ X \arrow[ur, "j"] \arrow[r, "i"'] & V, \end{tikzcd}$$

where i and j are closed immersions, and p is smooth. Functoriality then gives homomorphisms

$$I/I^2 \longrightarrow J/J^2, \qquad p^*(\Omega^1_{V/Y}) \longrightarrow \Omega^1_{V'/Y},$$

hence a diagram

$$\begin{tikzcd} I/I^2 \arrow[r] \arrow[d, "d"'] & J/J^2 \arrow[d, "d"] \\ i^*(\Omega^1_{V/Y}) \arrow[r] & j^*(\Omega^1_{V'/Y}). \end{tikzcd} \tag{2.2.1}$$

Since V and V' are smooth over Y, one obtains the exact sequence

$$0 \longrightarrow p^*(\Omega^1_{V/Y}) \longrightarrow \Omega^1_{V'/Y} \longrightarrow \Omega^1_{V'/V} \longrightarrow 0.$$

Since the modules of differentials considered are locally free on $\mathcal{O}_V$, applying the functor j^* gives the exact sequence

$$0 \longrightarrow i^*(\Omega^1_{V/Y}) \longrightarrow j^*(\Omega^1_{V'/Y}) \longrightarrow j^*(\Omega^1_{V'/V}) \longrightarrow 0. \tag{2.2.2}$$

Furthermore, j defines a section j' of $X' = V' \times_V X$:

$$\begin{tikzcd} X' \arrow[r, "i'"] \arrow[d, "p'"'] & V' \arrow[d, "p"] \\ X \arrow[ur, "j"] \arrow[r, "i"'] & V. \end{tikzcd}$$

Let I, J, K be the ideals defining i, j, j'. Then $j = i'j'$, and the immersions i', j, j' are regular; moreover i' is defined by the ideal $p^*(I)$, because p is flat, and its conormal sheaf is therefore

$$i'^*(p^*(I)) = p'^*(I/I^2).$$

One obtains the exact sequence

$$0 \longrightarrow I/I^2 \longrightarrow J/J^2 \longrightarrow K/K^2 \longrightarrow 0. \tag{2.2.3}$$

We shall show: (a) the diagram (2.2.1) is commutative; this will give a homomorphism of complexes θ from the complex $L_\bullet^{X/Y}$ defined by V to the complex $L_\bullet^{X/Y}$ defined by V'; (b) the homomorphism

$$K/K^2 \longrightarrow j^*(\Omega^1_{V'/V})$$

defined by (2.2.2), (2.2.3), and (2.2.1) is an isomorphism. This will imply that the mapping cylinder of θ is acyclic, hence that θ is a quasi-isomorphism, and therefore defines an isomorphism in the derived category.

The properties (a) and (b) are local on X. One may therefore suppose X, Y, V, V' affine. Put $Y = \mathrm{Spec}(A)$, $V = \mathrm{Spec}(B)$, $V' = \mathrm{Spec}(B')$, $X = \mathrm{Spec}(C)$, and keep the same letters for the corresponding homomorphisms and ideals. One has the diagram

$$\begin{tikzcd} B \otimes_A B \arrow[r, "p\otimes p"] \arrow[d, "\Delta"'] & B' \otimes_A B' \arrow[d, "\Delta"] & B' \otimes_B B' \arrow[l] \arrow[d] \\ B \arrow[r, "p"] \arrow[dr, "i"'] & B' \arrow[d, "j"] & B' \otimes_B C = B'/IB' \arrow[dl, "p'"] \\ & C & \end{tikzcd}$$

where the left and right displayed squares are denoted (1) and (2), respectively. The homomorphism $I/I^2 \to J/J^2$ is deduced from the homomorphism $IB' \to J$ and coincides with that deduced from $p : I \to J$; the homomorphism

$$i^*(\Omega^1_{V/Y}) \longrightarrow j^*(\Omega^1_{V'/Y})$$

is deduced from the commutative square (1). Hence property (a) is evident.

The homomorphism $K/K^2 \to j^*(\Omega^1_{V'/V})$ considered in (b) may be defined as follows. Let $\overline{x} \in K/K^2$, and let x be an element of J lifting $\overline{x}$ (such an element exists because $K = J/IB'$). To $\overline{x}$ one associates the image of x by the composite

$$J \xrightarrow{d} \Omega^1_{B'/A} \otimes_{B'} C \longrightarrow \Omega^1_{B'/B} \otimes_{B'} C.$$

On the other hand, let H be the augmentation ideal of

$$B' \otimes_B B' \longrightarrow B'.$$

The commutative square (2) gives a homomorphism $H \to K$, hence a homomorphism

$$H/H^2 \otimes_{B'} C \longrightarrow K/K^2.$$

But $H/H^2 = \Omega^1_{B'/B}$. It remains only to check that the homomorphisms thus constructed are inverse to each other, which is straightforward.

Definition 2.3. Let $f : X \to Y$ be a smoothable morphism. The complex $L_\bullet^{X/Y}$, defined up to canonical isomorphism in the derived category $D(X)$, is called the *relative cotangent complex* of X over Y (or of f).

Proposition 2.4. Let $f : X \to Y$ be a smoothable complete-intersection morphism. Then $L_\bullet^{X/Y}$ is perfect. More precisely, for every factorization of f as the composite of an immersion and a smooth morphism, the complex $L_\bullet^{X/Y}$ defined in 2.1 is strictly perfect.

Let $f = gi$, where g is smooth and i is an immersion. By 1.2, i is a regular immersion. If J is the ideal defining i, then J/J^2 is a locally free finite-type $\mathcal{O}_X$-module. Likewise, since $g : V \to Y$ is smooth, $i^*(\Omega^1_{V/Y})$ is locally free of finite type. This proves 2.4.

Corollary 2.5. Let $f : X \to Y$ be a smoothable complete-intersection morphism, and let $f = gi$ be a factorization of f through a smooth Y-scheme V. Let J be the ideal of $\mathcal{O}_V$ defining i. If $K^\bullet(X)$ denotes the Grothendieck group of locally free finite-type $\mathcal{O}_X$-modules (respectively of perfect complexes on X), the element $T_f \in K^\bullet(X)$ defined by

$$T_f = \mathrm{cl}(i^*\Omega^1_{V/Y}) - \mathrm{cl}(J/J^2)$$

respectively

$$T_f = \mathrm{cl}(L_\bullet^{X/Y})$$

does not depend on the chosen factorization of f. The value at a point x of X of the augmentation of T_f is precisely the relative virtual dimension of X over Y at x (1.9).

The element T_f is well defined in both cases by 2.4; its class in $K^\bullet(X)$ is independent of the factorization, by 2.2 in the second case and by the proof of 2.2 in the first. The last assertion is trivial.

Proposition 2.6. Let $f : X \to Y$ and $g : Y \to Z$ be two smoothable complete-intersection morphisms, and suppose in addition that the following condition is satisfied:

(G) There exists a smooth Y-scheme V factorizing f such that, for every closed immersion $Y \to V'$ into a smooth Z-scheme, one can find a smooth V'-scheme V'' with

$$V \simeq V'' \times_{V'} Y.$$

This condition is in particular satisfied if f is quasi-projective and if Y has an ample sheaf, by taking for V some $\mathbb{P}^r_Y$.

Then gf is complete-intersection and smoothable, and there is a distinguished triangle

$$\begin{tikzcd}
& L_\bullet^{X/Y} \arrow[dl] & \\
f^*L_\bullet^{Y/Z} \arrow[rr] & & L_\bullet^{X/Z} \arrow[ul]
\end{tikzcd}$$

in the derived category.

Let V' be a smooth Z-scheme factorizing g. Consider the commutative diagram

$$\begin{tikzcd}
X \arrow[r, "i"] \arrow[dr, "f"'] & V \arrow[r, "j'"] \arrow[d, "p"] & V'' \arrow[d, "p''"] \\
& Y \arrow[r, "j"] \arrow[dr, "g"'] & V' \arrow[d, "p'"] \\
& & Z.
\end{tikzcd}$$

Then the immersion j' is regular, since it is obtained from j by a flat base change. Hence $j'i$ is a regular immersion factorizing gf, which is therefore smoothable and complete-intersection. Moreover, one has the exact sequence of conormal sheaves

$$0 \longrightarrow i^*(J'/J'^2) \longrightarrow K/K^2 \longrightarrow I/I^2 \longrightarrow 0,$$

where I, J, J', K denote the sheaves defining respectively the immersions $i, j, j', j'i$. Since p'' is flat, $J' = p''^*(J)$, and the sequence can be written

$$0 \longrightarrow f^*(J/J^2) \longrightarrow K/K^2 \longrightarrow I/I^2 \longrightarrow 0. \tag{2.6.1}$$

Furthermore, since V' and V'' are smooth over Z, there is an exact sequence

$$0 \longrightarrow p''^*(\Omega^1_{V'/Z}) \longrightarrow \Omega^1_{V''/Z} \longrightarrow \Omega^1_{V''/V'} \longrightarrow 0.$$

Applying $(j'i)^*$, one obtains an exact sequence

$$0 \longrightarrow (p''ji)^*(\Omega^1_{V'/Z}) \longrightarrow (j'i)^*(\Omega^1_{V''/Z}) \longrightarrow (j'i)^*(\Omega^1_{V''/V'}) \longrightarrow 0.$$

Since

$$j'^*(\Omega^1_{V''/V'}) = \Omega^1_{V/Y}$$

by hypothesis (G), one obtains

$$0 \longrightarrow f^*(j^*\Omega^1_{V'/Z}) \longrightarrow (j'i)^*(\Omega^1_{V''/Z}) \longrightarrow i^*(\Omega^1_{V/Y}) \longrightarrow 0. \tag{2.6.2}$$

Combining (2.6.1) and (2.6.2), one is reduced to checking that the diagram

$$\begin{tikzcd}
0 \arrow[r] & f^*(J/J^2) \arrow[r] \arrow[d, "f^*(d)"'] & K/K^2 \arrow[r] \arrow[d, "d"] & I/I^2 \arrow[r] \arrow[d, "d"] & 0 \\
0 \arrow[r] & f^*(j^*\Omega^1_{V'/Z}) \arrow[r] & (j'i)^*\Omega^1_{V''/Z} \arrow[r] & i^*\Omega^1_{V/Y} \arrow[r] & 0
\end{tikzcd}$$

is commutative. This is verified locally without difficulty by returning to the definitions of the homomorphisms involved. The announced distinguished triangle follows from the resulting exact sequence of complexes.

Corollary 2.7. With the notation and hypotheses of 2.6, one has in $K^\bullet(X)$ the equality

$$T_{gf} = T_f + f^*(T_g).$$

3. Riemann–Roch theorem: statement

3.1. In no. 3, Y is a quasi-compact scheme having an ample sheaf.

Let $f : X \to Y$ be a projective complete-intersection morphism (1.1); suppose moreover that the relative virtual dimension of f (1.9) is constant on X. Since f is projective, there is a closed immersion $X \to P$, where P is a projective bundle over Y; since Y has an ample sheaf, one may suppose that $P = \mathbb{P}^r_Y$. Thus there is a factorization

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr, "f"'] & \mathbb{P}^r_Y \arrow[d, "g"] \\ & Y. \end{tikzcd}$$

Since f is complete-intersection, the immersion i is regular (1.2).

By 1.7, f is a perfect morphism. Moreover Y has an ample sheaf, and therefore so do X and P. One can therefore define an additive homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(Y)$$

(IV 2.12), the groups $K^\bullet$ defined in terms of locally free sheaves of finite type, or of perfect complexes, being identical. One can also define additive homomorphisms

$$i_* : K^\bullet(X) \longrightarrow K^\bullet(P), \qquad g_* : K^\bullet(P) \longrightarrow K^\bullet(Y),$$

and one has $f_* = g_* i_*$.

Let d be the relative virtual dimension of X over Y. The codimension of i is then $r - d$. By VII 4.6, the homomorphism i_* gives, after tensoring with $\mathbb{Q}$, inclusions

$$i_*(\operatorname{Fil}^k K^\bullet(X)_{\mathbb{Q}}) \subset \operatorname{Fil}^{k+r-d} K^\bullet(P)_{\mathbb{Q}}.$$

By VI 5.8, the homomorphism g_* gives inclusions

$$g_*(\operatorname{Fil}^{k'} K^\bullet(P)) \subset \operatorname{Fil}^{k'-r} K^\bullet(Y).$$

Composing, one obtains the following.

Proposition 3.2. Let $f : X \to Y$ be a projective complete-intersection morphism whose relative virtual dimension d is constant. Suppose that Y is quasi-compact and has an ample sheaf. Then for every $k \in \mathbb{Z}$,

$$f_*(\operatorname{Fil}^k K^\bullet(X)_{\mathbb{Q}}) \subset \operatorname{Fil}^{k-d} K^\bullet(Y)_{\mathbb{Q}}.$$

Consequently one obtains an additive homomorphism

$$f_*^{\mathrm{gr}} : \operatorname{Gr} K^\bullet(X)_{\mathbb{Q}} \longrightarrow \operatorname{Gr} K^\bullet(Y)_{\mathbb{Q}}$$

sending $\operatorname{Gr}^k K^\bullet(X)_{\mathbb{Q}}$ into $\operatorname{Gr}^{k-d} K^\bullet(Y)_{\mathbb{Q}}$.

3.3. Recall that in V 6.1 we defined a functor

$$A \longmapsto \operatorname{Ch}(A)$$

from the category of graded K-algebras to the category of augmented K-λ-algebras, where K is a binomial ring (V 2.7). If A is a graded K-algebra, then

$$\operatorname{Ch}(A) = K \times (1 + \widehat{A}^+).$$

For every augmented K-λ-algebra Λ we defined a K-λ-homomorphism, called the completed Chern class (V 6.7),

$$\widetilde{c} = (\varepsilon, c) : \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}_\Lambda).$$

Finally, we defined a ring homomorphism, functorial in A, called the Chern character (V 6.4),

$$\mathrm{ch} : \mathrm{Ch}(A) \longrightarrow K \oplus \widehat{A}^+_{\mathbb{Q}},$$

and a group endomorphism, called the Todd character (V 1.16),

$$\mathrm{Todd} : 1 + \widehat{A}^+ \longrightarrow 1 + \widehat{A}^+.$$

We shall apply these definitions to the augmented λ-rings $K^\bullet(X)$ and $K^\bullet(Y)$; the augmentation rings $H^0(X, \mathbb{Z})$ and $H^0(Y, \mathbb{Z})$ are binomial (V 2.8). To simplify notation, for every scheme Z and every $z \in K^\bullet(Z)$ we write

$$\mathrm{ch}(z) = \mathrm{ch}(\widetilde{c}(z)), \qquad \mathrm{Todd}(z) = \mathrm{Todd}(c(z)).$$

For every graded algebra A we regard the group $1 + \widehat{A}^+$ as contained in $\widehat{A}$. Finally, if $f : A \to B$ is additive and compatible with the gradings up to a shift, we shall still write f for the homomorphism $\widehat{A} \to \widehat{B}$ deduced from it.

3.4. Let X be a scheme. For every locally free finite-type $\mathcal{O}_X$-module E, denote by $\check{E}$ the class in $K^\bullet(X)$ of the dual $\check{E}$ of E. It is clear that the function $E \mapsto \check{E}$ is additive on the category of locally free finite-type $\mathcal{O}_X$-modules, and therefore defines an endomorphism of $K^\bullet(X)$, denoted $x \mapsto \check{x}$.

Proposition 3.5. Let X be a quasi-compact scheme. Then, for all $x \in K^\bullet(X)$ and all $i \geq 0$, one has

$$c^i(\check{x}) = (-1)^i c^i(x).$$

Equivalently,

$$c_t(\check{x}) = c_{-t}(x), \qquad c_t(x) = \sum_{i \geq 0} c^i(x) t^i.$$

By additivity one may suppose that x is the class of a locally free finite-type $\mathcal{O}_X$-module E. Moreover, after replacing X by the flag scheme of E, which gives an injection on the groups $\mathrm{Gr}\, K^\bullet$ (VI 5.5), one may suppose that E is an invertible sheaf. Then $\check{E} = E^{-1}$, so $\check{x} = x^{-1}$.

The $c^i(\check{x})$ and $c^j(x)$ are zero for $i, j \geq 2$. It is enough to prove

$$c^1(x^{-1}) = -c^1(x).$$

Now $c^1(x)$ is the class in $\mathrm{Gr}^1 K^\bullet(X)$ of $x - 1$, while $c^1(x^{-1})$ is that of $x^{-1} - 1$. The equality

$$x^{-1} - 1 = 1 - x + (x - 1)(1 - x^{-1})$$

gives

$$x^{-1} - 1 = 1 - x \pmod{\mathrm{Fil}^2 K^\bullet(X)},$$

and the result follows.

Theorem 3.6 (Riemann–Roch). Let $f : X \to Y$ be a projective complete-intersection morphism (1.1), whose relative virtual dimension d is constant (1.9). Suppose that Y is quasi-compact and admits an ample sheaf. With the notation of 3.3, for every $x \in K^\bullet(X)$ one has

$$\mathrm{ch}(f_*(x)) = f_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f)),$$

where $\check{T}_f$ denotes the dual element of the class of the relative cotangent complex $L_\bullet^{X/Y}$ (2.5 and 2.3).

In other words, the diagram

$$\begin{tikzcd}
K^\bullet(X) \arrow[r, "\mathrm{ch}\circ\widetilde{c}"] \arrow[d, "f_*"'] & \mathrm{Gr}\,K^\bullet(X)_{\mathbb{Q}} \arrow[d, "x\mapsto f_*^{\mathrm{gr}}(x\,\mathrm{Todd}(\check{T}_f))"] \\
K^\bullet(Y) \arrow[r, "\mathrm{ch}\circ\widetilde{c}"'] & \mathrm{Gr}\,K^\bullet(Y)_{\mathbb{Q}}
\end{tikzcd}$$

is commutative.

Continuation of the proof of Theorem 3.6

The proof of Theorem 3.6 will occupy the end of this exposé. Let us give some preliminaries.

Proposition 3.7. For arbitrary

$$x \in \mathrm{Gr}\,K^\bullet(X)_{\mathbb{Q}}, \qquad y \in \mathrm{Gr}\,K^\bullet(Y)_{\mathbb{Q}},$$

one has

$$f_*^{\mathrm{gr}}(x \cdot f^{\mathrm{gr}}(y)) = f_*^{\mathrm{gr}}(x)\,y.$$

By additivity, one may suppose x and y homogeneous; say

$$x \in \mathrm{Gr}^i\,K^\bullet(X)_{\mathbb{Q}}, \qquad y \in \mathrm{Gr}^j\,K^\bullet(Y)_{\mathbb{Q}}.$$

Let

$$x' \in \mathrm{Fil}^i\,K^\bullet(X)_{\mathbb{Q}}, \qquad y' \in \mathrm{Fil}^j\,K^\bullet(Y)_{\mathbb{Q}}$$

be elements lifting x and y. Then the projection formula gives

$$f_*(x' \cdot f^*(y')) = f_*(x')\,y'.$$

By 3.2, this equality is written in $\mathrm{Fil}^{i+j-d}\,K^\bullet(Y)_{\mathbb{Q}}$. One deduces from it an equality in $\mathrm{Gr}^{i+j-d}\,K^\bullet(Y)_{\mathbb{Q}}$, and one verifies immediately that this is the required equality.

Lemma 3.8. Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms satisfying the hypotheses of 3.6. Then $g \circ f$ also satisfies them. If x is an element of $K^\bullet(X)$, and if 3.6 is verified by f for the element x, and by g for the element $f_*(x)$, then it is verified by $g \circ f$ for the element x.

By hypothesis, one has

$$\mathrm{ch}(f_*(x)) = f_*^{\mathrm{gr}}\big(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f)\big), \tag{3.8.1}$$

and

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}}\big(\mathrm{ch}(f_*(x))\,\mathrm{Todd}(\check{T}_g)\big). \tag{3.8.2}$$

Combining (3.8.1) and (3.8.2), one obtains

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}}\Big(f_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f)) \cdot \mathrm{Todd}(\check{T}_g)\Big).$$

Applying 3.7, one finds

$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}}\Big(f_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_f) \cdot f^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_g)))\Big).$$

The functoriality of c and of Todd implies

$$f^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_g)) = \mathrm{Todd}(f^*(\check{T}_g)).$$

By 2.7, one has
$$\check{T}_{g\circ f} = \check{T}_f + f^*(\check{T}_g).$$
The multiplicativity of the Todd character therefore gives
$$\mathrm{Todd}(\check{T}_f)\, f^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_g)) = \mathrm{Todd}(\check{T}_{g\circ f}).$$
Thus finally
$$\mathrm{ch}(g_*(f_*(x))) = g_*^{\mathrm{gr}} f_*^{\mathrm{gr}}(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_{g\circ f})),$$
which is the announced result.

3.9. If one applies 3.8 to the factorization of f given in 3.1, one sees that, to prove the Riemann–Roch theorem, it is enough to prove it in the case of a regular immersion of constant codimension, and in the case of the structural morphism
$$\mathbb{P}^r_Y \longrightarrow Y.$$
Moreover, one may suppose that the immersion i satisfies the condition of VII 4.1. Indeed, let s be the section of
$$\mathbb{P}^2_{\mathbb{P}^r_Y}$$
introduced in VII 4.2. Then $s \circ i$ is a regular immersion satisfying the condition of VII 4.1 (VII 4.2), and by 3.8 it is enough to prove 3.6 for
$$s \circ i, \qquad \mathbb{P}^2_{\mathbb{P}^r_Y} \longrightarrow \mathbb{P}^r_Y, \qquad \mathbb{P}^r_Y \longrightarrow Y.$$
We shall therefore prove 3.6 by proving the following two particular cases:

i) f is a regular immersion of constant codimension, satisfying the condition of VII 4.1;

ii) f is the structural morphism $\mathbb{P}^r_Y \to Y$.

4. Riemann–Roch theorem: case of a regular closed immersion

4.1. Let $i : Y \to X$ be a regular closed immersion. The Riemann–Roch theorem for i is essentially relation VII 4.4, which makes it possible to compute the Chern classes of $i_*(x)$ from the $\gamma^p(N, x)$. More precisely, if one introduces the λ-ring $K^\bullet(Y)_{\mathbb{Q}N}$ defined in V 5.5, then the homomorphism
$$K^\bullet(Y)_{\mathbb{Q}N} \longrightarrow K^\bullet(X)_{\mathbb{Q}}$$
deduced from i_* is a λ-homomorphism by VII 2.8 and VII 4.3. The Riemann–Roch theorem is then reduced to an analogous formula concerning the additive homomorphism
$$K^\bullet(Y) \longrightarrow K^\bullet(Y)_{\mathbb{Q}N}$$
defined by $y \mapsto (0, y)$, that is, to a formal relation valid for every λ-ring. We shall not develop this proof here, in order to avoid introducing still more formalism on λ-rings (cf. RRR, I 1.5), and we shall resume the method of [1], which used blow-up schemes. This is the method already employed for the proof of VII 4.3, and also the method at the end of RRR.

As indicated in 3.9, one may restrict to the case of a regular immersion of constant codimension d satisfying the condition of VII 4.1; the case of a general regular immersion is then deduced from 3.8. In this paragraph we resume the notation of VII 3.1. Thus one has the Cartesian square

$$\begin{tikzcd} Y' \arrow[r, "j"] \arrow[d, "g"'] & X' \arrow[d, "f"] \\ Y \arrow[r, "i"'] & X, \end{tikzcd}$$

where X' is the X-scheme obtained by blowing up Y. We suppose moreover that X is quasi-compact and has an ample sheaf.

Proposition 4.2. The morphism f is projective, a complete-intersection morphism, and of virtual relative dimension zero. In addition one has the relation

$$f_*^{\mathrm{gr}} f^{\mathrm{gr}} = \mathrm{Id}_{\mathrm{Gr}\,K^\bullet(X)_{\mathbb{Q}}}.$$

The first assertion follows immediately from VII 1.8. The second relation is a consequence of the relation

$$f_* f^* = \mathrm{Id}_{K^\bullet(X)}$$

proved in VII 3.6.

Proposition 4.3. For every $y \in K^\bullet(Y)_{\mathbb{Q}}$, one has the relation

$$f_*^{\mathrm{gr}} i_*^{\mathrm{gr}}(y) = j_*^{\mathrm{gr}}(g^{\mathrm{gr}}(y)\, c^{d-1}(\check{F})),$$

where c^{d-1} denotes the $(d-1)$-st Chern class.

This relation is contained in VII 4.8, for it is nothing other than

$$v^{\mathrm{gr}} \circ u^{\mathrm{gr}} = 0.$$

Proposition 4.4. Let X be a quasi-compact scheme. If E is a locally free sheaf of finite type on X, of class E in $K^\bullet(X)$, and of rank n, then

$$\mathrm{ch}(\lambda_{-1}(E)) = c^n(\check{E})\,\mathrm{Todd}(\check{E})^{-1}.$$

The two sides of the relation to be proved are multiplicative. This is clear for $\mathrm{ch} \circ \lambda_{-1}$ and for Todd; it is also true for c^n, which is the class of $(-1)^n E \lambda_{-1}(E)$. Consequently, by passing to the flag bundle of E, one may suppose that E is a sum of classes of invertible sheaves, and reduce to proving the formula for an invertible sheaf. Let ξ be the class of such a sheaf. Then

$$\mathrm{ch}(\lambda_{-1}(\xi)) = \mathrm{ch}(1-\xi) = 1 - \mathrm{ch}(\xi).$$

Since $\widetilde{c}(\xi) = (1, 1 + c^1(\xi))$, the definition of ch gives

$$\mathrm{ch}(\lambda_{-1}(\xi)) = 1 - \exp(c^1(\xi)).$$

Moreover, ξ has augmentation 1; the Chern class which intervenes in formula 4.4 relative to ξ is therefore c^1. Taking into account the definition of Todd, and 3.5, the relation to be proved is written

$$1 - \exp(c^1(\xi)) = -c^1(\xi)\left(\frac{-c^1(\xi)}{1-\exp(c^1(\xi))}\right)^{-1},$$

and is therefore verified.

Remark 4.4.1. It is also possible to deduce 4.4 from a corresponding purely algebraic statement; cf. RRR, I 1.3.

4.5. We shall now show that, in order to prove the Riemann–Roch theorem in the case considered in this paragraph, one may reduce to the case where Y is a divisor of X, and where the element to which the formula is applied is the inverse image of an element of $K^\bullet(X)$.

By 3.2, the formula to be proved,

$$\mathrm{ch}(i_*(x)) = i_*^{\mathrm{gr}}\big(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_i)\big),$$

is equivalent to the formula

$$f^{\mathrm{gr}}\big(\mathrm{ch}(i_*(x))\big) = f^{\mathrm{gr}} i_*^{\mathrm{gr}}\big(\mathrm{ch}(x)\,\mathrm{Todd}(\check{T}_i)\big).$$

Taking account of the functoriality of ch, and of 4.3, it may also be written

$$\mathrm{ch}\big(f^*(i_*(x))\big) = j_*^{\mathrm{gr}}\big(\mathrm{ch}(g^*(x))\,\mathrm{Todd}(g^*(\check{T}_i))\,c^{d-1}(\check{F})\big),$$

or, by VII 3.4,

$$\mathrm{ch}\big(j_*(g^*(x)\lambda_{-1}(F))\big) = j_*^{\mathrm{gr}}\big(\mathrm{ch}(g^*(y))\,\mathrm{Todd}(g^*(\check{T}_i))\,c^{d-1}(\check{F})\big).$$

By definition,

$$T_i = -c^1(J/J^2) = -N, \qquad T_j = -c^1(J'/J'^2) = -L.$$

Moreover, the exact sequence

$$0 \longrightarrow F \longrightarrow g^*(N) \longrightarrow \mathcal{O}_{Y'}(1) \longrightarrow 0$$

gives the relation

$$g^*(T_i) = -(F + L),$$

hence

$$\mathrm{Todd}(g^*(\check{T}_i)) = \mathrm{Todd}(\check{T}_j)\,\mathrm{Todd}(\check{F})^{-1}.$$

Taking account of 4.4, the relation to be proved is therefore written

$$\mathrm{ch}\big(j_*(g^*(x)\lambda_{-1}(F))\big) = j_*^{\mathrm{gr}}\big(\mathrm{ch}(g^*(x)\lambda_{-1}(F))\,\mathrm{Todd}(\check{T}_j)\big).$$

Putting

$$z = g^*(x)\lambda_{-1}(F),$$

this relation is nothing but the Riemann–Roch theorem applied to the regular immersion of codimension 1, namely j, and to the element $z \in K^\bullet(Y')$.

Furthermore, since i was supposed to satisfy the hypothesis of VII 4.1, $\lambda_{-1}(F)$ is divisible by $1 - L$. Thus one can find z' such that $z = z'(1 - L)$, or still

$$z = j^*(j_*(z'))$$

(Exp. VII 2.7). We are therefore reduced to the case where Y is a divisor, and to an element x for which there exists y, with $x = i^*(y)$.

4.6. It remains to show the relation

$$\mathrm{ch}\big(i_*(i^*(y))\big) = i_*^{\mathrm{gr}}\big(\mathrm{ch}(i^*(y))\,\mathrm{Todd}(\check{T}_i)\big).$$

The projection formula gives

$$i_*(i^*(y)) = y\, i_*(1),$$

and from the exact sequence

$$0 \longrightarrow J \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0,$$

where J is here invertible, one obtains

$$i_*(1) = 1 - J.$$

Thus

$$\mathrm{ch}(i_*(i^*(y))) = \mathrm{ch}(y)\ \mathrm{ch}(1 - J).$$

Since J is invertible, one has

$$\widetilde{c}(J) = (1, 1 + c^1(J)), \qquad \mathrm{ch}(J) = \exp(c^1(J)).$$

Consequently

$$\mathrm{ch}(i_*(i^*(y))) = \mathrm{ch}(y)(1 - \exp(c^1(J))).$$

On the other hand, the functoriality of ch and the projection formula 3.7 imply

$$i_*^{\mathrm{gr}}(\mathrm{ch}(i^*(y))\,\mathrm{Todd}(\check{T}_i)) = \mathrm{ch}(y)\, i_*^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_i)).$$

Now

$$T_i = -c^1(J/J^2) = -i^*(J).$$

Hence

$$i_*^{\mathrm{gr}}(\mathrm{Todd}(\check{T}_i)) = i_*^{\mathrm{gr}}\, i^{\mathrm{gr}}(\mathrm{Todd}(-\check{J})) = i_*^{\mathrm{gr}}(1)\,\mathrm{Todd}(-\check{J}).$$

Since $i_*(1) = 1 - J$, $i_*^{\mathrm{gr}}(1)$ is the image in $\mathrm{Gr}^1 K^\bullet(X)$ of $1 - J$, hence it is $-c^1(J)$. Moreover

$$\mathrm{Todd}(-\check{J}) = \mathrm{Todd}(J)^{-1} = \frac{1 - \exp(c^1(J))}{-c^1(J)}.$$

Thus finally

$$i_*^{\mathrm{gr}}(\mathrm{ch}(i^*(y))\,\mathrm{Todd}(\check{T}_i)) = \mathrm{ch}(y)(1 - \exp(c^1(J))),$$

which proves the relation to be shown.

5. Riemann–Roch theorem: case of the structural morphism of a projective bundle

5.1. By 3.9, in order to prove the Riemann–Roch theorem, it remains to prove it for the structural morphism

$$f : \mathbb{P}^r_Y \longrightarrow Y,$$

where Y is a quasi-compact scheme admitting an ample sheaf. Put

$$P = \mathbb{P}^r_Y, \qquad \xi = c^1(\mathcal{O}_P(1)), \qquad E = (\mathcal{O}_Y)^{r+1}.$$

Proposition 5.2. On P one has the exact sequence

$$0 \longrightarrow \Omega^1_{P/Y} \longrightarrow f^*(E)(-1) \longrightarrow \mathcal{O}_P \longrightarrow 0,$$

i.e. one has an isomorphism

$$\Omega^1_{P/Y} \simeq F(-1), \qquad F = \mathrm{Ker}\big(f^*(E) \longrightarrow \mathcal{O}_P(1)\big).$$

Consider the vector bundle over Y defined by E, and denote by Z the complement of the zero section of this bundle. One has the commutative diagram

$$\begin{tikzcd} P \arrow[d, "f"'] & Z \arrow[l, "h"'] \arrow[dl, "g"] \\ Y & \end{tikzcd}$$

The morphism h identifies Z with the affine P-scheme defined by the sheaf of algebras

$$B = \bigoplus_{n \in \mathbb{Z}} \mathcal{O}_P(n);$$

B is a graded $\mathcal{O}_P$-algebra.

Since f, g, and h are smooth morphisms, one may write the exact sequence of sheaves of relative differentials. Since Z is affine over P, one may write this sequence in terms of B-modules; thus one obtains

$$0 \longrightarrow \Omega^1_{P/Y} \otimes_{\mathcal{O}_P} B \longrightarrow \Omega^1_{B/Y} \longrightarrow \Omega^1_{B/\mathcal{O}_P} \longrightarrow 0. \tag{5.2.1}$$

This exact sequence is an exact sequence of graded B-modules. Indeed, it follows from the commutative diagram

$$\begin{tikzcd} S(E) \arrow[rr] & & \bigoplus_{n\in\mathbb{Z}} S(E)(n) \simeq S(E)[T,T^{-1}] \\ & \mathcal{O}_X \arrow[ul] \arrow[ur] & \end{tikzcd}$$

where the arrows are algebra homomorphisms, compatible with the grading of $S(E)[T, T^{-1}]$ (grading defined by the degree in the indeterminate T), the two other algebras being supposed to be reduced to degree 0.

The module of differentials $\Omega^1_{B/\mathcal{O}_P}$ is isomorphic to

$$B(-1) \otimes_{\mathcal{O}_P} \mathcal{O}_P(1),$$

for one may define an $S(E)$-derivation

$$S(E)[T, T^{-1}] \longrightarrow S(E)[T, T^{-1}] \otimes_{S(E)} S(E)\, dT$$

satisfying the universal property which defines $\Omega^1_{B/\mathcal{O}_P}$, by setting

$$d(aT^n) = naT^{n-1} \otimes dT;$$

moreover, d is compatible with the gradings if the degree 1 is attributed to the factor $S(E)\, dT$.

Let V be the vector bundle defined by E over Y. Then $\Omega^1_{Z/Y}$ is the restriction to Z of $\Omega^1_{V/Y}$. But V is the affine Y-scheme defined by $S(E)$, so $\Omega^1_{V/Y}$ is defined by

$$S(E) \otimes_{\mathcal{O}_Y} E$$

(since E is locally free of finite type), and therefore $\Omega^1_{V/Y}$ is equal to $g^*(E)$. Consequently

$$\Omega^1_{B/Y} = f^*(E) \otimes_{\mathcal{O}_P} B.$$

To see its grading, one looks locally, and one sees at once that $f^*(E)$ must be considered as being of degree 1. Taking the terms of degree 0 in the exact sequence (5.2.1), one obtains the announced exact sequence.

Remark 5.2.1. Using the general results on the differential study of Hilbert functors ([2], 5), one directly obtains an isomorphism

$$\Omega^1_{P/Y} \xrightarrow{\sim} F(-1)$$

from the functorial characterization given of $g^*(\Omega^1_{P/Y})$ for every Y-morphism $g : Y' \to P$.

5.3. By 5.2, one has

$$T_f = c_1(\Omega^1_{P/Y}) = f^*(E)\xi^{-1} - 1.$$

Consequently

$$\widetilde{c}(T_f) = \widetilde{c}(E)\,\widetilde{c}(\xi^{-1}) - \widetilde{c}(1).$$

Since $E = (\mathcal{O}_Y)^{r+1}$, one has $E = r+1$, and hence

$$\widetilde{c}(E) = (r+1, 1).$$

Put

$$x = c^1_{\mathrm{Gr}^1 K^\bullet(P)}(\xi - 1) = c^1_{\mathrm{Gr}^1 K^\bullet(P)}(1 - \xi^{-1}) = c^1(\xi).$$

Since $\xi^{-1} = \check{\xi}$, one has

$$\widetilde{c}(\xi^{-1}) = (1, 1 - x).$$

By V (6.2.1), one deduces

$$\widetilde{c}(T_f) = (r, (1-x)^{r+1}).$$

Taking 3.5 into account, one finally obtains

$$\widetilde{c}(\check{T}_f) = (r, (1+x)^{r+1}).$$

By the definition of Todd, one therefore has

$$\operatorname{Todd}(\check{T}_f) = \left(\frac{x}{1 - \exp(-x)}\right)^{r+1}. \tag{5.3.1}$$

5.4. Using the basis of $K^\bullet(P)$ over $K^\bullet(Y)$ formed by the elements ξ^k, with $-r \le k \le 0$, one sees by additivity that it is enough to prove the theorem for elements of the form

$$f^*(a)\xi^k.$$

Then

$$\operatorname{ch}\bigl(f_*(f^*(a)\xi^k)\bigr) = \operatorname{ch}(af_*(\xi^k)) = \operatorname{ch}(a)\,\operatorname{ch}(f_*(\xi^k)).$$

On the other hand,

$$f_*^{\mathrm{gr}}\bigl(\operatorname{ch}(f^*(a)\xi^k)\operatorname{Todd}(\check{T}_f)\bigr) = f_*^{\mathrm{gr}}\bigl(f^{\mathrm{gr}}(\operatorname{ch}(a))\operatorname{ch}(\xi^k)\operatorname{Todd}(\check{T}_f)\bigr) = \operatorname{ch}(a)\,f_*^{\mathrm{gr}}\bigl(\operatorname{ch}(\xi^k)\operatorname{Todd}(\check{T}_f)\bigr).$$

Thus it is enough to prove the theorem for the ξ^k, with $-r \le k \le 0$. By VI (5.2.3), one has

$$f_*(\xi^k) = 0 \quad \text{for } -r \le k < 0, \qquad f_*(1) = 1.$$

Consequently

$$\operatorname{ch}(f_*(\xi^k)) = 0 \quad \text{if } -r \le k < 0, \qquad \operatorname{ch}(f_*(1)) = 1. \tag{5.4.1}$$

We now calculate the second member of the Riemann–Roch formula. For every k, one has

$$\widetilde{c}(\xi^k) = (\widetilde{c}(\xi))^k = (1, 1+x)^k = (1, 1+kx)$$

by V (6.2.1). Hence

$$\operatorname{ch}(\xi^k) = \exp(kx). \tag{5.4.2}$$

The second member of the Riemann–Roch formula is therefore

$$f_*^{\mathrm{gr}}\left(\exp(kx)\,\frac{x^{r+1}}{(1-\exp(-x))^{r+1}}\right). \tag{5.4.3}$$

Since x is the class in $\operatorname{Gr}^1 K^\bullet(P)$ of $1-\xi^{-1}$, $f_*^{\mathrm{gr}}(x^n)$ is, for $0 \le n \le r$, the class in $\operatorname{Gr}^{n-r} K^\bullet(Y)$ of $f_*((1-\xi^{-1})^n) = 1$. Thus

$$f_*^{\mathrm{gr}}(x^n) = 0 \quad \text{if } 0 \le n < r, \qquad f_*^{\mathrm{gr}}(x^r) = 1. \tag{5.4.4}$$

Finally recall that the elements

$$1, x, \dots, x^{r+1}$$

are linearly dependent over $\operatorname{Gr} K^\bullet(Y)$ (VI 5.6), and satisfy the equation

$$x^{r+1} = 0, \tag{5.4.5}$$

and that $1, \dots, x^r$ are linearly independent over $\operatorname{Gr} K^\bullet(Y)$.

One can therefore write the series

$$\varphi_k = \exp(kx)\,\frac{x^{r+1}}{(1-\exp(-x))^{r+1}}$$

in a unique way as a linear combination, with coefficients in $\operatorname{Gr} K^\bullet(Y)_{\mathbb{Q}}$, of the elements $1, \dots, x^r$. The image under f_*^{gr} of φ_k is then, by (5.4.4) and 3.7, its coefficient of x^r. Taking account of (5.4.1), the Riemann–Roch theorem for f is therefore equivalent to

$$\begin{cases} \text{for } -r \le k < 0, \text{ the coefficient of } x^r \text{ in } \varphi_k \text{ is } 0, \\ \text{for } k = 0, \text{ the coefficient of } x^r \text{ in } \varphi_k \text{ is } 1. \end{cases} \tag{RR}$$

This coefficient is the residue of the differential form

$$\frac{\exp(kx)\,dx}{(1-\exp(-x))^{r+1}}.$$

Putting $y = 1-\exp(-x)$, one is reduced to calculating the residue of

$$\frac{(1-y)^{-k-1}\,dy}{y^{r+1}},$$

that is, the coefficient of y^r in $(1-y)^{-k-1}$. For $-r \le k < 0$, it is zero, and for $k=0$, it is 1.

This completes the proof of the Riemann–Roch theorem.

Bibliography

[1] A. Borel and J.-P. Serre, *Le théorème de Riemann–Roch*, Bull. Soc. Math. de France, t. 86, 1958.

[2] A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique*, Sém. Bourbaki, 1960–61.

Exposé IX. Some computations of $K_\bullet$-groups

by P. Berthelot

This exposé corresponds to no oral exposé and will not be used in the sequel of the seminar. It contains a certain number of computations of the groups $K_\bullet$ for various fibrations: vector bundles, projective bundles, torsors under a linear group, and so on. Throughout the exposé, the base scheme will be assumed noetherian, and the notation $K_\bullet(X)$ will denote the Grothendieck group of coherent sheaves on a scheme X. Recall that, if X is noetherian, the homomorphism from the K of the category of coherent $\mathcal{O}_X$-modules to the K of the category of pseudo-coherent complexes on X is an isomorphism (IV 2.4).

The results set out here are the analogues of those one obtains for the Chow ring, and which are to be found in Grothendieck's exposés in the Chevalley seminar of 1958. The question also arises of determining to what extent statements of the same type can be obtained for the graded group associated with $K_\bullet$ for the filtration defined by the dimension of the support (cf. X). It does not seem that the answer is known, even for Proposition 1.1, the homotopy exact sequence, which is one of the properties most used in the computations which follow.

1. Vector bundles

Proposition 1.1 (homotopy exact sequence). Let X be a noetherian scheme, let X' be a closed subscheme of X, let U be the open complement of X' in X, let i be the immersion of X' in X, and let j be the immersion of U in X. Then the sequence

$$K_\bullet(X') \xrightarrow{i_*} K_\bullet(X) \xrightarrow{j^*} K_\bullet(U) \longrightarrow 0$$

is exact.

If F' is a coherent $\mathcal{O}_{X'}$-module, $i_*(F')$ is an $\mathcal{O}_X$-module whose support is contained in X'; consequently its restriction to U is zero, and $j^*i_*(\mathrm{cl}(F')) = 0$. Since the elements of the form $\mathrm{cl}(F')$ generate $K_\bullet(X')$, $j^*i_* = 0$. We therefore obtain a homomorphism

$$\rho : K_\bullet(X)/\operatorname{Im} K_\bullet(X') \longrightarrow K_\bullet(U).$$

We shall define an inverse homomorphism to ρ. Let G be a coherent $\mathcal{O}_U$-module. Since X is noetherian, there exists a coherent $\mathcal{O}_X$-module F whose restriction to U is isomorphic to G. If F' is a second coherent $\mathcal{O}_X$-module whose restriction to U is isomorphic to G, then in $K_\bullet(X)$ one has

$$\mathrm{cl}(F) - \mathrm{cl}(F') \in \operatorname{Im} K_\bullet(X').$$

To see this, consider G as a subsheaf of $j^*(F \oplus F')$ by means of the diagonal map. Then G extends to a subsheaf F'' of $F \oplus F'$, and one is reduced to proving that

$$\mathrm{cl}(F) - \mathrm{cl}(F'') \in \operatorname{Im} K_\bullet(X')$$

and likewise for F'. Let H and K be the kernel and cokernel of the composite homomorphism

$$F'' \longrightarrow F \oplus F' \longrightarrow F.$$

This homomorphism restricts on U to the identity of G, and hence H and K have support contained in X'. If I is the ideal of X' in X, it follows that there exist integers m and n such that $I^m H = 0$ and $I^n K = 0$ (EGA I 5.3.4). Thus one may dévissser H and K into extensions of modules annihilated by I, hence of $\mathcal{O}_{X'}$-modules, and

$$\mathrm{cl}(H) \in \operatorname{Im} K_\bullet(X'), \qquad \mathrm{cl}(K) \in \operatorname{Im} K_\bullet(X'),$$

which proves the assertion. Thus, to every coherent $\mathcal{O}_U$-module, assigning the class of one of its extensions to X defines a function on the category of coherent $\mathcal{O}_U$-modules with values in $K_\bullet(X)/\operatorname{Im} K_\bullet(X')$.

This function is additive. Indeed, let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

be an exact sequence of coherent $\mathcal{O}_U$-modules. Let F be an extension of G to X. There exists a submodule F' of F extending G', and F/F' extends G''. Hence the function factors through $K_\bullet(U)$, defining the desired homomorphism $K_\bullet(U) \to K_\bullet(X)/\operatorname{Im} K_\bullet(X')$. One verifies at once that it is inverse to ρ.

Proposition 1.2. Let X and S be noetherian schemes, and let $f : X \to S$ be a flat morphism. Let $(\xi_i)_{i\in I}$ be a family of elements of $K^\circ(X)$, and consider the homomorphism

$$\varphi : K_\bullet(S)^{(I)} \longrightarrow K_\bullet(X)$$

defined by

$$\varphi((x_i)_{i\in I}) = \sum_{i\in I} \xi_i\, f^*(x_i),$$

where f^* is the inverse-image homomorphism from $K_\bullet(S)$ to $K_\bullet(X)$ defined in IV 2.12, and where $K_\bullet(X)$ is considered as a $K^\circ(X)$-module by IV 2.10. Suppose that, for every point $s \in S$, the images of the ξ_i in $K_\bullet(X_s)$ by the homomorphism

$$K^\circ(X) \longrightarrow K^\circ(X_s) = K_\bullet(X_s)$$

generate $K_\bullet(X_s)$ as an abelian group. Then φ is surjective.

Let S' be a closed subscheme of S, and let U be the open complement of S' in S. Write X_U for the restriction of X over U, and $X_{S'}$ for the restriction of X over S'. We again denote by ξ_i the images of the ξ_i in $K^\circ(X_{S'})$ and $K^\circ(X_U)$, and write

$$\varphi_{S'} : K_\bullet(S')^{(I)} \to K_\bullet(X_{S'}), \qquad \varphi_U : K_\bullet(U)^{(I)} \to K_\bullet(X_U)$$

for the homomorphisms defined in the same way as φ.

Consider the diagram

$$\begin{array}{ccccccc} K_\bullet(S')^{(I)} & \xrightarrow{i_*} & K_\bullet(S)^{(I)} & \xrightarrow{j^*} & K_\bullet(U)^{(I)} & \to & 0 \\ {\scriptstyle\varphi_{S'}}\downarrow & & \downarrow{\scriptstyle\varphi} & & \downarrow{\scriptstyle\varphi_U} & & \\ K_\bullet(X_{S'}) & \xrightarrow{u_*} & K_\bullet(X) & \xrightarrow{v^*} & K_\bullet(X_U) & \to & 0, \end{array} \tag{1.2.1}$$

where i and j denote the immersions of S' and U into S, and u and v the immersions of $X_{S'}$ and X_U into X. We show that, if $\varphi_{S'}$ and φ_U are surjective, then so is φ.

The two horizontal rows are exact by 1.1. To prove that φ is surjective, it is therefore enough to prove that the two squares commute. For $x_i \in K_\bullet(S')$, one has

$$\varphi(i_*(x_i)) = \sum_i \xi_i f^* i_*(x_i),$$

whereas

$$u_*\varphi_{S'}((x_i)) = u_*\left(\sum_i \xi_i u^* f^*(x_i)\right) = \sum_i \xi_i u_*(u^* f^*(x_i)),$$

the last equality following from the projection formula IV 2.11.1.2. The equality $\varphi i_* = u_* \varphi_{S'}$ therefore follows from $f^* i_* = u_* u^* f^*$, which is true by IV 3.1.1 since i is proper and f is flat. Similarly, for $(x_i) \in K_\bullet(S)^{(I)}$,

$$\varphi_U(j^*(x_i)) = \sum_i \xi_i f_U^* j^*(x_i),$$

and

$$v^* \varphi((x_i)) = v^* \left(\sum_i \xi_i f^*(x_i) \right) = \sum_i \xi_i v^* f^*(x_i) = \sum_i \xi_i f_U^* j^*(x_i),$$

the middle equality being IV 2.12.2.

This result permits one to prove 1.2 by noetherian induction on the set of closed subschemes of S. If one assumes that, for every closed subscheme S' of S distinct from S, $\varphi_{S'}$ is surjective, the preceding argument shows that one may reduce to proving 1.2 for an irreducible affine base scheme, after possibly replacing S by an irreducible affine open. Moreover, the diagram

$$\begin{tikzcd} K_\bullet(S_{\mathrm{red}})^{(I)} \arrow[r] \arrow[d, "\varphi_{S_{\mathrm{red}}}"'] & K_\bullet(S)^{(I)} \arrow[d, "\varphi"] \\ K_\bullet(X_{S_{\mathrm{red}}}) \arrow[r] & K_\bullet(X) \end{tikzcd}$$

where $K_\bullet(X_{S_{\mathrm{red}}}) \to K_\bullet(X)$ is surjective by 1.1, shows that one may also assume S reduced.

We are thus on an integral affine base scheme. Put $S = \mathrm{Spec}(A)$, and let K denote the fraction field of A. If s is the generic point of S, then $\mathcal{O}_{S,s} = K$. Let $y \in K_\bullet(X)$. By hypothesis, $K_\bullet(X_s)$ is generated by the images of the ξ_i; if y_s is the image of y in $K_\bullet(X_s)$, then one can find integers n_i such that

$$y_s = \sum_i \xi_i n_i.$$

On the other hand $K = \varinjlim_f A_f$, where f runs through the elements of A. Hence $\mathrm{Spec}(K) = \varprojlim_f \mathrm{Spec}(A_f)$ and

$$X_s = X \times_S \mathrm{Spec}(K) = \varprojlim_f X \times_S \mathrm{Spec}(A_f)$$

by EGA IV 8.2.3 and 8.2.5. Since the schemes considered are noetherian and the transition morphisms affine and flat, one has, by IV 3.2.3,

$$K_\bullet(X_s) = \varinjlim_f K_\bullet(X \times_S \mathrm{Spec}(A_f)).$$

As inverse-image homomorphisms on the $K_\bullet$ are linear with respect to K° (IV 2.12.2), there exists a non-empty open $U = \mathrm{Spec}(A_f)$ of S such that in $K_\bullet(X_U)$ one has

$$y_U = \sum_i \xi_i n_i.$$

Let S' be a closed subscheme of S with support $S - U$. Then $\varphi_{S'}$ is surjective by the induction hypothesis, and consequently diagram (1.2.1) shows that

$$y - \sum_i \xi_i n_i \in \mathrm{Im}(\varphi).$$

Since $n_i = f^*(n_i)$, φ is surjective.

Corollary 1.3. Let $f : X \to S$ be a flat morphism of noetherian schemes. Suppose that, for every point s of S, the homomorphism

$$K^\circ(X) \longrightarrow K_\bullet(X_s)$$

is surjective. Then the homomorphism deduced from f^*,

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \longrightarrow K_\bullet(X),$$

is surjective.

Take for the family (ξ_i) the family of all elements of $K^\circ(X)$. The hypotheses of 1.2 are then satisfied, and hence

$$\varphi : K_\bullet(S)^{(K^\circ(X))} \longrightarrow K_\bullet(X)$$

is surjective. By definition of φ, this is equivalent to the surjectivity of the displayed tensor-product homomorphism.

Corollary 1.4. Let $f : X \to S$ be a flat morphism of noetherian schemes. Suppose that, for every point s of S, the canonical homomorphism $\mathbb{Z} \to K_\bullet(X_s)$ is an isomorphism. Then

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(X)$$

is surjective. If, moreover, f has a section g of finite Tor-dimension, then f^* is an isomorphism.

Take for the family (ξ_i) the family reduced to the single element $1 \in K^\circ(X)$. Then $\varphi = f^*$, and the conditions of 1.2 being fulfilled, f^* is surjective. If f admits a section g of finite Tor-dimension, one may define a homomorphism

$$g^* : K_\bullet(X) \longrightarrow K_\bullet(S)$$

(IV 2.12). Since $g^* f^* = (fg)^* = \mathrm{Id}$ by IV 2.12.1.1, f^* is injective, hence an isomorphism.

Remark 1.5. For g to be of finite Tor-dimension, it is necessary and sufficient that the fibres of X be regular at the points of $g(S)$, i.e. that f be regular at the points of $g(S)$, or smooth at these points if one assumes f of finite presentation.

These properties being local on X and S, one may suppose them affine: $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$. Then

$$\operatorname{tor.dim}_B(A) = \operatorname{dim.proj}_B(A) = \sup_{s \in S} \operatorname{dim.proj}_{B \otimes_A k(s)}(k(s)),$$

by EGA IV 12.3.2, where the finite-presentation hypothesis on B may be replaced by noetherian hypotheses. But $k(s)$ is isomorphic to the residue field of $\mathcal{O}_{X_s, g(s)}$, and by EGA 0_{IV} 17.2.5 one has

$$\operatorname{dim.proj}_{B \otimes_A k(s)}(k(s)) = \operatorname{dim.proj}_{\mathcal{O}_{X_s, g(s)}}(k(s)).$$

It follows from EGA 0_{IV} 17.2.11 that the latter is finite if and only if $\mathcal{O}_{X_s, g(s)}$ has finite cohomological dimension, i.e. is regular.

Proposition 1.6. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of finite type, let X be the vector bundle associated with E, and let f be the structural morphism of X. Then

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(X)$$

is an isomorphism.

Since f is smooth and admits a section, namely the zero section, it is enough, to prove 1.6, to show that for every $s \in S$ one has $K_\bullet(X_s) \simeq \mathbb{Z}$ (1.4). Thus one is reduced to the case where the base scheme is a field. In this case E is a direct sum of rank-one submodules, which permits one to decompose f into a sequence of rank-one fibrations, and hence to reduce to the case where X has rank one, namely $X = \operatorname{Spec}(k[T])$. The proposition then follows from the next lemma.

Lemma 1.7. Let A be a principal ideal domain and let $X = \operatorname{Spec}(A)$. Then the canonical homomorphism

$$\mathbb{Z} \longrightarrow K_\bullet(X)$$

is an isomorphism.

Since X is regular, the homomorphism $K^\circ(X) \to K_\bullet(X)$ is an isomorphism. Hence $K_\bullet(X)$ is generated by the classes of locally free $\mathcal{O}_X$-modules of finite type, i.e. by the classes of projective A-modules of finite type. Since A is principal, every projective module of finite type is free, which gives surjectivity. Injectivity follows from the fact that $K^\circ(X)$ is $\mathbb{Z}$-augmented.

Proposition 1.8. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank r, let X be the vector bundle associated with E, let f be the structural morphism of X, and let g be the zero section of X. Then the composite homomorphism

$$K_\bullet(S) \xrightarrow{g_*} K_\bullet(X) \xrightarrow{g^*} K_\bullet(S)$$

where g^* is defined by IV 2.12.1 and 1.5, is multiplication by the element $\lambda_{-1}(E)$ of $K^\circ(S)$, writing E for the class of E in $K^\circ(S)$ and putting

$$\lambda_{-1}(E) = \sum_{i=0}^{r} (-1)^i \lambda^i(E).$$

This statement is the analogue, for $K_\bullet$, of VII 2.7. Let F be a coherent $\mathcal{O}_S$-module, of class F in $K_\bullet(S)$. Then $g_*(F)$ is the class in $K_\bullet(X)$ of the complex $Rg_*(F) = g_*(F)$, hence of F considered as an $\mathcal{O}_X$-module through g. To compute $g^*(g_*(F))$, one must take the class of

$$Lg^*(F) = F \otimes^L_{\mathcal{O}_X} \mathcal{O}_S.$$

For this, consider the Koszul complex

$$0 \longrightarrow \Lambda^r(E) \otimes_{\mathcal{O}_S} S(E) \longrightarrow \Lambda^{r-1}(E) \otimes_{\mathcal{O}_S} S(E) \longrightarrow \cdots \longrightarrow E \otimes_{\mathcal{O}_S} S(E) \longrightarrow S(E) \longrightarrow 0,$$

where $S(E)$ is the symmetric algebra of E. This is a resolution of $\mathcal{O}_S$ over $S(E)$ (VI 1.11). The associated complex on X, denoted $L_\bullet$, has k-th term $\Lambda^k(f^*E)$ and is therefore a resolution of $\mathcal{O}_S$ by locally free $\mathcal{O}_X$-modules of finite type. Thus

$$L_\bullet \otimes_{\mathcal{O}_X} F = \mathcal{O}_S \otimes_{\mathcal{O}_X} F,$$

and hence

$$\begin{aligned} g^*(g_*(F)) = \operatorname{cl}(L_\bullet \otimes_{\mathcal{O}_X} F) &= \sum_{i=0}^{r} (-1)^i \operatorname{cl}(L_i \otimes_{\mathcal{O}_X} F) \\ &= \sum_{i=0}^{r} (-1)^i \operatorname{cl}(\Lambda^i(E) \otimes_{\mathcal{O}_S} F) = F\lambda_{-1}(E). \end{aligned}$$

The proposition follows by additivity.

2. Principal bundles under split tori

2.1. Let S be a noetherian scheme, and let T be a torus over S. Suppose T is split; thus one has an isomorphism $T \simeq D_S(M)$, where M is a free abelian group of rank r, or equivalently

$T \simeq \mathbb{G}_{m,S}^r$. Let P be a torsor under T. Then P is affine over S, defined by a graded $\mathcal{O}_S$-algebra A of type M. If

$$A = \bigoplus_{m \in M} A_m,$$

then A_m is an invertible $\mathcal{O}_S$-module for every m, and, for $m, m' \in M$, the homomorphism

$$A_m \otimes_{\mathcal{O}_S} A_{m'} \longrightarrow A_{m+m'}$$

defined by the multiplication of A is an isomorphism (SGA 3 VIII 4.1). One therefore defines a multiplicative homomorphism

$$\eta : M \longrightarrow K^\circ(S)$$

by assigning to $m \in M$ the class of A_m in $K^\circ(S)$.

Proposition 2.2. With the notation and hypotheses of 2.1, let $f : P \to S$ be the structural morphism of P. Then the homomorphism

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(P)$$

is surjective, and its kernel is the subgroup of $K_\bullet(S)$ generated by the subgroups $(\eta(m)-1)K_\bullet(S)$ for $m \in M$, or, equivalently, for m running through a system of generators of M.

First suppose $r = 1$, i.e. $T = \mathbb{G}_{m,S}$. Then $M = \mathbb{Z}$, and for every $m \in M$ one has

$$A_m \simeq A_1^{\otimes m}.$$

If $A_1 = L$, then $A = \bigoplus_{n \in \mathbb{Z}} L^{\otimes n}$, which shows that P identifies with the complement of the zero section in the bundle $X = \mathbb{V}(L)$. Considering S as a closed subscheme of X by the zero section, one has, by 1.1, the exact sequence

$$K_\bullet(S) \xrightarrow{g_*} K_\bullet(X) \longrightarrow K_\bullet(P) \longrightarrow 0,$$

where $g : S \to X$ is the zero section of X. Moreover f^* factors as

$$K_\bullet(S) \xrightarrow{f'^*} K_\bullet(X) \longrightarrow K_\bullet(P),$$

where f' is the structural morphism of X. By 1.6, f'^* is an isomorphism, inverse to g^*. Thus we have the exact sequence

$$K_\bullet(S) \xrightarrow{g^* g_*} K_\bullet(S) \xrightarrow{f^*} K_\bullet(P) \longrightarrow 0,$$

and by 1.8 one obtains

$$K_\bullet(P) \simeq K_\bullet(S)/\lambda_{-1}(L)K_\bullet(S) = K_\bullet(S)/(1-L)K_\bullet(S),$$

where L is the class of L in $K^\circ(S)$. Since, for every $n \in \mathbb{Z}$, $L^n - 1$ is a multiple of $1 - L$, this proves 2.2 when P has rank one.

Continue the proof by induction on the rank r of P. Suppose the result proved for torsors under split tori of rank $< r$. Write $M = M_1 \oplus M_2$, with $M_1 \simeq \mathbb{Z}^{r-1}$ and $M_2 \simeq \mathbb{Z}$. Put $A' = \bigoplus_{m \in M_1} A_m$ and $A'' = \bigoplus_{m \in M_2} A_m$, so that $A = A' \otimes_{\mathcal{O}_S} A''$. Let $P_1 = \mathrm{Spec}(A')$. Then P_1 is a torsor under $D_S(M_1)$, and P is, over P_1, a torsor under the group $\mathbb{G}_{m,P_1}$ with algebra the inverse image of A'' on P_1. By the preceding rank-one case, $K_\bullet(P)$ is the quotient of $K_\bullet(P_1)$ by the subgroup generated by the $(\eta(m_2)-1)K_\bullet(P_1)$ for $m_2 \in M_2$. By the induction hypothesis, $K_\bullet(P_1)$ is the quotient of $K_\bullet(S)$ by the subgroup generated by the $(\eta(m_1)-1)K_\bullet(S)$ for $m_1 \in M_1$. Taking account of the relation

$$\eta(m_1 + m_2) - 1 = \eta(m_1)\eta(m_2) - 1 = \eta(m_1)(\eta(m_2) - 1) + \eta(m_1) - 1,$$

we conclude that $K_\bullet(P)$ is the quotient of $K_\bullet(S)$ by the subgroup generated by the $(\eta(m)-1)K_\bullet(S)$ for $m \in M$, or equivalently for m running through a system of generators of M.

Corollary 2.3. With the notation and hypotheses of 2.1, suppose S regular. Then one has a canonical isomorphism

$$K^\circ(P) \simeq K^\circ(S)/\big((\eta(m)-1)_{m\in M}\big),$$

where $((\eta(m)-1)_{m\in M})$ is the ideal of $K^\circ(S)$ generated by the $\eta(m)-1$, for m running through a system of generators of M.

If S is regular, the canonical homomorphism $K^\circ(S) \to K_\bullet(S)$ is an isomorphism (IV 2.5). Likewise, P is then regular, and $K^\circ(P) \to K_\bullet(P)$ is an isomorphism. The corollary therefore follows immediately from 2.2.

3. Projective bundles and flag bundles

Proposition 3.1. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let $\pi = (p_1, \ldots, p_k)$ be a sequence of integers such that $\sum p_i = n$, and let $X = D_\pi(E)$ be the scheme of flags of type π in E. Then the canonical homomorphism

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \longrightarrow K_\bullet(X)$$

is an isomorphism.

We resume the notation of VI 4.5. Thus L_i denotes the canonical quotients of the inverse image of E on X, F_j denotes the kernel of $L_j \to L_{j-1}$, and $\lambda_i^{(j)}$ denotes the class in $K^\circ(X)$ of $\Lambda^i(F_j)$. Then the $\lambda_i^{(j)}$ generate $K^\circ(X)$ as a $K^\circ(S)$-algebra (VI 4.6). Likewise, the images of the $\lambda_i^{(j)}$ in $K^\circ(X_s)$ generate $K^\circ(X_s)$ as an algebra over $K^\circ(k(s)) = \mathbb{Z}$ for every point s of S. Since X_s is a flag scheme over a field, it is regular, and hence $K^\circ(X_s) \to K_\bullet(X_s)$ is an isomorphism. It follows that $K^\circ(X) \to K_\bullet(X_s)$ is surjective. By 1.3, this implies that

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \longrightarrow K_\bullet(X)$$

with $K^\circ(X)$ in the second factor is surjective.

To prove injectivity, one may first reduce to the case of the complete flag bundle $D(E)$. Indeed, suppose that

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(D(E)) \longrightarrow K_\bullet(D(E))$$

is injective. Since $K_\bullet(S) \to K_\bullet(D(E))$ factors through $K_\bullet(X)$, the corresponding homomorphism factors as

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \otimes_{K^\circ(X)} K^\circ(D(E)) \longrightarrow K_\bullet(X) \otimes_{K^\circ(X)} K^\circ(D(E)) \longrightarrow K_\bullet(D(E)).$$

The first homomorphism is injective. Since, by VI 4.3, 4.6 and 4.7, $K^\circ(D(E))$ is free over $K^\circ(X)$, this implies injectivity for

$$K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X) \to K_\bullet(X).$$

Thus assume $X = D(E)$. One proceeds by induction on the rank of E. The case $n = 1$ is trivial. Suppose the property proved for $n-1$. Introduce $P = \mathbb{P}(E)$, and let F be the kernel of the canonical homomorphism $p^*(E) \to \mathcal{O}_P(1)$, where p is the structural morphism of P. Then X identifies with the scheme of flags of F over P, and by the induction hypothesis it is enough to prove 3.1 for P. Taking VI 1.1 into account, the result to prove may therefore be stated as the following corollary.

Corollary 3.2. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let $X = \mathbb{P}(E)$ be the associated projective bundle, let f be the structural morphism of X, and let ξ be the class in $K^\circ(X)$ of the sheaf $\mathcal{O}_X(1)$. Then the homomorphism

$$\varphi : (K_\bullet(S))^n \longrightarrow K_\bullet(X)$$

defined by

$$\varphi((x_i)) = \sum_{i=0}^{n-1} \xi^{-i} f^*(x_i)$$

is an isomorphism.

We have already seen that this homomorphism is surjective, since the elements ξ^i, $i = 0, \dots, -n+1$, form a basis of $K^\circ(X)$ over $K^\circ(S)$. To see that it is injective, use the homomorphism f_*, defined because f is projective. By IV 2.12.4,

$$f_*\left(\sum_{i=0}^{n-1} \xi^{-i} f^*(x_i)\right) = \sum_{i=0}^{n-1} x_i f_*(\xi^{-i}) = x_0,$$

the last equality following from the fact that $f_*(\xi^i) = 0$ for $-n+1 \leq i < 0$ (VI 2.8). If $\varphi((x_i)) = 0$, what precedes shows that $x_0 = 0$. Applying the same reasoning to the elements $\xi^i \varphi((x_i))$, for $i = 1, \dots, n-1$, one obtains by induction that all the x_i are zero. Thus φ is injective.

3.3. Let S still be a noetherian scheme, and let E be a locally free $\mathcal{O}_S$-module of rank n. Write $V = \mathbb{V}(E)$ for the vector bundle associated with E, contravariantly in E (SGA 3 I 4.6), considered as a group scheme over S. Let P be a torsor under V; we wish to compute $K_\bullet(P)$.

Recall how one embeds P in a projective bundle over S. Torsors under V correspond to extensions F of E by $\mathcal{O}_S$. Indeed, if one has an extension

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{s} F \xrightarrow{u} E \longrightarrow 0, \tag{3.3.1}$$

one obtains a torsor under V by considering the homomorphism $\mathbb{V}(F) \to \mathbb{V}(\mathcal{O}_S)$ defined by s, the section of $\mathbb{V}(\mathcal{O}_S) = \mathrm{Spec}(\mathcal{O}_S[T])$ defined by $T = 1$, and taking

$$P = \mathbb{V}(F) \times_{\mathbb{V}(\mathcal{O}_S)} S,$$

the operations of V on P being induced by those of V on $\mathbb{V}(F)$.

Conversely, let P be a torsor under V. Denote by $\underline{P}$ the sheaf of germs of homomorphisms from S to P. It is a principal homogeneous sheaf under the sheaf of germs of homomorphisms from S to V, which is identified with the dual $\check{E}$ of E. In the classical way this yields an extension of $\mathcal{O}_S$ by $\check{E}$, and by passing to duals one obtains (3.3.1).

Thus let P be a torsor under V, corresponding to an extension F of E by $\mathcal{O}_S$. The algebra $A(P)$ of P over $\mathcal{O}_S$ is

$$S(F) \otimes_{S(\mathcal{O}_S)} \mathcal{O}_S,$$

that is,

$$S(F)/(s-1)S(F),$$

which identifies with the ring of homogeneous fractions of degree 0, $S(F)_{(s)}$. Consequently P identifies with the open $D_+(s)$ of $\mathbb{P}(F)$. The complement of P in $\mathbb{P}(F)$ is then $\mathrm{Proj}(S(F)/sS(F))$, which is identified, by (3.3.1), with $\mathbb{P}(E)$. Thus one has immersions

$$\begin{tikzcd} \mathbb{P}(E) \arrow[r, hook] \arrow[dr] & \mathbb{P}(F) & P \arrow[l, hook'] \arrow[dl] \\ & S & \end{tikzcd}$$

where $\mathbb{P}(E)$ is closed in $\mathbb{P}(F)$, and P is open and complementary to $\mathbb{P}(E)$.

Proposition 3.4. Let S be a noetherian scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let $V = \mathbb{V}(E)$ be the associated vector bundle, let P be a torsor under V, and let f be the structural morphism of P. Then the canonical homomorphism

$$f^* : K_\bullet(S) \longrightarrow K_\bullet(P)$$

is an isomorphism.

With the identifications of 3.3, one has the exact sequence

$$K_\bullet(\mathbb{P}(E)) \xrightarrow{i_*} K_\bullet(\mathbb{P}(F)) \longrightarrow K_\bullet(P) \longrightarrow 0, \tag{3.4.1}$$

where i denotes the closed immersion $\mathbb{P}(E) \to \mathbb{P}(F)$ defined by u. By 3.2,

$$K_\bullet(\mathbb{P}(E)) = K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(\mathbb{P}(E)), \qquad K_\bullet(\mathbb{P}(F)) = K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(\mathbb{P}(F)).$$

Since $\mathbb{P}(E)$ and $\mathbb{P}(F)$ are smooth, i is a regular immersion, and hence we have a homomorphism

$$i_* : K^\circ(\mathbb{P}(E)) \longrightarrow K^\circ(\mathbb{P}(F)).$$

The homomorphism between the $K_\bullet$ groups is then obtained by tensoring the homomorphism between the K° groups by $K_\bullet(S)$. It is enough, to see this, to show that for $a \in K_\bullet(S)$ and $b \in K^\circ(\mathbb{P}(E))$ one has

$$i_*(b\, h^*(a)) = i_*(b) g^*(a),$$

where g and h are the structural morphisms of $\mathbb{P}(F)$ and $\mathbb{P}(E)$. Since $h = gi$,

$$i_*(b\, h^*(a)) = i_*(b\, i^* g^*(a)) = g^*(a) i_*(b)$$

by IV 2.12.4.

From (3.4.1) one therefore gets

$$K_\bullet(P) \simeq K_\bullet(S) \otimes_{K^\circ(S)} \big(K^\circ(\mathbb{P}(F)) / \operatorname{Im} K^\circ(\mathbb{P}(E))\big).$$

Let ξ_F and ξ_E be the classes of the canonical invertible sheaves on $\mathbb{P}(F)$ and $\mathbb{P}(E)$. By VI 1.1, $K^\circ(\mathbb{P}(F))$ (respectively $K^\circ(\mathbb{P}(E))$) has as basis over $K^\circ(S)$ the elements ξ_F^k for $k = 0, \ldots, -n$ (respectively ξ_E^k for $k = 0, \ldots, -n+1$). The projection formula gives

$$i_*(\xi_E^k) = i_*(i^*(\xi_F^k)) = \xi_F^k i_*(1).$$

If I is the ideal of $\mathbb{P}(E)$ in $\mathbb{P}(F)$, then I is invertible and $i_*(1) = 1 - I$. Moreover I is defined by the ideal J of $S(F)$, kernel of $S(F) \to S(E)$, which is generated by a submodule of F isomorphic to $\mathcal{O}_S$. Hence

$$I \simeq \mathcal{O}_{\mathbb{P}(F)}(-1).$$

The image of $K^\circ(\mathbb{P}(E))$ is thus the $K^\circ(S)$-submodule generated by the $\xi_F^k - \xi_F^{k-1}$, for $k = 0, \ldots, -n+1$. Therefore

$$K^\circ(S) \longrightarrow K^\circ(\mathbb{P}(F)) / \operatorname{Im} K^\circ(\mathbb{P}(E))$$

is an isomorphism, and consequently so is $K_\bullet(S) \to K_\bullet(P)$.

4. Principal bundles under the groups $\mathrm{GL}(n)_S$

Proposition 4.1. Let S be a noetherian scheme, let $G = \mathrm{GL}(n)_S$, let P be a torsor under G, and let E be the locally free $\mathcal{O}_S$-module of rank n corresponding to P. Denote by E the class of E in $K^\circ(S)$, and let J be the λ-ideal generated by $E - n$. If f is the structural morphism of P, then f^*, defined because f is flat, is surjective and defines an isomorphism

$$K_\bullet(S)/JK_\bullet(S) \xrightarrow{\sim} K_\bullet(P).$$

Let $E_0 = (\mathcal{O}_S)^n$, with evident basis $e_1, \dots, e_n$, and let $E_0^{(i)}$ be the submodule of E_0 generated by $e_1, \dots, e_i$. The subfunctor of G which to every S-scheme S' assigns the subgroup of automorphisms of $E_{0,S'}$ leaving invariant the submodules $E_{0,S'}^{(i)}$ for all i is represented by a group subscheme B of G, a Borel subgroup of G. The quotient sheaf G/B for the fpqc topology is represented by the flag scheme of E_0, denoted X_0. Let $L_0^{(i)}$ be the canonical quotients of the inverse image of E_0 on X_0, let $G_0^{(i)}$ be the kernel of $E_{0,X_0} \to L_0^{(i)}$, and put

$$F_0^{(i)} = G_0^{(i-1)}/G_0^{(i)} \simeq \ker(L_0^{(i)} \to L_0^{(i-1)}).$$

Moreover, B is the semi-direct product of its unipotent radical U and of its maximal subtorus T leaving the e_i invariant. Let $Y_0 = G/U$. Then Y_0 is a torsor over X_0 under the group T_{X_0}. If S' is an S-scheme, the datum of an S'-point of Y_0 is equivalent to the datum, on some faithfully flat quasi-compact S'' over S', of an automorphism of $E_{0,S''}$ modulo the unipotents of $U_{S''}$; that is, of a flag of $E_{0,S''}$ with free submodules, and, for each i, of a class of free elements f_i^α on $\mathcal{O}_{S''}$ such that $f_i^\alpha - f_i^\beta \in G_{0,S''}^{(i+1)}$. Equivalently, it is the datum of a flag of $E_{0,S''}$ and, for every i, of a basis of $F_{0,S''}^{(i)}$, this datum being subject to the gluing condition on $S'' \times_{S'} S''$. By fpqc descent, this amounts to the same datum on S'. Thus Y_0 is the affine X_0-scheme defined by the algebra

$$A_0 = \bigoplus_{k_i \in \mathbb{Z}} (F_0^{(1)})^{\otimes k_1} \otimes \cdots \otimes (F_0^{(n)})^{\otimes k_n}.$$

The group G operates on E_0, on X_0 provided with the sheaves $L_0^{(i)}$, and on Y_0. Taking the product of these objects with P modulo G, one obtains the sheaf E associated with P, the scheme $X = D(E) = P/B$ of flags of E, and the scheme $Y = P/U$. We denote the canonical sheaves on X by the same letters as their analogues on X_0. Thus one has the diagram

$$\begin{tikzcd}
 & Y = P/U \arrow[dr] & \\
P \arrow[ur] \arrow[dr] & & X = P/B = D(E) \arrow[dl] \\
 & S &
\end{tikzcd}$$

We note that P is a torsor over Y with structural group U_Y, and that Y is a torsor over X with structural group T_X, defined by the $\mathcal{O}_X$-algebra

$$A = \bigoplus_{k_i \in \mathbb{Z}} (F^{(1)})^{\otimes k_1} \otimes \cdots \otimes (F^{(n)})^{\otimes k_n}.$$

The homomorphism $K_\bullet(S) \to K_\bullet(P)$ therefore factors as

$$K_\bullet(S) \longrightarrow K_\bullet(X) \longrightarrow K_\bullet(Y) \longrightarrow K_\bullet(P).$$

By 3.1, $K_\bullet(X) = K_\bullet(S) \otimes_{K^\circ(S)} K^\circ(X)$. Let ξ_i be the class of $F^{(i)}$ in $K^\circ(X)$. By 2.2, $K_\bullet(Y)$ is the quotient of $K_\bullet(X)$ by the submodule generated by the $(\xi_i - 1)K_\bullet(X)$. If I is the ideal of $K^\circ(X)$ generated by the $\xi_i - 1$, then

$$K_\bullet(Y) = K_\bullet(S) \otimes_{K^\circ(S)} (K^\circ(X)/I).$$

By VI 4.8, $K^\circ(X)$ is the $K^\circ(S)$-algebra generated by the $\xi_i - 1$, subject to the relations deduced from

$$\gamma_t(E-n) = \prod_{i=1}^{n} \gamma_t(\xi_i - 1).$$

Therefore $K^\circ(X)/I = K^\circ(S)/J$, where J is the ideal generated by the $\gamma^j(E-n)$ for $j = 1, \ldots, n$, hence the λ-ideal generated by $E - n$. Thus

$$K_\bullet(Y) \simeq K_\bullet(S)/JK_\bullet(S).$$

The demonstration is completed by the following lemma.

Lemma 4.2. Let S be a noetherian scheme, let B be a parabolic subgroup of a reductive S-group, and let P be a torsor under the unipotent radical U of B. Then one has an isomorphism

$$K_\bullet(S) \xrightarrow{\sim} K_\bullet(P).$$

Indeed, there exists a finite sequence of subgroup schemes of U,

$$U_0 = U \supset U_1 \supset U_2 \supset \cdots \supset U_p = e,$$

closed in B, finite because B is noetherian, and such that, for every i, U_i/U_{i+1} is isomorphic to an S-vector group (SGA 3 XXVI 2.1). Consequently $P \to S$ factors as a sequence of torsors under vector groups, and the result follows from 3.4.

Corollary 4.3. Let S be a noetherian scheme and let $G = \mathrm{GL}(n)_S$. Then the canonical homomorphism

$$K_\bullet(S) \longrightarrow K_\bullet(G)$$

is an isomorphism.

This follows immediately from 4.1, applied to $P = G$, since then $E = \mathcal{O}_S^n$.

Corollary 4.4. Under the hypotheses of 4.1, suppose S regular. Then one has an isomorphism

$$K^\circ(S)/J \xrightarrow{\sim} K^\circ(P).$$

If S is regular, G is also regular, and P as well. Thus $K^\circ(S) \simeq K_\bullet(S)$ and $K^\circ(P) \simeq K_\bullet(P)$, whence the result follows from 4.1.

Exposé X. Formalism of Intersections on Proper Algebraic Schemes

by O. Jussila
(with an appendix by A. Grothendieck)

1. Compatibility of the filtrations with the composition law

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X).$$

1.1

First let X be a noetherian scheme, and let $K_\bullet(X)$ be the group of classes of coherent $\mathcal{O}_X$-modules. For every coherent $\mathcal{O}_X$-module F, we denote its class in $K_\bullet(X)$ by $\mathrm{cl}_\bullet(F)$.

Definition 1.1.1. For every integer $n \in \mathbb{Z}$, let $\mathrm{Filt}_n(X)$ denote the subgroup of $K_\bullet(X)$ generated by the classes $\mathrm{cl}_\bullet(F)$ of coherent $\mathcal{O}_X$-modules F such that

$$\dim(\mathrm{Supp}(F)) \leq n.$$

This defines an increasing filtration $(\mathrm{Filt}_n(X))_{n\in\mathbb{Z}}$ of $K_\bullet(X)$ such that $\mathrm{Filt}_n(X) = 0$ for $n < 0$.

Proposition 1.1.2. Let X be a noetherian scheme, let F be a coherent $\mathcal{O}_X$-module such that $\dim(\mathrm{Supp}(F)) \leq n$, and, for $n \in \mathbb{Z}$, let $X(n)$ be the subset

$$\{x \in X\,;\, \dim(\overline{\{x\}}) = n\}$$

of X. For every $x \in X$, let $Z(x)$ denote the integral closed subscheme associated with $\overline{\{x\}}$. Then the length of the $\mathcal{O}_{X,x}$-module F_x is finite for every $x \in X(n)$ and is nonzero for only finitely many $x \in X(n)$, and one has

$$\mathrm{cl}_\bullet(F) = \sum_{x\in X(n)} \mathrm{length}(F_x)\,\mathrm{cl}_\bullet(\mathcal{O}_{Z(x)}) + w, \tag{1.1.3}$$

where $w \in \mathrm{Filt}_{n-1}(X)$.

By immediate induction on n, one obtains:

Corollary 1.1.4. For every noetherian scheme X and every integer n, $\mathrm{Filt}_n(X)$ is the subgroup of $K_\bullet(X)$ generated by the classes $\mathrm{cl}_\bullet(\mathcal{O}_Z)$, where Z ranges over the integral closed subschemes of X of dimension at most n.

Proof of 1.1.2. Let $X' = \mathrm{Supp}(F)$. Since X' is a noetherian space of dimension at most n, the elements of $X' \cap X(n)$ are maximal in X' and hence finite in number; moreover, $\mathrm{length}(F_x)$ is finite for every $x \in X' \cap X(n)$. This proves the first assertion.

Let $\mathcal{K} = \mathrm{Coh}(X)$ be the category of coherent $\mathcal{O}_X$-modules, and let $\mathcal{K}'$ be the subset of $\mathrm{Ob}(\mathcal{K})$ formed by the coherent $\mathcal{O}_X$-modules concentrated on X' that satisfy condition (1.1.3). Plainly $\mathcal{K}'$ contains the sheaves $\mathcal{O}_{Z(x)}$ for $x \in X'$. It is therefore enough to show that $\mathcal{K}'$ is an "exact" subset of $\mathrm{Ob}(\mathcal{K})$, for then the devissage lemma (EGA III 3.1.2) gives $F \in \mathcal{K}'$.

Thus let

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0 \tag{1.1.5}$$

be an exact sequence of $\mathcal{O}_X$-modules having two of its terms in $\mathcal{K}'$. The sheaves G', G, and G'' are then all concentrated on X'. Hence the lengths ℓ'_x, ℓ_x, and ℓ''_x of the $\mathcal{O}_{X,x}$-modules G'_x, G_x, and G''_x are finite for every $x \in X(n)$, and

$$\ell_x = \ell'_x + \ell''_x \qquad (x \in X(n)). \tag{1.1.6}$$

Assertion (1.1.3) for all the terms of (1.1.5) now follows immediately from the hypothesis and the equation

$$\mathrm{cl}_\bullet(G) = \mathrm{cl}_\bullet(G') + \mathrm{cl}_\bullet(G''). \tag{1.1.7}$$

It is known that $K_\bullet$ is a covariant functor on the category of locally noetherian schemes with respect to proper morphisms. The following proposition asserts that our filtration is functorial.

Proposition 1.1.8. Let $f : X' \to X$ be a proper morphism of locally noetherian schemes, and let

$$f_* : K_\bullet(X') \longrightarrow K_\bullet(X)$$

be the morphism induced by f. Then

$$f_*(\mathrm{Filt}_n(X')) \subset \mathrm{Filt}_n(X) \tag{1.1.9}$$

for every integer n.

Proof. If F is a coherent $\mathcal{O}_{X'}$-module, then by definition (IV 2.11),

$$f_*(\mathrm{cl}_\bullet(F)) = \sum_{i\in\mathbb{N}} (-1)^i \,\mathrm{cl}_\bullet(R^i f_*(F)).$$

The $\mathcal{O}_X$-modules $R^i f_*(F)$ are concentrated on $f(\mathrm{Supp}(F))$, and by EGA IV 5.4.1(ii),

$$\dim(\mathrm{Supp}(F)) \geq \dim(f(\mathrm{Supp}(F))),$$

which proves the assertion.

1.2

Let now X be an arbitrary scheme, and consider the group $K^\circ(X)$ of classes of locally free $\mathcal{O}_X$-modules (henceforth understood to be of finite type). It is an augmented pre-λ-algebra over the ring $H^0(X,\mathbb{Z})$ (V 3.9.1). The map $f \rightsquigarrow f \cdot 1 \in K^\circ(X)$ allows us henceforth to identify $H^0(X,\mathbb{Z})$ with a subring of $K^\circ(X)$. If E is a locally free $\mathcal{O}_X$-module, we denote its class in $K^\circ(X)$ by $\mathrm{cl}^\circ(E)$. The operators λ^k ($k \in \mathbb{N}$) are defined by

$$\lambda^k(\mathrm{cl}^\circ(E)) = \mathrm{cl}^\circ(\wedge^k_{\mathcal{O}} E) \tag{1.2.1}$$

and by

$$\lambda^k(x+y) = \sum_{\substack{i+j=k\\ i,j\in\mathbb{N}}} \lambda^i(x)\lambda^j(y). \tag{1.2.2}$$

(V 2.2). Recall that in the λ-filtration $(\mathrm{Filt}^i(X))_{i\in\mathbb{Z}}$ of $K^\circ(X)$, one has $\mathrm{Filt}^i(X) = K^\circ(X)$ for $i \leq 0$, while for $i \geq 1$, $\mathrm{Filt}^i(X)$ is the ideal generated by the products

$$P = \gamma^{i_1}(x_1)\cdots\gamma^{i_k}(x_k) \tag{1.2.3}$$

with $1 \leq i_1, \ldots, i_k$; $i_1 + \cdots + i_k \geq i$; and $x_1, \ldots, x_k \in I = \mathrm{Ker}(\varepsilon)$, where ε is the augmentation homomorphism (cf. Exposé V 3.10). It was proved in V 3.10 that $\mathrm{Filt}^i(X)$ is even the $H^0(X,\mathbb{Z})$-submodule, hence, when X is connected, the abelian subgroup generated by the products (1.2.3), where one may moreover suppose that the elements x_p ($p = 1, \ldots, k$) have the form

$$x_p = \mathrm{cl}^\circ(E_p) - n_p, \tag{1.2.4}$$

with E_p a locally free $\mathcal{O}_X$-module of rank $n_p \in H^0(X,\mathbb{Z})$.

1.3

Recall (IV 2.10) that the bifunctor

$$(E, F) \rightsquigarrow E \otimes_{\mathcal{O}_X} F,$$

where E is a locally free $\mathcal{O}_X$-module and F is a coherent $\mathcal{O}_X$-module, induces a composition law

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X) \tag{1.3.1}$$

that makes $K_\bullet(X)$ a unital $K^\circ(X)$-module. The key result of the present exposé is the following.

Theorem 1.3.2. Let X be a noetherian scheme. For all $i, j \in \mathbb{Z}$, the composition law (1.3.1) induces by restriction a composition law

$$\mathrm{Filt}^i(X) \times \mathrm{Filt}_j(X) \longrightarrow \mathrm{Filt}_{j-i}(X). \tag{1.3.3}$$

Corollary 1.3.3. If X is a noetherian scheme of dimension at most j, then the canonical homomorphism

$$\Theta_X : K^\circ(X) \longrightarrow K_\bullet(X),$$

defined by $\Theta_X(x) = x \cdot \mathrm{cl}_\bullet(\mathcal{O}_X)$ $(x \in K^\circ(X))$, induces for every $i \in \mathbb{Z}$ a homomorphism

$$\mathrm{Filt}^i(X) \longrightarrow \mathrm{Filt}_{j-i}(X).$$

Proof. First let us show that (1.3.3), for $i + j \leq k \in \mathbb{Z}$ and every integral closed subscheme of dimension at most j, implies Theorem 1.3.2 for $i + j \leq k$. By (1.1.4), it is enough to show that, for every $x \in \mathrm{Filt}^i(X)$ and every integral closed subscheme Z of dimension at most j, one has

$$x\,\mathrm{cl}_\bullet(\mathcal{O}_Z) \in \mathrm{Filt}_{j-i}(X).$$

To avoid confusion, let 1_Z denote the class of $\mathcal{O}_Z$ in $K_\bullet(Z)$. If $\alpha : Z \to X$ is the canonical immersion, the projection formula (IV …) gives

$$x\,\mathrm{cl}_\bullet(\mathcal{O}_Z) = x\alpha_*(1_Z) = \alpha_*(\alpha^*(x)1_Z).$$

Since $\alpha^*(x) \in \mathrm{Filt}^i(Z)$, (1.1.8) and (1.3.3), applied with $X = Z$ and $i + j \leq k$, give

$$\alpha_*(\alpha^*(x)1_Z) \in \mathrm{Filt}_{j-i}(X),$$

which proves the assertion.

Theorem 1.3.2 is trivial for $i + j \leq 0$. Suppose by induction that it has been proved for $i + j < k$ and for every noetherian scheme X. By the preceding remark, it remains to prove that, for every integral noetherian scheme X of dimension j and every product (1.2.3) satisfying (1.2.4), one has

$$P\,\mathrm{cl}_\bullet(\mathcal{O}_X) \in \mathrm{Filt}_{j-i}(X). \tag{1.3.4}$$

If $k > 1$ in (1.2.3), the assertion follows from the induction hypothesis and the associativity of the law (1.3.1). We may therefore suppose, since X is connected, that

$$x = P = \gamma^m(\mathrm{cl}^\circ(E) - n), \tag{1.3.5}$$

where $m \geq i$ and E is a locally free $\mathcal{O}_X$-module of rank $n > 0$. We distinguish two cases.

I. Suppose that there exist invertible $\mathcal{O}_X$-modules $L_1, \dots, L_n$ such that

$$\mathrm{cl}^\circ(E) = \mathrm{cl}^\circ(L_1) + \cdots + \mathrm{cl}^\circ(L_n) \in K^\circ(X).$$

It follows from (V) that x is the m-th elementary symmetric function in the $\mathrm{cl}^\circ(L_r) - 1$ $(r = 1, \ldots, n)$, and hence is a homogeneous polynomial of degree m with integral coefficients in the variables $(\mathrm{cl}^\circ(L_r) - 1)$ $(r = 1, \ldots, n)$. Applying again the induction hypothesis and the associativity of the law (1.3.1), we are reduced to proving that, for every invertible $\mathcal{O}_X$-module L, one has

$$(\mathrm{cl}^\circ(L) - 1)\,\mathrm{cl}_\bullet(\mathcal{O}_X) \in \mathrm{Filt}_{j-1}(X). \tag{1.3.6}$$

Since X is integral, L and $\mathcal{O}_X$ are isomorphic to submodules of the constant $\mathcal{O}_X$-module R_X of rational functions on X. The $\mathcal{O}_X$-module sum $F = \mathcal{O}_X + L$ is coherent, and one obtains

$$\begin{aligned}(\mathrm{cl}^\circ(L) - 1)\,\mathrm{cl}_\bullet(\mathcal{O}_X) &= \mathrm{cl}_\bullet(L) - \mathrm{cl}_\bullet(\mathcal{O}_X)\\ &= (\mathrm{cl}_\bullet(F) - \mathrm{cl}_\bullet(\mathcal{O}_X)) - (\mathrm{cl}_\bullet(F) - \mathrm{cl}_\bullet(L))\\ &= \mathrm{cl}_\bullet(F/\mathcal{O}_X) - \mathrm{cl}_\bullet(F/L).\end{aligned}$$

But the fibers at the generic point of the sheaves $F/\mathcal{O}_X$ and F/L vanish; their supports in X are therefore nowhere dense (since X is irreducible), and hence have dimension at most $j-1$. Assertion (1.3.6) follows immediately.

II. It remains to prove the general case. Introduce the flag bundle

$$g : D \longrightarrow X$$

of E over X, on which there exist invertible $\mathcal{O}_D$-modules $L_1, \ldots, L_n$ such that

$$g^*(\mathrm{cl}^\circ(E)) = \mathrm{cl}^\circ(L_1) + \cdots + \mathrm{cl}^\circ(L_n) \in K^\circ(D). \tag{1.3.7}$$

Let e be the generic point of X, and let $k(e)$ be its residue field. There exists a rational point u of the generic fiber D_e over $k(e)$,

because these points are in one-to-one correspondence with flags of type $(1, \ldots, 1)$ (n terms) in the vector space E_e of dimension $n > 0$ over $k(e)$. These rational points also correspond to germs of sections of g in a neighborhood of e, so there exists such a section $s : U \to D$ over an open subset U of X. Let $X' = \overline{s(U)}$, endowed with its reduced induced structure, and let $f : X' \to X$ be the restriction of g to X'. The morphism f is projective and, since it is an isomorphism over $s(U)$, it is birational.

Thus $\dim(X') \leq \dim(X)$ (EGA IV 5.6.6.1). On the other hand, $f(X') = g(X')$ is closed and contains U, so f is surjective and $\dim(X') \geq \dim(X)$ (EGA IV 5.4.1(ii)). The hypotheses of case I are therefore satisfied for X' and $x' = f^*(x) \in \mathrm{Filt}^i(X')$, and one obtains

$$f^*(x)\,\mathrm{cl}_\bullet(\mathcal{O}_{X'}) \in \mathrm{Filt}_{j-i}(X'). \tag{1.3.7'}$$

Consequently, by the projection formula and (1.1.8),

$$x f_*(\mathrm{cl}_\bullet(\mathcal{O}_{X'})) \in \mathrm{Filt}_{j-i}(X). \tag{1.3.8}$$

On the other hand, the canonical homomorphism

$$\mathcal{O}_X \longrightarrow f_*(\mathcal{O}_{X'})$$

is an isomorphism over the dense open subset U, and the $\mathcal{O}_X$-modules $R^i f_*(\mathcal{O}_{X'})$ $(i \geq 1)$ are concentrated on the nowhere-dense closed subset $X - U$. It follows that

$$f_*(\mathrm{cl}_\bullet(\mathcal{O}_{X'})) = \mathrm{cl}_\bullet(\mathcal{O}_X) + w, \tag{1.3.9}$$

where $w \in \mathrm{Filt}_{j-1}(X)$. By the induction hypothesis, $x \cdot w \in \mathrm{Filt}_{j-i-1}(X)$; hence, by (1.3.8) and (1.3.9),

$$x\,\mathrm{cl}_\bullet(\mathcal{O}_X) \in \mathrm{Filt}_{j-i}(X).$$

This proves the assertion.

Remark 1.4. Let X be a locally noetherian scheme. Observe that, in this section, we have used only the following properties of the "dimension function"

$$D(x) = D_Y(x) = \dim(\overline{\{x\}})$$

on proper X-schemes Y:

(i) For every proper X-scheme Y and every $x \in Y$, one has $D_Y(x) \in \mathbb{Z}$.

(ii) If $y \neq x$ is a specialization of x, then $D(y) < D(x)$.

(iii) For every X-morphism $f : Y_1 \to Y_2$ of proper X-schemes and every $y \in Y_1$, one has

$$D_{Y_2}(f(y)) \leq D_{Y_1}(y).$$

(iv) Equality holds in (iii) if f is an immersion, or if Y_1 is irreducible, f is birational and surjective, and y is the generic point of Y_1.

Thus all the results of this section, in particular (1.3.2) and (1.3.3), remain valid unchanged if "dimension" is replaced by a function D satisfying conditions (i)–(iv) above and the filtration of $K_\bullet(X)$ is defined using such a function D.

Example 1.5. Let S be a locally noetherian universally catenary scheme (for example, a subscheme of a Cohen–Macaulay scheme). Put

$$D_S(x) = n - \operatorname{codim}(\overline{\{x\}}, S) = n - \dim \mathcal{O}_{S,x} \qquad (x \in S), \tag{1.5.1}$$

and, for every proper S-scheme $s_Y : Y \to S$ and every $y \in Y$, also put

$$D_Y(y) = D_S(s_Y(y)) + \operatorname{deg.tr}_{k(s_Y(y))} k(y). \tag{1.5.2}$$

If $f : Y_1 \to Y_2$ is an S-morphism of proper S-schemes, the additivity of transcendence degrees and (1.5.2) immediately give

$$D_{Y_1}(y) = D_{Y_2}(f(y)) + \operatorname{deg.tr}_{k(f(y))} k(y) \qquad (y \in Y_1). \tag{1.5.3}$$

We show that the functions D_Y so defined satisfy conditions (i)–(iv) of 1.4. By (1.5.3), condition (ii) is the only nontrivial one. In view of (iv), we may plainly suppose that $Y = \overline{\{x\}}$. Factor the structural morphism s_Y through its image $s_Y(Y) = Z \hookrightarrow S$, say $s_Y = g \circ f$:

$$Y \xrightarrow{f} Z \xhookrightarrow{g} S,$$

where Y and Z are irreducible, f is proper and surjective, and g is a closed immersion. Put $x' = s_Y(x)$ and $y' = s_Y(y)$. From (1.5.2) and (1.5.1) one obtains

$$\begin{aligned} D_Y(x) - D_Y(y) = {} & \dim \mathcal{O}_{S,y'} - \dim \mathcal{O}_{S,x'} \\ & + \operatorname{deg.tr}_{k(x')} k(x) - \operatorname{deg.tr}_{k(y')} k(y). \end{aligned}$$

Since S is catenary, one has

$$\dim \mathcal{O}_{S,y'} - \dim \mathcal{O}_{S,x'} = \dim \mathcal{O}_{Z,y'} - \dim \mathcal{O}_{Z,x'}$$

(EGA 0_{IV} 14.3.2.1), whence

$$D_Y(x) - D_Y(y) = [\dim \mathcal{O}_{Z,y'} - \deg.\operatorname{tr}_{k(y')} k(y)] \\ - [\dim \mathcal{O}_{Z,x'} - \deg.\operatorname{tr}_{k(x')} k(x)]. \tag{1.5.4}$$

Since Z is universally catenary, EGA IV 5.6.5.1 applies and gives

$$C_1 = \dim \mathcal{O}_{Y,y} - e, \qquad C_2 = \dim \mathcal{O}_{Y,x} - e,$$

where C_1 and C_2 are the two terms on the right of (1.5.4). It follows immediately from (1.5.4) that

$$D_Y(x) - D_Y(y) = \dim \mathcal{O}_{Y,y} - \dim \mathcal{O}_{Y,x} > 0,$$

as required.

Thus (1.3.2) and (1.3.3) remain valid for a noetherian universally catenary scheme X of dimension n if the filtration of $K_\bullet(X)$ is defined by means of (1.5.1), according to the codimension of supports. This filtration coincides with that of (1.1.1) when, in addition, X is biequidimensional.

1.6

One may ask whether it is always possible, on a noetherian scheme X of finite dimension, to define a function D possessing property (i) of 1.4 and also the following property, stronger than (ii):

> (ii bis) If $x \in X$ and $y \neq x$ is an immediate specialization of x, then $D(x) = D(y) + 1$.

For this it is plainly necessary that X be catenary, and sufficient that X be biequidimensional (EGA 0_{IV} 14.3.3). There nevertheless exist noetherian schemes of finite dimension that are catenary and equidimensional but admit no such function D. To see this, modify the example of EGA IV 5.6.11 as follows. Take two copies Y_1 and Y_2 of the spectrum of the ring E of loc. cit., and let y_i and z_i ($i = 1, 2$) be the closed points of Y_i corresponding respectively to the maximal ideals $\mathfrak{m}$ and $\mathfrak{m}'$ of E, of heights 2 and 1. Recall that the residue fields of $\mathfrak{m}$ and $\mathfrak{m}'$ are isomorphic. One can therefore construct a scheme X by gluing Y_1 and Y_2, identifying y_1 with z_2 and y_2 with z_1. The schemes Y_1 and Y_2 may then be identified with the two irreducible components of X; inside Z, $Y_1 \cap Y_2 = \{y_1, y_2\}$ has dimension zero. Thus X is a noetherian, catenary, equidimensional scheme of dimension 2. If (ii bis) held on X, one would have

$$D(y_1) = D(z_1) - 1 = D(y_2) - 1,$$

and, on the other hand,

$$D(y_1) = D(z_2) = D(y_2) + 1,$$

which is a contradiction.

2. Snapper polynomials

2.1

We first recall the definition and a few elementary properties of polynomial maps.

Definition 2.1.1. Let f be a map from an abelian group A to an abelian group B. We say that f is a polynomial map if, for every sequence $x_1, \ldots, x_r$ ($r \geq 1$) of elements of A, there is a family

$$\big(y(x_1, \ldots, x_r; k_1, \ldots, k_r)\big)_{k_1,\ldots,k_r \geq 0}$$

of elements of B, almost all zero, satisfying the following condition: for every sequence $n_1, \ldots, n_r$ of integers, one has

$$f(n_1x_1 + \cdots + n_rx_r) = \sum_{k_1,\ldots,k_r \geq 0} n_1^{k_1} \cdots n_r^{k_r} y(x_1, \ldots, x_r; k_1, \ldots, k_r).$$

We say that f has degree $\leq m$ if, for every integer $r \geq 1$ and all $x_1, \ldots, x_r \in A$, one has

$$y(x_1, \ldots, x_r; k_1, \ldots, k_r) = 0$$

whenever $k_1 + \cdots + k_r > m$.

Lemma 2.1.2. Every homomorphism of abelian groups is a polynomial map of degree ≤ 1. If $h = f \circ g$, where f is a polynomial map of degree p and g is a polynomial map of degree q, then h is a polynomial map of degree $\leq pq$.

The proof is trivial.

Lemma 2.1.3. Let

$$0 \longrightarrow G' \longrightarrow G \xrightarrow{j} G'' \longrightarrow 0$$

be an exact sequence of abelian groups such that G' is p-divisible for some prime p and G'' is a torsion group. Then every polynomial map $f : G \to \mathbb{Q}$ that factors through $\mathbb{Z}$ is constant.

Proof. It is enough to show that $f(x) = f(0)$ for every $x \in G$.

(a) Suppose first that x is a torsion element, say $mx = 0$ ($m \in \mathbb{Z}$, $m \neq 0$). Then the polynomial $P_x(n) = f(nx) - f(0)$ has infinitely many roots, namely all integers divisible by m. Thus P_x is identically zero, and $f(x) = f(0)$.

(b) Suppose now that x is an element of infinite order in G, divisible by every power p^k ($k \in \mathbb{Z}$). Consider the polynomials P_k defined by

$$P_k(n) = f(nx/p^k) = \sum_{i \in \mathbb{N}} n^i y(x/p^k; i) \in \mathbb{Z} \qquad (n \in \mathbb{Z};\ y(x/p^k; i) \in \mathbb{Q}).$$

For every $n \in \mathbb{Z}$, one has the identity

$$P_k(n) = P_{k+1}(pn),$$

that is,

$$\sum_{i \in \mathbb{N}} n^i y(x/p^k; i) = \sum_{i \in \mathbb{N}} p^i n^i y(x/p^{k+1}; i).$$

Comparing coefficients gives

$$y(x/p^k; i) = p^i y(x/p^{k+1}; i) \qquad (i \in \mathbb{N}).$$

Every polynomial $P : \mathbb{Q} \to \mathbb{Q}$ is continuous for the topology induced by $\mathbb{R}$. Thus, for k sufficiently large,

$$|P_x(n/p^k) - P_x(0)| < 1.$$

But, by definition, $P_x(n/p^k)$ is an integer for all $n, k \in \mathbb{Z}$. It follows that $P_x(n/p^k) - P_x(0) = 0$ for sufficiently large k. Hence P_x is constant and, in particular, $P_x(1) = P_x(0)$, i.e. $f(x) = f(0)$.

The two cases above show that the restriction of f to G' is constant. We now show that f is constant on each coset of G' in G. For every $b \in G$, let $g_b : G' \to \mathbb{Q}$ be defined by

$$g_b(y) = f(b + y) \qquad (y \in G').$$

The map g_b is a polynomial map from G' to $\mathbb{Q}$ that factors through $\mathbb{Z}$, and is therefore constant by the preceding argument. Thus f is constant on the cosets of G' in G. Consequently one may write $f = h \circ j$, where $h : G'' \to \mathbb{Q}$ is a polynomial map that factors through $\mathbb{Z}$. Since G'' is a torsion group, h is constant. Therefore f is constant.

Lemma 2.1.4. Let A be a unital commutative ring and I a nilpotent ideal of A. Let P be the multiplicative subgroup of A^* consisting of the invertible elements of A of the form $x = 1 + y$ with $y \in I$, and let $i : P \to A$ be the natural injection.

Then the composite map

$$\alpha : P \xrightarrow{i} A \xrightarrow{j} A \otimes_{\mathbb{Z}} \mathbb{Q},$$

where $j(x) = x \otimes 1$, is a polynomial map. If $I^{m+1} = 0$, then α has degree $\leq m$.

Proof. Let $x_1, \ldots, x_r$ be a sequence of elements of P, with $x_i = 1 + y_i$ $(i = 1, \ldots, r)$. The binomial formula gives, for every $n_1, \ldots, n_r \in \mathbb{Z}$,

$$x_1^{n_1} \cdots x_r^{n_r} = \sum_{k_1, \ldots, k_r \geq 0} \binom{n_1}{k_1} \cdots \binom{n_r}{k_r} y_1^{k_1} \cdots y_r^{k_r}, \tag{2.1.5}$$

where the sum on the right is finite because the elements $y_1, \ldots, y_r$ are nilpotent. The coefficients $\binom{n_1}{k_1} \cdots \binom{n_r}{k_r}$ are polynomials with rational coefficients in $n_1, \ldots, n_r$, of degree $k_1 + \cdots + k_r$; hence α is indeed a polynomial map. If $I^{m+1} = 0$, then $y_1^{k_1} \cdots y_r^{k_r} = 0$ whenever $k_1 + \cdots + k_r > m$. Thus α has degree $\leq m$.

2.2

Now let X be a noetherian scheme and let $y \in \operatorname{Filt}_j(X)$ (1.1.1). The assignment $x \mapsto xy$ defines a homomorphism of $K^\circ(X)$-modules

$$\Theta_y : K^\circ(X) \longrightarrow K_\bullet(X),$$

which, by (1.3.2), factors through the canonical epimorphism

$$j : K^\circ(X) \longrightarrow K^\circ(X)/\operatorname{Filt}^{j+1}(X) = A.$$

Let $\beta_y : A \to K_\bullet(X)$ be the homomorphism so defined. If I denotes the ideal

$$I = \operatorname{Filt}^1(X)/\operatorname{Filt}^{j+1}(X)$$

of A, then $I^{j+1} = 0$. Keeping the notation of Lemma 2.1.4, one obtains a homomorphism of abelian groups

$$\gamma : \operatorname{Pic}(X) \longrightarrow P$$

that sends the class $\operatorname{cl}_P(L)$ of an invertible $\mathcal{O}_X$-module L in $\operatorname{Pic}(X)$ to the element $j(\operatorname{cl}^\circ(L))$ of P.

Lemmas 2.1.2 and 2.1.4 immediately give:

Proposition 2.2.1. For every noetherian scheme X and every $y \in \operatorname{Filt}_j(X)$ $(j \in \mathbb{Z})$, the composite map

$$\operatorname{Pic}(X) \longrightarrow K_\bullet(X) \otimes_{\mathbb{Z}} \mathbb{Q},$$

defined by

$$\operatorname{cl}_P(L) \longmapsto (\operatorname{cl}^\circ(L) y) \otimes 1,$$

is a polynomial map of degree $\leq j$.

2.3

If X is a proper scheme over a field k and $g : X \to \operatorname{Spec}(k)$ is the structural morphism, then the homomorphism

$$g_* : K_\bullet(X) \longrightarrow K_\bullet(\operatorname{Spec}(k)) \simeq \mathbb{Z}$$

sends the class $\mathrm{cl}_\bullet(F)$ of every coherent $\mathcal{O}_X$-module F to the Euler–Poincaré characteristic $\chi(F) \in \mathbb{Z}$ of F on X. Proposition 2.2.1 immediately gives:

Proposition 2.3.1. For every scheme X proper over k and every $y \in \mathrm{Filt}_j(X)$, the map

$$S_y : \operatorname{Pic}(X) \longrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q},$$

defined by

$$\mathrm{cl}_P(L) \longmapsto g_*(\mathrm{cl}^\circ(L)y) = \chi(\mathrm{cl}^\circ(L)y),$$

is a polynomial map of degree $\leq j$.

When $L_1, \ldots, L_n$ are invertible $\mathcal{O}_X$-modules and F is a coherent $\mathcal{O}_X$-module such that $\dim(\operatorname{Supp}(F)) = j$, Proposition 2.3.1 implies in particular that the Euler–Poincaré characteristic

$$\chi(L_1^{\otimes m_1} \otimes \cdots \otimes L_n^{\otimes m_n} \otimes F) \tag{2.3.2}$$

is a numerical polynomial in $m_1, \ldots, m_n \in \mathbb{Z}$, of degree $\leq \dim(\operatorname{Supp}(F))$, sometimes called the Snapper polynomial of F and the sequence $L_1, \ldots, L_n$.

2.4

Recall [2] that, for the fpqc topology on Sch/k, the Picard functor $\operatorname{Pic}_{X/k}$ is the sheaf of groups associated with the presheaf

$$T \longmapsto \operatorname{Pic}(X \times_k T) \qquad (T \in \operatorname{Ob}(\mathrm{Sch}/k)).$$

When k is algebraically closed, the canonical homomorphism

$$\operatorname{Pic}(X) \longrightarrow \operatorname{Pic}_{X/k}(\operatorname{Spec}(k))$$

is an isomorphism.

When X is proper over k, it is known [4] that the Picard functor is representable by a group scheme, locally of finite type over k, also denoted $\operatorname{Pic}_{X/k}$. The neutral component $\operatorname{Pic}^0_{X/k}$ of $\operatorname{Pic}_{X/k}$ is a connected commutative algebraic group over k.

Definition 2.4.1. Let X be a scheme proper over a field k, and let L be an invertible $\mathcal{O}_X$-module (respectively, let $x \in \operatorname{Pic}(X)$). We say that L (respectively, x) is τ-equivalent to zero if there is an integer $n \neq 0$ such that $L^{\otimes n}$ (respectively, nx) is algebraically equivalent to zero. The elements of $\operatorname{Pic}(X)$ algebraically equivalent (respectively, τ-equivalent) to zero form a subgroup $\operatorname{Pic}^0(X)$ (respectively, $\operatorname{Pic}^\tau(X)$) of $\operatorname{Pic}(X)$.

Lemma 2.4.3. For every proper scheme X over an algebraically closed field k, the group $\operatorname{Pic}^0(X)$ is divisible by every integer n prime to the characteristic of k.

Proof. Since k is algebraically closed, one has

$$\operatorname{Pic}^0(X) \simeq \operatorname{Pic}^0_{X/k}(k).$$

Since $\operatorname{Pic}^0_{X/k}$ is connected and of finite type, the n-th-power endomorphism is surjective when n is prime to the characteristic of k (SGA 3 XV 1.3 and VI 1.3.2). Since k is algebraically closed, the corresponding endomorphism of $\operatorname{Pic}^0_{X/k}(k)$ is surjective.

It follows from 2.4.3 that, in the exact sequence

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^\tau(X) \longrightarrow T \longrightarrow 0,$$

$\mathrm{Pic}^0(X)$ is p-divisible for some prime p and T is a torsion group. Thus Lemma 2.1.3 applies to the restriction of the polynomial map (2.3.1) to $\mathrm{Pic}^\tau(X)$. One obtains:

Proposition 2.4.4. The polynomial map S_y of (2.3.1) is constant on the cosets of $\mathrm{Pic}^\tau(X)$ in $\mathrm{Pic}(X)$.

Proof. If K is an algebraically closed extension of k, one has a commutative diagram

$$\begin{tikzcd}
\mathrm{Pic}(X) \arrow[r, "S_y"] \arrow[d] & K_\bullet(\mathrm{Spec}(k)) \arrow[r] & \mathbb{Z} \arrow[r, hook] & \mathbb{Q} \\
\mathrm{Pic}(X_K) \arrow[r, "\overline{S}_y"'] & K_\bullet(\mathrm{Spec}(K)) \arrow[ur] & &
\end{tikzcd}$$

where $\overline{S}_y$ is the polynomial map on $\mathrm{Pic}(X_K)$ corresponding to S_y. Hence one may suppose that k is algebraically closed. The assertion then follows immediately from Lemmas 2.1.3 and 2.4.3.

Remark 2.4.5. In Exposé XIII this result will be proved without using representability of the functor $\mathrm{Pic}_{X/k}$.

3. Projection formulas for the associated graded groups

3.1

Let X be a noetherian scheme. Denote by $\mathrm{Gr}^\circ(X)$ the graded group associated with the λ-filtration of $K^\circ(X)$, and by $\mathrm{Gr}_\bullet(X)$ the graded group associated with the filtration 1.1.1 of $K_\bullet(X)$. By Theorem 1.3.2, the composition law (1.3.1)

$$K^\circ(X) \times K_\bullet(X) \longrightarrow K_\bullet(X)$$

induces, on passing to the associated graded groups, composition laws

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_j(X) \longrightarrow \mathrm{Gr}_{j-i}(X) \qquad (i, j \in \mathbb{Z}). \tag{3.1.1}$$

Thus $\mathrm{Gr}_\bullet(X)$ becomes a module over the ring $\mathrm{Gr}^\circ(X)$, and even a graded module over the graded ring $\mathrm{Gr}^\circ(X)$, provided that the sign of the grading of $\mathrm{Gr}_\bullet(X)$ is changed by putting

$$\mathrm{Gr}^i_\bullet(X) = \mathrm{Gr}_{-i}(X) \qquad (i \in \mathbb{Z}).$$

Now let

$$f : X' \longrightarrow X$$

be a proper morphism of noetherian schemes. Then f induces a homomorphism

$$f^* : \mathrm{Gr}^\circ(X) \longrightarrow \mathrm{Gr}^\circ(X')$$

of graded rings (Exposé V), and, by Proposition 1.1.8, a homomorphism

$$f_* : \mathrm{Gr}_\bullet(X') \longrightarrow \mathrm{Gr}_\bullet(X)$$

of graded abelian groups. Passing the projection formula (IV 2.11) and (3.1.1) to the associated graded groups gives the graded projection formula

$$x \cdot f_*(y) = f_*(f^*(x) \cdot y) \qquad (x \in \mathrm{Gr}^\circ(X),\ y \in \mathrm{Gr}_\bullet(X')). \tag{3.1.2}$$

3.2

Let $f : X' \to X$ be a projective complete-intersection morphism (Exposé VIII), of virtual dimension r, between noetherian schemes, with X carrying an ample invertible $\mathcal{O}_X$-module. In Exposé VIII it was proved that f induces a homomorphism

$$f_* : \mathrm{Filt}^k(X')_{\mathbb{Q}} \longrightarrow \mathrm{Filt}^{k-r}(X)_{\mathbb{Q}} \qquad (k \in \mathbb{Z}), \tag{3.2.1}$$

which moreover satisfies the projection formula (Exposé IV):

$$f_*(f^*(x) \cdot y) = x \cdot f_*(y) \qquad (x \in K^\circ(X)_{\mathbb{Q}},\ y \in K^\circ(X')_{\mathbb{Q}}). \tag{3.2.2}$$

Passing to the associated graded groups in (3.2.1) and (3.2.2), one obtains a homomorphism of degree $-r$:

$$f_* : \mathrm{Gr}^\circ(X')_{\mathbb{Q}} \longrightarrow \mathrm{Gr}^\circ(X)_{\mathbb{Q}}, \tag{3.2.3}$$

and, by Theorem 1.3.2, a graded projection formula

$$f_*(f^*(x) \cdot y) = x \cdot f_*(y) \qquad (x \in \mathrm{Gr}^\circ(X)_{\mathbb{Q}},\ y \in \mathrm{Gr}^\circ(X')_{\mathbb{Q}}).$$

3.3

Finally, let $f : X' \to X$ be a morphism of finite Tor-dimension between noetherian schemes. It is known (Exposé IV) that f induces a homomorphism

$$f^* : K_\bullet(X) \longrightarrow K_\bullet(X')$$

such that, for every coherent $\mathcal{O}_X$-module F,

$$f^*(\mathrm{cl}_\bullet(F)) = \sum_i (-1)^i \,\mathrm{cl}_\bullet(\mathrm{Tor}_i^{\mathcal{O}_X}(F, \mathcal{O}_{X'})).$$

The $\mathcal{O}_{X'}$-modules $\mathrm{Tor}_i^{\mathcal{O}_X}(F, \mathcal{O}_{X'})$ are supported on $f^{-1}(\mathrm{Supp}(F))$. If the fibers of f are everywhere of dimension $\leq d$, then (EGA IV 5.6.7)

$$\dim(f^{-1}(\mathrm{Supp}(F))) \leq \dim(\mathrm{Supp}(F)) + d.$$

It follows that f induces, for every $k \in \mathbb{Z}$, a homomorphism

$$f^* : \mathrm{Filt}_k(X) \longrightarrow \mathrm{Filt}_{k+d}(X'),$$

hence a homomorphism of degree d of graded abelian groups.

Thus one has

$$f^* : \mathrm{Gr}_\bullet(X) \longrightarrow \mathrm{Gr}_\bullet(X'). \tag{3.3.1}$$

3.4

Suppose finally that the morphisms (3.2.3) and (3.3.1) are both defined for a morphism $f : X' \to X$ and that $r = d$. This is the case, for example, if X is quasi-compact with an ample invertible $\mathcal{O}_X$-module and if $X' = \mathbf{P}_X(E)$, where E is a locally free $\mathcal{O}_X$-module of rank $r+1$. The projection formula in the derived category then gives, for every $x \in \mathrm{Filt}^i(X')_{\mathbb{Q}}$ and $y \in \mathrm{Filt}_j(X)_{\mathbb{Q}}$,

$$f_*(x \cdot f^*(y)) = f_*(x) \cdot y \in \mathrm{Filt}_{j+r-i}(X)_{\mathbb{Q}}.$$

When X is noetherian, Theorem 1.3.2 yields the corresponding graded formula.

4. Intersection numbers

4.1

Let X be a proper scheme over a field k. Composing the product (3.1.1)

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathrm{Gr}_0(X) = \mathrm{Filt}_0(X) \qquad (i \in \mathbb{Z})$$

with the Euler–Poincaré characteristic

$$\chi : \mathrm{Filt}_0(X) \longrightarrow \mathbb{Z}$$

gives a pairing called the *intersection number*:

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \longrightarrow \mathbb{Z}. \tag{4.1.1}$$

We denote this pairing by

$$(x, y) \longmapsto \langle x, y\rangle \in \mathbb{Z} \qquad (i \in \mathbb{Z},\ x \in \mathrm{Gr}^i(X),\ y \in \mathrm{Gr}_i(X)).$$

Example 4.2. Suppose that X has dimension n. Let $y \in \mathrm{Gr}_n(X)$, for example $y = \mathrm{cl}_\bullet(\mathcal{O}_Z)$, where Z is an integral closed subscheme of X of dimension n. Let P also be a homogeneous polynomial of weight n,

$$P \in \mathbb{Z}[T_1, \ldots, T_n],$$

where T_i is assigned weight i ($i = 1, \ldots, n$); for example, P may contain a monomial $T_{i_1} \cdots T_{i_r}$ with $i_1 + \cdots + i_r = n$. Substituting for each variable T_i the Chern class

$$c_i(x) = \mathrm{cl}^i(\gamma^i(x - \varepsilon(x))) \in \mathrm{Gr}^i(X) \qquad (x \in K^\circ(X))$$

gives an element

$$P(c_1(x), \ldots, c_n(x)) \in \mathrm{Gr}^n(X).$$

The integer

$$P_y(x) = \langle P(c_1(x), \ldots, c_n(x)), y\rangle$$

is called the *Chern number* of x relative to y. When $y = \mathrm{cl}_\bullet(\mathcal{O}_X)$, the reference to y is omitted from this terminology.

4.3

Without any hypothesis on the dimension of X, consider invertible $\mathcal{O}_X$-modules $\mathcal{L}_1, \ldots, \mathcal{L}_d$ and a coherent $\mathcal{O}_X$-module $\mathcal{F}$ such that $\dim(\mathrm{Supp}(\mathcal{F})) \le d$. Put

$$L_i = \mathrm{cl}^\circ(\mathcal{L}_i) \quad (i = 1, \ldots, d), \qquad F = \mathrm{cl}_\bullet(\mathcal{F}).$$

Proposition 4.3.1. With the preceding notation, for every sequence of nonnegative integers $\alpha_1, \ldots, \alpha_d$ such that $\sum_{i=1}^d \alpha_i = d$, the coefficient of

$$\binom{n_1}{\alpha_1} \cdots \binom{n_d}{\alpha_d}$$

in the Snapper polynomial

$$\chi(\mathcal{L}_1^{\otimes n_1} \otimes \cdots \otimes \mathcal{L}_d^{\otimes n_d} \otimes \mathcal{F})$$

is equal to

$$\langle c_1(L_1)^{\alpha_1} \cdots c_1(L_d)^{\alpha_d}, |F| \rangle, \tag{4.3.2}$$

where $|F|$ is the class of F in $\mathrm{Gr}_d(X)$. Thus, in the ordinary expansion of this polynomial in monomials in the n_i $(i = 1, \dots, d)$, the coefficient of $n_1^{\alpha_1} \cdots n_d^{\alpha_d}$ is

$$\frac{1}{\alpha_1! \cdots \alpha_d!} \langle c_1(L_1)^{\alpha_1} \cdots c_1(L_d)^{\alpha_d}, |F| \rangle.$$

In particular, one obtains immediately:

Corollary 4.3.3. The coefficient of the monomial $n_1 n_2 \cdots n_d$ in the Snapper polynomial

$$\chi(\mathcal{L}_1^{\otimes n_1} \otimes \cdots \otimes \mathcal{L}_d^{\otimes n_d} \otimes \mathcal{F})$$

is equal to

$$\langle c_1(L_1) \cdots c_1(L_d), |F| \rangle = \big(c_1(L_1) \cdots c_1(L_d), |F|\big). \tag{4.3.4}$$

Proof. Put $L_i = 1 + M_i$ for $i = 1, \dots, d$. By the definitions, the integer (4.3.2) is equal to

$$\chi(M_1^{\alpha_1} \cdots M_d^{\alpha_d} F).$$

Substituting

$$A = K^\circ(X)/\operatorname{Filt}^{d+1}(X), \qquad I = \operatorname{Filt}^1(X)/\operatorname{Filt}^{d+1}(X)$$

in Lemma 2.1.4, formula (2.1.5) gives, since $M_i \in \operatorname{Filt}^1(X)$ for every $i = 1, \dots, d$, the congruence

$$L_1^{n_1} \cdots L_d^{n_d} \equiv \sum_{\substack{\alpha_1, \dots, \alpha_d \geq 0 \\ \alpha_1 + \cdots + \alpha_d \leq d}} \binom{n_1}{\alpha_1} \cdots \binom{n_d}{\alpha_d} M_1^{\alpha_1} \cdots M_d^{\alpha_d} \pmod{\operatorname{Filt}^{d+1}(X)}. \tag{4.3.5}$$

Hence the coefficient of $\binom{n_1}{\alpha_1} \cdots \binom{n_d}{\alpha_d}$ in $\chi(\mathcal{L}_1^{\otimes n_1} \otimes \cdots \otimes \mathcal{L}_d^{\otimes n_d} \otimes \mathcal{F})$ is $\chi(M_1^{\alpha_1} \cdots M_d^{\alpha_d} F)$, as required.

4.4

The integer (4.3.4) is sometimes called the *intersection number* of the invertible $\mathcal{O}_X$-modules $\mathcal{L}_1, \dots, \mathcal{L}_d$ with $\mathcal{F}$, and is denoted by

$$(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{F}) \qquad (\textit{cf.}\ [3]).$$

Expression (4.3.4) immediately gives the following properties.

1) The integer (4.3.4) is a symmetric d-linear function of the classes $L_1, \dots, L_d \in \operatorname{Pic}(X)$, is zero if $\dim(\operatorname{Supp}(\mathcal{F})) < d$, and is additive in $\mathcal{F}$, that is, in its class $\mathrm{cl}_\bullet(\mathcal{F}) = F$.

2) By Proposition 1.1.2, one obtains

$$(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{F}) = \sum_{x \in X(d)} \operatorname{length}_{\mathcal{O}_{X,x}}(\mathcal{F}_x)(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{O}_{Z(x)}). \tag{4.4.1}$$

3) Let Y be a closed subscheme of X, and let $\mathcal{G}$ be a coherent $\mathcal{O}_Y$-module such that $\mathcal{F} = f_*(\mathcal{G})$, where $f : Y \to X$ is the canonical immersion. Then

$$(f^*\mathcal{L}_1, \dots, f^*\mathcal{L}_d; \mathcal{G}) = (\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{F}).$$

More generally, the projection formula gives

$$\chi(f^*(x) \cdot y) = \chi\big(f_*(f^*(x) \cdot y)\big) = \chi(x \cdot f_*(y)) \qquad (x \in K^\circ(X),\ y \in K_\bullet(Y)) \tag{4.4.2}$$

for every k-morphism $f : Y \to X$ of proper k-schemes.

4) Suppose that, in the morphism $f : Y \to X$ above, X and Y are irreducible of dimension $\leq d$. If f has degree n, then

$$(f^*\mathcal{L}_1, \dots, f^*\mathcal{L}_d; \mathcal{O}_Y) = n(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{O}_X). \tag{4.4.3}$$

Recall the definition of the degree n of f [3]: it is zero unless

$$\dim(X) = \dim(Y) = \dim(f(Y)).$$

In the latter case, one puts

$$n = \frac{\mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{Y,y})}{\mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x})}, \tag{4.4.4}$$

where x is the generic point of X and y is the generic point of Y.

In the proof of (4.4.3), the case $\dim(X) = \dim(Y)$ is the only nontrivial one. If $\dim(f(Y)) < d$, then

$$R^i f_*(\mathcal{O}_Y) = 0 \tag{4.4.5}$$

for $i > 0$ in a neighborhood of x. Condition (4.4.5) also holds when $\dim(f(Y)) = d$, since the generic fiber then has dimension zero (EGA III 4.2.2 and EGA IV 5.5.6). Proposition 1.1.2 therefore gives the congruences

$$f_*(\mathrm{cl}_\bullet(\mathcal{O}_Y)) \equiv \mathrm{cl}_\bullet(f_*\mathcal{O}_Y) \equiv \mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{Y,y})\,\mathrm{cl}_\bullet(\mathcal{O}_{X_{\mathrm{red}}}) \pmod{\mathrm{Filt}_{d-1}(X)}.$$

By (4.4.2), it follows that

$$(f^*\mathcal{L}_1, \dots, f^*\mathcal{L}_d; \mathcal{O}_Y) = \mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{Y,y})(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{O}_{X_{\mathrm{red}}}).$$

On the other hand, item 2 gives

$$(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{O}_X) = \mathrm{length}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x})(\mathcal{L}_1, \dots, \mathcal{L}_d; \mathcal{O}_{X_{\mathrm{red}}}),$$

which proves (4.4.3).

4.5

The intersection product $\langle\, ,\, \rangle$ defines an equivalence relation called *numerical equivalence*.

Definition 4.5.1. Two elements x and y of $\mathrm{Gr}^i(X)$ ($i \in \mathbb{Z}$), respectively of $\mathrm{Gr}_i(X)$, are said to be *numerically equivalent* if they define the same homomorphism

$$\mathrm{Gr}_i(X) \longrightarrow \mathbb{Z}, \qquad \text{respectively} \qquad \mathrm{Gr}^i(X) \longrightarrow \mathbb{Z}.$$

Two invertible $\mathcal{O}_X$-modules are said to be numerically equivalent if their first Chern classes in $\mathrm{Gr}^1(X)$ are numerically equivalent.

Proposition 4.5.2. An invertible $\mathcal{O}_X$-module $\mathcal{L}$ is numerically equivalent to zero if and only if

$$\chi(\mathcal{L}^{\otimes n} \otimes_{\mathcal{O}_X} \mathcal{O}_Z) = \chi(\mathcal{O}_Z) \tag{4.5.3}$$

for every $n \in \mathbb{Z}$ and every integral closed curve $Z \subset X$.

This follows from Propositions 1.1.2 and 4.3.1.

Let $\mathrm{Pic}^N(X)$ denote the subgroup of $\mathrm{Pic}(X)$ consisting of the classes of invertible $\mathcal{O}_X$-modules numerically equivalent to zero. Proposition 2.4.4 gives:

Corollary 4.5.3.

$$\mathrm{Pic}^\tau(X) \subset \mathrm{Pic}^N(X).$$

It will be proved in Exposé XIII that in fact

$$\mathrm{Pic}^N(X) = \mathrm{Pic}^\tau(X).$$

5. The isomorphism $\mathrm{Pic}(X) \simeq \mathrm{Gr}^1(X)$

5.1

Let X be an arbitrary ringed topos. The ring $K^\circ(X)$ and the filtration $(\mathrm{Filt}^i(X))_{i\in\mathbb{Z}}$ can be defined exactly as for schemes. Likewise, define $\mathrm{Pic}(X)$ to be the group of isomorphism classes of invertible $\mathcal{O}_X$-modules. Let

$$i : \mathrm{Pic}(X) \longrightarrow K^\circ(X) \tag{5.1.1}$$

be the map sending the class $\mathrm{cl}_P(\mathcal{L}) \in \mathrm{Pic}(X)$ of an invertible $\mathcal{O}_X$-module $\mathcal{L}$ to $\mathrm{cl}^\circ(\mathcal{L}) \in K^\circ(X)$. This map is plainly a homomorphism from $\mathrm{Pic}(X)$ to the multiplicative group $K^\circ(X)^*$ of units of $K^\circ(X)$. Henceforth the group law of $\mathrm{Pic}(X)$ is written multiplicatively.

The map i has a left inverse defined by the determinant functor. If $\mathcal{E}$ is a locally free $\mathcal{O}_X$-module of rank n, put

$$\det(\mathcal{E}) = \bigwedge\nolimits^n \mathcal{E}.$$

If

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}'' \longrightarrow 0 \tag{5.1.2}$$

is an exact sequence of locally free $\mathcal{O}_X$-modules, there is a canonical isomorphism

$$\det(\mathcal{E}) \xrightarrow{\sim} \det(\mathcal{E}') \otimes_{\mathcal{O}_X} \det(\mathcal{E}''),$$

arising from (6.2.9) and the fact that (5.1.2) is locally split. One therefore obtains a homomorphism of abelian groups

$$\det : K^\circ(X) \longrightarrow \mathrm{Pic}(X) \tag{5.1.3}$$

which is the required left inverse of i. In particular, i is injective, and henceforth $\mathrm{Pic}(X)$ is identified with its image in $K^\circ(X)^*$.

5.2

Define another map

$$\delta : \mathrm{Pic}(X) \longrightarrow \mathrm{Filt}^1(X) \tag{5.2.1}$$

by putting, for every $L \in \mathrm{Pic}(X)$,

$$\delta(L) = 1 - L^{-1}.$$

Remarks 5.2.2.

a) Suppose that X is the locally ringed topos of Zariski sheaves on a scheme S. If D is an effective Cartier divisor on S and

$$L = \mathrm{cl}^\circ(\mathcal{O}_S(D)),$$

then

$$\Theta_S(\delta(L)) = \mathrm{cl}_\bullet(\mathcal{O}_D), \tag{5.2.3}$$

where $\Theta_S : K^\circ(X) \to K_\bullet(S)$ is the canonical homomorphism of 1.2.3. This follows immediately from the exact sequence

$$0 \longrightarrow \mathcal{O}_S(-D) \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

b) Under the hypotheses of 5.1 and 5.2, one has, for every $L \in \mathrm{Pic}(X)$,

$$\delta(L) \equiv L - 1 \pmod{\mathrm{Filt}^2(X)}, \tag{5.2.4}$$

which justifies the notation $c_1(L)$ introduced below. Indeed,

$$(L-1) - (1-L^{-1}) = (L-1)(1-L^{-1}) \in \mathrm{Filt}^2(X).$$

5.3

Composing δ with the canonical epimorphism

$$\mathrm{Filt}^1(X) \longrightarrow \mathrm{Gr}^1(X)$$

gives the *first Chern class* on $\mathrm{Pic}(X)$:

$$c_1 : \mathrm{Pic}(X) \longrightarrow \mathrm{Gr}^1(X). \tag{5.3.1}$$

Theorem 5.3.2. The map c_1 in (5.3.1) is an injective homomorphism, and the homomorphism

$$\det : \mathrm{Filt}^1(X) \longrightarrow \mathrm{Pic}(X)$$

induces on passing to the quotient a left inverse

$$\alpha : \mathrm{Gr}^1(X) \longrightarrow \mathrm{Pic}(X)$$

of c_1. If every section n of the constant sheaf $\mathbb{Z}_X$ is bounded—for example, if X is quasi-compact, or if the final object e of X is a finite disjoint sum of connected objects—then c_1 is an isomorphism and α is its inverse.

Proof. The map c_1 is a homomorphism, since for all $L, L' \in \mathrm{Pic}(X)$ one has

$$(1-(LL')^{-1}) - (1-L^{-1}) - (1-L'^{-1}) = -(1-L^{-1})(1-L'^{-1}) \in \mathrm{Filt}^2(X).$$

For the moment, admit that

$$\det(\mathrm{Filt}^2(X)) = \{1\} \subset \mathrm{Pic}(X),$$

so that α is well defined. Then

$$\alpha \circ c_1 = \mathrm{id}_{\mathrm{Pic}(X)},$$

because, by 5.2.2 b),

$$\det(1-L^{-1}) = \det(L^{-1})^{-1} = L$$

for every $L \in \mathrm{Pic}(X)$. Next we show that

$$c_1 \circ \alpha = \mathrm{id}_{\mathrm{Gr}^1(X)}$$

when every section n of $\mathbb{Z}_X$ is bounded. Since $\mathrm{Filt}^1(X)$ is the abelian group generated by the elements $\mathrm{cl}^\circ(\mathcal{E}) - n$, where $\mathcal{E}$ is a locally free $\mathcal{O}_X$-module of rank n, it is enough, in view of 5.2.2 b), to show that

$$\det(\mathrm{cl}^\circ(\mathcal{E})) - \mathrm{cl}^\circ(\mathcal{E}) + n - 1 \in \mathrm{Filt}^2(X). \tag{5.3.3}$$

Since n is bounded, the final object e of X decomposes as a finite disjoint sum $\coprod_{i=1}^p e_i$ of subobjects such that n is constant on each subtopos X/e_i. Since the functor

$$Y \longmapsto K^\circ(X/Y) \qquad (Y \in \mathrm{Ob}(X))$$

takes finite disjoint sums to finite direct products, one may suppose that n is constant on X, say $n \in \mathbb{Z}$. Put $E = \mathrm{cl}^\circ(\mathcal{E})$. Then (cf. Exposé V):

$$\det(E) = \lambda^n(E) = \gamma^n((E-n)+1) = \sum_{0\leq i\leq n} \gamma^i(E-n).$$

This proves (5.3.3), since its left-hand side is

$$\sum_{2\leq i\leq n} \gamma^i(E-n).$$

It remains to prove the following lemma.

Lemma 5.3.4. For every ringed topos X and every $x \in \mathrm{Filt}^2(X)$,

$$\det(x) = 1.$$

Proof. Let $\varepsilon : K^\circ(X) \to \Gamma(\mathbb{Z}_X)$ be the augmentation homomorphism. Formula (6.2.8), together with 6.5.3 below, gives by linearity

$$\det(xy) = \det(x)^{\varepsilon(y)} \det(y)^{\varepsilon(x)} \qquad (x, y \in K^\circ(X)). \tag{5.3.5}$$

It is therefore enough to show that, for every $k > 1$,

$$\det\bigl(\gamma^k(E-n)\bigr) = 1,$$

where, as above, $E = \mathrm{cl}^\circ(\mathcal{E})$ and $n = \varepsilon(E)$. For every $x \in K^\circ(X)$,

$$\gamma^k(x) = \lambda^k(x + k - 1),$$

and hence

$$\begin{aligned}\det\bigl(\gamma^k(E-n)\bigr) &= \det\bigl(\lambda^k(E-n+k-1)\bigr)\\ &= \det\left(\sum_{p+q=k} \binom{-n+k-1}{p} \lambda^q(E)\right).\end{aligned}$$

But (6.2.10) and 6.5.3 give

$$\det\bigl(\lambda^q(E)\bigr) = \det(E)^{\binom{n-1}{q-1}} \qquad (q > 0).$$

It follows that

$$\det\bigl(\gamma^k(E-n)\bigr) = \det(E)^r,$$

where

$$r = \sum_{\substack{p+q=k\\ q\geq 1}} \binom{-n+k-1}{p}\binom{n-1}{q-1}.$$

Now r is a section of $\mathbb{Z}_X$, everywhere equal to the coefficient of t^{k-1} in $(1+t)^{k-2}$, and is therefore zero. This proves the lemma.

6. Appendix: Calculation of Determinants of Locally Free Sheaves

6.1

First let $X = (\mathcal{C}, \mathcal{O})$ be an arbitrary ringed topos. For every positive section $n \in \Gamma(X, \mathbb{Z}_X)$ of the constant sheaf and every $\mathcal{O}$-module $M \in \mathrm{Mod}_{\mathcal{O}}(X)$, one can define the $\mathcal{O}$-modules

$$M^n, \qquad \bigotimes^n M, \qquad \bigwedge^n M, \quad \text{etc.} \tag{6.1.1}$$

The functors $(\ )^n$, $\bigotimes^n$, and $\bigwedge^n$ carry free modules, respectively locally free modules, to free modules, respectively locally free modules.

More generally, let $n_1, \dots, n_r$ be sections of $\mathbb{Z}_X$ and let T be an r-variable multifunctor, covariant for definiteness,

$$T : \mathrm{Mod}^{n_1}_{\mathcal{O}}(X) \times \cdots \times \mathrm{Mod}^{n_r}_{\mathcal{O}}(X) \longrightarrow \mathrm{Mod}_{\mathcal{O}}(X),$$

where $\mathrm{Mod}^n_{\mathcal{O}}(X)$ denotes the category fibered over X of locally free $\mathcal{O}$-modules of rank n. Put

$$M = T(\mathcal{O}^{n_1}, \dots, \mathcal{O}^{n_r})$$

and

$$G = \mathrm{GL}(n_1) \times \cdots \times \mathrm{GL}(n_r), \qquad \mathrm{GL}(n_i) = \mathrm{Aut}_{\mathcal{O}}(\mathcal{O}^{n_i}).$$

It follows immediately that T defines a representation

$$\alpha_T : G \longrightarrow \mathrm{Aut}_{\mathcal{O}}(M). \tag{6.1.2}$$

If M is itself locally free of rank m, then for every composite functor $T' \circ T$, where

$$T' : \mathrm{Mod}^m_{\mathcal{O}}(X) \longrightarrow \mathrm{Mod}_{\mathcal{O}}(X),$$

one has

$$\alpha_{T' \circ T} = \alpha_{T'} \circ \alpha_T. \tag{6.1.3}$$

Conversely, from a representation $\alpha = \alpha_T : G \to \mathrm{Aut}_{\mathcal{O}}(M)$ one recovers $T(E_1, \dots, E_r)$

for every sequence $E_1, \dots, E_r$ of locally free $\mathcal{O}$-modules of ranks $n_1, \dots, n_r$, by gluing the free module $T(\mathcal{O}^{n_1}, \dots, \mathcal{O}^{n_r})$ on local trivializing charts through α. More precisely, the sequence $E_1, \dots, E_r$ determines a right G-torsor $P = P_{E_1, \dots, E_r}$ and canonical isomorphisms

$$E_i \xrightarrow{\sim} P \wedge^G \mathcal{O}^{n_i} \qquad (i = 1, \dots, r), \tag{6.1.3}$$

where $\wedge^G$ denotes the contracted product and G acts on $\mathcal{O}^{n_i}$ through the i-th projection $G \to \mathrm{GL}(n_i)$. Every representation

$$\alpha : G \longrightarrow \mathrm{Aut}_{\mathcal{O}}(M)$$

therefore defines a multifunctor $T = T_\alpha$ such that

$$T_\alpha(E_1, \dots, E_r) = P_{E_1, \dots, E_r} \wedge^G M. \tag{6.1.4}$$

Moreover, for every composite representation $\alpha = \alpha_1 \circ \alpha_2$, one has

$$T_\alpha = T_{\alpha_1} \circ T_{\alpha_2}. \tag{6.1.5}$$

6.2

For every locally free $\mathcal{O}$-module E of rank n, write

$$\det(E) = \bigwedge^n E.$$

With the notation of 6.1, the functor $\det(T(E_1, \dots, E_r))$ is determined, up to canonical isomorphism, by the corresponding representation

$$\det \circ \alpha_T : G \longrightarrow \mathbb{G}_{\mathrm{m}} = \mathrm{GL}(1). \tag{6.2.1}$$

Suppose now that X is the ringed topos of sheaves on the ringed site $(\mathrm{Sch}\,/S)$ endowed with the étale topology. Then $\det(T(E_1, \dots, E_r))$, or equivalently the representation $\det \circ \alpha_T$, can be calculated by means of the following proposition.

Proposition 6.2.2. For every positive section n of $\mathbb{Z}$, the étale sheaf of groups $\mathrm{Der}(\mathrm{GL}(n))$ of commutators of $\mathrm{GL}(n)$ is represented by

$$\mathrm{SL}(n) = \mathrm{Ker}\big(\det : \mathrm{GL}(n) \longrightarrow \mathbb{G}_{\mathrm{m}}\big).$$

Corollary 6.2.3. Every representation

$$\alpha : G = \mathrm{GL}(n_1) \times \cdots \times \mathrm{GL}(n_r) \longrightarrow \mathbb{G}_{\mathrm{m}} \tag{6.2.4}$$

is of the form

$$\alpha(Y)(s) = \det(s_1)^{m_1} \cdots \det(s_r)^{m_r}, \tag{6.2.5}$$

where $Y \in \mathrm{Ob}(\mathrm{Sch}\,/S)$, $s = (s_1, \dots, s_r) \in G(Y)$, and $m_1, \dots, m_r$ are sections of $\mathbb{Z}$ uniquely determined by α.

Corollary 6.2.6. With the notation of 6.1,

$$\det\big(T_\alpha(E_1, \dots, E_r)\big) = \det(E_1)^{\otimes m_1} \otimes \cdots \otimes \det(E_r)^{\otimes m_r}, \tag{6.2.6}$$

where $m_1, \dots, m_r$ are the sections of $\mathbb{Z}$ uniquely determined by α.

Remark 6.2.7. The exponents $m_1, \dots, m_r$ in (6.2.5) and (6.2.6) can be determined by successively substituting in (6.2.5)

$$s_i = \lambda\,\mathrm{id}_{\mathcal{O}_Y^{n_i}} \quad (\lambda \in \mathbb{G}_{\mathrm{m}}(Y)), \qquad s_j = \mathrm{id}_{\mathcal{O}_Y^{n_j}} \quad (j \neq i),$$

for $i = 1, \dots, r$. Then

$$\det(s_i) = \lambda^{n_i}, \qquad \alpha(Y)(s) = \lambda^{n_i m_i}.$$

For example, (6.2.6) gives

$$\det(E_1 \otimes E_2) = \det(E_1)^{\otimes n_2} \otimes \det(E_2)^{\otimes n_1}, \tag{6.2.8}$$

$$\det(E_1 \oplus E_2) = \det(E_1) \otimes \det(E_2), \tag{6.2.9}$$

$$\det\Big(\bigwedge^m E_1\Big) = \det(E_1)^{\otimes \binom{n_1-1}{m-1}}. \tag{6.2.10}$$

6.3

Let us first show how 6.2.3 follows from 6.2.2. It is plainly enough to prove (6.2.5) for

$$s_{(i)} = (1, \dots, s_i, \dots, 1) \qquad (i = 1, \dots, r).$$

Thus one may suppose that $r = 1$ and $G = \mathrm{GL}(n_1)$. By 6.2.2, the representation α factors through $\mathrm{GL}(n_1)/\mathrm{SL}(n_1)$; equivalently, there is a commutative diagram

$$\begin{tikzcd} \mathrm{GL}(n_1) \arrow[r, "\alpha"] \arrow[d, "j"'] & \mathbb{G}_{\mathrm{m}} \\ \mathrm{GL}(n_1)/\mathrm{SL}(n_1) \arrow[ur, "\beta"'] & \end{tikzcd}$$

in which j is the canonical epimorphism and β is a representation of $\mathrm{GL}(n_1)/\mathrm{SL}(n_1)$ in $\mathbb{G}_{\mathrm{m}}$. The homomorphism

$$\det : \mathrm{GL}(n_1) \longrightarrow \mathbb{G}_{\mathrm{m}}$$

has a section. Consequently, the monomorphism

$$\mathrm{GL}(n_1)/\mathrm{SL}(n_1) \longrightarrow \mathbb{G}_{\mathrm{m}}$$

induced by det is an isomorphism. On the other hand, it is known (SGA 3 VIII 1.3) that every representation of $\mathbb{G}_{\mathrm{m}}$ in itself is of the form

$$s \longmapsto s^m,$$

where m is a section of $\mathbb{Z}$. The assertion follows.

Corollary 6.2.6 is simply a reformulation of 6.2.3. Notice that 6.2.3 also follows immediately from SGA 3 XXII 6.2.1, but we have wished to give an elementary proof here by means of 6.2.2.

6.4. Proof of 6.2.2

It is enough to prove that, for every $Y \in \mathrm{Ob}(\mathrm{Sch}/S)$, every $y \in Y$, and every $s \in \mathrm{SL}(n)(Y)$, there is an étale neighborhood $Y' \to Y$ of y such that the restriction of s to Y' is a product of commutators in $\mathrm{GL}(n)(Y')$. Since the question is local, one may suppose that Y is affine with ring A and that the section n is constant on Y. The case $n = 1$ is trivial; hence assume $n \geq 2$. Let $\mathfrak{p}$ be the prime ideal corresponding to y, and write $s = (s_{ij})_{i,j=1,\ldots,n}$. For every $\lambda \in A$ and every $h, k \in \{1, \ldots, n\}$ with $h \neq k$, let $c_{h,k,\lambda}$ be the matrix $C = (c_{ij})$ defined by

$$c_{hk} = \lambda, \qquad c_{ij} = \delta_{ij} \quad \text{for } (i,j) \neq (h,k),$$

where δ_{ij} is the Kronecker symbol.

Thus $c_{h,k,\lambda} = I + \lambda E_{hk}$. For every matrix a, left multiplication by $c_{h,k,\lambda}$ adds λ times row k of a to row h, whereas right multiplication adds λ times column h of a to column k. In particular,

$$\det(c_{h,k,\lambda}) = 1,$$

and

$$c_{h,k,\lambda} c_{h,k,\mu} = c_{h,k,\lambda+\mu} \tag{6.3.1}$$

for all $\lambda, \mu \in A$. Hence one obtains a representation

$$c_{h,k} : \mathbb{G}_{\mathrm{a}} \longrightarrow \mathrm{SL}(n) \qquad (h,k = 1, \ldots, n,\ h \neq k).$$

No invertibility assumption is needed in the row-reduction step. We reduce to invertible parameters before the commutator step.

The proof has two steps. First, we show that s is a product of matrices $c_{h,k,\lambda}$ in a sufficiently small Zariski neighborhood of y. Then we show that these matrices are products of commutators in $\mathrm{GL}(n)$ in a sufficiently small étale neighborhood of y.

For the first step, one must transform s into the identity matrix $I = (\delta_{ij})$ in a Zariski neighborhood of y by multiplying on the left and right by suitable matrices $c_{h,k,\lambda}$. We first make $s_{11} = 1$. If $s_{11} \neq 1$ and $s_{21} \in \mathfrak{p}$, then, since $\det(s) = 1 \notin \mathfrak{p}$, there is an index i for which $s_{i1} \notin \mathfrak{p}$. Replacing s by $c_{2,i,1}s$ replaces s_{21} by $s_{21} + s_{i1}$, which is not in $\mathfrak{p}$. Thus, after localizing, one may suppose that s_{21} is invertible.

If $s_{11} - 1 \in \mathfrak{p}$, replacing s by $c_{1,2,1}s$ replaces $s_{11} - 1$ by

$$(s_{11} + s_{21}) - 1 \notin \mathfrak{p}.$$

After a further localization, one may therefore suppose that $s_{11} - 1$ is invertible. Then

$$\lambda = \frac{1 - s_{11}}{s_{21}}$$

is invertible, and left multiplication by $c_{1,2,\lambda}$ makes the new entry s_{11} equal to 1.

Assume now that $s_{11} = 1$. Left multiplication by $c_{i,1,-s_{i1}}$ for $i > 1$ clears the remaining entries in the first column, and right multiplication by $c_{1,i,-s_{1i}}$ clears the remaining entries in the first row. Repeating this procedure on the lower-right block transforms s, in a Zariski neighborhood U of y, into a matrix $s' = (s'_{ij})$ satisfying

$$s'_{ij} = \delta_{ij} \qquad \text{whenever } i < n \text{ or } j < n.$$

By construction, $s' = c_1 s c_2$ on U, where c_1 and c_2 are products of matrices $c_{h,k,\lambda}$. Hence

$$\det(s') = \det(s) = 1,$$

so $s'_{nn} = 1$ and therefore $s' = I$. Since $c_{h,k,\lambda}^{-1} = c_{h,k,-\lambda}$, this proves that s is itself a product of elementary matrices of this form.

After shrinking about y, a parameter $a \notin \mathfrak{p}$ is a unit. If $a \in \mathfrak{p}$, then $a = 1 + (a - 1)$, and both 1 and $a - 1$ are units near y. By (6.3.1),

$$c_{h,k,a} = c_{h,k,1} c_{h,k,a-1}$$

Thus it remains only to treat invertible parameters.

It remains to prove that these elementary matrices are products of commutators in an étale neighborhood of y. For invertible λ and μ, the two matrices

$c_{h,k,\lambda}$ and $c_{h,k,\mu}$ are conjugate. More precisely, there is an invertible matrix $a = (a_{ij})$ such that

$$a\, c_{h,k,\lambda} = c_{h,k,\mu}\, a.$$

Indeed, the diagonal matrix defined by

$$a_{hh} = \lambda^{-1}, \qquad a_{kk} = \mu^{-1}, \qquad a_{ii} = 1 \quad (i \neq h, k)$$

has this property; recall that λ and μ are invertible and that $h \neq k$. Thus, for every homomorphism

$$\beta : \mathrm{GL}(n)(U) \longrightarrow B$$

to an abelian group B, written additively, one has

$$\beta(c_{h,k,\lambda}) = \beta(c_{h,k,\mu}).$$

If $\lambda + \mu$ is invertible, this equality and (6.3.1) imply

$$\beta(c_{h,k,\lambda}) = 0.$$

It follows that $c_{h,k,\lambda}$ belongs to the commutator subgroup. If the residue field $k(y)$ is not the prime field $\mathbb{F}_2 = \{0,1\}$, then, for every invertible $\lambda \in A$, there are a Zariski neighborhood $U = \operatorname{Spec}(B)$ of y and an invertible $\mu \in B$ such that $\lambda + \mu$ is invertible in B. Hence $c_{h,k,\lambda}$ is a product of commutators on U.

If $k(y) = \mathbb{F}_2$, then

$$V = \operatorname{Spec}\bigl(A[T]/(T^2 + T + 1)\bigr)$$

is an étale neighborhood of y whose residue field is different from $\mathbb{F}_2$. This completes the proof.

6.5. Remarks

6.5.1. If no residue field of S is isomorphic to $\mathbb{F}_2$, the preceding proof shows that 6.2.2 remains valid for the ringed topos of Zariski sheaves on S. By contrast, already for $S = \operatorname{Spec}(\mathbb{F}_2)$ one has

$$\mathrm{SL}(2, \mathbb{F}_2) = \mathrm{GL}(2, \mathbb{F}_2),$$

which is different from the commutator subgroup of $\mathrm{GL}(2, \mathbb{F}_2)$. The latter has order 3, whereas $\mathrm{SL}(2, \mathbb{F}_2)$ has order 6 (cf. [1]).

6.5.2. On an arbitrary locally ringed topos X, it is not true in general that every representation $\mathbb{G}_{\mathrm{m}} \to \mathbb{G}_{\mathrm{m}}$ is given by a

power. For example, on the topos (Ens) ringed by the field $\mathbb{Q}$, the representation $\mathbb{G}_{\mathrm{m}} \to \mathbb{G}_{\mathrm{m}}$ defined by

$$r \longmapsto |r| \qquad (r \in \mathbb{Q}^*)$$

is not a power. In particular, 6.2.3 does not remain valid for an arbitrary locally ringed topos X and an arbitrary representation $\alpha : G \to \mathbb{G}_{\mathrm{m}}$.

6.5.3. On the other hand, 6.2.3 remains valid, on an arbitrary ringed topos $X = (\mathcal{C}, \mathcal{O})$, for the representations associated with formulas (6.2.8), (6.2.9), and (6.2.10), because these representations are *algebraic* in the following sense.

Let e be the final object of X, put $S = \operatorname{Spec}(\mathcal{O}(e))$, and consider the functor

$$u : \mathcal{C} \longrightarrow (\mathrm{Sch}\,/S)$$

which sends $Y \in \mathrm{Ob}(\mathcal{C})$ to the affine S-scheme $\operatorname{Spec}(\mathcal{O}(Y))$. For every positive integer n, the presheaf $\mathrm{GL}(n)_X$ on $\mathcal{C}$ is obtained, by composition with u, from the corresponding presheaf $\mathrm{GL}(n)_S$ on $(\mathrm{Sch}\,/S)$. A representation on X

$$\alpha : \mathrm{GL}(n) \longrightarrow \mathbb{G}_{\mathrm{m}} \tag{6.5.4}$$

is said to be *algebraic* if it is induced via u by a representation

$$\alpha_S : \mathrm{GL}(n)_S \longrightarrow \mathbb{G}_{\mathrm{m},S} \tag{6.5.5}$$

of group presheaves on $(\mathrm{Sch}\,/S)$; that is, if

$$\alpha(s) = \alpha_S(s)$$

for every $Y \in \mathrm{Ob}(\mathcal{C})$ and every $s \in \mathrm{GL}(n)(Y) = \mathrm{GL}(n)_S(u(Y))$. The representation α in (6.5.4) is of course completely determined by α_S. It follows immediately that 6.2.3 remains valid in an arbitrary ringed topos whenever α is algebraic. In particular, formulas (6.2.8), (6.2.9), and (6.2.10) remain valid on every ringed topos.

7. Appendix: Specialization in Intersection Theory

by A. Grothendieck

Lemma 7.1. Let $i : Y \to X$ be a regular closed immersion (VII 1.4) of bounded codimension. Then i is a perfect morphism (VII 1.9), so the inverse-image homomorphism (IV 2.12)

$$i^* : K_\bullet(X) \longrightarrow K_\bullet(Y)$$

is defined. Let N be the conormal sheaf of i. It is a locally free sheaf of finite type on Y (VII 1.6), and hence one can form

$$\lambda_{-1}(N) \in K^\circ(Y).$$

Consider also the direct-image homomorphism (IV 2.11)

$$i_* : K_\bullet(Y) \longrightarrow K_\bullet(X).$$

Then

$$i^* i_*(x) = x\,\lambda_{-1}(N) \qquad (x \in K_\bullet(Y)),$$

where the product on the right is taken with respect to the $K^\circ(Y)$-module structure on $K_\bullet(Y)$ defined in IV 2.10.

The proof is, essentially without change, the same as that of the corresponding formula for $K^\circ(X)$ and $K^\circ(Y)$ established in VII 2.7.

Corollary 7.2. Let $f : X \to S$ be flat, where S is a trait, that is, the spectrum of a discrete valuation ring A. Let s be the closed point of S, let t be its generic point, and let $i : X_s \to X$ be the canonical inclusion. Then i has finite Tor-dimension, so (IV 2.12) the homomorphism

$$i^* : K_\bullet(X) \longrightarrow K_\bullet(X_s)$$

is defined.

On the other hand, (IV 2.11) gives a homomorphism

$$i_* : K_\bullet(X_s) \longrightarrow K_\bullet(X).$$

The composite

$$i^* i_* : K_\bullet(X_s) \longrightarrow K_\bullet(X) \longrightarrow K_\bullet(X_s)$$

is zero.

Indeed, $i : X_s \to X$ is a regular immersion of codimension 1. Its conormal sheaf N is free of rank 1, because X_s is defined by the global equation $f^*(g) = 0$, where g is a uniformizer of A. Thus $\lambda_{-1}(N) = 0$, and the assertion follows from 7.1.

For the rest of this appendix, suppose that X is noetherian and flat over S. Apply the exact homotopy sequence (IX 1.1) to X, with the open subscheme X_t and complementary closed subscheme X_s:

$$K_\bullet(X_s) \xrightarrow{i_*} K_\bullet(X) \xrightarrow{j^*} K_\bullet(X_t) \longrightarrow 0,$$

where $j : X_t \to X$ is the open immersion. Corollary 7.2 gives:

Proposition 7.3. With the notation of 7.2, and assuming X noetherian, there is a unique group homomorphism

$$\sigma_\bullet : K_\bullet(X_t) \longrightarrow K_\bullet(X_s), \tag{7.3.1}$$

called the *specialization homomorphism*, for which the diagram

$$\begin{tikzcd} K_\bullet(X) \arrow[r, "j^*"] \arrow[d, "i^*"'] & K_\bullet(X_t) \arrow[dl, "\sigma_\bullet"] \\ K_\bullet(X_s) & \end{tikzcd} \tag{7.3.2}$$

commutes.

Regard the two sides of (7.3.1) as $K^\circ(X)$-modules via the ring homomorphisms

$$K^\circ(X) \xrightarrow{j^*} K^\circ(X_t), \qquad K^\circ(X) \xrightarrow{i^*} K^\circ(X_s) \qquad \text{(IV 2.7 b)}$$

and via the $K^\circ(X_t)$- and $K^\circ(X_s)$-module structures on $K_\bullet(X_t)$ and $K_\bullet(X_s)$, respectively (IV 2.10). Then $\sigma_\bullet$ is a homomorphism of $K^\circ(X)$-modules.

This last assertion follows because $i^* : K_\bullet(X) \to K_\bullet(X_s)$ is $K^\circ(X)$-linear, as is the epimorphism $j^* : K_\bullet(X) \to K_\bullet(X_t)$, by IV 2.12.2.

We now review other properties of the specialization homomorphism.

7.4. Compatibility with filtrations

Endow the groups $K_\bullet$ with the filtration $\mathrm{Filt}_\bullet K_\bullet$ by the dimension of supports (1.1.1). If f is of finite type, and more generally if X is catenary, then the specialization homomorphism is compatible with these filtrations and consequently induces a homomorphism, again called specialization,

$$\sigma_\bullet : \mathrm{Gr}_\bullet(X_t) \longrightarrow \mathrm{Gr}_\bullet(X_s). \tag{7.4.1}$$

To prove this, by 1.14 one is reduced to showing that, if Y_t is an integral closed subscheme of dimension d of X_t, then $\sigma_\bullet(\mathrm{cl}_\bullet(\mathcal{O}_{Y_t}))$ has filtration at most d. Let Y be the schematic closure of Y_t in X. Then Y is flat over S (EGA IV 2.8.1), and one readily obtains

$$i^*(\mathrm{cl}_\bullet(\mathcal{O}_Y)) = \mathrm{cl}_\bullet(\mathcal{O}_{Y_s}).$$

It follows that

$$\sigma_\bullet(\mathrm{cl}_\bullet(\mathcal{O}_{Y_t})) = \mathrm{cl}_\bullet(\mathcal{O}_{Y_s}).$$

Since X is catenary, and hence Y is catenary, one checks that

$$\dim Y_s \leq \dim Y_t.$$

When f is of finite type, this follows immediately, after reducing to the case in which X is integral, from EGA IV 7.1.13. This proves the assertion.

7.5. "Conservation of number"

Suppose now that f is proper. Then X_t and X_s are proper over $k(t)$ and $k(s)$, respectively, and one has the homomorphisms defined in 2.3,

$$\chi_t : K_\bullet(X_t) \longrightarrow \mathbb{Z}, \qquad \chi_s : K_\bullet(X_s) \longrightarrow \mathbb{Z}.$$

For every $x \in K_\bullet(X_t)$ one has

$$\chi_s(\sigma_\bullet(x)) = \chi_t(x). \tag{7.5.1}$$

This is the *conservation-of-number theorem*; equivalently, the diagram

$$\begin{tikzcd} K_\bullet(X_t) \arrow[r, "\sigma_\bullet"] \arrow[dr, "\chi_t"'] & K_\bullet(X_s) \arrow[d, "\chi_s"] \\ & \mathbb{Z} \end{tikzcd} \tag{7.5.2}$$

commutes.

To prove (7.5.1), one is again reduced to the case $x = \mathrm{cl}_\bullet(\mathcal{O}_{Y_t})$, where Y_t is an integral closed subscheme of X_t. With the notation of the proof of 7.4, it remains to prove

$$\chi(Y_s) = \chi(Y_t),$$

which is well known (EGA III 7.9.4, or Exposé III 5.7).

7.6

It is immediate that

$$\sigma_\bullet(\mathrm{cl}_\bullet(\mathcal{O}_{X_t})) = \mathrm{cl}_\bullet(\mathcal{O}_{X_s}). \tag{7.6.1}$$

Since $\sigma_\bullet$ is $K^\circ(X)$-linear, the following diagram

$$\begin{tikzcd} & K^\circ(X) \arrow[dl, "j^*"'] \arrow[dr, "i^*"] & \\ K^\circ(X_t) \arrow[d] & & K^\circ(X_s) \arrow[d] \\ K_\bullet(X_t) \arrow[rr, "\sigma_\bullet"'] & & K_\bullet(X_s) \end{tikzcd} \tag{7.6.2}$$

commutes.

The vertical arrows are the canonical homomorphisms $K^\circ \to K_\bullet$ (IV 2.5.1).

7.7. Specialization for $K^\bullet$ and $\mathrm{Gr}^\bullet$

When X_t and X_s are regular (for example, when f is smooth), the two preceding vertical arrows are isomorphisms (IV 2.5). The specialization homomorphism can then be interpreted as a homomorphism

$$\sigma^\bullet : K^\bullet(X_t) \longrightarrow K^\bullet(X_s), \tag{7.7.1}$$

which is equivalently the unique homomorphism making the diagram

$$\begin{tikzcd} K^\bullet(X) \arrow[r, "j^*"] \arrow[d, "i^*"'] & K^\bullet(X_t) \arrow[dl, "\sigma^\bullet"] \\ K^\bullet(X_s) \end{tikzcd} \tag{7.7.2}$$

commute.

In this diagram i^* and j^* are ring homomorphisms (IV 2.7 b), and j^* is surjective. Hence (7.7.1) is a ring homomorphism. For the same reason, if X is separated, it is even a homomorphism of λ-rings. Indeed, a regular separated noetherian scheme is divisorial (II 2.2.7.1), so its $K^\bullet$ is canonically identified with its "naive" $K^\bullet$ (IV 2.9(ii)), which carries a natural λ-ring structure (VI 3.2). Moreover, the regularity of X_t and X_s implies that X is regular (EGA 0_{IV} 17.1.8), and

$$i^* : K^\bullet(X) \longrightarrow K^\bullet(X_s), \qquad j^* : K^\bullet(X) \longrightarrow K^\bullet(X_t)$$

are λ-homomorphisms. Therefore the same is true of (7.7.1), which is obtained from i^* by passage to the quotient.

Since X is regular, hence normal, its irreducible components are its connected components, and they dominate S. Consequently the homomorphism

$$H^0(X, \mathbb{Z}) \longrightarrow H^0(X_t, \mathbb{Z})$$

induced by $X_t \to X$ is bijective. The homomorphism induced by $X_s \to X$ can therefore also be regarded as a specialization homomorphism

$$H^0(X_t, \mathbb{Z}) \longrightarrow H^0(X_s, \mathbb{Z}). \tag{7.7.3}$$

It follows immediately from the definitions that (7.7.1) is compatible, via (7.7.3), with the augmented-algebra structures of its source and target over $H^0(X_t, \mathbb{Z})$ and $H^0(X_s, \mathbb{Z})$, respectively, defined in IV 2.8.

Still assuming that X is separated, (7.7.1) induces a homomorphism of graded rings

$$\sigma^\bullet : \mathrm{Gr}^\bullet(X_t) \longrightarrow \mathrm{Gr}^\bullet(X_s). \tag{7.7.4}$$

Using the Chern-class construction of V 6, one obtains the commutative diagram

$$\begin{tikzcd}
K^\bullet(X_t) \arrow[r, "c_{X_t}"] \arrow[d] & \mathrm{Ch}\ \mathrm{Gr}^\bullet(X_t) \arrow[d] \\
K^\bullet(X_s) \arrow[r, "c_{X_s}"'] & \mathrm{Ch}\ \mathrm{Gr}^\bullet(X_s).
\end{tikzcd}$$

7.8. Specialization for Pic

Retain the hypotheses of 7.7: X is separated, and X_t and X_s are regular. Using the canonical isomorphism (5.3.2)

$$\mathrm{Gr}^1 \simeq \mathrm{Pic},$$

the homomorphism (7.7.4) gives a canonical homomorphism

$$\mathrm{Pic}(X_t) \longrightarrow \mathrm{Pic}(X_s). \tag{7.8.1}$$

An equivalent description is as follows. Since X is regular, its local rings are factorial by the Auslander–Buchsbaum theorem (EGA IV 21.11.1). Thus every invertible sheaf L_t on X_t extends to an invertible sheaf L on X (EGA IV 21.6.11). The image of $\mathrm{cl}(L_t) \in \mathrm{Pic}(X_t)$ under (7.8.1) is $\mathrm{cl}(L_s) \in \mathrm{Pic}(X_s)$, where $L_s = L|_{X_s}$.

In fact, assume only that the local rings of X are factorial and that X_s is locally integral, without assuming regularity or separatedness. Then $\mathrm{cl}(L_s)$ is independent of the chosen extension L of L_t to an invertible sheaf on X, and this gives (7.8.1) directly. To prove the independence, it is enough to show that

$$L_t \simeq \mathcal{O}_{X_t} \quad \Longrightarrow \quad L_s \simeq \mathcal{O}_{X_s}.$$

The hypothesis means that $L \simeq \mathcal{O}_X(D)$ for a divisor D on X whose support is contained in X_s (EGA IV 21.2.11). One reduces at once to the case in which X_s is integral. Then $D = nX_s$, and $D = \mathrm{div}(g^n)$, where g is a uniformizer of A. Hence $\mathcal{O}_X(D) \simeq \mathcal{O}_X$, and a fortiori $L_s \simeq \mathcal{O}_{X_s}$.

7.9. Specialization and numerical equivalence

Suppose that f is proper and that X_s and X_t are regular. Consider the two specialization homomorphisms (7.4.1) and (7.7.4):

$$\begin{aligned} \sigma_\bullet &: \mathrm{Gr}_\bullet(X_t) \longrightarrow \mathrm{Gr}_\bullet(X_s), \\ \sigma^\bullet &: \mathrm{Gr}^\bullet(X_t) \longrightarrow \mathrm{Gr}^\bullet(X_s). \end{aligned} \tag{7.9.1}$$

They are induced by the homomorphism

$$\sigma^\bullet : K^\bullet(X_t) \longrightarrow K^\bullet(X_s). \tag{7.9.2}$$

Since (7.9.2) is compatible with the ring structures, formula (7.5.1) gives

$$\langle \sigma^\bullet(y_t), \sigma_\bullet(x_t)\rangle = \langle y_t, x_t\rangle\,, \qquad x_t \in \mathrm{Gr}_\bullet(X_t), \quad y_t \in \mathrm{Gr}^\bullet(X_t), \tag{7.9.3}$$

where the pairings between $\mathrm{Gr}^\bullet$ and $\mathrm{Gr}_\bullet$ are those of 4.1. It follows formally that, if $x \in \mathrm{Gr}_\bullet(X_t)$ (respectively, $x \in \mathrm{Gr}^\bullet(X_t)$) and $\sigma_\bullet(x)$ (respectively, $\sigma^\bullet(x)$) is numerically equivalent to zero (4.5.1), then x is numerically equivalent to zero. The converse holds if (7.9.2), or, equivalently, either of the homomorphisms in (7.9.1) obtained from it by passage to the associated graded groups, is surjective modulo torsion. This condition occurs only in exceptional cases.

7.9.4. It is plausible, at least when f is proper and smooth, that the converse just considered always holds: if x is numerically equivalent to zero, then so is $\sigma_\bullet(x)$ (respectively, $\sigma^\bullet(x)$). This would imply that the homomorphisms (7.9.1) induce monomorphisms on the corresponding groups modulo numerical equivalence,

$$\mathrm{Gr}_\bullet^{\mathrm{num}} \qquad \text{and} \qquad \mathrm{Gr}^\bullet_{\mathrm{num}}\,.$$

At least the preceding result shows that $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_t)$ is isomorphic to a quotient of a subgroup of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_s)$; in particular, its rank is at most that of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_s)$. The same holds for $\mathrm{Gr}^\bullet_{\mathrm{num}}$, and also for $K^\bullet_{\mathrm{num}}$.

7.10. Compatibility with direct and inverse images

Let

$$g : X \longrightarrow Y$$

be a morphism of noetherian schemes flat over S. Denote by $\sigma_{X,\bullet}$ and $\sigma_{Y,\bullet}$ the specialization homomorphisms for X and Y, and by i_X, j_X (respectively, i_Y, j_Y)

the inclusions of the special and generic fibers into the ambient schemes. Write

$$g_t : X_t \longrightarrow Y_t, \qquad g_s : X_s \longrightarrow Y_s$$

for the induced morphisms. They give the Cartesian squares

$$\begin{tikzcd} X_t \arrow[r, "g_t"] \arrow[d, "j_X"'] & Y_t \arrow[d, "j_Y"] \\ X \arrow[r, "g"] & Y \end{tikzcd} \qquad \begin{tikzcd} X_s \arrow[r, "g_s"] \arrow[d, "i_X"'] & Y_s \arrow[d, "i_Y"] \\ X \arrow[r, "g"] & Y. \end{tikzcd} \tag{7.10.0}$$

First suppose that g is proper. Then the diagram

$$\begin{tikzcd} K_\bullet(X_t) \arrow[r, "g_{t*}"] \arrow[d, "\sigma_{X,\bullet}"'] & K_\bullet(Y_t) \arrow[d, "\sigma_{Y,\bullet}"] \\ K_\bullet(X_s) \arrow[r, "g_{s*}"'] & K_\bullet(Y_s) \end{tikzcd} \tag{7.10.1}$$

commutes. This generalizes 7.5: in the present notation take $Y = S$ (equivalently, take the scheme denoted by X in (7.3.2) to be S); the specialization homomorphism for S is the identity on $\mathbb{Z}$. If X and Y satisfy the conditions of 7.7, that is, if their fibers are regular, then g_t and g_s automatically have finite Tor-dimension, and (7.10.1) may be read as a commutative square whose four terms are $K^\bullet$-groups. We leave this reformulation to the reader.

Using the definitions of $\sigma_{X,\bullet}$ and $\sigma_{Y,\bullet}$, the commutativity of (7.10.1) reduces to that of the two front faces of the prism

$$\begin{tikzcd}
K_\bullet(X_t) \arrow[rr, "g_{t*}"] & & K_\bullet(Y_t) & \\
& K_\bullet(X) \arrow[ul, "j_X^*"] \arrow[dl, "i_X^*"'] \arrow[rr, "g_*"] & & K_\bullet(Y) \arrow[ul, "j_Y^*"] \arrow[dl, "i_Y^*"'] \\
K_\bullet(X_s) \arrow[rr, "g_{s*}"'] & & K_\bullet(Y_s). &
\end{tikzcd} \tag{*}$$

These two faces are the base-change squares obtained from (7.10.0), so IV 3 applies. It remains only to verify that they correspond to the pairs (X, Y_t) and (X, Y_s) of schemes over Y, and that these pairs are Tor-independent over Y. This is clear for the first pair because $Y_t \to Y$ is flat. It is also immediate for the second, because $Y_s \to Y$ is a regular immersion of codimension 1 and remains so after base change by $X \to Y$.

Now drop the properness assumption and instead suppose that g has finite Tor-dimension. Then g_t and g_s also have finite Tor-dimension. This is immediate for g_t; for g_s it follows from the Tor-independence of X and Y_s over Y just noted. Hence one can form the square

$$\begin{tikzcd}
K_\bullet(Y_t) \arrow[r, "g_t^*"] \arrow[d, "\sigma_{Y,\bullet}"'] & K_\bullet(X_t) \arrow[d, "\sigma_{X,\bullet}"] \\
K_\bullet(Y_s) \arrow[r, "g_s^*"'] & K_\bullet(X_s),
\end{tikzcd} \tag{7.10.2}$$

whose horizontal arrows are those defined in IV 2.12. If X and Y satisfy the conditions of 7.7, that is, if their fibers are regular, then Y is regular and g automatically has finite Tor-dimension; the square may then be interpreted as a commutative square of $K^\bullet$-rings.

To prove the commutativity of (7.10.2), it is again enough to prove that of the two front faces of the prism analogous to $(*)$, with the direct-image homomorphisms g_{t*}, g_*, g_{s*} replaced by the inverse-image homomorphisms g_t^*, g^*, g_s^*. Their commutativity follows

immediately from that of the corresponding squares in (7.10.0), using the transitivity of inverse images (IV 2.12).

Remark 7.10.3. When g is proper, the commutativity of (7.10.1) immediately implies that of the following diagram, obtained by passage to the associated graded groups:

$$\begin{tikzcd}
\mathrm{Gr}_\bullet(X_t) \arrow[r, "g_{t*}"] \arrow[d, "\sigma_{X,\bullet}"'] & \mathrm{Gr}_\bullet(Y_t) \arrow[d, "\sigma_{Y,\bullet}"] \\
\mathrm{Gr}_\bullet(X_s) \arrow[r, "g_{s*}"'] & \mathrm{Gr}_\bullet(Y_s).
\end{tikzcd} \tag{7.10.1 bis}$$

Likewise, if X and Y are separated and have regular fibers, the commutativity of (7.10.2) implies that of

$$\begin{tikzcd} \mathrm{Gr}^\bullet(Y_t) \arrow[r, "g_t^*"] \arrow[d, "\sigma_Y^\bullet"'] & \mathrm{Gr}^\bullet(X_t) \arrow[d, "\sigma_X^\bullet"] \\ \mathrm{Gr}^\bullet(Y_s) \arrow[r, "g_s^*"'] & \mathrm{Gr}^\bullet(X_s). \end{tikzcd} \tag{7.10.2 bis}$$

If X and Y are moreover quasi-projective and smooth over S, then g_t^* and g_s^* are compatible with the filtrations by codimension of supports. Thus one obtains a commutative diagram of the same form as (7.10.2 bis), with $\mathrm{Gr}^\bullet$ replaced by $\mathrm{Gr}^\bullet_{\mathrm{top}}$ (Exposé 0, 2.1).

Finally, if X and Y are quasi-projective and separated over S and have regular fibers, then g_t and g_s are automatically complete-intersection morphisms (VIII 1.1). Using VIII 3.2, one obtains a commutative diagram analogous to (7.10.1 bis), with $\mathrm{Gr}_\bullet$ replaced by $\mathrm{Gr}^\bullet(-)_{\mathbb{Q}}$.

7.11. For a scheme T over an algebraically closed field k, one can define the subgroup of $K^\bullet(T)$ consisting of elements algebraically equivalent to zero as the subgroup of elements of the form

$$E(x_1) - E(x_0), \tag{7.11.1}$$

where x_0 and x_1 are two k-rational points of a proper, regular, connected algebraic curve C over k, where $E \in K^\bullet(T \times_k C)$, and where, for every k-rational point x of C, one sets

$$E(x) = i_x^*(E) \in K^\bullet(T), \qquad \text{where} \qquad i_x : T \longrightarrow T \times_k C$$

is the morphism $t \mapsto (t, x)$ defined by x. That the elements of the form (7.11.1) constitute a subgroup of $K^\bullet(T)$ follows by a standard argument, using a product $C \times_k C'$, the two points (x_1, x_1') and (x_0, x_0') of that product, and the fact that one can always find an irreducible algebraic curve through any two points of an irreducible quasi-projective algebraic scheme over k (here $C \times_k C'$). (A reader unwilling to admit this fact may define algebraic equivalence by the group generated by the elements of the form (7.11.1).)

The elements algebraically equivalent to zero form an ideal of $K^\bullet(T)$, as one sees at once. This ideal is moreover stable under the operations λ^i whenever they are defined, that is, when one works with the "naive" $K^\bullet$; it is also contained in the augmentation ideal of $K^\bullet(T)$, as is immediate. One can therefore define a quotient ring

$$K^\bullet_{\mathrm{alg}}(T), \tag{7.11.2}$$

which is even a λ-ring when one works with naive $K^\bullet$-groups.

In the same way one can define a subgroup of elements algebraically equivalent to zero in $K_\bullet(T)$, taking the element E above in $K_\bullet(T \times_k C)$ and observing that

$i_x : T \to T \times_k C$ is a perfect morphism: it is obtained, by flat base change along $T \times_k C \to C$, from the morphism $j_x : x \to C$, which is perfect because C is regular. One thereby obtains a quotient group

$$K_\bullet^{\mathrm{alg}}(T) \tag{7.11.3}$$

of $K_\bullet(T)$. This group is naturally a module over $K^\bullet_{\mathrm{alg}}(T)$, since the definitions show at once that the product of an element of $K^\bullet(T)$ with an element of $K_\bullet(T)$ is algebraically equivalent to zero if either factor is, using (IV 2.12.2) for i_x.

When one works with naive $K^\bullet$, define the ideal in $\mathrm{Gr}^\bullet(T)$ of elements algebraically equivalent to zero to be the ideal generated by the $c^i(x)$, where $i \geq 1$ and $x \in K^\bullet(T)$ is algebraically equivalent to zero. The quotient graded ring

$$\mathrm{Gr}^\bullet_{\mathrm{alg}}(T) \tag{7.11.4}$$

is then precisely the Chern ring associated with the λ-ring $K^{\bullet}_{\mathrm{alg}}(T)$.

Finally, when T is locally noetherian, define the graded subgroup of $\mathrm{Gr}_{\bullet}(T)$ consisting of elements algebraically equivalent to zero by requiring its component of dimension i to be the image in $\mathrm{Gr}_i(T)$ of the subgroup of $\mathrm{Fil}_i\, K_{\bullet}(T)$ formed by the elements algebraically equivalent to zero in $K_{\bullet}(T)$. It follows at once that the quotient graded group

$$\mathrm{Gr}^{\mathrm{alg}}_{\bullet}(T) \tag{7.11.5}$$

is a module over the ring $\mathrm{Gr}^{\bullet}_{\mathrm{alg}}(T)$, by passage to the quotient from the module structure of $\mathrm{Gr}_{\bullet}(T)$ over $\mathrm{Gr}^{\bullet}(T)$ (3.1).

7.11.6. The preceding definitions would still make sense without assuming k algebraically closed, provided that C above is taken geometrically connected over k. It is not clear, however, that these definitions would be very useful. In particular, it does not appear that, for the group $\mathrm{Pic}(T)$, one would

recover the classical "geometric" notion of algebraic equivalence considered in 2.4.1. We therefore prefer to say that an element of $K^{\bullet}(T)$ is algebraically equivalent to zero if its image in $K^{\bullet}(T \otimes_k \bar{k})$ is so, where $\bar{k}$ is an algebraic closure of k, and to proceed in the same way when defining algebraic equivalence in $K_{\bullet}(T)$, $\mathrm{Gr}^{\bullet}(T)$, and $\mathrm{Gr}_{\bullet}(T)$.

7.12. With these definitions in place, I claim that the *specialization homomorphism* (7.3.1) *carries elements algebraically equivalent to zero to elements algebraically equivalent to zero*, and consequently induces, by passage to the quotient, a homomorphism

$$K^{\mathrm{alg}}_{\bullet}(X_t) \longrightarrow K^{\mathrm{alg}}_{\bullet}(X_s). \tag{7.12.1}$$

To see this, one reduces immediately to the case in which A is complete and $k(s)$ is algebraically closed, using the straightforward compatibility made explicit in 7.15. It is enough to show that $\sigma_{\bullet}$ carries an element of the form (7.11.1) in $K_{\bullet}(X_t)$ to an element algebraically equivalent to zero in $K_{\bullet}(X_s)$. Here $T = X_t$, and accordingly we write C_t, E_t in place of C, E above. Since C_t is projective over $k(t)$, EGA IV 2.8.5 gives a projective flat scheme C over S whose generic fiber is C_t. Using Abhyankar's resolution of singularities for arithmetic surfaces [6], one sees that C can even be chosen regular. (N.B. Flatness is preserved under resolution, by EGA IV 14.3.8.) Since C is proper over S, the $k(t)$-rational points x_{0t} and x_{1t} of C_t extend to sections x_0, x_1 of C over S. Since C is regular, it is well known that C is smooth over S at the points of $x_0(S)$ and $x_1(S)$ (EGA IV 17.12.1(c)$\Rightarrow$(a), and 19.1.1). Consequently,

the immersions $x_i : S \to C$ $(i = 0, 1)$ are regular and remain regular immersions $x_i : s \to C_s$ after base change by $s \to S$. Note that, by the exact homotopy sequence (IX 1.1), the element

$$E_t \in K_{\bullet}(X_t \times_{k(t)} C_t)$$

comes from an element $E \in K_{\bullet}(X \times_S C)$. Thus the element $E_t(x_{1t}) - E_t(x_{0t})$ of $K_{\bullet}(X_t)$ comes from the element $E(x_1) - E(x_0)$ of $K_{\bullet}(X)$, and its image in $K_{\bullet}(X_s)$ under (7.3.1) is therefore

$$i^*(E(x_1) - E(x_0)). \tag{*}$$

For a section x of C over S, of course, we set

$$E(x) = i_x^*(E), \qquad \text{where} \qquad i_x : X \longrightarrow X \times_S C$$

is defined as above. It may be viewed as obtained from x by the flat base change $X \times_S C \to C$; it is therefore a regular immersion and, a fortiori, a perfect morphism, so that the expression

$E(x)$ is well defined. For the same reason $E_s(x_s)$ is defined. By transitivity of inverse images (IV 2.12) in the following commutative diagram of perfect morphisms,

$$\begin{tikzcd} X \arrow[r, "i_x"] & X \times_S C \\ X_s \arrow[u, "i"] \arrow[r, "i_{x_s}"] & X_s \times_{k(s)} C_s \arrow[u] \end{tikzcd},$$

the expression $(*)$ is equal to

$$E_s(x_{1s}) - E_s(x_{0s}). \tag{**}$$

Because $k(s)$ is algebraically closed, the reduced irreducible components of C_s are geometrically integral over $k(s)$, and their intersection points are rational over $k(s)$. It follows readily that $(**)$ is algebraically equivalent to zero in $K_\bullet(X_s)$, using the connectedness

of C_s, which follows from the geometric connectedness of C_t.

From (7.12.1), using 7.4, one obtains a homomorphism

$$\mathrm{Gr}^{\mathrm{alg}}_\bullet(X_t) \longrightarrow \mathrm{Gr}^{\mathrm{alg}}_\bullet(X_s). \tag{7.12.2}$$

If X_s and X_t are moreover regular, then (7.12.1) can also be interpreted as a ring homomorphism

$$K^\bullet_{\mathrm{alg}}(X_t) \longrightarrow K^\bullet_{\mathrm{alg}}(X_s), \tag{7.12.3}$$

which, if X is also separated, is a homomorphism of λ-rings and therefore induces a homomorphism

$$\mathrm{Gr}^\bullet_{\mathrm{alg}}(X_t) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{alg}}(X_s). \tag{7.12.4}$$

7.12.5. Let T be a smooth proper algebraic scheme over a field k. It is easy to verify that an element $x \in K^\bullet(T)$ for which there is an integer $n \geq 1$ such that nx is algebraically equivalent to zero is numerically equivalent to zero. It is plausible that the converse is always true, at least when T is projective. In considering this question, one may plainly assume that k is algebraically closed and replace $K^\bullet(T)$ by $\mathrm{Gr}^\bullet(T)$ or $\mathrm{Gr}_\bullet(T)$ without changing the conjecture. One obtains an equivalent conjecture by working with the Chow ring $A(T)$ (cf. XIV 4.1), namely the more or less classical conjecture that, if Z is an algebraic cycle on T numerically equivalent to zero, then there is an integer $n \geq 1$ such that nZ is algebraically equivalent to zero. If this conjecture holds for X_t, the compatibility of (7.3.1) with algebraic equivalence would imply that this specialization homomorphism, when $f : X \to S$ is projective and smooth, is also compatible with numerical equivalence, as conjectured in 7.9.

7.13. Relations with ℓ-adic cohomology

Recall (SGA 5 VII) that, for every locally free module of finite type E on a scheme T, and every prime number ℓ distinct from the residue characteristics of T, one defines ℓ-adic Chern classes (for $i \geq 0$)

$$c^i(E) \in H^{2i}(T, \mathbb{Z}_\ell(i)),$$

satisfying the additivity formula

$$c(E) = c(E')c(E'')$$

when E is an extension of a locally free module E' by a locally free module E''. It follows from this additivity that the total Chern class $c(E) = \sum_i c^i(E)$ arises from a homomorphism

$$c : K^\bullet(T) \longrightarrow \widehat{G}(H^{2*}(T, \mathbb{Z}_\ell(*))) = 1 + \prod_{i \geq 1} H^{2i}(T, \mathbb{Z}_\ell(i)), \tag{7.13.0}$$

where the target is a group under the product explained in V 1, and where $K^\bullet(T)$ is the "naive" $K^\bullet$. One obtains canonical maps

$$c^i : K^\bullet(T) \longrightarrow H^{2i}(T, \mathbb{Z}_\ell(i)). \tag{7.13.1}$$

Now let X be a proper flat scheme over S, with regular fibers, and suppose that S is henselian. Assume that ℓ is different from the residue characteristic of S. One can then form the diagram

$$\begin{tikzcd} K^\bullet(X_t) \arrow[r, "c^i"] \arrow[d, "\sigma"'] & H^{2i}(X_t, \mathbb{Z}_\ell(i)) \\ K^\bullet(X_s) \arrow[r, "c^i"'] & H^{2i}(X_s, \mathbb{Z}_\ell(i)) \arrow[u, "\sigma^{-1}"'] \end{tikzcd}, \tag{7.13.2}$$

where the horizontal arrows are the Chern-class maps for X_t and X_s, respectively, and where σ^{-1} is the composite homomorphism

$$H(X_s) \underset{\sim}{\overset{i^*}{\longleftarrow}} H(X) \xrightarrow{j^*} H(X_t)\,;$$

the arrows displayed here are restriction homomorphisms, and one uses the fact that i^* is an isomorphism by the proper base change theorem (SGA 4 XII 5.5). I claim that the square (7.13.2) is commutative. The proof is essentially the same as the proof of the commutativity of (7.10.2): it is enough to use the compatibility of the formation of Chern classes with inverse images for the morphisms $j : X_t \to X$ and $i : X_s \to X$.

7.13.3. The square (7.13.2) is obtained by passage to the projective limit from the analogous squares in which cohomology with coefficients in the ℓ-adic sheaf $\mathbb{Z}_\ell(i)$ is replaced by cohomology with coefficients in the torsion sheaves $\mu_n^{\otimes i}$, where n runs through the powers of ℓ. Thus the commutativity of (7.13.2) merely expresses the commutativity of the squares involving Chern classes modulo n, for every power n of ℓ.

Choose an algebraic closure $\bar{K}$ of $K = k(t)$, and hence an algebraic closure $\bar{k}$ of $k = k(s)$, namely the residue field of the integral closure of A in $\bar{K}$. For every finite subextension K_α of $\bar{K}$, one can consider the analogous commutative diagram, for fixed n, in the situation obtained from $f : X \to S$ by the base change $S_\alpha \to S$, where S_α is the normalization of S in K_α. By Krull–Akizuki and the henselian hypothesis on S, the scheme S_α is indeed a trait. Here one must moreover assume that the fibers of X over S are geometrically regular, that is, that X is smooth over S, so that this hypothesis is preserved by base change. As α varies, one obtains an inductive system of commutative squares. Passing to the inductive limit and using SGA 4 VII 5.8 and IV 3.2, one obtains a commutative square (cf. 7.16):

$$\begin{tikzcd} K^\bullet(X_{\bar{t}}) \arrow[r, "c^i"] \arrow[d, "\sigma"'] & H^{2i}(X_{\bar{t}}, \mu_n^{\otimes i}) \arrow[d, "\sigma'"] \\ K^\bullet(X_{\bar{s}}) \arrow[r, "c^i"'] & H^{2i}(X_{\bar{s}}, \mu_n^{\otimes i}) \end{tikzcd},$$

where $X_{\bar{t}}$ and $X_{\bar{s}}$ are the geometric fibers of X. Moreover, the specialization homomorphism σ^{-1} for cohomology is now known to be an isomorphism (SGA 4 XVI 2.2). Hence the commutativity of the preceding square can also be expressed as the commutativity of the square obtained by reversing that arrow. Passing to the projective limit over n in the resulting commutative squares gives a commutative square

$$
\begin{tikzcd}
K^{\bullet}(X_{\bar{t}}) \arrow[r] \arrow[d, "\sigma"'] & H^{2i}(X_{\bar{t}}, \mathbb{Z}_{\ell}(i)) \arrow[d, "\sim"'] \\
K^{\bullet}(X_{\bar{s}}) \arrow[r] & H^{2i}(X_{\bar{s}}, \mathbb{Z}_{\ell}(i))
\end{tikzcd}
\quad . \tag{7.13.4}
$$

7.13.5. Another way to express this commutativity is to say that, if $x_t \in K^{\bullet}(X_t)$ and x_s is its image in $K^{\bullet}(X_s)$, then $c^i(x_{\bar{t}})$ and $c^i(x_{\bar{s}})$ are the values at the geometric points $\bar{t}$ and $\bar{s}$, respectively, of a section of $R^{2i}f_*(\mathbb{Z}_{\ell}(i))$. This section is easily interpreted: take an element $x \in K^{\bullet}(X)$ inducing x_t. Then $c^i(x) \in H^{2i}(X, \mathbb{Z}_{\ell}(i))$ defines the required section of $R^{2i}f_*(\mathbb{Z}_{\ell}(i))$.

7.13.6. Let T be a scheme. By passage to the associated graded groups, the homomorphism (7.13.0) induces homomorphisms

$$\psi_i : \mathrm{Gr}^i(T) \longrightarrow H^{2i}(T, \mathbb{Z}_{\ell}(i)),$$

the image under ψ_i of an element $x \in \mathrm{Filt}^i K^{\bullet}(T)$ being, by definition, $c^i(x)$. (The compatibility of (7.13.0) with the λ-filtration on $K^{\bullet}(T)$ and the degree filtration on the target follows readily from V 6.6.4 and from the fact that the c^i define a Chern theory on $K^{\bullet}(T)$ in the sense of V 6.13, when $K^{\bullet}(T)$ is regarded as a λ-algebra augmented over the binomial ring $H^0(T, \mathbb{Z})$.) One thereby obtains maps

$$\mathrm{cl} : \mathrm{Gr}^i(T) \longrightarrow H^{2i}(T, \mathbb{Q}_{\ell}(i)) \overset{\mathrm{dfn}}{=} H^{2i}(T, \mathbb{Z}_{\ell}(i)) \otimes_{\mathbb{Z}_{\ell}} \mathbb{Q}_{\ell}, \tag{7.13.7}$$

by the formula

$$\mathrm{cl}(x^{(i)}) = \frac{1}{(-1)^{i-1}(i-1)!}\, \psi_i(x^{(i)}), \tag{7.13.8}$$

compare XIV 5.1 and 5.2. The commutativity of (7.13.2) and (7.13.4) immediately implies that of the analogous diagrams

$$
\begin{tikzcd}
\mathrm{Gr}^i(X_t)_{\mathbb{Q}} \arrow[r, "\mathrm{cl}"] \arrow[d, "\sigma_{\mathbb{Q}}"'] & H^{2i}(X_t, \mathbb{Q}_{\ell}(i)) \\
\mathrm{Gr}^i(X_s)_{\mathbb{Q}} \arrow[r, "\mathrm{cl}"'] & H^{2i}(X_s, \mathbb{Q}_{\ell}(i)) \arrow[u, "\sigma'"']
\end{tikzcd}
\quad , \tag{7.13.9}
$$

and

$$
\begin{tikzcd}
\mathrm{Gr}^i(X_{\bar{t}})_{\mathbb{Q}} \arrow[r, "\mathrm{cl}"] \arrow[d] & H^{2i}(X_{\bar{t}}, \mathbb{Q}_{\ell}(i)) \arrow[d, "\sim"] \\
\mathrm{Gr}^i(X_{\bar{s}})_{\mathbb{Q}} \arrow[r, "\mathrm{cl}"'] & H^{2i}(X_{\bar{s}}, \mathbb{Q}_{\ell}(i))
\end{tikzcd}
\quad . \tag{7.13.10}
$$

Remark 7.13.11. The commutativity of the last two diagrams can be expressed by saying that specialization of algebraic cycles is compatible with formation of the associated ℓ-adic cohomology classes. This statement is not very satisfactory in the form just given, because we had to replace the coefficients $\mathbb{Z}_{\ell}(i)$ by $\mathbb{Q}_{\ell}(i)$, that is, to neglect torsion. To obtain a statement that does not neglect torsion, it seems that one will be forced to work with the Chow ring, since it is on that ring that the homomorphism cl is naturally defined with values in $H^{2*}(-, \mathbb{Z}_{\ell}(*))$; compare XIV 4.1. Unfortunately, no theory of specialization for the Chow ring has yet been developed (cf. 7.14) that would allow one to write commutative squares analogous to (7.13.9) and (7.13.10) in terms of Chow rings and cohomology with coefficients in $\mathbb{Z}_\ell(*)$. All one can say when $f : X \to S$ is smooth and proper is that there is a commutative square

$$\begin{array}{ccc} Z^i(X_t) & \xrightarrow{\mathrm{cl}} & H^{2i}(X_t, \mathbb{Z}_\ell(i)) \\ {\scriptstyle\sigma}\downarrow & & \uparrow \\ Z^i(X_s) & \xrightarrow[\mathrm{cl}]{} & H^{2i}(X_s, \mathbb{Z}_\ell(i)) \end{array}, \tag{7.13.12}$$

analogous to (7.13.9), together with a corresponding square analogous to (7.13.10), which we leave to the reader to write down. Here Z^i denotes the group of algebraic cycles of codimension i, modulo no relation at all; the horizontal arrows are the associated-cohomology-class homomorphisms defined in SGA 5 IV; and σ is the specialization homomorphism for algebraic cycles, associating with a cycle Z_t the cycle Z_s, where Z is the schematic closure of Z_t in X (cf. 7.14). This commutativity follows from the following fact, proved in SGA 5 IV. If Z is a closed subscheme of X of codimension at least i on every fiber, one can associate with it a cohomology class

$$\mathrm{cl}(Z) \in H^{2i}(X, \mathbb{Z}_\ell(i))$$

whose formation commutes with every base change and whose restriction to X_t (respectively, X_s) is therefore $\mathrm{cl}(Z_t)$ (respectively, $\mathrm{cl}(Z_s)$). Of course, once it is known that the homomorphism σ in (7.13.12) carries cycles rationally equivalent to zero to cycles rationally equivalent to zero, with X projective over S, one can replace the Z^i by the A^i in this diagram and obtain a diagram of the desired kind in terms of the Chow rings of X_t and X_s.

Remark 7.13.13. Let T be a smooth projective scheme over an algebraically closed field. It follows immediately from the formalism developed in SGA 5 IV that, if Z is a cycle of codimension i on T whose image in $H^{2i}(T, \mathbb{Q}_\ell(i))$ is zero, then Z is numerically equivalent to zero. One of the standard conjectures on algebraic cycles asserts that the converse is also true. By VII 4.11, one obtains an equivalent conjecture by replacing the group of algebraic cycles with $\mathrm{Gr}^\bullet(T)_\mathbb{Q}$, using the definition (7.13.8), or with $K^\bullet(T)$, by stating that an element of $K^\bullet(T)$ is numerically equivalent to zero if and only if its augmentation and its Chern classes in the groups $H^{2i}(T, \mathbb{Q}_\ell(i))$ vanish.

It follows immediately from the commutativity of (7.13.10) that, if $X_{\bar{t}}$ satisfies the preceding conjecture, then the specialization homomorphism

$$\mathrm{Gr}^\bullet(X_{\bar{t}}) \longrightarrow \mathrm{Gr}^\bullet(X_{\bar{s}})$$

(and, a fortiori, $\mathrm{Gr}^\bullet(X_t) \to \mathrm{Gr}^\bullet(X_s)$) is compatible with numerical equivalence, as conjectured in 7.9. The conjecture used here (numerical equivalence implies $\mathbb{Q}_\ell$-cohomological equivalence) is weaker than the conjecture considered in 7.12.5.

7.14. Specialization homomorphism for groups of cycles and cycle classes

If T is a noetherian scheme, denote by $Z_i(T)$ (respectively, $Z^i(T)$) the group of algebraic cycles on T that are purely of dimension i (respectively, purely of codimension i). When $f : X \to S$ is of finite type, one defines a specialization homomorphism

$$\sigma_i : Z_i(X_t) \longrightarrow Z_i(X_s) \tag{7.14.1}$$

as follows. For every prime cycle Z_t of X_t, purely of dimension i and identified with an integral subscheme of X_t, define $\sigma_i(Z_t)$ to be the cycle associated with the closed subscheme Z_s of X_s, where Z is the schematic closure of Z_t in X. This Z is a closed subscheme of X flat over S

(EGA IV 2.8), with generic fiber equal to the original Z_t, and it satisfies $\dim Z_s = \dim Z_t$ (EGA IV 7.1.13). When X_s is locally equidimensional, that is, when all irreducible components passing through a given point have the same dimension, the specialization homomorphism just defined immediately induces homomorphisms

$$\sigma^i : Z^i(X_t) \longrightarrow Z^i(X_s). \tag{7.14.1 bis}$$

Now suppose that f is quasi-projective, so that rational equivalence of cycles is available on X_t and X_s (cf. [5], where the theory is developed over an algebraically closed field). It seems very plausible that (7.14.1) carries cycles rationally equivalent to zero to cycles rationally equivalent to zero and therefore induces a corresponding homomorphism on the "Chow" groups of cycle classes,

$$\sigma_i : A_i(X_t) \longrightarrow A_i(X_s), \tag{7.14.2}$$

or, when X_s is locally equidimensional,

$$\sigma^i : A^i(X_t) \longrightarrow A^i(X_s). \tag{7.14.2 bis}$$

One might try to prove this by a method paralleling that of 7.12, but here resolution of singularities and reduction to an algebraically closed residue field would be unnecessary, since $C = \mathbb{P}^1_S$ is available as a "parameter curve" along which to deform cycles. A difficulty nevertheless arises. With the notation of 7.12, E must now denote a cycle on $\mathbb{P}^1_S \times_S X$ for which $E_t(x_{1t}) - E_t(x_{0t})$ is "defined," but it does not follow that the cycle $E_s(x_{1s}) - E_s(x_{0s})$ is also defined. As was to be expected, one is thus reduced to formulating a statement of the kind of Chow's moving lemma in order to define (7.14.2). Observe, however, that the existence of (7.4.1) shows that (7.14.1) is compatible with the equivalence relation on cycles defined by the surjective homomorphism

$$Z_\bullet(T) \longrightarrow \mathrm{Gr}_\bullet(T),$$

which is only slightly coarser than linear equivalence. Moreover, if T is quasi-projective and smooth over a field, this relation differs from linear equivalence only by torsion (XIV 4.1, 4.2). In that case one therefore obtains a homomorphism analogous to (7.14.2), or better to (7.14.2 bis), with both sides tensored by $\mathbb{Q}$.

The most natural way to define (7.14.2) directly would seem to be to proceed as in 7.3. This would require a theory of Chow groups A_i (or of the Chow ring in the regular case) for schemes more general than schemes quasi-projective over a field, broad enough to include X, together with a contravariant law for these groups making it possible to define the restriction homomorphism to X_s,

$$A_i(X) \longrightarrow A_i(X_s).$$

Compare XIV 8. Once such a sufficiently general theory of Chow groups and rings is available, it should be possible to transcribe into Chow theory all the results stated above for the specialization homomorphism in K-theory.

7.15. Varying S

Let S' be a second trait, and let

$$S' \longrightarrow S$$

be a morphism of traits, that is, a dominant local morphism, carrying the closed point to the closed point and the generic point to the generic point. Denote the closed point (respectively, the generic point) of S' by s' (respectively, t'), and set $X' = X \times_S S'$. Suppose that X' is noetherian, as is the case, for example, when X is of finite type over S. There is a commutative diagram

$$\begin{tikzcd}
X & X_t \arrow[l, hook', "j_X"'] & & \\
& & X' \arrow[ull] & X'_{t'} \arrow[l, hook', "j_{X'}"'] \arrow[ull] \\
X_s \arrow[uu, hook, "i_X"] & & & \\
& & X'_{s'} \arrow[ull] \arrow[uu, hook, "i_{X'}"'] &
\end{tikzcd},$$

in which the oblique arrows are flat morphisms. One can therefore take the inverse-image homomorphisms on $K_\bullet$ for every arrow of the diagram (IV 2.12). Applying Definition 7.3, one finds that the diagram

$$\begin{tikzcd}
K_\bullet(X_t) \arrow[r, "\sigma_X"] \arrow[d] & K_\bullet(X_s) \arrow[d] \\
K_\bullet(X'_{t'}) \arrow[r, "\sigma_{X'}"'] & K_\bullet(X'_{s'})
\end{tikzcd} \tag{7.15.1}$$

is commutative, where the vertical arrows are the inverse-image homomorphisms. One immediately obtains an analogous commutative diagram with $\mathrm{Gr}_\bullet$ in place of $K_\bullet$. When X_s and X_t are regular, one may likewise write $K^\bullet$ and $\mathrm{Gr}^\bullet$ in place of $K_\bullet$ and $\mathrm{Gr}_\bullet$.

7.16. Passage to geometric fibers

Let $\bar{K}$ be an algebraic closure of $K = k(t)$, expressed as the inductive limit of its finite subextensions K_i, and let $\bar{A} = \varinjlim_i \bar{A}_i$ be the normalization of A in $\bar{K}$, the inductive limit of its normalizations $\bar{A}_i$ in the K_i. Choose a maximal ideal $\bar{\mathfrak{m}}$ of $\bar{A}$; it induces a maximal ideal $\mathfrak{m}_i$ on each $\bar{A}_i$. Set $B_i = (\bar{A}_i)_{\mathfrak{m}_i}$. By Krull–Akizuki, B_i is again a discrete valuation ring. Denote its spectrum by S_i and its closed and generic points by s_i and t_i, respectively. The residue field $\bar{k} = \bar{A}/\bar{\mathfrak{m}}$ is an algebraic closure of $k = k(s)$ and is the inductive limit of the residue fields $k_i = \bar{A}_i/\mathfrak{m}_i = k(s_i)$ of the B_i. Set $\bar{t} = \mathrm{Spec}(\bar{K})$ and $\bar{s} = \mathrm{Spec}(\bar{k})$.

Assume now that X is of finite type over S. Then $X_{\bar{t}}$ and $X_{\bar{s}}$ are of finite type over $\bar{t}$ and $\bar{s}$, respectively, and hence are noetherian. By IV 3.2.3,

$$K_\bullet(X_{\bar{t}}) \xleftarrow{\sim} \varinjlim_i K_\bullet(X_{t_i}), \qquad K_\bullet(X_{\bar{s}}) \xleftarrow{\sim} \varinjlim_i K_\bullet(X_{s_i}).$$

On the other hand, 7.15 gives a homomorphism from the inductive system of the $K_\bullet(X_{t_i})$ to that of the $K_\bullet(X_{s_i})$. Passing to the limit gives a canonical homomorphism

$$K_\bullet(X_{\bar{t}}) \longrightarrow K_\bullet(X_{\bar{s}}), \tag{7.16.1}$$

again, as expected, called the *specialization homomorphism*. By 7.4, it induces a homomorphism $\mathrm{Gr}_\bullet(X_{\bar{t}}) \to \mathrm{Gr}_\bullet(X_{\bar{s}})$, using the fact (whose verification is left to the reader) that the functor $\mathrm{Gr}_\bullet$, like $K_\bullet$, commutes with the limits occurring here. Similarly, when f is smooth, one may interpret (7.16.1) as a homomorphism on $K^\bullet$; it is then a ring homomorphism, and even a λ-ring homomorphism if X is separated. In that case, it therefore induces a homomorphism on $\mathrm{Gr}^\bullet$.

7.16.2. We shall not review here, for geometric fibers, the extension of all the properties of specialization homomorphisms proved in the preceding sections for the "arithmetic" fibers; they all follow immediately from the latter by passage to the limit. Let us mention only the following interesting consequence of 7.9.4. When f is smooth and proper, $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_{\bar{t}})$ is isomorphic to a quotient of a subgroup of $\mathrm{Gr}_\bullet^{\mathrm{num}}(X_{\bar{s}})$; in particular, its rank is bounded above by that of the latter group. This is therefore a semi-continuity result for the rank of the group of cycles modulo numerical equivalence.

7.17. Case of an arbitrary base

Let

$$f : X \longrightarrow S$$

be a flat and quasi-compact morphism of noetherian preschemes, and let s, t be two points of S such that s is a specialization of t. We wish to relate the groups $K_\bullet$, etc., for the two fibers X_t and X_s, as well as for the corresponding geometric fibers. For this, note (EGA II 7.1.9) that there exists a morphism

$$g : S' \longrightarrow S,$$

where S' is a trait and g carries the closed point s' to s and the generic point t' to t; one may even assume that $k(t) \xrightarrow{\sim} k(t')$. Suppose that $X' = X \times_S S'$ is noetherian. This is the case if f is of finite type or if S' is essentially of finite type over S (EGA IV 1.3.8), as one may assume when S itself is universally Japanese (EGA 0_{IV} 23.1.1). The specialization homomorphism for X' then defines a homomorphism

$$\sigma_\bullet : K_\bullet(X_t) \xrightarrow{\sim} K_\bullet(X_{t'}) \longrightarrow K_\bullet(X_{s'}). \tag{7.17.1}$$

Thus one obtains a specialization homomorphism on $K_\bullet(X_t)$, but with values not in $K_\bullet(X_s)$, but in $K_\bullet(X_{s'})$, where $X_{s'}$ is obtained from X_s by an extension $k(s) \to k(s')$ of the base fields (which may be assumed to be of finite type if S is universally Japanese). Passing to geometric fibers as explained in 7.16, one likewise obtains a homomorphism

$$\sigma_\bullet : K_\bullet(X_{\bar{t}}) \longrightarrow K_\bullet(X_{\overline{s'}}). \tag{7.17.2}$$

Here again, the "geometric" specialization homomorphism takes its values not in $K_\bullet(X_{\bar{s}})$, but in $K_\bullet(X_{\overline{s'}})$, where $X_{\overline{s'}}$ is obtained from $X_{\bar{s}}$ by the extension of algebraically closed fields $k(\bar{s}) \to k(\overline{s'})$.

7.17.3. In general, the homomorphism $K_\bullet(X_{\bar{s}}) \to K_\bullet(X_{\overline{s'}})$ is not an isomorphism, because of the possible presence of "continuous" components in $K_\bullet(X_{\bar{s}})$. It is, however, easy to verify that it is an isomorphism modulo algebraic equivalence (7.11), and also modulo numerical equivalence (4.5.1) if f is smooth and proper. Consequently, using 7.12, one finds that (7.17.2) induces a specialization homomorphism

$$K_\bullet^{\mathrm{alg}}(X_{\bar{t}}) \longrightarrow K_\bullet^{\mathrm{alg}}(X_{\bar{s}}); \tag{7.17.3.1}$$

and, if f is proper and smooth, provided that the specialization homomorphism (7.17.2) is compatible with numerical equivalence (cf. 7.9.4), one likewise obtains a homomorphism

$$K_\bullet^{\mathrm{num}}(X_{\bar{t}}) \longrightarrow K_\bullet^{\mathrm{num}}(X_{\bar{s}}). \tag{7.17.3.2}$$

We leave to the reader the task of developing, if needed, the corresponding constructions for $\mathrm{Gr}_\bullet$, $K^\bullet$, and $\mathrm{Gr}^\bullet$, and the properties of these specialization homomorphisms that follow immediately from those developed in the preceding sections. Let us mention only that, by 7.16.2, $\mathrm{Gr}^{\mathrm{num}}_\bullet(X_{\bar{t}})$ is isomorphic to a quotient of a subgroup of $\mathrm{Gr}^{\mathrm{num}}_\bullet(X_{\bar{s}})$ when f is proper and smooth; in particular, its rank is again bounded above by the rank of the latter group.

7.17.4. When S is regular at s and t is a maximal point, one can construct S' by blowing up s in the localization of S at s, and then localizing at the generic point of the fiber of s in the blow-up. One then obtains a *pure extension* $k(s')$ of $k(s)$. Now it is not difficult to show that, if k' is a pure extension of a field k, then for every noetherian scheme T over k the canonical homomorphism

$$K_\bullet(T) \longrightarrow K_\bullet(T \otimes_k k')$$

is an isomorphism: represent k' as the inductive limit of the coordinate rings of nonempty affine open subsets of schemes of the form $\operatorname{Spec} k[T_1, \dots, T_n]$, and use IX 1.6. It follows that (7.17.1) may be viewed as a canonical homomorphism

$$K_\bullet(X_t) \longrightarrow K_\bullet(X_s). \tag{7.17.4.1}$$

One should note, however, that this does not yield a homomorphism $K_\bullet(X_{\bar{t}}) \to K_\bullet(X_{\bar{s}})$: after passing to a finite extension K_i of $K = k(t)$ and normalizing S in K_i, the regularity hypothesis on S is lost.

Another way to define (7.17.4.1) is to reduce to the case in which S is local with closed point s (which is immediate), and to consider an increasing sequence of regular closed subschemes S_i of S, with generic points t_i, such that

$$\dim S_i = i, \qquad 0 \leq i \leq n = \dim S.$$

For every i, $0 \leq i \leq n-1$, the local ring $\mathcal{O}_{S_{i+1},t_i}$ is a discrete valuation ring with residue field $k(t_i)$ and fraction field $k(t_{i+1})$. There is therefore a specialization homomorphism

$$K_\bullet(X_{t_{i+1}}) \longrightarrow K_\bullet(X_{t_i}),$$

and the composite of these homomorphisms for $0 \leq i \leq n-1$ defines (7.17.4.1). By an argument analogous to that of 7.3, one can show that this is the same homomorphism as the one defined above by means of a blow-up. The advantage of the second method is that one even has homomorphisms of the form (7.16.1):

$$K_\bullet(X_{\bar{t}_{i+1}}) \longrightarrow K_\bullet(X_{\bar{t}_i}),$$

with suitable coherent choices of algebraic closures $k(\bar{t}_i)$ of the $k(t_i)$. Their composite defines a homomorphism

$$K_\bullet(X_{\bar{t}}) \longrightarrow K_\bullet(X_{\bar{s}}), \tag{7.17.4.2}$$

which probably depends essentially on the choice of the S_i, that is, of the t_i.

7.17.5. The construction of a specialization homomorphism (7.17.4.2) can be generalized to the case in which S is no longer assumed regular at s and t is no longer assumed to be a maximal point, provided that resolution of singularities is available for the integral closed subscheme T of S with generic point t, or merely for its localization at s. This hypothesis holds, in particular, if $\mathcal{O}_{S,s}$ is excellent of characteristic zero, by Hironaka. Indeed, one can then find a morphism

$$S' \longrightarrow \operatorname{Spec}(\mathcal{O}_{T,s})$$

with S' regular. Taking a closed point of the fiber S'_s, one can apply the preceding construction to $X' = X \times_S S'$ over S', its generic point t', and its closed point s'. This yields a homomorphism of the form (7.17.4.2), since $k(s')$ is a finite extension of $k(s)$ and therefore has the same algebraic closure.

7.17.6. In all the constructions considered in 7.17, one has reduced the construction of a specialization homomorphism to the case in which the base is a trait. It follows that the properties of the specialization homomorphism reviewed in the preceding sections for a trait as base extend automatically to the case of a more general base. Of course, the methods of definition considered apply equally well to $\mathrm{Gr}_\bullet$, to $K^\bullet$ when f is smooth, and to $\mathrm{Gr}^\bullet$ when f is also separated.

7.17.7. Finally, note that the definition of a homomorphism (7.17.3.1) or (7.17.3.2), or of a homomorphism (7.17.4.2), depends essentially on certain arbitrary choices: a suitable trait over S in 7.17.3, a sequence (S_i) in 7.17.4, and, in addition, the choice of a resolution of singularities in 7.17.5. The same is true of the variants for $\mathrm{Gr}_\bullet$, etc. Obvious examples can be constructed with s, say, an ordinary double point of an algebraic curve and X projective and smooth over S (for example, an abelian scheme of relative dimension 2), in which the specialization homomorphism on $\mathrm{Gr}^1 = \mathrm{Pic}$, and even on $\mathrm{Gr}^1_{\mathrm{alg}} = \mathrm{NS}$ (the Néron–Severi group), actually depends on the trait S' chosen over S, or, what amounts to the same thing here, on the choice of a point above s in the normalization of S.

7.18. Appendix to the Appendix: base-field extension for certain functors with values in commutative groups

We spell out here a technical point raised in 7.17.4.

7.18.1. Let k be a field, let C be the category of integral k-algebras and flat homomorphisms of such algebras, and let F be a functor on C with values in the category (Ens) of sets. Suppose that F commutes with filtered inductive limits. Given an extension K of k, we seek criteria for

$$F(k) \longrightarrow F(K)$$

to be injective (respectively, surjective).

7.18.2. Suppose that one of the following two conditions holds:

a) The field k is separably closed, and K/k is a separable extension.

b) The field k is infinite, and K is a pure extension of k.

Then the map $F(k) \to F(K)$ is injective.

Indeed, K is the filtered inductive limit of its finite-type k-subalgebras, with flat transition morphisms. Using this fact and the commutation of F with filtered inductive limits, one is reduced in case a) to proving the following: if A is an integral k-algebra of finite type whose fraction field K is separable over k, then $F(k) \to F(A)$ is injective. The hypotheses on k and K imply that there is a k-homomorphism $A \to k$; the composite

$$F(k) \longrightarrow F(A) \longrightarrow F(k)$$

is therefore the identity, and the assertion follows. In case b), passage to the limit similarly reduces the question to an integral finite-type k-algebra A whose fraction field is a pure extension of k. Since k is infinite, the hypothesis again implies the existence of a k-homomorphism $A \to k$, and one concludes as above.

7.18.3. Suppose now that F takes values in the category (Ab) of abelian groups, and suppose that for every finite extension k' of k a norm homomorphism

$$N_{k'/k} : F(k') \longrightarrow F(k)$$

has been defined, such that the composite

$$F(k) \longrightarrow F(k') \xrightarrow{N_{k'/k}} F(k)$$

is multiplication by $[k':k]$. Under these conditions:

(i) for every extension K of k, the kernel of $F(k) \to F(K)$ is a torsion group;

(ii) for every pure extension K of k, the homomorphism $F(k) \to F(K)$ is injective. (Note that k is not assumed infinite.)

For (i), it is enough to prove that every element x in the kernel of $F(k) \to F(A)$, where A is an integral k-algebra of finite type, is a torsion element. Using a k-homomorphism $A \to k'$, where k' is a finite k-algebra, one reduces to the case in which $A = k'$ is a finite extension of k. The norm property above then gives $[k':k]x = 0$, proving that x is torsion. One even has $x = 0$ when the greatest common divisor of the degrees of the finite extensions k' of k that are quotients of A is 1. This is the case when the fraction field K of A is a pure extension of k, as is easily checked (only the case in which k is finite requires a new verification). This proves (ii).

7.18.4. Suppose that, for every nonempty affine open subset U of the affine space $\operatorname{Spec} k[T_1, \dots, T_n]$, if A is the ring of U, the homomorphism $F(k) \to F(A)$ is surjective (compare IX 1.6). Using the commutation of F with filtered inductive limits, one then finds that $F(k) \to F(K)$ is surjective for every pure extension K of k. If, in addition, k is infinite, or if norm homomorphisms can be defined as in 7.18.3, this homomorphism is therefore bijective.

7.18.5. As an example, let X be a scheme of finite type over k, and set

$$F(A) = K_\bullet(X \otimes_k A),$$

which is a functor on the category C by IV 2.12. By IV 3.2.3, it commutes with filtered inductive limits. Moreover, if k' is a finite extension of k, the projection

$$p : X \otimes_k k' \longrightarrow X$$

is finite, flat, and of finite presentation, and therefore defines a direct image homomorphism

$$p_* = N_{k'/k} : K_\bullet(X \otimes_k k') \longrightarrow K_\bullet(X).$$

The condition on the norm homomorphisms in 7.18.3 is immediately verified. Finally, if A is the affine algebra of an open subset U of the affine space $\operatorname{Spec} k[T_1, \dots, T_n]$, then $F(k) \to F(A)$ identifies with the map

$$K_\bullet(X) \longrightarrow K_\bullet(X \times_k U),$$

which is surjective for X noetherian, by IX 1.6. The results proved above therefore give:

(i) If K/k is a pure extension of k, then $K_\bullet(X) \to K_\bullet(X_K)$ is injective, and is even bijective if X is noetherian.

(ii) If k is separably closed and K/k is a separable extension, then $K_\bullet(X) \to K_\bullet(X_K)$ is injective.

(iii) For every extension K of k, the kernel of $K_\bullet(X) \to K_\bullet(X_K)$ is a torsion group.

These results remain valid if one assumes only that X is noetherian, provided one restricts to extensions K of k of finite type (so that X_K is still noetherian). One obtains essentially the same results for $K^\bullet$ in place of the functor $K_\bullet$, with the sole difference that in (i) bijectivity cannot be asserted in general, because there is no homotopy theorem for $K^\bullet$ in place of $K_\bullet$. To conclude bijectivity, one would have to assume that X is noetherian and regular; then the groups $K^\bullet$ and $K_\bullet$ agree for X and, respectively, for X_K.

One obtains essentially the same results for $\mathrm{Gr}^\bullet$ in place of $K_\bullet$ or $K^\bullet$, with the same reservation concerning the bijectivity assertion in (i). If one restricts to $\mathrm{Gr}^1 = \mathrm{Pic}$, that assertion is valid when X is noetherian and normal: in that case, for every open subset U of an affine space over k, one sees at once that

$$\mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X \times_k U)$$

is surjective (EGA $\mathrm{Err}_{\mathrm{IV}}$ 53, 21.4.13).

Bibliography

[1] Dickson, E. *Linear groups*, Dover, New York, 1958.

[2] Grothendieck, A. TDTE, V, VI; Séminaire Bourbaki 232, 236.

[3] Kleiman, S. *Towards a numerical theory of ampleness*, Ann. of Math. 84 (1966), pp. 293–344.

[4] Murre, J. *On contravariant functors from the category of preschemes over a field into the category of abelian groups*, Publications Mathématiques de l'IHES 23 (1964).

[5] Chevalley, C. *Les classes d'équivalence rationnelles, I et II*, Séminaire à l'ENS 1958, exposés 2 et 3.

[6] S. Abhyankar, *Resolution of singularities of arithmetical surfaces*, in: *Arithmetical Algebraic Geometry*, Purdue University, Harper–Row, New York, 1965.

Exposé XII. A relative representability theorem for the Picard functor

by M. Raynaud (written up by S. Kleiman)

Warning

The purpose of Exposés XII and XIII is to prove the nine finiteness theorems for the Picard functor stated by A. Grothendieck in the comments (pp. C 07–C 11) to his *Fondements de la Géométrie Algébrique* (cited as FGA). In the first exposé, the first finiteness theorem is proved in the following form, which is more precise than the statement in FGA. Let X and Y be proper schemes over an integral noetherian base S, and let $f : X \to Y$ be a surjective S-morphism. Then there is a nonempty open subset V of S over which the induced morphism $f^* : \mathrm{Pic}_{Y/S} \to \mathrm{Pic}_{X/S}$ is representable by a quasi-affine morphism of finite presentation. In the second exposé, the remaining assertions are proved.

The two exposés differ notably in their proof techniques. In the first, Oort's dévissage, non-flat descent, and the representability of unramified functors are used in an essential way. In the second, one first reduces to the case in which the schemes X/S under consideration are projective and flat with geometrically integral fibers, or, when necessary, normal or smooth, always by applying the first finiteness theorem to a suitable morphism $X' \to X$. One then uses the tool of (b)-sheaves, forged by I. Castelnuovo and sharpened and used by T. Matsusaka, D. Mumford, Y. Nakai, and S. Kleiman. In particular, this tool makes it possible to dispense with Chow coordinates, the Frobenius morphism, and the Lefschetz theorems used in Grothendieck's original proofs.

The present Exposé XII was presented orally by M. Raynaud following Grothendieck; the editor wishes to thank him for communicating his personal notes.

1. Statement of the main theorem and applications

Theorem 1.1. Let S be a noetherian scheme, let X and Y be two proper S-schemes, and let $f : X \to Y$ be a morphism. Suppose that S is integral and that f is surjective. Then there is a nonempty open subset V of S such that the morphism

$$f^* : \mathrm{Pic}_{Y/S}|_V \longrightarrow \mathrm{Pic}_{X/S}|_V$$

between the Picard functors (localized for the fppf topology) is representable by a quasi-affine morphism of finite presentation.

Corollary 1.2. Let X be a proper scheme over a noetherian scheme S, and suppose that S is integral. Then there is a nonempty open subset V of S such that the Picard functor $\mathrm{Pic}_{X/S}|_V$ is representable by a V-group scheme that is a sum of quasi-projective open subschemes of finite type, and hence, *a fortiori*, is separated over V.

Indeed, recall the following fact (FGA no. 232). Let S be a scheme and let X be a projective S-scheme, of finite presentation and integral over S (that is, flat with geometrically integral fibers). Then the Picard functor is represented by an S-group scheme $\mathrm{Pic}_{X/S}$, an increasing union of quasi-projective open subschemes of finite presentation. Consideration of the "Hilbert polynomials" decomposes $\mathrm{Pic}_{X/S}$ here as a sum of open subschemes $\mathrm{Pic}^q_{X/S}$, $q \in \mathbb{Q}[t]$. In the following exposé it is proved, under the additional hypothesis that S is noetherian, that the

$\mathrm{Pic}^q_{X/S}$ are of finite type over S. Thus 1.2 is known when X is in addition projective and integral over S; the general case will be deduced using the following lemma.

Lemma 1.3. Let X be of finite type over an integral noetherian scheme S. Then there are a nonempty open subset V of S, an integral scheme S', and a finite flat morphism $S' \to V$ such that every irreducible component X'_i of $X_{S'}$, endowed with the induced reduced structure, is integral over S'.

Indeed, there is a finite extension L of the function field $k(S)$ such that every irreducible component of $(X_L)_{\mathrm{red}}$ is geometrically integral (EGA IV 4.6.8). Let S_1 be an integral S-scheme such that $k(S_1) = L$ and such that there are a nonempty open subset V_1 of S and a finite flat morphism $h : S_1 \to V_1$. There is a nonempty open subset S_2 of S_1 such that every irreducible component of X_{S_2}, endowed with the induced reduced structure, is integral over S_2 (EGA IV 9.7.7). Set

$$V = V_1 \setminus h(S_1 \setminus S_2), \qquad S' = h^{-1}(V).$$

We now prove 1.2. Let $h : S' \to V_1$ and the X'_i be as in 1.3. For each i, let $f_i : Z_i \to X'_i$ be a Chow presentation, with Z_i integral and f_i birational. After shrinking V and S', one may suppose that Z_i is integral over S' (EGA IV 9.7.7). Let

$$f : Z = \coprod_i Z_i \longrightarrow X' = X_{S'}$$

be the canonical morphism. Then $\mathrm{Pic}_{Z/S'} = \prod_i \mathrm{Pic}_{Z_i/S'}$ is representable by a group scheme that is a sum of quasi-projective open subschemes of finite type. Finally, 1.1 says that, after shrinking V and S', the morphism

$$f^* : \mathrm{Pic}_{X'/S'} \longrightarrow \mathrm{Pic}_{Z/S'}$$

is representable by a quasi-affine morphism. Consequently, $\mathrm{Pic}_{X'/S'}$ is representable by a group scheme that is a sum of quasi-projective open subschemes of finite type; by finite flat descent (SGA 1 VIII 7.6 and EGA II 6.6.1), the same is true of $\mathrm{Pic}_{X/S}|_V$.

Lemma 1.4. Let k be a field, let P and Q be k-group schemes locally of finite type, and let $f : P \to Q$ be a quasi-affine morphism. Then f is affine.

Indeed, let $R = \mathrm{Ker}(f)$. Then R is a quasi-affine k-group scheme of finite type; by SGA 3 $\mathrm{VI_B}$ 11.11, R is affine. Thus $P \to P/R$, which makes P an R-torsor over P/R, is affine. Moreover, by SGA 3 I 4.2, $P/R \to Q$ is a closed immersion, hence affine; therefore the composite $P \to P/R \to Q$ is affine.

Corollary 1.5. Let k be a field.

a) If X is a proper k-scheme, then $\mathrm{Pic}_{X/k}$ is representable (J. P. Murre).

b) If $f : X \to Y$ is a surjective morphism between two proper k-schemes, then the induced morphism

$$f^* : \mathrm{Pic}_{Y/k} \longrightarrow \mathrm{Pic}_{X/k}$$

is affine.

Corollary 1.6. Let S be an integral noetherian scheme, and let $f : X \to Y$ be a surjective S-morphism between two proper S-schemes.

a) If Y is normal over S (that is, flat with geometrically normal fibers), then, after shrinking S, $f^* : \mathrm{Pic}_{Y/S} \to \mathrm{Pic}_{X/S}$ is quasi-finite.

b) If $\mathrm{Pic}_{Y/S}$ is essentially proper over S (that is, if it satisfies the valuative criterion of properness; for example, this holds if Y is smooth over S), then, after shrinking S, $f^* : \mathrm{Pic}_{Y/S} \to \mathrm{Pic}_{X/S}$ is finite (and, in particular, affine).

Indeed, in (a), for every $s \in S$, $R_s = \mathrm{Ker}(f_s^*)$ is an affine algebraic group; its identity component is a proper algebraic group over $k(s)$ (FGA 236, Theorem 2.1 (iii)), so R_s is finite. In (b), f^* is both proper and quasi-affine, hence finite.

Remark 1.7. It is not known whether 1.1 remains valid when "quasi-affine" is replaced by "affine," as 1.5 and 1.6 might suggest.

2. First reductions

2.1. Apply 1.3 to X/S and Y/S. One obtains a nonempty open subset V of S, an integral scheme S', and a finite flat morphism $S' \to V$ such that every irreducible component X_i' of $X_{S'}$ and Y_j' of $Y_{S'}$, endowed with the induced reduced structure, is integral over S'. By fpqc descent (EGA IV 2.7.1 (xiv) and SGA 1 VIII 7.9), one may replace S by S'. It then follows from the commutative diagram

$$\begin{tikzcd} X_{\mathrm{red}} \arrow[r, "f_{\mathrm{red}}"] \arrow[d, "i"'] & Y_{\mathrm{red}} \arrow[d, "j"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

which implies the commutative diagram

$$\begin{tikzcd} \mathrm{Pic}_{X_{\mathrm{red}}/S} & \mathrm{Pic}_{Y_{\mathrm{red}}/S} \arrow[l] \\ \mathrm{Pic}_{X/S} \arrow[u] & \mathrm{Pic}_{Y/S} \arrow[l] \arrow[u] \end{tikzcd}$$

and from elementary sorites concerning morphisms of functors representable by quasi-affine morphisms (EGA II 5.1.10 (ii), (v)) that it is enough to prove the following two assertions:

I. Theorem 1.1 when f is an immersion (necessarily nilpotent).

II. Theorem 1.1 when X and Y are reduced and every irreducible component X_i of X and Y_i of Y, endowed with the induced reduced structure, is integral over S.

2.2. Consider assertion II. For every i, let $j = \varphi(i)$ be an index such that $f(X_i) \subset Y_j$, and put $\Sigma_j = \coprod_{\varphi(i)=j} X_i$. Then one has the two diagrams

$$\begin{tikzcd} Y & \coprod_j Y_j \arrow[l] \\ X \arrow[u] & \coprod_j \Sigma_j \arrow[l] \arrow[u] \end{tikzcd}$$

and

$$\begin{tikzcd} \mathrm{Pic}_{Y/S} \arrow[r, "①"] \arrow[d, "②"'] & \mathrm{Pic}_{\coprod_j Y_j/S} = \prod_j \mathrm{Pic}_{Y_j/S} \arrow[d, "④"] \\ \mathrm{Pic}_{X/S} \arrow[r, "③"'] & \mathrm{Pic}_{\coprod_j \Sigma_j/S} = \prod_j \mathrm{Pic}_{\Sigma_j/S}\,. \end{tikzcd}$$

To establish II, it is enough to see that, after shrinking S, ①, ③, and ④ are quasi-affine.

Consider ④ first. Replacing Y by Y_j and X by Σ_j, one is reduced to the case in which Y is integral over S. Since f is surjective, one of the X_i, say X_0, dominates Y, and one has the two sequences

$$X_0 \hookrightarrow \coprod_i X_i \longrightarrow Y$$

and

$$\mathrm{Pic}_{X_0/S} \longleftarrow \mathrm{Pic}_{\coprod_i X_i/S} \longleftarrow \mathrm{Pic}_{Y/S}\,.$$

Thus, if the $\mathrm{Pic}_{X_i/S}$, and hence also $\mathrm{Pic}_{\coprod_i X_i/S}$, are known to be representable and the composite morphism is quasi-affine, then ④ is quasi-affine. In view of the proof of 1.2, proving ④ is therefore reduced to proving the following assertion:

II$'$. Theorem 1.1 when X and Y are integral over S.

For ① and ③, one must treat the case of a morphism $\coprod_i X_i \to X$, where X is reduced and the X_i are the irreducible components of X and are integral over S. One may suppose that X has connected fibers. Indeed, let η be a geometric generic point of S. Then the irreducible components of X_η are the $(X_i)_\eta$, and, after shrinking S, one may suppose that $X_i \cap X_j = \varnothing$ if and only if $(X_i)_\eta \cap (X_j)_\eta = \varnothing$. Let $Y_k = \bigcup_{i\in I(k)} X_i$ be the connected components of X. Then f decomposes as

$$f = \coprod_k f_k : \coprod_k \left(\coprod_{i\in I(k)} X_i \right) \longrightarrow \coprod_k Y_k,$$

and the assertion follows. Finally, an immediate induction reduces us to the case of a morphism $X_1 \amalg X_2 \to X$. After shrinking S, ① and ③ then follow from EGA III 7.8.6 and the following proposition.

Proposition 2.3. Let S be a scheme, let $g : Y \to S$ be a proper morphism, and let Y_1, Y_2 be two closed subschemes defined by the ideals I_1, I_2. Suppose that:

a) $Y, Y_1, Y_2, Y_1 \cap Y_2$ are faithfully flat and of finite presentation over S;

b) $g_*\mathcal{O}_Y = \mathcal{O}_S$ and $(g_i)_*\mathcal{O}_{Y_i} = \mathcal{O}_S$ universally, where $g_i = g|_{Y_i}$, $i = 1, 2$;

c) $I_1 \cap I_2 = 0$.

Then

$$\mathrm{Pic}_{Y/S} \longrightarrow \mathrm{Pic}_{(Y_1 \amalg Y_2)/S}$$

is representable by an affine morphism of finite presentation.

First note that the hypotheses are stable under every base change. For (c), this follows from the two exact sequences

$$0 \longrightarrow I_1 + I_2 \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{O}_{Y_1\cap Y_2} \longrightarrow 0$$

and

$$0 \longrightarrow I_1 \cap I_2 \longrightarrow I_1 \oplus I_2 \longrightarrow I_1 + I_2 \longrightarrow 0.$$

The first shows that $I_1 + I_2$ is flat over S, and the second then shows that $I_1 \cap I_2$ is stable under base change.

The assertion is local for the fppf topology. After the base change $Y_1 \cap Y_2 \to S$, one may therefore suppose that $Y_1 \cap Y_2$ has a section α, which defines sections α_i of Y_i and β of Y. After a suitable base change, one is reduced to showing that every point $\mu : S \to \mathrm{Pic}_{(Y_1 \amalg Y_2)/S}$ has as inverse image a subfunctor of $\mathrm{Pic}_{Y/S}$ that is representable by an affine scheme over S. Now μ is defined by an invertible sheaf L_i on Y_i, furnished with a rigidification

$$u_i : \alpha_i^* L_i \xrightarrow{\sim} \mathcal{O}_S, \qquad i = 1, 2.$$

Likewise, a point $\lambda : S \to \mathrm{Pic}_{Y/S}$ corresponds to an invertible sheaf L on Y furnished with a rigidification $u : \beta^* L \xrightarrow{\sim} \mathcal{O}_S$. The image of λ in $\mathrm{Pic}_{(Y_1 \amalg Y_2)/S}$ is μ if and only if there are isomorphisms of rigidified sheaves $f_i^* L \simeq L_i$, where $f_i : Y_i \hookrightarrow Y$ is the inclusion, for $i = 1, 2$. These isomorphisms define an isomorphism

$$\varphi : L_1|_{Y_1 \cap Y_2} \xrightarrow{\sim} L_2|_{Y_1 \cap Y_2}$$

such that $u_1 = u_2 \circ \varphi$.

Conversely, if φ is such an isomorphism, there is a rigidified invertible sheaf L on Y obtained by gluing along φ. Explicitly, if $f_{12} : Y_1 \cap Y_2 \hookrightarrow Y$ denotes the common restriction of f_1 and f_2, then L is the kernel of the canonical homomorphism

$$(f_1)_* L_1 \oplus (f_2)_* L_2 \longrightarrow (f_{12})_* (L_2|_{Y_1 \cap Y_2})$$

defined by φ. This L then corresponds to a point λ of $\mathrm{Pic}_{Y/S}$ mapping to μ.

Finally, let I be the functor that assigns, to an S-scheme T, the set of isomorphisms

$$\varphi : (L_1)_T|_{(Y_1 \cap Y_2)_T} \xrightarrow{\sim} (L_2)_T|_{(Y_1 \cap Y_2)_T}$$

such that $(u_1)_T = (u_2)_T \circ \varphi$. We have just established an isomorphism between the inverse image of μ in $\mathrm{Pic}_{Y/S}$ and the functor I. It is therefore enough to apply the following lemma (with $X = Y_1 \cap Y_2$).

Lemma 2.4. Let X be a flat, proper scheme of finite presentation over S. Let L, M be two invertible sheaves on X, and let $\mathrm{Isom}(L, M)$ be the functor of isomorphisms $L_T \to M_T$, for T/S. Then:

i) $\mathrm{Isom}(L, M)$ is representable by an affine S-scheme of finite presentation.

ii) If, moreover, L and M are furnished with rigidifications (relative to a given section of X), the subfunctor of $\mathrm{Isom}(L, M)$ formed by the isomorphisms compatible with these rigidifications is representable by a closed subscheme of $\mathrm{Isom}(L, M)$.

Indeed, (ii) is clear. For (i), recall that the functors $T \mapsto \mathrm{Hom}(L_T, M_T)$ and $T \mapsto \mathrm{Hom}(M_T, L_T)$ are representable by vector bundles $\mathbb{V}(E)$ and $\mathbb{V}(F)$ (EGA III 7.7.8). Thus, if $u \in \mathrm{Hom}(L_T, M_T)$ and $v \in \mathrm{Hom}(M_T, L_T)$, the conditions $u \circ v = 1$ and $v \circ u = 1$ define a closed subscheme Z of $\mathbb{V}(E) \times_S \mathbb{V}(F)$, and it is clear that $Z \simeq \mathrm{Isom}(L, M)$.

2.5. Finally, let us reduce assertion II$'$ to the following assertion:

II*. Theorem 1.1 when X and Y are integral over S and the sequence

$$\mathcal{O}_Y \longrightarrow f_* \mathcal{O}_X \rightrightarrows g_* \mathcal{O}_{X \times_Y X} \tag{*}$$

(where $g : X \times_Y X \to Y$ is the structural morphism) is exact.

For this, it is enough to prove that there is a factorization of f

$$X \xrightarrow{f_0} Y_0 \xrightarrow{f_1} Y_1 \longrightarrow \cdots \xrightarrow{f_n} Y_n = Y$$

such that every f_i satisfies (*) and, after possibly restricting S, every Y_i is integral over S.

For this purpose, put $Y_0 = \mathrm{Spec}(f_*\mathcal{O}_X)$, with f_0 arising from the Stein factorization of f (EGA III 4.3.3), so that $f_{0*}\mathcal{O}_X \xleftarrow{\sim} \mathcal{O}_{Y_0}$; hence f_0 does satisfy (*). To factor $Y_0 \to Y$ in the same way, one uses the following lemma.

Lemma 2.6. Let Y be a noetherian scheme, and let $f : X \to Y$ be a finite epimorphism (that is, $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is injective). Then f factors into a finite sequence of finite morphisms

$$X \xrightarrow{\sim} Y_0 \xrightarrow{f_1} Y_1 \longrightarrow \cdots \longrightarrow Y_n = Y$$

that are effective epimorphisms (that is, for each i the sequence

$$\mathcal{O}_{Y_i} \longrightarrow (f_i)_*\mathcal{O}_{Y_{i-1}} \rightrightarrows (g_i)_*(\mathcal{O}_{Y_{i-1}} \otimes_{\mathcal{O}_{Y_i}} \mathcal{O}_{Y_{i-1}})$$

is exact).

Put $A = \mathcal{O}_Y$, $B_0 = f_*\mathcal{O}_X$, and define recursively

$$B_i = \mathrm{Ker}\big(B_{i-1} \rightrightarrows B_{i-1} \otimes_A B_{i-1}\big).$$

One checks at once that

$$B_{i-1} \otimes_A B_{i-1} \xrightarrow{\sim} B_{i-1} \otimes_{B_i} B_{i-1}.$$

It remains to prove that $B_i = A$ for all sufficiently large i. Indeed, consider the decreasing sequence of B_0-modules $M_i = B_i/A$, and put $S_i = \mathrm{Supp}(M_i)$. The decreasing sequence of closed subsets S_i of Y is stationary; let S_∞ be its stationary value. We show that $S_\infty = \varnothing$. Otherwise, let s be a maximal point of S_∞. After replacing Y by $\mathrm{Spec}\,\mathcal{O}_{Y,s}$, we may suppose that Y is local with closed point s. By the choice of s, M_i is artinian for all sufficiently large i, and therefore the sequence (M_i) is stationary. Thus $B_i = B_{i+1}$ for some i. It follows that $\mathrm{Spec}(B_i) \to Y$ is a monomorphism, hence a closed immersion (Nakayama). Since $A \subset B_i$, one has $A = B_i$, contrary to $s \in S_\infty$.

3. Proof of I: Oort's dévissage

Proposition 3.1. (Oort, *Bull. soc. math. France*, vol. 90 (1962)). Let S be a locally noetherian scheme, let $f : X \to S$ be a proper morphism, and let $X' \subset X$ be a closed subscheme. Suppose that:

i) X' is defined by an ideal I contained in the nilradical N of X, and $NI = 0$.

ii) X' is flat over S and cohomologically flat over S in dimension 0 (EGA III 7.8.1).

iii) I is flat over S and cohomologically flat over S in dimensions 0 and 1, so that (by EGA III 7.4.6) there are a coherent $\mathcal{O}_S$-module Q and, functorially in the coherent $\mathcal{O}_S$-module M, an isomorphism

$$R^2 f_*(I \otimes f^*M) \simeq \mathrm{Hom}_{\mathcal{O}_S}(Q, M).$$

iv) The cokernel E of $f_*\mathcal{O}_{X'} \to R^1 f_*(I)$ is a locally free $\mathcal{O}_S$-module.

v) The homomorphism

$$(f_*\mathcal{O}_X)_{\mathrm{red}} \longrightarrow (f_*\mathcal{O}_{X'})_{\mathrm{red}}$$

is surjective.

Then the canonical morphism

$$\mathrm{Pic}_{X/S} \longrightarrow \mathrm{Pic}_{X'/S}$$

is representable by an affine morphism of finite presentation. More precisely, there is a canonical exact sequence of fppf sheaves

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \mathrm{Pic}_{X/S} \longrightarrow \mathrm{Pic}_{X'/S} \xrightarrow{u} \mathbb{V}(Q),$$

and, if $P = \mathrm{Ker}(u)$, the monomorphism $P \hookrightarrow \mathrm{Pic}_{X'/S}$ is representable by a closed immersion and $\mathrm{Pic}_{X/S}$ is a torsor over P under the group $\mathbb{V}(E^\vee)$ (that is, the sequence of fppf sheaves

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \mathrm{Pic}_{X/S} \longrightarrow P \longrightarrow 0$$

is exact).

For an S-scheme T, consider the following two diagrams with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_T & \longrightarrow & \mathcal{O}_{X_T} & \longrightarrow & \mathcal{O}_{X'_T} & \longrightarrow & 0 \\ & & \| & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & I_T & \xrightarrow{\alpha} & \mathcal{O}^*_{X_T} & \longrightarrow & \mathcal{O}^*_{X'_T} & \longrightarrow & 0, \end{array} \qquad \alpha(i) = 1 + i,$$

and

$$\begin{array}{ccccccccc} (f_T)_*\mathcal{O}_{X'_T} & \xrightarrow{\delta_T} & R^1(f_T)_*I_T & & & & & & \\ \uparrow & & \| & & & & & & \\ (f_T)_*\mathcal{O}^*_{X'_T} & \xrightarrow[\delta_T^*]{} & R^1(f_T)_*I_T & \to & R^1(f_T)_*\mathcal{O}^*_{X_T} & \to & R^1(f_T)_*\mathcal{O}^*_{X'_T} & \to & R^2(f_T)_*I_T. \end{array}$$

Note carefully that the left-hand square in this diagram is *not necessarily commutative.* Nevertheless,

$$\textbf{3.2.} \qquad \mathrm{im}\,\delta_T^* = \mathrm{im}\,\delta_T.$$

Indeed, we may plainly suppose that S and T are affine. Choose an affine covering $\{V_i\}$ of X_T. View $\mathcal{O}_{X'_T}$ as its direct image under the closed immersion $X'_T \hookrightarrow X_T$. Let $g \in \Gamma(X_T, \mathcal{O}^*_{X'_T})$, and let g_i be a lift of $g|_{V_i}$ in $\Gamma(V_i, \mathcal{O}^*_{X_T})$. Then $\delta_T^*(g)$ is the class of the 1-cocycle

$$\delta^*_{ij} = g_i g_j^{-1} - 1,$$

whereas $\delta_T(g)$ is the class of

$$\delta_{ij} = g_i - g_j = g_j \delta^*_{ij} = g\delta^*_{ij}.$$

Consequently,

$$\delta_T(g) = g\,\delta_T^*(g),$$

taking into account the action of $\Gamma(X_T, \mathcal{O}_{X'_T})$ on I_T, and hence on $H^1(X_T, I_T)$.

By (ii),

$$\Gamma(X_T, \mathcal{O}_{X'_T}) = \Gamma(X, \mathcal{O}_{X'}) \otimes_S \Gamma(T, \mathcal{O}_T),$$

and hence $g = \sum g_\beta \otimes t_\beta$, with $g_\beta \in \Gamma(X, \mathcal{O}_{X'})$ and $t_\beta \in \Gamma(T, \mathcal{O}_T)$. By (v), the map

$$(f_*\mathcal{O}_X)_{\mathrm{red}} \longrightarrow (f_*\mathcal{O}_{X'})_{\mathrm{red}}$$

is surjective. Thus $g_\beta = v(h_\beta) + n_\beta$, where $h_\beta \in \Gamma(X, \mathcal{O}_X)$, $v : \Gamma(X, \mathcal{O}_X) \to \Gamma(X, \mathcal{O}_{X'})$, and $n_\beta \in \mathrm{nilrad}\big(\Gamma(X, \mathcal{O}_{X'})\big)$.

By (i), $(n_\beta \otimes t_\beta)I_T = 0$. Therefore $\delta_T(g) = h\,\delta_T^*(g)$, where

$$h = \sum h_\beta \otimes t_\beta \in \Gamma(X_T, \mathcal{O}_{X_T}).$$

It follows that

$$\delta_T^*(g) = h^{-1}\delta_T(g) = \delta_T(h^{-1}g),$$

and hence $\operatorname{im}\delta_T^* \subset \operatorname{im}\delta_T$.

Finally, let $g = \sum g_\beta \otimes t_\beta \in \Gamma(X_T, \mathcal{O}_{X'_T})$. Write $g_\beta = v(h_\beta) + n_\beta$, with $n_\beta \in \operatorname{nilrad}\bigl(\Gamma(X, \mathcal{O}_{X'})\bigr)$. Then

$$\begin{aligned} \delta_T(g) &= \delta_T\Bigl(\sum n_\beta \otimes t_\beta\Bigr) \\ &= \delta_T\Bigl(1 + \sum n_\beta \otimes t_\beta\Bigr) \\ &= \delta_T^*\Bigl(1 + \sum n_\beta \otimes t_\beta\Bigr), \end{aligned}$$

whence $\operatorname{im}\delta_T \subset \operatorname{im}\delta_T^*$.

By 3.2, (iii), and (iv), there is an exact sequence, functorial in T,

$$0 \longrightarrow \mathbb{V}(E^\vee)(T) \longrightarrow R^1(f_T)_*\mathcal{O}_{X_T}^* \longrightarrow R^1(f_T)_*\mathcal{O}_{X'_T}^* \longrightarrow \mathbb{V}(Q)(T),$$

which, on passing to the associated fppf sheaves, gives an exact sequence

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \operatorname{Pic}_{X/S} \longrightarrow \operatorname{Pic}_{X'/S} \xrightarrow{u} \mathbb{V}(Q).$$

If $P = \operatorname{Ker}(u)$, the monomorphism $P \hookrightarrow \operatorname{Pic}_{X'/S}$ is obtained by base change from the unit section $S \hookrightarrow \mathbb{V}(Q)$. Since $\mathbb{V}(Q)$ is representable and separated, it follows that $P \hookrightarrow \operatorname{Pic}_{X'/S}$ is representable by a

closed immersion.

Finally,

$$0 \longrightarrow \mathbb{V}(E^\vee) \longrightarrow \operatorname{Pic}_{X/S} \xrightarrow{w} P \longrightarrow 0$$

is a torsor for the fppf topology. A fortiori, by fpqc descent from a "splitting" of the torsor, w is representable by a faithfully flat affine morphism of finite presentation (see Oort, *Commutative group schemes*, Proposition 17.4).

Corollary 3.3. Under the hypotheses of 3.1:

i) If

$$\operatorname{Pic}_{(\mathrm{zar})(X'/S)} \longrightarrow \operatorname{Pic}_{(\mathrm{fppf})(X'/S)}$$

is an isomorphism, then

$$\operatorname{Pic}_{(\mathrm{zar})(X/S)} \longrightarrow \operatorname{Pic}_{(\mathrm{fppf})(X/S)}$$

is an isomorphism.

ii) If the S-scheme T is affine and

$$\operatorname{Pic}(X'_T) \longrightarrow \operatorname{Pic}^{(T)}_{(\mathrm{zar})(X'/S)}$$

is surjective, then

$$\operatorname{Pic}(X_T) \longrightarrow \operatorname{Pic}^{(T)}_{(\mathrm{zar})(X/S)}$$

is surjective.

Indeed, (i) follows by applying the five lemma to the following diagram of sheaves for the Zariski topology:

$$\begin{tikzcd}
0 \arrow[r] \arrow[d, "S"] & \mathbb{V}(E^\vee) \arrow[r] \arrow[d, "S"] & \mathrm{Pic}_{(\mathrm{zar})(X/S)} \arrow[r] \arrow[d] & \mathrm{Pic}_{(\mathrm{zar})(X'/S)} \arrow[r] \arrow[d, "S"] & \mathbb{V}(Q) \arrow[d, "S"] \\
0 \arrow[r] & \mathbb{V}(E^\vee) \arrow[r] & \mathrm{Pic}_{(\mathrm{fppf})(X/S)} \arrow[r] & \mathrm{Pic}_{(\mathrm{fppf})(X'/S)} \arrow[r] & \mathbb{V}(Q).
\end{tikzcd}$$

Assertion (ii) follows from the following diagram with exact rows:

$$\begin{tikzcd}
H^1(X_T, I_T) \arrow[r] \arrow[d] & H^1(X_T, \mathcal{O}^*_{X_T}) \arrow[r] \arrow[d] & H^1(X_T, \mathcal{O}^*_{X'_T}) \arrow[r] \arrow[d] & H^2(X_T, I_T) \arrow[d] \\
H^0(T, R^1(f_T)_* I_T) \arrow[r] & H^0(T, R^1(f_T)_* \mathcal{O}^*_{X_T}) \arrow[r] & H^0(T, R^1(f_T)_* \mathcal{O}^*_{X'_T}) \arrow[r] & H^0(T, R^2(f_T)_* I_T),
\end{tikzcd}$$

because the outside vertical arrows are isomorphisms when T is affine.

Corollary 3.4. Let k be a field, and let X be a proper k-scheme. Suppose that $\mathrm{Pic}_{X_{\mathrm{red}}/k}$ possesses a universal invertible sheaf. (For example, descent theory shows that this is the case if every connected component X'_i of X_{red} has a rational point; in particular, if k is algebraically closed, or more generally if the greatest common divisor of the degrees of the 0-cycles on every X'_i is 1.) Then, over every affine open subset V of $\mathrm{Pic}_{X/k}$, there is a universal invertible sheaf on $X \times_k V$.

Finally, we prove assertion I in the following more precise form.

Proposition 3.5. Let S be an integral noetherian scheme, and let $f : X \hookrightarrow Y$ be a nilpotent immersion of proper S-schemes. Then there is a nonempty open subset V of S such that the morphism

$$f^* : \mathrm{Pic}_{Y/S}|_V \longrightarrow \mathrm{Pic}_{X/S}|_V$$

is representable by an affine morphism of finite presentation. In fact, $N = \mathrm{Ker}(f^*)$ is an affine group scheme with unipotent fibers (in the sense of SGA 3 XVII).

Indeed, X is defined by a nilpotent ideal I. The ideal I has a filtration by ideals

$$I_0 = I \supset I_1 \supset \cdots \supset I_n = 0$$

such that $NI_{i-1} \subset I_i$ for $1 \leq i \leq n$, where N is the nilradical of $\mathcal{O}_Y$ (for example, $I_i = N^i I$). Let X_i be the closed subscheme of Y defined by I_i. By induction it is enough to study the immersions $X_{i-1} \hookrightarrow X_i$. We may therefore suppose that $NI = 0$; this is condition (i) of 3.1, and we are reduced finally to the case in which the other conditions (ii)–(v) are satisfied as well. By the generic flatness theorem (EGA IV 6.9.1), applied in the notation of 3.1 to $\mathcal{O}_{X'}$, the $R^i f_*(I)$, and E, and taking account of the Künneth relations of EGA III 6.9.9, after restricting S we may suppose that conditions (ii), (iii), and (iv) hold.

There is a finite extension K of $k(S)$ such that $(f_K)_*\mathcal{O}_{X'_K}$ has all its residue field extensions trivial. If S' is an integral scheme, finite and flat over an open subset V of S, with function field $k(S') = K$, there is an open subset V' of S' such that

$$[(f_{S'})_*\mathcal{O}_{X'_{S'}}]_{\mathrm{red}} \simeq [(f_{S'})_*\mathcal{O}_{X_{S'}}]_{\mathrm{red}}$$

over V'. After restricting S and S' further, we may suppose that $S = V$ and $S' = V'$. Since the assertion to be proved is local for the fpqc topology, we may replace S by S'. Finally, apply 3.1.

Remark 3.6. It follows in the same way, taking account of the exact sequences

$$\mathrm{Pic}_{X_i/S} \longrightarrow \mathrm{Pic}_{X_{i-1}/S} \longrightarrow \mathbb{V}(Q_i),$$

that, for every $s \in S$,

$$\operatorname{coker}(\operatorname{Pic}^0_{Y_s/s} \longrightarrow \operatorname{Pic}^0_{X_s/s})$$

is a unipotent group.

4. Proof of II^*: the most delicate part of the proof

Recall assertion II^*: Theorem 1.1 when X and Y are integral over S and the sequence

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \rightrightarrows g_*\mathcal{O}_{X\times_Y X}$$

is exact. Applying the generic flatness theorem, EGA IV 6.9.1, and EGA III 7.8.8, we are reduced to proving the following proposition.

Proposition 4.1. Let S be an integral noetherian scheme, let $\lambda : X \to S$ and $\mu : Y \to S$ be proper morphisms of finite presentation, let $f : X \to Y$ be a surjective S-morphism, and let $g : X \times_Y X \to Y$ be the canonical morphism. Suppose that:

1) X, Y, and $X \times_Y X$ are flat over S.
2) $\lambda_*\mathcal{O}_X = \mathcal{O}_S$ and $\mu_*\mathcal{O}_Y = \mathcal{O}_S$ universally.
3) For all $i, j \geq 0$, the modules $R^i f_*\mathcal{O}_X$ and $R^j g_*\mathcal{O}_{X\times_Y X}$ are flat over S.
4) The sequence
$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \overset{\pi_1}{\underset{\pi_2}{\rightrightarrows}} g_*\mathcal{O}_{X\times_Y X} \tag{*}$$
is exact, and $\operatorname{coker}(\pi_1 - \pi_2)$ is an $\mathcal{O}_Y$-module flat over S.

Then

$$f^* : \operatorname{Pic}_{Y/S} \longrightarrow \operatorname{Pic}_{X/S}$$

is representable by a quasi-affine morphism of finite presentation.

First note that (1)–(4) are stable under every base change $T \to S$. Only the stability of (3) and (4) is not evident. By the Künneth formula (EGA III 6.9.9), (1) and (3) imply that the formation of $R^i f_*\mathcal{O}_X$ and $R^i g_*\mathcal{O}_{X\times_Y X}$ commutes with base change $T \to S$; this immediately implies the stability of condition (3). Moreover, the formation of the cokernel of

$$\pi_1 - \pi_2 : f_*\mathcal{O}_X \longrightarrow g_*\mathcal{O}_{X\times_Y X}$$

also commutes with base change. Since it is flat over S by (4), $\operatorname{im}(\pi_1 - \pi_2)$ is flat over S and commutes with base change. Hence $\operatorname{Ker}(\pi_1 - \pi_2)$ commutes with base change $T \to S$ and is therefore universally equal to $\mathcal{O}_Y$.

The stability of the hypotheses implies two things:

a. Since the assertion to be proved is local for the fppf topology, by descent we may suppose that X has a section α; then Y has the section $\beta = f \circ \alpha$.

b. To prove the assertion, it is enough to consider an S-point of $\operatorname{Pic}_{X/S}$, that is, by (a) and (2), an invertible sheaf L on X endowed with a rigidification u relative to α, and to show that the fppf sheaf Z, the inverse image of this point in $\operatorname{Pic}_{Y/S}$, is representable by a quasi-affine S-scheme of finite presentation.

Lemma 4.2. Under the hypotheses of 4.1, f is a descent morphism for the category of locally free sheaves; that is, for every pair L, M of locally free sheaves on Y, the diagram

$$\mathrm{Hom}_Y(L, M) \to \mathrm{Hom}_X(f^*L, f^*M) \rightrightarrows \mathrm{Hom}_{X\times_Y X}(g^*L, g^*M)$$

is exact.

Indeed, if $N = \mathrm{Hom}_{\mathcal{O}_Y}(L, M)$, one must show that the sequence

$$\Gamma(Y, N) \longrightarrow \Gamma(X, f^*N) \rightrightarrows \Gamma(X \times_Y X, g^*N)$$

is exact, or equivalently (by EGA 0_{I} 5.4.10.1) that the sequence

$$\Gamma(Y, N) \longrightarrow \Gamma(Y, f_*\mathcal{O}_X \otimes_{\mathcal{O}_Y} N) \rightrightarrows \Gamma(Y, g_*\mathcal{O}_{X\times_Y X} \otimes_{\mathcal{O}_Y} N)$$

is exact. But the sequence (*)

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \rightrightarrows g_*\mathcal{O}_{X\times_Y X}$$

is exact by hypothesis. Applying the left exact functor $\Gamma(N \otimes -)$ gives the lemma, since N is flat over Y.

Let L be an invertible sheaf on X, endowed with a rigidification u relative to α. By 4.2 there is an equivalence between the category C of pairs (M, φ) consisting of an invertible sheaf M on Y and an isomorphism $\varphi : f^*M \xrightarrow{\sim} L$, and the category $\mathrm{Deff}(S)$ of effective descent data on L relative to f. If $f^*M \simeq L$, then, since L is rigidified relative to α, M is canonically endowed with a rigidification relative to β. Consequently, C is equivalent to the discrete category defined by the set of classes of β-rigidified invertible sheaves (M, v) on Y such that

$$f^*(M, v) \simeq (L, u).$$

But this is precisely the set $Z(S)$ of S-points of the inverse image in $\mathrm{Pic}_{Y/S}$ of the S-point of $\mathrm{Pic}_{X/S}$ defined by (L, u). Moreover, this description of $Z(S)$ is functorial in S. Hence Z identifies with Deff, and we may forget the rigidifications from now on.

There is a canonical monomorphism from Deff into the functor Rec of gluing data on L. By definition, Rec is the functor $\mathrm{Isom}(p_1^*L, p_2^*L)$, where the $p_i : X \times_Y X \to X$ are the projections; by 2.4 this functor is represented by an affine scheme I of finite presentation over S. It therefore remains to represent the monomorphism

$$i : \mathrm{Deff} \hookrightarrow \mathrm{Isom}(p_1^*L, p_2^*L)$$

by a morphism of finite presentation, which will necessarily be quasi-affine (by EGA IV 8.11.2).

Let us note in passing that, if desired, one may restrict $\mathrm{Isom}(p_1^*L, p_2^*L)$ by imposing the cocycle condition, but this is not needed below.

After the base change $I \to S$, we are thus reduced to the following problem: if L is an invertible sheaf on X, endowed with a gluing datum

$$\varphi : p_1^*L \xrightarrow{\sim} p_2^*L,$$

represent the subfunctor of S on which this datum becomes effective. We have the following lemma.

Lemma 4.3. Under the hypotheses of 4.1, let L be an invertible sheaf on X, endowed with a gluing datum

$$\varphi : p_1^*L \xrightarrow{\sim} p_2^*L.$$

This datum is effective if and only if the following five conditions hold.

1) For every $i \geq 0$, $R^i f_*(L)$ is flat over S.

2) For every $j \geq 0$, $R^j g_*(p_1^* L)$ is flat over S.

3) The cokernel of

$$\pi_1 - \pi_2 : f_*(L) \longrightarrow g_*(p_1^* L) \simeq g_*(p_2^* L)$$

is flat over S, where the latter isomorphism is $g_*(\varphi)$.

4) The kernel N of $\pi_1 - \pi_2 : f_*(L) \to g_*(p_1^* L)$ is invertible on Y.

5) The composite morphism

$$f^* N \longrightarrow f^* f_* L \longrightarrow L$$

is an isomorphism.

Indeed, these conditions are sufficient, since (4) and (5) alone imply effectivity. Conversely, let M be invertible on Y, and suppose that there is an isomorphism $f^*M \simeq L$ compatible with the gluing datum. We may cover Y by open subsets V over which M is isomorphic to $\mathcal{O}_Y$. The descent datum on L, restricted over $f^{-1}(V)$, is therefore isomorphic to the trivial descent datum on $\mathcal{O}_X|_{f^{-1}(V)}$. Finally, conditions (1)–(5) are local on Y, and the hypotheses of 4.1 show that they hold for $L = \mathcal{O}_X$; hence they hold for L.

It remains to represent conditions (1)–(5), one after another, by monomorphisms of finite presentation. For (1) and (2), we apply the following lemma.

Lemma 4.4. Let S be a scheme, let $\lambda : X \to S$ and $\mu : Y \to S$ be proper morphisms of finite presentation, and let $f : X \to Y$ be an S-morphism. Let F be a sheaf on X, of finite presentation and flat over S. For an integer n, let S_n be the subfunctor of S on which $R^i f_*(F)$ is flat over S for every $i \geq n$; more precisely, for a morphism $\phi : T \to S$,

$$S_n(T) = \begin{cases} \{\phi\}, & \text{if } R^i(f_T)_*(F_T) \text{ is flat for every } i \geq n, \\ \varnothing, & \text{otherwise.} \end{cases}$$

Then $S_n \hookrightarrow S$ is representable by a surjective monomorphism of finite presentation; moreover, for $i \geq n$, the formation of $R^i(f_{S_n})_*(F_{S_n})$ commutes with base changes $T \to S_n$. In addition, S_n coincides with the functor S'_n on which the functors

$$M \longmapsto R^i(f_V)_*\big(F_V \otimes \lambda_V^* M\big), \qquad i \geq n,$$

are exact, where M ranges over the quasi-coherent $\mathcal{O}_V$-modules, for every affine open subset V of S.

Indeed, the assertion is local on S, so we may suppose that S is quasi-compact. The relative dimension of X/Y is then bounded, say by r, and, for $n > r$,

$$R^n f_*(F_V \otimes \lambda_V^* M) = 0.$$

The assertion therefore holds for $n > r$.

Proceeding by descending induction, suppose it is known for $n+1$. After replacing S by $S_{n+1} = S'_{n+1}$, we may suppose that $R^i f_*(F)$ is flat over S for $i \geq n+1$. Since F is flat over S, EGA III 6.9.8 then shows that the formation of $R^n f_*(F)$ commutes with every base change. Consequently, the representability of S_n follows from the next lemma (see J. P. Murre, *Representation of unramified functors*, Séminaire Bourbaki, May 1965, p. 294-11, Theorem 2).

Lemma 4.5. Let $X \to S$ be a proper morphism of finite presentation, and let F be a sheaf of finite presentation on X. Then the subfunctor S_F of S on which F is flat over S is representable by a surjective monomorphism of finite presentation (which, of course, is not an immersion in general).

To represent condition (3) of 4.3, note that, by (1) and (2), the formation of $f_*(L)$ and $g_*(p_1^* L)$ commutes with the base changes $I \to S$; hence so does the formation of $\operatorname{Coker}(\pi_1 - \pi_2)$. We then apply 4.5.

To represent (4), note that, by (3), and because $f_*(L)$ and $g_*(p_1^* L)$ are flat and commute with base change,

$$N = \operatorname{Ker}(\pi_1 - \pi_2)$$

also commutes with base change. We then apply the following lemma.

Lemma 4.6. Let $\lambda : X \to S$ be a proper morphism of finite presentation, and let $u : F \to G$ be a homomorphism of $\mathcal{O}_X$-modules of finite presentation, with G flat over S. Then the subfunctor S_1 of S on which u is an isomorphism is representable by an open subscheme of S, of finite presentation.

As usual, we reduce to the case where S is noetherian, using EGA IV 8 and 11.2.6; this relieves us of the need to consider finite presentation. First let us represent the functor on which u is an epimorphism, that is, on which $\operatorname{Coker} u$ is zero. Since the formation of $\operatorname{Coker} u$ commutes with base change, this functor is represented by the open subset

$$S - \lambda(\operatorname{Supp}(\operatorname{Coker} u))$$

of S. We may therefore suppose that u is already an epimorphism. Using now the hypothesis that G is flat over S, we find that the formation of $\operatorname{Ker} u$ commutes with base change; consequently, the functor under consideration is represented by the open subset

$$S - \lambda(\operatorname{Supp}(\operatorname{Ker} u))$$

of S.

Exposé XIII. Finiteness Theorems for the Picard Functor

by S. Kleiman

1. The (b)-sheaves

Let k be an algebraically closed field and X a projective k-scheme, endowed with an ample invertible sheaf $\mathcal{O}_X(1)$.

Definition 1.1 (Mumford). Let $m \in \mathbb{Z}$. A coherent sheaf F on X is said to be *m-regular* if, at every point of the support of F, $\mathcal{O}_X(1)$ is generated by its global sections and if, for every $q \geq 1$,

$$H^q(F(m-q)) = 0.$$

Lemma 1.2. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence of coherent sheaves on X.[24] If F is m-regular, then so is G.

Indeed, we have the corresponding exact sequence

$$H^q(F(m-q)) \longrightarrow H^q(G(m-q)) \longrightarrow H^{q+1}(F(m-q-1)).$$

Proposition 1.3. If a sheaf F on X is m-regular, then, for every $n \geq m$,

(i) F is n-regular.

(ii) The map

$$H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) \longrightarrow H^0(F(n+1))$$

is surjective.

(iii) $F(n)$ is generated by $H^0(F(n))$.

[24] Here, as below in analogous situations, it is understood that $F(-1) \to F$ is defined by tensoring with a given section of $\mathcal{O}_X(1)$.

The proof is by induction on the dimension s of the support of F; if $s = -1$, the assertion is trivial. Choose a section $\sigma \in H^0(\mathcal{O}_X(1))$ which vanishes at no point of the finite set $\operatorname{Ass}(F)$. Then σ defines an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0,$$

and G is m-regular by 1.2, with support of dimension $s-1$.

Consider the corresponding exact sequence

$$H^q(F(n-q-1)) \longrightarrow H^q(F(n-q)) \longrightarrow H^q(G(n-q)).$$

By the induction hypothesis, $H^q(G(n-q)) = 0$ for $n \geq m$, so the vanishing of $H^q(F(n-q-1))$ implies that of $H^q(F(n-q))$. But $H^q(F(m-q)) = 0$, which proves (i).

To prove (ii), we chase through the following commutative diagram:

$$\begin{tikzcd}
 & & H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) \arrow[r] \arrow[d] & H^0(G(n)) \otimes H^0(\mathcal{O}_X(1)) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & H^0(F(n)) \arrow[r] \arrow[ur] & H^0(F(n+1)) \arrow[r] & H^0(G(n+1)) \arrow[d] & \\
 & & & 0 &
\end{tikzcd}$$

Here the oblique arrow is defined by $\varphi \mapsto \varphi \otimes \sigma$; the horizontal rows are exact, the first by the n-regularity of F and by (i), already proved, and the second column is exact by the induction hypothesis.

Finally, (iii) follows from the commutative diagram

$$\begin{tikzcd}
H^0(F(n))_X \otimes H^0(\mathcal{O}_X(1))_X \arrow[r, "\alpha_n"] \arrow[d] & H^0(F(n)(1))_X \arrow[d, "\beta_{n+1}"] \\
{[H^0(F(n))_X](1)} \arrow[r, "\beta_n \otimes \mathrm{id}"] & F(n)(1).
\end{tikzcd}$$

Here, if M is a vector space over k, we write M_X for $M \otimes_k \mathcal{O}_X$. For $n \geq m$, α_n is surjective by (ii); therefore, if β_{n+1} is surjective, so is β_n. By Serre's theorem, β_{n+1} is surjective for $n \gg 0$; this proves (iii).

Proposition 1.4. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence on X. If G is m-regular, then:

(i) $H^q(F(n)) = 0$ for $q \geq 2$ and $n \geq m - q$.

(ii) $h^1(F(n-1)) \geq h^1(F(n))$ for $n \geq m - 1$.

(iii) $H^1(F(n)) = 0$ for $n \geq (m-1) + h^1(F(m-1))$.

In particular, F is $[m + h^1(F(m-1))]$-regular.

Indeed, in the exact sequence

$$H^{q-1}(G(n)) \longrightarrow H^q(F(n-1)) \longrightarrow H^q(F(n)) \longrightarrow H^q(G(n)),$$

the two outer groups are zero for $q \geq 2$ and $n \geq m - (q-1)$, by 1.3(i). Thus

$$H^q(F(m-q)) \xrightarrow{\sim} H^q(F(m-q+1)) \xrightarrow{\sim} H^q(F(m-q+2)) \xrightarrow{\sim} \cdots ;$$

but, by Serre's theorem, $H^q(F(n)) = 0$ for $n \gg 0$, proving (i).

For $n \geq m - 1$, consider the exact sequence

$$0 \to H^0(F(n-1)) \to H^0(F(n)) \xrightarrow{\alpha_n} H^0(G(n)) \to H^1(F(n-1)) \to H^1(F(n)) \to 0.$$

It implies (ii). Now suppose that α_n is surjective, and consider the commutative diagram

$$\begin{tikzcd}
H^0(F(n)) \otimes H^0(\mathcal{O}_X(1)) \arrow[r, "\alpha_n \otimes \mathrm{id}"] \arrow[d] & H^0(G(n)) \otimes H^0(\mathcal{O}_X(1)) \arrow[r] \arrow[d, "\beta_n"] & 0 \\
H^0(F(n+1)) \arrow[r, "\alpha_{n+1}"] & H^0(G(n+1)). &
\end{tikzcd}$$

If $n \geq m$, then β_n is surjective by 1.3(ii), and hence α_{n+1} is surjective. Consequently,

$$H^1(F(n-1)) \xrightarrow{\sim} H^1(F(n)) \xrightarrow{\sim} \cdots \xrightarrow{\sim} 0.$$

On the other hand, if α_n is not surjective, then $h^1(F(n-1)) > h^1(F(n))$. Thus the function $n \longmapsto h^1(F(n))$ decreases *strictly*, starting from $h^1(F(m-1))$, until it becomes zero; this proves (iii).

Definition 1.5. Let F be a coherent sheaf on X, let $r \geq \dim \operatorname{Supp}(F)$, and let $(b) = (b_0, \ldots, b_r)$ be a sequence of $r+1$ integers. Then F is said to be a (b)-*sheaf* if it satisfies either of the following two equivalent sets of conditions.

(1.5.1)

(i) At every point of the support of F, $\mathcal{O}_X(1)$ is generated by $H^0(\mathcal{O}_X(1))$.

(ii) $h^0(F(-1)) \leq b_0$.

(iii) If $r \geq 1$, there is a section $\sigma \in H^0(\mathcal{O}_X(1))$ which defines an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

such that G is a $(b_1, \ldots, b_r)$-sheaf.

(1.5.2) There is an F-regular sequence of r sections $\sigma_1, \ldots, \sigma_r \in H^0(\mathcal{O}_X(1))$ such that, for $i = 0, \ldots, r$, if F_i denotes the restriction of F to the zero scheme of $\sigma_1, \ldots, \sigma_i$, then

$$h^0(F_i(-1)) \leq b_i.$$

Proposition 1.6. Let F be a (b)-sheaf on X. Then:

(i) For every sufficiently general sequence σ of r sections $\sigma_1, \ldots, \sigma_r \in H^0(\mathcal{O}_X(1))$, if $F_{\sigma,i}$ denotes the restriction of F to the zero scheme of $\sigma_1, \ldots, \sigma_i$, then $h^0(F_{\sigma,i}(-1)) \leq b_i$, for $0 \leq i \leq r$.

(ii) Every subsheaf G of F is a (b)-sheaf.

Indeed, let $S' = \mathbb{A}_k^N$ be the affine space whose k-points correspond to the sequences σ, and let T be the open subset of S' corresponding to the F-regular sequences. For fixed i, the sheaves $F_{\sigma,i}(-1)$ corresponding to the k-points of T are contained in a flat family over T, and T is nonempty by hypothesis. Thus (i) follows from upper semicontinuity of the function

$$\sigma \longmapsto h^0(F_{\sigma,i}(-1))$$

(EGA III 7.7.5). Finally, since every sufficiently general sequence is G-regular and is such that $G_{\sigma,i}$ is a subsheaf of $F_{\sigma,i}$, for every i, (ii) follows from (i).

Lemma 1.7. Let

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

be an exact sequence on X. If the Hilbert polynomial of F is written in the form

$$\chi(F(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i},$$

then

$$\chi(G(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

Indeed,

$$\begin{aligned} \chi(G(n)) &= \chi(F(n)) - \chi(F(n-1)) \\ &= \sum_{i=0}^{r} a_i \left[\binom{n+i}{i} - \binom{n-1+i}{i} \right] \\ &= \sum_{i=1}^{r} a_i \binom{n+i-1}{i-1}, \end{aligned}$$

which proves the assertion.

Proposition 1.8. Let F be a (b)-sheaf on X, let $s = \dim \operatorname{Supp}(F)$, and let

$$\chi(F(n)) = \sum_{i=0}^{s} a_i \binom{n+i}{i}$$

be the Hilbert polynomial of F. Then:

(i) For $n \geq -1$,

$$h^0(F(n)) \leq \sum_{i=0}^{s} b_i \binom{n+i}{i}.$$

(ii) $a_s \leq b_s$, and F is also a $(b_0, \ldots, b_{s-1}, a_s)$-sheaf.

Indeed, we argue by induction on s. If $s = 0$, the assertion is clear, since $a_0 = h^0(F) = h^0(F(-1)) \leq b_0$. If $s \geq 1$, we have an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

with G a $(b_1, \dots, b_r)$-sheaf and $\dim \operatorname{Supp}(G) = s-1$. Then

$$h^0(F(n)) - h^0(F(n-1)) \le h^0(G(n)),$$

whereas

$$h^0(G(n)) \le \sum_{i=0}^{s-1} b_{i+1} \binom{n+i}{i}$$

by induction on s, and $h^0(F(-1)) \le b_0$. Assertion (i) follows by induction on $n \ge -1$. Finally, by 1.7 and induction on s, $a_s \le b_s$, and G is a $(b_1, \dots, b_{s-1}, a_s)$-sheaf; this proves (ii).

Definition 1.9. The following polynomials with nonnegative rational coefficients, defined recursively, are called the *(b)-polynomials*:

$$\begin{cases} P_{-1} = 0, \\ P_r(X_0, \dots, X_r) = P_{r-1}(X_1, \dots, X_r) + \displaystyle\sum_{i=0}^{r} X_i \binom{P_{r-1}(X_1, \dots, X_r) - 1 + i}{i}. \end{cases}$$

Remark 1.10. For the (b)-polynomials, induction on $r \ge t$ immediately gives

$$P_r(X_0, \dots, X_t, 0, \dots, 0) = P_t(X_0, \dots, X_t).$$

Theorem 1.11. *(The main theorem on (b)-sheaves.)* Let F be a (b)-sheaf on X, and let

$$\chi(F(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$$

be the Hilbert polynomial of F. Let $(c) = (c_0, \dots, c_r)$ be a sequence of integers satisfying $(c) \ge (b) - (a)$, and let $m = P_r(c)$ be the integer given by the r-th (b)-polynomial (1.9). Then $m \ge 0$, and F is m-regular. In particular, if $s = \dim \operatorname{Supp}(F)$, then F is

$$P_{s-1}(c_0, \dots, c_{s-1})\text{-regular.}$$

Indeed, we argue by induction on r. If $r = 0$, the assertion is clear, since $m = 0$ and F is n-regular for every n. If $r \geq 1$, there is an exact sequence

$$0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$$

in which G is a $(b_1, \ldots, b_r)$-sheaf. By 1.7 and the induction hypothesis, if $n = P_{r-1}(c_1, \ldots, c_r)$, then $n \geq 0$ and G is n-regular. Hence, by 1.4, F is $[n+h^1(F(n-1))]$-regular, and $h^q(F(n-1)) = 0$ for $q \geq 2$. Therefore

$$h^1(F(n-1)) = h^0(F(n-1)) - \chi(F(n-1)),$$

which, by 1.8(i), is at most

$$\sum_{i=0}^{r}(b_i - a_i)\binom{n-1+i}{i},$$

since $b_i \geq 0$. Finally, by 1.3(i), F is

$$\left[n + \sum_{i=0}^{r} c_i \binom{n-1+i}{i}\right]\text{-regular.}$$

This proves the first assertion; the second follows from 1.8(ii) and 1.10.

1.12. Let S be a noetherian scheme, and let X be an S-scheme of finite type. Let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibers of X/S. In other words, for every point $s \in S$ and every extension K of $k(s)$, coherent sheaves F_K are specified on the algebraic scheme X_K; and F_K and $F'_{K'}$ are declared to lie in the same class if there are $k(s)$-homomorphisms from K and K' into a common extension K'' of $k(s)$ such that

$$F_{K''} = F_K \otimes_K K'' \quad \text{and} \quad F'_{K''} = F'_{K'} \otimes_{K'} K''$$

are isomorphic on $X_{K''}$. The family $\mathcal{F}$ is said to be *bounded by the coherent sheaf* F *on* $X_T = X \times_S T$, where T is of finite type over S, if $\mathcal{F}$ is contained in the family of classes of the sheaves $F_{k(t)}$, with $t \in T$. The family $\mathcal{F}$ is said to be *bounded* if such T and F exist.

Suppose in addition that X/S is projective and endowed with a (relatively) ample invertible sheaf $\mathcal{O}_X(1)$. Then $\mathcal{F}$ is called a (b)*-family*, for a sequence of integers $(b) = (b_0, \ldots, b_r)$, if every class in $\mathcal{F}$ is represented by an F_K, with K algebraically closed, which is a (b)-sheaf.

Theorem 1.13. *Let S be a noetherian scheme, and let X be a projective S-scheme endowed with an ample sheaf $\mathcal{O}_X(1)$ such that the induced sheaves $\mathcal{O}_{X_s}(1)$, for $s \in S$, are generated by their $H^0(\mathcal{O}_{X_s}(1))$. Let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibers of X/S. Then the following conditions are equivalent:*

- (i) *$\mathcal{F}$ is bounded. Moreover, if all the $F_K \in \mathcal{F}$ are locally free of rank p, then $\mathcal{F}$ may be assumed to be bounded by a sheaf F on some X_T, with F locally free of rank p.*
- (ii) *The set of Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{F}$, is finite, and there exists a sequence of integers (b) such that $\mathcal{F}$ is a (b)-family.*
- (iii) *The set of Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{F}$, is finite, and there exists an integer m such that all the $F_K \in \mathcal{F}$ are m-regular.*
- (iv) *The set of Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{F}$, is finite, and $\mathcal{F}$ is contained in the family of classes of quotients of sheaves of the form E_K, where E is a coherent sheaf on some X_T. Moreover, one may assume that $T = S$ and that E has the form $\mathcal{O}_X(-m)^{\oplus M}$.*
- (v) *$\mathcal{F}$ is contained in the family of classes of cokernels of homomorphisms of the form $E'_K \to E_K$, where E' and E are coherent sheaves on some X_T. Moreover, one may assume that $T = S$ and that E' and E have the forms $\mathcal{O}_X(-m')^{\oplus M'}$ and $\mathcal{O}_X(-m)^{\oplus M}$, respectively.*

Indeed, suppose that $\mathcal{F}$ is bounded by a sheaf F on some X_T. After partitioning T into pieces and applying the generic flatness theorem (EGA IV 6.9.1), one may assume that F is flat over T. Then, the number of polynomials $\chi(F(n))$ is at most the number of connected components of T (EGA III 7.9.5). It is also an elementary lemma that, if $t \in T$ and F_t is locally free of rank p, then F is locally free of rank p over an open neighborhood of t in T (see EGA IV 11.3.10, with $X = Y$ and $f = \mathrm{id}$). The hypotheses and EGA III 7.7.6 show, moreover, that after partitioning T still more finely one can find a sequence σ of sections

$$\sigma_1, \ldots, \sigma_r \in H^0(\mathcal{O}_X(1))$$

that is F-regular. Thus (ii) follows from the upper semicontinuity of the function

$$t \longmapsto h^0(F_{t,i}(-1)),$$

where, for $t \in T$ and $0 \leq i \leq r$, $F_{t,i}$ denotes the restriction of F_t to the zero scheme of $\sigma_1, \ldots, \sigma_i$ (EGA III 7.7.5).

The implication (ii) $\Rightarrow$ (iii) follows immediately from 1.11. The implication (iii) $\Rightarrow$ (iv) follows from 1.3 (iii), on taking

$$M = \max(\chi(F_K(m))), \qquad E = \mathcal{O}_X(-m)^{\oplus M}.$$

Suppose that $\mathcal{F}$ satisfies (iv). There are exact sequences

$$0 \longrightarrow F'_K \longrightarrow E_K \longrightarrow F_K \longrightarrow 0,$$

and the set of polynomials $\chi(F_K(n))$ is finite. By hypothesis, the family of classes of the E_K is bounded. Hence, by the already proved implication (i) $\Rightarrow$ (ii), the set of polynomials $\chi(E_K(n))$ is finite, and there exists a sequence of integers (b) such that all the E_K, with K algebraically closed, are (b)-sheaves. It follows that the set of polynomials $\chi(F'_K(n))$ is finite and, by 1.6, all the F'_K are (b)-sheaves. Applying the already proved implication (ii) $\Rightarrow$ (iv) to the family of classes of the F'_K, one obtains (v).

Finally, suppose that $\mathcal{F}$ satisfies (v), and let us prove that $\mathcal{F}$ is bounded. One may assume that E (respectively E') has the form $\mathcal{O}_X(-m)^{\oplus M}$. Indeed, by (i) $\Rightarrow$ (iv), take a surjection

$$L = \mathcal{O}_{X_T}(-m)^{\oplus M} \longrightarrow E.$$

After partitioning T into pieces so that E becomes flat over T, formation of the exact sequence

$$0 \longrightarrow I \xrightarrow{u} L \longrightarrow E \longrightarrow 0$$

commutes with passage to the fibers. By (i) $\Rightarrow$ (iii) and (i) $\Rightarrow$ (iv), choose m_1 sufficiently large that all the groups

$$\mathrm{Ext}^1(\mathcal{O}_{X_K}(-m_1), I_K) = H^1(I_K(m_1))$$

vanish and that there exist surjections

$$L_1 = \mathcal{O}_{X_T}(-m_1)^{\oplus M_1} \longrightarrow E', \qquad \alpha : L_2 = \mathcal{O}_{X_T}(-m_1)^{\oplus M_2} \longrightarrow I.$$

The maps

$$\mathrm{Hom}(L_{1,K}, L_K) \longrightarrow \mathrm{Hom}(L_{1,K}, E_K)$$

are then surjective. Let $\beta : E'_K \to E_K$ be a homomorphism, and let γ be the composite

$$L_{1,K} \longrightarrow E'_K \xrightarrow{\beta} E_K.$$

Then γ comes from a homomorphism $\delta : L_{1,K} \to L_K$, and

$$(\delta, u_K\alpha) : L_{1,K} \oplus L_{2,K} \longrightarrow L_K$$

and $\beta : E'_K \to E_K$ have the same cokernel. Consequently, $\mathcal{F}$ is contained in the family of classes of cokernels of homomorphisms of the form

$$\mathcal{O}_{X_K}(-m_1)^{\oplus(M_1+M_2)} \longrightarrow \mathcal{O}_{X_K}(-m)^{\oplus M}.$$

Still after partitioning T into pieces, one may assume in addition that X_T (and hence $\mathcal{H}om_{\mathcal{O}_{X_T}}(E', E)$) and all the

$$R^q(f_T)_*\mathcal{H}om_{\mathcal{O}_{X_T}}(E', E),$$

where $f_T : X_T \to T$ is the structural morphism, are flat over T, so that formation of

$$G = (f_T)_*\mathcal{H}om_{\mathcal{O}_{X_T}}(E', E)$$

commutes with every base change $T' \to T$ (EGA III 6.9.9.2). Let

$$R = \mathbb{V}(G^*).$$

Then X_R canonically carries a *universal* exact sequence

$$E'_R \longrightarrow E_R \longrightarrow F \longrightarrow 0,$$

and F bounds $\mathcal{F}$.

2. Several technical lemmas

2.1. Let k be a field and X a proper k-scheme. Let $L_1, \ldots, L_s$ be invertible sheaves on X, let $c_1(L_i)$ be the Chern class of L_i, and let Y be a closed subscheme of X of dimension $\leq s$. Recall that the notation

$$\langle c_1(L_1) \cdots c_1(L_s) \cdot Y \rangle$$

denotes the intersection number, which may be calculated as the coefficient of the monomial $n_1 \cdots n_s$ in the Snapper polynomial

$$\chi\Big(L_1^{\otimes n_1} \otimes \cdots \otimes L_s^{\otimes n_s} \otimes \mathcal{O}_Y\Big)$$

(X 4.3.1).

2.2. From now on, suppose that k is algebraically closed, that X is projective and endowed with a very ample invertible sheaf $H = \mathcal{O}_X(1)$, and that X is integral of dimension r.

Let L be an invertible sheaf on X. The integer

$$\langle c_1(L) \cdot c_1(H)^{r-1} \rangle$$

is called the *degree* of L (relative to H) and is denoted $d(L)$.

In what follows we consider an invertible sheaf L and the Hilbert polynomials

$$\chi(L(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}, \qquad \chi(\mathcal{O}_X(n)) = \sum_{i=0}^{r} e_i \binom{n+i}{i}.$$

Lemma 2.3. *One has:*

(i) $a_r = e_r = d(H)$.

(ii) *If* $r \geq 1$, *then* $a_{r-1} - e_{r-1} = d(L)$.

Indeed, (i) and (ii) follow immediately by setting $m = 0, 1$ in the following polynomial:

$$\chi(L^{\otimes m} \otimes H^{\otimes n}) = \sum_{i+j \leq r} f_{ij} \binom{m+i}{i} \binom{n+j}{j}.$$

Lemma 2.4. *If* $d(L) \leq 0$ *and* $h^0(L) \neq 0$, *then* $L \simeq \mathcal{O}_X$; *hence* $d(L) = 0$ *and* $h^0(L) = 1$.

Indeed, since X is integral, there is a divisor $Y \geq 0$ such that $L \simeq \mathcal{O}_X(Y)$. If $Y > 0$, one would have

$$0 \geq \langle c_1(L) \cdot c_1(H)^{r-1} \rangle = \langle c_1(H)^{r-1} \cdot Y \rangle \geq 1.$$

Lemma 2.5. *If* $d(L) \leq 0$, *then* L *is a* $(0, \dots, 0, a_r)$*-sheaf (with* $r - 1$ *zeros).*

Indeed, if $r = 0$, then L is plainly an a_0-sheaf. If $r \geq 1$, then by 2.4 one has $h^0(L(-1)) = 0$, since $d(L(-1)) < 0$. Let Y be a hyperplane section of X, and let

$$0 \longrightarrow L(-1) \longrightarrow L \longrightarrow L_Y \longrightarrow 0$$

be an exact sequence defined by Y. If $r = 1$, then by 1.7 one has $h^0(L_Y(-1)) = a_1$; hence L is a $(0, a_1)$-sheaf. If $r \geq 2$, one may take Y integral by Bertini's theorem and may thus assume inductively that L_Y is a $(0, \dots, 0, a_r)$-sheaf; the assertion follows.

Lemma 2.7. *If* $r = 2$ *and* $d(L) \leq 0$, *then:*

(i) *For every* n,
$$h^1(L(n)) \leq a_1(a_1 + 1)a_2 - a_0.$$

(ii) $a_0 \leq a_1(a_1 + 1)$.

Let Y be an integral hyperplane section of X. By 1.7 and 2.6, L_Y is a $(0, a_2)$-sheaf with Hilbert polynomial

$$\chi(L_Y(n)) = a_1 + a_2 \binom{n+1}{1}.$$

Thus, by 1.11, $-a_1 \geq 0$, and L_Y is $[-a_1]$-regular.

Consider the following exact sequence:

$$H^0(L_Y(n)) \longrightarrow H^1(L(n-1)) \longrightarrow H^1(L(n)) \longrightarrow H^1(L_Y(n)).$$

For $n \geq -a_1 - 1$, one has $h^1(L(-a_1 - 2)) \geq h^1(L(n))$, since $h^1(L_Y(n)) = 0$ by 1.3 (i). For $n \leq -1$, one has $h^1(L(n-1)) \leq h^1(L(-1))$, since $h^0(L_Y(n)) = 0$ by 2.4. Finally, for every n, $h^0(L_Y(n-1)) \leq h^0(L_Y(n))$, since there is an injection $L_Y(n-1) \to L_Y(n)$, and

$$h^0(L_Y(-a_1 - 1)) = a_2 \binom{-a_1}{1} + a_1,$$

since $h^1(L_Y(-a_1 - 1)) = 0$. Hence, for $n \leq -a_1 - 2$,

$$h^1(L(n)) - h^1(L(n+1)) \leq -a_2 a_1 + a_1.$$

It follows that, for every n,

$$h^1(L(n)) \leq h^1(L(-a_1-2)) + (-a_1-1)(-a_2a_1+a_1). \tag{2.7.1}$$

Since L_Y is $[-a_1]$-regular, one has $h^2(L(-a_1-2))=0$ by 1.4; hence

$$h^1(L(-a_1-2)) = h^0(L(-a_1-2)) - \chi(L(-a_1-2)).$$

But

$$h^0(L(-a_1-2)) \leq a_2\binom{-a_1}{2}$$

by 1.8 (i), since L is a $(0,0,a_2)$-sheaf by 2.5. Therefore

$$0 \leq h^1(L(-a_1-2)) \leq -a_1(-a_1-1) - a_0. \tag{2.7.2}$$

Finally, 2.7.2 gives (ii), and 2.7.1 together with 2.7.2 gives (i).

Lemma 2.8. *Suppose that $r=2$ and $d(L)\leq 0$. Put $\delta = d(L)$ and $\beta = \delta(e_2-1)+e_1$. Then:*

$$a_0 \geq \langle c_1(L)^2\rangle + 2e_0 + \sum_{i=1}^{2}(2e_i-a_i)\binom{\delta-1+i}{i} + \beta(\beta+1).$$

Indeed, write

$$\chi(L^{\otimes m}(\delta-1)) = \alpha_2m^2+\alpha_1m+\alpha_0.$$

Comparison of polynomials gives

$$2\alpha_2 = \langle c_1(L)^2\rangle, \qquad \alpha_0 = \sum_{i=0}^{2} e_i\binom{\delta-1+i}{i},$$

and

$$\alpha_2+\alpha_1+\alpha_0 = \sum_{i=0}^{2} a_i\binom{\delta-1+i}{i}.$$

Write

$$\chi(L^{-1}(\delta)(n)) = \sum_{i=0}^{2}\beta_i\binom{n+i}{i}.$$

Then $\beta_0 = \alpha_2-\alpha_1+\alpha_0$, and by 2.3,

$$\beta_1 = d(L^{-1}(\delta)) + e_1 = \beta.$$

Thus

$$a_0 = 2\alpha_2 + 2\alpha_0 - \sum_{i=1}^{2} a_i\binom{\delta-1+i}{i} - \beta_0.$$

Since $d(L^{-1}(\delta))\leq 0$, one has $\beta_0 \leq \beta(\beta+1)$ by 2.7 (ii); whence the assertion.

Lemma 2.9. *Suppose that $r\geq 2$, that $d(L)\leq 0$, and that X is normal. Then:*

(i) *$h^1(L(-n)) = 0$ for $n \geq 2+h_2(a_{r-2},a_{r-1},a_r)$;*

(ii) *$h^1(L(n)) \leq h_r(a_{r-2},a_{r-1},a_r)$ for every n;*

(iii) *$|a_i| \leq g_{r-i}(a_{r-2},a_{r-1},a_r)$ for $i=0,\ldots,r$, where $h_2(X_0,X_1,X_2), h_3(X_0,X_1,X_2),\ldots$ and $g_0(X_0,X_1,X_2), g_1(X_0,X_1,X_2),\ldots$ are universal polynomials with coefficients in $\mathbb{Q}$, independent of both k and X.*

Indeed, first consider the case $r = 2$. Put

$$g_i(X_0, X_1, X_2) = X_{2-i}^2 \quad (i = 0, 1, 2), \qquad h_2(X_0, X_1, X_2) = X_1(X_1 + 1)X_2 - X_0.$$

Then (iii) is clear, and (ii) is 2.7 (i).

Let $X \subset P = \mathbb{P}_k^N$, and put

$$\omega = \mathcal{E}xt^{N-2}_{\mathcal{O}_P}(\mathcal{O}_X, \Omega^N_{P/k}).$$

Then

$$h^1(L(-n)) = h^1(\omega \otimes L^{-1}(n)),$$

and ω is invertible outside the finite set Z of singular points of X (see FGA 149 § 6). Let Y be a normal hyperplane section of X not containing Z. Then $\omega_Y(1) \simeq \Omega^1_{Y/k}$, and there is an exact sequence

$$0 \longrightarrow \omega \otimes L^{-1}(-1) \longrightarrow \omega \otimes L^{-1} \longrightarrow \Omega^1_{Y/k} \otimes L_Y^{-1}(-1) \longrightarrow 0.$$

Now

$$h^1(\Omega^1_{Y/k} \otimes L_Y^{-1}(n)) = h^0(L_Y(-n))$$

for every n, and $h^0(L_Y(-n)) = 0$ for $n \geq 1$ by 2.5. Thus $\Omega^1_{Y/k} \otimes L_Y^{-1}(-1)$ is 3-regular and, by 1.4, $\omega \otimes L^{-1}$ is

$$\big[3 + h^1(\omega \otimes L^{-1}(2))\big]\text{-regular.}$$

Consequently, $h^1(L(-n)) = 0$ for $n \geq 2 + h^1(L(-2))$; this proves (i).

Now suppose that $r \geq 3$, and argue by induction on r. Let Y be a normal hyperplane section of X, and consider the exact sequence

$$H^0(L_Y(-n)) \longrightarrow H^1(L(-n-1)) \longrightarrow H^1(L(-n)) \longrightarrow H^1(L_Y(-n)). \tag{2.9.1}$$

By 1.7,

$$\chi(L_Y(n)) = \sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

Thus, for $n \geq 2 + h_2(a_{r-2}, a_{r-1}, a_r)$, the induction hypothesis gives $h^1(L_Y(-n)) = 0$; and since $n \geq 2$ by (ii) for $r = 2$, 2.4 gives $h^0(L_Y(-n)) = 0$. Hence

$$h^1(L(-n-1)) = h^1(L(-n)).$$

But for $n \gg 0$, one has $h^1(L(n)) = 0$ (see FGA 149 § 6); hence (i).

As induction hypothesis, suppose that the polynomials g_i and h_i have been defined for $i \leq r-1$. For every n, 2.9.1 gives

$$h^1(L(n)) \leq h^1(L(n-1)) + h_{r-1}(a_{r-2}, a_{r-1}, a_r).$$

Therefore, by (i) already proved, it follows that for every $m \geq -2$,

$$h^1(L(n)) \leq \big[2 + h_2(a_{r-2}, a_{r-1}, a_r) + m\big] h_{r-1}(a_{r-2}, a_{r-1}, a_r) \tag{2.9.2}$$

for every $n \leq m$.

By 2.5, L_Y is a $(0, \ldots, 0, a_r)$-sheaf. Put

$$c_i = g_{r-i}(a_{r-2}, a_{r-1}, a_r) \quad (i = 1, \ldots, r), \qquad m = P_{r-2}(c_1, \ldots, c_{r-1}).$$

By 1.11, $m \geq 0$, and L_Y is m-regular. Hence, by 1.4 (ii),

$$h^1(L(m-2)) \geq h^1(L(n)) \qquad \text{for every } n \geq m-1.$$

Thus, if polynomials are defined by

$$M = P_{r-2}(g_{r-1}, \dots, g_1), \qquad h_r = (h_2 + M)h_{r-1},$$

then (ii) follows from 2.9.2.

Finally, since L_Y is m-regular, one has $h^q(L(m)) = 0$ for $q \geq 2$ by 1.4 (i). Hence

$$\sum_{i=0}^{r} a_i \binom{m+i}{i} = h^0(L(m)) - h^1(L(m)).$$

But, by 1.8 (i),

$$h^0(L(m)) \leq a_r \binom{m+r}{r}.$$

Therefore, if one defines

$$g_r = \sum_{i=1}^{r} g_{r-i} \binom{M+i}{i} + a_r \binom{M+r}{r} + h_r,$$

then (iii) follows.

Remark 2.10. For fixed m, consider an equality of polynomials in n:

$$\sum_{i=0}^{r} a_{m,i} \binom{n+i}{i} = \sum_{j=0}^{r} a_j \binom{m+n+j}{j}.$$

One readily finds the formula

$$a_{m,i} = \sum_{j=0}^{r-i} a_{j+i} \binom{m-1+i}{j}$$

by induction on r, reducing $a_{m,i}$ to $a_{m,0}$, and then setting $n = -1$.

Similarly, one finds a polynomial with rational coefficients

$$P_i^{(m,r)}(X_r, \dots, X_i),$$

increasing in X_i, such that in the equality

$$\sum_{i=0}^{r} a_i^{(m)} \binom{n+i}{i} = \sum_{j=0}^{r} a_j \binom{mn+j}{j},$$

one has

$$a_i^{(m)} = P_i^{(m,r)}(a_r, \dots, a_i).$$

Lemma 2.11. *Let S be a noetherian scheme and X a projective S-scheme endowed with a very ample sheaf $H = \mathcal{O}_X(1)$. Suppose that the fibers of X/S are geometrically integral and all have the same dimension r. Let $\mathcal{L}$ be a family of classes of invertible sheaves L_K on the fibers of X/S. For $\mathcal{L}$ to be bounded, it is necessary and sufficient that all the coefficients a_i in the Hilbert polynomials*

$$\chi(L_K(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$$

of the $L_K \in \mathcal{L}$ remain bounded. Moreover, if the fibers of X/S are also geometrically normal, then it is enough that the three coefficients a_r, a_{r-1}, a_{r-2} remain bounded.

Indeed, let

$$\delta = \sup\{d(L_K)\}.$$

If $\delta > 0$, one may replace $\mathcal{L}$ by the family of classes of the $L_K(-\delta)$, by 2.3 and 2.10; thus one may assume $\delta \leq 0$. Then $\mathcal{L}$ is a (b)-family by 2.5, and is therefore bounded if all the coefficients a_i remain bounded, by 1.13. If the fibers of X/S are geometrically normal, it is enough for this purpose that the three coefficients a_r, a_{r-1}, a_{r-2} remain bounded, by 2.9.

3. General finiteness theorems

Lemma 3.1. *Let S be a scheme and X a quasi-compact and quasi-separated S-scheme. Let $\underline{\mathrm{Pic}}_{X/S}$ be the Picard functor, localized for the fppf topology.*[25] *Then, for every filtered inverse system (Z_α) of quasi-compact and quasi-separated S-schemes with affine transition morphisms, the canonical map*

$$\varinjlim_{\alpha} \underline{\mathrm{Pic}}_{X/S}(Z_\alpha) \longrightarrow \underline{\mathrm{Pic}}_{X/S}\left(\varprojlim_{\alpha} Z_\alpha\right) \tag{3.1.1}$$

is bijective.

Consequently (EGA IV 8.14.2), *if the Picard functor is represented by an S-scheme* $\mathrm{Pic}_{X/S}$, *then* $\mathrm{Pic}_{X/S}$ *is locally of finite presentation.*

Indeed, an element λ_α of $\underline{\mathrm{Pic}}_{X/S}(Z_\alpha)$ is represented by an invertible sheaf L'_α on $X_{Z'_\alpha}$, where $Z'_\alpha \to Z_\alpha$ is an fppf morphism. Put

$$Z = \varprojlim_{\alpha} Z_\alpha.$$

The image λ of λ_α in $\underline{\mathrm{Pic}}_{X/S}(Z)$ is represented by the invertible sheaf L' on $X_{Z'}$ obtained by inverse image from L'_α, where

$$Z' = Z'_\alpha \times_{Z_\alpha} Z.$$

If λ is the neutral element of $\underline{\mathrm{Pic}}_{X/S}(Z)$, there exists an fppf morphism $Z'' \to Z'$ such that the inverse image of L' on $X_{Z''}$ is isomorphic to $\mathcal{O}_{X_{Z''}}$. By EGA IV 8.8.2, 8.10.5 (vi), 11.2.6 (ii), 8.5.2 (i), and 8.5.2.4, there exist $\beta \geq \alpha$ and an fppf morphism

$$Z''_\beta \longrightarrow Z'_\beta, \qquad Z'_\beta = Z'_\alpha \times_{Z_\alpha} Z_\beta,$$

such that the inverse image of L'_α on $X_{Z''_\beta}$ is isomorphic to $\mathcal{O}_{X_{Z''_\beta}}$. The image of λ_α is therefore already the neutral element in $\underline{\mathrm{Pic}}_{X/S}(Z_\beta)$. Thus 3.1.1 is injective.

The surjectivity of 3.1.1 is proved similarly.

Proposition 3.2. *Let S be a noetherian scheme and X a proper S-scheme. Suppose that the Picard functor is represented by an S-scheme* $\mathrm{Pic}_{X/S}$. *Then:*

(i) $\mathrm{Pic}_{X/S}$ *is locally of finite type over S.*

(ii) *Subsets Λ of* $\mathrm{Pic}_{X/S}$ *correspond bijectively to families $\mathscr{L}$ of classes of invertible sheaves on the fibers of X/S.*

(iii) *A subset Λ of* $\mathrm{Pic}_{X/S}$ *is quasi-compact if and only if the corresponding family $\mathscr{L}$ is bounded.*

[25] Cf. SGA 3 IV 6.3 and FGA no. 232 for the definition of $\underline{\mathrm{Pic}}_{X/S}$; here, as indicated, the fpqc topology should be replaced by the fppf topology.

Indeed, (i) follows from 3.1. Assertion (ii) follows from the following fact: if s is a point of S and K is an algebraically closed extension of $k(s)$, then the K-points of $\mathrm{Pic}_{X/S}$ correspond bijectively to the isomorphism classes of invertible sheaves on X_K.

For (iii), suppose that the subset Λ of $\mathrm{Pic}_{X/S}$ is quasi-compact. Then Λ is contained in an open subscheme U of finite type, and the inclusion $U \to \mathrm{Pic}_{X/S}$ corresponds to an invertible sheaf L on some X_T, where $T \to U$ is a surjective (flat) morphism of finite type. Finally $T \to S$ is of finite type, so L bounds $\mathscr{L}$.

Conversely, suppose that $\mathscr{L}$ is bounded by a coherent sheaf L on some X_T, with T of finite type over S. One may assume that L is invertible (1.13 (i)), so that L defines a morphism

$$f : T \longrightarrow \mathrm{Pic}_{X/S} .$$

The image of f contains Λ and is quasi-compact, hence noetherian by (i). It follows that Λ is quasi-compact, proving (iii).

By 3.2, the usual definition of morphisms of finite type between two Picard schemes generalizes as follows.

Definition 3.3. *Let X and Y be proper over a noetherian scheme S. A morphism between the Picard functors*

$$\underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is said to be of finite type if every bounded family of classes of invertible sheaves on the fibers of X/S has as inverse image a bounded family of classes of invertible sheaves on the fibers of Y/S.

Lemma 3.4. *Let X, Y, X', Y' be proper over a noetherian scheme S, and let $T \to S$ be surjective and of finite type. Then:*

(i) *A morphism $\Phi : \underline{\mathrm{Pic}}_{Y/S} \to \underline{\mathrm{Pic}}_{X/S}$ is of finite type if and only if $\Phi_T : \underline{\mathrm{Pic}}_{Y_T/T} \to \underline{\mathrm{Pic}}_{X_T/T}$ is of finite type.*

(ii) *In a commutative diagram*

$$\begin{array}{ccc} \underline{\mathrm{Pic}}_{Y'/S} & \xrightarrow{\Phi'} & \underline{\mathrm{Pic}}_{X'/S} \\ {\scriptstyle \Phi_Y}\uparrow & & \uparrow{\scriptstyle \Phi_X} \\ \underline{\mathrm{Pic}}_{Y/S} & \xrightarrow[\Phi]{} & \underline{\mathrm{Pic}}_{X/S} \end{array}$$

in which Φ' and Φ_Y are of finite type, Φ is of finite type.

Suppose that Φ_T is of finite type. Let $\mathscr{L}$ be a family on the fibers of X/S, bounded by an invertible sheaf L on some X_R. The family $\mathscr{L}_T = \mathscr{L} \times_S T$ on the fibers of X_T/T is bounded by the inverse image of L on $X_{R \times_S T}$. If $\Phi_T^{-1}(\mathscr{L}_T)$ is bounded by an invertible sheaf M on some Y_Q, then M also bounds $\Phi^{-1}(\mathscr{L})$. The converse in (i), and assertion (ii), are proved similarly.

Theorem 3.5. *Let S be a noetherian scheme, let X and Y be proper S-schemes, and let $f : X \to Y$ be a morphism. If f is surjective, then the induced morphism between the Picard functors*

$$f^* : \underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is of finite type.

By 3.4 (i), one may assume that S is reduced and irreducible. There is a nonempty open subset $U \subset S$ such that $\underline{\mathrm{Pic}}_{Y/S}$ and $\underline{\mathrm{Pic}}_{X/S}$ are representable and the restriction of f^* is quasi-affine

(XII 1.1), hence of finite type by 3.2 (i). Give $T = S - U$ its induced reduced structure. By noetherian induction, one may suppose that

$$\underline{\mathrm{Pic}}_{Y_T/T} \longrightarrow \underline{\mathrm{Pic}}_{X_T/T}$$

induced by f^* is of finite type. The assertion follows from 3.4 (i), applied to $U \amalg T \to S$.

Theorem 3.6. *Let S be a noetherian scheme, let X be a proper S-scheme, and let $\underline{\mathrm{Pic}}_{X/S}$ be its Picard functor. For every nonzero integer n, the n-th power homomorphism*

$$\Phi_n : \underline{\mathrm{Pic}}_{X/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is a morphism of finite type.

Let $f : X' \to X$ be a Chow presentation: X' is projective over S, and f is surjective. If the theorem is known for X', then 3.5 and 3.4 (ii) imply it for X. We may suppose X/S projective.

There are a nonempty open $U \subset S$, an integral S', and a finite surjective morphism $S' \to U$, such that every irreducible component X'_i of $X_{S'}$, with its induced reduced structure, is integral over S' (XII 1.3). Give $T = S - U$ its induced reduced structure. By noetherian induction, the theorem may be assumed for X_T/T. If it is assumed for all X_i/S', it follows for $\coprod_i X_i/S'$ by the following easy lemma, then for $X_{S'}/S'$ by 3.5 and 3.4 (ii), and finally for X/S by 3.4 (i).

Lemma 3.7. *Let $X = \coprod_i X_i$ be of finite type over a noetherian scheme S, and let $\mathcal{F}$ be a family of classes of coherent sheaves on the fibers of X/S. Then $\mathcal{F}$ is bounded if and only if, for every i, the induced family $\mathcal{F}_i$ on the fibers of X_i/S is bounded.*

In the proof of 3.6 we reduced to X projective and integral over S; endow X/S with a very ample sheaf $\mathcal{O}_X(1)$. Let $\mathscr{L}$ be bounded on the fibers of X/S, and let L on some X_T, with T of finite type over S, bound it. After partitioning T, suppose L flat over T; after taking connected components, suppose T connected. Then

$$P(q, m) = \chi(L_t^{\otimes q}(m))$$

is independent of $t \in T$ (EGA III 7.9.4).

Put $\mathscr{L}_n = \Phi_n^{-1}(\mathscr{L})$. It consists of classes of invertible sheaves M_K whose n-th power class belongs to $\mathscr{L}$. Write

$$\chi(M_K^{\otimes p}(m)) = \sum_{i=0}^{r} a_i(p) \binom{m+i}{i},$$

where $a_i(p)$ has degree at most $r - i$ in p. Since $\chi(M_K^{\otimes qn}(m)) = P(q, m)$, every $a_i(qn)$, and hence every polynomial $a_i(p)$, is independent of M_K. Thus $\mathscr{L}_n$ has one Hilbert polynomial $\chi(M_K(m))$, and is bounded by 2.11.

Theorem 3.8. *Let S be noetherian, let Y be a projective S-scheme with a relatively ample invertible sheaf $\mathcal{O}_Y(1)$, let X be the zero scheme of a section φ of $\mathcal{O}_Y(1)$, and let $f : X \to Y$ be the inclusion. Assume*

that for every $s \in S$ and every irreducible component Y_s^i of Y_s, the irreducible components of $X_s \cap Y_s^i = V(\varphi|_{Y_s^i})$ have dimension at least 2 (as happens when the irreducible components of the fibers of Y have dimension at least 3). Then

$$f^* : \underline{\mathrm{Pic}}_{Y/S} \longrightarrow \underline{\mathrm{Pic}}_{X/S}$$

is of finite type. Moreover, if S is integral, there is a nonempty open $U \subset S$ such that the restricted Picard functors are representable and

$$\underline{\mathrm{Pic}}_{Y/S}|_U \longrightarrow \underline{\mathrm{Pic}}_{X/S}|_U$$

is quasi-affine (and of finite type).

For the first assertion, noetherian induction permits replacing S by any nonempty open. One reduces to S integral and Y integral over S, replacing X by $X \times_Y Y'$ when Y is replaced by Y'. One then reduces to Y normal over S, using the following lemma in place of XII 1.3.

Lemma 3.9. *Let S be integral noetherian and X an S-scheme integral over S. There are a nonempty open $U \subset S$, an integral scheme S' finite faithfully flat over U, an S'-scheme X' normal and integral over S', and a finite surjective S'-morphism $X' \to X_{S'}$.*

Let η be the generic point of S, and let K be an algebraic closure of $k(\eta)$. The scheme $\mathrm{Spec}(K)$ is the limit of the filtered inverse system (S_α) of integral affine schemes with finite faithfully flat morphisms $S_\alpha \to U_\alpha$, where U_α is a nonempty open of S. The normalization of X_K descends to an S_α-scheme X' for some α, and the finite surjective map $X'_K \to X_K$ descends to a finite surjective S_α-morphism $X' \to X_{S_\alpha}$ (EGA IV 8.8.2, 8.10.5). Since X'_K is normal and integral, there is an open $S' \subset S_\alpha$ such that $X'_{S'}$ is normal and integral over S' (EGA IV 9.9.5).

Remark 3.10. Retain 3.9 and suppose there is a resolving morphism $X'_K \to X_K$ (EGA IV 7.9.1), as happens when $\dim(X_K) \le 2$, by Abhyankar. A similar argument gives a finite faithfully flat $S' \to U$, a scheme X' smooth over S', and a proper surjective birational morphism $X' \to X_{S'}$.

In the proof of 3.8 we have reduced to S integral and Y normal and integral over S. If $\varphi = 0$, then $X = Y$. Otherwise, after restricting S, suppose that $\varphi_s \neq 0$ for every $s \in S$. Thus φ_s is $\mathcal{O}_{Y_s}$-regular, and the dimension r of Y_s is at least 3.

Let $\mathscr{L}$ be bounded on the fibers of X/S, and let M_K be invertible on a geometric fiber Y_K, with

$$\chi(M_K(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}.$$

The Hilbert polynomial of f^*M_K is

$$\sum_{i=0}^{r-1} a_{i+1} \binom{n+i}{i}.$$

As the class of M_K varies through $(f^*)^{-1}(\mathscr{L})$, that is, as

the class of f^*M_K varies through $\mathscr{L}$, the coefficients a_r, a_{r-1}, a_{r-2} remain bounded by 1.13.

Let $H = \mathcal{O}_Y(m)$ be a very ample multiple of $\mathcal{O}_Y(1)$. In the Hilbert polynomials relative to H,

$$\chi(M_K(mn)) = \sum_{i=0}^{r} a_i^{(m)} \binom{n+i}{i},$$

the coefficients $a_r^{(m)}, a_{r-1}^{(m)}, a_{r-2}^{(m)}$ remain bounded by 2.10. Consequently $(f^*)^{-1}(\mathscr{L})$ is bounded by 2.11, proving the first part of 3.8.

For the second part, generic representability and separatedness of the Picard functors were established in XII 1.2. An argument like XII 2, using 3.9 and XII 1.1, reduces to Y normal over S. It is enough for $\underline{\mathrm{Pic}}_{Y/S} \to \underline{\mathrm{Pic}}_{X/S}$ to be separated and quasi-finite (EGA IV 8.12.8), which can be checked on geometric fibers and follows from:

Lemma 3.11. *Let k be algebraically closed, let Y be normal and projective with an ample sheaf $\mathcal{O}_Y(1)$, and let X be the zero scheme of a section φ. Suppose* $\dim(Y) \ge 3$ *(respectively,* $\dim(Y) \ge 2$*). Then*

$$\mathrm{Pic}_{Y/k} \longrightarrow \mathrm{Pic}_{X/k} \quad \text{(respectively, } \mathrm{Pic}^\tau_{Y/k} \longrightarrow \mathrm{Pic}^\tau_{X/k}\text{; see 4.1)}$$

is of finite type and has a finite kernel N, in general nonreduced.

Assume $\varphi \neq 0$; it is regular because Y is integral. The map is of finite type by 3.8 (respectively, 4.7 (iii)). Its kernel N is an algebraic group contained in $\mathrm{Pic}^{\tau}_{Y/k}$, hence is the same

in both cases. Since $\mathrm{Pic}^{\tau}_{Y/k}$ is proper because Y is normal (FGA no. 236, 2.13), and N is closed (SGA 3 VI_B 1.9.2), it is enough to show that N contains no subgroup of type μ_ℓ $(= \mathbb{Z}/\ell\mathbb{Z})$, with $(\ell, \mathrm{char}(k)) = 1$.

Consider Kummer theory, arising from

$$0 \longrightarrow \mu_\ell \longrightarrow \mathbb{G}_m \xrightarrow{x \mapsto x^\ell} \mathbb{G}_m \longrightarrow 0$$

on Y (respectively, on X):

$$\begin{array}{ccccccccc} & & & & H^1(Y,\mu_\ell) & \longrightarrow & H^1(Y,\mathbb{G}_m) & \xrightarrow{\ell} & H^1(Y,\mathbb{G}_m) \\ & & & & \downarrow & & \downarrow & & \downarrow \\ H^0(X,\mathbb{G}_m) & \xrightarrow{\ell} & H^0(X,\mathbb{G}_m) & \longrightarrow & H^1(X,\mu_\ell) & \longrightarrow & H^1(X,\mathbb{G}_m) & \xrightarrow{\ell} & H^1(X,\mathbb{G}_m). \end{array}$$

By Hilbert's Theorem 90, $H^1(Y, \mathbb{G}_m) = \mathrm{Pic}_{Y/k}(k)$ (respectively, $H^1(X, \mathbb{G}_m) = \mathrm{Pic}_{X/k}(k)$). Since X is proper and integral, $H^0(X, \mathbb{G}_m) = k^*$, so $H^1(X, \mu_\ell) \to H^1(X, \mathbb{G}_m)$ is injective.

It remains to show that

$$H^1(Y, \mathbb{Z}/\ell\mathbb{Z}) \longrightarrow H^1(X, \mathbb{Z}/\ell\mathbb{Z})$$

is injective. This follows from $\pi_1(X, \xi) \to \pi_1(Y, \xi)$ surjective; equivalently (SGA 1 V 6.9), every connected étale cover $\widetilde{Y}$ of Y with group $\mathbb{Z}/\ell\mathbb{Z}$ has connected restriction $\widetilde{X} = X \times_Y \widetilde{Y}$. This is a consequence of Bertini and Zariski connectedness (SGA 1 X 2.10).

Remark 3.12. Retain the hypotheses of 3.11.

(i) The kernel N is finite *unipotent* (SGA 3 XVII 1.1), since it contains no subgroup of multiplicative type μ_n, for any n (SGA 3 XVII 4.6.1 (vi)). This is proved similarly, using instead

of Kummer theory the following fact of Grothendieck (SGA 1 XI, last page): if X is proper over k, with $H^0(X, \mathcal{O}_X) = k$, then there is an isomorphism of k-group schemes

$$\underline{\mathrm{Hom}}_{\mathrm{grp}}(\mu_n, \underline{\mathrm{Pic}}_{X/k}) \simeq \underline{H^1(X, \mathbb{Z}/n\mathbb{Z})}.$$

Consequently, when k is algebraically closed, its k-points give

$$\mathrm{Hom}_{\mathrm{grp}}(\mu_n, \underline{\mathrm{Pic}}_{X/k}) \simeq H^1(X, \mathbb{Z}/n\mathbb{Z}).$$

(ii) If k has characteristic zero, then $N = 0$, since every finite unipotent group is zero (SGA 3 XVII 1.7). This approximately recovers Kodaira's vanishing theorem in Mumford's form [2].

Theorem 3.13. *Let S be noetherian and X a projective S-scheme with a relatively ample invertible sheaf $H = \mathcal{O}_X(1)$. Suppose the geometric fibers of X/S are integral of dimension r. For a family $\mathscr{L}$ of classes of invertible sheaves on the fibers, the following are equivalent:*

(i) *$\mathscr{L}$ is bounded.*

(ii) *In $\chi(L_K(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$, the coefficient a_{r-1} remains bounded and a_{r-2} remains bounded below.*

(iii) *The degree*

$$d(L_K) = \langle c_1(L_K)c_1(H_K)^{r-1}\rangle$$

remains bounded and $\langle c_1(L_K)^2 c_1(H_K)^{r-2}\rangle$ *remains bounded below.*

(iv) *There is an integer* N *such that*

$$H^{-N} < \mathscr{L} < H^N;$$

that is, $H_K^{\otimes N} \otimes L_K$ *and* $H_K^{\otimes N} \otimes L_K^{-1}$ *are ample for every* $L_K \in \mathscr{L}$.

Suppose $\mathscr{L}$ is bounded by L on some X_T. Then (iv) holds by EGA II 4.6.12. Taking L flat over T, the polynomial $\chi(L_t^{\otimes m}(n))$ depends only on the connected component of T (EGA III 7.9.4). Hence (ii) holds, and 2.1 gives (iii).

The families of classes of H_K and $\mathcal{O}_{X_K}$ are bounded, so they satisfy (iii) and (ii), respectively. If $\mathscr{L}$ satisfies (iv), then

$$\langle [Nc_1(H_K) + c_1(L_K)]c_1(H_K)^{r-1}\rangle > 0,$$

so $-Nd(H_K) < d(L_K) < Nd(H_K)$. Moreover,

$$\langle [Nc_1(H_K) + c_1(L_K)]^2 c_1(H_K)^{r-2}\rangle > 0,$$

whence

$$\langle c_1(L_K)^2 c_1(H_K)^{r-2}\rangle > -2Nd(L_K) - N^2 d(H_K).$$

Thus (iii) holds.

Finally, suppose (ii), respectively (iii), holds. By 2.10 and 2.3, replace H by a multiple and assume it very ample. Partition S so there is a linear section Y of X, integral over S, with two-dimensional fibers. By 3.8, replace $X, \mathscr{L}$ by $Y, \mathscr{L}|_Y$; 1.7, respectively the projection formula, preserves the relevant condition. Thus $r = 2$. Put

$$\delta = \sup\{d(L_K)\}.$$

If $\delta > 0$, replace $\mathscr{L}$ by the classes of $L_K(-\delta)$, using 2.3 and 2.10, so that $\delta \leq 0$.

If a_1 remains bounded and a_0 bounded below in $\sum_{i=0}^{2} a_i\binom{n+i}{i}$, all three coefficients remain bounded by 2.3 and 2.7. Thus $\mathscr{L}$ is bounded by 2.11, so (ii) implies

(i). On the other hand, (iii) implies (ii) by 2.3 and 2.8.

4. Finiteness theorems for $\mathrm{Pic}^{\tau}_{X/S}$

4.1. Let S be a scheme and G a commutative group functor over S. Denote by G^0, respectively G^τ, the subgroup functor whose T-points $f : T \to G$ satisfy: for every algebraically closed field K and K-point t of T, there are a connected K-scheme Z, two K-points z, z_0, and $g : Z \to G$, with $g(z) = f(t)$ and $g(z_0) = 0$ in $G(K)$; respectively, there is a nonzero n such that $nf(t) \in G^0(K)$.

Formation of G^0, respectively G^τ, commutes with base change, and $G \to G'$ induces $G^0 \to G'^0$, respectively $G^\tau \to G'^\tau$. If S is a field and G is represented by a group scheme locally of finite type, G^0 is represented by the connected component of the identity, an open and closed finite-type geometrically irreducible subscheme (SGA 3 VI$_A$ 2.4).

Lemma 4.2. *Let* $\Phi : G \to G'$ *be a morphism of commutative group functors over* S. *Suppose that for every algebraically closed* K *over* S, G_K, G'_K *are represented by group schemes locally of finite type and* Φ_K *by a finite-type (equivalently quasi-compact) morphism. Then*

$$\Phi^{-1}(G'^\tau) = G^\tau.$$

The inclusion $\Phi^{-1}(G'^{\tau}) \supset G^{\tau}$ has already been noted. Conversely, $g \in G(K)$ belongs to $G^{\tau}(K)$ if and only if the set of multiples ng, $n \in \mathbb{N}$, is quasi-compact. Since Φ_K is quasi-compact, it is enough that the set $\Phi(ng) = n\Phi(g)$ have this property. This proves the lemma.

Lemma 4.3. *Let X, Y be proper over S, and let $f : X \to Y$ be surjective (respectively, the inclusion of the zero scheme of a section φ of an ample $\mathcal{O}_Y(1)$). Then*

$$(f^*)^{-1}\underline{\mathrm{Pic}}^{\tau}_{X/S} = \underline{\mathrm{Pic}}^{\tau}_{Y/S}.$$

This follows formally from 4.2, XII 1.2, 3.1, and 3.5 (respectively, 3.8).

4.4. Let k be algebraically closed, X proper over k, and L invertible on X. The sheaf L is *algebraically equivalent* to zero if there are a connected finite-type k-scheme T, an invertible sheaf M on $X_T = X \times_k T$, and two k-points t, t_0, such that $L \simeq M_t$ and $\mathcal{O}_X \simeq M_{t_0}$. It is *$\tau$-equivalent* to zero if some nonzero tensor power is algebraically equivalent to zero. Equivalently, the point λ corresponding to L belongs to $\underline{\mathrm{Pic}}^{0}_{X/k}(k)$, respectively $\underline{\mathrm{Pic}}^{\tau}_{X/k}(k)$.

Recall that L is *numerically equivalent* to zero if

$$\langle c_1(L) \cdot Y \rangle = 0$$

for every closed integral curve $Y \subset X$ (see 2.1 or X 4.5).

Proposition 4.5 (Functorial behavior). *Let $f : X' \to X$ be a morphism between proper k-schemes, and let L be invertible on X. Then:*

(i) *If L is τ-equivalent to zero (respectively, numerically equivalent to zero), then so is f^*L.*

(ii) *Conversely, if f is surjective and f^*L is τ-equivalent to zero (respectively, numerically equivalent to zero), then so is L.*

The assertions for τ-equivalence follow from 4.1 and 4.3. Let Y' be a closed integral curve in X', and let $Y = f(Y')$, with its induced reduced structure. If $\dim(Y) = 1$, the projection formula (X 4.4.4) gives

$$\langle c_1(f^*L) \cdot Y' \rangle = [k(Y') : k(Y)]\, \langle c_1(L) \cdot Y \rangle. \tag{4.5.1}$$

If $\dim(Y) = 0$, then $(f^*L)_{Y'} \simeq \mathcal{O}_{Y'}$. This proves (i) for numerical equivalence.

Conversely, for a closed integral curve $Y \subset X$, choose a point $y' \in f^{-1}(y)$ algebraic over the function field $k(y)$ of the generic point y, and let Y' be its closure. Then $f(Y') = Y$, and 4.5.1 proves (ii).

Theorem 4.6 (The numerical characterization of $\mathrm{Pic}^{\tau}_{X/k}$). *Let k be algebraically closed, X proper over k, and L invertible on X. The following are equivalent:*

(i) *L is τ-equivalent to zero.*

(ii) *For every coherent F on X,*

$$\chi(F \otimes L) = \chi(F).$$

(iii) *L is numerically equivalent to zero.*

If X/k is projective with an ample $H = \mathcal{O}_X(1)$, these are also equivalent to:

(iv) *$L^{\otimes m}(1)$ is ample for every integer m.*

(v) *If X is irreducible, then*

$$\chi(L^{\otimes m}(n)) = \chi(\mathcal{O}_X(n))$$

for every m, n.

(vi) *If X is irreducible of dimension $r \geq 2$, then*

$$\langle c_1(L)c_1(H)^{r-1}\rangle = 0, \qquad \langle c_1(L)^2 c_1(H)^{r-2}\rangle = 0.$$

For (i) $\Rightarrow$ (ii), devissage (EGA III 3.1) and 4.5 reduce to X integral and $F = \mathcal{O}_X$. It remains to show that $\chi(L^{\otimes m})$ is constant. Replace L by a nonzero tensor power and suppose it algebraically equivalent to zero. If M on $X \times_k T$, with T connected of finite type, deforms L to $\mathcal{O}_X$, then $\chi(M_t^{\otimes m})$ is independent of $t \in T$ (EGA III 7.9.5).

For a closed integral curve Y, take $F = \mathcal{O}_Y$ in (ii). The Riemann formula

$$\chi(L_Y) = \langle c_1(L) \cdot Y\rangle + \chi(\mathcal{O}_Y)$$

shows that (ii) implies (iii).

Let $f : X' \to X$ be a Chow presentation. If (iii) implies (i) for X', then 4.5 gives it for X. Hence assume X projective with an ample H. Passing through the finite surjection $\coprod_i X_i \to X$, where the X_i are the reduced irreducible components, reduces (iii) $\Rightarrow$ (i),(iv) to X integral (4.5, EGA III 2.6.2).

For irreducible X, (ii) implies (v), (iii) implies (vi), and 2.1 gives (v) implies (vi); condition (vi) is unchanged on X_{red}. It remains to prove (vi) implies (i),(iv), and (iv) implies (i), for X integral. Under (vi), respectively (iv), the family of classes $L^{\otimes m}$, $m \in \mathbb{N}$, satisfies 3.13 (iii), respectively (iv), and is bounded. Hence some invertible $\mathcal{M}$ on X_T, with T of finite type over k, bounds it. Choose distinct p, q such that $L^{\otimes p}$ and $L^{\otimes q}$ correspond to the same connected component of T. Then $L^{\otimes(p-q)}$ is algebraically equivalent to zero, proving (i).

Condition (vi) also gives an N such that $L^{\otimes m} \otimes H^{\otimes N}$ is ample for every m. Therefore $(L^{\otimes m} \otimes H)^{\otimes N}$ is ample, so $L^{\otimes m} \otimes H$ is ample. Thus (vi) implies (iv).

Theorem 4.7. *Let S be noetherian and X proper over S. Then:*

(i) $\underline{\mathrm{Pic}}^{\tau}_{X/S} \to \underline{\mathrm{Pic}}_{X/S}$ *is representable by an open immersion.*

(ii) *If X is projective and integral over S, this map*
is also representable by a closed immersion.

(iii) $\underline{\mathrm{Pic}}^{\tau}_{X/S}$ *is of finite type over S; that is, the family of classes τ-equivalent to zero on the fibers is bounded.*

For (i), respectively (ii), take an arbitrary $T \to \underline{\mathrm{Pic}}_{X/S}$ and show that the inverse image of $\underline{\mathrm{Pic}}^{\tau}_{X/S}$ is an open, respectively open and closed, subscheme of T. Locally, $T = \mathrm{Spec}(B)$ over $\mathrm{Spec}(A) \subset S$, and T is the limit of $T_\alpha = \mathrm{Spec}(B_\alpha)$, for the finite-type A-subalgebras $B_\alpha \subset B$. By 3.1 the morphism descends to some T_α, so assume T of finite type over S.

The morphism corresponds to an fppf $R \to T$ and an invertible sheaf L on X_R. By fppf descent replace T by R, then base-change along $R \to S$. Compatibility with base change (4.1) reduces to a section $S \to \underline{\mathrm{Pic}}_{X/S}$ defined by L on X.

Let S_0 be the set of $s \in S$ such that L_s is τ-equivalent to zero. It must be open, and under (ii) also closed. Since S is noetherian, for $s \in S_0$ it suffices that

(a) every generalization of s lies in S_0;

(b) every specialization sufficiently near s lies in S_0;

(c) under (ii), every specialization of s lies in S_0.

Let $f : X' \to X$ be a Chow presentation. If (a),(b),(c), and (iii) hold for X', then 4.3 and 3.5 give them for X; assume X projective over S.

Let H be ample for X/S. If $(L^{\otimes m} \otimes H)_s$ is ample, then it remains ample at every generalization of s (EGA III 4.7.1). The equivalence 4.6 (i) $\iff$ (iv) gives (a).

For (b),(c), and (iii), reduce to S integral, with s generic in (b),(c); for (iii), apply 3.4 (i) to $\coprod_i S_i \to S$, where S_i are the reduced irreducible components.

There are an integral S' and an fppf $S' \to S$ such that every reduced irreducible component X'_i of $X_{S'}$ is integral over S' (XII 1.3). It is enough to prove (b),(iii) for $X_{S'}/S'$, by descent and by noetherian induction with 3.4 (i). The surjection $\coprod_i X'_i \to X_{S'}$, together with 4.3, 3.5, and 3.7, reduces (b),(iii) to (c),(iii) for each X'_i/S'.

Thus we are reduced to proving (c) and (iii), assuming that S is integral with generic point s, and that X is projective and integral over S. For a very ample $\mathcal{O}_X(1)$ for X/S, the values $\chi(L_t^{\otimes m}(n))$ and $\chi(\mathcal{O}_{X_t}(n))$, for every m, n, are independent of $t \in S$ (EGA III 7.9.4). Hence (c) follows from 4.6 (i) $\iff$ (v), which also says that the family of classes of invertible sheaves τ-equivalent to zero on the fibers of X/S has only one

Hilbert polynomial. Thus (iii) follows from 2.11.

Remark 4.8. Let X be proper and integral over a noetherian scheme S. Without the projectivity hypothesis on X/S, it is unknown whether $\underline{\mathrm{Pic}}^{\tau}_{X/S} \to \underline{\mathrm{Pic}}_{X/S}$ remains a *closed* immersion, except when X is normal over S and $\underline{\mathrm{Pic}}_{X/S}$ is *representable* (FGA 236, 2.3).

5. Finiteness theorems of "Néron–Séveri" type

Theorem 5.1. *Let X be proper over noetherian S. Then the Néron–Séveri groups $\underline{\mathrm{Pic}}_{X_K/K}(K)/\underline{\mathrm{Pic}}^0_{X_K/K}(K)$ of the geometric fibers X_K/K of X/S are finitely generated, and their ranks and the orders of their torsion subgroups $\underline{\mathrm{Pic}}^{\tau}_{X_K/K}(K)/\underline{\mathrm{Pic}}^0_{X_K/K}(K)$ remain bounded.*

By noetherian induction, replace S by an arbitrary nonempty open of S_{red}, and suppose $\underline{\mathrm{Pic}}^{\tau}_{X/S}$ represented by a finite-type S-scheme (XII 1.2, 4.7 (iii)). The order of $\underline{\mathrm{Pic}}^{\tau}_{X_K/K}(K)/\underline{\mathrm{Pic}}^0_{X_K/K}(K)$ is the number of connected components of $(\underline{\mathrm{Pic}}^{\tau}_{X/S})_K$, constant over a nonempty open of S (EGA IV 9.7.9).

It remains to bound the rank of $\underline{\mathrm{Pic}}_{X_K/K}(K)/\underline{\mathrm{Pic}}^{\tau}_{X_K/K}(K)$. Reduce as in 4.7 (iii) to S integral and X projective and integral. Let r be the common fiber dimension of X/S. For $r = 0$, respectively 1, the rank is 0, respectively 1. For $r \geq 2$, reduce to $r = 2$ and X smooth. After restricting S, take a linear section Y of X, still

integral over S, with two-dimensional fibers, and replace X by Y using 4.3. Then use 4.3 and 3.10, together with resolution of surface singularities, and replace S by a flat cover of a nonempty open, to make X smooth over S.

By 4.6 (i) $\iff$ (iii) and finiteness and specialization for étale cohomology (SGA 4 XVI 2.2), Theorem 5.1 follows from:

Proposition 5.2. *Let k be algebraically closed and X proper smooth over k. The group $C^p_{\mathrm{num}}(X)$ of codimension-p algebraic cycles modulo numerical equivalence is free of rank at most b^{2p}, the $2p$-th Betti number.*

Disregarding the action of roots of unity, étale cohomology $H^*(X, \mathbb{Q}_\ell) =_{\mathrm{dfn}} \left[\varprojlim_\nu H^*(X, \mathbb{Z}/\ell^\nu\mathbb{Z})\right] \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, where $(\ell, \mathrm{char}(k)) = 1$, is a Weil cohomology: it satisfies Poincaré duality and the Künneth formula, and receives a functorial, nonzero homomorphism from the ring of algebraic cycles modulo rational equivalence $C^*_{\mathrm{rat}}(X)$ (see X 7.13).

Choose $a_1, \dots, a_m \in C^{r-p}_{\mathrm{rat}}(X)$, $r = \dim X$, whose images form a basis for the $\mathbb{Q}_\ell$-subspace of $H^{2(r-p)}(X, \mathbb{Q}_\ell)$ spanned by algebraic cycles. Define

$$\alpha : C^p_{\mathrm{rat}}(X) \longrightarrow \mathbb{Z}^m.$$

It is defined by $\alpha(b) = (\langle b \cdot a_1\rangle, \dots, \langle b, a_m\rangle)$, where $\langle\ \rangle$ denotes the intersection number. Its image identifies with $C^p_{\mathrm{num}}(X)$.

Remark 5.3. If X is a proper smooth surface over algebraically closed k, then its second Betti number is $b^2(X) = c_2(X) - 2 + 2b^1(X)$, where $b^1(X) = 2\dim \mathrm{Pic}^0_{X/k}$, and $c_2(X)$, the second Chern class, is the Euler-Poincaré characteristic, equivalently the self-intersection of the diagonal in $X \times X$. Thus $b^2(X)$ has a cohomology-independent interpretation. In characteristic > 0, the inequality $\mathrm{rank}\, C^1_{\mathrm{num}}(X) \le b^2(X)$ in this form is due, before étale cohomology, to Igusa, who used deep results on the tame fundamental group of an algebraic curve minus some points.

Theorem 5.4. *Let k be algebraically closed and X proper over k. There are finitely many closed integral curves $Y_1, \dots, Y_p$ in X such that a subset Λ of $\mathrm{Pic}_{X/k}$ is quasi-compact if and only if the intersection numbers $\langle\lambda \cdot Y_i\rangle$, for $\lambda \in \Lambda$, remain bounded. Moreover, p may be taken to be the rank of the Néron–Séveri group $\mathrm{Pic}_{X/k}(k)/\mathrm{Pic}^0_{X/k}(k)$.*

Indeed, $\mathrm{Pic}^\tau_{X/k}$ is an open quasi-compact subgroup of $\mathrm{Pic}_{X/k}$ (4.7). Thus Λ is quasi-compact if and only if the corresponding set of points in $\left[\mathrm{Pic}_{X/k}(k)/\mathrm{Pic}^\tau_{X/k}(k)\right] \otimes_{\mathbb{Z}} \mathbb{Q}$ is finite. By 4.6 (i) $\Longleftrightarrow$ (iii), τ-equivalence equals numerical equivalence. Choose curves whose images form a basis of the dual vector space; the assertion follows.

6. Appendix: Study of (b)-sheaves on $P = \mathbf{P}^N_k$.

6.1. Let k be an algebraically closed field and let $P = \mathbf{P}^N_k$. Let F be a coherent sheaf on P, with Hilbert polynomial

$$\chi(F(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i},$$

and let

$$(c) = (c_0, \dots, c_r)$$

be a sequence of integers with $(c) \ge (a)$. Put

$$a_{m,i} = \sum_{j=0}^{r-i} a_{j+i} \binom{m-1+j}{j} \quad \text{and} \quad c_{m,i} = \sum_{j=0}^{r-i} c_{j+i} \binom{m-1+j}{j}.$$

Finally, for every integer $q \ge 1$, denote by N_q the largest coherent subsheaf of F whose support has dimension at most $q-1$, and put $F_q = F/N_q$.

Lemma 6.2. *Under the conditions of 6.1, consider an exact sequence*

$$0 \longrightarrow I \longrightarrow \mathcal{O}_P^{\oplus M} \longrightarrow F(m) \longrightarrow 0$$

with $m \ge 0$. Then I is p-regular, where $p = P_r(c_{m,0}, \dots, c_{m,r})$, and P_r is the r-th (b)-polynomial of 1.9.

Indeed, $\mathcal{O}_P$ is plainly a $(0, \dots, 0, 1)$-sheaf ($N+1$ terms), so I is a $(0, \dots, 0, M)$-sheaf by 1.6. By 2.10, $\chi(I(n)) = M\binom{n+N}{N} - \sum_{i=0}^{r} a_{m,i}\binom{n+i}{i}$. The assertion follows from 1.11 and 1.10.

Proposition 6.3. *Under the conditions of 6.1, suppose that F is a (b)-sheaf. Then, by means of universal polynomials in the a_i, the b_i, and N, one can calculate integers m, M, m_1, M_1 for which there is an exact sequence*

$$\mathcal{O}_P(-m_1)^{\oplus M_1} \longrightarrow \mathcal{O}_P(-m)^{\oplus M} \longrightarrow F \longrightarrow 0.$$

Moreover, N enters only into the calculation of M_1.

Indeed, take $m = P_{r-1}(c_0, \ldots, c_{r-1})$, with $c_i = b_i - a_i$, and $M = \sum_i a_i \binom{m+i}{i}$, so that, by 1.11 and 1.3, there is an exact sequence

$$0 \longrightarrow I \longrightarrow \mathcal{O}_P^{\oplus M} \longrightarrow F(m) \longrightarrow 0.$$

Likewise, taking account of 6.2, take $m_1 = m + p$, where $p = P_r(a_{m,0}, \ldots, a_{m,r})$, and $M_1 = M\binom{p+N}{N} - \sum_i a_i \binom{m_1+i}{i} \quad (= \chi(I(p)))$.

Theorem 6.4. *Under the conditions of 6.1, suppose that F is a quotient of $\mathcal{O}_P(-m)^{\oplus M}$, with $m \geq 0$. Put $b_i = P_{r-i}(c_{m,i}, \ldots, c_{m,r})$ for $i = 0, \ldots, r$, where P_s is the s-th (b)-polynomial of 1.9; put $b = b_1 + m - 1$, and put $B = \sum_{i=1}^{r} a_i \binom{b+i}{i}$. Then:*

(i) *F is b-regular.*

(ii) $-B \leq a_0 = h^0(F(b)) - B$

(iii) *F is a $(b_0, \ldots, b_r)$-sheaf.*

Indeed, choose a section σ of $\mathcal{O}_P(1)$ for which one obtains the following commutative diagram with exact rows and columns:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & I(-1) \arrow[r] \arrow[d] & I \arrow[r] \arrow[d] & J \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{O}_P(-1)^{\oplus M} \arrow[r] \arrow[d] & \mathcal{O}_P^{\oplus M} \arrow[r] \arrow[d] & \mathcal{O}_Y^{\oplus M} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & F(m-1) \arrow[r] \arrow[d] & F(m) \arrow[r] \arrow[d] & G(m) \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Since $\chi(G(n)) = \sum_{i=0}^{r-1} a_{i+1}\binom{n+i}{i}$ by 1.7, J is b_1-regular by 6.2; hence, by 1.4(i), $H^q(I(p)) = 0$ for $q \geq 2$ and $p \geq b_1 - q$. The exact sequence $H^q(\mathcal{O}_P^{\oplus M}(p)) \longrightarrow H^q(F(m+p)) \longrightarrow H^{q+1}(I(p))$ then shows that $F(m)$ is $(b_1 - 1)$-regular; this proves (i).

Assertion (i) implies $0 \leq h^0(F(b)) = \chi(F(b)) = \sum_{i=0}^{r} a_i \binom{b+i}{i}$, which proves (ii). By 2.10 and 1.9,

$$\chi(F(b)) = \sum_{i=0}^{r} a_{m,i} \binom{b_1 - 1 + i}{i} \leq b_1 + \sum_{i=0}^{r} c_{m,i} \binom{b_1 - 1 + i}{i} = b_0.$$

Thus $h^0(F(-1)) \leq h^0(F(b)) \leq b_0$. By induction on r, one may suppose that G is a $(b_1, \ldots, b_r)$-sheaf, so (iii) follows from 1.6.2.

Lemma 6.5. *Under the conditions of 6.1, suppose that F has no associated prime cycle of dimension* 0.

(i) *If $h^0(F) \geq 1$, then $h^0(F(-1)) \leq h^0(F) - 1$.*

(ii) *$H^0(F(-n)) = 0$ for $n \geq h^0(F)$.*

(iii) *Let $0 \to F(-1) \to F \to G \to 0$ be exact, and let $n_0 \geq 0$ satisfy $H^0(G(-n_0)) = 0$. Then $H^0(F(-n_0)) = 0$ and $h^0(F) \leq n_0 \cdot h^0(G)$.*

For (i), let $0 \neq \sigma \in H^0(F)$. The section σ defines a nonzero subsheaf $F' = \mathcal{O}_X \cdot \sigma$ of F, so $s = \dim \operatorname{Supp}(F') \geq 1$. Consider the commutative diagram with exact rows and columns

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & F'(-1) \arrow[r] \arrow[d] & F' \arrow[r] \arrow[d] & G' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & F(-1) \arrow[r] & F \arrow[r] & G \arrow[r] & 0.
\end{tikzcd}$$

Since $s \geq 1$, one has $G' \neq 0$; hence F' is not contained in $F(-1)$. In other words, the section σ is not in the image of the injection $H^0(F(-1)) \to H^0(F)$, which proves (i).

Assertion (ii) follows immediately from (i), bearing in mind that $h^0(F) = 0$ plainly implies $h^0(F(-1)) = 0$, since $h^0(F(-1)) \leq h^0(F)$. In (iii) one obtains $0 \leq h^0(F(n)) - h^0(F(n-1)) \leq h^0(G(n))$. Thus, for $n \geq n_0$, $h^0(F(-n_0)) = h^0(F(-n))$, since $h^0(G(-n_0)) = 0$; hence $h^0(F(-n_0)) = 0$ by (ii). Consequently, for $p \geq 1$, $h^0(F(p-n_0)) \leq h^0(G(1-n_0)) + \cdots + h^0(G(p-n_0)) \leq p \cdot h^0(G(p-n_0))$, which gives (iii).

Proposition 6.6. *Under the conditions of 6.1, suppose that F is a $(b_0, \ldots, b_r)$-sheaf. Then F_q is a $\big((b_{q-1})^q, \ldots, (b_{q-1})^2, b_{q-1}, b_q, \ldots, b_r\big)$-sheaf.*

Indeed, let σ be a "general" section of $\mathcal{O}_X(1)$. It defines a commutative diagram with exact rows and columns:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & N_q(-1) \arrow[r] \arrow[d] & N_q \arrow[r] \arrow[d] & G' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & F(-1) \arrow[r] \arrow[d] & F \arrow[r] \arrow[d] & G \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & F_q(-1) \arrow[r] \arrow[d] & F_q \arrow[r] \arrow[d] & G'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

where G is a $(b_1, \ldots, b_r)$-sheaf by 1.6(i), and where $G'' = G_{q-1}$ (a depth question).

Suppose first that $q = 1$. Then $\dim \operatorname{Supp}(N_q) = 0$, so $H^1(N_q(-1)) = 0$ and $h^0(F_q(-1)) \leq h^0(F(-1)) \leq b_0$. Moreover, $G' = 0$, and hence $G \simeq G''$. Thus F_1 is indeed a $(b_0, \ldots, b_r)$-sheaf.

In addition, 6.5(ii) shows that $H^0(F_1(-b_0)) = 0$. If $q \geq 2$, argue by induction on q. One may suppose that G_{q-1} is a $((b_{q-1})^{q-1}, \ldots, b_{q-1}, \ldots, b_r)$-sheaf and that $H^0(G_{q-1}(-b_{q-1})) = 0$. Lemma 6.5 then gives $H^0(F_q(-b_{q-1})) = 0$ and $h^0(F_q(-1)) \leq (b_{q-1})^q$, which proves the assertion.

Theorem 6.7. *There are two sequences $\{A_i(X_0, \ldots, X_i; Y)\}$ and $\{A_i^{(q)}(X_0, \ldots, X_q; Y)\}$ of polynomials having the following properties. Assume the conditions of 6.1 and suppose that F is a quotient of $\mathcal{O}_P(-m)^{\oplus M}$, with $m \geq 0$.*

(i) *If F is a $(b_0, \ldots, b_r)$-sheaf, then, for $i = 0, \ldots, r$,*

$$|a_i| \leq A_{r-i}(b_i, \ldots, b_r; m).$$

(ii) *If $\chi(F_q(n)) = \sum_{i=0}^r a_i^{(q)} \binom{n+i}{i}$ is the Hilbert polynomial of F_q, then, for $i = 0, \ldots, q-1$,*

$$|a_i^{(q)}| \leq A_{r-i}^{(r-q)}(c_{q-1}, \ldots, c_r; m).$$

(Notice that $a_{q-1}^{(q)} \leq a_{q-1}$ and $a_q^{(q)} = a_q, \ldots, a_r^{(q)} = a_r$, since $\dim \operatorname{Supp}(N_q) \leq q-1$.*)*

To prove (i), use induction on r and apply 6.4 to an exact sequence $0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$ in which G is a $(b_1, \ldots, b_r)$-sheaf. By induction on r, one may suppose that $A_0, \ldots, A_{r-1}$ have been defined, taking 1.7 into account. Then 6.4(ii), together with 1.8, displays the definition of A_r. Finally, (ii) follows formally from 6.4(iii), 6.6, and assertion (i), applied to F_q in place of F.

Corollary 6.8. (Grothendieck). *Let X be projective over a noetherian scheme S, let $\mathcal{O}_X(1)$ be very ample for X/S, and let $\mathcal{J}$ be a family of classes of coherent sheaves on the fibers of X/S. Suppose that:*

(a) *There is a coherent sheaf E on X such that $\mathcal{J}$ is contained in the family of classes of quotients of sheaves of the form E_K.*

(b_q) *In the Hilbert polynomials $\chi(F_K(n))$, for $F_K \in \mathcal{J}$, the coefficients in degrees at least $q-1$ remain bounded above.*

Then the $(F_K)_q$ (as F_K ranges over $\mathcal{J}$) form a bounded family; moreover, the coefficients of $\chi(F_K(n))$ in degrees at least $q-2$ remain bounded below.

Indeed, one may plainly suppose that $X = \mathbf{P}_S^N$ and $E = \mathcal{O}_X(-m)^{\oplus M}$. The first assertion follows formally from 6.7(ii) and 1.13. The second follows by induction, using an exact sequence $0 \longrightarrow F(-1) \longrightarrow F \longrightarrow G \longrightarrow 0$ and 6.4(ii).

Definition 6.9. *Let k be a field. We call a* special positive k-cycle of dimension r *any projective k-scheme X, equipped with a very ample sheaf $\mathcal{O}_X(1)$, which is a union of closed subschemes X_j of dimension r, each X_j being obtained by base change* $\operatorname{Spec}(k) \to \operatorname{Spec}(k_j)$ *as the inverse image of an integral k_j-scheme X'_j (equipped with $\mathcal{O}_{X'_j}(1)$). We call the* degree *of X the coefficient a_r in the Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_i a_i \binom{n+i}{i}$.*

Lemma 6.10. *Let k be an algebraically closed field, and let X be a special positive k-cycle of dimension r and degree d. Then $\mathcal{O}_X$ is a $(0, \ldots, 0, d)$-sheaf.*

Indeed, view X as a subscheme of $\mathbf{P}_k^N$. After replacing k by the algebraic closure of a purely transcendental extension, which is permissible by 1.6(i), one may suppose that k contains $r(N+1)$ elements algebraically independent over all the k_j. The r successive "general" hyperplane sections of X defined by these elements are then always special positive k-cycles of degree d.

Arguing by induction on r, it remains only to observe that, when $r = 0$, one has $h^0(\mathcal{O}_X(-1)) = d$, while, when $r \geq 1$, one has $h^0(\mathcal{O}_X(-1)) = 0$. For the latter point, the injection $\mathcal{O}_X \longrightarrow \coprod_j \mathcal{O}_{X_j}$ allows one to suppose that X itself comes from an integral scheme X'. But then $h^0(\mathcal{O}_X(-1)) = h^0(\mathcal{O}_{X'}(-1)) = 0$, because $H^0(\mathcal{O}_{X'})$ is a field.

Corollary 6.11. (i) *Let r, d be two integers, put $c_i = A_{r-i}(0, \ldots, 0, d; 0)$ for $i = 0, \ldots, r$ (6.7(i)), and put $p = P_r(c_0, \ldots, c_r)$ (1.9). Let k be an algebraically closed field, and let X be a special positive k-cycle of dimension at most r and degree at most d, with Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$. Then, for $i = 0, \ldots, r$,*

$$|a_i| \leq c_i.$$

Moreover, X embeds as a subscheme of $\mathbf{P}_k^N$, with $N = d(r+1) - 1$, defined by (at most) $\binom{N+p}{N}$ equations of degree p.

(ii) (Chow) *Let S be a noetherian scheme. As K ranges over the algebraically closed extensions of the $k(s)$, for $s \in S$, the special positive K-cycles of bounded dimension and degree form a "bounded family" (either with the X regarded as subschemes of a fixed $\mathbf{P}_S^N$, or in an evident abstract sense).*

Indeed, in (i), the first assertion follows from 6.7(i); the second follows from 6.10 and 1.8(i), and from 6.2, applied to $0 \longrightarrow I \longrightarrow \mathcal{O}_P \longrightarrow \mathcal{O}_X \longrightarrow 0$, $P = \mathbf{P}_k^N$, together with 1.3. Finally, (ii) follows from (i) and 1.13.

Remark 6.12. Let k be an algebraically closed field and let X be a projective k-scheme, purely of dimension r, with no embedded associated prime cycle. If arbitrary nilpotent elements are allowed in $\mathcal{O}_X$, the coefficients in the Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_{i=0}^{r} a_i \binom{n+i}{i}$ can no longer be bounded in terms of a_r alone.

For example, let Z be a smooth curve of degree d in $\mathbf{P}_k^3$. For every $n \gg 0$, there is a smooth surface Y of degree n containing Z. Then Z is a Cartier divisor on Y; put $\mathcal{O}_{X_n} = \mathcal{O}_Y / \mathcal{O}_Y(-2Z)$. There is an exact sequence $0 \longrightarrow \mathcal{O}_{X_n}(-Z) \longrightarrow \mathcal{O}_{X_n} \longrightarrow \mathcal{O}_Z \longrightarrow 0$ and the formula $\langle Z^2 \rangle_Y = 2p_a(Z) - 2 - (n-4)d$. It follows that $a_0 = \chi(\mathcal{O}_{X_n}(-1)) \longrightarrow \infty$ as $n \longrightarrow \infty$, although $a_1 = 2d$.

Corollary 6.13. *Under the conditions of 6.1, consider an exact sequence*

$$\mathcal{O}_P(-m_1)^{\oplus M_1} \longrightarrow \mathcal{O}_P(-m)^{\oplus M} \xrightarrow{\alpha} F \longrightarrow 0$$

with $m_1 \geq m \geq 0$. Then, for $i = 0, \ldots, N$,

$$|a_i| \leq A_{N-i}(0, \ldots, 0, M; m_1) + M \binom{m}{N-i}.$$

Indeed, let $I = \operatorname{Ker}(\alpha)$. Plainly, I is a $(0, \ldots, 0, M)$-sheaf, and $\chi(I(n)) = \sum_{i=0}^{N} \left[M \binom{-m-1+N-i}{N-i} - a_i \right] \binom{n+i}{i}$. The assertion follows from 6.7(i) and the formula $\binom{-m-1+Q}{Q} = (-1)^Q \binom{m}{Q}$.

Corollary 6.14. (I. Hermann). *For every $N \geq 0$, there is a polynomial $R_N(m)$ such that, for every field k_0 and every ideal I of the polynomial ring $k_0[T_1, \ldots, T_N]$ generated by elements of degree at most m, the radical $\sqrt{I}$ is generated by elements of degree at most $R_N(m)$.*

Indeed, let k be the algebraic closure of k_0. After introducing an auxiliary indeterminate, let Y (respectively X) be the closed subscheme of $P = \mathbf{P}_k^N$ defined by the homogenized ideal of I (respectively of $\sqrt{I}$). In the Hilbert polynomial $\chi(\mathcal{O}_Y(n)) = \sum_{i=0}^{N} b_i \binom{n+i}{i}$, one has $|b_i| \leq c_i$, where $c_i = A_{N-i}(0, \ldots, 0, 1; m)$, by 6.13.

Let $X = \bigcup X^q$ be the decomposition by dimension. Then $\deg(X^q) \leq e_q$, where $e_q = P_{N-q}(c_q, \ldots, c_N)$. Indeed, after cutting X (respectively Y) by a "general" linear subspace of codimension q, one may suppose that $q = 0$. Then $\deg(X^q) = h^0(\mathcal{O}_{X^q})$ and $h^0(\mathcal{O}_{X^q}) \leq h^0(\mathcal{O}_Y)$, and 6.4(iii) gives $h^0(\mathcal{O}_Y) \leq e_q$.

Consequently, $\mathcal{O}_{X^q}$ is a $(0, \ldots, 0, e_q)$-sheaf by 6.10. It follows from the injection $\mathcal{O}_X \longrightarrow \prod_q \mathcal{O}_{X^q}$ that $\mathcal{O}_X$ is an $(e_0, \ldots, e_N)$-sheaf. Hence, in the Hilbert polynomial $\chi(\mathcal{O}_X(n)) = \sum_{i=0}^N a_i \binom{n+i}{i}$, one has $|a_i| \leq f_i$, where $f_i = A_{N-i}(e_i, \ldots, e_N; 0)$, by 6.7. Finally, the ideal $\sqrt{I}$ of X is $R_N(m)$-regular, where $R_N(m) = P_N(f_0, \ldots, f_N)$, and the assertion follows from 1.3.

7. Appendix: The Hodge index theorem

Here we recover and complete some earlier results by a direct argument.

Theorem 7.1. *Let X be an irreducible surface, proper over an algebraically closed field. Suppose that there is an invertible sheaf H on X such that $\langle c_1(H)^2\rangle > 0$ (for example, H ample). Then an invertible sheaf L on X satisfying*

$$\langle c_1(L) \cdot c_1(H)\rangle = 0, \qquad \langle c_1(L)^2\rangle \geq 0,$$

is numerically equivalent to 0. *Consequently, by 4.6, the intersection form on $A^1_{\mathrm{num}}(X)_{\mathbb{Q}} = [\operatorname{Pic}(X)/\operatorname{Pic}^\tau(X)] \otimes \mathbb{Q}$ is nondegenerate of type* $(1, \rho - 1)$.

7.1.1. First suppose that X is reduced (hence integral), that H is ample, that $\langle c_1(L) \cdot c_1(H)\rangle = 0$, and that $\langle c_1(L)^2\rangle > 0$. Arguing by contradiction, choose $m > 0$ such that $H_1 = L \otimes H^{\otimes m}$ is ample, and choose n such that, on putting $N = L^{\otimes n} \otimes H^{-1}$, one has $\langle c_1(N) \cdot c_1(H_1)\rangle = n\langle c_1(L)^2\rangle - m\langle c_1(H)^2\rangle > 0$. One also has $\langle c_1(N) \cdot c_1(H)\rangle = -\langle c_1(H)^2\rangle < 0$ and $\langle c_1(N)^2\rangle = n^2\langle c_1(L)^2\rangle + \langle c_1(H)^2\rangle > 0$, contradicting the following lemma.

Lemma 7.1.2. *Let X be an integral surface, projective over an algebraically closed field. For an invertible sheaf N on X such that $\langle c_1(N)^2\rangle > 0$, the following conditions are equivalent:*

(i) for every ample sheaf H, one has $\langle c_1(N) \cdot c_1(H)\rangle > 0$;

(i′) there is an ample sheaf H such that $\langle c_1(N) \cdot c_1(H)\rangle > 0$;

(ii) there is an integer $n > 0$ such that $H^0(X, N^{\otimes n}) \neq 0$.

Indeed, (ii) implies that there is an effective divisor D such that $N^{\otimes n} \simeq \mathcal{O}_X(D)$, and $\langle c_1(N)^2\rangle > 0$ implies $D > 0$; hence (i).

To see that (i′) implies (ii), one may plainly suppose that $H = \mathcal{O}_X(1)$ is very ample. There is then a closed integral locally principal curve Y such that $H = \mathcal{O}_X(Y)$. By the Riemann theorem, (i′) implies $H^1(N_Y^{\otimes n}(p)) = 0$ for $p \geq 0$ and $n \geq n_0$, where $n_0 = 2h^1(\mathcal{O}_Y) - 1$. Since $H^2(N^{\otimes n}(p)) = 0$ for $p \gg 0$, the exact sequence

$$H^1(N_Y^{\otimes n}(p)) \longrightarrow H^2(N^{\otimes n}(p-1)) \longrightarrow H^2(N^{\otimes n}(p)) \longrightarrow 0$$

gives $H^2(N^{\otimes n}) = 0$ for $n \geq n_0$. Thus, for $n \gg 0$,

$$h^0(N^{\otimes n}) = \chi(N^{\otimes n}) + h^1(N^{\otimes n}) \geq \chi(N^{\otimes n}) = \frac{1}{2}\langle c_1(N)^2\rangle n^2 + \cdots > 0,$$

which completes the proof of the lemma.

7.1.3. Now suppose in 7.1 that there is an invertible sheaf M on X such that $\langle c_1(L) \cdot c_1(M)\rangle \neq 0$, and argue by contradiction. Replace X by the underlying reduced scheme of a Chow presentation of X, which is permissible by the projection formula X 4.4.4, and let G be ample on X. Choose $m > 0$ such that $H_1 = G^{\otimes m} \otimes M$ is ample and $\langle c_1(L) \cdot c_1(H_1)\rangle \neq 0$. Choose p, q, with $q \neq 0$, such that, on putting $L_1 = L^{\otimes p} \otimes H^{\otimes q}$, one has $\langle c_1(L_1) \cdot c_1(H_1)\rangle = p\langle c_1(L) \cdot c_1(H_1)\rangle + q\langle c_1(H) \cdot c_1(H_1)\rangle = 0$. Then one also has $\langle c_1(L_1)^2\rangle = p^2\langle c_1(L)^2\rangle + q^2\langle c_1(H)^2\rangle > 0$. This is the case 7.1.1, with $L = L_1$ and $H = H_1$.

7.1.4. Finally, consider the general case of 7.1. Still arguing by contradiction, suppose that L is not numerically equivalent to 0. Replace X by the normalization of X_{red}, so that X has

only finitely many singular points; this reduction is justified by the projection formula X 4.4.4 and by 4.5(ii). There is therefore a closed integral curve Y on X such that $\langle c_1(L) \cdot Y\rangle \neq 0$, and Y is locally principal outside a finite set Z. Let $f : X' \to X$ be the blow-up of the ideal $I \subset \mathcal{O}_X$ defining Y. The ideal $\mathcal{M} = I\mathcal{O}_{X'}$ is invertible and defines a subscheme of X' which is the union of two subschemes Y' and Y'' such that $f(Y') = Y$ and $f(Y'') \subset Z$. By the projection formula X 4.4.4, $\langle c_1(f^*L) \cdot c_1(\mathcal{M})\rangle = -\langle c_1(f^*L) \cdot Y'\rangle - \langle c_1(f^*L) \cdot Y''\rangle = -\langle c_1(L) \cdot Y\rangle - 0 \neq 0$. Thus one may replace X, L, H by X', f^*L, f^*H. This reduces the problem to the preceding case and completes the proof of 7.1.

Remark 7.2. It is not known whether an irreducible proper surface over an algebraically closed field which possesses an invertible sheaf H satisfying $\langle c_1(H)^2\rangle > 0$ is necessarily projective. Nor is it known whether, for every irreducible proper surface, the intersection form on $A^1_{\mathrm{num}}(X)_{\mathbb{Q}}$ is nondegenerate. It follows from 7.1 that, when it is degenerate, it is necessarily negative (non-definite).

Corollary 7.3. *Under the conditions of 7.1, let M be an invertible sheaf on X which is not numerically equivalent to H, and put $d = \langle c_1(M) \cdot c_1(H)\rangle$ and $h = \langle c_1(H)^2\rangle$. Then $\langle c_1(M)^2\rangle < d^2/h$.*

Indeed, let $L = M^{\otimes h} \otimes H^{\otimes(-d)}$. Since $\langle c_1(L) \cdot c_1(H)\rangle = 0$, Theorem 7.1 gives $\langle c_1(L)^2\rangle < 0$, which proves the assertion.

Corollary 7.4. *Let X be an irreducible scheme of dimension $r \geq 2$, projective over an algebraically closed field, and let H be an ample sheaf on X. Then:*

(i) *An invertible sheaf L on X is numerically equivalent to 0 if and only if*

$$\langle c_1(L) \cdot c_1(H)^{r-1}\rangle = 0, \qquad \langle c_1(L)^2 \cdot c_1(H)^{r-2}\rangle \geq 0;$$

(ii) *For every invertible sheaf M on X which is not numerically equivalent to H,*

$$\langle c_1(M)^2 \cdot c_1(H)^{r-2}\rangle < \frac{\langle c_1(M) \cdot c_1(H)^{r-1}\rangle^2}{\langle c_1(H)^r\rangle}.$$

Assertion (ii) follows immediately from (i), by the argument of 7.3. To prove (i), let Y be a closed integral curve in X; we wish to prove that $\langle c_1(L) \cdot Y\rangle = 0$. Embed X in $\mathbf{P}^N_k$ by means of $H^{\otimes n}$, with $n > 0$. Let $I = \bigoplus I_\nu$ be the homogeneous ideal of Y, and let ν_0 be an integer such that I_{ν_0} generates I_ν for $\nu \geq \nu_0$. Then the hypersurfaces Z of degree ν_0 containing Y have no common points outside Y. By induction on $r = \dim X$ and Bertini's theorem, one finds $r-2$ hypersurfaces $Z_1, \ldots, Z_{r-2}$ of degree ν_0 containing Y such that $X' = Z_1 \cap \cdots \cap Z_{r-2} \cap X$ is irreducible. Since $\langle c_1(L) \cdot Y\rangle = \langle c_1(L|_{X'}) \cdot Y\rangle$, it is enough to prove that $L|_{X'}$ is numerically equivalent to 0 on X'. By the projection formula, $\langle c_1(L|_{X'}) \cdot c_1(H|_{X'})\rangle = 0$ and $\langle c_1(L|_{X'})^2\rangle \geq 0$, so the assertion follows from 7.1.

Corollary 7.5. *Let k be an algebraically closed field, let Y be a projective k-scheme equipped with an ample sheaf H, and let X be the zero scheme of a section φ of H. Suppose that, for every irreducible component Y^i of Y, the irreducible components of $X \cap Y^i = V(\varphi|_{Y^i})$ have dimension at least 2. Then the morphism $A^1_{\mathrm{num}}(Y) \longrightarrow A^1_{\mathrm{num}}(X)$ is injective.*

Indeed, one may plainly suppose that Y is integral, and the assertion follows from 7.4(i) by the projection formula.

Bibliography

[1] S. Abhyankar, "Resolution of singularities of arithmetical surfaces," in *Arithmetical Algebraic Geometry*, Purdue University, Harper & Row, New York, 1965.

[2] D. Mumford, "Pathologies III," *Amer. J. Math.*, 1967, pp. 94–103. *Amer. J. Math.*, 1967, pp. 94–103.

[3] A. Grothendieck and J. Dieudonné, *EGA, Éléments de Géométrie Algébrique*, Publ. Math. Inst. des Hautes Études Sci., 1960.

[4] A. Grothendieck, *FGA, Fondements de la géométrie algébrique*, extracts from the Séminaire Bourbaki 1957–1962.

[5] A. Grothendieck et al., *SGA, Séminaire de géométrie algébrique*, 1960–66, I.H.E.S.

[6] O. Jussila, "Formalisme des intersections sur les schémas algébriques propres," Exposé X in this seminar.

[7] M. Raynaud, "Un théorème de représentabilité relative sur le foncteur de Picard," Exposé XII in this seminar.

Exposé XIV. Open problems in intersection theory

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In the present exposé we review some of the problems connected with questions encountered in this seminar which deserve to be studied. We leave aside the problems connected with Serre's *local intersection theory* [14], which we have not had to use in the seminar, and refer in particular to *loc. cit.* for the two fundamental conjectures which remain open in that direction (*loc. cit.*, Chapter V, §4). We also pass over the problems, very important for arithmetic applications, connected with the *structure of groups of cycle classes*, especially on a smooth projective algebraic scheme. For discussion of these problems, we refer to S. Kleiman, "Algebraic cycles and the Weil conjectures," in *Dix exposés sur la cohomologie des schémas* (Masson–North Holland Publishing Co.). In no. 4 of the present exposé we also give some supplements to the seminar, concerning in particular the *Chow ring*, which had not found a place in the preceding exposés.

1. The operations Λ^i in the derived category $D(X)$[26]

Let X be a locally ringed topos, and let $K^\bullet(X)$ be the ring of classes of perfect complexes on X (Exposé IV). One would like to define on $K^\bullet(X)$ a λ-ring structure (Exposé V), functorial in X, such that the natural homomorphism

$$K^\circ_{\mathrm{naive}}(X) \longrightarrow K^\bullet(X)$$

(where the first member is the ring of classes of locally free modules on X, equipped with the λ-structure defined in Exposé V) is a λ-homomorphism. For this purpose, one would have to define functors, in general nonadditive, of "exterior power"

$$\Lambda^i : D^-(X) \longrightarrow D^-(X),$$

[26] Cf. the note on p. 23.

transforming pseudo-coherent (respectively perfect) complexes into pseudo-coherent (respectively perfect) complexes and having the formal properties with respect to exact triangles needed to define a λ-structure on $K^\bullet(X)$ by the formula

$$\lambda^i(\mathrm{cl}(L^\bullet)) = \mathrm{cl}(\Lambda^i L^\bullet). \tag{1.1}$$

When X is of characteristic zero, that is, when its structure sheaf is a sheaf of $\mathbb{Q}$-algebras, the problem is solved simply by putting

$$\Lambda^i L^\bullet = \left({}^{\otimes i} L^\bullet\right)^{\mathrm{alt}}, \tag{1.2}$$

where the i-th tensor power, in the sense of derived categories, of $L^\bullet$ is regarded as an object of $D^-(\mathcal{O}_i)$, with $\mathcal{O}_i = \mathcal{O}_X[\mathfrak{S}_i]$ the algebra of the symmetric group $\mathfrak{S}_i$ over $\mathcal{O}_X$, and where the superscript alt denotes passage to the "alternating part under the action of $\mathfrak{S}_i$," an operation which makes sense because of the characteristic-zero hypothesis. A formula of this type was already used in [8] (cited below as [RRR]) in the first proof of the Riemann–Roch theorem.

Without a characteristic hypothesis, when $L^\bullet$ is zero in degrees > 0, Dold–Puppe [6] gives a canonical homotopy from $L^\bullet$ to the complex associated with a semisimplicial module $L'^\bullet$ canonically attached to $L^\bullet$. For $L^\bullet$ flat, say, a fairly natural candidate for $\Lambda^i L^\bullet$ would be the complex associated with the semisimplicial module $\Lambda^i L'^\bullet$, whose degree-n component is $\Lambda^i L'_n$. But to extend the definition to complexes of $D^-(X)$ which are no longer assumed to be zero in degrees > 0, one would have to study the behavior of the preceding operation with respect to translation $L^\bullet \mapsto L^\bullet[1]$, or *suspension* in Dold–Puppe terminology [6]. Unfortunately, there does not seem to be an isomorphism $\Lambda^i(L^\bullet[1]) \simeq \Lambda^i(L^\bullet)[i]$ which would make the desired extension trivial.

2. The Riemann–Roch formula without projectivity hypotheses

2.1. Once one has a natural λ-structure on the $K^\bullet(X)$, it is possible to formulate the Riemann–Roch theorem, in the form contemplated in Exposé VIII, for any proper complete-intersection morphism $f : X \to Y$, and even, more generally, for a relative scheme X over a locally ringed topos Y, with X still subject to the corresponding relative conditions. The Todd class of f is defined by means of the class in $K^\bullet(X)$ of the relative cotangent complex T_f constructed in Exposé 0, 4.4; that complex is perfect here by the local hypothesis on f. It seems clear that a proof of the Riemann–Roch formula in this general case will require essentially new ideas, and it is possible that these same ideas could provide the key

to a proof of the analogous formula in the complex analytic (or rigid-analytic!) case. Recall that, in the complex analytic case, only the case of a locally free sheaf F on a compact nonsingular variety X, over a one-point Y, is currently known, by the Atiyah–Singer index formula; the case in which F is only assumed coherent still seems to escape presently known techniques.

2.2. As an essential ingredient of the Riemann–Roch formula, recall also the *finiteness problem* raised in Exposé III: if $f : X \to Y$ is a proper morphism and $L^\bullet$ is a complex on X which is pseudo-coherent relative to Y, is $Rf_*(L^\bullet)$ pseudo-coherent? A general proof of this conjecture will probably use algebraic ideas which will yield a proof of Grauert's "finiteness theorem" [7] under appreciably more general conditions than those of *loc. cit.*, working from the outset over topologically ringed spaces (or topoi) in $\mathbb{C}$-algebras Y more general than analytic spaces. This would include the cases in which Y is a locally compact space with the sheaf of continuous complex-valued functions, a C^m-manifold with the sheaf of complex-valued C^m-functions, a real analytic space, and so forth.

2.3. Finally, in these questions the properness hypothesis on f should be replaced by the hypothesis that the supports of the cohomology sheaves $H^i(L^\bullet)$ of the complexes under consideration on X are proper over Y.

3. A Riemann–Roch formula "without denominators" for an immersion

3.1. Let $f : Y \to X$ be a regular closed immersion (Exposé VII), with conormal sheaf $\underline{N}$. For simplicity, suppose that X admits an ample invertible sheaf. The Riemann–Roch formula for $y \in K^\bullet(Y)$,

$$\mathrm{ch}_X(f_*(y)) = f_*(\mathrm{ch}_Y(y)\,\mathrm{Todd}(-\check{\underline{N}})),$$

together with the general formula (Exposé VII)

$$f_*(y')f_*(y'') = f_*(y'y''c_d(\underline{N})), \qquad y', y'' \in \mathrm{Gr}^\bullet(Y)_{\mathbb{Q}},$$

allows one, assuming f has codimension d, to express the Chern classes of $f_*(y)$ in $\mathrm{Gr}^\bullet(X)_{\mathbb{Q}}$ in the form

$$c_i(f_*(y)) = f_*(P_i((c_\alpha(y))_\alpha, (c_\beta(\underline{N}))_\beta)), \tag{3.1}$$

where P_i is a universal polynomial in the $c_\alpha(y)$ and the $c_\beta(\underline{N})$, a polynomial which a priori has rational coefficients. In fact, one finds in [RRR] that these polynomials have *integer coefficients*. This raises the question of the validity of a Riemann–Roch formula "without denominators."

In fact, such a formula for $c_i(f_*(y))$ cannot in general be sought in $\mathrm{Gr}^\bullet(X)$ itself, since the homomorphism $f_* : K^\bullet(Y) \longrightarrow K^\bullet(X)$ is not itself compatible with the λ-filtrations, modulo a shift by d (4.6), but is compatible only up to torsion. Thus it is not possible to define a homomorphism $f_* : \mathrm{Gr}^\bullet(Y) \longrightarrow \mathrm{Gr}^\bullet(X)$ inducing the natural homomorphism $f_* : \mathrm{Gr}^\bullet(Y)_{\mathbb{Q}} \longrightarrow \mathrm{Gr}^\bullet(X)_{\mathbb{Q}}$ and making it possible to interpret (3.1) as a formula in $\mathrm{Gr}^\bullet(X)$ rather than only in $\mathrm{Gr}^\bullet(X)_{\mathbb{Q}}$.

When X is regular, however, so that $K^\bullet(X) \simeq K_\bullet(X)$ (IV 2.5), one can equip $K^\bullet(X)$ with the *topological filtration* by codimension of supports, inherited from the corresponding filtration of $K_\bullet(X)$. It is plausible that this filtration is compatible with the ring structure. This is proved at least in [RRR] when X is smooth over a field k, using the theory of the Chow ring [4]. When this is so, the associated graded

$$\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$$

is a *graded ring*; and, if Y is also regular, one has precisely what is needed to obtain a natural homomorphism of degree d of graded groups

$$f_* = f_*^{\mathrm{top}} : \mathrm{Gr}^\bullet_{\mathrm{top}}(Y) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X).$$

Since the λ-filtration of $K^\bullet(X)$ is finer than the topological filtration by codimension just considered (X 1.3.3), one obtains a canonical homomorphism of graded groups (and even of graded rings),

$$\mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X),$$

which is, moreover, an isomorphism modulo torsion (Exposé VIII). This makes it possible, as in [RRR], to consider a theory of Chern classes on X with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and likewise on Y, and thus to interpret (3.1) as a formula in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$. In this form, the formula is proved in [RRR] when X is of *characteristic zero*, at least if X is smooth over a field (an assumption used only to ensure that the topological filtration of $K^\bullet(X)$ is compatible with the ring structure).

The characteristic-zero hypothesis is used essentially to have formulas of the type (1.1), (1.2), and essentially the same proof should apply without a characteristic hypothesis once the question raised in no. 1 has been answered affirmatively.

3.2. When X is smooth over a field, one also has the Chow ring

$$A(X),$$

and a theory of Chern classes on X with values in $A(X)$ [9], finer than the theory with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ just introduced. The latter Chern classes are the images of the Chow-style Chern classes under the canonical surjective homomorphism (cf. 4.1)

$$\varphi : A(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X).$$

This homomorphism is an isomorphism modulo torsion, but not an isomorphism in general (4.7). That said, if X and Y are smooth, the inclusion $f : Y \to X$ still defines a homomorphism of degree d of graded groups

$$f_* : A(Y) \longrightarrow A(X),$$

which makes it possible to interpret (3.1) as a formula in $A^i(X)$. This formula reduces to $0 = 0$ for $i < d$ and is therefore of interest only for $i > d$. Even for $i = d = 2$, this formula has not been proved at present (cf. 4.3), nor has the less precise formula deduced from it by replacing rational equivalence of algebraic cycles by algebraic equivalence. Unlike the analogous formula in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, there does not seem to be any convincing reason to suppose that the formula without denominators should be true in $A(X)$. A good way to test it, given our almost total ignorance of the nature of the groups $A^i(X)$[27] for $i > 2$, seems to be to test the formulas that can be deduced from it in groups $A^1(Z) = \mathrm{Pic}(Z)$, for example by applying to both sides of (3.1) $g_* : A(X) \longrightarrow A(Z)$, where $g : X \to Z$ is a proper morphism to a smooth quasi-projective Z of dimension $\dim X - i + 1$.

3.3. Finally, to give a meaning to (3.1), one may also use the cohomological theory of Chern classes (SGA 5 VII), whenever one has *Gysin homomorphisms*

$$f_* : H^p(Y, \mu_n^{\otimes i}) \longrightarrow H^{p+2d}(X, \mu_n^{\otimes(i+d)}).$$

This holds at least when Y and X are both smooth over a scheme S, by the theorem of relative cohomological purity (SGA 4 XVI 3.7); conjecturally, it also holds when Y and X are both regular, by the theorem of absolute cohomological purity, proved at least for excellent X of characteristic zero (SGA 4 XIX 3.2).

When X and Y are smooth over the field of complex numbers, one may also use the integral Chern class on the locally compact space $X(\mathbb{C})$ of complex points of X. Formula (3.1) was proved in this case by Atiyah and Hirzebruch [1], who more generally treat embeddings of complex analytic varieties (working with naive K'). It follows, by the Lefschetz principle and the general results of SGA 4, that the ℓ-adic formula (3.1) holds whenever Y and X are smooth over an algebraically closed field of characteristic zero. This makes the cohomological formula (3.1) extremely plausible. Nevertheless, it should be noted that, at present, it has not been proved even when X and Y are smooth over a field k of characteristic zero, if k is not algebraically closed.

3.4. One may also work with the cohomological theory of Chern classes, in the de Rham or Hodge style [13]. The question of the validity of (3.1) in this setting arises only when X is not of characteristic zero, since one knows in any case (no. 6) that the formula in question is true modulo torsion. For smooth quasi-projective schemes over a field k, the de Rham or Hodge

[27] The formula, as well as the variant mentioned in 3.3, has since been proved by P. Jouanolou (Nov. 1969). Cf.

version of (3.1) would moreover follow from the version using cycle classes modulo algebraic equivalence. Thus the de Rham or Hodge cohomological version provides another accessible way to test the validity of the cycle-class version of (3.1).

3.5. The proof [1] of the transcendental cohomological version of (3.1) is essentially transcendental in nature and uses tubular neighborhoods of Y in X. This suggests that a proof of a formula of type (3.1) may be tied to the systematic consideration of $K^\bullet$-groups and cohomology groups with support in Y, which might provide an adequate technical means of *localization around* Y.

4. Relations between $K^\bullet(X)$ and the Chow ring $A(X)$

4.1. Suppose for simplicity that X is a smooth quasi-projective scheme over a field k (see no. 8 for a more general case), so that the Chow ring $A(X)$ of cycle classes modulo rational equivalence is available [4]. Two variants of this ring, defined under appreciably more general conditions, are $\mathrm{Gr}^\bullet(X)$, the graded ring associated with the λ-filtration of $K^\bullet(X)$, and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, the graded ring associated with the filtration of $K^\bullet(X) \simeq K_\bullet(X)$ by codimension of supports. There are theories of Chern classes with values in each of these three rings (see [9] for the case of $A(X)$, and this seminar for $\mathrm{Gr}^\bullet(X)$, which maps into $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$). There is also an ℓ-adic cohomological theory (SGA 5 VII). Thus, when ℓ is prime to the characteristic of k,

$$c_i(E) \in H^{2i}(X, \mathbf{Z}_\ell(i)),$$

where E is a vector bundle on X, or more generally any element of $K^\bullet(X)$. These theories are related by the following canonical homomorphisms, which are compatible with the multiplicative structures:

$$\begin{tikzcd} H^{2*}(X, \mathbf{Z}_\ell(*)) & \\ A(X) \arrow[u, "\gamma"] \arrow[r, "\varphi"] & \mathrm{Gr}^\bullet_{\mathrm{top}}(X) \\ & \mathrm{Gr}^\bullet(X) \arrow[u, "j"'] \end{tikzcd} \tag{4.1}$$

where

$$H^{2*}(X, \mathbf{Z}_\ell(*)) = \bigoplus_i H^{2i}(X, \mathbf{Z}_\ell(i)).$$

The homomorphism j comes from the fact that the λ-filtration of $K^\bullet(X)$ is finer than the topological filtration by codimension of support (Expose X), and it is an isomorphism modulo torsion (Expose VII 4.11). The homomorphism φ, defined for example in [RRR], associates to the class of the codimension-i cycle carried by a closed subscheme $Y \subset X$ the element $\mathrm{cl}^i(\mathcal{O}_Y)$ of

$$\mathrm{Filt}^i_{\mathrm{top}} K^\bullet(X) / \mathrm{Filt}^{i+1}_{\mathrm{top}} K^\bullet(X).$$

It is surjective, and we shall see in 4.2 that it is an isomorphism modulo torsion. Hence, modulo torsion, the three rings $A(X)$, $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and $\mathrm{Gr}^\bullet(X)$ are canonically isomorphic; consequently, the Riemann–Roch formulas expressed using any one of these rings after tensoring with $\mathbb{Q}$ are equivalent.

The graded ring $\mathrm{Gr}^\bullet(X)$, used in this seminar chiefly as a substitute for the Chow ring, has the considerable advantage of being defined for every locally ringed topos, and in particular for every scheme, whether regular or not. Its disadvantage (4.6) with respect to $A(X)$ and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ is that it is not covariant as it stands for proper morphisms, even between smooth quasi-projective schemes over a field; it acquires that covariance only after tensoring with $\mathbb{Q}$.

Another advantage of $A(X)$ over its competitors $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ and $\mathrm{Gr}^\bullet(X)$ is the homomorphism

$$A(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*)),$$

which permits cohomological Chern classes to be interpreted as images of Chern classes with values in $A(X)$. In general this homomorphism does not factor through $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ (4.7), and it is at least doubtful that there is in general a homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*))$$

carrying Chern classes to Chern classes (see no. 5). These difficulties disappear after tensoring with $\mathbb{Q}$, and one obtains a homomorphism

$$\mathrm{Gr}^\bullet_{\mathrm{top}}(X) \longrightarrow H^{2*}(X, \mathbf{Q}_\ell(*)) = H^{2*}(X, \mathbf{Z}_\ell(*)) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell \tag{4.2}$$

that carries Chern classes to Chern classes.

4.2. To show that the surjective homomorphism φ in (4.1) is an isomorphism modulo torsion, one uses the Chern homomorphism obtained from the theory of Chern classes [9],

$$K^\bullet(X) \longrightarrow \mathrm{Ch}(A(X)),$$

where the target is the Chern ring of $A(X)$ as defined in Expose V. This homomorphism is compatible with the topological filtration of $K^\bullet(X)$ and the natural filtration of $\mathrm{Ch}(A(X))$, and therefore induces a homomorphism of graded groups

$$\psi : \mathrm{Gr}^\bullet_{\mathrm{top}}(X) \longrightarrow A(X). \tag{4.3}$$

One proves the formulas

$$\begin{cases} \psi\varphi(x) = (-1)^{i-1}(i-1)!\, x \qquad (\text{mod torsion}), \quad x \in A^i(X), \\ \varphi\psi(x) = (-1)^{i-1}(i-1)!\, x \qquad (\text{mod torsion}), \quad x \in \mathrm{Gr}^i_{\mathrm{top}}(X). \end{cases} \tag{4.4}$$

The first formula implies the second because φ is surjective. These formulas show that φ and ψ are isomorphisms modulo torsion and, up to the factor $(-1)^{i-1}(i-1)!$ in degree i, are inverse to one another.

To prove the first formula in (4.4), one may suppose that x is represented by an irreducible closed subscheme Y of X of codimension i. The formula is then equivalent to

$$c_i(\mathcal{O}_Y) = (-1)^{i-1}(i-1)!\ \mathrm{cl}(Y) \qquad (\text{mod torsion}). \tag{4.5}$$

On the left, $\mathcal{O}_Y$ is regarded as a coherent sheaf on X and hence defines an element of $K^\bullet(X)$; on the right, $\mathrm{cl}(Y)$ denotes the class of Y in $A^i(X)$. Since $A^i(X)$ is unchanged upon deleting from X a closed subset of codimension $> i$, one reduces to the case where Y is regular, and therefore smooth over the base field k if k is perfect. In that case, (4.5) is precisely the Riemann–Roch formula for the inclusion $Y \to X$ and the element $1 \in K^\bullet(Y)$, written using Chern classes with values in the Chow ring $A(X)$. Under the present hypotheses, this formula is proved in [2] by a method very close to the one used in this seminar for the analogous formula with values in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$.

Thus, at least when k is perfect, (4.4) follows, and so does the fact that φ is an isomorphism modulo torsion. This in turn implies, a posteriori, that Riemann–Roch with denominators for a proper morphism of smooth algebraic schemes over k is essentially the same whether it is written using $A(X)$, as in [2], or using $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, as in this seminar. The case of an arbitrary k reduces easily to the algebraically closed case: for a finite extension k'/k of degree n, the kernel of the natural map

$$A(X) \xrightarrow{f_X^*} A(X \otimes_k k')$$

is annihilated by n. Indeed, the trace homomorphism

$$f_{X*} : A(X \otimes_k k') \longrightarrow A(X)$$

satisfies $f_{X*} f_X^* = n \ \mathrm{id}$.[28]

4.3. It seems quite plausible that formulas (4.4) hold exactly as stated, and not merely modulo torsion; equivalently, that formula (4.5) holds not merely modulo torsion. This question is a very special case of the question whether the denominator-free Riemann–Roch formula (3.1) is valid in $A(X)$ for an immersion $Y \to X$, namely the case $i = d$. This special case may reasonably be expected to hold. Indeed, we already noted in no. 3 that, when k has characteristic zero, the denominator-free Riemann–Roch formula is valid at least in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$. Consequently the two sides of (4.5) have the same image in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$; equivalently, the second formula in (4.4) holds. This result will extend automatically to arbitrary characteristic once

the question raised in no. 1 has been satisfactorily resolved. As evidence in support of formula (4.5) in $A(X)$ itself, let us note that it is clear a priori that the restriction of the left-hand side to $X - Y$ is zero. Thus, since Y is assumed irreducible, the left-hand side must a priori be an integral multiple of $\mathrm{cl}(Y)$:

$$c_i(Y) = n(Y)\,\mathrm{cl}(Y).$$

This already proves the exact formula (4.5) whenever $\mathrm{cl}(Y)$ is not a torsion element of $A^i(X)$. It is immediate that this is the case if X is projective (and not merely quasi-projective) over the field k: indeed, if $\xi \in A^1(X)$ is a polarization of X, then $\deg(Y \cdot \xi^i) > 0$. This proves the desired formula in that case. It follows that the formula is still valid whenever X is isomorphic to an open subscheme of a smooth projective scheme over k, as will be the case when an adequate solution of the problem of resolution of singularities is available. In particular, this holds when k has characteristic zero, by Hironaka [11].

The validity of the exact formula (4.5), and hence of (4.4), would imply that the kernel of

$$\varphi^i : A^i(X) \longrightarrow \mathrm{Gr}^i_{\mathrm{top}}(X)$$

is annihilated by $(i-1)!$, and in particular *that* $\varphi^2 : A^2(X) \to \mathrm{Gr}^2_{\mathrm{top}}(X)$ *is an isomorphism.* (That φ^1 is an isomorphism already follows immediately from X 5.3.2.) Unless I am mistaken, the assertion that φ^2 is an isomorphism has not yet been proved, even in the special case $\dim X = 3$. It was therefore in any event premature for the editor to assert the validity of (4.4) and (4.5) in [9, no. 4, 3°].

4.4. At present, the principal advantage of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ (or of $\mathrm{Gr}^\bullet(X)$) over the Chow ring $A(X)$ is that it is better suited to explicit calculations of certain intersection formulas between cycles in cases of excess intersection, that is, intersections of excessively large dimension. Indeed, the definition of the intersection class by means of Chow's lemma, which allows a cycle to be moved sufficiently within its class, is poorly suited to calculation. A striking example in this regard is the following fundamental formula [Exposé VII 2.7], for a closed immersion $f : Y \to X$ of smooth quasi-projective schemes over k:

$$f^* f_*(y) = y\, c_d(\underline{N}), \tag{4.6}$$

[28] Thus the formula is established, thanks to P. Jouanolou, for X smooth and quasi-projective over a field (see the note on p. 5).

a formula valid in $\mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$ for $y \in \mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$, where $\underline{N}$ is the conormal sheaf

of Y in X, assumed everywhere to have rank d. In the cited passage, this formula follows from a very simple computation of the sheaves $\underline{\mathrm{Tor}}$. By 4.2, the preceding formula implies the same formula in $A(Y)$, but only modulo torsion. It is quite remarkable that formula (4.6) in $A(Y)$ itself has not yet been proved,[29] nor even the special case obtained by taking $y = 1$, which gives the self-intersection of Y:

$$f^*(\mathrm{cl}(Y)) = c_d(\underline{N}), \tag{4.7}$$

even when k is the field of complex numbers.

An analogous problem arises when one attempts to compute the Chow ring of the variety X' obtained by blowing up X along Y, by imitating the method used to compute $\mathrm{Gr}^\bullet(X')_{\mathbb{Q}}$ [Exposé VII 4.8], which also works (without tensoring with $\mathbb{Q}$) for the computation of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X')$. With the notation of the cited passage, the key formula for this computation is

$$f^*(i_*(y)) = j_*(g^*(y)c_{d-1}(\check{\underline{F}})). \tag{4.8}$$

At present, this formula has been established in $\mathrm{Gr}^\bullet_{\mathrm{top}}(X')$ (for $y \in \mathrm{Gr}^\bullet_{\mathrm{top}}(Y)$) by a computation of $\underline{\mathrm{Tor}}$ (Exposé VII 3.4), while its validity in $A(X')$ (for $y \in A(Y)$) remains to be proved.

Let us note that even the ℓ-adic analogues of formulas (4.6), (4.7), and (4.8) have not yet been proved; when y is the cohomology class of an algebraic cycle, formulas (4.6) and (4.8) are in any event true modulo torsion.

4.5. In this section and the following ones, we shall give examples for which, in diagram (4.1), the homomorphism j, respectively the homomorphism φ, is not an isomorphism. For a given X, it is plainly equivalent to say that

$$j : \mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X)$$

is an isomorphism, a monomorphism, or an epimorphism. Moreover, the image of j is the subring of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ generated by the Chern classes of locally free sheaves on X. If $K^\bullet(X)$ is generated as a $\mathbb{Z}$-algebra by the classes of invertible sheaves on X, one sees at once that the subring in question is also the subring generated by $\mathrm{Gr}^1_{\mathrm{top}}(X)$, the degree-one component of $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$.

Now take for X a flag variety

$$X = G/B,$$

where G is a simply connected semisimple algebraic group over the algebraically closed field k, and B is a Borel subgroup [5]. When $k = \mathbb{C}$, the Bruhat cell decomposition of X [5, Exposé 13] implies that the canonical homomorphism

$$K^\bullet(X) \longrightarrow K(X(\mathbb{C})) \tag{*}$$

is an isomorphism, as is the canonical homomorphism

$$A(X) \longrightarrow H^*(X(\mathbb{C}), \mathbb{Z}), \tag{**}$$

where $X(\mathbb{C})$ is the compact space of complex points of X, endowed with its usual topology induced by that of $\mathbb{C}$. On the other hand, L. Hodgkin [12] proved that $K(X(\mathbb{C}))$ is generated by the classes of invertible sheaves on X. The same is therefore true of $K^\bullet(X)$ itself; moreover, it is not difficult to deduce, by a specialization argument using semisimple groups split over $\mathbb{Z}$ (SGA 3), that the result remains valid over every algebraically closed field k. The cell decomposition of X (or, when $k = \mathbb{C}$, the bijectivity of $(**)$) also readily implies that $A(X)$ is torsion-free. Hence, by 4.2, the homomorphism

$$\varphi : A(X) \longrightarrow \mathrm{Gr}^\bullet_{\mathrm{top}}(X)$$

[29] Cf. the note on p. 23.

is an isomorphism. It follows that j is an isomorphism if and only if $A(X)$ is generated by its elements of degree 1, or, when $k = \mathbb{C}$, if and only if the integral cohomology of $X(\mathbb{C})$ is generated by its elements of degree 2, i.e. if the classifying space $B_{G(\mathbb{C})}$ is without torsion. It is known [15, p. 217, Theorem B, (1) $\Leftrightarrow$ (2), and 2.5] that this holds if and only if G is without torsion, equivalently if G is isomorphic to a product of groups $\mathrm{SL}(n)$ and $\mathrm{Sp}(m)$. Thus it suffices to take for G a group $\mathrm{SO}(n)$ with $n \geq 7$, or a simple exceptional group, to obtain an example in which j is not an isomorphism.

4.6. In this example there can be no doubt that the good ring is the torsion-free ring

$$A(X) \simeq \mathrm{Gr}^\bullet_{\mathrm{top}}(X),$$

whose basis consists of the cells in the Bruhat decomposition and which, when $k = \mathbb{C}$, is also isomorphic to the integral cohomology of $X(\mathbb{C})$. By contrast, $\mathrm{Gr}^\bullet(X)$ appears as a ring having torsion of a rather pathological nature. This example therefore confirms the view that the invariant $\mathrm{Gr}^\bullet(X)$ is hardly interesting

except modulo torsion, i.e. after tensoring with $\mathbb{Q}$. At the same time, this example shows that $\mathrm{Gr}^\bullet(X)$ is covariant with respect to proper morphisms only after tensoring with $\mathbb{Q}$, even if one restricts to closed immersions of smooth quasi-projective schemes. Indeed, if X is such that j is not an isomorphism, there is an integral closed subscheme Y of X such that

$$\mathrm{cl}^\bullet(Y) \in K^\bullet(X)$$

does not lie in $\mathrm{Filt}^i(X)$, where $i = \mathrm{codim}(Y, X)$. We may choose Y of minimal dimension, so that

$$\mathrm{Filt}^{i+1}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X).$$

If Z is the singular locus of Y, and $X' = X \setminus Z$, $Y' = Y \setminus Z$, it follows that

$$\mathrm{cl}^\bullet(Y') \in K^\bullet(X')$$

does not lie in $\mathrm{Filt}^i(X')$. Here one uses the fact that $K^\bullet(X) \to K^\bullet(X')$ is surjective and compatible with the augmentations, and that its kernel is

$$\mathrm{Im}(K(Z) \longrightarrow K(X)) \subset \mathrm{Filt}^{i+1}(X) = \mathrm{Filt}^{i+1}_{\mathrm{top}}(X).$$

Now consider the inclusion

$$f : Y' \longrightarrow X',$$

which is a closed immersion of codimension i between smooth quasi-projective schemes over k, *such that*

$$f_* : K^\bullet(Y') \longrightarrow K^\bullet(X')$$

is not compatible with the λ-filtrations, even up to a shift by i, since $f_*(1) \notin \mathrm{Filt}^i(X')$.

4.7. Retaining the notation of 4.5, note that Exposé IX shows that Hodgkin's theorem quoted above is equivalent to the isomorphism

$$K^\bullet(G) \simeq \mathbb{Z}, \qquad \text{hence} \qquad \mathrm{Gr}^\bullet_{\mathrm{top}}(G) \simeq \mathbb{Z}.$$

The homomorphism $K^\bullet(G/B) \to K^\bullet(G)$ is a priori surjective, and its kernel is the ideal generated by the elements $\mathrm{cl}^\bullet(L) - 1$, where L ranges over the invertible sheaves on $X = G/B$ associated with representations $B \to \mathbb{G}_\mathrm{m}$ (which in fact yield all invertible sheaves on X [5, Exposé 15]). An analogous argument shows that

$$A(G/B) \longrightarrow A(G)$$

is also surjective, and that its kernel is the ideal generated by the elements of $A^1(G/B)$ defined by representations $B \to \mathbb{G}_{\mathrm{m}}$, hence by all of $A^1(G/B)$.[30] Thus $A(G)$ is reduced to $A^0(X) = \mathbb{Z}$ if and only if G is "without torsion", i.e. if and only if G is isomorphic to a product of groups $\mathrm{Sl}(n)$ and $\mathrm{Sp}(m)$. Thus, in this case and only in this case, the homomorphism

$$\varphi : A(G) \longrightarrow \mathrm{Gr}^{\bullet}_{\mathrm{top}}(G)$$

is an isomorphism. Consequently, if G has torsion, the underlying scheme provides an example of a non-bijective φ.

4.8. One likewise finds an example in which the homomorphism γ of (4.1)

$$\gamma : A(G) \longrightarrow H^{2*}(G, \mathbf{Z}_\ell(*)) \tag{4.9}$$

does not factor through $\mathrm{Gr}^{\bullet}_{\mathrm{top}}(G)$, or, what amounts to the same thing here, an example in which this homomorphism is nonzero in degrees > 0. Restricting (as one may, thanks to suitable specialization theorems) to the case $k = \mathbb{C}$, one is reduced to studying the homomorphism

$$A(G) \longrightarrow H^*(G(\mathbb{C}), \mathbb{Z}), \tag{4.10}$$

which is moreover obtained, by passage to the quotient, from the canonical homomorphism

$$A(G/B) \simeq H^*(G/B(\mathbb{C}), \mathbb{Z}) \longrightarrow H^*(G(\mathbb{C}), \mathbb{Z}).$$

It is tempting to conjecture that (4.10) is in fact injective, since it is also known that the torsion primes for $G(\mathbb{C})$ are exactly those dividing the order of the finite group

$$A^+(G) = \coprod_{i>0} A^i(G).$$

If this were so, whenever G had torsion, (4.10) would not factor through φ; hence, for a suitable ℓ (more precisely, for ℓ dividing the order of $A^+(G)$, i.e. for ℓ a torsion prime of G), the same would hold for (4.9). We can at least assert that, for the group $G = G_2$ over $\mathbb{C}$, (4.10) is nonzero in degree > 0: more precisely, there exists an element of $A^3(G)$ whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ has order 2. It follows that (4.9) does not factor through φ when $\ell = 2$. The fact used about G_2, which amounts to saying that there exists in $H^6(G/B(\mathbb{C}), \mathbb{Z})$ an element whose image in $H^6(G(\mathbb{C}), \mathbb{Z})$ is nonzero, is found in [3], via the familiar dictionary between complex semisimple algebraic groups and compact Lie groups; it was kindly communicated to me by J.-P. Serre.

5. Relations between $\mathrm{Gr}^{\bullet}(X)$ and $H^{2*}(X, \mathbf{Z}_\ell(*))$

5.1. The example given in 4.8 does not exclude the possibility that, for an arbitrary scheme X, one might find a ring homomorphism

$$g' : \mathrm{Gr}^{\bullet}(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*)),$$

such that, for every locally free module E on X, g^i sends the Chern class $c_i(E)$ of Exp. V to the cohomological Chern class of SGA 5 VII. Of course, ℓ here denotes a prime distinct from the residual characteristics of X. Since the ring $\mathrm{Gr}^{\bullet}(X)$ is generated by the $c^i(E)$, one sees in any case that g' is uniquely determined by the requirement that it be compatible with Chern classes.

5.2. Using the Chern homomorphism

$$K^{\bullet}(X) \longrightarrow \mathrm{Ch}(H^{2*}(X, \mathbf{Z}_\ell(*))),$$

[30] Cf. IX 3.4, 2.2.

one obtains, by passing to the associated gradeds respectively for the filtration on the first member and the filtration by degree on the second, a homomorphism of graded groups (but not of rings!)

$$\psi : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Z}_\ell(*)),$$

which satisfies

$$\psi(c_i(E)) = (-1)^{i-1}(i-1)!\, c_i^\ell(E),$$

where $c_i^\ell(E)$ denotes the ℓ-adic Chern class. Dividing the components ψ^i by the indicated factors therefore gives a ring homomorphism

$$g_{\mathbb{Q}} : \mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbf{Q}_\ell(*)) = H^{2*}(X, \mathbf{Z}_\ell(*)) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell,$$

sending Chern classes to Chern classes, and such that the image of the first member is contained in that of $H^2(X, \mathbf{Z}_\ell(*))$. Consequently, the question of the existence of g' has an affirmative answer provided $H^{2*}(X, \mathbf{Z}_\ell(*))$ is replaced by its quotient by its torsion subgroup. The first doubtful case for defining the components g^i of the hypothetical homomorphism g is that of g^3; a priori one succeeds only in defining $2g^3 = \psi^3$. The answer to the question of the existence of g^3 would be negative if, for example, one could find three invertible sheaves L_1, L_2, L_3 on X satisfying

$$(L_1 - 1)(L_2 - 1)(L_3 - 1) = 0 \quad \text{in } K^\bullet(X),$$

but such that

$$c_1(L_1)c_1(L_2)c_1(L_3) \neq 0 \quad \text{in } H^6(X, \mu_2^{\otimes 3}).$$

5.3. When X is smooth and quasi-projective over a field k, one may similarly ask whether there exists a ring homomorphism

$$g : \mathrm{Gr}^\bullet(X) \longrightarrow A(X)$$

taking values in the Chow ring $A(X)$ and sending Chern classes to Chern classes. The preceding observations may be repeated in this connection; they show in particular how to construct $(-1)^{i-1}(i-1)!g^i$ directly when g^i itself is unavailable.

5.4. Let us note, to conclude this section, that essentially the same question arises if, instead of a scheme X, one considers a compact topological space (say) X. One can still consider the associated graded $\mathrm{Gr}^\bullet(X)$ of $K^\bullet(X)$ endowed with its λ-filtration, and seek a ring homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow H^{2*}(X, \mathbb{Z})$$

sending Chern classes to Chern classes.

6. Cohomological Riemann–Roch theorem and Gysin homomorphism

6.1. For simplicity, we shall confine ourselves here to schemes X admitting an ample invertible module, so that there is no need to distinguish between the naive $K^\bullet(X)$ and the other one. Let us begin with the Riemann–Roch formula "with denominators", for a proper complete-intersection morphism $f : X \longrightarrow Y$. For a prime ℓ prime to the residual characteristics of Y, one can define the ℓ-adic Chern classes of a locally free module E,

$$c_i(E) \in H^{2i}(X, \mathbf{Z}_\ell(i))$$

(SGA 5 VII), corresponding to a homomorphism

$$K^\bullet(X) \longrightarrow \mathrm{Ch}(H^{2*}(X, \mathbf{Z}_\ell(*))). \tag{6.1}$$

The question therefore arises of giving a cohomological version of the Riemann–Roch formula. The only ingredient missing in order to give meaning to the formula is a Gysin homomorphism

$$H^n(X, \mathbf{Q}_\ell(i)) \longrightarrow H^{n-2d}(Y, \mathbf{Q}_\ell(i-d)), \tag{6.2}$$

where d is the virtual relative dimension of f. Assuming this homomorphism to be defined, and taking into account the Riemann–Roch formula in the form of Exp. VIII, with values in $\mathrm{Gr}^\bullet(Y)_\mathbb{Q}$, one sees that the validity of the ℓ-adic Riemann–Roch formula for the morphism f is exactly equivalent to the commutativity of the square

$$\begin{tikzcd} \mathrm{Gr}^\bullet(X)_\mathbb{Q} \arrow[r] \arrow[d] & H^{2*}(X, \mathbf{Q}_\ell(*)) \arrow[d] \\ \mathrm{Gr}^\bullet(Y)_\mathbb{Q} \arrow[r] & H^{2*}(Y, \mathbf{Q}_\ell(*)), \end{tikzcd} \tag{6.3}$$

where the horizontal arrows are obtained from (6.1) by passage to the associated gradeds and division, so as to send Chern classes to Chern classes, while the vertical arrows are respectively the arrow of Exp. VIII and the Gysin homomorphism in cohomology (assumed given).

6.2. Note that the Gysin homomorphism is defined in all cases if Y is smooth over a field k (SGA 5 IV), using the theorem of relative cohomological purity over k. The same construction applies when Y is excellent, regular, and of characteristic zero, thanks to the theorem of absolute cohomological purity of SGA 4 XIX; and it will automatically apply to every excellent regular Y, without restriction on the characteristic, once the theorem of absolute cohomological purity has been proved in full generality. Finally, the same construction applies, thanks to the theorem of relative cohomological purity (SGA 4 XVI 3.7), when X and Y are both smooth over the same scheme S and f is an S-morphism. It is natural to conjecture that, in each of these cases in which the Gysin homomorphism is defined, diagram (6.3) is indeed commutative. The only case in which this assertion is currently verified is that in which X and Y are both smooth over a field k; it then follows from the commutativity of the analogous diagram

$$\begin{tikzcd} A(X) \arrow[r] \arrow[d] & H^{2*}(X, \mathbf{Z}_\ell(*)) \arrow[d] \\ A(Y) \arrow[r] & H^{2*}(Y, \mathbf{Z}_\ell(*)), \end{tikzcd} \tag{6.4}$$

involving the rings $A(X)$ and $A(Y)$ of cycle classes modulo rational equivalence, whose commutativity was established in SGA 5 IV.

6.3. Note that in each of the cases considered above in which one succeeds in defining a Gysin homomorphism, one in fact obtains a construction for cohomology with coefficients in $\mathbf{Z}_\ell(*)$, and not only in $\mathbf{Q}_\ell(*)$. This makes it possible, when f is an *immersion*, to pose the question of the validity of the Riemann–Roch formula "without denominators", as explained in no. 3. When X and Y are both smooth over a field k, the commutativity of (6.4) shows that it would suffice to prove the formula without denominators in the Chow ring $A(Y)$; cf. (3.3).

6.4. When no regularity hypothesis is made on Y, and even if X and Y are of finite type over $\mathbb{C}$, one no longer sees a plausible definition of a cohomological Gysin homomorphism (6.2), even for a regular immersion of codimension 1. Can one nevertheless find one with reasonable properties, in particular making square (6.3) commutative? The question is evidently connected with that of defining, for X a complete intersection of codimension i in Y, "its cohomology class"

$$\gamma(X) \in H^{2i}(Y, \mathbf{Z}_\ell(i)),$$

in such a way as to turn intersections (without excess components) into cup products. This latter question has an affirmative solution at least modulo torsion, i.e. in $H^{2*}(Y, \mathbf{Q}_\ell(*))$, by defining

$$\gamma(X) = \frac{1}{(-1)^{i-1}(i-1)!}\, c_i(\mathcal{O}_X).$$

But I do not know whether $c_i(\mathcal{O}_X)$ is divisible by $(i-1)!$ in $H^{2i}(Y, \mathbf{Z}_\ell(i))$. In particular, if Y is, say, a projective scheme of dimension 3 over the field of complex numbers and X is reduced to a (singular) point of Y, is $c_3(\mathcal{O}_X)$ divisible by 2 in $H^6(Y, \mathbb{Z}_2(3))$, and in particular is the Chern class zero in $H^6(Y, \mathbb{Z}/2\mathbb{Z})$? Let us also mention, along the same lines, the following question: let Y be a projective (singular) scheme of dimension 4, and X a closed subscheme of dimension 1 which is globally an intersection of three positive divisors D_1, D_2, D_3. Consider the product of their Chern classes modulo 2,

$$c_1(D_1)c_1(D_2)c_1(D_3) \in H^6(Y, \mathbb{Z}/2\mathbb{Z});$$

is this element of H^6 independent of the particular choice of the D_i whose intersection is X?

7. Chern classes of perfect complexes

7.1. When one poses the question of developing a cohomological Riemann–Roch formula, with or without denominators, outside the rather artificial hypothesis that an ample invertible module exists on the schemes under consideration, one is necessarily faced with the question of defining, for a perfect complex of modules $L^\bullet$ on a scheme X, the Chern classes

$$c_i(L^\bullet) \in H^{2i}(X, \mathbf{Z}_\ell(i)),$$

generalizing the Chern classes of locally free modules, SGA 5 VII. The same question arises, moreover, for general locally ringed topoi (compare [10]), as well as for perfect complexes on paracompact spaces ringed by continuous complex-valued functions, working then with ordinary integral cohomology. For a compact space, it is known that the $K^\bullet(X)$ constructed with perfect complexes is the same as the "naive" one constructed with locally free modules (Exp. IV), and in that case one is trivially reduced to the classical definition. But already for a locally compact space the problem posed seems nontrivial, since the integral cohomology of such a space is not always the inverse limit of the cohomologies of its compact subsets.

7.2. More precisely, when Y is a closed subset of X such that $L^\bullet|_{X-Y}$ is zero (in the sense of derived categories), i.e. such that the cohomology sheaves of $L^\bullet$ are supported in Y, one would like to define Chern classes *with support in* Y

$$c_i^Y(L^\bullet) \in H^{2i}_Y(X, \mathbf{Z}_\ell(i)) \quad \text{respectively} \quad H^{2i}_Y(X, \mathbb{Z}),$$

which would recover the "ordinary" (sic) Chern classes of 7.1 by means of the canonical homomorphism

$$H^*_Y(X, -) \longrightarrow H^*(X, -).$$

Even in the case where the latter reduce to the Chern classes of locally free modules (X a scheme with an ample invertible module, or a compact space), the construction of these Chern classes "with supports" has not been carried out and will probably only be possible together with that of the general Chern classes postulated in 7.1.[31]

[31] Added in October 1968. The problems raised here were solved in the framework of Hodge cohomology by L. Illusie; cf. [13].

It is probable that Chern classes with supports will have a role to play in the proof of the cohomological Riemann–Roch formula without denominators (cf. 3.5). On the other hand, they should make it possible to give an improved version of the Riemann–Roch formula (with or without denominators) by introducing families of supports into that formula.

8. Chow ring of regular schemes

The problem is to define rational equivalence of cycles and to study the resulting ring of classes (in particular its functorial properties) for a regular noetherian scheme X admitting an ample invertible module, thereby generalizing the known theory when X is smooth over a field [4]. The key technical result will again be a "moving lemma", ensuring that a cycle can be varied within its class so as to meet a given cycle in the proper dimension.

The problem under consideration is quite natural a priori, especially since we have seen (4.5 to 4.8) that the substitutes $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$ for the Chow ring $A(X)$ are hardly satisfactory when one does not confine oneself to a theory "modulo torsion". When X is obtained from a smooth projective algebraic scheme Y over a field k by taking the local scheme at the vertex of a projective cone C and removing the origin, one finds (assuming that $A(X)$ is indeed the inductive limit of the Chow rings of $U - \{O\}$, where U ranges over the open neighborhoods in C of the vertex O) that $A(X)$ is canonically isomorphic to the quotient ring

$$A(Y)/\xi A(Y),$$

where $\xi \in A^1(Y)$ is induced by the chosen projective embedding of Y. This shows that knowing the Chow ring $A(X)$ (which is of a nonclassical type) is equivalent to knowing $A(Y)/\xi A(Y)$, which might be called the "primitive part" of the Chow ring $A(Y)$ (in a sense suggested by the Lefschetz–Hodge cohomological theory of nonsingular projective algebraic varieties). This example shows that the "local Chow rings" of local rings with isolated

singularities are geometric invariants that are by no means trivial and will undoubtedly have a role to play in the "arithmetic" study of these local rings.

Of course, once one has a reasonable notion of a Chow ring for a regular noetherian scheme admitting an ample invertible module, one should generalize to this case the relations established or conjectured in no. 4 between this ring and $\mathrm{Gr}^\bullet_{\mathrm{top}}(X)$, and formulate in terms of this ring the question of the "Riemann–Roch formula without denominators" stated in no. 3.

Bibliography of Exposé XIV

[1] M. F. Atiyah and F. Hirzebruch, *The Riemann–Roch theorem for analytic embeddings*, Topology, vol. 1, pp. 151–166 (1962).

[2] A. Borel and J.-P. Serre, *Le théorème de Riemann–Roch*, Bull. Soc. Math. France, vol. 86, pp. 97–136 (1958).

[3] R. Bott and H. Samelson, Amer. Journ. of Math., vol. 80, p. 1004 (1958).

[4] C. Chevalley, *Les classes d'équivalence rationelle*, I and II, Séminaire à l'ENS 1958, Exposés 2 and 3.

[5] C. Chevalley, *Classification des groupes de Lie algébriques*, Séminaire à l'ENS 1956/58.

[6] A. Dold and D. Puppe, *Homologie nicht-additiver Funktoren. Anwendungen*, Annales de l'Inst. Fourier, t. XI, pp. 201–312 (1961).

[7] H. Grauert, *Ein Theorem der analytischen Garbentheorie*, Pub. Math. no. 5, pp. 5–64 (1960).

[8] A. Grothendieck, *Classes de faisceaux et théorème de Riemann–Roch*, mimeographed notes prepared by the Institute for Advanced Study, 1957 (the work cited as [RRR] in the present Exposé and reproduced in our seminar as an appendix to Chapter 0).

[9] A. Grothendieck, *La théorie des classes de Chern*, Bull. Soc. Math. France, vol. 86, pp. 137–154 (1958).

[10] A. Grothendieck, *Classes de Chern et représentations linéaires des groupes discrets*, in: *Dix exposés sur la cohomologie des schémas*, Masson–North Holland Pub. Cie.

[11] H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero*, Ann. of Math. 79, pp. 109–326 (1964).

[12] L. Hodgkin, thesis (forthcoming?).

[13] L. Illusie, work in preparation.

[14] J.-P. Serre, *Algèbre Locale, Multiplicités*, Lecture Notes in Math. no. 11 (1965), Springer.

[15] A. Borel, *Sous-groupes commutatifs et torsion des groupes de Lie compacts connexes*, Tohoku Math. Journal, vol. 13, pp. 216–240 (1961).

[16](*) P. Deligne, Exposé XVIII in *Théorie des topos et cohomologie étale des schémas (SGA 6)*, by M. Artin, A. Grothendieck, and J. L. Verdier.

[17](*) U. Mausin, *Correspondances, motifs et transformations monoïdales* (in Russian). Mat. Sborik, t. 77 (119), no. 4, 1968.

(*) Added in February 1969.

Note for page 1

The theory of the operations Λ^i in $D^-(X)$, and of the operations λ^i in $K^\bullet(X)$, was written down by P. Deligne in 1968. He proves that $K^\bullet(X)$ thereby becomes a pre-λ-ring and, when $\mathcal{O}_X$ is a sheaf of algebras over a field, that $K^\bullet(X)$ is even a λ-ring. It is plausible that a refinement of Deligne's method would make it possible to dispense with the restrictive hypothesis on $\mathcal{O}_X$.

Notes for page 11

(*) (April 1969) D. Mumford has pointed out to us a proof of (4.6) in the *Chow ring* (without disregarding torsion), which he obtained in a seminar at Harvard in 1959! The reader will find the proof of this beautiful result in SGA 5 VII 9. This result obviously contains the corresponding cohomological result mentioned in note (**). The cohomological result, however, is proved by J.-P. Jouanolou under more general conditions (without a quasi-projective hyperplane, and in the relative smooth case over an arbitrary base).

(**) (January 1969) J.-P. Jouanolou has just proved these formulas for every y, without disregarding torsion; cf. SGA 5 VII § 4.9. Deligne and Manin [17] had previously observed independently, and by (easy) arguments of a rather different nature, that it was enough to establish the formulas for "algebraic" y, and even for $y = 1$ in [16], in order to conclude the same relation in the general case.

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